PEDIATRICS — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations + TO lecture notes · 113 chapters (1 from TO notes) –

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Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer · "TO – verify" = TO figure that differs from standard teaching

ACCESSORY NAVICULAR

– Pediatrics · 1 / 112 –

1 Basics

Epidemiology

Incidence	normal variant in **up to 12%** of population (Orthobullets) · **2–21%** (OB explanation) · **majority asymptomatic**
Sex	more commonly **symptomatic in females**

Genes & lab markers

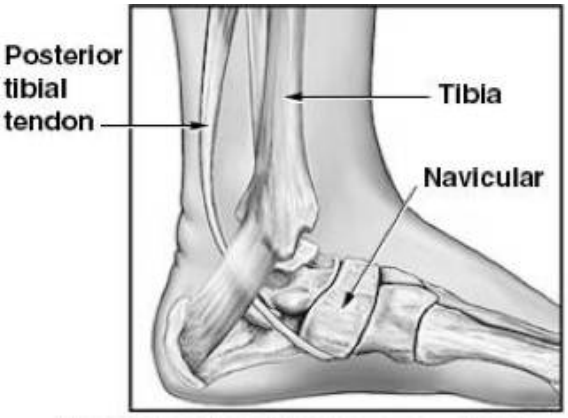
Inheritance	**autosomal dominant**

Pathoanatomy

Lesion	**plantar medial enlargement** of the navicular ✗ Wrong options: dorsal medial · dorsal lateral · plantar lateral · central
Forms	separate accessory bone **or** completely ossified extension of navicular
Origin	tuberosity develops from a **secondary ossification centre that fails to unite** in childhood · accessory navicular does **not ossify before age 8**
Normal navicular	single ossification centre · ossifies **age 3 (girls) / 5 (boys)** · fuses **~13 yr**
Associated	**flat feet (pes planus)** · **posterior tibial tendon insufficiency** (rare)

Anatomy & tibialis posterior insertion

Tibialis posterior	inserts on **medial tuberosity of navicular** · innervated by **tibial nerve** ✗ Wrong options: deep/superficial peroneal · medial/lateral plantar
PTT function	**inverts hindfoot + assists ankle plantarflexion** ✗ Wrong options: dorsiflexion = tibialis anterior · plantarflexion = gastrocsoleus · eversion = peroneus brevis



Tibialis posterior inserting on navicular tuberosity

Other insertions (distractors)	FDL → bases of 2nd–5th distal phalanges · FHL → base of great-toe distal phalanx · PB → base 5th MT · PL → plantar medial cuneiform + lateral base 1st MT

– – Active space – –

3 Presentation

Symptoms	usually **asymptomatic** · **medial arch pain**, worse with overuse/after practice · may follow a perceived **inversion injury / ankle sprain**
Pain source	repeated **microfracture at the synchondrosis** or **PTT insertional inflammation**
Exam	firm, **tender medial-plantar navicular prominence** · ± swelling · pain with **resisted inversion**
Rigid arch supports	rub on the prominence → **worsen pain** → use soft, cushioned inserts

4 Diagnosis & Classification

Key radiology findings

Radiographs	**AP, lateral, external oblique** · **best seen on external oblique view**
Finding	bony enlargement or accessory ossicle at medial navicular · no periosteal reaction / destruction / matrix ✗ Wrong options: biopsy · MRI + chest CT
MRI	only to evaluate **other pathology** · not needed for work-up ✗ Wrong options: "obtain MRI" as next step



Accessory navicular – external oblique view

View (TO)	**reverse (internal) oblique** shows accessory navicular · often mistaken for navicular fracture

Radiographic classification

Type	Description	Image
1	**Sesamoid** in substance of tibialis posterior insertion	
2	Separate accessory bone attached via **synchondrosis**	
3	**Complete bony enlargement** (cornuate navicular)	

5 Management

Symptomatic accessory navicular

- First line → **activity + shoe modification, orthotics, NSAIDs, PT**
- Failed above → **short cast immobilization**
- Failed extended nonop → **excision through synchondrosis**
  - Flatfoot is primary problem → **add MCO + PTT reattachment**

Nonoperative care – orthotics & casting

First line	**activity restriction, shoe modification, non-narcotic analgesics/NSAIDs**, PT (✓ even after acute sprain)
UCBL orthosis	**inverts heel** during walking → offloads prominence → ↓ symptoms
Orthotic rule	must **offload** the accessory navicular or it **exacerbates** symptoms
Cast immobilization	**next step** after failed activity + shoe modification ✗ Wrong options: continued activity modification · MRI · ORIF
Outcome	most become **asymptomatic at skeletal maturity**

Surgery – excision ± PTT advancement; with flatfoot correction

Indication	recalcitrant pain after **extended nonoperative management**
Procedure	medial approach (distal talus → medial cuneiform) · reflect PTT / **tendon-splitting** approach · excise ossicle **through the synchondrosis** · **trim navicular body** flush with medial border of medial cuneiform

– – Active space – –

PTT advancement (Kidner)	Orthobullets: **do NOT advance PTT** (no added benefit, ↑ morbidity) · OB: **simple excision = Kidner** in outcomes · ✓ "split PTT and excise prominence" ✗ Wrong options: PTT advancement · hindfoot osteotomy · osteosynthesis of ossicle (does not shrink prominence)

Flatfoot is the primary problem (flexible, hindfoot valgus, symptomatic Type II) – **MCO + excision + PTT reattachment** · otherwise flatfoot **not** corrected at the same time
✗ Wrong options: isolated excision (fails) · FDL transfer · lateral closing wedge · subtalar fusion

6 Complications

Most common	**persistent medial prominence and pain** → navicular body **not trimmed enough**

X vs Y

	Simple excision	Kidner (+ PTT advancement)	MCO + excision + PTT reattachment
Use / evidence	Isolated ossicle · relieves pain · **= Kidner**	Historical · **no added benefit**, ↑ morbidity	**Flexible flatfoot + hindfoot valgus** primary

How it's asked

"Accessory navicular excision – tendon elevated is innervated by?"	→ **tibial nerve**
"Function of muscle attached to the ossicle?"	→ **inverts hindfoot, assists plantarflexion**
"Rigid arch insoles make medial foot pain worse – which tendon inserts there?"	→ **posterior tibialis**
"Underlying anatomy?"	→ **plantar medial enlargement of navicular**
"Failed 6 months activity + shoe modification – next?"	→ **cast immobilization** ✗ Wrong options: MRI · continued modification
"Symptomatic after sprain / incidental ossicle – initial treatment?"	→ **activity modification, orthotics (UCBL), PT** ✗ Wrong options: cast 6 weeks · excision + PTT advancement
"Failed orthotics + casting – next?"	→ **excise medial prominence including synchondrosis** ✗ Wrong options: ORIF / bone stimulator for "nonunion"
"Best surgical procedure?"	→ **split PTT, excise prominence** ✗ Wrong options: Kidner advancement (no outcome difference)
"Symptomatic accessory navicular + flexible flatfoot with hindfoot valgus?"	→ **MCO + excision + PTT reattachment**

"Most common complication of excision?"	**→ persistent prominence/pain — inadequate trimming**	– – Active space – –

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ACHONDROPLASIA

– Pediatrics · 2 / 112 –

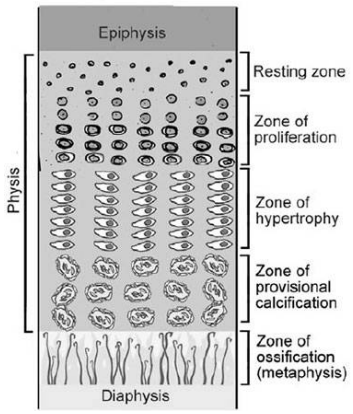
1 Basics

Epidemiology

Most common skeletal dysplasia	**80–90% of all dwarfism** (TO) ✗ Wrong options: diastrophic · MED · nail-patella
Incidence	**1 in 15,000–30,000** (Orthobullets) · **1/30,000** live births (OB)
Sex / timing	no sex predominance · diagnosed in utero or early infancy · short stature apparent at birth
Risk factor	**advanced paternal age** (>50 yr: 1 in 1,875; >35 yr still ↑)

FGFR3 genetics, inheritance & affected physeal zone (proliferative)

Inheritance	**sporadic 80%** (de novo) · **autosomal dominant 20%** · **100% penetrance** · once present → **AD transmission**, 50% risk per child (TO) · "most common inheritance pattern" → ✓ **AD** ✗ Wrong options: AR · X-linked · mitochondrial
Gene	**FGFR3** (fibroblast growth factor **receptor** 3, not FGF-3 the ligand) on **chromosome 4p**
Mutation	point mutation **Gly → Arg** in **transmembrane domain** · **G380R** (OB) = nucleotide **Gly1138Arg** (Orthobullets) · always the same site ✗ Wrong options: COL1A1 (>200 sites)
Effect	**gain of function** ("turned on") → ↑ **tyrosine kinase** activity → ↑ **inhibition of chondrocyte proliferation + differentiation** (TO)
Zone	**proliferative zone** of physis (TO) · **quantitative cartilage defect** · defect of **endochondral** ossification
Intramembranous ossification normal	normal skull vault, clavicles, diaphyseal **width** (TO)
Mechanism detail	mutation stabilises FGFR3 dimers → escapes **ubiquitin** degradation → persistent signalling
Pathways	**STAT1** (inhibits chondrocyte proliferation) · **MAPK** · PLCγ · PI3K-AKT
Modulator	**C-type natriuretic peptide (CNP)** → NPR-B → ↑ cGMP → ↓ FGFR3 activity → basis of **vosoritide**



Physeal zones — achondroplasia = proliferative zone

Genes & lab markers

Condition	Gene / defect
Achondroplasia	**FGFR3** gain of function, chr 4p, ↑ tyrosine kinase
Hypochondroplasia	FGFR3 **Asn540Lys**
Thanatophoric dysplasia	FGFR3 **Lys650Glu** — lethal (respiratory)
SADDAN	FGFR3 **Lys650Met**
Apert syndrome	**FGFR2**
Pseudoachondroplasia	**COMP**, chr **19**
MED	**COMP** (chr 19)
SED · Kniest	**COL2A1** (type II collagen)
Osteogenesis imperfecta	**Type I collagen** (COL1A1 / COL1A2)
Diastrophic dysplasia	**SLC26A2 (DTDST)**, chr **5** — sulfate transport, **AR**
Cleidocranial dysplasia	**RUNX2 (CBFA1)**
Fibrous dysplasia	**GNAS** activating → ↑ cAMP
Marfan	**Fibrillin**
Gaucher	**Glucocerebrosidase**
Jansen metaphyseal chondrodysplasia	**PTHrP**
Schmid metaphyseal chondrodysplasia	**Type X collagen** (AD)
Larsen syndrome	**Filamin B** (AD form)

– – Active space – –

Physeal zone associations (distractors)

Zone	Pathology
Epiphyseal bone	**MED**
Reserve / resting	**Gaucher**, **diastrophic dysplasia**, pseudoachondroplasia
Proliferative	**Achondroplasia** ✓
Hypertrophic	**SCFE**, physeal fractures, enchondroma, rickets
Primary spongiosa	**Metaphyseal corner fractures** (abuse), **scurvy**
Secondary spongiosa	SCFE in **renal osteodystrophy**

Labelled-figure questions letter varies (B or C in different stems) → always choose the **proliferative zone**

3 Presentation

Clinical features – rhizomelia, frontal bossing, trident hand

Rhizomelic dwarfism **proximal** segment shortest (humerus < forearm, femur < tibia) · **disproportionate** short stature · **normal trunk** (TO)

Height adult ~**50 in** (male 4′4″, female 4′) · 6–7 SD below normal

Face **macrocephaly**, **frontal bossing**, **midface hypoplasia**, depressed nasal bridge

Hands brachydactyly · **trident hand** = all fingers same length, divergent **middle and ring** fingers (TO)

Upper limb limited elbow extension · **radial head subluxation/dislocation** · posterior humeral bowing

Lower limb **genu varum** · **hypermobile hips and knees** · muscular hypotonia · long fibulae
✗ Wrong options: genu valgum

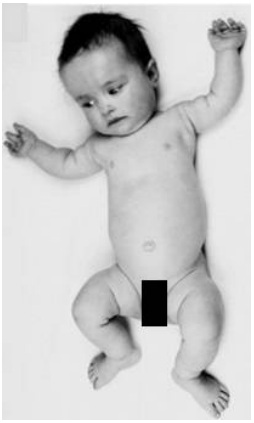
Spine **thoracolumbar kyphosis** · **lumbar hyperlordosis** (up to 80%) · **lumbar stenosis** · **foramen magnum stenosis**

Neuro / development **normal intelligence** · **delayed motor milestones**

ENT **otitis media 90% before age 2** · hearing loss · tonsillar hypertrophy · obesity

NOT a feature **cervical spine instability** (→ pseudoachondroplasia) · **atlantoaxial instability** (→ Morquio MPS IVA, Down) · dural ectasia (→ NF, Marfan) · cauliflower ear (→ diastrophic)

– – Active space – –



Achondroplasia clinical features



Trident hand

Dwarfism by segment

Type	Segment short	Example
Rhizomelic	**Proximal** (humerus, femur)	**Achondroplasia**
Mesomelic	Middle (forearm, leg)	**Leri-Weill dyschondrosteosis**
Mesoacromelic	Middle + distal	**Ellis–van Creveld**
Acromelic	Distal (hands, feet)	**Peripheral dysostosis**

Category (TO)	Examples
Short trunk	Kniest, metatrophic dysplasia, **SED**
Proportionate	Diastrophic, cleidocranial dysplasia, mucopolysaccharidoses
Disproportionate (short limb)	**Achondroplasia**, metaphyseal chondrodysplasias

4 Diagnosis & Classification

Diagnosis	clinical + **skeletal survey** in most · **FGFR3 molecular testing** only if atypical / to distinguish similar disorders
Prenatal	US 2nd/3rd trimester: short long bones, wide metaphyses, **↑ biparietal diameter (>95th centile)** · cell-free fetal DNA
Sleep study	**all infants** (screen for apnea)

– – Active space – –

Key radiology findings

Views	**skeletal survey** · **AP pelvis + femur** (most diagnostic features)
Spine	**↓ interpedicular distance L1 → S1/L5 (pathognomonic)** · **short pedicles** · posterior vertebral scalloping · T12/L1 apical wedging · bullet-shaped vertebrae in infants · hyperlordosis with flat sacrum
Pelvis	**champagne-glass pelvis** (wider than deep) · **squared ("ping-pong paddle") iliac wings** · **narrow sciatic notch** · flat horizontal acetabula
Limbs	short thick bones · metaphyseal flaring · **inverted-V distal femoral physis** · proximal femoral radiolucency · **epiphyses normal**
MRI	FMS: obliterated posterior subarachnoid space, cord "nicking/waisting", T2 signal change, syrinx · also lumbar stenosis



Champagne-glass pelvis



Short pedicles, posterior scalloping

5 Management

Thoracolumbar kyphosis – natural history

Most common dysplasia with TLK	**achondroplasia** ✗ Wrong options: Trevor · MED (normal spine) · Larsen (cervical kyphosis) · diastrophic (cervical)
Cause	flexible, forward slouching with unsupported sitting · 50–100% born with TLK

Natural history	**resolves by 12–18 months** (OB) · **1.5–2 years** (TO) as child **walks** and gains trunk strength · resolution 90% by 18 months · ~70% by age 3 (OB) · prevalence ~87% at 1–2 yr → ~10% at 5–10 yr

Thoracolumbar kyphosis in achondroplasia

- Flexible (most) → **observe; no unsupported sitting**
- Fixed >30° / anterior wedging → **TLSO until walking**
- Neuro deficit / failed brace / >45–60° (>50° TO) → **posterior fusion ± anterior decompression**
  (delay to age ≥4 for instrumentation)

Observation	firm-backed seating · avoid C-position · **no sitting >60°**
Bracing (TLSO)	**fixed component >30°** (prone lateral) · substantial anterior wedging · posterior apical displacement · worn until independent walking · mean residual 8°
Fusion	neuro compromise · kyphosis **>45–60°** (Orthobullets) · fixed **>50°** (TO) · failed brace · **wait until age 4** · corpectomy if severe anterior cord impingement

Foramen magnum stenosis – sleep apnea

FMS → central sleep apnea → SIDS (2–5% mortality in 1st year) (TO)

Central vs obstructive	**adenotonsillar hypertrophy → obstructive** apnea, no hypotonia · **FMS → central** apnea + **hypotonia**
Signs	clonus, **hypotonia**, hyperreflexia, clumsiness, **dysphagia** (lower CN), weakness
Work-up	**sleep study all infants** · **MRI/CT** if abnormal exam or sleep study

Foramen magnum stenosis

- No central apnea, no compression, neuro intact → **observe, repeat MRI 6–12 mo**
- Central apnea / cord compression / neuro deficit → **foramen magnum decompression + upper cervical laminectomy ± duraplasty**
  (intra-op US; 10% recurrence)

Lumbar stenosis

Prevalence	37–89% · **25% need surgery** · disability in **3rd–4th decade**
Causes	short thick pedicles · ↓ interpedicular distance L1→L5 · hyperplastic disc + ligamentum flavum
First line	weight loss, PT, corticosteroid injections
Surgery	severe neurogenic claudication · failed nonop · neuro symptoms at rest → **multilevel laminectomy + fusion**
Technique	**3 levels above block → at least S2** · **no sublaminar wires / laminar hooks** (narrow canal) · pedicle screws preferred but pedicles **cranially directed, ~10 mm shorter**, smaller sagittal diameter
Post-laminectomy kyphosis	**common** despite >50% facet preservation → **fuse if ≥5 levels or skeletally immature**

– – Active space – –

Operative criteria

Problem	Surgery when
TLK – brace	Fixed **>30°**, anterior wedging
TLK – fusion	Neuro deficit · **>45–60°** (Orthobullets) / **>50°** (TO) · failed brace · age ≥4
FMS	**Central sleep apnea** · cord compression · weakness, hyperreflexia, clonus
Lumbar stenosis	Severe claudication · failed nonop · neuro symptoms at rest
Genu varum	Progressive severe deformity · **fibular thrust**
Upper-limb lengthening	**Bilateral** humeral shortening preventing **perineal / head hygiene** (TO)

Extremities & stature

Genu varum	bracing generally unsuccessful · **tibial ± femoral osteotomy** at CORA for progressive deformity / fibular thrust · TO: osteotomy > growth modulation
Growth hormone	controversial · +6–8 cm over 5 yr
Vosoritide	**CNP analogue** · daily injection · age **≥5 yr with open physes**
Lower-limb lengthening	very controversial · early adolescence or two stages (age 7 + 12) · gain ~20.5 cm

Upper-limb lengthening

Indication	**bilateral affection** with humeral shortening limiting **perineal hygiene** (TO) ✗ Wrong options: unilateral LLD (upper limb non-weight-bearing, cosmetic)
Gain	humerus ~10 cm
Complications	**25–43%**: fracture, early consolidation, nonunion, malalignment, stiffness, infection (TO)

6 Complications

Hydrocephalus	shunt in **5%**
ENT	otitis media 90% < age 2 · **50% need tympanostomy tubes**
Spinal surgery	recurrent FMS · CSF leak · malpositioned screw (abnormal pedicles) · neuro injury (overcorrection, wires/hooks) · **post-laminectomy kyphosis**
Life expectancy	~**61 yr** (10 yr less) · highest mortality **<4 yr** (FMS)

X vs Y

	Achondroplasia	Pseudoachondroplasia	Hypochondroplasia
Gene	**FGFR3** (chr 4p)	**COMP** (chr 19)	FGFR3 (Asn540Lys)
Face	**Frontal bossing**, midface hypoplasia	**Normal face/skull**	Milder
Onset	**At birth**	Normal at birth, diagnosed **2–3 yr**	Often **puberty** (no growth spurt)
Spine	**Spinal stenosis**, ↓ interpedicular distance	**No stenosis** (normal interpedicular distance); **C1–C2 instability** (odontoid hypoplasia)	Craniocervical issues less common
Epiphyses	**Normal**	**Epiphysis + metaphysis** involved, early hip dysplasia	Similar X-ray to achondroplasia
Other	Normal IQ	–	Taller; seizures/cognitive issues more common

	FMS	Adenotonsillar hypertrophy
Apnea	**Central**	**Obstructive**
Hypotonia	**Yes**	No
Test	Sleep study → **MRI/CT**	Sleep study

How it's asked

"Achondroplasia – which zone of the physis?"	→ **zone of proliferation** ✗ Wrong options: hypertrophy · resting · provisional calcification
"Receptor defective and zone affected?"	→ **FGFR3, proliferative zone** ✗ Wrong options: FGFR2 = Apert · COMP
"FGFR3 mutation is associated with?"	→ **achondroplasia** ✗ Wrong options: Marfan · Gaucher · fibrous dysplasia · diastrophic
"G380R transmembrane FGFR3 – which radiograph?"	→ **achondroplasia** ✗ Wrong options: rickets cupping · osteopetrosis · OI · cleidocranial
"Mother and daughter, short stature, frontal bossing – gene?"	→ **FGFR3** ✗ Wrong options: COMP chr 19 · RUNX2/CBFA1
"AD, chr 4p, ↑ tyrosine kinase in proliferative zone?"	→ **achondroplasia** ✗ Wrong options: pseudoachondroplasia · Larsen · diastrophic · Kniest
"Defect in rhizomelic dwarf with champagne pelvis?"	→ **FGFR3** ✗ Wrong options: type I/II collagen · COMP · sulfate transport

– – Active space – –

"Achondroplasia caused by abnormality of?"	→ **FGFR3** ✗ Wrong options: PTHrP = Jansen · BMP · TGF-β · IGF-BP
"Genetic defect?"	→ **FGFR3 receptor** ✗ Wrong options: FGF-3 · COL1A1 · trisomy 21
"Most common inheritance pattern?"	→ **autosomal dominant** (even though 80% sporadic)
"Pedigree consistent with rhizomelic dwarfism?"	→ **autosomal dominant pedigree**
"Abnormal tyrosine phosphorylation + FMS in toddler?"	→ **gain-of-function FGFR3** ✗ Wrong options: COMP = MED · COL2A1 = SED · PTHrP = Jansen · type X = Schmid
"Rhizomelic dwarfism refers to?"	→ **short proximal limb segments**
"FGFR3 dwarfism — all EXCEPT?"	→ **cervical spine instability** (pseudoachondroplasia)
"Extremity radiographs in G380R child?"	→ **disproportionate, proximal segment short** ✗ Wrong options: proportionate · mesomelic · acromelic
"Most common skeletal dysplasia?"	→ **achondroplasia**
"9-month-old with back deformity, champagne pelvis — management?"	→ **resolves by 1.5–2 years** (TO) ✗ Wrong options: laminectomy + fusion · growing rods
"Natural history of infant TLK?"	→ **spontaneous resolution by 12–18 months**
"Most common dysplasia with thoracolumbar kyphosis?"	→ **achondroplasia**
"Associated with achondroplasia?"	→ **(central) sleep apnea** (TO) ✗ Wrong options: genu valgum · LLD · atlantoaxial instability · dural ectasia · cauliflower ear
"Hypotonic 3-year-old with central apnea — work-up for?"	→ **foramen magnum stenosis** ✗ Wrong options: C1–C2 instability · adenotonsillar hypertrophy · tethered cord
"Indication for upper-limb lengthening?"	→ **bilateral affection** (perineal hygiene) (TO)

ADOLESCENT BLOUNT'S DISEASE

– Pediatrics · 3 / 112 –

1 Basics

Definition & epidemiology

Definition	progressive pathologic **genu varum** centred at proximal tibia, onset **>10 years** · idiopathic proximal tibia vara
Risk factors	**obesity** (causative) · **African-American** descent · more common in males (OB)
Associated	**hypertension** (often undiagnosed, under-treated, modifiable) · vitamin D deficiency prevalence = healthy children ✗ Wrong options: rickets
Onset vs diagnosis	deformity often begins 4–10 yr · diagnosed as adolescent form **after age 10** (OB)

Pathophysiology

Cause	**dyschondrosis of medial proximal tibial physis** (posteromedial) · **mechanical overload** in **genetically susceptible** child
Deformity	genu varum + procurvatum + internal tibial torsion
Associated deformity	**distal femoral varus** + **distal tibial valgus** (TO) ✗ Wrong options: compensatory distal femoral valgus (that's infantile)

3 Presentation

Hallmark	**genu varum**, usually **unilateral** · obesity · "waddle"
Other	LLD secondary to deformity · mild–moderate **MCL laxity** · medial knee pain

4 Diagnosis & Classification

Diagnosis – obesity & late-onset tibia vara

Diagnosis	clinical genu varum + radiographic **↑ metaphyseal–diaphyseal angle**
Differentials ruled out	congenital hypothyroidism (presents earlier) · metabolic syndrome (normal A1c, TGs) · primary varus gonarthrosis (adult) · vitamin D deficiency/rickets (**bilateral** femoral + tibial bowing)
Physiologic	genu varum normal **<2 years**

Key radiology findings

Views	**standing long-cassette AP both lower limbs**, **patellae forward** (internal tibial torsion common)

– – Active space – –

Findings	narrowing of tibial epiphysis · **widening of medial tibial physis** · occasional widening of lateral distal femoral physis · **metaphyseal beaking less common** than infantile
Metaphyseal–diaphyseal (Drennan) angle	line through metaphyseal beaks vs perpendicular to tibial axis · **>16° abnormal**
Tibiofemoral angle	femoral vs tibial longitudinal axes



Metaphyseal–diaphyseal (Drennan) angle

Classification	**no radiographic classification** for adolescent form (Langenskiöld = infantile) · severity graded mild / moderate / severe by **mechanical axis deviation (MAD)**

5 Management

Rule	**surgery** (TO) · observation/bracing **unlikely to work** – progressive, never resolves spontaneously · untreated → medial pain, altered kinematics, **early arthritis** ✗ Wrong options: orthotics

Adolescent Blount

- Mild–moderate, growth remaining (>2 yr) → **lateral tibial ± fibular hemiepiphysiodesis**
- Severe / near or at maturity (<2 yr growth) → **proximal tibial osteotomy**
  - No LLD → **focal dome osteotomy at CORA**
  - LLD corrected by opening wedge → **medial opening wedge HTO + plate**
  - LLD too large → **osteotomy + external fixator (gradual)**

Distal femoral varus ≥8° → distal femoral osteotomy or epiphysiodesis

Operative criteria

– – Active space – –

Situation	Treatment
Mild–moderate + significant growth remaining	**Lateral tibia + fibula hemiepiphysiodesis** (8-plate / staple / permanent)
Closed physes / insufficient growth (e.g. 15 yr ♀, triradiate fused)	**Osteotomy** (✗ Wrong options: growth modulation)
Varus + LLD 2 cm, physes closed	**Medial opening wedge HTO + plate**
LLD not correctable by opening wedge	Osteotomy + **external fixator**
Distal femoral varus **≥8°**	Distal femoral osteotomy / epiphysiodesis
Goal	**Neutral mechanical axis** (✗ Wrong options: overcorrection into valgus (that's infantile))

Treatment – hemiepiphysiodesis (failure) & osteotomy

Principle
Hueter-Volkmann law: lateral tether slows lateral growth → medial "catches up"
✗ Wrong options: flexible nails = secondary healing · VDRO · Ilizarov distraction · lag screw = primary healing

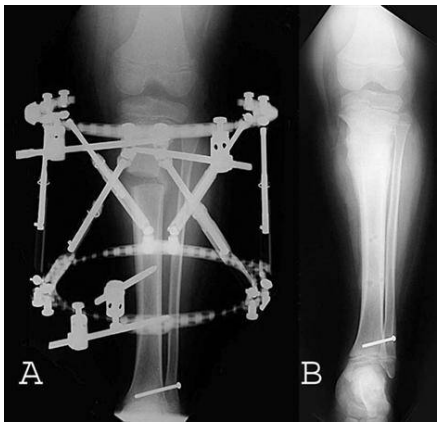
Transient hemiepiphysiodesis
8-plate or staple · removable · needs significant growth · watch for physeal arrest or **rebound** growth

Permanent hemiepiphysiodesis
lateral physeal ablation · overcorrection uncommon · no rotational correction · does not preclude later osteotomy

Residual deformity
up to **25%** need formal osteotomy

Acute HTO + internal fixation
closing, opening, dome, serrated, inclined osteotomies · immediate correction · risks **neurologic injury** (acute lengthening) + **compartment syndrome** → consider **prophylactic fasciotomy** · PWB 6–8 wk

Gradual correction (Taylor Spatial Frame / Ilizarov)
12–18 wk · **limits NV compromise + compartment syndrome** · all planes + length · cons: pin-site infection, duration, bulk



Taylor spatial frame for adolescent Blount

– – Active space – –

Paley osteotomy rules (tibia vara)

Rule	Osteotomy vs ACA/CORA	Result
1	Osteotomy **through ACA-CORA**	Pure angulation, **no translation**
2	ACA through CORA, osteotomy elsewhere	Angulation + **bone-end translation**, axes **collinear**
3	Osteotomy through ACA, CORA at another level	No bone-end translation, axes **parallel but translated**

Best osteotomy — **focal dome centred on CORA** → restores alignment, minimal axis translation + shortening, least bone-end translation, most bone contact
✗ Wrong options: intra-articular (chondral/cruciate risk) · horizontal (translates ends + axes)

6 Complications

Most common complication of growth modulation — **implant failure** (screw breakage) (TO)
✗ Wrong options: recurrence (infantile / genu valgum) · complete physeal arrest (not seen) · failure of correction (severe physeal disease, rare)

Commonest failure site — **tibial metaphyseal screw**

Hardware-failure risk factor	✓ ↑ risk	✗ distractor
Deformity	**Greater severity**	–
Onset	**Earlier** (younger)	Later onset
Screws	**Cannulated**	Solid screws
BMI	**Higher**	Lower BMI
Metal	**Titanium** → use **stainless steel**	–
Construct	Staple = screw-plate (equal)	Staple constructs

Other — undercorrection (too little growth) · **rebound deformity** · NV injury, compartment syndrome (acute osteotomy) · pin-site infection (frame)

X vs Y

	Infantile Blount	Adolescent Blount
Age	**2–5 yr**	**>10 yr**
Frequency	More common	Less common
Side	**~50% bilateral**	**Usually unilateral**
Risks	Early walking, large stature, obesity	**Obesity**
Classification	**Langenskiöld**	**None**
Location	Physeal / epiphyseal	Metaphyseal
Severity	More severe physeal change	Milder
Femur	**Compensatory distal femoral valgus**	**Distal femoral varus** (+ distal tibial valgus)
Natural history	Stage II–IV may resolve (before 4 yr)	**Progressive, never resolves**
Treatment	**Bracing** + surgery	**Surgery only**
Osteotomy goal	Overcorrect into valgus	**Neutral axis**

How it's asked

"Feature of adolescent Blount?"	→ **associated distal femoral varus** (TO) ✗ Wrong options: bilaterality · self-resolution · bracing first line
"Obese hypertensive 10-year-old with unilateral bowing?"	→ **adolescent Blount** ✗ Wrong options: metabolic syndrome · vitamin D deficiency
"15-year-old, fused triradiate, varus + 2 cm LLD?"	→ **medial opening wedge osteotomy** (TO) ✗ Wrong options: hemiepiphysiodesis · bracing · lateral closing wedge
"Closed physes, varus + 2 cm LLD?"	→ **medial opening wedge HTO + plate** ✗ Wrong options: medial closing wedge + ex-fix (worsens varus + LLD)
"Most common complication of growth modulation?"	→ **implant failure** (TO)
"Osteotomy with least translation, most contact?"	→ **focal dome at CORA**
"Which procedure uses the Hueter-Volkmann law?"	→ **hemiepiphysiodesis**
"Broken 8-plate at 4 weeks – contributor?"	→ **increased deformity severity** ✗ Wrong options: later onset · solid screws · lower BMI · staples
"Broken titanium plate screws – what may have prevented it?"	→ **stainless steel implants** ✗ Wrong options: cannulated screws · waiting for maturity
"Why gradual frame correction?"	→ **↓ neurovascular injury + compartment syndrome**

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ANKLE FRACTURES - PEDIATRIC

– Pediatrics · 4 / 112 –

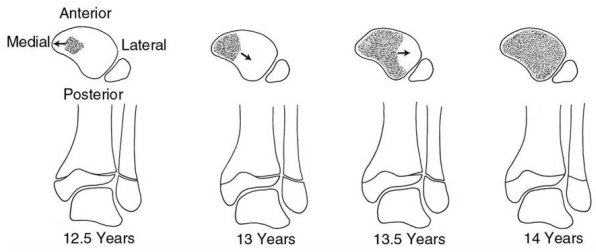
1 Basics

Epidemiology

Incidence	**25–40% of all physeal injuries** (**2nd most common**) · 5% of pediatric fractures
Demographics	males **2:1** · age **8–15 yr**
Risk factors	sports · ↑ BMI
Patterns	SH I, II · **SH III → Tillaux**, medial malleolus · **SH IV → triplane**, medial malleolus

Anatomy – physes

Distal tibial physis	**35–40% of tibial growth**, 15–20% of limb growth · **3–4 mm/yr** · growth until **14 yr girls / 16 yr boys** · closes over **18 months**
Closure order	**central → anteromedial → posteromedial → lateral** (basis of Tillaux, triplane)



Distal tibial physeal closure: central → anteromedial → posteromedial → lateral

Distal fibular physis	closes **12–24 months after** distal tibia
Ligaments	origins **distal to physes** → physis fails first

Ligament	Attachments	Note
AITFL	**Chaput** (anterolateral tibial epiphysis) → **Wagstaffe** (anterior fibula)	Avulses **Tillaux** / transitional fractures
PITFL	**Volkmann** tubercle → posterior fibula	–
Inferior transverse	Posterior fibula across posterior tibial plafond	Posterior labrum
Deep deltoid	–	Primary restraint to lateral talar shift

2 Mechanism

Mechanism	direct trauma · **twisting about a planted foot**

High-energy abduction injuries (e.g. skateboarding) → ↑ physeal closure risk

3 Presentation

Symptoms pain, inability to bear weight

Exam swelling, ecchymosis, deformity if displaced · **distal fibular physeal tenderness = nondisplaced SH I**

4 Diagnosis & Classification

Key radiology findings

Views **AP, mortise, lateral** · full-length tibia to rule out **Maisonneuve**

CT displacement (**best post-reduction**), articular step-off, pre-op planning

Physeal widening after reduction → **interposed periosteum**

Salter-Harris classification

Type	Freq.	Description	Note
I	15%	Through physis	Nondisplaced: fibular tenderness only
II	**45%**	Physis + metaphysis (**Thurston-Holland**)	Most common
III	25%	Physis + epiphysis	**Tillaux**, medial malleolus · ↑ arrest
IV	25%	Metaphysis + physis + epiphysis	**Triplane**, medial malleolus shear · ↑ arrest
V	1%	Crush	Diagnosed late at growth arrest
VI	Rare	Perichondral ring	Lawnmower / iatrogenic

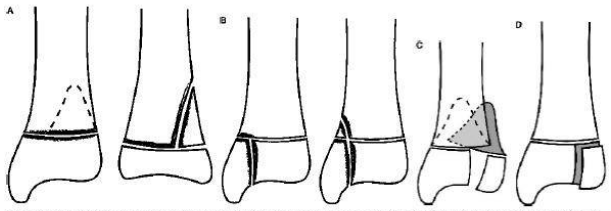


FIGURE 1. Modified SH fracture classification. A: SH II (AP/lateral views). B: Medial malleolar fractures: SH III (left) and SH IV (right). C: Triplane (SH IV). D: Tillaux (SH III).

Salter-Harris classification

Dias & Tachdjian (mechanism) classification

Mechanism	Injury
Supination–inversion	Gr 1: **fibular SH I/II** avulsion · Gr 2: + **medial malleolus SH III/IV**
Supination–plantarflexion	Tibial SH I/II, epiphysis displaced **posteriorly**, posterior Thurston-Holland · **no fibular fracture** · hard to see on AP
Supination–external rotation	Gr 1: tibial SH II, fragment **posteromedial**, line runs proximal-medial on AP · Gr 2: + **low spiral fibula**
Pronation/eversion–ER	Tibial SH I/II + **transverse fibula**, displaced **laterally** · ± diastasis
Axial compression	**SH V** – diagnosed late

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5 Management

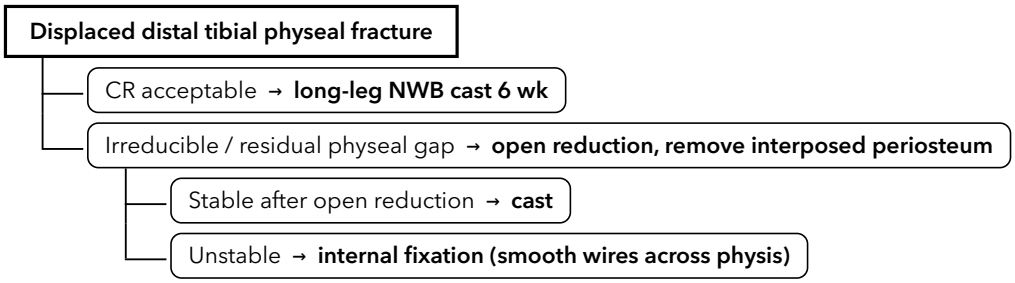
Reduction (TO) — **75% traction, 25% manipulation** · avoid repeated manipulation · step-off **< 1–2 mm** · in adolescents **joint congruity before physeal preservation** · **look at the fibula first**: no fibular fracture → think transitional; isolated fibula fracture may hide distal tibial physeal injury · SH II most common, **III ~25%, IV ~25%** (TO) · ankle = **most common lower-limb physeal injury** (15–20% of physeal injuries)

Operative criteria

Fracture	Nonop	CRPP	ORIF
Isolated distal fibula	**<2 mm** → boot / NWB SLC 4 wk · >2 mm acceptable CR → NWB cast 6 wk	>2 mm, failed CR, **>2 yr growth**	>2 mm, failed CR, **<2 yr growth**
Distal tibia SH I/II	Acceptable CR → NWB cast 6 wk	Unacceptable CR + **>2 yr growth**	Unacceptable CR + **<2 yr growth**
Distal tibia SH III/IV	–	**<2 mm** after CR	**>2 mm** after CR

Acceptable SH I/II tibia — **no varus · <10° valgus · <10° recurvatum/procurvatum · <3 mm physeal widening**

Acceptable tibial reduction — **<2 mm** · confirm on **mortise**



Irreducible fractures – interposed periosteum & open reduction

Closed reduction — adequate sedation + relaxation · **max 2 attempts** (repeated attempts injure physis)

Block to reduction — **periosteum on the tension side** · also tendon, NV structures
✗ Wrong options: compression side · comminution · unrecognised fracture line



Periosteum entrapped on the tension side

Failed CR under sedation → ORIF
✗ Wrong options: repeat CR under GA · cast and follow-up · admit

– – Active space – –

Why open	interposed periosteum → **premature physeal closure** (makes bone, cartilage or fibrous tissue in physis) ✗ Wrong options: "to prevent nonunion" (nonunion rare)
SH I lower extremity	growth problems are **common** ✗ Wrong options: "rare"

Salter-Harris III/IV medial malleolus – ORIF parallel to physis

SH III/IV medial malleolus	large **cartilaginous** portion (bigger than X-ray fragment) → articular incongruity → **ORIF avoiding physis and joint** ✗ Wrong options: casting · CAM walker
Fixation	**transepiphyseal, parallel to physis** · 2 parallel epiphyseal screws (medial malleolus shear) · cannulated screws parallel to physis (Tillaux, triplane) · transverse metaphyseal screw if large fragment · **smooth K-wires** if crossing physis ✗ Wrong options: screw crossing physis
Follow-up	long-term (high growth-arrest rate)

6 Complications

Physeal arrest risk – residual gap

Rate	premature physeal closure **8–50%** after distal tibial physeal fractures
Highest growth disturbance	**medial malleolus SH IV**

Risk factor for premature physeal closure	
Residual physeal gap – greatest risk	**>3 mm** (Orthobullets, OB) · **>2 mm** (OB) – read the options
Initial displacement	**+15% per 1 mm**
Entrapped periosteum	Yes
High-energy (abduction, skateboarding)	Yes (✗ Wrong options: football)
SH III / IV	Yes (all types at risk)
Age >8 yr	**NOT a risk factor**
Number of reductions, ORIF vs closed	not associated if anatomic

Partial arrest	**distal fibular arrest → valgus** · **medial tibial arrest → varus** (continued fibular growth)
Complete arrest	**LLD**

– – Active space – –

Physeal bar – resection, osteotomy & fibular epiphysiodesis

Distal tibial physeal bar

- <50% area + >2 yr growth → **bar resection + interposition (fat)**
  (add osteotomy if angulation >10–20°)
- >50% + ≥2 yr growth → **complete epiphysiodesis + ipsilateral fibular epiphysiodesis**
- Skeletally mature with deformity → **osteotomy**

Bar resection	**<50% physis**, **>2 yr (or 2 cm) growth**, angulation **<20°** (Orthobullets) · interpose fat, cranioplast, bone wax, cartilage, muscle, silicone · via **metaphyseal window** ✗ Wrong options: through the joint (bar not visible)
Osteotomy	angulation **>10–20°** (won't remodel after resection) → **osteotomy + bar resection through osteotomy site**
Fibular epiphysiodesis	bar **>50%** + ≥2 yr growth + open fibular physis → prevents **varus** + lateral impingement
Bar types	**peripheral** (angular deformity, resectable) · **elongated** (SH III/IV, middle) · **central** (tents articular surface) · small **<25% central** = best prognosis
LLD	bar resection if <50% + >2 yr growth · **contralateral epiphysiodesis** near maturity

Rotational malunion

Unrecognised deformity	**external rotation** through physis, noticed months after walking (↑ foot-progression angle) ✗ Wrong options: internal rotation · procurvatum/recurvatum (seen and corrected initially) · valgus
Prevention / treatment	correct rotation at casting · **derotational osteotomy**
Other malunions	anterior angulation / plantarflexion after **supination–plantarflexion SH II** · **valgus** after **ER SH II**

Others

Arthritis	articular step-off **>2 mm**
Extensor retinacular syndrome	posteriorly displaced fractures
Reflex sympathetic dystrophy	more common in **girls** · PT, counselling, drugs, sympathetic block

X vs Y

	Distal fibular arrest	**Medial distal tibial arrest**
Deformity	**Valgus**	**Varus**
Prevention	–	**Fibular epiphysiodesis** (bar >50%)

– – Active space – –

Scenario	Fibular epiphysiodesis?
13 yr ♂, **>50% bar**, 5 months post-fracture	✓ **Yes**
10 yr ♀, <50% bar, ≥2 yr growth	✗ Wrong option: bar resection
13 yr ♀, <50% central bar, <2 yr growth	✗ Wrong option: nothing (little growth)
15 yr ♀, 25° varus 2 yr later	✗ Wrong option: osteotomy (mature)
10 yr ♂, SH IV with >3 mm residual	✗ Wrong option: repeat reduction

How it's asked

"Supination-inversion fracture, CR under sedation unchanged — next?"	→ **ORIF** ✗ Wrong options: CR under GA · cast
"Physeal widening after two CR attempts?"	→ **open reduction, remove periosteum, smooth wires — to ↓ premature physeal closure** ✗ Wrong options: to prevent nonunion
"Block to reduction?"	→ **periosteum on the tension side**
"NOT a risk factor for premature physeal closure?"	→ **age >8 years**
"Greatest risk of premature physeal closure?"	→ **residual gap after CR** ✗ Wrong options: initial displacement · attempts · ORIF
"6-year-old SH IV medial malleolus?"	→ **ORIF parallel to physis** ✗ Wrong options: screw crossing physis
"12-year-old SH III/IV medial malleolus?"	→ **ORIF avoiding physis and joint**
"SH IV 2 yr ago, 30° varus, small central bar?"	→ **osteotomy + bar resection + fat through osteotomy** ✗ Wrong options: resection alone · arthroscopic through joint · epiphysiodesis
"Fibular epiphysiodesis indicated when?"	→ **>50% distal tibial bar with ≥2 yr growth**
"Deformity unrecognised until months after walking?"	→ **external rotation**

– – Active space – –

ANTEROLATERAL BOWING & CONGENITAL PSEUDOARTHROSIS OF TIBIA

– Pediatrics · 5 / 112 –

1 Basics

Epidemiology

Incidence	extremely rare · **1:140,000–190,000** (Orthobullets) · **1 in 250,000** live births (TO)
Continuum	**anterolateral bowing → congenital pseudarthrosis of tibia (CPT)** = same disease ✗ Wrong options: calcaneovalgus · fibular hemimelia · CTEV · CVT
Pathophysiology	pseudarthrosis is **not congenital** · develops **post-natally** from fracture nonunion

NF1 association & natural history (fracture/pseudarthrosis)

NF1	**most common association** · in **50–55%** of anterolateral bowing/CPT (Orthobullets, TO) · 50–90% in large series (TO)
Reverse	only **6–10% of NF1** patients have anterolateral bowing (TO: **~6%**) ✗ Wrong options: 20% · 50% · NF2
Fibrous dysplasia	**15%**
First sign	presents in **1st year** → may be **first recognised manifestation of NF1** · NF does **not** change union rate or outcome (TO)
NF1 genetics	**neurofibromin**, **chromosome 17**, usually **spontaneous** mutation · café-au-lait spots
NF1 orthopaedic manifestations	**scoliosis** (dystrophic + non-dystrophic), **CPT**, soft-tissue tumours · **nerve sheath tumours** (neurofibromas) → "not associated with NF1?" → **none of the above** (TO)
Natural history	does **not** self-resolve · **spontaneous fracture** at site of bowing → nonunion · once fractured, low chance of spontaneous healing ✗ Wrong options: resolution by 2 yr · no change · spread

Three types of tibial bowing

Bowing	Cause	Natural history
Anterolateral	**NF1 / CPT**	**Fracture → pseudarthrosis**; no remodelling
Posteromedial	**Physiologic** (intrauterine packing) · **calcaneovalgus** foot	Resolves over **5–7 yr** · **LLD 3–4 cm** · rarely fractures
Anteromedial	**Fibular hemimelia (deficiency)**	–
Posterolateral	No specific association	–

Describe by apex (TO)	apex on AP (medial / lateral) + apex on lateral (anterior / posterior) · **anterolateral → CPT** · posteromedial → physiological · **anteromedial ("intramedial") → fibular hemimelia**

Pathobiology (TO)

Periosteal disease (TO)	**not a simple nonunion** · **fibrous hamartoma replaces periosteum** → compression, osteolysis, vascular constriction · bone viable but atrophic · canal obliterated by sclerosis · **↓ BMP** from osteocytes · **↑ osteoclast** activity
Pathomechanics (TO)	anterolateral diaphyseal angulation · **proximal migration of fibula** · small cross-sectional area at CPT site

3 Presentation

Onset	bowing in **first year** of life
Exam	anterolateral deformity · **café-au-lait spots** + other NF stigmata
Fibula	dysplastic fibula / fibular pseudarthrosis in **two-thirds** (TO) · **one-third** (other TO explanation) – read the options
Clinical (TO)	bow at **junction of middle and distal third** · usually unilateral · **skin dimple over apex** · fibular dysplasia · ankle valgus · LLD · always head-to-toe exam for NF1 (café-au-lait "coast of Maine", axillary freckling, Lisch nodules, optic glioma)

4 Diagnosis & Classification

Key radiology findings

Radiographs	anterolateral bow · **intramedullary sclerosis / narrowed or obliterated canal** at apex (**prefracture stage**) · cystic change · hourglass constriction · tapered, sclerotic ends



Anterolateral bowing of tibia

– – Active space – –



Congenital pseudarthrosis of tibia

Treatment-guiding criteria

Classifications Boyd, Andersen, Crawford: **none guide management or predict outcome**

Two practical criteria **fracture present or absent · age at fracture: early onset <4 yr, late onset >4 yr**

Boyd classification (TO)

Type	Feature	Prognosis
I	Anterior bowing + tibial **defect at birth**	–
II	Anterior bowing + **hourglass constriction**; fracture **<2 yr** ("high-risk tibia"), sclerotic, canal obliterated	**Most common**, **NF-associated**, **poorest**
III	**Congenital cyst** (middle/distal third junction)	Recurrence less than II
IV	**Sclerotic segment**, no narrowing; stress fracture	**Good**
V	**Dysplastic fibula**	Good if confined to fibula
VI	Intraosseous **neurofibroma / schwannoma**	Extremely rare

Boyd II onset fracture **< 2 yr** (TO table above) · "usually presents **after 2 yr**" (other TO lecture) – read the options

Crawford classification (TO) – all anterolateral

Type	Feature	Prognosis
I	**Preserved canal**, cortical thickening at apex	**Good**, may never fracture
II	Thinned canal, cortical thickening, trabeculation defect	–
III	**Cystic** lesion	**Early fracture** → early treatment
IV	Established pseudarthrosis (tibia ± fibula)	–

Andersen dysplastic, cystic, sclerotic, clubfoot types · **El-Rosasy–Paley–Herzenberg** – I atrophic mobile, no surgery · II atrophic mobile, previous surgery · III broad ends, stiff · **Choi fibula** – A (fibula intact) · B (fibular pseudarthrosis, B1–B3)

Crawford – other TO versions	lecture A: I non-dysplastic bowing with **medullary sclerosis** · **IIa failure of tubularisation** · IIb cystic · IIc frank pseudarthrosis · lecture B: I bowing no fracture · II cystic · III pseudarthrosis · IV sclerotic segment – read the options
Paley III	broad ends, stiff (TO lecture A) · **fracture without frank mobility** (TO lecture B)

Congenital pseudarthrosis of fibula (TO)

Type (TO)	Feature
1	Fibular bowing
2	Fibular pseudarthrosis, **no ankle deformity**
3	Fibular pseudarthrosis + **ankle valgus**
4	Fibular pseudarthrosis + **tibial involvement**

Rx (TO) long-term bracing · **tibiofibular synostosis (Langenskiöld procedure)**

5 Management

Anterolateral bowing / CPT

- Bowing, no fracture, ambulatory → **brace (clamshell / PTB / AFO anterior shell) to maturity**
  (TO: 8-plate guided growth for cystic, unfractured tibia)
- Fracture or pseudarthrosis → **resect, realign, graft, IM rod ± frame / vascularised fibula / 4-in-1**
- Multiple failed unions, severe LLD → **Syme / Boyd amputation**

Bracing to prevent fracture

Indication	ambulatory (weight-bearing) child with **bowing, no fracture/ pseudarthrosis**
Brace	**clamshell** orthosis · **patellar tendon bearing (PTB)** orthosis · **AFO with anterior shell** · until **skeletal maturity**
Goal	**prevent further bowing + fracture** ✗ Wrong options: correct deformity (bowing is not brace-correctable)
Contraindicated	**osteotomy for bowing alone** (→ nonunion) ✗ Wrong options: observation · IM rod · vascularised fibula · PT
Posteromedial bowing case	"pseudarthrosis, brace" is the **false** statement · observation only, LLD, calcaneovalgus association (TO)
Bracing (TO)	**prevents fracture, does NOT correct deformity** · infants before walking: education + protection enough · clamshell (circumferential) to maturity

Growth modulation for deformity

Unfractured anterolateral bowing with cystic change	**distal tibial anterolateral 8-plate (guided growth)** → gradual cyst resolution + correction (TO, Laine 2020) ✗ Wrong options: osteotomy + IM nail/plate (nonunion) · 4-in-1 (for established pseudarthrosis)

– – Active space – –

Surgical indications & NF1 association

Surgical indication — **pseudarthrosis or fracture** (TO)
✗ Wrong options: bowing >45° · failure of brace to correct

Goals — resect pseudarthrosis to normal bone · correct alignment · bone graft + stabilise · **intramedullary splinting**

Aims (TO) — union · prevent refracture · LLD · ankle valgus prevention

Principles (TO) — **complete excision of fibrous hamartoma to normal fat planes** + abnormal periosteum · **recanalise medullary canal** · **autogenous cancellous graft** (stem cells) · **rigid IM fixation** (↓ refracture) · angular correction at CPT site mandatory · **periosteal grafting** · **large cross-sectional union** · tibiofibular **cross-union prevents refracture**

Technique (TO)	Union	Note
IM rod + graft	**~85%**	**First line**, any age · retain rod to maturity · rod crossing ankle ↓ valgus
Vascularised fibula	**90–95%**	**Refracture up to ⅓** (no permanent IM fixation)
4-in-1 / cross-union (Choi, popularised by Paley)	–	Excise fibrous tissue · acute docking + compression · **tibiofibular cross-union** · circumferential corticocancellous graft · 3-layer graft: **periosteum + cancellous graft + BMP-2** · telescoping rod ± Ilizarov · **IV zoledronate** pre + post-op

Technique	Key point
IM rod + bone graft	Resect, **shorten tibia**, rod, graft
Free vascularised fibula (Farmer)	**Contralateral** fibula (ipsilateral involved) · when IM nailing + graft fails
Ilizarov / circular frame	Lengthening or bone transport
4-in-1 (Choi)	Resect both sclerotic ends + cortical onlay grafts → **tibiofibular cross-union** · for Choi **B2/B3** fibula · + Ilizarov lengthening if shortening >2–3 cm (TO)
Fibula	Resect fibular pseudarthrosis (↑ union) · osteotomise intact fibula

Amputation — after **multiple failed unions**, severe LLD, dysfunctional deformity · **Syme or Boyd** (ankle disarticulation) preferred over amputation at pseudarthrosis

Amputation (TO) — multiple failed surgeries · **shortening > 7.5 cm** · nonfunctional limb



Vascularised fibular graft

6 Complications

– – Active space – –

Refracture	**≥50%** even after union · decreases with age, ceases after maturity
Refracture (TO)	**most common** complication · **14–60%** · prevent with orthosis + retained IM rod to maturity · others: ankle valgus (fibular growth disturbance), LLD, infection, ankle stiffness / OA
Other	**valgus** deformity · **LLD ~5 cm** at maturity · no treatment gives predictable results

X vs Y

	Anterolateral	Posteromedial	Anteromedial
Association	**NF1**, fibrous dysplasia	**Calcaneovalgus foot**	**Fibular deficiency**
Fracture risk	**High**	Rare	–
Treatment	**Brace** until maturity	**Observe**	Per fibular deficiency
Sequel	Pseudarthrosis, refracture	**LLD 3–4 cm**	LLD

How it's asked

"All true about CPT EXCEPT?"	→ **20% of NF patients have pseudarthrosis** (✓ anterior angulation commonest, ✓ NF commonest association, ✓ dysplastic fibula) (TO)
"Nonunited tibial fracture + café-au-lait spots – prior deformity?"	→ **anterolateral bowing** (TO) ✗ Wrong options: anteromedial (fibular deficiency) · posteromedial (physiologic)
"Not associated with NF1?"	→ **none of the above** (nerve sheath tumour, CPT, scoliosis all associated) (TO)
"True statement about NF1 and CPT?"	→ **6% of NF1 have CPT** (TO) ✗ Wrong options: 50% of NF1 · NF2
"Anterolateral bowing is associated with?"	→ **congenital pseudarthrosis of tibia**
"5-month-old, anterolateral bow with medullary sclerosis – natural history?"	→ **spontaneous fracture**
"Chromosome 17 mutation, pseudarthrosis after casting – bowing type?"	→ **anterolateral**
"6-month-old posteromedial bow – NOT true?"	→ **"pseudarthrosis, protective bracing"** (TO)
"2-year-old ambulatory, prefracture CPT?"	→ **PTB orthosis** ✗ Wrong options: osteotomy + rod (nonunion)
"12-year-old NF, anterolateral bow, intact cortices?"	→ **AFO with anterior shell** (brace to maturity)

– – Active space – –

"Anterolateral bow with cysts, no fracture — ideal surgery?"	→ **growth modulation (8-plate)** (TO)
"Indication for surgery in CPT?"	→ **pseudarthrosis (or fracture)** (TO) ✗ Wrong options: bowing >45° · brace failure
"CPT is primarily a disease of?"	→ **periosteum (fibrous hamartoma)** (TO)
"First-line operative fixation for CPT?"	→ **IM rod + bone graft** (TO)

ARTHROGRYPOSIS

– Pediatrics · 6 / 112 –

1 Basics

Definition & epidemiology

Definition	**non-progressive** congenital disorder · **multiple rigid joints** in all 4 limbs · a physical finding in ~300 disorders (TO)
Incidence	**1:3000** live births (TO)
Course	contractures **do not worsen with growth** ✗ Wrong options: "progressive, peak at maturity"

Etiology

Mechanism	**fetal akinesia** (↓ movement in utero) → **symmetric** contractures · movement needed from **8 weeks' gestation** (TO)
Neurogenic vs myopathic	**neurogenic 90%**, myopathic 10% (Orthobullets) · neurologic **70–80% / 80%** (TO)
Pathophysiology	maternal antibodies vs fetal **acetylcholine receptors** → ↓ anterior horn cells · anterior horn cell disease (Werdnig-Hoffmann, SMA)
Other causes (TO)	connective tissue (diastrophic dysplasia, Larsen, Beals) · myopathies · uterine space limits (oligohydramnios, fibroids, multiple pregnancy) · vascular compromise · maternal disease (diabetes, MS, myasthenia, myotonic dystrophy, rubella, CMV, toxoplasma)

No genetic predisposition for classic form (OB)

Classification

Type	Description
I	Single localised deformity (e.g. forearm pronation)
II	**Full expression** – absent shoulder muscles, thin limbs, elbows extended, wrists flexed + ulnar deviated, intrinsic-plus hands, adducted thumbs, **no flexion creases**
III	Type II + **polydactyly** + non-neuromuscular systems

By neurological function (TO)	normal: **amyoplasia**, distal arthrogryposis, fetal crowding, systemic tissue disorder, multiple pterygium · abnormal neurological function
Amyoplasia	**most common recognisable form** · **sporadic**, symmetric · normal/above-average IQ · **midline facial hemangioma** · hip involvement ~**80%**

Distal arthrogryposis & related syndromes (TO)

Distal arthrogryposis	hands + feet, large joints spared · ulnar-deviated fingers at MCP · "cup-like" palm, single crease · flexed adducted thumb · equinovarus or vertical talus · **better prognosis**
Freeman-Sheldon	**common variant of type 2 distal arthrogryposis** · **whistling face** (pursed lips, deep eyes) ✗ Wrong options: Larsen · Escobar

– – Active space – –

Pterygium syndromes	skin webs across joints · **Escobar** = multiple pterygium (neck webs, vertebral anomalies) · **popliteal pterygium** variant
Larsen	**filamin B** · hypermobility · **congenital dislocation of multiple joints** (anterior knee dislocation) · **cervical kyphosis/injury** · normal IQ
Bruck syndrome	**OI + congenital contractures**

3 Presentation

Clinical characteristics

Region	Classic finding
Shoulder	**Adducted, internally rotated** (absent deltoid)
Elbow	**Extended**, no flexion creases (deficient biceps, brachialis) · rarely flexed
Wrist	**Flexed + ulnar deviated** (TO)
Hand	**Intrinsic plus** · IP flexion + relative MCP extension · **adducted thumb / thumb-in-palm**
Hip	**Flexed, abducted, externally rotated** · subluxation / **teratologic dislocation**
Knee	Extended (classic) but **most often flexed** · weak quads · dimple over patella
Foot	**Clubfoot (most common deformity)** · vertical talus rarer
Spine	**C-shaped neuromuscular** scoliosis – **33%** (Orthobullets) · 28–67% / 10–30% (TO) · long thoracolumbar curves

Normal	**intelligence**, facies, **sensation**, viscera
Other	**fractures 25%** · limbs cylindrical/fusiform with ↓ skin creases · **nonambulatory 25%**



Classic arthrogryposis – IR shoulders, extended elbows, flexed wrists

4 Diagnosis & Classification

Diagnosis & upper-limb pattern

Diagnosis	**clinical**: multiple symmetric contractures (e.g. stiff fingers with few creases, IR arms, flexed-ER hips, stiff clubfeet) → **arthrogryposis** ✗ Wrong options: Larsen · CP · Down
Studies at 3–4 months	neurologic studies · enzyme tests · **muscle biopsy** · genetic testing

Prenatal US (TO) lack of movement, fixed flexion, abnormal position, ↑ nuchal translucency, polyhydramnios-type signs, thin ribs, diaphyseal fractures

– – Active space – –

5 Management

Principle multidisciplinary · **stretching + casting first** → soft-tissue release / osteotomy · correct alignment, preserve ROM, tendon transfers for active motion

Operative criteria

Region	Indication / timing
Upper limb surgery	After age **4** (independent eating) (Orthobullets) · after ambulation, **3–4 yr**; not >8 yr (TO)
Elbow extension contracture	**Triceps V-Y lengthening + posterior capsulectomy at 1.5–3 yr**
Unilateral hip dislocation	**Open reduction** ≥6 mo (Orthobullets) · **6–9 mo** (OB) · **before 1 yr** (OB)
Knee flexion contracture	**>30°** → hamstring release at **6–9 mo** (Orthobullets) · not corrected by 6–12 mo (TO)
Hip flexion contracture	**>45°** → surgical release (TO)
Knee extension contracture	Failed casting by **6–12 mo** → **quadricepsplasty** (TO)
Foot	Around time of standing (↓ recurrence) (TO)

Upper limb – passive manipulation & casting

Goals drive electric chair, use computer devices · **one elbow extended** (positioning, **perianal care**) + **one flexed** (**feeding**)

First line **passive manipulation + serial casting** (13-month-old, good triceps)
✗ Wrong options: observation · EMG/NCS · osteotomy · soft-tissue release + transfer

Deformity	Procedure
Elbow extension	**Triceps V-Y lengthening + posterior capsulectomy** (passive flexion) · active flexion: **bipolar pectoralis major / latissimus transfer**, long-head triceps transfer, **Steindler flexorplasty**
Stiff flexed elbow	Not severe → **no surgery** (TO)
Wrist palmar flexion + ulnar deviation	**FCU release/lengthening/transfer to wrist extensors** · **dorsal carpal closing-wedge osteotomy** · fusion = salvage · ideal 5–20° dorsiflexion
Thumb-in-palm, syndactyly	**Z-plasty**, adductor release, syndactyly release
Finger deformity	**PIP arthrodesis**
Shoulder IR	External rotation humeral osteotomy (TO)

Hip dislocation – open reduction (not Pavlik)

Teratologic hip **68–80%** · dislocates **early in utero** → stiff, rigid

– – Active space – –

Pavlik / abduction brace	**unlikely to succeed** → US still shows **dislocation (α ~40°, β unmeasurable)** · Pavlik in dislocated hip prevents posterior wall development
Closed reduction	**rarely successful** (skin/skeletal traction not helpful)

Arthrogrypotic hip dislocation

- Unilateral → **open reduction (prevent pelvic obliquity, scoliosis)**
  - <1 yr (6–9 mo) → **medial approach ± femoral shortening**
  - Older child → **anterior approach + femoral shortening ± pelvic osteotomy**
- Bilateral → **observe (controversial; many ambulate)**
  - Good arc of motion (e.g. 80°) → **bilateral medial open reduction (OB)**



Teratologic hip dislocation in arthrogryposis

Approach	**medial** (good results 80%; Szoke) · **anteromedial or anterolateral** both good (OB) · anterior reserved for **older children**
Prerequisite	reasonable flexion/extension arc + active lower-limb movement
Risk	resulting **hip flexion contracture worsens gait**
Sequence conflict	knee releases **before** hip reduction to help maintain reduction (Orthobullets; TO: correct knees before hip) · **"hip first, knee second"** (other TO explanation) – read the options

Knee contractures

Flexion contracture	serial splinting/casting → **hamstring release** (>30°, early) · ambulation possible with residual **15–20°** · **supracondylar femoral extension osteotomy** at maturity · guided growth with osteotomies (may not fix poor quads) · Ilizarov for severe
Extension contracture / dislocation	early stretching + casting · **open quadriceps tenotomy at 4–6 mo** · late **V-Y quadricepsplasty** · reduce knee dislocation first

Clubfoot & lower-limb deformity management

Most common deformity in AMC	**talipes equinovarus** (TO) ✗ Wrong options: CVT · knee flexion · IP flexion
Newborn	start **serial stretching + splinting** immediately (TO) ✗ Wrong options: wait for weight gain · wait 1 month

Ponseti — **effective but more casts** than idiopathic · **high relapse** (27% at 2 yr) (TO)
✗ Wrong options: as effective · not effective · not used

– – Active space – –

Arthrogrypotic clubfoot

- Early → **Ponseti casting (more casts)**
- Rigid / failed Ponseti → **posteromedial + posterolateral release**
- Recurrence after release → **talectomy (child) / triple arthrodesis (adolescent)**
  (alternatives: Verebelyi-Ogston talar decancellation, lateral column shortening, circular frame)

Congenital vertical talus

CVT in arthrogryposis — **soft-tissue release first** · **talectomy** if recurrence

CVT associations — **hip dislocation 43–50%** (✓ "congenital hip dislocation") · myelomeningocele, arthrogryposis, **trisomy 18**, Cri-du-Chat · familial **HOXD10** · 60–85% have associated anomalies
✗ Wrong options: amniotic band · fibular hemimelia · PFFD · polydactyly

Spine — multilevel fusion for arthrogrypotic curves

X vs Y

	Arthrogryposis (amyoplasia)	Larsen	Distal arthrogryposis
Joints	**Stiff, rigid**, all 4 limbs	**Hypermobile**, multiple **dislocations**	Hands + feet; large joints spared
Gene	Sporadic	**Filamin B**	Variants incl. Freeman-Sheldon
Hallmark	IR shoulders, extended elbows, flexed-ulnar wrists	Anterior knee dislocation, **cervical kyphosis**	Ulnar-deviated fingers, cup palm
IQ	Normal	Normal	Normal

How it's asked

"All characteristic of arthrogryposis EXCEPT?"	→ **progressive contractures peaking at maturity**
"Arthrogrypotic newborn, Pavlik 4 weeks — US at 1 month?"	→ **α 40°, β unmeasurable** (still dislocated)
"Amyoplasia, unilateral irreducible hip?"	→ **delayed open reduction ± pelvic/femoral osteotomy** (6–9 mo) ✗ Wrong options: Pavlik · semi-rigid brace · traction · closed reduction + spica
"10-month-old arthrogrypotic, bilateral dislocations, 80° arc?"	→ **bilateral open reduction, medial approach** ✗ Wrong options: traction + CR · anterior approach

– – Active space – –

"6-month-old, rigid unilateral dislocation — initial management?"	→ **open reduction (anteromedial/ anterolateral)** ✗ Wrong options: Pavlik · observe · wait >1 yr · spica
"13-month-old, little elbow motion, good triceps?"	→ **passive manipulation + serial casting**
"Rigid rocker-bottom foot — associated condition?"	→ **congenital hip dislocation**
"Stiff fingers, IR arms, flexed-ER hips, stiff clubfeet — diagnosis?"	→ **arthrogryposis** (TO) ✗ Wrong options: Larsen · CP · Down
"Common variant of distal arthrogryposis?"	→ **Freeman-Sheldon** (TO)
"Classical arthrogryposis commonly involves?"	→ **wrist flexed with ulnar deviation** (TO) ✗ Wrong options: thumb abduction · flexed elbows · finger extension
"Most common deformity in AMC?"	→ **talipes equinovarus** (TO)
"10-day-old with AMC — advice for lower limbs?"	→ **serial stretching and splinting** (TO)
"Ponseti in arthrogrypotic clubfoot?"	→ **effective but more casts needed** (TO)

BECKWITH-WIEDEMANN SYNDROME

– Pediatrics · 7 / 112 –

1 Basics

Epidemiology

Incidence	**1 in 13,700** births · M:F **1:1**
Type	**overgrowth syndrome**

Genes & lab markers

Locus	**chromosome 11p15.5**
Mechanism	overactivity of **IGF-2** (growth factor) and/or loss of active **CDKN1C** (cell-proliferation inhibitor)
Inheritance	**sporadic 85%** · **AD with incomplete penetrance 15%**
Lab	**neonatal hypoglycemia** (✗ Wrong options: hyperglycemia) from **pancreatic islet cell hypertrophy** (✗ Wrong options: pancreatic insufficiency) → repeated bouts → **spasticity**, altered development

3 Presentation

Diagnostic criteria

Major	Minor
Overgrowth (large for gestational age, gigantism)	**Hemihypertrophy** – **10–20%** (Orthobullets) · **~15%** (OB)
Abdominal wall defects incl. **omphalocele**	**Ear anomalies** (linear lobe creases)
Macroglossia	**Neonatal hypoglycemia**
	Nephromegaly

Exam macroglossia · midface hypoplasia · infraorbital creases · anterior linear ear-lobe creases · **visceromegaly** · exophthalmos · **hemihypertrophy** (whole side incl. foot)

– – Active space – –



Beckwith-Wiedemann – macroglossia

No skin markings absence of nevi / vascular markings excludes **NF, Proteus, Klippel-Trenaunay** as causes of hemihypertrophy



Hemihypertrophy in Beckwith-Wiedemann

4 Diagnosis & Classification

Diagnosis **physical exam + genetic testing**

Key radiology findings

Radiographs **comparison view** of affected vs contralateral limb (LLD)

Bone age inaccurate when very young · later useful to **time epiphysiodesis**
✗ Wrong options: as initial study at age 2

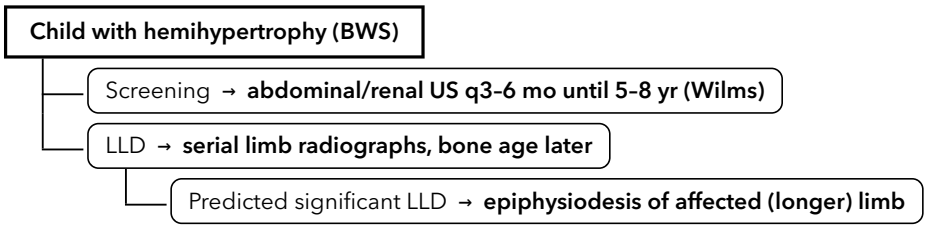
5 Management

Wilms tumour screening

Predisposition **embryonal tumours**: **Wilms tumour** (most common), **hepatoblastoma**, adrenal neoplasm · hemihypertrophy from BWS → **40% malignancy risk** (OB)

Screening **renal + abdominal ultrasound every 3–6 months until 5–8 years** (Orthobullets) · until **~7 years** (OB) · intervals ≤4 months ↓ late-stage disease

Tumour side may be **contralateral** to hemihypertrophy



– – Active space – –

Operative criteria

Epiphysiodesis	**growth arrest of the affected (overgrown) limb** for LLD
Multidisciplinary	orthopaedic, abdominal and overgrowth issues

How it's asked

"Large-for-dates neonate with hypoglycemia, now right hemihypertrophy + 2 cm LLD — next imaging?"	→ **renal and abdominal ultrasonography** ✗ Wrong options: left-wrist bone age · MRI spine/brain · hip US
"11p15.5 mutation, hemihypertrophy, macroglossia, omphalocele — screen every 3–6 months for?"	→ **Wilms tumor** ✗ Wrong options: hearing loss · hyperglycemia · pancreatic insufficiency · vision loss
"Cause of infantile hypoglycemia in BWS?"	→ **pancreatic islet cell hypertrophy**
"Treatment of BWS limb overgrowth?"	→ **epiphysiodesis** of affected limb

– – Active space – –

BIPARTITE PATELLA

– Pediatrics · 8 / 112 –

1 Basics

Epidemiology

Incidence	**2–3%** (Orthobullets) · **~2%** (TO) · "1 in 20" (OB)
Sex	**no sex predilection** (Orthobullets) · **M:F 9:1** (TO) – read the options
Bilateral	**50%** (Orthobullets, OB) · "most patients" (TO)
Location	**superolateral (Saupe type III)** most common

Etiology & anatomy

Nature	**normal variant**, **failure of fusion** of accessory ossification centre · often confused with **patella fracture**
Ossification	patella = **largest sesamoid**, single nucleus · ossifies 4–5 yr ♂ / 3 yr ♀ · **accessory centre at 8–12 yr** · fragment joined by **fibrocartilage**
Painful bipartite patella	trauma disrupts **fibrocartilaginous zone** → cannot unite (no arterial penetration to fragment) · **vastus lateralis traction** → separation/ nonunion
Blood supply	predominantly **distal → proximal** · 6 arteries: superior + inferior medial/ lateral geniculates (popliteal), supreme geniculate (SFA), anterior tibial recurrent
Associated	**nail-patella syndrome**

Histology

Interposed tissue	**fibrocartilage** > fibrous > hyaline · **avascular**
Adjacent bone	scalloped, osteoclasts, **well vascularised**

3 Presentation

Symptoms	mostly **asymptomatic, incidental** · only **2% symptomatic**: anterior knee pain after direct trauma (fall, kick) or overuse (cycling, hill climbing) · worse squatting, jumping, stairs · giving way
Exam	tenderness over fragment (superolateral) · hematoma · quad inhibition · palpable defect / large patella

4 Diagnosis & Classification

Key radiology findings

AP knee	**best view** to see bipartite patella
Skyline	prone (non-weight-bearing) vs **squatting (weight-bearing)** → ↑ **separation** of fragments on squatting view

Contralateral knee	similar findings in 50%
Bipartite vs fracture	**superolateral** · **smooth, rounded borders** · often bilateral
MRI	painful bipartite patella → **edema** around fragment = pain source
Bone scan	equivocal X-ray: ↑ **uptake superolateral** · controversial
CT	shows fragment, **not edema**



Bipartite patella type III – AP

Saupe classification

Type	Incidence	Location	Image
I	5%	**Inferior pole** (transverse split)	
II	20%	**Lateral margin** (longitudinal)	
III	**75%**	**Superolateral pole** (oblique)	

5 Management

Bipartite patella

- Asymptomatic / benign exam → **reassure, symptomatic Rx, RTP as tolerated**
- Symptomatic → **rest, NSAIDs, isometric quads, brace 30° flexion, ± steroid ≥6 months**
- Failed >6 mo / displaced / significant impairment → **excision of fragment**
  - Superolateral fragment → **± lateral release or subperiosteal VL release**

– – Active space – –

Location & nonoperative care

Incidental with benign exam	**reassurance + observation** · **symptomatic treatment, return to play as tolerated** ✗ Wrong options: MRI · ORIF · cast 6 wk · immobilise 120° (that's quad contusion)
Nonop ≥6 months	rest/sport restriction · NSAIDs · **isometric quadriceps** strengthening · **brace in 30° flexion** · local corticosteroid · less successful in young athletes (compliance)

Operative criteria

Indication for surgery
Failed nonop **>6 months**
Displaced fragment needing reduction
Direct trauma causing onset of pain
Significant impairment of daily activities

Procedure	**Note**
Open excision	**Most common**, good results · poor if **large articular** fragment (PF incongruity)
Arthroscopic excision	Faster recovery, spares quad tendon
Lateral retinacular release	Superolateral fragment – removes **VL traction**
Subperiosteal VL release	Superolateral; avoids long lateral release; arthroscopic = quicker, less effusion
ORIF	**Large fragments** · controversial

6 Complications

PF maltracking	excision of large fragment / lateral release → PF degeneration
Others	**effusion** (arthrocentesis) · persistent pain (after lateral release) · quad weakness · osteonecrosis

Differential – dorsal defect of patella; double-layered patella (MED)

Double-layered patella	**pathognomonic of MED** · two segments split by **coronal septum** · anterior in quad/patellar tendons · posterior articulates · painful snapping, locking, dislocation → **fusion** of segments (best) or resection (TO) ✗ Wrong options: CDP · SED · achondroplasia
Dorsal defect of patella	round, lytic, well-defined **subchondral** lesion, **superolateral** quadrant, sclerotic rim, no matrix, no periosteal reaction · MRI: T2 bright, no marrow edema · usually incidental, **self-limited** → activity modification · rarely curettage + graft (TO) ✗ Wrong options: OCD · chondroblastoma · Brodie abscess

X vs Y

	Bipartite patella	Patella fracture	Dorsal defect of patella
Site	**Superolateral**	Any	Superolateral, **subchondral dorsal**
Edges	**Smooth, rounded**	Sharp	Sclerotic rim, lytic
Bilateral	50%	No	–
Treatment	Symptomatic	Per fracture	Self-limited

How it's asked

"Knee pain after fall, benign exam – which X-ray warrants reassurance?"	→ **bipartite patella** ✗ Wrong options: patellar dislocation · patella alta/tendon rupture · Segond · GCT
"Helmet to knee, benign exam, bipartite patella on X-ray – next?"	→ **symptomatic treatment, RTP as tolerated** ✗ Wrong options: MRI · ORIF
"Most common location?"	→ **superolateral** (Saupe III)
"Double-layered patella is pathognomonic of?"	→ **MED** (TO)
"12-year-old, superolateral dorsal subchondral lucency with sclerotic rim?"	→ **dorsal defect of patella** (TO) ✗ Wrong options: OCD · chondroblastoma · Brodie abscess

– – Active space – –

BLADDER EXSTROPHY

– Pediatrics · 9 / 112 –

1 Basics

Epidemiology

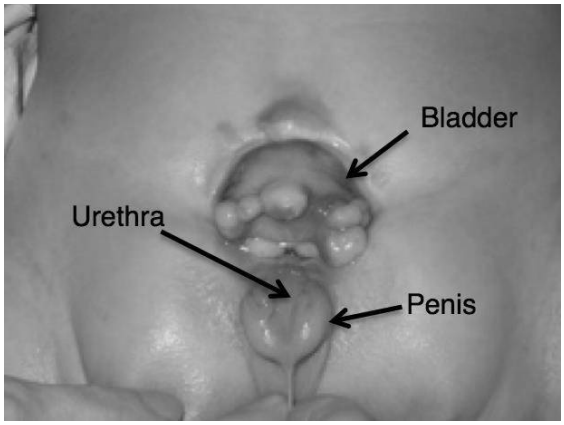
Classic exstrophy	**1/40,000**
Cloacal exstrophy	**1/200,000** (intestinal tract also involved)

Etiology

Pathology	abnormal anterior rupture of the **cloacal membrane** early in embryonic period · inhibited mesenchymal ingrowth into abdominal wall · **altered sclerotome migration** to anterior elements (pubis)
Result	failure of abdominal wall closure → **posterior bladder wall protrudes** through lower abdomen

3 Presentation

GU	**exposed bladder** · urinary tract infections · epispadias (males)
Musculoskeletal	acetabula **~12° retroverted** (no pubis to tether anterior ring → posterior elements retrovert) · **waddling gait** with **external foot progression**



Bladder exstrophy – exposed bladder

4 Diagnosis & Classification

Key radiology findings

AP pelvis	**pubic rami diastasis** · **shortened pubic rami** · **acetabular retroversion**

Acetabular retroversion

Most common acetabular finding	**retroversion** ✗ Wrong options: anteversion · protrusio · neutral · hypoplasia

– – Active space – –



Pubic diastasis in bladder exstrophy

5 Management

Goals close abdominal wall · **urinary continence** · normal renal function · functional genitalia · reconstruction depends on a **stable anterior pelvic ring**

All cases **surgery**, staged and multidisciplinary (urology, general surgery)

Stage	Procedure	Age
I	**Primary bladder closure**	Newborn
II	**Epispadias repair** (males)	1–2 yr
III	**Bladder neck reconstruction**	4 yr
Any stage	**Pelvic osteotomies** – ↓ tension on bladder + abdominal wall → ↓ **dehiscence**	–

Pelvic osteotomy fixation by age

- Newborn → **none (skin traction, hips flexed 90°)**
- Young child → **external fixation**
- Age >8 → **plate fixation to augment diastasis correction**

6 Complications

Recurrent pubic diastasis common **with or without osteotomy** · does not affect activity level

Anterior innominate osteotomy wound dehiscence · **transient femoral nerve palsy**

How it's asked

"Most common acetabular finding in bladder exstrophy?"	→ **retroversion**
"Pelvic radiograph findings?"	→ **pubic diastasis, short pubic rami, acetabular retroversion**
"Why pelvic osteotomy?"	→ **↓ tension on bladder/abdominal closure → ↓ dehiscence**
"Gait in bladder exstrophy?"	→ **waddling, external foot progression**

– – Active space – –

BOTH BONE FOREARM FRACTURE - PEDIATRIC

1 Basics

Epidemiology

Incidence	**~40% of all pediatric fractures** · males > females
Location	distal physis 14% · **distal metaphysis 60%** · midshaft 20% · proximal third 4%
Peak	during peak bone turnover (remodelling mismatch)
Associated	**floating elbow** (ipsilateral supracondylar) **15%** · nerve injury **1%**, most often **median**

Anatomy

Bowing	physiologic **apex-lateral radial bow** · apex-posterior ulnar bow
Deforming forces	**biceps + supinator** flex/supinate proximal fragment · **pronator teres + quadratus** pronate distal fragment · **brachioradialis** dorsiflexes + radially deviates distal fragment
Periosteum	intact on **concave** side · interosseous membrane taut in **neutral to slight supination**

2 Mechanism

Mechanism	fall from height, sport, **playground (monkey bars)**

3 Presentation

Symptoms	forearm pain, refuses to use arm
Exam	swelling, deformity, ecchymosis · **subtle poke-hole open wounds** · check elbow + wrist · NV exam · **compartment syndrome**

4 Diagnosis & Classification

Types	**incomplete**: greenstick, torus, **plastic deformation** · **complete** · proximal / middle / distal third · apex volar or dorsal

Assessing rotational alignment

View	Landmarks 180° apart when aligned
AP	Radial styloid + bicipital tuberosity
Lateral	Ulnar styloid + coronoid

Other clues fragment **diameters** and **cortical thickness** should match

– – Active space – –



Rotational alignment — bicipital tuberosity vs radial styloid

Key radiology findings

Views **AP + lateral forearm** + orthogonal **elbow and wrist**

Findings both bones fractured · one fracture + **plastic deformation** of the other · atypical bowing without fracture = plastic deformation

5 Management

Acceptable alignment & remodelling by age

Age	Angulation	Malrotation	Bayonet apposition
<10 yr	**<15°**	**<45°**	**Yes** if <1 cm short (100% translation OK)
≥10 yr	**<10°**	**<30°**	**No**
<2 yr growth left	**0°**	**0°**	**No** – adult tolerances

By level (girls ≤8, boys ≤10) **distal third 20° · middle 15° · proximal 10°** · remodelling ↓ distal → proximal

Bayonet acceptable (esp. distal radius) if angulation ≤20° and ≥2 yr growth remain

8-year-old, 12° dorsal angulation at 1 week → acceptable → **no sequelae**
✗ Wrong options: ↓ rotation · malunion · nonunion

– – Active space – –

Operative criteria

Indication	Treatment
Unacceptable alignment after CR (beyond table)	Fixation
Bayonet apposition **>10 yr**	Fixation
Both-bone fracture **<13 yr**	**Elastic IM nails**
Both-bone fracture **>13 yr** (near maturity)	**ORIF (plate)**
Open fracture, **refracture**	ORIF
Comminuted / segmental, highly displaced	ORIF (relative)
Malunion with callus + lost rotation (missed follow-up)	**ORIF** (open callus removal) (✗ Wrong options: brace · recast · repeat CR · cast wedging)

Pediatric both-bone forearm fracture

- Acceptable after CR → **cast, weekly X-rays**
- Unacceptable / lost reduction
  - <13 yr, simple midshaft → **elastic IM nails**
  - >13 yr, comminuted, open, refracture → **ORIF**

Casting technique – cast index & loss of reduction

Closed reduction traction (finger traps), thumb pressure, rotation, **3-point bending** · **apex volar → pronate** · **apex dorsal → supinate**

Plastic deformation accept if **≤20°** · reduce with **slow steady 3-point bending** (abrupt force → fracture)

Short vs long arm cast (distal BBFF) – **no difference** in loss of reduction · SAC fewer complications
✗ Wrong options: more swelling, pain, remanipulation

Cast index **lateral inner width ÷ AP inner width** · **>0.8 → loss of reduction**
✗ Wrong options: pain · physeal arrest · nonunion · swelling

Ideal cast index (TO) **0.7** (sagittal ÷ coronal width)

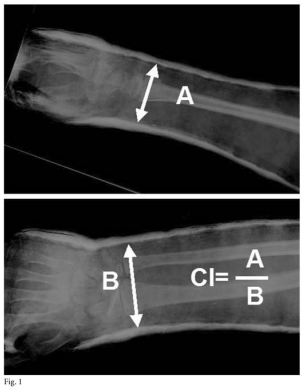


Fig. 1
The cast Index (CI) is determined by dividing the sagittal cast width (A) by the coronal cast width (B) at the fracture site.

Cast index

Remanipulation risk — **initial fracture displacement** (angulation + translation) · same-level radius + ulna fractures · cast index >0.8 · residual deformity
✗ Wrong options: mechanism · distal location · below-elbow cast · cast index <0.7

Swelling — bivalve cast (compartment syndrome)

Operative fixation – intramedullary nails vs plates

Ulnar nail — **olecranon tip or through anconeus** (protect ulnar nerve)

Ulnar nail entry (TO) — just distal to olecranon apophysis · radius just proximal to physis

Radial nail — just proximal to **radial styloid** or dorsal distal radius proximal to physis · dorsal central start risks **EPL**

Technique — reduce easiest bone first · **open the fracture after 2–3 failed passes** · nail removal at 3–4 months

	Elastic IM nails	ORIF plate
Time	**Shorter**	Longer
Blood loss	**Less**	More
Union, **radial bow**, rotation	**Equal**	Equal
Refracture	**Lower** (no screw holes)	Higher after plate removal
Best for	Midshaft, transverse/oblique, minimal comminution, **acute** (pre-callus)	Comminution, older, open

TO points — midshaft acceptable angulation: infants **< 25°**, children **< 10°** · displacement up to **90%** in young children · **any rotational deformity → reduce** (rotation never remodels) · above-elbow cast, **3-point moulding** · **plastic deformation does not remodel** → reduce if visible deformity or ↓ pronation-supination

6 Complications

Complication – loss of forearm rotation

Most common long-term after IM fixation — **loss of pronation/supination** (soft-tissue injury, stripping)
✗ Wrong options: infection · malunion · refracture · nonunion

Refracture — **5–10%** · risks: **plate removal**, **greenstick** pattern → **ORIF**

Malunion — loss of rotation · from initial reduction or delayed diagnosis → **corrective osteotomy**

Compartment syndrome — high energy · **multiple reduction / nail-passage attempts → fasciotomy**

Synostosis — rare · head injury, high energy · resection rarely helps

How it's asked

"14-year-old both-bone fracture – acceptable angulation, malrotation, bayonet?" → **0°, 0°, none**

– – Active space – –

"8-year-old, 12° angulation after loss of reduction — outcome?"	→ **no additional sequelae**
"7-year-old acceptable alignment at 1 week?"	→ **observation** ✗ Wrong options: repeat CR · ORIF · IMN
"Short vs long arm cast after anatomic CR of distal BBFF?"	→ **no difference in outcomes**
"Cast index 0.95 — risk?"	→ **loss of reduction**
"Factor predisposing to remanipulation?"	→ **initial fracture displacement**
"IMN vs plate — true statement?"	→ **radial bow restoration similar** ✗ Wrong options: IMN higher nonunion · more blood loss · longer surgery
"Missed follow-up, 6 weeks, healed malaligned, lost pronosupination?"	→ **ORIF radius + ulna** ✗ Wrong options: cast wedging · repeat CR
"Radiographic rotational alignment?"	→ **AP: radial styloid + biceps tuberosity 180° apart**
"Most common long-term complication after IM fixation?"	→ **loss of forearm rotation**

– – Active space – –

BRACHYMETATARSIA

– Pediatrics · 11 / 112 –

1 Basics

Epidemiology

Sex	**female:male 25:1**
Location	**4th metatarsal** most common · **often bilateral**

Etiology

Cause	**premature epiphyseal (physeal) closure** of metatarsal → congenital hypoplasia of ≥1 metatarsal
Associated	**Down** · **Turner** · **Larsen** · Albright · diastrophic dwarfism

3 Presentation

Symptoms	present since birth · **discomfort with shoe wear** (e.g. heels) · forefoot pain

4 Diagnosis & Classification

Key radiology findings

Views	**weight-bearing AP + lateral**
Findings	short metatarsal(s) · **broken metatarsal parabola**



Brachymetatarsia of 4th metatarsal

5 Management

Initial management – shoe modification

First line	**extra-depth or extra-wide shoes** → generally improves symptoms ✗ Wrong options: immobilisation + NSAIDs · MTP fusion · ray resection · lengthening first

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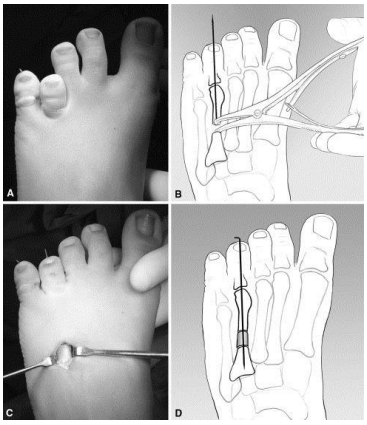
Ineffective taping, manipulative reduction

Symptomatic brachymetatarsia

- First → **extra-depth / extra-wide shoes**
- Persistent symptoms in older child → **metatarsal lengthening**
  - Very rarely → **ray amputation**

Operative criteria

Procedure	Indication
Metatarsal lengthening	**Persistent symptoms in older child** after shoe modification
Amputation (ray resection)	**Very rarely**, persistent symptoms
Extensor tenotomy + capsulotomy	**None** – insufficient correction
Fusion (MTPJ)	**Not indicated** – complications, ineffective



Metatarsal lengthening

How it's asked

"18-year-old, congenital short 4th metatarsal, new forefoot pain in heels – next step?"	→ **extra-depth shoe modification** ✗ Wrong options: 4th MTPJ fusion · ray resection · lengthening
"Most commonly affected metatarsal?"	→ **4th**
"Cause?"	→ **premature physeal closure**
"Failed shoe modification?"	→ **metatarsal lengthening**

CALCANEOVALGUS FOOT

– Pediatrics · 12 / 112 –

1 Basics

Epidemiology

Incidence	mild form up to **40% of newborns** · true deformity **1 in 1,000**
Demographics	**females**, **first-born**, young mothers

Etiology

Cause	positional, **intrauterine "packaging"** → benign **soft-tissue contracture** · hindfoot **eversion + dorsiflexion**

3 Presentation

Exam	excessively dorsiflexed hindfoot, **passively correctable** · **dorsum of foot rests on anterior tibia**
Associated	**posteromedial tibial bowing** (almost always some calcaneovalgus)



Calcaneovalgus foot – dorsum against tibia

4 Diagnosis & Classification

Diagnosis	clinical: calcaneovalgus foot **without congenital deformity or dislocation**

Key radiology findings

Not required if foot flexible enough to exclude CVT

AP + lateral tibia	look for **posteromedial bowing**
Plantarflexion lateral	distinguishes from **CVT** · before navicular ossifies (**age 3**) use **1st metatarsal** as proxy · calcaneovalgus: **1st MT lines up with talus** · CVT: talar axis passes **plantar** to 1st MT on standard + plantarflexion laterals
Apex of deformity	calcaneovalgus: **ankle joint** · posteromedial bowing: **distal tibia**

– – Active space – –

Diagnosis & posteromedial tibial bowing

Posteromedial bowing — intrauterine positioning · named for apex (posterior + medial) · **"PosteroMedial = Probably Mild"** · bowing improves; follow for **LLD**



Posteromedial bowing — apex at distal tibia

5 Management

Calcaneovalgus foot

- Plantarflexes beyond neutral → **observation + parental passive stretching**
- Cannot plantarflex beyond neutral → **serial casting**
  (resolves spontaneously by 3–6 months)

Outcome — **resolves by 3–6 months** · severity of initial deformity **does not predict** outcome

6 Complications

LLD — when associated with **posteromedial bowing** · LLD correction = **most common surgery** for posteromedial bowing

Flexible flatfoot — years after resolution

X vs Y

	Calcaneovalgus	Congenital vertical talus
Flexibility	**Flexible**, passively correctable	**Rigid**
Hindfoot	**Calcaneus (dorsiflexion)**	**Equinus** + valgus
Midfoot	Normal	**Dorsal TN dislocation**, rocker-bottom, medial talar head
Plantarflexion lateral	**1st MT lines up with talus**	Talus plantar to 1st MT
Treatment	Stretching ± casting	Reverse Ponseti / surgery

– – Active space – –

Paralytic calcaneovalgus (L5 spina bifida)	
Overactive / spared	**Dorsiflexors (L4, L5)**, **evertors (S1)**
Weak	**Plantarflexors (S1, S2)**, **invertors (L5)**

How it's asked

"Newborn, dorsum of foot resting on tibia — associated X-ray?"	→ **posteromedial bowing of tibia** ✗ Wrong options: fibular hemimelia · anterolateral bowing · clubfoot · tibial deficiency
"Apex posteromedial tibia, dorsiflexed foot correctable to neutral?"	→ **calcaneovalgus** ✗ Wrong options: cavovarus · clubfoot · skewfoot · CVT (not correctable)
"Foot cannot plantarflex past neutral?"	→ **serial casting**
"Natural history?"	→ **resolves by 3–6 months**

– – Active space – –

CAVOVARUS FOOT IN PEDIATRICS & ADULTS

– Pediatrics · 13 / 112 –

1 Basics

Deformity components

Components	**cavus** (high arch) · **plantarflexed 1st ray + forefoot pronation** · **hindfoot varus** · forefoot adduction

Etiology

Neurologic	**~67% (two-thirds)** · identifying the neuro cause is critical

Unilateral → rule out **tethered cord / spinal cord tumour** · **Bilateral** → most commonly **Charcot-Marie-Tooth (CMT)**, often hereditary/congenital

Idiopathic	subtle, bilateral (**subtle cavus**, no weakness)
Traumatic	talus fracture malunion · compartment syndrome · crush injury
Conditions with cavovarus	**CMT** · **cerebral palsy** · **Friedreich ataxia** · spinal cord lesions · polio · amniotic band syndrome

Pathomechanics (PL vs TA, PB vs PTT) & Charcot-Marie-Tooth

Pair	Strong (spared)	Weak	Result
PL vs TA	**Peroneus longus** (hypertrophic)	**Tibialis anterior**	**Plantarflexed 1st ray** (cavus) – primary deformity
PTT vs PB	**Tibialis posterior**	**Peroneus brevis**	**Hindfoot varus** + forefoot pronation
Extrinsics vs intrinsics	**Long flexors + extensors**	**Lumbricals, interossei** (first affected)	**Claw toes** (intrinsic minus)
EHL recruited for weak TA	**EHL**	TA	**1st MTP hyperextension**, windlass drives 1st MT down

Sequence (forefoot-driven)	plantarflexed 1st MT + pronated forefoot → medial forefoot strikes first → **subtalar joint supinates** for tripod contact → **compensatory hindfoot varus** → flexible at first, **rigid with time**
Forefoot position	**pronation** ✗ Wrong options: supination (the EXCEPT answer)
CMT genetics	HMSN · **PMP22 duplication, chromosome 17** · weak TA + PB, **normal PL + PT**
Overactive muscle with cavus	**posterior tibialis** ✗ Wrong options: TA (causes DF + inversion) · toe flexors/extensors
Gait effect	transverse tarsal joint **locks in stance** → lateral column overload

3 Presentation

History	recurrent **ankle sprains**, lateral ankle pain (peroneal pathology) · **lateral foot pain** (5th MT stress fracture) · **calluses under 1st + 5th MT heads** · **plantar fasciitis**
Exam	**peek-a-boo heel** (varus heel seen from front) · prominent 1st MT fat pad · **Silfverskiöld** (gastrocnemius tightness common) · unstable gait, ↑ double-limb stance
Neuro clues	**1st dorsal interosseous wasting** of hand → **CMT** · **scoliosis** → CMT · **spinal dysraphism** · high-steppage gait
Tripod (TO)	foot stands on **1st MT head, 5th MT head, calcaneus** · CMT: intrinsics weaken first; PL overpowers weak TA → 1st ray drops → hindfoot varus
Cavus mimicking equinus (TO)	hide the forefoot: heel may be neutral · lateral X-ray **high calcaneal pitch with normal tibiocalcaneal angle = cavus** · TAL would cause calcaneus · CORA at navicular → osteotomise **medial cuneiform** (plantar opening wedge for cavus, medial opening for adductus)

Coleman block test – hindfoot flexibility & forefoot-driven deformity

Used for	**cavovarus foot** (TO) ✗ Wrong options: CVT · planovalgus · metatarsus adductus
Tests	**hindfoot flexibility** ✗ Wrong options: forefoot flexibility · ankle instability · Achilles tightness (that's Silfverskiöld)
Technique	**1–1.5 inch block under lateral foot + heel** → 1st ray drops off medially → removes forefoot contribution
Flexible	heel corrects to **neutral or valgus** → forefoot-driven → **forefoot procedures** only
Rigid	heel **stays in varus** → **lateralising calcaneal osteotomy** + forefoot procedures



Coleman block test – flexible hindfoot corrects

Other signs (distractors)	"too many toes" = forefoot abduction in flatfoot · inability to single-heel-rise = PTT insufficiency

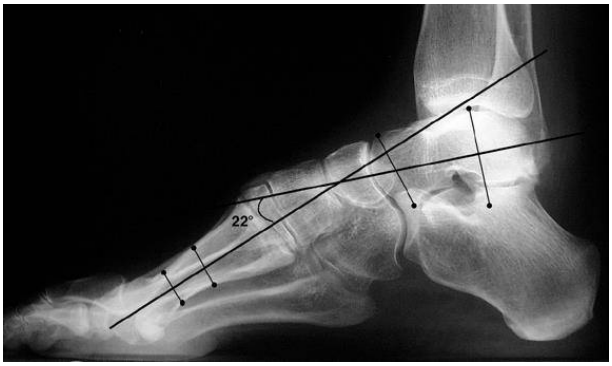
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4 Diagnosis & Classification

Key radiology findings

Views standing AP + lateral ankle · standing AP, lateral, oblique foot

View	Measurement	Cavovarus value
AP foot	**Talocalcaneal (Kite) angle**	**<20°** (nl 20–45°) → hindfoot varus
AP foot	Talonavicular coverage	**>7°** overcoverage → forefoot adduction
AP foot	Metatarsal overlap	Forefoot pronation
Lateral foot	**Meary's (talo–1st MT) angle**	**>4° apex dorsal** (1st MT line passes dorsal to talus)
Lateral foot	**Calcaneal pitch**	**>30°**
Lateral foot	**Sinus tarsi see-through, double talar dome**	Hindfoot ER relative to cassette
Lateral foot	Bell-shaped cuboid · ↑ 5th MT base–medial cuneiform distance	–
Oblique	MT stress fractures, CN coalition	–



Meary's angle – apex dorsal break

Calcaneal pitch **> 30°** (Orthobullets) · **> 25–30°** (TO) – read the options

Half of cavus = CMT (TO) treat as neuromuscular until proven otherwise (TO)

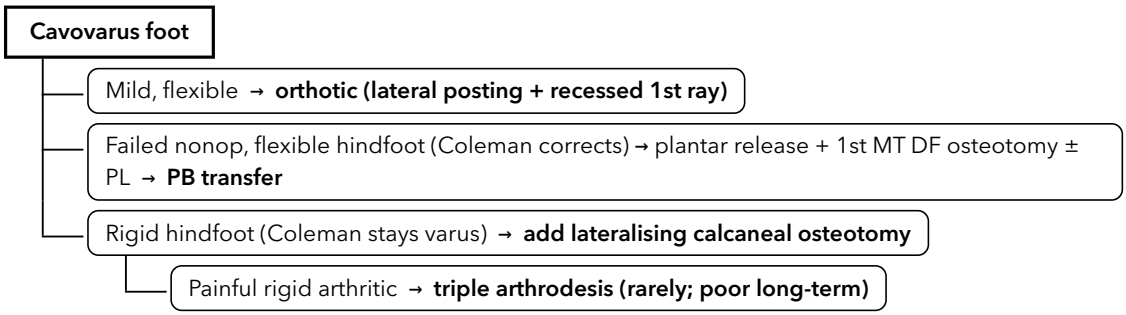
Work-up – electrodiagnostic studies

CMT diagnostic algorithm **neurologic exam → EMG/NCS → genetic testing** (genetic first if family member has known CMT)

Bilateral cavovarus, lateral pain, hand intrinsic wasting, family history → electrodiagnostic studies next

✗ Wrong options: muscle biopsy (dystrophies) · nerve biopsy (only if inconclusive) · CK/aldolase · echo

5 Management



Orthoses – lateral hindfoot posting & recessed first ray

Supple adult cavovarus	**full-length semi-rigid orthotic with 1st-ray recess + lateral hindfoot posting/wedge** · not indicated in children (Orthobullets) ✗ Wrong options: medial posting (flatfoot) · walker boot · accommodative · lace-up
Other orthotic matches	hallux rigidus → **Morton's extension** · midfoot arthritis → rocker-bottom shoe · flexible planovalgus → **medial posting** · rigid cavovarus → orthoses don't work
More severe	**SMO** · **AFO** if dynamic equinus (equinocavovarus) · lace-up brace / high-top if rigid bracing not tolerated · accommodative shoes rarely enough

Surgery – first metatarsal dorsiflexion osteotomy, plantar release, calcaneal osteotomy, EHL transfer

Problem	Procedure
Cavus	**Plantar fascia release** · Steindler stripping (short flexors off calcaneus)
Plantarflexed 1st ray	**PL → PB transfer** (↓ 1st-ray PF without losing eversion) · **1st MT dorsiflexion osteotomy**
Flexible hindfoot varus (Coleman corrects)	**1st MT DF osteotomy** ± plantar release – calcaneus spared
Rigid hindfoot varus (Coleman fails)	**Lateralising (lateral slide / Dwyer closing-wedge) calcaneal osteotomy** + 1st MT DF osteotomy
PTT overpull / TA weakness	**PTT transfer to dorsum**
Equinus	**Gastrocnemius recession** (less weakness, with plantar release) · **TAL staged weeks after** plantar release
Chronic lateral ankle instability	**Broström** (fails unless deformity corrected)
Clawed hallux	**Modified Jones: EHL → 1st MT neck + IP fusion**
Lesser toe clawing	Extensors → 2nd–5th MT necks
Painful rigid arthritic hindfoot	Triple arthrodesis – **almost never** (poor long-term)

Clawed hallux distractors	✗ Wrong options: EHL to proximal phalanx / FHB tenotomy (worsen MTP hyperextension) · original Jones (EHL to neck only – inadequate) · FHL tenotomy alone · MTP + IP fusion (stiff ray) · Keller · metatarsal head resection
Flatfoot procedures (distractors)	✗ Wrong options: medial calcaneal slide · lateral column lengthening · FDL transfer · 1st TMT fusion in plantarflexion

– – Active space – –

Equinus – gastrocnemius recession (Silfverskiöld)

Silfverskiöld	DF ↑ with **knee flexion** → **isolated gastrocnemius** tightness → **gastrocnemius recession** · tight in both positions → **Achilles/soleus** → TAL
Diabetic equinus after casting	toe-drag in swing + DF improves with knee flexion → **gastrocnemius recession** ✗ Wrong options: Thompson positive · too-many-toes · fixed dorsiflexion

6 Complications

Ankle instability	lateral ligament reconstruction **fails** if cavovarus not corrected · → **varus ankle arthritis**
Stress fractures	**5th MT base (Jones)** · 4th MT · **navicular** · **medial malleolus** (↑ medial compression; ↑ nonunion → cast NWB)
Others	**hallux sesamoiditis** (plantarflexed 1st MT) · **peroneal tendon pathology** (**PB** most common) · **plantar fasciitis** · metatarsalgia, calluses
NOT caused by cavovarus	**hallux valgus** (→ planovalgus)

X vs Y

	Flexible hindfoot	Rigid hindfoot
Coleman block	Heel → **neutral / valgus**	Heel **stays varus**
Driver	**Forefoot** (plantarflexed 1st ray)	Fixed hindfoot
Orthotic	**Lateral posting + recessed 1st ray**	Not effective
Surgery	**1st MT DF osteotomy + plantar release** ± PL→PB	**+ lateralising calcaneal osteotomy**

How it's asked

"Coleman block test assesses?"	→ **cavovarus foot** · **hindfoot flexibility** (TO)
"Had lateralising calcaneal + 1st MT DF osteotomy – preop finding?"	→ **heel stays in varus with block under lateral foot**
"Calluses under 1st + 5th MT heads; heel corrects on block?"	→ **flexible hindfoot varus**
"Adult cavovarus not correcting on Coleman?"	→ **1st MT DF osteotomy + lateral displacement calcaneal osteotomy** ✗ Wrong options: LCL · medial slide
"First-ray plantarflexion: weak X, strong Y, transfer Y to Z?"	→ **TA weak, PL strong, PL → PB**
"Meary's line dorsal to talus, neuromuscular disorder – overactive muscle?"	→ **posterior tibialis**
"PMP22 duplication – EXCEPT?"	→ **forefoot supination** (it's pronation)

– – Active space – –

"Potential result of cavovarus EXCEPT?"	→ **hallux valgus** (✓ instability, 5th MT fracture, peroneal subluxation, plantar fasciitis)
"PMP duplication chr 17 — relative strengths?"	→ **weak PB, normal PT**
"CMT child warned of higher risk of?"	→ **medial malleolus stress fracture**
"Bilateral cavovarus, hand wasting, family history — next diagnostic step?"	→ **electrodiagnostic studies**
"Orthotic for flexible adult cavovarus?"	→ **lateral hindfoot posting with recessed 1st ray**
"Pictured orthotic suits?"	→ **supple adult pes cavus**
"Coleman → 3° valgus, failed orthotic — part of surgery?"	→ **1st ray DF osteotomy + plantar fascia release** ✗ Wrong options: triple · subtalar fusion · medial slide
"Fixed clawed hallux?"	→ **EHL to MT neck + IP fusion (modified Jones)**
"HMSN 12-year-old, block test corrects heel?"	→ **plantar release + 1st MT osteotomy ± tendon transfers** ✗ Wrong options: calcaneal osteotomy · triple
"Clawed hallux in cavovarus, painless MTP/IP motion?"	→ **1st IP fusion + EHL transfer**
"Hindfoot corrects on Coleman, no arthritis — include?"	→ **1st ray DF osteotomy**
"PL→PB, Broström, lateralising calcaneal osteotomy — supporting exam?"	→ **↑ inversion vs other side + no change on Coleman**
"Equinus — when is gastrocnemius recession right?"	→ **DF improves with knee flexion**

– – Active space – –

CEREBRAL PALSY - FOOT CONDITIONS

– Pediatrics · 14 / 112 –

1 Basics

Foot deformity — **most common deformity in CP** · forms: **equinus**, **hallux valgus**, **equinoplanovalgus**, **equinocavovarus**

Deformity	Typical CP type	Muscle imbalance
Equinus	All; **isolated in hemiplegia**	Spastic/contracted **gastrocsoleus** vs dorsiflexors
Equinoplanovalgus	**Spastic diplegia + quadriplegia**, bilateral	**Spastic peroneals + heel cord**, **weak PTT**, lax ligaments
Equinocavovarus	**Spastic hemiplegia**	**Invertors (PTT ± TA) overpower peroneals**
Hallux valgus	**Diplegics** with equinus + planovalgus	**Adductor hallucis overactivity** + external forces

Lever-arm dysfunction — planovalgus and cavovarus both disrupt push-off / second rocker

3 Presentation

Equinus

Symptoms — shoe wear, **tripping** (poor clearance), instability

Exam — forefoot calluses · **toe walking, no heel strike** · compensatory **knee hyperextension** · late equinoplanovalgus

True equinus gait — toe walking with **lack of heel strike** at initial contact · **plantarflexion moment** in stance
✗ Wrong options: ↑ DF in stance/swing · DF moment

Silfverskiöld — DF ↑ with **knee flexed → gastrocnemius** tight · equal → **Achilles** tight · DF is **worse with knee extended** when gastroc spastic (crosses knee)

Planovalgus trap — uncontrolled subtalar joint **overestimates** ankle DF → **correct heel valgus first** before testing for Achilles contracture

Equinoplanovalgus

Findings — bilateral · **valgus heel** · **talar head callus** in arch · **midfoot break** · **hallux valgus develops over time**

Equinocavovarus

Findings — lateral-border / **5th MT calluses** (ulcers) · in-toeing · foot drop if TA weak · supination on TA firing = TA is main source

4 Diagnosis & Classification

– – Active space – –

Key radiology findings

Deformity	Views	Findings
Equinus	Not required unless hindfoot malalignment	–
Equinoplanovalgus	**WB AP ankle** (rule out **ankle valgus**) + foot	**↓ calcaneal pitch** · **negative talo–1st MT angle** · talar head uncovering
Equinocavovarus	WB foot + ankle	**Metatarsal overlap** · **↑ calcaneal pitch**
Hallux valgus	WB series	↑ HVA, IMA, DMAA, HVI



Equinoplanovalgus in CP

5 Management

Equinus – botulinum toxin, serial casting & gastrocnemius recession

Treatment	Indication
Serial casting	Mild, dynamic, young · **unbraceable foot** (can't reach neutral) after botox → cast, then brace
Botulinum toxin A (gastrocnemius)	Mild spasticity, delays surgery · **prevents presynaptic release of acetylcholine (ACh)** · not more often than **every 3 months**
Hinged / articulated AFO	Mild, passively correctable, no myostatic contracture · not if excessive midstance DF
Solid AFO	Mild–moderate, partly correctable, mild contracture · not if heavy, ≥12 yr, rigid malalignment
Gastrocnemius recession	**Silfverskiöld positive** (isolated gastroc) · less immobilisation, **higher recurrence**
TAL	Gastrocsoleus (Achilles) tight · rarely isolated **except in hemiplegia** · walking cast 4–6 wk

Botox mechanism distractors ✗ Wrong options: binds nicotinic receptor (curare, succinylcholine) · destroys receptor (myasthenia) · sarcolemma instability (dystrophy)

Surgery timing delay equinus surgery until **≥4 yr** (OB) · **≥6 yr** to prevent recurrence (OB) – read the options

Hemiplegic 3-year-old, toe-walking, DF short of neutral after botox → serial casting of affected side

– – Active space – –

✗ Wrong options: bilateral AFOs (unaffected side toe-walks in compensation) · heel-cord lengthening · repeat botox at 1 month · baclofen pump (diplegia)

Gastrocnemius recession technique	posterior/posteromedial incision · protect **sural nerve** · incise aponeurosis transversely or U-shape · cast in neutral

Equinovarus – split tibialis posterior transfer

Flexible equinocavovarus (hemiplegia)

- PTT EMG continuous → **SPOTT (split PTT to peroneus brevis), age 4–7**
- PTT EMG out-of-phase → **whole PTT transfer to dorsum (lateral cuneiform)**
- Spastic TA (supination on TA firing) → **SPLATT ± TAL + PTT lengthening**
- Fixed deformity → **add calcaneal osteotomy / arthrodesis**

SPOTT	**passively correctable** · **age 4–7** · **spastic hemiplegia** · **continuous PTT EMG** · posterior half rerouted behind tibia (anterior to NV bundle) → woven into **PB** · more consistent than whole transfer ✗ Wrong options: athetosis (unpredictable) · rigid deformity · quadriplegic/diplegic valgus
SPLATT	passively correctable + **spastic TA** · lateral half → **cuboid** tunnel · not for **weak TA / foot drop**
Bony	**lateral slide or lateral closing-wedge calcaneal osteotomy** (fixed varus) · **triple arthrodesis** (severe fixed; recurs if soft tissues not balanced)
Bracing	supple only · rarely successful, worsens calluses

Equinoplanovalgus

Flexible → bracing

Pain / pressure sores despite bracing → soft tissue + bone

Soft tissue	Bone
TAL · **peroneus brevis lengthening** · **PTT advancement**	**Medial slide calcaneal osteotomy** · **lateral column lengthening (Evans)** (trapezoidal graft anterior to middle facet) · **Grice** (extra-articular subtalar, **does not affect tarsal growth**) · **subtalar arthroereisis** (plug; breakdown)



Grice extra-articular subtalar arthrodesis

Hallux valgus – first MTP arthrodesis

Painless, shoes OK → observe

Painful (failed accommodative shoes) → 1st MTP arthrodesis – **highest success** in ambulatory + nonambulatory CP

✗ Wrong options: double/crescentic MT osteotomy · modified McBride · night orthotics (recurrence unacceptably high)

Valgus interphalangeus → **Akin** proximal phalanx osteotomy

– – Active space – –

6 Complications

Planovalgus surgery	**overcorrection into varus (most common)** · **sural nerve injury** · **overlengthened lateral column → lateral forefoot overload** · CC joint destabilisation
Cavovarus	**recurrence** if all components not addressed

X vs Y

	Equinoplanovalgus	Equinocavovarus
CP type	**Diplegia / quadriplegia**	**Hemiplegia**
Imbalance	Spastic peroneals, weak PTT	**PTT ± TA > peroneals**
Callus	**Talar head**	**Lateral border / 5th MT**
Calcaneal pitch	↓	↑
Surgery	LCL, medial slide, Grice, arthroereisis	**SPOTT / SPLATT**, lateral calcaneal osteotomy

How it's asked

"5-year-old hemiplegic, flexible equinocavovarus, lateral ulcers, continuous PTT EMG?"	→ **split posterior tibial tendon transfer** ✗ Wrong options: PTT to lateral cuneiform (out-of-phase) · triple · calcaneal slide (fixed)
"Best candidate for SPOTT?"	→ **6-year-old spastic hemiplegic, flexible equinovarus** ✗ Wrong options: athetosis · rigid
"Spastic diplegic with true equinus gait — expected?"	→ **toe walking, no heel strike**
"Botulinum toxin mechanism?"	→ **prevents presynaptic ACh release**
"3-year-old hemiplegic, DF −5° after botox 1 month ago?"	→ **serial casting** of affected side
"Isolated gastrocnemius recession best when?"	→ **DF 15° knee flexed vs neutral knee extended** ✗ Wrong options: too-many-toes · Thompson · equal DF · no single heel rise
"15-year-old CP community ambulator, painful hallux valgus?"	→ **1st MTP arthrodesis**

– – Active space – –

CEREBRAL PALSY - GAIT DISORDERS

– Pediatrics · 15 / 112 –

1 Basics

Gait disorder — **primary reason for orthopaedic consultation** in CP

Ambulatory prognosis & milestones

Independent walking — expected **12–18 months**
✗ Wrong options: towers of 4 blocks = 2–3 yr · transfers toys = 6–9 mo · jumping = 2–3 yr · overhand throw = 18–24 mo

Good prognostic sign for walking — **positive parachute reaction** (reaches toward floor when lowered prone)

Poor signs — **Moro beyond 6 months** · **asymmetric tonic neck reflex**, **extensor thrust**, **absent foot placement** (abnormal at any age)

Pathophysiology – deviations

Level	Definition	Examples
Primary	Direct CNS insult	**Spasticity**, weakness, ↓ proprioception
Secondary	Growth-related from abnormal loading	**Myostatic contractures**, **femoral anteversion**, hip subluxation, equinoplanovalgus
Tertiary	**Compensations** for secondary deviations	–

Modern approach — define deviation, **plane** (sagittal, coronal, transverse) and **level** (hip, knee, ankle)

4 Diagnosis & Classification

Gait patterns – crouch, jump & stiff-knee (rectus femoris)

Pattern	Hip	Knee	Ankle	Note
True equinus	Normal	Normal / hyperextended	**Equinus** (foot < plantigrade)	Isolated, **hemiplegics** · GRF anterior to knee
Jump gait	**Flexed**	**Flexed**	**Equinus**	+ anterior pelvic tilt · diplegia
Apparent equinus	Flexed	Flexed	**Plantigrade-capable** but toe-walks	Proximal deviation (knee flexion)
Crouch gait	**Flexed**	**Flexed**	**Excess dorsiflexion (calcaneus)**	GRF **posterior to knee** · diplegia, quadriplegia at puberty
Stiff-knee gait	–	**↓ swing flexion**	–	**Rectus femoris** out of phase
Asymmetric	Each limb in a different group			–

Crouch causes — **iatrogenic isolated TAL in jump gait** · pubertal growth spurt → ↓ strength-to-weight · breaks plantarflexion–knee-extension couple → very inefficient, tiring

Stiff-knee gait — **rectus femoris overactive in pre-swing + early swing** · EMG: quadriceps active from **terminal stance through swing** · compensation from hip deviation → surgery won't help
✗ Wrong options: adductor magnus · gluteus medius · sartorius · TFL

Descriptive classifications — fail to classify **up to 40%** of patterns

		Sagittal Kinematics				Treatment	
Sagittal Gait Pattern		Pelvic Tilt	Hip Flex/Ext	Knee Flex/Ext	Ankle DF/PF	Orthosis	Surgery
Group I	True Equinus	Normal or anterior	Normal	Normal or recurvatum	Equinus	Hinged AFO or posterior leaf spring AFO	Gastroc lengthening
Group II	Jump Gait	Normal or anterior	Normal or flexed	Flexed	Equinus	Hinged AFO or posterior leaf spring AFO	Gastroc, hamstring & psoas lengthening; transfer of rectus femoris to semitendinosus
Group III	Apparent Equinus	Normal or anterior	Flexed	Flexed	Normal	Solid AFO	Hamstring & psoas lengthening
Group IV	Crouch Gait	Anterior, normal or posterior	Flexed	Flexed	Calcaneus	Ground reaction AFO	Hamstring & psoas lengthening; correction of medial femoral torsion, lateral tibial torsion and stabilization of the foot
Group V	Asymmetrical Gait	Combination of 2 of the above patterns				Combination	Combination

Sagittal gait patterns in spastic diplegia

Key radiology findings & gait analysis

Quantitative gait analysis — **more accurate than observation** · physical exam (spasticity, contractures, torsion) · **kinetics** (forces, moments) · **kinematics** (3-plane motion) · **pedobarography** · **dynamic EMG** · video

Planes — sagittal: pelvic tilt, hip/knee/ankle flexion · coronal: pelvic obliquity, hip ab/adduction · **transverse = least reliable** (rotation, foot progression)

– – Active space – –

5 Management

Equinus management – AFO, serial casting & gastrocnemius recession

Orthosis	Indication
Solid AFO	Flexible equinus — ankle passively correctable to neutral (subtalar neutral) · stance + swing control · excessive midstance DF
Posterior leaf-spring (PLSO) / hinged	**Absent heel strike + excessive swing plantarflexion + some passive DF (~5°)**
Floor-reaction AFO	**Crouch** gait
KAFO	Knee instability in minimally ambulatory/standers
SMO	Flexible **coronal** hindfoot supination/pronation

3-year-old diplegic (PVL, walked at 2), mild contractures, gastroc tightness → ankle bracing + therapy

✗ Wrong options: percutaneous/open TAL (surgery not indicated at this age) · steroids (dystrophy)

Botulinum toxin A temporise / primary · **hamstring spasticity without fixed deformity** in ambulators

Silfverskiöld + popliteal angle — choosing surgery

DF knee extended	DF knee flexed	Popliteal angle	Procedure
−5°	**+20°**	**25° (normal)**	**Gastrocnemius recession only** ✓
−5°	+20°	55°	Gastroc recession **+ hamstring lengthening**
−10°	−10°	25°	**TAL** (Achilles)
−20°	−20°	55°	TAL + hamstring lengthening
+20°	+20°	55°	Hamstring lengthening only

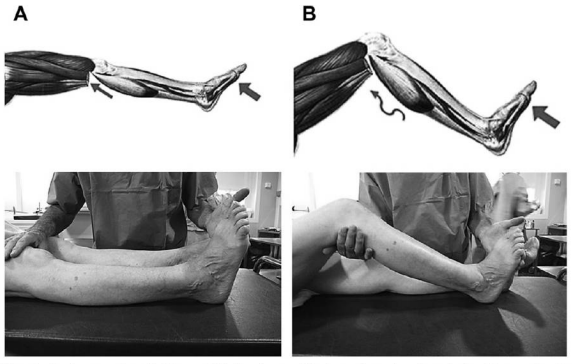


Fig. 1. Gastrocnemius tightness. (*A*) Equinus with knee in extension, but with moderate strength applied. (*B*) Equinus disappears when the knee is flexed. (*Adapted from* Barouk S. Forefoot reconstruction. New York: Springer-Verlag; 2003.)

Silfverskiöld test

Normal popliteal angle ~25° · >50° = significant hamstring tightness

Gastrocnemius recession **Silfverskiöld positive** · protect **sural nerve** (superficial to fascia) · divide tendon only · goal **10° DF**

TAL rigid (not correctable to neutral), true equinus, Silfverskiöld negative · **contraindicated in spastic diplegia** → weakness, **calcaneus/crouch**

Operative criteria – SEMLS

SEMLS — **gold standard** · correct all primary + secondary deviations at multiple levels in **one** operation (avoids iatrogenic worsening) · AFOs + aggressive PT · **~1 year recovery**

Deviation	SEMLS procedure / criterion
↑ Femoral anteversion (lever-arm)	**External rotation proximal femoral osteotomy**
Hip flexion contracture	**Intramuscular psoas lengthening**
Mild knee flexion (<5° fixed), young	**Medial hamstring fractional lengthening** (✗ Wrong options: lateral (excess weakness))
Knee flexion **10–25°**, **≥2 yr growth**	**Guided growth** (anterior distal femur)
Knee flexion **10–30°** + quad lag, at/near maturity	**Distal femoral extension osteotomy** ± patellar tendon advancement
Stiff-knee gait	**Distal rectus femoris transfer** posterior to knee axis (better than release)
Equinus	TAL or gastrocnemius recession
Flatfoot	Reconstruction

Selective dorsal rhizotomy

Permanent tone reduction — **selective posterior (dorsal) rhizotomy**
✗ Wrong options: tone-reducing AFOs · botox (wears off 3–6 mo via axonal sprouting) · single intrathecal baclofen injection (pump gives prolonged reduction) · fractional lengthening

Rhizotomy drawbacks — excess weakness · **hip dislocation** · **spinal deformity**

6 Complications

Recurrent hamstring contracture — esp. jump gait

Worsening crouch — isolated / over-lengthened **Achilles**

Patella alta — multiple releases **without gait analysis** · hard to treat

Knee pain — TAL without gait analysis

X vs Y

	Jump gait	Crouch gait
Ankle	**Equinus**	**Excess dorsiflexion / calcaneus**
Hip + knee	Flexed	Flexed
GRF at knee	Anterior (early)	**Posterior**
Cause	Spastic gastroc + hamstrings + psoas	**Isolated TAL**, growth spurt weakness
Orthosis	–	**Floor-reaction AFO**

– – Active space – –

How it's asked

"Crouch gait combination?"	**→ hip flexion, knee flexion, ankle dorsiflexion**
"Stiff-knee gait in spastic diplegia — overactive muscle?"	**→ rectus femoris**
"Equinus, knee + hip flexion contractures, anterior pelvic tilt?"	**→ jump gait** ✗ Wrong options: crouch (calcaneus) · apparent equinus
"Posterior leaf-spring AFO appropriate for?"	**→ absent heel strike, swing plantarflexion, 5° passive DF**
"3-year-old hemiplegic, AFO not fitting, DF −10° after botox?"	**→ serial casting** ✗ Wrong options: repeat botox <3 mo · baclofen pump · heel-cord lengthening
"3-year-old diplegic toe-walker, mild contractures?"	**→ ankle bracing and therapy**
"Hemiplegic, ONLY gastrocnemius recession needed?"	**→ DF −5° extended, +20° flexed, popliteal angle 25°**
"Milestone at 12–18 months?"	**→ walks without assistance**
"Good prognostic sign for walking at 22 months?"	**→ positive parachute reaction**
"Permanent decrease in tone?"	**→ selective posterior rhizotomy**

CEREBRAL PALSY - GENERAL

– Pediatrics · 16 / 112 –

1 Basics

Definition & epidemiology

Definition	**non-progressive UMN disease** (static encephalopathy) from **injury to immature brain** · onset **before age 2** · musculoskeletal picture **worsens with growth** (bone grows faster than spastic muscle)
Incidence	**2–3 per 1000** · **most common cause of chronic childhood disability**
Risk factors	**prematurity (most common)** (Orthobullets) · **prenatal causes most common (70%)** (TO) – TORCH, toxins, Rh (kernicterus), malformations · perinatal: anoxia, traumatic delivery, chorioamnionitis · postnatal: meningitis, encephalitis, HIE, **NAT**
Brain MRI	**periventricular leukomalacia (PVL) most frequent (56%)** (preterm, spastic diplegia) · grey matter 18% · malformations 9%
Manifestations	primary: abnormal tone, **spasticity**, loss of motor control, impaired balance · secondary: **contractures** (dynamic → static), hip subluxation, spine deformity, foot deformity, gait disorders, **fractures**

Classification & ambulation prognosis

Physiologic	Feature
Spastic	**Most common (80%)** · velocity-dependent ↑ tone, hyperreflexia, clonus, Babinski · agonist-antagonist co-contraction · **most amenable to surgery**
Athetoid (dyskinetic)	Slow writhing involuntary movements · normal reflexes · contractures uncommon
Choreiform	Continuous purposeless distal movements
Rigid	Lead-pipe/cogwheel, no hyperreflexia · most hypertonic
Ataxic	Cerebellar · wide-based, drunken gait, tremor
Hypotonic	Precedes spastic/ataxic for 2–3 yr
Mixed	Spastic + athetoid, total body

Anatomic (geographic)	Feature
Diplegia	**Legs > arms** · usually **ambulatory** · near-normal IQ · **most common** (TO)
Hemiplegia	One side, **arm > leg** · ~50% mentally retarded (TO)
Quadriplegia	All four, **nonambulatory**, low IQ, ↑ mortality
Paraplegia	Legs only, arms spared
Double hemiplegia	All four, **arms > legs**
Monoplegia	One limb (rare)
Total body	+ absent head control, drooling, dysphagia

– – Active space – –

Lower > upper limb → diplegia
✗ Wrong options: paraplegia (arms spared) · double hemiplegia (arms > legs)

GMFCS (best motor-function score)

Level	Ability (age 6–12)
I	Walks without limits · runs, jumps · stairs without rail
II	Walks with limits – difficulty uneven ground, inclines, long distances · stairs with rail · minimal run/jump
III	**Walks with handheld mobility device** (walker, crutches) · wheeled mobility for community/long distance
IV	Self-mobility limited · **powered mobility** / pushed wheelchair
V	**Transported in manual wheelchair** · dependent in all care

Best score for motor function — **GMFCS** · EDACS = eating/drinking
✗ Wrong options: MACS (upper limb/hands) · CFCS (communication) · PEDI

Motor plateau — levels I–II by ~5 yr · III by ~8 · IV–V by ~7, may decline in adolescence

Ambulation prognosis

Most reliable predictor — **independent sitting by age 2** (OR 26:1)
✗ Wrong options: creeping · rolling · CP pattern · motor type

Poor prognosis — no sitting by **4 yr** · not walking by **8 yr** (without severe contracture) → unlikely ever

Bleck poor signs — asymmetric tonic neck reflex · **persistent Moro** · **strong extensor thrust** on vertical suspension · **persistent neck-righting reflex** · **absent parachute reaction after 11 months** → "all EXCEPT" = **presence of parachute reaction** (TO)

3 Presentation

History — perinatal, development, function (sitting, standing, hands, communication, hearing, vision)

Spine — scoliosis flexibility, pelvic obliquity · **hamstring contracture → ↓ lordosis** · **hip flexion contracture → ↑ lordosis**

Hip — flexion + **adduction** contractures · dislocation may be missed as "contracture" → thigh length sitting, Galeazzi

Foot — equinovarus, planovalgus · toe walking, knee hyperextension for heel contact · Silfverskiöld

– – Active space – –



Equinus feet in CP

Foot & knee deformity patterns (recurvatum, foot deformities)

Common CP foot	**equinocavovarus (hemiplegia)** · **equinoplanovalgus (diplegia in crouch)** · **hallux valgus** · **isolated cavus is NOT typical** → think **spinal dysraphism / CMT** (TO)
Knee recurvatum causes (TO)	quad–hamstring **co-spasticity** (quads stronger) · **weakened hamstrings** (over-lengthening/transfer) · gastrocsoleus weakness · **ankle equinus** · **quadriceps weakness** is NOT a cause
Recurvatum work-up	**prone rectus test** (quad spasticity) · SLC/AFO at neutral: still recurvatum → not from equinus · treat with long-leg braces locked in 20° flexion

4 Diagnosis & Classification

Key radiology findings

Baseline	**AP + lateral hips** · **standing spine** films

Hip surveillance (TO)

Reimers migration percentage (MP)	% of femoral head lateral to **Perkin line** (Hilgenreiner line through triradiates) · supine, hips together, patellae forward

MP	Meaning
<30%	Stable — monitor
>30% (surveillance) / **>33%** (Reimers)	**Displacement / subluxation**
>60%	High risk of dislocation · **soft tissue alone fails**
>100%	**Dislocation**
Rise **>10%/yr**	Red flag — rapid progression

– – Active space – –

GMFCS	Hip displacement risk	X-ray frequency
I	~0–5% (≈1%)	Age 2, then if gait changes
II	~15%	Age 2, then 2-yearly to 10
III	~35%	Age 2, then yearly to 10
IV	~70%	Age 2, then 6–12-monthly
V	**~90%**	Age 2, then **6-monthly**

Clinical red flags passive **abduction <30°** (knee extended) · new pain with dressing/hygiene/sitting · new LLD

Hip at risk flexion contracture **>20°** + abduction **<30°** · ↑ neck-shaft angle (**coxa valga**) · ↑ **femoral anteversion** · acetabular dysplasia (**posterosuperior** in CP vs anterolateral in DDH)

Dislocation by type **70–90% occur in quadriplegia**

5 Management

Spasticity management – botulinum toxin, baclofen (withdrawal) & SDR contraindications

Agent	Mechanism	Use / note
Botulinum toxin A	Binds **motor nerve terminal** → enters → cleaves **SNARE** → **blocks presynaptic ACh release** (flaccid chemical denervation)	**Dynamic** contracture, too young for surgery · little benefit in **static** contracture · lasts **2–3 mo** (Orthobullets) / **3–4 mo** (OB) / 3–6 mo (OB) → reverses by axonal sprouting · inject belly near nerve entry
Baclofen	**GABA-B agonist** (spinal cord, central)	**Intrathecal** preferred (oral → sedation, poor bioavailability) · **nonambulatory GMFCS IV–V**
Lidocaine	Na^+ channel block	–
Acetylcholinesterase	Degrades ACh	–
Pyridostigmine / neostigmine	AChE inhibitors → ↑ ACh	–
Trihexyphenidyl	Muscarinic antagonist	–
Gabapentin	Ca^{2+} channel	–
Methacholine	Direct cholinergic agonist	–

Botox distractors ✗ Wrong options: destroys ACh receptor (myasthenia gravis) · binds muscle receptor (curare, succinylcholine)

Baclofen withdrawal (pump failure) rebound rigidity, myoclonus, **hyperthermia**, **rhabdomyolysis → ↑ creatinine**, **↑ K^+**, **↑ CPK**, multi-organ failure
✗ Wrong options: ↓ K^+ · ↓ CPK · hypothermia · atony

Spasticity grading (TO) **R1** = angle of muscle catch · **R2** = passive ROM · **R2 − R1 >20°** = significant **dynamic** component → **botox**

Phelps test hip abduction ↓ only with knee extended → **gracilis** spasticity

5-yr GMFCS III diplegic, jump gait, DF R1 −10/R2 +20, popliteal R1 80/R2 40, abduction 45° flexed / 20° extended → botox to gastrocnemius + gracilis + semitendinosus +

semimembranosus (TO)
✗ Wrong options: surgery · adductor release (abduction already 45°)

Selective dorsal rhizotomy

Procedure	section **L2–S1 dorsal rootlets** with abnormal response to stimulation
Ideal candidate	**spastic diplegia**, ambulatory, **age 4–8** (Orthobullets, OB) / **3–8** (TO), GMFCS II–III, voluntary + trunk control, pure spasticity, **no fixed contractures**, preterm, motivated family · improves **one GMFCS level** (TO)
Contraindications	**athetosis** (unpredictable) · **nonambulatory spastic quadriplegia** (spinal deformity) · hemiplegia not recommended (TO) ✗ Wrong options: "hemiplegics benefit most before 5"
Complications	**hip subluxation/dislocation** · lumbar **hyperlordosis** (esp. >60° preop) · scoliosis · spondylolysis/listhesis · weakness

Athetoid 5-year-old – contraindicated? → SDR
✗ Wrong options: botox · UE tendon transfer · SEMLS · intrathecal baclofen

Operative principles

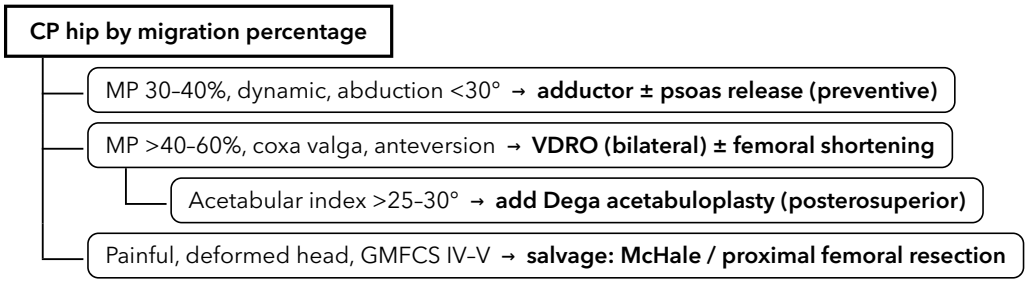
Muscle behaviour	Procedure
Continuously active	**Tenotomy** (hip adductor) or **lengthening** (Achilles, hamstring)
Out of phase	**Tendon transfer** (rectus femoris, tibialis posterior)

Soft-tissue surgery	age **3–5** with spasticity + voluntary control
Bony surgery	later childhood/adolescence
SEMLS	one event, gait-lab guided · simple lengthenings "uncover" other contractures and worsen gait

Upper extremity surgery – voluntary control predicts outcome

Voluntary motor control	best predicts improvement after **wrist/hand tendon transfers** ✗ Wrong options: botox · SDR · femoral/tibial derotation osteotomy

Hip subluxation in GMFCS V – adductor release



Adductor release	goal **>60° abduction hips flexed, 45° extended** · bilateral (avoid **windswept** pelvis) · then abduction splinting 6 mo · avoid excessive release in IV/V → abduction contracture
Psoas	flexion contracture **15–30° → intramuscular recession over pelvic brim** · **>30°** → add rectus, sartorius, TFL, anterior glutei release · ambulators: fractional at brim (complete lesser-trochanter release → flexor void, crouch)

– – Active space – –

VDRO — neck-shaft angle **110–115°** · correct anteversion · **shorten 1–3 cm** if tight (↓ AVN, sciatic tension)
✗ Wrong options: 105° → abductor weakness, Trendelenburg

Acetabuloplasty — **Dega / San Diego** (incomplete trans-iliac, hinge on triradiate) → **deeper posterior graft** for **posterosuperior** coverage · Pemberton = volume-reducing, open triradiate
✗ Wrong options: Salter (anterolateral redirection, leaves posterior uncovered) · extra varus · isolated psoas tenotomy

GMFCS V, painful sitting, hygiene problems, coxa valga + unilateral dislocation, failed therapy → bilateral adductor release + bilateral VDRO ± pelvic osteotomy
✗ Wrong options: observe · soft tissue alone (MP >60% fails) · shortening alone · unilateral VDRO (asymmetry)

5-yr GMFCS IV, MP 18% → 31% in 1 yr, abduction 25° → 6-monthly X-rays + refer for preventive soft-tissue surgery (TO)
✗ Wrong options: annual surveillance · VDRO now · discharge at 5

Operative criteria

Situation	Choice
Open reduction of dislocated hip – all 4	Moderately mature intellect · at least **sitting potential** · minimal/corrected **pelvic obliquity** · ideally **unilateral**
GMFCS IV, MP 100%, deformed head	**Femoral head resection** (McHale: resection + subtrochanteric valgus osteotomy + interposition) over reconstruction (TO)
Schanz osteotomy	Pelvic-support for sitting platform
THA	Selected higher-function painful arthritic hips
Hamstring lengthening	Popliteal angle **>45–50°** · **medial** preferred (lateral → ↑ ext. tibial torsion, lordosis)
DFEO – "rule of 15"	Fixed knee flexion **>15°** (TO) · guided growth/soft tissue below
Guided growth (anterior 8-plate)	FKFD with growth left (<11–12 yr)

Crouch gait & knee surgery (DFEO, patellar advancement)

PFKE couple — **plantarflexion–knee extension couple**: competent gastrocsoleus stops tibia falling forward · failure = **iatrogenic over-lengthened Achilles / triceps surae weakness** → crouch (TO)
✗ Wrong options: rectus spasticity · psoas contracture · hamstring shortening

Apparent equinus (toe-walking, ankle neutral, knee 35° + hip 20° flexion, **patella alta** Insall-Salvati 1.4) → **DFEO + patellar tendon advancement** (TO)
✗ Wrong options: TAL (causes severe crouch) · isolated hamstring lengthening · gastroc recession + AFO

Patellar tendon advancement/ plication — **mandatory with DFEO** in chronic crouch to restore quad lever arm (TO)

Post-DFEO anterior knee pain / extensor shift — **unaddressed patella alta** with shortened femur (TO)

GRAFO (floor-reaction AFO) shifts GRF **anterior to knee → knee extension moment** (TO)

– – Active space – –

Hamstring activity throughout stance on EMG → hamstring lengthening likely to fail (TO)

Post-op early weight bearing · night knee immobilisers for years

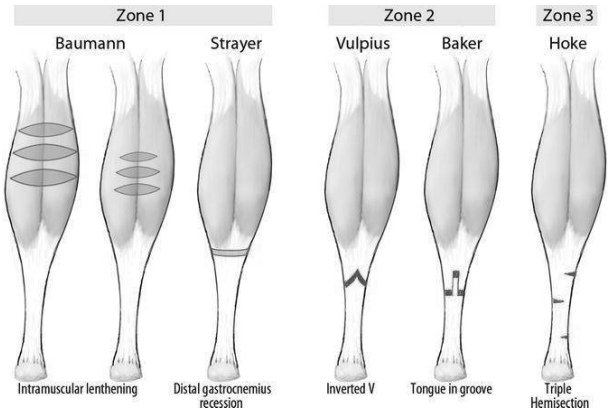
Stiff-knee gait – rectus femoris transfer

Rectus femoris transfer distal rectus → gracilis/semitendinosus · **removes swing-phase restraint → ↑ peak knee flexion** (TO)
✗ Wrong options: ↑ extension moment · pure hip extensor

Stiff knee + internal rotation in swing + anteversion (hemiplegia Winters IV, double-bump rectus EMG) → femoral derotation osteotomy + rectus femoris transfer (TO)

Equinus – gastrocnemius recession & lengthening zones

Zone	Procedure	Structures
Zone 1	**Strayer** (transverse, above junction)	**Isolated gastrocnemius** – safest vs crouch
Zone 1	Baker (tongue-in-groove)	Gastroc + partial soleus
Zone 2	**Vulpius** (inverted V at MTJ)	Gastroc + often soleus fascia
Zone 3	**Hoke triple hemisection / TAL**	**Tendo-Achilles** (TO)
Intramuscular	Baumann	Multiple deep gastroc recessions



Gastrosoleus lengthening zones

Diplegia → usually **isolated gastroc** contracture · **hemiplegia → gastrosoleus** tight (TO)

8-yr diplegic, −10° knee extended, +15° knee flexed → Strayer (TO)
✗ Wrong options: percutaneous TAL (crouch) · Vulpius (weakens soleus) · SPLATT (varus)

– – Active space – –

Neuromuscular scoliosis surgery – evaluation, outcomes & complications



Neuromuscular scoliosis in CP

Pre-op work-up	nutrition (weight) · pulmonary (PFT/sleep study/clinical) · swallow + **GERD** (aspiration) · **valproate** may impair coagulation (↓ vWF) · **MRI NOT routine** (only if neuro change or rapid progression)
Most predictable outcome	**↑ caregiver satisfaction** (perceived comfort) ✗ Wrong options: cognition · nutrition · pain · mobility
Respiratory complication risk factor	**gastrostomy tube** · resp. complications **15–28%** · NMS has highest complication rate (congenital 10.6%, idiopathic 6.3%) ✗ Wrong options: lymphocytes <1000 · albumin <3.5 · reactive airway · Hb <8

Wound drainage 3 weeks after T2–pelvis fusion → culture + I&D + IV antibiotics + retain instrumentation
✗ Wrong options: oral abx · aspiration · hardware removal → rapid progression

Bone health – bisphosphonates for fragility fractures

Fragility fractures	**non-ambulators**, low BMD (immobility, nutrition, **anticonvulsants**, contractures) · prevalence 6–50% · annual 4% → **7% after a fracture**
IV pamidronate	**>3 fractures + DEXA Z-score <−2 SD** (Orthobullets) · prevents **refracture** ✗ Wrong options: LLD · nonunion · malunion · contracture
Mechanism	**nitrogen-containing** (pamidronate, alendronate) inhibit **farnesyl pyrophosphate synthase** (mevalonate pathway) → osteoclast apoptosis · non-N (clodronate, etidronate) → toxic ATP analogues
CROW walker	**Charcot foot** (TO) ✗ Wrong options: CP equinus · polio · spina bifida

X vs Y

	Botulinum toxin A	Baclofen	SDR
Site	Peripheral NMJ, presynaptic	Central spinal GABA-B	Dorsal rootlets L2–S1
Duration	Temporary (months)	Continuous via pump	**Permanent**
Best for	Dynamic, focal, young	**Nonambulatory GMFCS IV–V**	**Ambulatory spastic diplegia 4–8**
Avoid	Static contracture	Abrupt withdrawal	**Athetosis**, quadriplegia

	CP hip	DDH
Deficiency	**Posterosuperior**	Anterolateral
Pelvic osteotomy	**Dega / San Diego**	Salter, Pemberton
Driver	Spastic adductors + psoas	Capsular laxity

How it's asked

"Poor prognostic signs for walking — all EXCEPT?"	→ **presence of parachute reaction after 11 months** (TO)
"GMFCS IV at 6–12 years?"	→ **powered mobility** (TO)
"Lower limbs more affected than upper?"	→ **diplegia** (TO)
"Best score for motor function in CP?"	→ **GMFCS** (TO)
"7-year-old: walks level surfaces, rail on stairs, device for uneven/community, can't run?"	→ **GMFCS III** (TO)
"6-year-old, mild 4-limb spasticity, legs > arms, hip dysplasia?"	→ **diplegia**
"Most prognostic for walking?"	→ **independent sitting by age 2**
"SDR — all true EXCEPT?"	→ **"spastic hemiplegics benefit most before 5"** (TO)
"Botulinum toxin A action?"	→ **binds motor terminal receptors, prevents ACh release** (TO)
"Jump gait GMFCS III, R2−R1 >20°, Phelps positive?"	→ **botox gastrocnemius + gracilis + medial hamstrings** (TO)
"Athetoid diplegic — contraindicated?"	→ **selective dorsal rhizotomy**
"Decreases synaptic ACh by blocking presynaptic release?"	→ **botulinum toxin A** ✗ Wrong options: lidocaine · AChE · baclofen · pyridostigmine
"Inhibits presynaptic vesicle docking (SNARE)?"	→ **botulinum toxin A** ✗ Wrong options: trihexyphenidyl · gabapentin · neostigmine
"Baclofen pump failure sign?"	→ **elevated serum creatinine**
"Serial casting + drug blocking presynaptic ACh release?"	→ **botulinum toxin A** ✗ Wrong options: methacholine
"Voluntary control predicts improvement after?"	→ **wrist/hand tendon transfer**
"Femoral head resection over reconstruction when?"	→ **GMFCS IV, MP 100%, deformed head** (TO)
"GMFCS IV, MP 18% → 31% in a year?"	→ **6-monthly films + refer for soft-tissue surgery** (TO)
"After VDRO head still uncovered posterosuperiorly?"	→ **Dega acetabuloplasty, deeper posteriorly** (TO)

– – Active space – –

"GMFCS V, painful sitting, dislocated hip?"	→ **bilateral adductor release + bilateral VDRO ± pelvic osteotomy**
"Pre-op CP scoliosis — EXCEPT?"	→ **routine spinal MRI**
"Infection 3 weeks after fusion?"	→ **I&D, cultures, IV antibiotics, retain hardware**
"Most predictable outcome of CP scoliosis fusion?"	→ **improved caregiver satisfaction**
"Post-op pneumonia risk factor?"	→ **gastrostomy tube**
"Bisphosphonate after CP fragility fracture prevents?"	→ **refracture**
"Apparent equinus, knee 35° flexion, patella alta?"	→ **DFEO + patellar tendon advancement** (TO)
"With supracondylar extension osteotomy, mandatory add-on?"	→ **patellar tendon distalisation/ plication** (TO)
"Crouch with failed PFKE couple — cause?"	→ **over-lengthened Achilles / triceps surae weakness** (TO)
"Restore knee extension moment arm in crouch with FKFD?"	→ **DFEO + patellar advancement** (TO)
"Anterior knee pain after DFEO?"	→ **patella alta not addressed** (TO)
"GRAFO function?"	→ **knee extension moment via floor reaction** (TO)
"Hamstring EMG active through stance?"	→ **hamstring lengthening likely to fail** (TO)
"Rule of 15?"	→ **DFEO when FKFD >15°** (TO)
"Stiff knee + IR in swing + anteversion?"	→ **FDRO + rectus femoris transfer** (TO)
"Rectus femoris transfer objective?"	→ **↑ peak knee flexion in swing** (TO)
"Hoke lengthening — zone?"	→ **zone 3** (TO)
"Diplegic dynamic equinus, Silfverskiöld positive, avoid crouch?"	→ **Strayer** (TO)
"Foot deformity NOT common in CP?"	→ **cavus** (TO)
"Knee recurvatum NOT caused by?"	→ **quadriceps weakness** (TO)
"CROW walker suits?"	→ **Charcot foot** (TO)

CEREBRAL PALSY - HIP CONDITIONS

– Pediatrics · 17 / 112 –

1 Basics

Hip surveillance – risk by GMFCS level & pathogenesis

Incidence progressive subluxation in up to **50% of spastic quadriparesis** · rare in ambulators

GMFCS	Risk of subluxation by age 7 (Orthobullets, OB)	Hip displacement risk (TO surveillance table)
I	**<10%**	~0–5%
II	**10–15%**	~15%
III (walks with assistive devices)	**30–40%**	~35%
IV	**50–60%**	~70%
V	**70–80%**	**~90%**

Pathogenesis hips **normal at birth** · spastic **adductors (longus, brevis, magnus, gracilis) + iliopsoas (psoas major, iliacus)** → scissoring → coxa valga, anteversion, acetabular dysplasia → subluxation → dislocation
✗ Wrong options: sartorius · gluteus minimus · obturator externus · piriformis · rectus femoris

Direction dislocation **posterior + superior (>95%)** · acetabular deficiency **posterosuperior**

Degeneration femoral head cartilage erosion → **pain** · **50–75%** of dislocated hips become painful (OB) · pain by **2nd–3rd decade**

Windswept hips one hip **abducted**, other **adducted** → pelvic obliquity · **adducted hip on high side** at risk of dislocation · infrapelvic obliquity → compensatory **scoliosis**

3 Presentation

Symptoms hip/groin pain · **difficulty sitting** · **perineal care / hygiene / diaper** difficulty

Exam unreliable alone · ↓ ROM, pain with motion · lever-arm gait dysfunction · thigh-length discrepancy ("**pseudo-Galeazzi**")

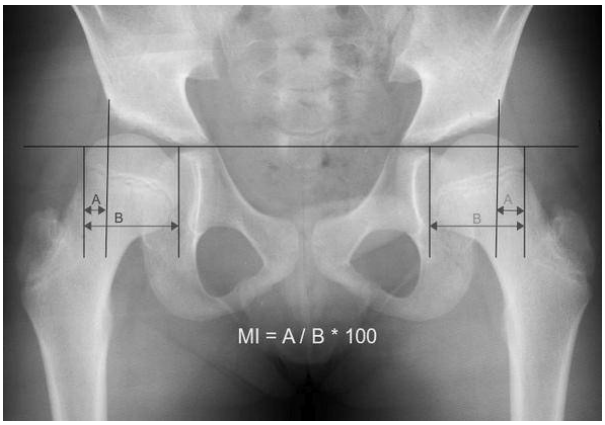
4 Diagnosis & Classification

Key radiology findings

Views **AP pelvis** + frog lateral if possible

– – Active space – –

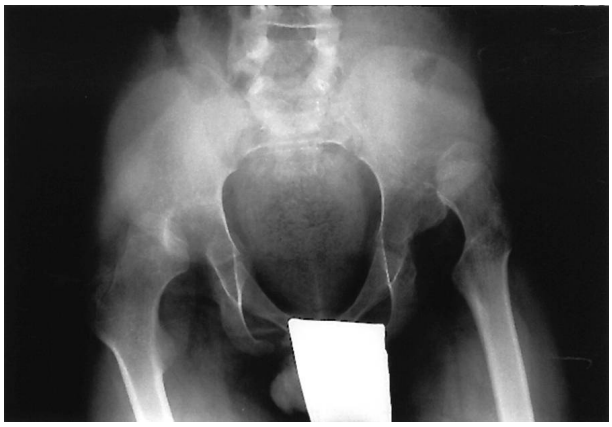
Reimers migration index	% of femoral head **uncovered** lateral to Perkin line · **most accurate method** to identify and monitor stability · **<33% at risk · >33% subluxated · >100% dislocated**
Natural history	hips **will dislocate** untreated if MI **>60–70%**
Acetabular index	**>25°** (OB) / >27° (OB) → acetabular dysplasia → **add acetabuloplasty**
CT	**operative planning** (3D acetabular correction) · CT scanogram for femoral version before derotation



Reimers migration index

Stages of hip deformity

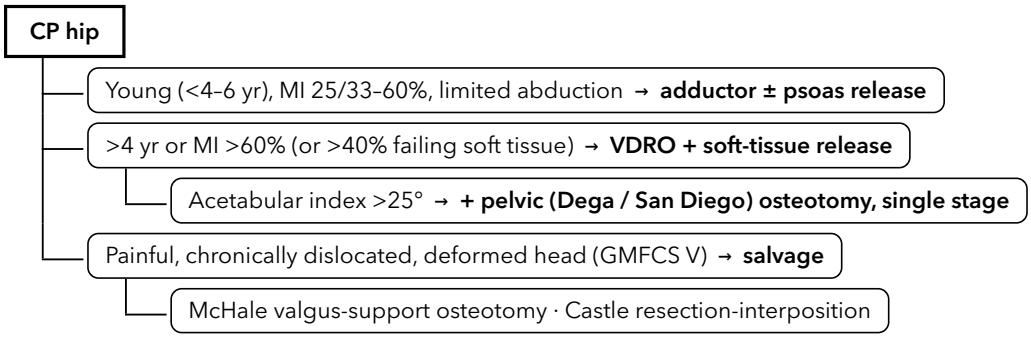
Stage	Characteristics	Treatment
Hip at risk	Abduction **<45°** + partial uncovering · **MI <33%**	**Botox (<3 yr)** to delay · **adductor ± psoas release, hamstring lengthening (3–4 yr)** · **avoid obturator neurectomy**
Subluxation	**MI >33%**	Adductor tenotomy if abduction restricted · persistent → **VDRO (5–6 yr)** · pelvic osteotomy if acetabular insufficiency
Spastic dislocation	**MI >100%**	**Open reduction + VDRO + femoral shortening + pelvic osteotomy**
Windswept	One abducted, one adducted	Brace adducted hip ± tenotomy · release abduction contracture of abducted hip



Spastic hip dislocation

5 Management

Nonop observation (mild) · **PT never shown to prevent subluxation** · **abduction bracing alone** does not ↓ dislocation, may cause **windswept** deformity



Operative criteria

Procedure	Indication
Adductor + psoas release	Age **<6** with MI **>25/33–60%** + limited abduction (Orthobullets) · **<4** with MI **>40%** (OB) · progressive subluxation **<8 yr** · adjunct to bony surgery
Psoas lengthening	Only with **hip flexion contracture (positive Thomas)**
VDRO ± Dega + soft tissue	Age **>4** or MI **>60%** · treat all at **single stage**
Add acetabuloplasty	**Acetabular index >25°** (✗ Wrong options: age <6 · GMFCS V · neck-shaft angle (VDRO fixes that) · prior botox)
Proximal femoral resection / McHale	Painful, chronically dislocated, **deformed head**, nonambulatory (GMFCS V)
Castle	Chronic dislocation, **adults**, can't stand to transfer (GMFCS V)
THA	Ambulatory or can **stand to transfer** · 85% 10-yr survival
Hip arthrodesis	Young patients
Girdlestone	**No longer done** – uniformly painful (proximal migration, no interposition)

Soft-tissue releases – adductor tenotomy indications & failure risk, obturator neurectomy

3-yr diplegic, abduction 20°, MI 20%, Thomas negative → bilateral adductor releases only · one of few surgeries in CP <4 yr
✗ Wrong options: psoas (no flexion contracture) · VDRO · pelvic osteotomy

Technique adductor longus tenotomy → sequentially **gracilis**, **adductor brevis** · psoas at **lesser trochanter (nonambulatory)** or **pelvic brim MTJ (ambulatory)** · goal **>45° abduction**

Failure risk **preop MI >40%** · acetabular index **>27°**
✗ Wrong options: lack of bracing · no neurectomy · age

Obturator nerve anterior branch lies on **adductor brevis/magnus** after longus release (Ludloff) · supplies **adductor longus, brevis, gracilis** · **pinch neurectomy → weak hip adduction** · neurectomy **does not improve** results and risks **abduction contracture** – avoid (Orthobullets, OB) · sometimes used in nonambulatory spastic adduction (OB)

– – Active space – –

6-yr nonambulatory quadriplegic, subluxation >50%, abduction 20° → open adductor tenotomy + iliopsoas release + bilateral VDRO (too old for soft tissue alone)
✗ Wrong options: percutaneous tenotomy · oral baclofen · phenol · neurectomy

Reconstruction – VDRO with pelvic (Dega) osteotomy

VDRO	shortening varus derotation → corrects **coxa valga + anteversion**
Dega	acetabular reshaping, incomplete trans-iliac · **improves posterior coverage** (neuromuscular hip needs posterior) · hinges on triradiate → **triradiate must be open** · bone graft placed · no fixation needed · **San Diego** = alternative ✗ Wrong options: salvage (shelf/Chiari are salvage)

Bilateral MI 70%/60%, AI 30° → combined bilateral VDRO + pelvic osteotomies + soft-tissue releases – lowest resubluxation
✗ Wrong options: soft tissue alone · VDRO alone · pelvic alone · closed reduction

8-yr wheelchair-bound, hygiene problems, abduction 5°/15°, bilateral dislocation + dysplasia → bilateral adductor release + varus femoral osteotomies + acetabuloplasties
✗ Wrong options: botox + PT · Pavlik · abductor release + valgus osteotomy

Salvage – proximal femoral resection

GMFCS V	no antigravity head/trunk control, transported

15-yr GMFCS V, painful sitting, flattened head on CT → proximal femoral resection
✗ Wrong options: open reduction (degenerate head stays painful)

Resection over reconstruction → GMFCS V, MI 100% + head deformity
✗ Wrong options: GMFCS IV MI 50% (reconstruct) · MI 25% (analgesia) · MI 100% normal head (osteotomy)

McHale (valgus support)	resect head/neck (ligamentum teres left, sutured to psoas/capsule) + **subtrochanteric valgus closing-wedge osteotomy → lesser trochanter articulates with acetabulum** · relieves pain, ↑ abduction
Castle	resect at **lesser trochanter** level · **vastus oversewn** over femur, abductors/psoas/capsule over acetabulum → large soft-tissue interposition (differs from Girdlestone)



McHale procedure

6 Complications

Osteonecrosis	**1–11%**
Heterotopic ossification	**radiation on POD 2–3** better than NSAIDs
Insufficiency fractures	**4–29%** · **distal femur after spica**

Abduction contracture **anterior obturator neurectomy**

– – Active space – –

X vs Y

	Spastic (CP) hip	DDH
At birth	**Normal**	Abnormal
Recognised	**After age 2**	1st year
Detection	**Radiographs**	Exam
Cause	Spastic muscles, pelvic obliquity	Breech, laxity, acetabular growth
Progression	**Common**	Rare
Pain	**2nd–3rd decade**	4th–5th decade
Deficiency	**Posterosuperior**	Anterior
Early Rx	**Muscle lengthening**	Pavlik / closed reduction
Late Rx	**Osteotomies, often without open reduction**	Reduction, often without osteotomy (<18 mo)
Salvage	**Castle resection-interposition**	THA

How it's asked

"GMFCS level with 30–40% risk of subluxation by age 7?"	**→ level III (walks with assistive devices)**
"GMFCS V dislocated hip — responsible muscles?"	**→ gracilis and psoas major**
"After adductor tenotomy, predictor of failure?"	**→ preop MI >40%** ✗ Wrong options: AI <27° · no bracing · no neurectomy · age
"8-yr wheelchair-bound, hygiene problems, bilateral dysplastic dislocated hips?"	**→ bilateral adductor release + varus osteotomies + acetabuloplasties**
"6-yr quadriplegic, subluxation >50%, abduction 20°?"	**→ open adductor tenotomy + iliopsoas release + bilateral VDRO**
"3-yr diplegic, abduction 20°, MI 20%, negative Thomas?"	**→ bilateral adductor releases**
"Pinch neurectomy during medial approach weakens?"	**→ hip adduction**
"Primary benefit of Dega in CP?"	**→ posterior coverage**
"MI 70%/60%, AI 30° — least resubluxation?"	**→ bilateral VDRO + pelvic osteotomies + soft-tissue release**
"Why add acetabuloplasty to VDRO?"	**→ acetabular index >25°**
"15-yr nonambulatory, painful sitting, flattened head?"	**→ GMFCS V — proximal femoral resection**
"Femoral head resection rather than reconstruction?"	**→ 17-yr GMFCS V, MI 100%, head deformity**

– – Active space – –

CEREBRAL PALSY - UPPER EXTREMITY DISORDERS

– Pediatrics · 18 / 112 –

1 Basics

Who — **hemiplegia** and **quadriplegia**

Typical posture — **shoulder IR · elbow flexion · forearm pronation · wrist flexion + ulnar deviation · thumb-in-palm** · finger flexion

Wrist flexion cause — weak wrist extensors · spastic wrist flexors · volar capsule contracture → weak grip, hygiene problems

Hygienic vs functional procedures

	Hygienic	Functional
Patient	↓ mental + physical function	**Voluntary control**, **IQ ≥50–70**, **good sensibility** (position sense without vision)
Examples	Releases, **wrist arthrodesis + PRC**	**Tendon transfers** (FCU → ECRB), PT rerouting

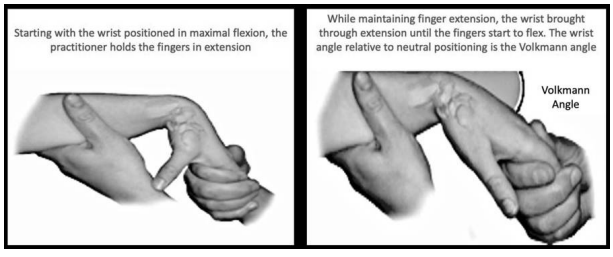
Voluntary control — best predictor of improvement after **hand/wrist tendon transfer**

3 Presentation

Volkmann angle

Technique — **fingers extended**, wrist from maximal flexion → extend wrist · angle short of full extension = **Volkmann angle**
✗ Wrong options: fingers flexed (must stretch extrinsic flexors)

Meaning — high Volkmann angle → **extrinsic finger flexor contracture** → correcting wrist alone gives **clenched fist** unless finger flexors also lengthened



Measuring Volkmann angle

Volkmann angle	Severity / treatment
~15°	Mild → **FCU or FCR lengthening**
~25°, correctable wrist	**FCU + FDS/FDP tenotomy + ECU → ECRB transfer**
Cannot reach within 45° of neutral	Severe → **wrist arthrodesis + PRC**
90° + palmar wounds (fingers-in-palm)	**PRC + wrist fusion + FDS → FDP**

Distractors locked IR shoulder + extended elbow = **arthrogryposis** · unstable thumb CMC = **Blauth IIIB hypoplasia** (pollicise) · constricting bands = constriction band syndrome

– – Active space – –

4 Diagnosis & Classification

House classification (thumb-in-palm)

Type	Deformity	Treatment
I	**1st MC adduction** contracture	Adductor release ± 1st dorsal interosseous release · **Z-plasty** 1st web
II	+ **MP flexion** contracture	Adductor release
III	+ **unstable / hyperextensible MP**	Adductor release + **MP fusion or capsulodesis**
IV	+ **MP + IP flexion** contractures	Adductor release + **FPB + FPL release/lengthening**



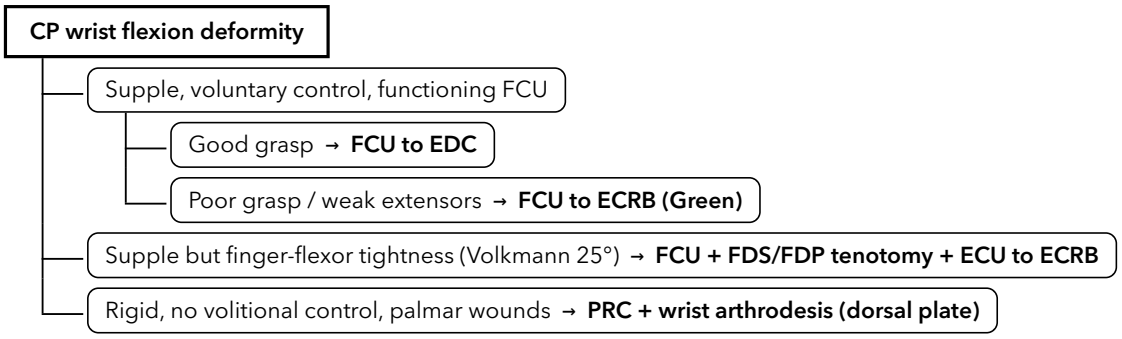
Thumb-in-palm deformity

- - Active space - -

5 Management

Operative criteria

Deformity	Procedure	Indication
Shoulder IR	Derotational humeral osteotomy and/or subscapularis + pectoralis lengthening	**>30°** contracture interfering with hand function
Elbow flexion	**Lacertus fibrosus release**, biceps + brachialis lengthening, brachioradialis origin release	Elbow flexion contracture
Forearm pronation	**Pronator teres release / rerouting** to anterolateral (makes it a **supinator**) · or FCU → ECRB	Pronation deformity · risk: **supination deformity** (worse than pronation)
Wrist flexion, mild	**FCU or FCR lengthening**	Good finger extension, little wrist-flexor spasticity
Wrist flexion, functional	**FCU → ECRB (Green)** · FCU → EDC	Supple wrist, voluntary control, IQ ≥50–70, sensibility · **better long-term than botox/ therapy**
Wrist flexion, spastic FCU	**ECU → ECRB**	Avoids hyperextension from transferring spastic FCU
Wrist flexor weakening	**Flexor-pronator slide** (from medial epicondyle)	Severe fixed wrist + finger flexion (also post-compartment syndrome)
Fixed, non-supple	**Wrist arthrodesis ± PRC**	No volitional control · hygiene + function
Swan neck	Correct wrist flexion first	Intrinsic tightness + extensor overpull



Wrist flexion deformity – tendon transfers vs arthrodesis (Volkmann angle, FCU to ECRB, pronator teres rerouting)

Green transfer (flexor carpi ulnaris → extensor carpi radialis brevis, FCU → ECRB)	supple wrist, **weak/absent wrist extensors**, resting ulnar deviation · improves **extension + supination** · **most effective for restoring supination** · not if no functioning **FCR** or **rigid wrist**
Pronator teres rerouting	converts PT to **supinator** · very functional patient with **no active but full passive supination** + good sensibility → **PT transfer** ✗ Wrong options: anconeus, brachialis, brachioradialis, supinator
PT to ECRB	powers wrist extension when wrist flexors can't be used

Distractors ✗ Wrong options: PT → FCU (↑ pronation + ulnar deviation) · botox (inferior to transfer) · PT/brachioradialis rerouting (less effective than Green for supination)

Wrist arthrodesis best for deformity **not passively correctable** · **with PRC** (↓ soft-tissue tension, provides graft) · dorsal plate · **union >95%**, high satisfaction
✗ Wrong options: <75% union · higher failure in spastic CP · only after failed Green

X vs Y

	FCU → ECRB (Green)	Wrist arthrodesis + PRC
Wrist	**Supple**	**Rigid**
Control	**Voluntary**, good sensibility	Lacks volitional control
Goal	Function – extension + supination	Hygiene, position, fingers out of palm
Contraindication	Rigid wrist, no FCR	–

How it's asked

"Exam missed before wrist fusion + PRC?"	→ **fingers extended, wrist can't extend to within 45° of neutral (Volkmann)**
"Functioning flexors, flexible wrist, absent extensors – best result?"	→ **FCU to ECRB**
"No active, full passive supination, good sensibility – transfer?"	→ **pronator teres**
"Best restoration of supination?"	→ **FCU to ECRB (Green)** ✗ Wrong options: PT → FCU · PT/BR rerouting · fusion
"True of wrist arthrodesis?"	→ **best when deformity not passively correctable**
"Correctable wrist, Volkmann 25°, lasting correction?"	→ **FCU + FDS/FDP tenotomy with ECU transfer** ✗ Wrong options: PT · botox · FCU tenotomy alone · fusion
"Rigid wrist fused in neutral with dorsal scar – indication?"	→ **Volkmann 90° with palmar wounds** ✗ Wrong options: GMFCS V alone · supple ulnar deviation

– – Active space – –

CHARCOT-MARIE-TOOTH DISEASE

– Pediatrics · 19 / 112 –

1 Basics

Epidemiology

Incidence	**1 in 2500** · no ethnic predisposition · **most common inherited neuromuscular disease** seen by orthopaedists
Sex	more common in **males**; females often worse (OB)
Name	**HMSN** (hereditary motor sensory neuropathy) · peroneal muscular atrophy · most common HMSN (others: Dejerine-Sottas)

Genetics & pathophysiology (PMP22 duplication, myelin, autosomal dominant)

Most common	**autosomal dominant duplication on chromosome 17p12 → PMP22** (peripheral myelin protein 22) expressed in **Schwann cells** · PMP duplication in **~70%** (OB)
Structure affected	**myelin sheath** (CMT1) ✗ Wrong options: perineural vessels · cell body · dendrite · NMJ

PMP22 deletion (same region) → **HNPP** (hereditary neuropathy with liability to pressure palsies)

Type	Inheritance	Pathology	NCV	Note
CMT1	**AD**	**Demyelinating**	**Slowed** (<38 m/s)	**50%** (Orthobullets, OB) · 80% (OB) · 1A = PMP22 duplication · onset 1st–2nd decade · **cavus** foot · motor > sensory · normal lifespan
CMT2	AD (AR-CMT2 subtype)	**Axonal – Wallerian degeneration**, intact myelin	**Normal** (>45 m/s), ↓ CMAP/SNAP amplitudes	~20% · onset 2nd decade+ · **less disabled** · flaccid foot
CMT4	**AR**	Demyelinating	Slowed	–
CMTX	**X-linked**	–	–	–

Genes & lab markers

Disorder	Defect (distractors)
CMT	**PMP22**, chr 17
Friedreich ataxia	**GAA repeat**, chr 9 (frataxin)
Duchenne / Becker	**Dystrophin** (X-linked recessive)
Limb-girdle	Sarcoglycan / dystroglycan
Myotonic dystrophy	Myotonin
SMA	Survival motor neuron protein
Marfan	**FBN1**, chr 15 (planovalgus)
OI	Type I collagen
Hurler (MPS I)	α-L-iduronidase
MED	COMP

HMSN look-alikes ALS (UMN + LMN death) · Refsum (branched-chain fatty acid oxidation) · **Dejerine-Sottas** (severe, rapidly progressive, early) · Parkinson (midbrain)

Deformity mechanism – plantar-flexed first ray (PL over TA) & PTT

Strong (preserved)	Weak	Result
Peroneus longus	**Tibialis anterior**	**Plantarflexed 1st ray = initial deformity** (cavus)
Tibialis posterior	**Peroneus brevis**	**Hindfoot varus**, foot adduction
Long toe extensors (recruited for DF)	**Intrinsics**	**Claw toes**

Predominant deforming muscles **posterior tibialis + peroneus longus**

Weakest first **peroneals** first + most profound (Orthobullets) · initial presentation = **weak tibialis anterior** (OB) → foot drop, later fixed equinus

Sequence plantarflexed 1st MT + pronated forefoot → **secondary hindfoot varus** → later fixed + equinus

Bilateral progressive cavovarus → HMSN/CMT (78% of bilateral cavovarus)
✗ Wrong options: DMD (equinovarus) · CP (equinus/equinovalgus) · myelomeningocele (clubfoot/CVT/calcaneus) · Werdnig-Hoffmann

3 Presentation

Motor distal weakness + atrophy · **high-steppage gait** (foot drop) · tripping, falls · **chronic ankle instability / sprains** · **"inverted champagne bottle"** legs · lateral foot pain · lateral shoe wear

Sensory stocking-glove loss · ↓ proprioception → imbalance · burning/numbness

Foot **bilateral symmetric cavovarus** + claw/hammer toes · flexible → rigid

Reflexes **hyporeflexia/areflexia** (absent ankle + knee jerks)

– – Active space – –

Hand	**intrinsic wasting** (1st dorsal interosseous) · **weak pinch** (grasp, dexterity spared)
Spine	scoliosis on Adams test



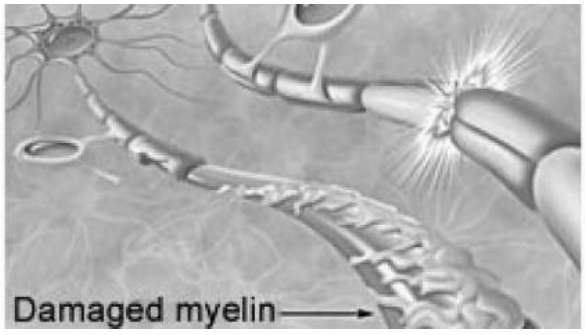
Pes cavus in CMT

Coleman block	corrects → **forefoot-driven** · doesn't → **hindfoot-driven/rigid**

4 Diagnosis & Classification

Diagnosis & imaging

NCS/EMG	classifies demyelinating vs axonal · **demyelinating cut-off 38 m/s** · axonal **>45 m/s** with ↓ CMAP/SNAP · **prolonged distal latencies** in peroneal, ulnar, median nerves
Genetic testing	**PCR for PMP22** · chr 17 duplication
Radiograph	**high-arched cavus foot** (↑ calcaneal pitch) ✗ Wrong options: accessory navicular · CN coalition · ball-and-socket ankle (fibular hemimelia)
Pathology	simultaneous demyelination + remyelination ("onion bulb")



CMT1 demyelination

5 Management

Orthoses – lateral forefoot & heel posting

Flexible cavovarus → lateral forefoot + lateral heel posting (forefoot post lets 1st MT flex; heel post corrects varus) · (✗ Wrong options: medial heel wedge (worsens) · medial flare · SACH) · metatarsal pad = pain relief only

Lateral hindfoot posting suits **flexible cavovarus**
✗ Wrong options: accessory navicular · middle-facet coalition · flexible flatfoot · metatarsus adductus

Other accommodative shoes (mild) · semi-rigid insole with 1st-ray recess + lateral wedge (flexible adult) · SMO · **AFO for foot drop** (best if equinus dynamic) · lace-up brace

24-yr HMSN, corrects with Coleman, no arthritis → custom orthotic with lateral forefoot + heel posting first
✗ Wrong options: medial posting · osteotomies

Surgery – flexible (PL to PB, first MT osteotomy, PTT transfer, calcaneal osteotomy) vs rigid (triple arthrodesis)

CMT cavovarus – failed orthoses

- Flexible (Coleman corrects) → plantar release + PL → **PB + 1st MT DF osteotomy ± PTT to dorsum**
- Partial / no correction → **+ lateralising calcaneal osteotomy (proximal translation)**
- Rigid + arthritic (TC, CC) → **triple arthrodesis**
  (poor long-term in CMT – last resort)

Procedure	Indication
Plantar release (fascia ± Steindler)	Cavus
PL → PB transfer	Plantarflexed 1st ray · ↓ 1st-ray PF, restores eversion
1st MT dorsiflexion osteotomy (or 1st MC dorsal closing wedge)	**Flexible hindfoot** (Coleman normal)
PTT transfer through IOM to dorsum (lateral foot)	**Flexible deformity, TA weak, PTT strong** → ↓ varus + gives DF · **improved outcomes**
Gastrocnemius recession / TAL	Equinus (gastroc recession less weakness, with plantar release)
Lateralising calcaneal osteotomy (± proximal translation)	**Hindfoot not correcting on Coleman** · proximal shift improves triceps lever arm
Triple arthrodesis	**Rigid + arthritic** hindfoot · **poor outcomes** in CMT (Wetmore & Drennan)
Jones (EHL → 1st MT neck + IP fusion)	Symptomatic claw hallux failing nonop

Distractors ✗ Wrong options: PB → PL (wrong direction) · SPLATT (TA weak) · TA → PB (TA weak) · PTT → TA tendon (out-of-phase; go **lateral**) · FDL transfer (PTT is the deforming force) · medializing calcaneal osteotomy (↑ varus) · LCL (flatfoot) · botox (weakens push-off) · ankle/subtalar fusion in supple young foot

25-yr chr-17 duplication, progressive foot drop, supple joints, equinus, strong PF + inversion → PTT transfer to dorsum + gastrocnemius lengthening

23-yr, partial Coleman correction, 1st MT callus, weak TA + PB → PL → PB + 1st MC dorsal closing wedge + lateralising calcaneal osteotomy with proximal translation

– – Active space – –

Claw toes

Mechanism	weak DF → **toe extensor recruitment** + weak intrinsics → claw · flexible → rigid
Treatment	**Jones procedure** (EHL to 1st MT neck + IP arthrodesis) for symptomatic claw hallux

Associated hip dysplasia

Prevalence	**<10%** (Orthobullets) · **6–8%** (OB) · often **asymptomatic → screen** · presents in adolescence in ambulators

Cavus foot + hip dysplasia in a teen → CMT
✗ Wrong options: OI · Down (planovalgus) · achondroplasia · cleidocranial

Treatment	**pelvic osteotomy** · ↑ **sciatic nerve palsy** risk · outcomes comparable to DDH

Scoliosis

Prevalence	**10–20%**
Distinguishing	**left thoracic** + **kyphotic** curve (vs idiopathic)
Bracing	rarely effective · **fusion if >50°** and progressive

6 Complications

Complication – transfer lesion after bunionette resection

5th MT head resection in CMT → transfer lesion to 4th MT head (lateral overload of cavovarus)
✗ Wrong options: fixed 5th claw (toe flail) · ulcer (no bone) · peroneal atrophy · Charcot

Rigid deformity	untreated progression → arthritis

X vs Y

	CMT1	CMT2
Lesion	**Demyelination**	**Axonal (Wallerian)**
NCV	**↓ (<38 m/s)**	**Normal (>45)**, ↓ amplitude
Onset	1st–2nd decade	2nd decade+
Disability	More	**Less**
Foot	**Cavus**	Flaccid

	CMT	Friedreich ataxia
Gene	PMP22, chr 17	GAA repeat, chr 9
Foot	Cavovarus	Cavovarus
Extra	Hand intrinsic wasting, areflexia	Ataxia, cardiomyopathy, progressive scoliosis

How it's asked

"5-yr boy, bilateral cavus, PMP duplication chr 17?"	→ **CMT** ✗ Wrong options: myelodysplasia · CP · polio
"Features of most common HMSN?"	→ **myelin/axon mutation → leg atrophy, loss of sensation/ proprioception in early adulthood** ✗ Wrong options: ALS · Refsum · Dejerine-Sottas · Parkinson
"PMP22 mutation disorder?"	→ **CMT** ✗ Wrong options: DMD · limb-girdle · myotonic · SMA
"PMP22 chr 17p12 — initial presentation?"	→ **weak tibialis anterior**
"Tripping, hand weakness, high arches, areflexia, paternal family history — protein?"	→ **PMP-22** ✗ Wrong options: dystrophin · COMP
"Cause of CMT cavovarus?"	→ **autosomal-dominant myelin sheath disease**
"CMT type I is a disorder of?"	→ **myelin sheath**
"Weak eversion, ↓ NCV in evertor nerve, chr 17 duplication?"	→ **CMT** ✗ Wrong options: Friedreich (chr 9)
"Plantarflexed 1st ray is initial deformity in?"	→ **CMT** ✗ Wrong options: oblique talus · Köhler · calcaneovalgus
"Cavus deformity imbalance?"	→ **PL overpowering TA**
"32-yr, lateral foot pain, awkward gait — initial deforming element?"	→ **plantar flexion of the first ray**
"Predominant functional muscles?"	→ **posterior tibialis + peroneus longus**
"Common CMT foot?"	→ **plantarflexed 1st MT + hindfoot varus**
"Progressive bilateral cavovarus most common in?"	→ **HMSN**
"17-yr with familial cavus — also evaluate for?"	→ **acetabular dysplasia**
"15-yr runner, hip dysplasia + cavus?"	→ **CMT**
"Lateral forefoot + heel posts suit?"	→ **CMT** ✗ Wrong options: PTTI stage II/III · Charcot · hallux rigidus
"HMSN, flexible on Coleman — initial management?"	→ **orthotic with lateral forefoot + heel posting**
"Lateral hindfoot posting most useful for?"	→ **flexible cavovarus foot**
"Orthosis for flexible CMT deformity?"	→ **lateral heel + forefoot posting**
"42-yr CMT, rigid on Coleman, TC + CC arthritis?"	→ **triple arthrodesis**

– – Active space – –

"14-yr, correctable hindfoot — procedure with improved outcomes?"	**→ PTT through IOM to dorsum** ✗ Wrong options: PB → PL · SPLATT · triple · LCL
"25-yr foot drop, supple, equinus, strong PF/inversion?"	**→ PTT to dorsum + gastrocnemius lengthening**
"Partial Coleman correction, 1st MT calluses?"	**→ PL → PB + 1st MC dorsal closing wedge + lateralising calcaneal osteotomy**
"Lateral calcaneal slide chosen — exam?"	**→ cavovarus not correcting on Coleman**
"After 5th MT head resection in CMT?"	**→ transfer lesion to 4th MT head**
"↑ peroneal latency + chr 17 duplication — radiograph?"	**→ high-arched cavus foot**

CHRONIC RECURRENT MULTIFOCAL OSTEOMYELITIS (CRMO)

– Pediatrics · 20 / 112 –

1 Basics

Epidemiology

Incidence	**1 in 250,000** · prevalence 1 in 160,000
Age	children/adolescents · **peak 10 years**
Sex	**girls > boys**
Site	**tubular long bones** (metaphyses) + **clavicle**

Definition & associations

Also called	chronic **non-bacterial** osteomyelitis · sterile, **rheumatologic** bone inflammation · multifocal, recurrent over years
Associations	**pustulosis palmoplantaris** (red patches + pustules on palms/soles → desquamation days later) · **SAPHO** (synovitis, acne, pustulosis, hyperostosis, osteitis) · **psoriasis** · **inflammatory bowel disease (Crohn's)**
Syndromic forms	**Majeed syndrome** · **DIRA** (deficiency of IL-1 receptor antagonist)

3 Presentation

Symptoms	localised pain + swelling, often **worse at night** · episodic fever, malaise · partial relief with NSAIDs
Exam	focal tenderness

4 Diagnosis & Classification

Diagnostic criteria

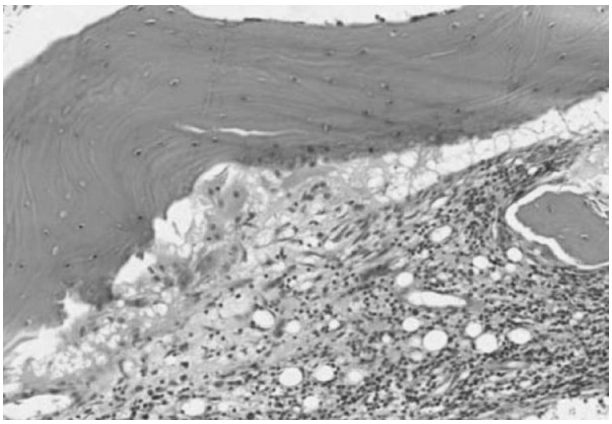
Criterion
Multiple sites of apparent osteomyelitis
Pathology + cultures negative
No response to antibiotics

Diagnosis of exclusion	biopsy essential

Genes & lab markers

Labs	**WBC normal** · **ESR / CRP mildly elevated**
Biopsy	inflammation, **no atypia** (✗ Wrong options: osteosarcoma), **no small round blue cells** (✗ Wrong options: Ewing) · cultures negative

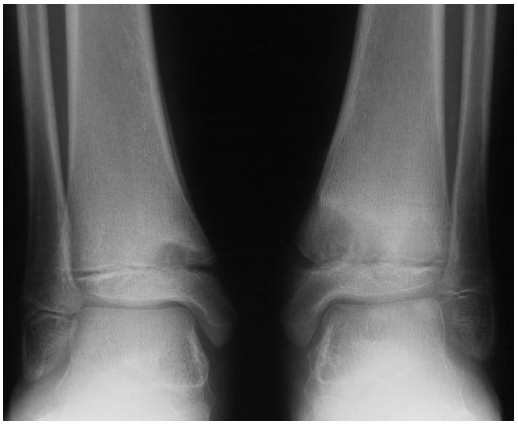
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CRMO biopsy – bland inflammation

Key radiology findings

X-ray	**eccentric lytic metaphyseal lesions** abutting physis with **surrounding sclerosis + new bone**
Bone scan	finds other sites
MRI	if malignancy suspected (soft-tissue involvement)



CRMO of distal tibiae

5 Management

Symptomatic during exacerbations	**NSAIDs** · corticosteroids · **pamidronate** (symptoms, vertebral remodelling) ✗ Wrong options: antibiotics (no response)

6 Complications

Growth disturbance	rare
Natural history	remissions + exacerbations · resolves in **3–5 years**

X vs Y

	CRMO	Infectious osteomyelitis	Osteoid osteoma
Sites	**Multiple**	–	**Solitary**
Cultures	**Negative**	Infection evident	–
Antibiotics	**No response**	Respond	–
X-ray	Metaphyseal lysis + sclerosis	–	**Nidus**
NSAIDs	Help	–	Help

How it's asked

"9-yr girl with Crohn's, night pain in forearm + clavicle, normal WBC, ↑ ESR/CRP, bland biopsy?"	**→ chronic recurrent multifocal osteomyelitis** ✗ Wrong options: osteosarcoma · Ewing · osteomyelitis · osteoid osteoma
"CRMO criteria?"	**→ multifocal, culture-negative, no antibiotic response**
"Skin association?"	**→ palmoplantar pustulosis / SAPHO**
"Treatment?"	**→ NSAIDs ± pamidronate**

– – Active space – –

CLAVICLE SHAFT FRACTURE - PEDIATRIC

– Pediatrics · 21 / 112 –

1 Basics

Epidemiology

Incidence	**15% of pediatric upper-extremity injuries** · shaft = **80%** of pediatric clavicle injuries
Birth fractures	**0.5%** normal deliveries · **1.6%** breech · **90% of obstetric fractures**
No trauma history	→ think **congenital pseudarthrosis of clavicle** (usually **right**, except dextrocardia)

Anatomy

Deforming forces	**medial fragment → posterosuperior** (**sternocleidomastoid**) · **lateral fragment → inferomedial** (**pectoralis** + arm weight)
Open fractures	buttonhole through **platysma**
AC stability	AC ligament (AP stability; **superior** strongest, then posterior) · CC ligaments trapezoid + **conoid (strongest)** (superior/inferior stability) · deltoid + trapezius dynamic

2 Mechanism

Mechanism	**FOOSH** or **direct blow to lateral shoulder** · birth trauma

3 Presentation

Exam	pain, deformity · **skin tenting** (impending open fracture) · NV exam (rare NV / brachial plexus injury)

4 Diagnosis & Classification

Key radiology findings

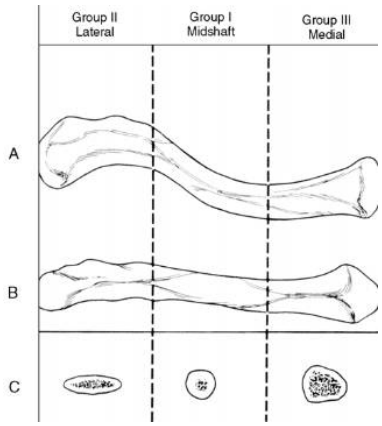
Views	**upright (sitting/standing) AP bilateral shoulders** · **15° cephalic tilt (Zanca)** for superior/inferior displacement · ± 5–10 lb weight in hand



Midshaft clavicle fracture — upright AP

Allman classification

Type	Location
I	**Middle third (most common)**
II	Lateral third, distal to CC ligaments
III	Medial (proximal) third



Allman classification

5 Management

Nonoperative care even when displaced

Newborn birth fracture — observation, careful lifting · union ~**1 week**

<12 years — **sling / shoulder immobiliser / figure-of-eight** (no difference sling vs figure-of-eight) · **no reduction attempted** · ROM at 2–4 wk, strengthening 6–10 wk · callus bump fades 6–12 mo

Displacement / shortening in <12 yr — **100% displaced + 12 mm short at 10 yr → nonoperative because of age**
✗ Wrong options: plate for displacement · high nonunion risk

Adolescent — **14 yr, completely displaced, 24 mm short** → sling → **union without significant shortening** (remodelling)
✗ Wrong options: malunion with shortening · nonunion

Symptomatic malunion — only **10–20%**
✗ Wrong options: 50%

– – Active space – –

Operative criteria

Absolute	Controversial
Open fracture	**Shortening >2 cm** – not an indication <12 yr; debated in adolescents
Skin at risk from tenting	–
Subclavian artery/vein injury	–

Pediatric clavicle shaft fracture

- Newborn → **observe, careful handling**
- <12 yr (even 100% displaced, shortened) → **sling**
- Open / tenting / vascular injury → **ORIF (precontoured plate)**
  (adolescent >2 cm shortening – controversial)

ORIF precontoured plate (± recon / locking / IM rod) · sling 7–10 d · strengthening ~6 wk · sport ~3 mo

6 Complications

Nonoperative **nonunion / malunion rare**

Operative **hardware prominence (up to 59%)** → plate removal · **anterior chest wall numbness** (supraclavicular nerves) · refracture · infection/dehiscence ~5%

TO points 10–15% of paediatric fractures · **90% midshaft** · simple sling · **figure-of-8 no advantage** · posteriorly displaced **proximal (medial)** fracture → tracheo-oesophageal compression

How it's asked

"10-yr, 100% displaced, 12 mm short, skin intact – true?"	→ **nonoperative because of age** ✗ Wrong options: plate · nonunion risk · 50% symptomatic malunion
"14-yr, 24 mm shortened midshaft, sling – long-term?"	→ **union without significant shortening**
"Absolute indications for ORIF?"	→ **open fracture, tenting skin, subclavian vessel injury**
"Medial fragment displaced by?"	→ **sternocleidomastoid (posterosuperior)**
"Commonest complication of plating?"	→ **hardware prominence**

CLEIDOCRANIAL DYSPLASIA (DYSOSTOSIS)

– Pediatrics · 22 / 112 –

1 Basics

Epidemiology

Incidence **~1 in 1,000,000**

RUNX2 mutation, inheritance & features

Defect **intramembranous ossification** → failure of **midline structures** (clavicles, skull, pubic symphysis, face)

Hallmark **hypoplastic or absent clavicles**

Genes & lab markers

Inheritance **autosomal dominant** · ~**two-thirds familial**

Gene **RUNX2 = CBFA1 (core-binding factor α1)** · **heterozygous loss of function** · transcription factor for **osteoblast differentiation**

No commercial test many mutations → diagnosis = family history + clinical/radiographic features

Gene (distractors)	Disorder
RUNX2 / CBFA1	**Cleidocranial dysplasia** ✓
SOX9	**Campomelic dysplasia**
cAMP / GNAS → c-fos	**Fibrous dysplasia**
BMP-2	– (not CCD)
COL1A1 / COL1A2	OI
DTDST	Diastrophic dysplasia
FGFR2	Apert (complex syndactyly)
FGFR3	Achondroplasia 10
Fibrillin-1	Marfan
Dystrophin	Duchenne (calf pseudohypertrophy)

3 Presentation

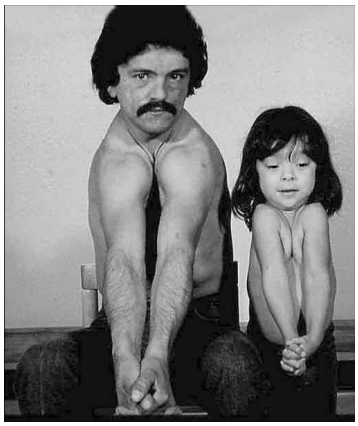
Body **proportionate dwarfism** · short stature

Shoulders **hypermobile, droopy** · can **approximate shoulders in front of chest** · narrow chest
✗ Wrong options: ectodermal dysplasia · MED · glenoid dysplasia

Skull/face **frontal bossing**, large head · **delayed closure of fontanelles + sutures** · **wormian bones** · hypertelorism, midface hypoplasia

– – Active space – –

Teeth	**delayed eruption of permanent teeth** (Orthobullets) · delayed primary teeth (OB) · **supernumerary teeth** → crowding (TO)
Hands	**short middle phalanges of 3rd–5th fingers** (Orthobullets) · short terminal phalanges (TO)
Hips/pelvis	**coxa vara** (± coxa valga) · **delayed ossification of pubis** (wide symphysis) · lateral notching of capital femoral epiphysis
Other	scoliosis, kyphosis, **genu valgum**, pes planus, syringomyelia · **normal intellect**

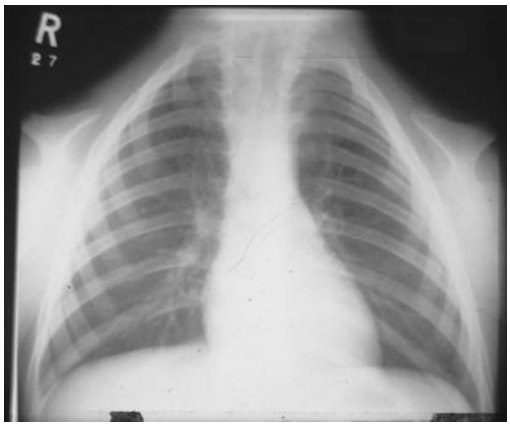


Cleidocranial dysplasia — shoulders approximated

4 Diagnosis & Classification

Key radiology findings

View	Look for
AP chest	**Absent / hypoplastic clavicles**
Lateral skull	**Delayed suture closure**, wormian bones
AP pelvis	**Coxa vara**, **unossified pubis**
AP hands	Short middle phalanges



Absent clavicles



Coxa vara in cleidocranial dysplasia

5 Management

Clavicles — **observe** · most manifestations need no intervention

Operative criteria

Intertrochanteric (reorienting valgus) osteotomy — **coxa vara with neck-shaft angle <100°**

Craniosynostosis – trigonocephaly

Trigonocephaly — **metopic suture** fusion (triangular "keel" forehead) · metopic is **first suture to close** (3–8 months) (TO)
✗ Wrong options: unilateral/bilateral coronal · sagittal

Shape	Suture
Scaphocephaly / dolichocephaly (boat)	**Sagittal — most common (~50%)**
Brachycephaly	Bicoronal ± bilambdoid
Frontal plagiocephaly	Unilateral coronal (harlequin eye)
Occipital plagiocephaly	Unilateral lambdoid
Trigonocephaly	**Metopic**
Oxycephaly / turricephaly	Sagittal + coronal + lambdoid
Cloverleaf	Intrauterine multi-suture (most severe)

Craniosynostosis — M:F **3:1** · 1 in 2000–2500 · coronal ~20%, lambdoid ~5%, metopic ~5% · syndromic ~8–10%: **Apert, Crouzon, Carpenter** · anterior fontanelle closes **18–24 mo**

– – Active space – –

X vs Y

	Cleidocranial dysplasia	SED tarda	SCFE / hypothyroid
Clavicles	**Absent/hypoplastic**	Normal	Normal
Hip	**Coxa vara**, unossified pubis	Coxa vara, poorly ossified epiphyses	Slipped epiphysis
Ossification defect	**Intramembranous**	Endochondral (COL2A1)	–

How it's asked

"Shoulder approximation in front of chest?"	→ **cleidocranial dysplasia** (TO)
"Heterozygous loss-of-function gene causing CCD?"	→ **RUNX2** ✗ Wrong options: BMP-2 · cAMP · c-fos · SOX-9
"8-yr, extreme shoulder motion, frontal bossing, abnormal clavicles + coxa vara?"	→ **cleidocranial dysplasia** ✗ Wrong options: SED tarda · hypothyroidism · SCFE · renal osteodystrophy
"Droopy shoulders, absent clavicles, delayed pubic ossification – inheritance?"	→ **autosomal dominant**
"Mutated core-binding factor A1 – clinical photo?"	→ **shoulders touching (CCD)** ✗ Wrong options: marfanoid · rhizomelia · calf pseudohypertrophy · cavovarus
"Defective intramembranous ossification – gene?"	→ **RUNX2/CBFA1** ✗ Wrong options: COL1A1 · DTDST · FGFR2 · SOX9
"Coxa vara in CCD – when operate?"	→ **neck-shaft angle <100°**
"Trigonocephaly results from?"	→ **metopic suture fusion** (TO)

CLUBFOOT (CONGENITAL TALIPES EQUINOVARUS)

– Pediatrics · 23 / 112 –

1 Basics

Epidemiology

Most common musculoskeletal birth defect	**1:1000** (1:250 in some; highest in **Hawaiians, Maoris**) · TO: 1.2/1000
Sex	**M:F 2:1** ✗ Wrong options: equal
Side	**half bilateral** ✗ Wrong options: majority unilateral
Isolated	**80%** idiopathic · 20% with syndromes / neuromuscular conditions

Associated findings, genetics & prognosis (limb hypoplasia, neurological cause, prenatal diagnosis)

Familial	**25%** · unaffected parents with one child → **2.5–6.5%** recurrence · monozygotic > dizygotic concordance
Family risk (TO)	**sibling 2–5%** · parent + sibling affected **up to 25%** · M:F **2.5:1** (TO) · arrest at **5th week** (one TO lecture) vs 6–8 wk (another) – read the options
Gene	**PITX1–TBX4** transcription pathway (hindfoot/limb development) · shared pathway with CVT ✗ Wrong options: PI3K-AKT achondroplasia · COL2A1 Kniest · FGFR2 Apert · RUNX2 CCD
Inheritance (TO)	**polygenic multifactorial** · **CASP10** gene implicated in idiopathic clubfoot
Aetiology theories (TO)	mechanical (intrauterine) · neuromuscular · **vascular (hypoplastic dorsalis pedis, medial compartment ischaemia)** · germ plasm defect · genetic
Syndromic associations (TO)	arthrogryposis · Larsen · **Freeman-Sheldon** · diastrophic dysplasia · myelomeningocele · amniotic band
Theories (TO lecture 2)	**arrest of embryonic development at 6–8 wk** (foot normally in equinus, cavus, adduction) · **fibroblastic proliferation** (↑ collagen in ligaments, tendons, capsules) · **germ plasm theory = most accepted** – defect in **cartilage anlage of talus** → dysmorphic talar neck, navicular subluxes medially · AD with incomplete penetrance described
Deforming structures (TO)	medial: **deltoid, spring ligament, TN capsule, tibialis posterior, FHL, FDL** · posterior: **TA, posterior ankle + subtalar capsules** · joints: talocalcaneal, TN, CC
Associations	**arthrogryposis** · **diastrophic dysplasia** · **myelodysplasia** · **tibial hemimelia** · **amniotic band** (Streeter; hand anomalies) · Pierre Robin · Opitz · **Larsen** · prune-belly
Vascular	**anterior tibial artery hypoplasia/absence** common · **posterior tibial artery** predominant supply

– – Active space – –

Limb	**small foot + calf, short tibia** = **intrinsic hypoplasia** of the limb · calf girth difference in unilateral cases ✗ Wrong options: tethered cord · scoliosis · contralateral hemihypertrophy
Walking	clubfoot does **not** delay walking · not walking by **18–24 months** → **neurologic disorder** ✗ Wrong options: hip dysplasia · in-toeing · LLD · foot
Prognosis	with treatment → **walks and runs normally** ✗ Wrong options: prosthesis

Pathoanatomy – CAVE

Component	Tight structures
Cavus (midfoot)	Intrinsics, **FHL, FDL** · plantarflexed 1st ray
Adductus (forefoot)	**Tibialis posterior**
Varus (hindfoot)	**Tendoachilles, tibialis posterior, tibialis anterior**
Equinus	**Tendoachilles**

Bony	medial spin of mid/forefoot around hindfoot · **talar neck medially + plantarly deviated** · calcaneus varus, rotated medially under talus · **navicular + cuboid medially displaced** · forefoot **pronated relative to hindfoot** (whole foot supinated)
Bony (TO)	talus: **short neck**, medially deviated, broad superior surface · calcaneus small, medial, varus · navicular medial, **hypoplastic**
Contracted ligaments (TO)	spring · deltoid · calcaneofibular · subtalar ligaments · capsules of ankle, subtalar, TN joints
Acetabulum pedis (TO)	**TN joint**; talus has **no muscle attachments**, mostly cartilage → stays still while calcaneo-pedal unit rotates around talar head
Wrung-towel biomechanics (TO)	**forefoot pronation maintains hindfoot varus** · **calcaneal eversion only after forefoot supination** → correct in supination, then abduction

3 Presentation

Exam	small foot/calf · **medial + posterior skin creases** · hindfoot **equinus + varus**, midfoot **cavus**, forefoot **adduction** · **rigid** equinus resisting passive correction (vs positional deformity)
General exam (TO)	spine + neuro · **sacral dimple, sinus, hairy patch** · hips, knees, upper limbs (syndromic) · **toe-to-shin (dorsiflexion) test** rules out positional foot

Atypical (complex) clubfoot (TO)

Features (TO)	**short, stubby foot** · severe cavus with **transverse plantar crease** · all metatarsals flexed · rigid equinus · **hyperextended short 1st toe** · fibrotic TA
Modified Ponseti (TO)	**two-thumb technique** · **heel pull-down** manoeuvre · max abduction **only 20°** · cast in **100–110° knee flexion**



Clubfoot – CAVE

4 Diagnosis & Classification

Diagnosis **clinical**

Pirani score (TO)

Pirani (0–6; each sign 0 / 0.5 / 1)	Signs
Midfoot score (0–3)	**Medial crease** · **curved lateral border** · **talar head coverage**
Hindfoot score (0–3)	**Posterior crease** · **rigid equinus** · **empty heel**

Midfoot contracture score ≈ 0 before tenotomy (TO) · **Dimeglio score** – older and neglected cases (TO)

Types of clubfoot (TO)

Type	Definition (TO)
Untreated (preferred to "neglected")	Never treated, presents after walking age (**> ~18 mo–2 yr**)
Recurrent	Corrected, then returns · brace failure · short TA (**equinus recurs FIRST / most often**) · overactive tibialis anterior
Resistant	A component cannot be corrected during treatment (TN won't reduce) · atypical clubfoot · tarsal coalition · syndromic
Atypical / complex	Severe cavus, all MTs flexed → **modified Ponseti**
Syndromic	Arthrogryposis, myelomeningocele, diastrophic → rigid, **recurrence is the disease**

Untreated vs neglected (TO lecture 2) untreated: never treated, any age · **neglected: no treatment after 18 months** (walking age; not 6 or 9 mo) · **relapse** = deformity returns **before** treatment completed · **recurrence** = returns **after** completion

Pirani (TO lecture 2) scored in the **maximally corrected position** · given as **0–10** with a forefoot component (adduction rigidity, long-flexor contracture) (TO – verify: standard Pirani is 0–6, midfoot + hindfoot)

– – Active space – –

Dimeglio classification (TO)

Dimeglio (0–20) grades **equinus, varus, adduction, rotation** (calcaneopedal block) · preferred in **older, neglected, rigid** feet

Dimeglio score (TO)	Type
1–4	Soft-soft (benign)
5–9	Soft-stiff
10–14	Stiff-soft
≥ 15	Stiff-stiff (very severe)

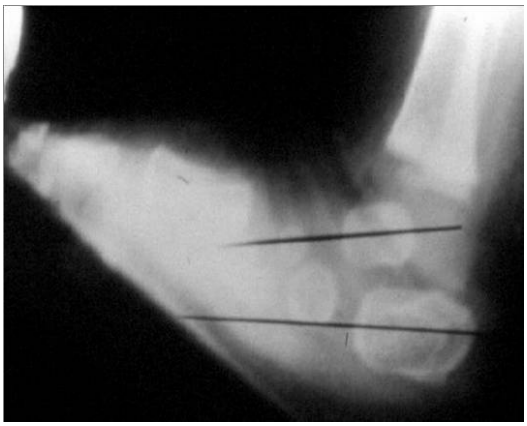
Key radiology findings

Usually not needed · simulate weight-bearing

View	Clubfoot finding
Dorsiflexion lateral (Turco)	**Hindfoot parallelism** · **talocalcaneal angle <25°** · flat talar head (older)
AP	**Kite (talocalcaneal) angle <20°** (normal 20–40°) · **negative talus–1st MT angle** (normal 0–20°) · parallelism
Lateral talocalcaneal angle	**< 25°** (Orthobullets) · **< 35°** (TO) – read the options
Talus–1st MT angle (TO)	**> 15°** (TO – verify: Orthobullets gives a negative angle, normal 0–20°)
Indications (TO)	Not routine – **hemimelias**, **tarsal coalition**, **flat-top talus** (outliers)
Older-child signs (TO)	**Talocalcaneal index (AP + lateral TC) < 40** · **tibiotalar angle negative** · talus–1st MT negative

Corrected TN joint after PMR (AP) **talar axis passes through centre of navicular and 1st MT shaft** (TO)
✗ Wrong options: points at 1st MT base only (residual adduction) · Kite <15° (that's deformity) · lateral parallelism

Prenatal US from **12 weeks** · **1st trimester** dx → associated anomalies very common · 2nd trimester → true clubfoot · **3rd trimester → ↑ false positives** (crowding)



Hindfoot parallelism on lateral

Clubfoot vs CVT on X-ray clubfoot: **parallel** talus + calcaneus · CVT: no parallelism, navicular (unossified) doesn't align with talus on plantarflexion

5 Management

Gold standard — **Ponseti** (**>90%** avoid comprehensive release) · start **as early as possible** (first 1–2 weeks); older infants (e.g. 9 months) still treated with Ponseti

French method — daily PT manipulation + taping/splints · 1–3 mo · family-dependent · rarely used in US

Ponseti technique – correction sequence, supination/abduction around talus & casting errors

Principle — **abduct the foot, rotating it laterally around a fixed talus** · **counterpressure on lateral talar head** (fulcrum)
✗ Wrong options: push against calcaneocuboid joint (Kite → rocker bottom) · manipulate heel directly

Order — **C → A → V → E**

Step	Manoeuvre
1 **Cavus** (first cast)	**Forefoot SUPINATION** – elevate 1st MT, align forefoot with hindfoot (✗ Wrong options: pronation → ↑ cavus, midfoot break)
2 Adductus + varus (together)	**Abduct** foot around talar head; supination gradually ↓ to neutral
3 Equinus (last)	Only when heel neutral/valgus + foot **≥60° abducted → percutaneous Achilles tenotomy** → cast **3 weeks** in max DF (15°)

Casts — **long-leg (toe-to-groin), knee 90°**, changed **weekly** · 5–6 casts usually · plaster
✗ Wrong options: short-leg (slips, doesn't control tibial torsion)

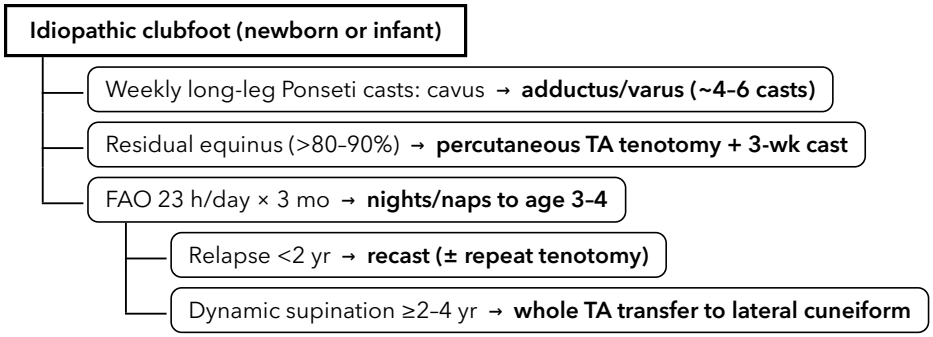
Final cast — max abduction **~70°** (hypercorrection) + DF 15° · end-point before tenotomy: TN reduced + **60° passive abduction** (TO)
✗ Wrong options: **30° abduction** (the false statement)

Manipulated in — **supination + abduction** (TO)
✗ Wrong options: pronation · adduction · dorsiflexion first

Casting details (TO) — manipulate **1–3 min** before each cast · change every **5–7 days** · tenotomy if **< 15–20° DF** · final cast **3 wk** in max DF + abduction

End-point of abduction (TO) — **TN joint reduction** (navicular covers talar head; head no longer felt), not an angle · usually **50–70°**, atypical foot as little as **20°** · no tenotomy before foot is **past the midline**

Cavus (TO) — no separate cast needed: keep forefoot **supinated** (1st ray raised) and abduct in the same cast → fewer casts · **never press on the calcaneus**



– – Active space – –

Percutaneous Achilles tenotomy for residual equinus

Rate	**80–90%** (Orthobullets) · **70–75%** / 70% (OB) · **>90%** (OB) – "most/almost all"
When	residual rigid equinus after cavus/adductus/varus corrected (~6–8 wk) · **under local anaesthetic** in clinic ✗ Wrong options: more casting · fiberglass · FAO first · TA transfer
Why	bracing a foot in equinus fails (slips, blisters) · prevents rocker bottom
Technique	**medial → lateral** (protect **posterior tibial artery**) · bleeding from **peroneal artery + lesser saphenous vein** (TO)
Open Z-lengthening of TA in varus foot	**distal horizontal limb medial** (corrects residual varus) (TO)
Equinus is NOT corrected by casting (TO)	tenotomy to reach **15–20° DF** · completion: sudden DF, palpable gap, **"pop"** · GA in older children
Spurious dorsiflexion (TO)	through lax midfoot · **lock subtalar by inverting** and judge true ankle DF · < ~**20°** real DF → tenotomy
Tenotomy level (TO)	**~1 cm above the palpable tuberosity** · too low → apophyseal damage, bony bump
Any age (TO)	first-time percutaneous tenotomy heals even at **8–10 yr** (intact paratenon; US-confirmed) · **repeat tenotomy rarely works** (unless ~10° needed) → **open Z-lengthening**

Ponseti casting & bracing protocol

Foot abduction orthosis (FAO / Denis Browne / Mitchell / Dobbs / Iowa / Steenbeek)	**critical**; noncompliance = biggest recurrence risk

Parameter	Value
Wear	**23 h/day × 3 months → nights + naps until age 3–4** (✗ Wrong options: 23 h for 12 months (TO false statement))
Affected foot ER/ abduction	**60–70°** (Orthobullets 60° · OB 70° · TO 70°)
Normal foot (unilateral)	**30–40°**
Dorsiflexion	**10–15°** (TO answer: **70° abduction/ER + 15–20° DF**)
Bar width	Heels at **shoulder width** (~1 inch wider)
Shoes	Straight medial border, low posterior cut, heel cup

After walking age	"daytime AFO + night bar" stated as **FALSE** (TO)
Bracing duration	nights + naps to **3–4 yr** (Orthobullets) · **4 yr (4–5 yr)** (TO) – read the options · braces: Steenbeek · Denis Browne · **John Mitchell** (TO)
"CTEV shoes" (TO)	straight inner border shoes are **not part of Ponseti** · walking children need no special shoes

Recurrence – bracing compliance & risk factors

Most dependent on	**brace (FAO) compliance** ✗ Wrong options: maternal age · family history · walking age · BMI
Risk factors	**noncompliance** + **parental education ≤ high school** ✗ Wrong options: race · sex

Relapse <2 yr → repeat manipulation + casting · **>2 yr** → casting, then **TA transfer** ± repeat TAL / gastroc recession

Highest recurrence (TO)	during the **bracing phase** · **equinus recurs first** · **recurrent cavus → think neurological** (tethered cord, dysraphism, CP, CMT) → image spine
Relapse / late presentation (TO)	**Ponseti first at any age** (even untreated at 5 yr) · **6 mo–2 yr hardest** to cast · limit recurrent casting to ~**3 casts**, then correct the residue surgically · **JESS** if child cannot comply with casting

Dynamic supination – tibialis anterior transfer

Cause	incomplete navicular reduction → **TA becomes a supinator** (medial overpull vs weak peroneals) · ~**one-third** of patients · begins **3–5 yr** (older than 12–18 mo)
Gait	**supination in swing**, lands on **lateral border** · forefoot inverts on active DF · foot passively plantigrade
Dynamic supination vs relapse (TO)	dynamic: foot **normal when standing**, problem only in gait · static deformity = **relapse** · **30–50%** at 4–5 yr (TO) · physiotherapy first (peronei) → TATT **> 4 yr**
In-toeing after clubfoot (TO)	internal tibial torsion, mostly resolves · surgery only after **8–10 yr**
Treatment	**whole TA transfer to lateral (3rd) cuneiform** (Garceau principle) · or split TA to cuboid / PB · Point = lateral cuneiform (more common) or cuboid · TA harvested from medial cuneiform + 1st MT base
Distractors	✗ Wrong options: more casting (to 60° abduction) · PTT through IOM (poor in clubfoot; for paralysed TA) · TA lengthening · TA to calcaneus (myelodysplasia calcaneus) · midtarsal osteotomy (fixed) · floor-reaction AFO · cuboid wedge

TA transfer criteria	
Rate	**10–50%** (Orthobullets) · **10–20% / 15–20%** (OB)
Age	**2–5 yr** (Orthobullets) · **~4 yr** (OB) · lateral cuneiform **ossified** – ≥2, usually ≥3 (TO) (✗ Wrong options: <1 yr · 12–15 yr)
TA strength	**≥ grade 4**
Subtalar joint	**Supple** – **rigidity = contraindication** (✗ Wrong options: "rigidity supplements")
Fixed hindfoot varus	Not a TA transfer case → osteotomy

– – Active space – –

Surgical release – outcomes & late deformities (dorsal bunion, residual cavus, rocker bottom)

Procedure	Indication
Posteromedial release + tendon lengthening	Resistant/recurrent after Ponseti · **rocker bottom from failed casting** · syndromic failures · at **9–10 mo** (non-syndromic) · <6 mo surgery → poorer results
Medial column lengthening / lateral column shortening / **cuboid decancellation**	>2–3 yr with release · **3–10 yr** recurrent "bean-shaped" foot
Talectomy	Severe rigid recurrent clubfoot in **arthrogryposis**, 6–10 yr
Multiplanar supramalleolar osteotomy	Salvage, older (8–10 yr), complex rigid or **insensate** foot
Ring fixator (TSF / Ilizarov / JESS)	Complex resistant/neglected · **high recurrence after removal**
Triple arthrodesis	**Almost never** · not in insensate feet (ulcers)

Outcome — **extent of soft-tissue release correlates inversely** with long-term function · stiffness + pain common
✗ Wrong options: subtalar/ankle function · cast duration

Master knot of Henry release — frees **FDL/FHL tether** to spring ligament + sustentaculum → allows **TN reduction** (TO)

Neglected clubfoot >3 yr — Ilizarov distraction histogenesis → **better joint mobility, ↓ recurrence** vs open PMR (TO)

JESS — distract **CC + TN joints (correct midfoot) before equinus** → avoids rocker bottom (TO)

PMR post-op — TN K-wire 6 wk · cast change at 2 wk · then FAO

À la carte surgery by residual component (TO)

Component left	Procedure (TO)
Equinus	1st time: percutaneous tenotomy · recurrent / scarred: **open TA Z-lengthening ± posterior capsulotomy** (ankle + subtalar)
Adductus (short medial, long lateral column)	**Superficial → deep plantar-medial release** · **lateral column shortening**: CC resection/fusion = **Dillwyn Evans**, cuboid decancellation, cuboid closing wedge · **double tarsal osteotomy** (medial cuneiform opening + cuboid closing wedge) for **"kidney-bean" foot**
Varus (hindfoot)	**Dwyer lateral closing-wedge calcaneal osteotomy** – min age **3–4 yr**
Cavus	**Plantar fascia release** → medial cuneiform plantar opening wedge or MT dorsal closing wedge (CORA near medial cuneiform)
Dynamic supination (> 4 yr)	**TA transfer to lateral cuneiform**
All components, rigid, older child	**Shingade (modified Japas) osteotomy** at midtarsal apex + TAL · PMSTR (older option)
Severe neglected / syndromic / rigid recurrence	**JESS / Ilizarov** gradual correction · **talectomy** salvage

Evans trap (TO) — clubfoot **Dillwyn Evans SHORTENS** lateral column (CC resection/fusion) · adult flatfoot **Evans LENGTHENS** it (calcaneal lengthening)

Dwyer vs midfoot osteotomy (TO) CORA of adductus is in the **midfoot** · correcting at calcaneus → **Z-shaped (skew) foot**

Lichtblau (TO) for long lateral column · no long-term stiffness · complication **Z / skew foot**

Superficial plantar-medial release (TO) open **laciniate ligament** · free 3 origins of **abductor hallucis** from plantar vessels/nerves · divide plantar fascia · deep: TN capsule, **long plantar + spring** ligaments, Z-lengthen tibialis posterior

PMSTR (TO) **Turco · Cincinnati** (circumferential) · releases **knot of Henry** (FHL × FDL) · **> 50% midfoot arthritis at 10–15 yr** → now avoided except posterior release

PMSTR (TO lecture 2) after failed Ponseti (**up to 10–11 casts**) · incisions: **Turco**, **Cincinnati**, **Carroll dual incision** · releases **3 tendons** (TP, FHL, FDL) + **3 ligaments** (deltoid, spring, TN) + **3 capsules** (ankle, subtalar, TN) ± TA Z-lengthening · lateral column shortening: **Lichtblau**, **Evans**, cuboid decancellation, CC fusion

Age (TO lecture 2)	Approach
< 4 yr	Casting + soft-tissue release
4–8 yr	Selective osteotomies
≥ 10 yr	**Triple arthrodesis** (rigid feet)

Triple arthrodesis **almost never** (Orthobullets) · **after 10 yr in rigid feet** (TO) – read the options

JESS (TO) **BB Joshi** external stabilisation system · gradual soft-tissue distraction without osteotomy, Ponseti order (equinus last)

Talectomy (TO) rigid uncorrectable recurrence, esp. **arthrogryposis** → tibia on calcaneus · **triple arthrodesis** (TO) – painful arthritic foot after PMSTR

Syndromic (TO) higher threshold for surgery · aim **braceable plantigrade** foot · lifelong (often 24 h) bracing

6 Complications

Rocker bottom **correcting equinus before heel varus** → DF through **midfoot** · failed casting with rocker bottom → **PMR + TAL**
✗ Wrong options: under-corrected pronation → residual cavus · over-corrected adductus → abduction · below-knee casts

Residual cavus (after surgery) **navicular pinned dorsally subluxated** (22%) · insufficient plantar release
✗ Wrong options: inadequate TAL → equinus · LCL → worsens adductus

Pes planus overcorrection (extensive subtalar release)

Intoeing **internal tibial torsion** / talar IR in mortise

Talar osteonecrosis vascular insult → collapse

Dorsal bunion **strong TA overpowering weak peroneus longus** (PL = **superficial peroneal nerve**) → elevated 1st ray · weak gastrocsoleus after release → **FHB/FHL** compensate → plantarflexed hallux · inadvertent PL lengthening · Rx: **TA lengthening/transfer + FHL transfer to 1st MT head (reverse Jones)** ± plantarflexion osteotomy

NOT a complication **tarsal tunnel syndrome** (TO)

– – Active space – –

Complications (TO) **flat-top talus** · dorsal bunion · rocker bottom · residual deformity · early arthritis



Dorsal bunion after clubfoot surgery

X vs Y

	Clubfoot	CVT	Metatarsus adductus	Calcaneovalgus
Hindfoot	**Equinus + varus**	Equinus + valgus	**Normal**	Calcaneus (DF) + valgus
Midfoot	Cavus, adductus	**Dorsal TN dislocation**, rocker	Adducted forefoot only	Normal
Flexibility	Rigid	Rigid	Flexible / stiff	Flexible
Lateral X-ray	**Talocalcaneal parallelism**	Vertical talus	Normal	–
Treatment	**Ponseti**	Reverse Ponseti	Observe / stretch	Stretch

	Ponseti	Kite
Fulcrum	**Lateral talar head**	Calcaneocuboid joint
Result	Correction	**Rocker bottom**
Order (TO)	**Cavus, adductus, varus together**; equinus last	Sequential (adduction before equinus)
Midtarsal joints (TO)	Mobile, plantigrade	**Rigid**

How it's asked

"Ponseti – all true EXCEPT?"	→ **pronation in initial cast** (✓ weekly long-leg casts, ✓ tenotomy before final cast, ✓ abduction with talar counterpressure, ✓ adduction before equinus)
"First component addressed?"	→ **cavus**
"9-month-old with talocalcaneal parallelism – first line?"	→ **serial casting**

"3-month-old with rocker bottom during Ponseti — error?"	→ **corrected equinus before heel varus**
"Key Ponseti concepts — EXCEPT?"	→ **last cast at 30° abduction** (it's ~70°)
"True of Ponseti?"	→ **metatarsal heads turned into supination with first cast** ✗ Wrong options: tenotomy in 20% · short-leg cast
"Foot manipulated in?"	→ **supination and abduction**
"Healthy newborn with clubfoot — best treatment?"	→ **Ponseti casting** ✗ Wrong options: parental stretching · AFO · TA transfer · short-leg cast
"Initial phase described to parents?"	→ **weekly long-leg casts for cavus/adductus/varus, then tenotomy + final cast**
"9-month-old untreated bilateral clubfoot — true?"	→ **abduct foot rotating laterally around fixed talus** ✗ Wrong options: 50% success
"After 6 weeks Ponseti, rigid 10° plantarflexion?"	→ **percutaneous Achilles release under local**
"Recurrence most dependent on?"	→ **brace compliance**
"Recurrence risk factors?"	→ **noncompliance + parental education ≤ high school**
"3-yr, lands on lateral border, supinates in swing, DF 20°?"	→ **tibialis anterior transfer** (TO)
"Muscle causing swing-phase dynamic deformity?"	→ **tibialis anterior**
"Destination of transferred tendon?"	→ **lateral cuneiform**
"TA transfer appropriate for?"	→ **3-yr whose foot supinates on dorsiflexion**
"TA transfer to lateral cuneiform — EXCEPT?"	→ **subtalar rigidity supplements the transfer**
"Age for TA transfer?"	→ **~4 years**
"Walks on lateral border, forefoot inverts on DF, plantigrade passively?"	→ **lateral transfer of anterior tibialis** ✗ Wrong options: PTT through IOM
"Supination in swing, 20° DF?"	→ **TA transfer to dorsolateral foot**
"Z-lengthening TA in CTEV — distal limb?"	→ **medial** (TO)
"Late deformity after clubfoot surgery — EXCEPT?"	→ **tarsal tunnel syndrome** (TO)
"Master knot of Henry release purpose?"	→ **free long flexors to reduce TN joint** (TO)
"JESS — prevent rocker bottom by?"	→ **distract CC + TN before equinus** (TO)
"Ilizarov vs PMR in neglected clubfoot >3 yr?"	→ **better mobility, less recurrence with frame** (TO)

– – Active space – –

"AP sign of reduced TN after PMR?"	→ **talar axis through navicular centre and 1st MT shaft** (TO)
"Residual cavus after release — technical error?"	→ **navicular pinned dorsally subluxated**
"Dorsal bunion — weak muscle's nerve?"	→ **superficial peroneal**
"Failed casting at 4 months with which deformity → PMR?"	→ **rocker-bottom**
"Long-term impairment after release correlates with?"	→ **extent of soft-tissue release**
"Release complications — EXCEPT?"	→ **tarsal tunnel syndrome** (✓ dorsal bunion, talar AVN, rigid planus, intoeing)
"Dorsal bunion + TA/PL imbalance — original condition?"	→ **congenital talipes equinovarus**
"16-yr dorsal bunion after childhood foot surgery — original deformity?"	→ **talipes equinovarus**
"Photograph consistent with clubfoot?"	→ **equinovarus with cavus + adductus** ✗ Wrong options: skewfoot · MTA · calcaneovalgus · CVT
"24-month-old with clubfoot not walking?"	→ **neurologic disorder**
"10-yr after clubfoot surgery, smaller limb — why?"	→ **intrinsic hypoplasia**
"Prenatal clubfoot — true?"	→ **will walk and run with treatment**
"Limb-development transcription factor + clubfoot?"	→ **PITX1–TBX4**
"Ponseti — all EXCEPT?"	→ **brace 23 h/day for 12 months** (it's 3 months) (TO)
"Latest Ponseti casting?"	→ **weekly long-leg cast → tenotomy → 3-week cast** (TO)
"False statement?"	→ **daytime AFO + night bar after walking age** (TO)
"Ideal FAO position?"	→ **70° abduction/ER + 15–20° DF** (TO)
"End-point of Ponseti casting?"	→ **easy reduction of the TN joint** (not abduction angle) (TO)
"Commonest component to recur?"	→ **equinus**; recurrent cavus → **neurological cause** (TO)
"Gene associated with idiopathic clubfoot?"	→ **CASP10** (TO)
"Most accepted theory of CTEV?"	→ **germ plasm theory** (talar cartilage anlage defect) (TO)
"Neglected clubfoot — age cut-off?"	→ **18 months** (TO)

CONGENITAL CURLY TOE

– Pediatrics · 24 / 112 –

1 Basics

Deformity	**flexion + varus of the interphalangeal joints** · toe curls under its medial neighbour
Location	**lateral three toes** (3rd + 4th most often) · usually **bilateral**
Cause	contracture of **FDL** or **FDB** (most common)

3 Presentation

Symptoms	usually **asymptomatic** · if symptomatic: pain, shoe-wear difficulty, **nail problems**
Natural history	**~25% improve spontaneously** (Sweetnam) · more likely to **persist in older child (e.g. 6 yr)**



Congenital curly toes

4 Diagnosis & Classification

Diagnosis	clinical

Key radiology findings

Radiographs	**not indicated**

5 Management

Congenital curly toe

- Infant / asymptomatic → **observation**
- Symptomatic (pain, shoes, nail) after age 3 → **flexor tenotomy**

Observation	first line (same success as strapping, less painful) · **toe strapping/taping not beneficial** (no change or temporary)

– – Active space – –

Operative criteria

Indication	Procedure
Severe toe or **nail-bed** deformity, symptomatic, **>3 yr**	**Flexor tenotomy** (FDL ± FDB) — **85–90%** effective
–	Tenotomy **= FDL transfer to extensor hood** in efficacy
Rarely	PIP arthrodesis

Techniques open tenotomy of both FDB slips + FDL · tenotomy of one FDB slip · FDL → extensor hood transfer

How it's asked

"8-month-old with curly toes, first surgeon offered FDB tenotomy + volar plate release?"	→ **observation** ✗ Wrong options: FDL transfer · FDB tenotomy · taping
"Symptomatic curly toe in a 6-year-old?"	→ **flexor tenotomy** ✗ Wrong options: observe · tape 8 wk · transfer (no better) · PIP fusion
"Cause?"	→ **FDL / FDB contracture**

CONGENITAL DISLOCATION OF PATELLA

– Pediatrics · 25 / 112 –

1 Basics

Definition	rare, **irreducible lateral (posterolateral) patellar dislocation** · present at birth (often missed) or reducible at birth → subluxation → **fixed dislocation in childhood**

Pathoanatomy

Bone	**small or absent patella** · **hypoplastic trochlea** · **external tibial torsion**
Soft tissue	**thick, tight lateral structures (ITB, retinaculum)** · **tight quadriceps** (superior subluxation, patella alta)
Associated	**Larsen** · arthrogryposis · diastrophic dysplasia · **nail-patella** · **Down** · Ellis–van Creveld

Pathomechanics

Quadriceps action	laterally displaced extensor mechanism → **quadriceps becomes a knee flexor** ✗ Wrong options: extension · internal rotation (it's external) · varus (it's valgus) · posterior translation
Patella anatomy	largest sesamoid · ossifies **3 yr (♀) / 4–5 yr (♂)** · blood supply **distal → proximal** (geniculates, supreme geniculate, anterior tibial recurrent)

3 Presentation

History	associated syndromes · **delayed walking** (mimics **cerebral palsy**)
Exam	**genu valgum** · **knee flexion contracture** · external tibial torsion · **"smiley face" kneecaps** · prominent condyles · small, laterally placed patella hard to palpate · limited active extension
Pain	all patients **≥7 yr** had pain (Gordon & Schoenecker)



Smiley-face knees in congenital patellar dislocation

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

<3 yr	patella unossified → radiographs unhelpful → **ultrasound or MRI**
>3 yr	**AP, lateral, sunrise** · dislocated patella · hypoplastic trochlea



Sunrise view – laterally dislocated patella

5 Management

Observation	**not recommended** (long-term functional deficit, malalignment, pain)
Surgery early	allows trochlear remodelling

Congenital patellar dislocation
- Early surgical reduction (Andrish)
  - Soft tissue: lateral release + VL advancement + quad lengthening + MPFL (semitendinosus)
  - Distal realignment if needed → **Roux-Goldthwait (open apophysis)**

Andrish soft-tissue steps	lengthen lateral retinaculum · detach VL from septum, advance proximally · release distal patellomeniscal ligaments · **lengthen quadriceps tendon, shorten patellar tendon** (alta) · **MPFL reconstruction** (semitendinosus rerouted through MCL to patella)
Bony	distal realignment rarely needed · tubercle transfer limited by apophysis → **Roux-Goldthwait** (lateral half patellar tendon medialised)

6 Complications

Recurrence

X vs Y

	Congenital dislocation	Recurrent patellar instability
Age	Birth / early childhood	Older child, activity
Patella	**Fixed, irreducible**	Reducible; normal between episodes
Associated	Syndromes (Down, nail-patella, Larsen)	Ligamentous laxity
Treatment	**Surgical reduction**	Nonop → MPFL

How it's asked

"10-day-old, laterally dislocated patella on US — quadriceps now does?"	**→ flexes the knee**
"Imaging under age 3?"	**→ ultrasound / MRI**
"Treatment?"	**→ early surgical reduction (lateral release, quad lengthening, MPFL)**
"Distal realignment with open apophysis?"	**→ Roux-Goldthwait**

– – Active space – –

CONGENITAL DISLOCATION OF THE KNEE

– Pediatrics · 26 / 112 –

1 Basics

Spectrum	**positional contracture → rigid dislocation** · tibia anterior on femur
Structural	**quadriceps contracture** (main) · anterior subluxation of hamstrings · **absent suprapatellar pouch** · tight collaterals · ACL abnormalities
Associated syndromes	**myelomeningocele** · **arthrogryposis** · **Larsen** (clubfeet + hip, knee, elbow dislocations)
Genu recurvatum (TO)	hyperextension **> 5°** · causes: ligament laxity · **congenital knee dislocation** · quadriceps weakness · physeal injury · associations: arthrogryposis, **absent cruciates**, **hypoplastic patella** · Rx serial manipulation + casting
Associated deformities	**DDH – 50%** (Orthobullets) · ipsilateral hip dislocation **70–100%** (OB) · **clubfoot** · metatarsus adductus

3 Presentation

Hyperextended knee at birth · anterior tibial translation on X-ray



Congenital knee dislocation

4 Diagnosis & Classification

Tarek classification

Grade	Radiology	Passive flexion	Treatment
I	Simple recurvatum	**>90°**	Manipulation + casting
II	Subluxation / dislocation	**30–90°**	Casting if <1 mo; **PQR** if seen later
III	Dislocation	**<30°**	**Surgery** (open V-Y quadricepsplasty)

Table 1 Tarek CDK grading system

Grade	Range of passive flexion	Radiology
GI	>90°	Simple recurvatum
GII	30–90°	Subluxation/dislocation
GIII	<30°	Dislocation

Tarek grades

5 Management

Congenital knee dislocation

- Newborn (Tarek I, early II) → **weekly long-leg manipulation + flexion casting**
  (knee + hip dislocated → treat KNEE first, then Pavlik)
- Failed to gain 30° flexion by 3 mo / grade III / recurrent → **surgical release**
  - Grade II (later) → **percutaneous quadriceps recession**
  - Older, failed casting, grade III → **open V-Y quadricepsplasty / open reduction**

Serial casting vs quadricepsplasty/open reduction

Knee + ipsilateral hip dislocated (2-day-old) → manipulate + cast knee first – **can't fit Pavlik on a hyperextended knee**
✗ Wrong options: Pavlik first · spica · delay · quad lengthening

Casting failure — **90° not attained in 4 weeks** (OB) · **30° not gained after 3 months** (Orthobullets) → surgery

6-month-old, failed 3 mo casting, flexion 15°, hyperextension 25° (recurrent grade III) → open V-Y quadricepsplasty + above-knee cast
✗ Wrong options: cast in extension · more flexion casts · ACL division alone · percutaneous recession (for grade II at presentation)

7-month-old, can't flex >30° after casting, hips normal → open reduction – ideally **~6 months, before age 1**
✗ Wrong options: observe · more casting · closed reduction · wait for walking

Operative criteria

Indication	Procedure
<30° flexion after 3 months casting	Soft-tissue release
Tarek II after 1 month, III, recurrent	Release
Goal	**90° flexion**
Options	**Percutaneous quadriceps recession (PQR)** · **V-Y quadricepsplasty / Z-lengthening** · anterior capsule release · posterior transposition of hamstrings · collateral mobilisation
Post-op	Cast **45–60° flexion × 3–4 weeks**

ACL division + quad release — only fair results

– – Active space – –

How it's asked

"2-day-old hyperextended knee + subluxated ipsilateral hip — next?"	→ **manipulation and casting of the knee**
"6-month-old, failed casting, recurrent grade III?"	→ **open V-Y quadricepsplasty + above-knee cast**
"7-month-old can't flex past 30° after casting?"	→ **open reduction**
"Indication for surgery?"	→ **failure to gain 30° flexion after 3 months casting**

CONGENITAL HALLUX VARUS (ATAVISTIC GREAT TOE)

– Pediatrics · 27 / 112 –

1 Basics

Definition	rare idiopathic **varus (medial) deviation of the great toe at the MTP joint** · noticed **after walking age** · few degrees → 90°
Pathoanatomy	**firm, fibrous, band-like abductor hallucis** · **short, thick 1st metatarsal** · **longitudinal epiphyseal bracket (LEB)** of 1st MT or proximal phalanx · skeletal dysplasia (**diastrophic dwarfism**)
Associated	**polydactyly** (often)
Pathoanatomy (TO)	short thick 1st MT · **fibrous medial band** · **rudimentary duplicated great toe** · hallux varus = deformity in **phalanges/MTP** vs metatarsus primus varus = deformity in **metatarsal**
Modified Farmer (TO)	release fibrous band + create **syndactyly/synostosis between 1st and 2nd toes**

Longitudinal epiphyseal bracket

LEB (delta phalanx / bracket epiphysis) – anomalous **C-shaped secondary ossification centre** along the **medial diaphysis** → blocks transverse/longitudinal growth → progressive shortening + angulation

Frequency	2–14% of congenital hand/foot deformities · **bilateral ~75%** · hand → clinodactyly, foot → **hallux varus**

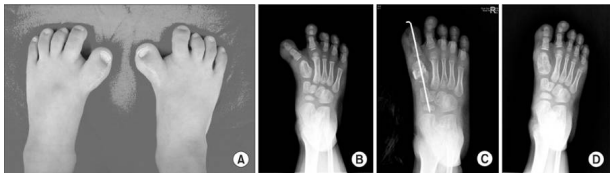
3 Presentation

Symptoms	cosmetic · **difficult shoe wear** · usually painless

4 Diagnosis & Classification

Key radiology findings

Views	AP, lateral, oblique foot
Findings	**short, thick 1st metatarsal** · bracket (if unossified → **ultrasound** shows cartilaginous C-shaped structure)



Congenital hallux varus – short thick 1st MT

Differentials	**metatarsus adductus** · **metatarsus primus varus** (1st MT varus, **no MTP deformity**)

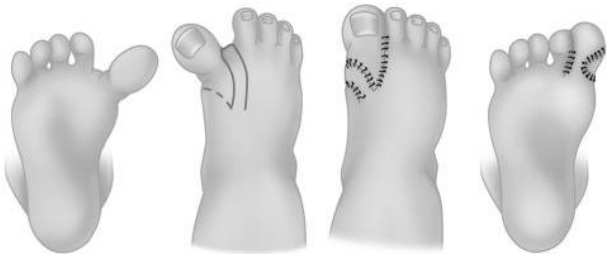
– – Active space – –

5 Management

Observation rarely; deformity **worsens with age** → surgery often in infancy

Operative criteria

Situation	Procedure
Mild, resistant	**Abductor hallucis release**
Epiphyseal bracket	**Excision of central bracket (physiolysis) ± PMMA / fat interposition – early, ideally <1 yr, before ossification** · growth often resumes · may need later realignment/lengthening
Moderate–severe	**Farmer – syndactylise hallux to 2nd toe** to hold correction
Older, ossified bracket, big deformity	Distraction / angular osteotomy



Farmer procedure – syndactyly to 2nd toe

Longitudinal epiphyseal bracket – excision

9-month-old, bilateral hallux varus, unossified C-shaped bracket on US → immediate excision of preossified bracket ± PMMA interposition
✗ Wrong options: observe until ossified then osteotomy · wait until 3–4 yr · ex-fix distraction · serial casting (soft-tissue deformities only)

6 Complications

Recurrence · incomplete correction

How it's asked

"Infant hallux varus with preossified bracket?"	**→ early bracket excision ± PMMA**
"Muscle causing atavistic great toe?"	**→ tight band-like abductor hallucis**
"X-ray finding?"	**→ short, thick 1st metatarsal**
"Moderate–severe deformity – procedure that holds correction?"	**→ Farmer (syndactyly to 2nd toe)**

CONGENITAL POPLITEAL WEBBING

– Pediatrics · 28 / 112 –

1 Basics

Also called **popliteal pterygium syndrome** · affects face, limbs, genitalia

Incidence rare, **1:300,000** · **M = F**

Genes & lab markers

Inheritance **autosomal dominant** (or sporadic – **advanced paternal age**)

Gene **IRF6** – transcription factor for head/face, skin, genital development

Anatomy of the web

Structure	Position
Subcutaneous fibrous band	Superficial, along **free edge** (ischium → calcaneus)
Sciatic / peroneal nerve	**Deep to band**, ~halfway between free edge and apex · often **shortened**
Popliteal artery/vein	Deep in popliteal space

3 Presentation

Web behind knee from **ischial tuberosity to heel** · usually bilateral · limits extension, abduction, rotation → **knee flexion contracture**

Other **cleft lip/palate** · syndactyly · triangular skin fold over hallux nail · **ankyloblepharon** (eyelids), **syngnathia** (jaws) · hypoplastic labia / **cryptorchidism, bifid scrotum** · cognitive delay · spina bifida



Popliteal pterygium

4 Diagnosis & Classification

Key radiology findings

X-ray (AP/lateral) knee flexion contracture

– – Active space – –

MRI — **pre-op planning** to locate neurovascular structures (wide variation, injury risk)



MRI of popliteal pterygium

5 Management

Femoral shortening with soft-tissue release

Contracture	Treatment
<20°	Nonoperative
20–40°	Hamstring lengthening + post-op splinting
Up to 60°	**Popliteal Z-plasties** + post-op serial casting (avoid NV tension)
>60°	**Femoral shortening osteotomy + soft-tissue releases** (Z-plasty, band excision, hamstring lengthening, posterior capsulotomy) · or gradual ex-fix — **recurs fast** without formal soft-tissue procedures + splinting

4-month-old, bilateral 100° fixed contractures → femoral shortening + soft-tissue releases
✗ Wrong options: serial casting · PT/splints · soft tissue alone · frame without release

Technique — resect/tenotomise fibrous band · **Z-plasty** skin · sciatic nerve grafting or tendon lengthening if needed

How it's asked

"100° popliteal pterygium contractures — most likely to succeed?"	**→ femoral shortening osteotomy + soft-tissue releases**
"Gene?"	**→ IRF6, autosomal dominant**
"Pre-op imaging?"	**→ MRI to locate sciatic nerve and vessels**
"Where is the sciatic nerve?"	**→ deep to fibrous band, halfway between free edge and apex**

CONGENITAL PSEUDOARTHROSIS OF CLAVICLE

– Pediatrics · 29 / 112 –

1 Basics

Definition — **failure of fusion of medial and lateral ossification centres** of the clavicle

Right-sided middle third

Side — **right ~90%** · **left only with dextrocardia / situs inversus** · **bilateral <10%** (Orthobullets) / ~10% (TO) – bilateral linked to **bilateral cervical ribs**

Site — **middle third**
✗ Wrong options: lateral third · left · both equally

Cause — extrinsic compression / pulsation of **subclavian artery** (higher on right)

Genetics — no clear involvement

3 Presentation

Exam — **painless, nontender mass** on clavicle · present at birth, **no callus**, no trauma history · ± scapular winging



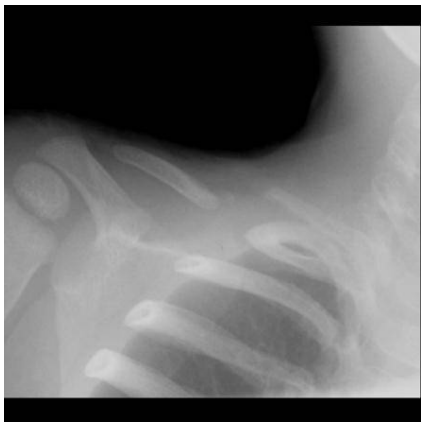
Clavicular pseudarthrosis – painless bump

4 Diagnosis & Classification

Key radiology findings

AP clavicle — **rounded, sclerotic bone ends** at pseudarthrosis · no callus (vs fracture)

– – Active space – –



Congenital pseudarthrosis of clavicle

5 Management

Observation minimal symptoms / cosmetic only

Operative criteria

ORIF + iliac crest bone graft at 3–6 yr pain · functional impairment · cosmesis · union usually achieved · **avoid bone-graft substitutes** (↑ nonunion)

X vs Y

	Congenital pseudarthrosis	Birth fracture	Cleidocranial dysplasia
Side	**Right**, middle third	Either	Bilateral
Pain	**Painless**	Painful	Painless
X-ray	**Rounded sclerotic ends**, no callus	Callus by ~1 wk	Absent / hypoplastic clavicles
Other	–	–	Skull, pubis, teeth

How it's asked

"Congenital pseudarthrosis of clavicle – side?"	→ **right** (TO)
"Almost always presents?"	→ **right side, middle third**
"Painless clavicle mass with rounded sclerotic ends – treatment?"	→ **observe; ORIF + iliac crest graft at 3–6 yr if symptomatic**

CONGENITAL VERTICAL TALUS

– Pediatrics · 30 / 112 –

1 Basics

Epidemiology

Incidence	rare, **1:150,000** · **M:F 2:1** · **50% bilateral**
Also called	congenital convex pes valgus · "teratologic dorsolateral dislocation of the talocalcaneonavicular joint"

Pathoanatomy

Rigid	**irreducible dorsolateral navicular dislocation** on talar neck · **vertical talus** · calcaneal **eversion** + equinus · **attenuated spring ligament**
Contractures	**Achilles** + **peroneals** (hindfoot equinovalgus) · **EDL, EHL, tibialis anterior** (forefoot DF + abduction)
Tendon displacement	**peroneus longus + tibialis posterior** shift to act as **dorsiflexors**

Associations – neuromuscular/genetic conditions & neural axis work-up

~50% associated with **neuromuscular disease or chromosomal/syndromic** conditions (>50% / 60–85% in other OB explanations)

Associated	Not associated
Myelomeningocele / spinal dysraphism, **diastematomyelia**	**Amniotic band syndrome** (TO, OB)
Arthrogryposis	Posteromedial tibial bowing (→ calcaneovalgus)
Congenital hip dislocation (43–50%)	Chromosome **22q11** deletion
Cerebral palsy · **spinal muscular atrophy** · NF	–
Chromosomal deletions, trisomy 18, Cri-du-Chat	–
Cleft lip/palate, CHD, chr 12 duplication (less often)	–

Highest association → neuromuscular disease

Genes & lab markers

Familial	up to **20%** · **HOXD10** (transcription factor) in familial/"idiopathic" cases · shared pathway with clubfoot

3 Presentation

Rigid rocker-bottom foot	**convex plantar surface** ✗ Wrong options: "concave" (the false statement)
Hindfoot	**fixed equinus + valgus** (heel in **equinus**; ✗ Wrong options: calcaneus · equinovarus · supple)

– – Active space – –

Midfoot	rigid dorsiflexion (dislocated navicular) · **prominent talar head** in medial plantar arch
Forefoot	**abducted + dorsiflexed**
Gait	**"peg-leg" / calcaneal gait** (weak push-off, excess heel contact)
Neuro exam	always (frequent neuromuscular cause)
CVT features (TO)	detected at birth · convex sole with **deep dorsolateral crease** · rounded medial + plantar prominence (talar head) · forefoot dorsiflexed at midtarsal joint · with growth: **"hourglass" talus**, callosities under anterior calcaneus + medial border, severe forefoot abduction, **heel off ground in stance** · ↑↑ Meary angle (TO)



Rigid rocker-bottom foot

4 Diagnosis & Classification

Diagnosis – forced plantar-flexion lateral radiograph; oblique talus

Diagnostic test	**lateral X-ray in maximum (forced) plantar flexion → persistent dorsal TN dislocation** ✗ Wrong options: MRI · US in DF · max-DF lateral · CT

Key radiology findings

View	Finding
Neutral lateral	**Vertical talus**, dorsal navicular · talar axis passes **plantar to 1st MT–cuneiform axis**
Forced plantarflexion lateral	TN **stays dislocated** · **Meary's angle >20°** · **TAMBA >35°** (talar axis–metatarsal base angle, Hamanishi)
Forced dorsiflexion lateral	**Fixed equinus**
AP	**Talocalcaneal angle >40°** (normal 20–40°)

Before navicular ossifies (~age 3)	use **1st metatarsal** as navicular proxy (✓ the true statement)

Radiograph details (TO) DF lateral: **calcaneal pitch stays negative** (fixed equinus) · AP: talus–1st MT and calcaneus–4th MT **> ~30°** abduction, Kite > 40° · incidence ~1 in 10,000 (TO) · 1st MT surrogate **< 2 yr** (TO)

– – Active space – –

(TO)	CVT	Oblique talus	Flexible flatfoot
PF stress view	**No reduction**	Reduces	Reduces
TAMBA	**> 35°**	< 35°	< 35°
Calcaneum	Locked	Locked	**Unlocked**

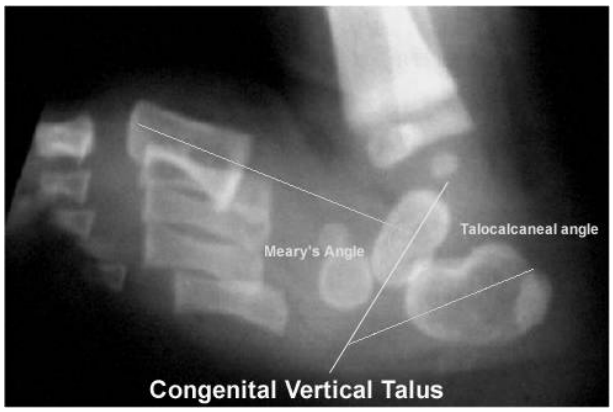
MRI **neuraxial imaging** to exclude neurologic disorder

Rigid flatfoot in infant (no plantarflexion, no subtalar motion, vertical talus not reducing) → neurologic work-up incl. neural-axis imaging first
✗ Wrong options: urgent surgery · foot MRI · reassurance

4-month-old arthrogrypotic with suggestive X-rays → obtain forced plantarflexion radiograph first
✗ Wrong options: cast or operate before confirming



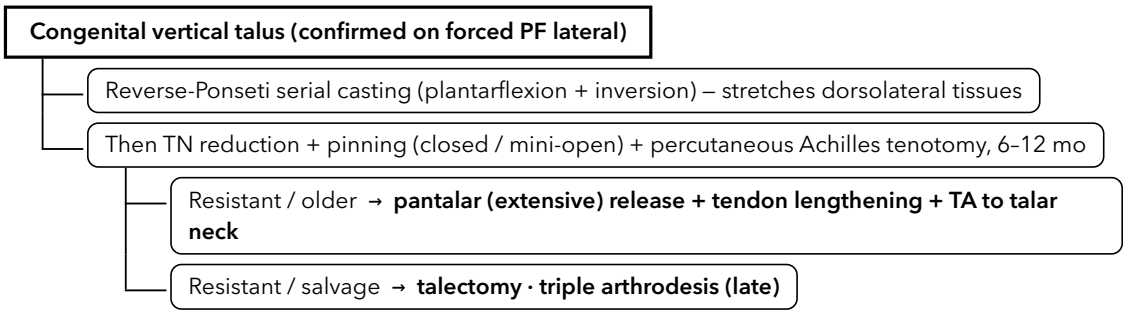
Forced plantarflexion lateral — persistent dislocation



Meary and talocalcaneal angles

– – Active space – –

5 Management



Treatment – reverse Ponseti casting then talonavicular pinning/Achilles tenotomy or open release

Casting purpose — **stretch dorsolateral soft tissues before surgery** · plantarflexion + supination/inversion until 1st MT aligns with talus
✗ Wrong options: casting alone corrects (that's oblique talus)

Initial management at 10 months — **manipulation + casting**
✗ Wrong options: shoes/scaphoid pad · immediate open reduction · LCL · observe

After full casting course → TN reduction + pinning (± Achilles tenotomy)
✗ Wrong options: gastroc recession alone · observe · tenotomy alone · isolated tendon reconstruction

Standard of care (Dobbs) — **serial casting → mini-open/percutaneous TN reduction + K-wire + percutaneous Achilles tenotomy** (TO)
✗ Wrong options: casting alone · open extensive release first · subtalar fusion

Toddler, irreducible CVT → casting then open reduction + Achilles lengthening
✗ Wrong options: Ponseti alone · talectomy · subtalar fusion

Operative criteria

Item	Criterion
Timing	**6–12 months** (Orthobullets) · **before 2 years** (TO)
<2 yr	Release + tendon lengthening + TN reduction + fixation (neural tube defect: + tendon transfers)
>2 yr	Extensive release + tendon transfers · **bony procedures >4 yr**
Missed / >3 yr	Less predictable → **triple arthrodesis** salvage
Options (TO)	**Open reduction 1–4 yr** · **Grice-Green extra-articular subtalar fusion** · triple arthrodesis in older children
Approach with least AVN	**Dorsal** (vs posteromedial) (TO)
Extensive release components	Lengthen **peroneals, Achilles, toe extensors** · reduce + pin TN · **spring ligament reconstruction** · **TA transfer to talar neck**

6 Complications

Missed CVT — reconstruction unpredictable after **age 3** → triple arthrodesis

Untreated painful rigid foot, midfoot calluses, significant disability

Talar AVN less with dorsal approach

X vs Y

	Congenital vertical talus	Congenital oblique talus	Calcaneovalgus
TN joint	**Fixed dorsal dislocation**	Subluxation	Normal
Forced PF lateral	**Stays dislocated**, 1st MT dorsal to talar axis	**Reduces** – talus parallels 1st MT	Normal
Hindfoot	**Equinus**-valgus	–	**Calcaneus** (DF)
Rigidity	**Rigid**	Flexible	Flexible
Treatment	**Cast → TN pin + tenotomy**	**Stretching** / casting; rarely pin + TAL	Stretch
Associations	Neuromuscular / syndromic ~50%	–	Posteromedial bowing

Short-foot exercises strengthen intrinsics, raise medial arch → **flexible flatfoot** (TO)
✗ Wrong options: plantar fasciitis · tennis elbow · post-triple arthrodesis

How it's asked

"NOT true of CVT?"	→ **rocker bottom with concave surface** (it's convex) (✓ TC angle >40°, ✓ calcaneal gait, ✓ surgery before 2 yr) (TO)
"Heel position in CVT?"	→ **equinus** (TO)
"Diagnostic criterion?"	→ **PF stress view TAMBA >35°** (TO)
"13-month-old, TN reduces on plantarflexion lateral?"	→ **congenital oblique talus**
"NOT characteristic of CVT?"	→ **posteromedial tibial bowing** association
"9-month-old rocker-bottom feet – accurate statement?"	→ **1st MT–talus relationship used before navicular ossifies** ✗ Wrong options: reduces with PF · equinovarus · talectomy usually
"Best study to confirm?"	→ **lateral X-ray in maximum plantar flexion**
"4-month-old arthrogrypotic, prominent talar head – next?"	→ **forced plantarflexion radiograph**
"NOT associated with CVT?"	→ **amniotic band syndrome** (TO, OB)
"Rocker-bottom feet – high association?"	→ **neuromuscular disease** ✗ Wrong options: 22q11 deletion
"7-month-old rigid flatfeet, no subtalar motion – next?"	→ **neurologic work-up with neural-axis imaging**

– – Active space – –

"Approach with least AVN?"	→ **dorsal** (TO)
"Ideal management at 10 months?"	→ **casting then mini-open reduction** (TO)
"Why manipulate and cast first?"	→ **stretch dorsolateral soft tissue before surgery**
"Irreducible CVT in toddler?"	→ **casting then open reduction + Achilles lengthening**
"10-month-old CVT – initial management?"	→ **manipulation and cast**
"Which deformity gets serial casting then surgery?"	→ **CVT**
"After casting, next step?"	→ **TN reduction and pinning**
"7-month-old, talus–1st MT collinear on PF view – treatment?"	→ **stretching** (oblique talus) (TO)
"Short-foot exercises indicated for?"	→ **flat foot** (TO)

DEVELOPMENTAL COXA VARA

– Pediatrics · 31 / 112 –

1 Basics

Epidemiology

Incidence	**1 in 25,000** (US) · **M = F** · presents **from walking age to 6 yr**
Bilateral	**1 in 3** (Orthobullets) · **30–50%** (OB)
Genetics	no clear inheritance
TO lecture	"often **autosomal dominant**" · associations: **cleidocranial dysostosis**, metaphyseal dysplasia, spondylometaphyseal dysplasia – read the options
Fairbank's triangle (TO)	triangular inferior-neck (osteoid) fragment = inverted-Y defect

Pathophysiology

Defect	**proximal femoral physis / ossification defect of inferior femoral neck** → ↓ neck-shaft angle, **vertical physis**
Mechanics	vertical physis → ↑ **shear** across physis + ↑ compression of inferomedial neck → progression
Associated	femoral neck **stress fractures** · LLD · **early OA**

Etiologies of coxa vara

Type	Examples
Developmental	This topic
Congenital	Congenital short femur, **PFFD**
Acquired	**SCFE**, infection, **Perthes**, trauma
Dysplasia	**OI**, Jansen, Schmid, **SED**
Other	**Cretinism** (hypothyroid)

3 Presentation

Symptoms	**painless** · **waddling / Trendelenburg gait** (abductor weakness from shortened lever)
Exam	**LLD**, limb shortening · **high-riding greater trochanter** · ↑ lumbar lordosis · non-tender ↓ hip ROM (esp. abduction, IR)

Difficulty cross-leg sitting (TO) · **HE angle, not neck-shaft angle, defines coxa vara** (NSA unreliable – physeal slip) (TO)

- - Active space - -

4 Diagnosis & Classification

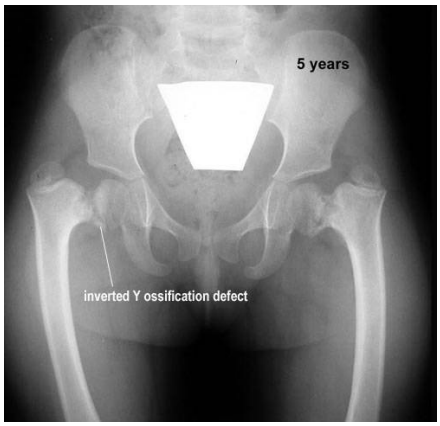
Key radiology findings

Views **AP hip in internal rotation** + lateral

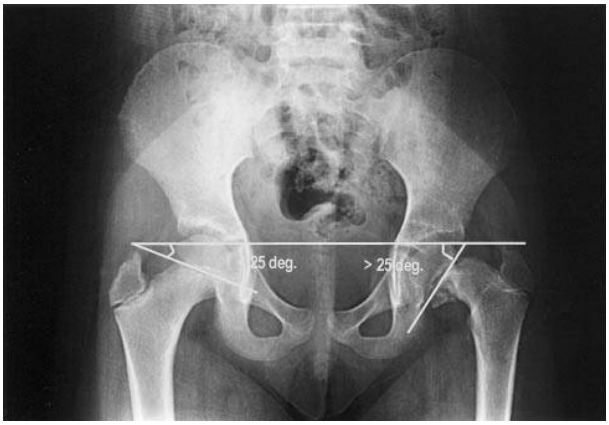
Finding	Value
Neck-shaft angle	**<120°** (varus)
Neck / physis	**Short neck, VERTICAL physis** (✗ Wrong options: horizontal)
Hilgenreiner–epiphyseal (HE) angle	↑ (normal **<25°**) – Hilgenreiner line vs proximal femoral physis
HE angle – TO lecture	Normal "~45°" (TO – verify: Orthobullets normal < 25°; 45° is the observe/progress threshold)
Inverted-Y lucency	**Triangular metaphyseal fragment of INFERIOR neck** (✗ Wrong options: superior · upright Y)
Version	↓ **femoral anteversion** (✗ Wrong options: increased)

Southwick angle (epiphyseal-diaphyseal on frog lateral) – ↑ in **SCFE**, not DCV

CT (3D) surgical planning, bone stock, deformity orientation



Inverted-Y ossification defect of inferior neck



Hilgenreiner–epiphyseal angle

5 Management

Hilgenreiner-epiphyseal angle & valgus osteotomy

HE angle	Action
<45°	**Unlikely to progress → observe**
45–60°	May progress → close follow-up · **operate if symptomatic** (limp, progression) (Orthobullets) or **neck-shaft <100°** (OB)
>60°	**Valgus (derotation) osteotomy**
Neck-shaft angle progressively **<110°**	Osteotomy (Orthobullets)

Developmental coxa vara

- HE <45° → **observe**
- HE 45–60° → **follow; osteotomy if symptomatic / progressive / NSA <100°**
- HE >60° → **valgus derotation osteotomy (overcorrect HE <38°)**
  ± adductor tenotomy, GT epiphysiodesis / transfer

Operative criteria

Valgus (derotation) intertrochanteric osteotomy — **overcorrect** → physis **horizontal**, **HE angle <38°** · correct LLD + version · restore abductor tension · blade plate · direct lateral approach · **adductor tenotomy** · spica / abduction pillow **4–6 wk**

Bilateral disease → bilateral valgus osteotomies
✗ Wrong options: varus osteotomy · night abduction brace · fixing the metaphyseal fragment

Greater trochanter — **epiphysiodesis** (prevent GT overgrowth) · **distal-lateral transfer** (tip at level of head centre)

3 goals (TO) — correct HE angle · **distalise GT** · prevent recurrence (epiphysiodesis)

Intertrochanteric osteotomy principles (TO) — most versatile level (iliopsoas stabilises proximal fragment) · CORA: **varus deformity → centre of head**, valgus deformity → base of GT · **valgus osteotomy needs adequate adduction**, varus needs abduction · fixation: varus **90° blade plate**, valgus **double-angle plate**

Powell osteotomy (TO) — high valgus osteotomy, fixation **crosses trochanteric physis** → vertical physis made horizontal → compression + fusion · **wedge angle = HE angle − 16°** · others: **Mosher** (multiple cuts), **Pauwels** intertrochanteric valgus

Target HE — **< 38°** (Orthobullets) · **~25–30°** (TO) – read the options

6 Complications

Loss of correction / recurrence (undercorrection) · **premature proximal femoral physeal closure** · proximal femoral overgrowth · **acetabular dysplasia**

– – Active space – –

X vs Y

	Developmental coxa vara	SCFE
Key angle	**Hilgenreiner–epiphyseal ↑** (AP)	**Southwick (epiphyseal-diaphyseal) ↑** (frog lateral)
Defect	Inferior-neck ossification defect, **vertical physis**	Through **hypertrophic zone**
Version	**↓ anteversion**	–

How it's asked

"Radiographic feature of developmental coxa vara?"	**→ increased Hilgenreiner–epiphyseal angle** ✗ Wrong options: ↑ anteversion · Southwick · horizontal physis · superior-neck Y
"4-year-old waddling, HE >60° bilaterally?"	**→ bilateral proximal femoral valgus osteotomies**
"HE angle that can be observed?"	**→ <45°**
"Goal of osteotomy?"	**→ overcorrect to HE <38° (horizontal physis)**

– – Active space – –

DEVELOPMENTAL DYSPLASIA OF THE HIP (DDH)

– Pediatrics · 32 / 112 –

1 Basics

Epidemiology

Most common orthopaedic disorder in newborns	**dysplasia 1:100 · dislocation 1:1000** (TO answer: **1 in 1000**)
Sex	**female 6:1** (Orthobullets) · 8:1 (OB)
Race	↑ **Native Americans, Laplanders** (swaddling hips extended + adducted) · **rare in African Americans**
Side	**left 60%** (left occiput anterior → left hip adducted against maternal sacrum) · **bilateral 20%**
Side order (TO)	**left > bilateral > right**
Incidence by age (TO)	**~1/100 at birth** (newborn laxity, maternal relaxin) → **~1/1000 at 6 weeks** and stays → reason screening US is at **6 wk, not birth**
"5 Fs" (TO)	**Female (5–9×)** · **First born** · **Fair** complexion · **Fetal malpresentation** (breech) · **Family history**

Risk factors & examination (Ortolani/Barlow, limited abduction)

Risk factors	**breech (highest relative risk, frank > footling)** · **female** (maternal relaxin/ oestrogen laxity) · **firstborn** (tight uterus) · **family history** · **oligohydramnios** · macrosomia · limited abduction · talipes · swaddling ✗ Wrong options: maternal diabetes · twins · metatarsus adductus (lesser)
Packaging associations	**torticollis (20%)** · **metatarsus adductus (10%)** · **congenital knee dislocation** · ↑ type III collagen conditions
Rates (other TO lecture)	torticollis **15–20%** · metatarsus adductus **1.5–10%** · CTEV

Spectrum & pathoanatomy

Entity	Feature
Dysplasia	Shallow acetabulum (**anterior/anterolateral deficiency**; CP = posterosuperior)
Subluxation	Partial contact
Dislocation	No contact
Teratologic	Dislocated **in utero**, **irreducible** at birth, **pseudoacetabulum** · arthrogryposis, myelomeningocele, Larsen, Ehlers-Danlos
Late (adolescent) dysplasia	Reduced, stable, dysplastic

Chronic changes	**limbus** (thickened rim cartilage) · ↑ femoral + acetabular **anteversion** · flat head · oblique roof · thick medial wall · secondary blocks (below)

– – Active space – –

Pathology (TO) small head with **delayed ossification** · **coxa valga** · ↑ anteversion · **limbus = inverted hypertrophied labrum** · **neolimbus = hypertrophied posterolateral acetabular cartilage** · pulvinar · hourglass capsule · iliopsoas

Clunk (TO) produced by the **neolimbus, not the labrum** · **click** = tendon snapping, harmless

Blocks to reduction

Extra-articular	Intra-articular
Iliopsoas → hourglass capsule	**Inverted labrum** (labrum deepens acetabulum 20–50%)
Adductor longus	**Inverted limbus** (fibrous response to superior migration)
–	**Transverse acetabular ligament** (contracted)
–	**Pulvinar** (fibrofatty; regresses after reduction)
–	**Hypertrophied ligamentum teres**

3 Presentation

Age	Key findings
<3 months	**Barlow** (adduct + depress flexed hip → **dislocates** a dislocatable hip, "clunk of exit") · **Ortolani** (abduct + elevate GT → **reduces** a dislocated hip, "clunk of entry") · Galeazzi · clicks nonspecific
3–12 months	**Limited abduction = most sensitive** · LLD · **Klisic** (GT–ASIS line points below umbilicus = dislocated; detects bilateral)
>12 months / walking	Limited abduction · **Galeazzi** · **Trendelenburg** gait · **toe-walking** (short side) · pelvic obliquity · ↑ lordosis (bilateral) · **Ortolani negative**

Barlow/Ortolani rarely positive after **~3 months** (8–12 weeks) as contractures develop

Ortolani-positive hips (rest outside, reducible) are **more resistant to treatment** than Barlow-positive (rest inside, dislocatable) (TO)

Other signs (TO) wide perineum · **supratrochanteric shortening** · **telescopy** · ↓ abduction **and IR** · **vascular sign of Narath** (absent/weak femoral pulse) · asymmetric folds **not reliable** · **Klisic = most reliable in bilateral DDH**

Swaddling (TO) swaddle so baby can kick freely (legs straight = risk)

Limited abduction present in ~70% of dysplastic infants but also **46% of normal** · **unilateral = specific (90%), sensitivity 70%** · **bilateral = not specific** · limited abduction with normal US → **observe**
✗ Wrong options: adductor tenotomy · ↑ risk



Galeazzi sign

4 Diagnosis & Classification

Classification by phase — subluxable (Barlow-suggestive) · dislocatable (**Barlow +**) · dislocated (**Ortolani +** reducible early; Ortolani − when irreducible)

Screening — **exam of all infants until walking** · **US at 4–6 weeks (AAP: 6 wk)** for **risk factors** (breech, family history) or abnormal exam · universal US **not cost-effective** in US · US before 4–6 wk → spurious

Clinical exam = gold standard screening (TO)

Indian DDH screening pathway (POSI + IOA + IAP) (TO) — clinical exam at birth + vaccination visits → suspected DDH (Ortolani +, Galeazzi/LLD, ↓ abduction, limp, uncertain exam) → **< 6 wk: schedule US at 6 wk · 6–14 wk: US · > 14 wk (or no quality US): AP pelvis X-ray** → abnormal → paediatric orthopaedic referral

Classification (TO)	Modality	Best use
Graf	Ultrasound	Screening, < ~4–6 mo
Tönnis	X-ray	After head ossification
Ferretti	MRI	Pre-op, non-ossified hips
Kashiwagi	MRI	Acetabular morphology

Key radiology findings

Ultrasound — **birth to 4 months** · also to **monitor Pavlik** (in harness) · femoral head bisected by iliac line

– – Active space – –

Radiograph primary after **4–6 months** (ossific nucleus visible by ~6 months in most)

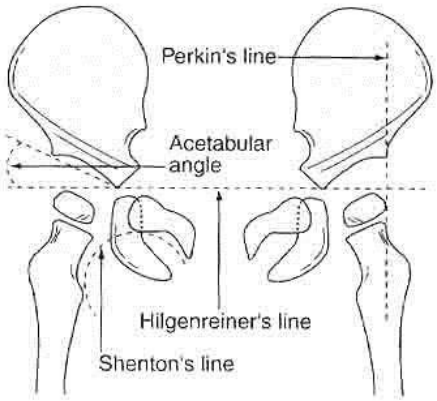
Radiographic line	Normal
Hilgenreiner	Horizontal through triradiates · head **inferior**
Perkin	⊥ Hilgenreiner at lateral acetabular margin · head **medial**
Shenton	Inferior neck → superior obturator foramen · **continuous arc**
Acetabular index	**<25°** after 6 mo · mean 27.5°, **>35° dysplastic** (OB)
Acetabular index (TO)	Normal **< 20°** · **30° = dysplasia** (TO) – read the options
Von Rosen view (TO)	Hips abducted + internally rotated
No ossific nucleus (TO)	Use **proximal femoral metaphyseal beak / H point** · dislocated head in **upper outer quadrant**
CE angle (Wiberg)	**<20° abnormal** · reliable only **>5 yr**
Teardrop	Present in all normal hips by **18 months** · absent before reduction of chronic dislocation · **appearance after reduction = good prognosis** · formed by quadrilateral surface + cotyloid fossa

Dislocated hip X-ray **delayed ossification** of affected head · head superolateral · broken Shenton · no teardrop
✗ Wrong options: AI 20° · CEA 40° · alpha angle 60° (that's FAI cam)

Klein's line SCFE (distractor)

False-profile view best for **anterior acetabular deficiency**
✗ Wrong options: inlet, Judet, AP, frog

Post-reduction in spica **MRI (safest + accurate, no radiation)** · CT historically (selective cuts)
✗ Wrong options: US · X-ray · arthrogram through cast



Hilgenreiner, Perkin, Shenton lines

Ultrasound – Graf classification, alpha angle & anatomy

Alpha angle **lateral ilium vs bony acetabular roof** (acute angle) · **normal ≥60°** (by 6–8 wk) · defines Graf type

Beta angle ilium vs **labrum** (echogenic triangle) · **normal <55°** · subtype

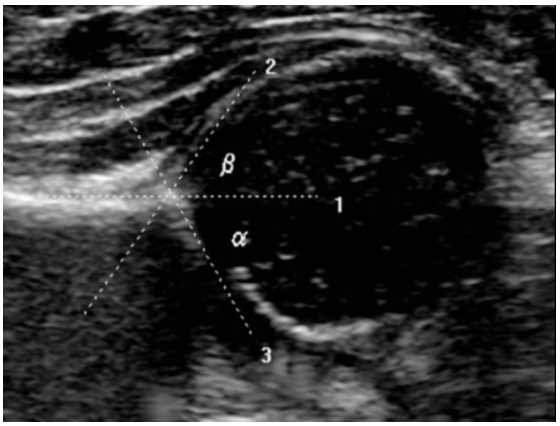
Standard Graf plane (TO) **horizontal (straight) iliac wing** · lower limb of os ilium · mid bony roof · **labrum** · linear probe over GT, coronal · small tilts change angles · stress US before and at end of treatment · ↓ alpha or ↑ beta = poor prognosis

3-wk-old, α 55°, clicky hip (TO) → rescan at 6 wk (normal 62° at 6 wk) · treating now = over-treatment

US anatomy ilium · femoral head · triradiate · **labrum** · ischium · normal head coverage **≥50%**

Graf	Alpha	Beta	Meaning	Treatment
I	**>60°**	<55°	Normal	None
II	**43–60°**	55–77°	Immature / delayed ossification	Variable – **IIa (<3 mo): observe, repeat US**; high risk → repeat US 3 wk or Pavlik
III	**<43°**	>77°	Subluxated	**Pavlik**
IV	Unmeasurable	Unmeasurable	Dislocated	Pavlik / closed vs open reduction

4-week-old stable Graf IIa (α 56°, 48% coverage) → repeat US at 10–12 weeks – >90% immature hips normalise
✗ Wrong options: Pavlik · night brace · MRI



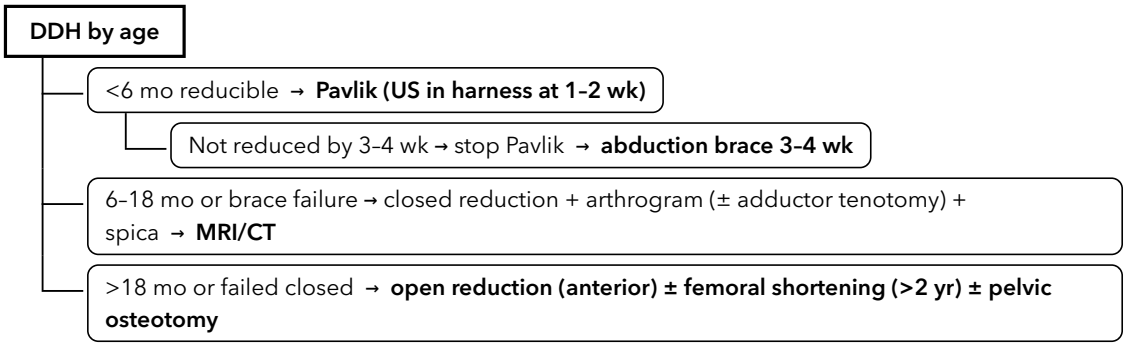
Hip ultrasound – alpha and beta angles

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5 Management

Operative criteria

Age	Treatment
<6 months, reducible	**Pavlik harness** (✗ Wrong options: teratologic, spina bifida, spasticity (needs normal muscle))
Pavlik failed (3–4 wk)	**Semi-rigid abduction brace** 2–4 wk (weekly US)
6–18 months or brace failure	**Closed reduction + arthrogram ± adductor tenotomy + spica**
>18 months (18–24) / failed closed	**Open reduction + spica**
>2 yr	Open reduction **+ femoral shortening** ± VDRO (femoral side: anteversion, valga; best <4 yr)
>2 yr, ↑ AI, mostly >4 yr	**+ pelvic osteotomy** (AI >28° after 4 yr)
Bilateral **≥6 yr** / unilateral **>8 yr**	Often **leave unreduced** · epiphysiodesis for LLD



Pavlik harness – indications, positioning & monitoring

Position	anterior straps: **hip flexion 90–100°** (prevent extension) · posterior straps: prevent adduction · **abduction 20–40°**, within **safe zone** · avoid flexion >120° or abduction >60–70°
Wear	**full-time (23 h/day)** ≥6 wk / until stable · wean 6–8 wk
Pavlik (TO)	first choice **6 wk–3 mo** (best < 6 mo) · size by **chest circumference at nipple level** · weekly US · continue **6 wk after hip stable** · **safe cone** (3D) concept · rigid abduction orthosis standard in Germany · no role for triple / abduction diapers
Femoral nerve palsy (TO)	flexion **> ~110°**, baby stops moving leg → **remove harness 1 wk**, reapply with less flexion or rigid orthosis
Monitoring	**US in the harness at 1–2 weeks** (7–10 d) then every 4–6 wk ✗ Wrong options: immediate US · night-time only · US out of harness · MRI/CT/X-ray

Ortolani-positive newborn / 2-week-old → Pavlik (✗ Wrong options: observe · closed reduction) · **6-week-old, α 40°, 10% coverage, Ortolani + → full-time Pavlik, check in 1 week** (✗ Wrong options: 3-month follow-up · double diapers)

4-month-old, located but AI 35–40° bilaterally → Pavlik
✗ Wrong options: observe · triple diapers · reduction

Well-positioned head at 2 wk → continue current position
✗ Wrong options: increase to 70° abduction

Success **~90%** (depends on age at start, time in harness)

Pavlik complications – femoral nerve palsy & osteonecrosis

Position error	Complication
Excessive flexion (>120°)	**Femoral nerve palsy** – no active knee extension, ↓ quads/patellar reflex → **loosen anterior (flexion) strap**
Excessive abduction (>60°)	**AVN** – impinges **posterosuperior retinacular branch of MFCA**
Prolonged wear with unreduced hip	**Pavlik disease** – erosion of posterior/posterolateral acetabulum

Femoral nerve palsy risk factors **dislocated hip at start (independent)** · flexion >120° · **older / heavier** infants (>6 mo) · severe dysplasia → reduce flexion immediately; if persists → reduction + cast/rigid orthosis
✗ Wrong options: younger/smaller

Femoral nerve location lateral to femoral artery, anterior to iliopsoas



Pavlik harness

Pavlik failure – abduction brace, stopping the harness & risk factors

Stop Pavlik if hip not reduced by **3–4 weeks** (Orthobullets) · **2–3 wk** / 2 wk (OB) → prevents **Pavlik disease**
✗ Wrong options: increase abduction to 80° · continue and recheck · accept

Next **semi-rigid / rigid abduction orthosis** 2–4 wk with weekly US (applied in office, no anaesthesia) · **93% success** protocol (Swaroop-Mubarak) · then closed ± open reduction
✗ Wrong options: immediate arthrogram/open reduction · pelvic osteotomy

Pavlik failure risk factors **Graf IV** (most) · **Ortolani-negative** · **start >7 weeks** · **right-sided**
✗ Wrong options: sex · unilateral

Closed reduction & spica – arthrogram, safe zone & post-reduction imaging

4-month-old failed Pavlik 3 wk + brace 3 wk → closed reduction + arthrogram ± adductor tenotomy + spica
✗ Wrong options: harness at 75° · osteotomies

– – Active space – –

Technique	GA · Ortolani reduction (gentle; force → AVN) · **arthrogram** confirms concentric reduction
Medial dye pool	**<5 mm** good (5–6 mm acceptable) · **>6 mm** (OB) / **>7 mm** (Orthobullets) → failure + **AVN** · 17-month-old with **8 mm** pool → **inappropriate**
Safe zone	arc between max abduction and redislocation adduction · **≥25° arc** · **adductor tenotomy** if excessive abduction needed
Ramsey safe zone (TO)	**narrow (< 20°) → adductor (longus) tenotomy ± iliopsoas release** · abduction **> 35–40°** ↑ AVN (TO) · more abduction needed to stay in = ↑ AVN
Arthrogram (TO)	essential when ossific nucleus absent · head points to **triradiate** · **coverage > 50% favourable, < 30% poor** · overlap **A good / B borderline / C poor**
Timing (TO)	closed reduction **6–18 mo** · AVN risk ↑ after 6 mo but still standard · **18 mo–3 yr: closed reduction contraindicated** → open
Spica	**"human position"**: **100° flexion, 45° abduction (<55°)**, neutral rotation · 3 months · change at 6 wk
Post-op	**MRI or CT** in cast (✗ Wrong options: X-ray before discharge) · CT shows posterior subluxation → **urgent repeat closed reduction + recast** (✗ Wrong options: wait 6 wk · osteotomy at 8 mo)
Arthrogram signs	**hourglass capsule = iliopsoas constriction** · inverted limbus (rim) · pulvinar · TAL



Hourglass capsule on arthrogram

Open reduction ± femoral shortening/pelvic osteotomy (older children)

Indications	**>18 months** · failed closed reduction · **reduction only in >60° abduction** (e.g. 70–80°) · unstable/persistent subluxation
Approach	**anterior (Smith-Petersen)**: most common, **>12 mo**, allows **capsulorrhaphy**, less MFCA risk · **medial**: <12 mo, addresses blocks directly, less blood loss, **no capsulorrhaphy, ↑ AVN**

Medial approach	Interval
Ludloff	Pectineus ↔ adductor longus/brevis (Orthobullets)
Weinstein (anteromedial)	NV bundle ↔ pectineus
Ferguson (posteromedial)	Adductor longus ↔ gracilis; deep brevis ↔ magnus

Artery at risk with psoas tenotomy (medial) — **MFCA** – usually from **profunda femoris** (sometimes femoral artery), passes between **psoas and pectineus** to posterior neck
✗ Wrong options: superior gluteal · LFCA ascending branch · SFA · internal iliac branch

Post-op position — **30° flexion, 30° abduction, 30° IR**

Femoral shortening — **>2 yr → ↓ tension, ↓ AVN + redislocation** · VDRO corrects anteversion/valga

Intra-op instability (TO) — **flexion instability → acetabular osteotomy · abduction + IR instability → VDRO**

Age (TO)	Treatment
3–10 yr	Open reduction + **bony procedure mandatory** · **< 4 yr VDRO** preferred · **> 4 yr acetabular osteotomy** preferred
> 10 yr	Limited reconstructive role · natural history often better · painful: **tectoplasty** (salvage)

Approach (TO) — medial for **non-walkers**, **not used after ~18 mo** · anterior for walkers (e.g. 3-yr bilateral waddling)

26-month-old ambulatory, irreducible, severe dysplasia, ↑ IR/↓ ER (anteversion) → open reduction + iliac osteotomy + femoral shortening derotation + spica · 22-mo, mild dysplasia, normal rotation → open reduction + spica

2-yr, dislocated + valgus neck → open reduction + femoral shortening
✗ Wrong options: Pavlik · brace · closed reduction · **abductor** tenotomy

20-month-old, concentric only at 70–80° abduction → open capsulorrhaphy ± femoral ± pelvic osteotomy
✗ Wrong options: closed reduction (AVN) · hamstring/gastroc releases (spastic)

Residual dysplasia – pelvic osteotomies (Salter, Chiari, shelf, PAO)

Osteotomy	Requirement / indication	Key feature
Salter	Young, **open triradiate**, concentric	Single iliac cut to notch · **hinges on pubic symphysis** · anterolateral coverage · lengthens ≤1 cm
Triple (Steele)	Older child (stiff symphysis), open triradiate	Salter + superior + inferior rami
PAO (Ganz)	**Triradiate CLOSED** (adolescent/adult)	Pubis, ilium, ischium cuts · 3D correction · **posterior column intact** · early WB
Pemberton	**Open triradiate**, moderate–severe, most versatile	**Hinges at triradiate** (posteriorly) · **↓ acetabular volume** · no fixation · bilateral one sitting OK
Dega	**Neuromuscular (CP)**, posterior deficiency	Incomplete trans-iliac, hinge triradiate · ↓ volume · anterior/central/**posterior** coverage
Dial	Rare, difficult	Spherical, intra-articular, teardrop stays
Shelf (Staheli)	**Salvage, >8 yr · no concentric reduction needed**	Extra-articular buttress · **fibrocartilage**
Chiari	**Salvage – incongruous / non-concentric hip** (CEA ≥−10°)	Iliac osteotomy, medialise acetabulum · **fibrocartilage**

4-yr after closed reduction, AI 35°, normal femur → iliac (Pemberton/Dega) osteotomy
✗ Wrong options: PAO (open triradiate) · + femoral osteotomy

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15-yr soccer player, closed triradiate → PAO (Ganz) · 20-yr → **PAO**

Painful incongruous hip on abduction-IR view → Chiari

Reorientation osteotomies require concentric reduction — all except **shelf**

Pelvic osteotomy groups (TO)

Group (TO)	Cuts	Volume	Fixation	Examples
Redirectional	**Complete**	Unchanged	**Always required**	Salter · double / **triple** innominate · **PAO**
Reshaping (acetabuloplasty)	**Incomplete**, hinge	**↓ volume**	Inherently stable, none	**Pemberton** · **Dega** · **San Diego**
Augmentation (salvage)	Extra-articular	↑ volume	–	**Shelf** (Staheli slotted, Wilson, Wainwright, Japanese tectoplasty) · **Chiari**

Salter (TO) — DDH **after 2 yr** · anterolateral coverage · corrects AI by **10–15°** · limited correction (~20–25°) · best **< 8 yr** · complications: loss of correction, triradiate closure, redislocation, AVN

Pemberton (TO) — starts **above AIIS** · hinges at triradiate (+ symphysis) · anterior + lateral coverage · **for DDH**

Dega (TO) — **neuromuscular hips** · cut from roof to triradiate, hinges only at triradiate · **posterior coverage** · age **2–12 yr**

San Diego (TO) — **most versatile** · anterior + posterior coverage · graft position decides direction · global deficiency

Triple (TO) — older children; triple allows max mobility · **Steel** limited by ligament (sacrospinous/sacrotuberous) constraints · **PAO** – preserves posterior column, single incision, preferred adolescent dysplasia

Chiari (TO) — uncontainable head · symptomatic subluxation · **CEA < 10°** · contraindicated: advanced OA, labral tear, bilateral

Shelf (TO) — ↑ CEA, **↓ Sharp angle**, prevents lateral subluxation · **fibrocartilage** · steps: arthrogram → adductor ± psoas tenotomy → slot above acetabulum → iliac crest graft → reattach rectus

Adolescent parameter (TO)	Values
LCEA	**< 20° dysplasia · 20–25° borderline · 25–40° normal · > 40° pincer**
Acetabular inclination (Tönnis)	**< 3° pincer · 3–13° normal · > 13° dysplasia**
Acetabular index	Age-dependent · **~25° at 1 yr → ~10° by adolescence**

Femoral version (TO) — assess before pelvic surgery (CT, **Murphy protocol**) · excess anteversion → add derotation / varus osteotomy

Residual dysplasia after reduction (TO) — observe to **4–5 yr** (many remodel) · operate only if persists **beyond 5 yr**

Pemberton true statement — **hinges at triradiate** (TO)
✗ Wrong options: can't do bilateral · needs pins · volume unchanged

Teratologic & neuromuscular hips

– – Active space – –

Teratologic (amyoplasia newborn, stiff clubfeet, Ortolani –, pseudoacetabulum) → no hip treatment/imaging for months · open reduction at **≥6 months**
✗ Wrong options: Pavlik · rigid brace · neonatal CT

CP hip deficiency **posterosuperior** (adductor + iliopsoas pull)

6 Complications

AVN all treatments · risks: **forceful/extreme abduction (>60°)** · **failed prior closed treatment** · repeat surgery · medial dye pool >7 mm · signs: no ossific nucleus growth 1 yr after reduction, broad neck, dense/fragmented head

AVN (TO) **4–10%** · more after closed reduction in infants · better termed **proximal femoral growth disturbance**: coxa valga / vara, **coxa breva**, coxa magna, eccentric epiphysis · classifications: **Bucholz-Ogden**, **Kalamchi-MacEwen**

Kalamchi-MacEwen (TO)	Feature
I	"Normal" (TO – verify: usual type I = changes in ossific nucleus only)
II	**Lateral physeal arrest**
III	**Complete (central) physeal arrest**
IV	**Epiphyseal + physeal involvement** (total)

Recurrence ~**10%** → follow to maturity

Transient femoral nerve palsy Pavlik hyperflexion

X vs Y

	Barlow	Ortolani
Manoeuvre	**Adduct + push posterior**	**Abduct + lift GT**
Result	**Dislocates** dislocatable hip	**Reduces** dislocated hip
Sound	Clunk of exit	Clunk of entry

	Typical DDH	Teratologic
Timing	Postnatal / perinatal	**In utero**
Reducible	Yes (Ortolani +)	**No**
Pavlik	Yes	**No**
Associations	Packaging	Arthrogryposis, myelomeningocele, Larsen

How it's asked

"Incidence of DDH?"	→ **1 in 1000** (TO)
"Ideal time for screening US?"	→ **6 weeks** (TO)
"Cause of the clunk in Ortolani/Barlow?"	→ **neolimbus** (TO)

– – Active space – –

"DDH >12 months — finding NOT expected?"	→ **positive Ortolani**
"3-month-old, bilateral limited abduction, normal US — true?"	→ **unilateral limited abduction is specific; bilateral is not**
"Manoeuvre shown — reducing the hip?"	→ **Ortolani**
"Highest risk factor?"	→ **frank breech**
"18-month-old toe-walking, short left leg, limited abduction — X-ray?"	→ **delayed ossification of left femoral head**
"Alpha angle defined as?"	→ **lateral ilium vs bony acetabular roof** (TO)
"Follow-up imaging in Pavlik?"	→ **ultrasound** (TO)
"Labrum on US?"	→ **echogenic triangle at acetabular rim**
"Which angle is alpha?"	→ **ilium + bony acetabulum**
"Breech baby US at 6 weeks — left α <60°?"	→ **right normal, left dysplastic**
"28-day-old breech, α 48°, β 72°?"	→ **Graf II — repeat US in 3 weeks or Pavlik**
"Structure B on infant US?"	→ **labrum**
"4-week-old, α 56°, stable?"	→ **repeat US at 10–12 weeks**
"Which figure is Perkin's line?"	→ **vertical line at lateral acetabular margin**
"True of pediatric hips?"	→ **ossific nucleus visible by 6 months** ✗ Wrong options: anteversion 10–20° (it's 30–40°) · SCFE in proliferative zone · separate physes
"Teardrop — true?"	→ **present by 18 months in normal hips**
"Line #1 along femoral neck to obturator?"	→ **Shenton's line**
"2-week-old Ortolani +?"	→ **Pavlik harness**
"Pavlik started — imaging at 2 weeks?"	→ **ultrasound**
"Pavlik placed — next?"	→ **full-time, US in harness at 7–10 days**
"Good position on 2-week US?"	→ **continue harness**
"6-week-old, α 40°, 10% coverage?"	→ **full-time Pavlik, follow-up 1 week**
"4-month-old located, AI 35–40°?"	→ **Pavlik harness**
"Can't kick, knee flexed, Pavlik flexion 125°?"	→ **excessive hip flexion → femoral nerve palsy**
"Diminished knee extension in Pavlik — fix?"	→ **loosen anterior flexion strap**
"No active knee extension at 120° flexion?"	→ **improper harness positioning**
"Structure responsible for no knee extension?"	→ **femoral nerve**

"Position–complication pairs?"	→ **flexion → femoral nerve palsy; abduction → osteonecrosis**
"Femoral nerve palsy risk factor?"	→ **dislocated hip at diagnosis**
"Pavlik 3 weeks, still dislocated at 4 weeks of age?"	→ **semi-rigid abduction brace**
"Semi-rigid brace after Pavlik failure?"	→ **can achieve stable hip without surgery**
"8-week-old, 4 weeks Pavlik, still dislocated?"	→ **rigid abduction orthosis + weekly US**
"Persistent dislocation at 3 weeks in harness?"	→ **remove harness** (avoid acetabular deformity)
"Dislocated after 3 weeks Pavlik (1-week-old start)?"	→ **switch to abduction orthosis**
"Graf IV – biggest Pavlik failure risk?"	→ **Graf classification**
"Failed Pavlik + brace at 4 months?"	→ **closed reduction + arthrogram ± adductor tenotomy + spica**
"Best imaging after closed reduction in spica?"	→ **MRI**
"Stable in 50° abduction, <5 mm pool?"	→ **spica + post-reduction CT/MRI**
"CT shows posterior subluxation in spica?"	→ **urgent repeat closed reduction + recast**
"Inappropriate age-based treatment?"	→ **17 months, closed reduction with 8 mm dye pool**
"Arrows on arthrogram (hourglass)?"	→ **capsule constricted by iliopsoas**
"Inverted limbus on arthrogram?"	→ **rim structure pointed at acetabular edge**
"20-month-old, reduced only at 70–80° abduction?"	→ **open capsulorrhaphy ± femoral ± pelvic osteotomy**
"24-month-old dislocated?"	→ **open reduction ± pelvic/femoral osteotomy + spica**
"26-month-old complete dislocation?"	→ **open reduction + femoral shortening + capsulorrhaphy + spica**
"2-yr dislocated, valgus neck?"	→ **open reduction + femoral shortening**
"Artery at risk with medial psoas tenotomy?"	→ **MFCA (from profunda femoris)**
"Open reduction + iliac osteotomy + femoral derotation shortening fits?"	→ **26-month ambulatory, severe dysplasia, ↑ IR**
"Pemberton – true?"	→ **hinges at triradiate cartilage** (TO)
"Most versatile reshaping pelvic osteotomy?"	→ **San Diego** (TO)
"Pelvic osteotomies that always need fixation?"	→ **redirectional (Salter, triple, PAO)** (TO)
"15-yr, closed triradiate?"	→ **Ganz PAO**
"4-yr residual dysplasia AI 35°?"	→ **iliac (Pemberton/Dega) osteotomy**

– – Active space – –

"Incongruous painful hip?"	→ **Chiari**
"No concentric reduction needed?"	→ **Staheli shelf**
"View for anterior deficiency?"	→ **false profile**
"4-yr residual left dysplasia after closed reduction?"	→ **left pelvic osteotomy**
"20-yr symptomatic dysplasia?"	→ **PAO**
"CP dislocated hip – deficient quadrant?"	→ **posterior-superior**
"Newborn amyoplasia, teratologic dislocation?"	→ **no hip treatment/imaging for months**

DIASTROPHIC DYSPLASIA

– Pediatrics · 33 / 112 –

1 Basics

Epidemiology rare · **Finland** (carrier **1 in 70** Finns)

SLC26A2 sulfate transporter & phenotype

Genes & lab markers

Inheritance **autosomal recessive** (both parents carriers)

Gene **DTDST = SLC26A2**, **chromosome 5q** → defective **sulfate transporter** → **undersulfation of cartilage proteoglycan** → abnormal cartilage matrix (ear, trachea, ligaments)

Ossification failure of secondary ossification centres (OB)

No commercial test many mutations → clinical + radiographic + family history · prenatal US possible

Gene (distractors)	Disorder
SLC26A2 / DTDST	**Diastrophic dysplasia** ✓
FGFR3	Achondroplasia, hypochondroplasia
FGFR2	Apert (craniosynostosis + complex syndactyly)
CBFA-1 / RUNX2	Cleidocranial dysplasia (osteocalcin)
Gs α (GNAS)	Fibrous dysplasia
COL1A1/2	OI (blue sclera, bowing)
COL5A1/2	Ehlers-Danlos (classic)
Lysosomal storage	Hunter, Hurler
Lipid storage	Gaucher

– – Active space – –

3 Presentation

"Twisted dwarf" **short-limbed** (rhizomelic) dwarfism, diagnosed at birth

Feature	Note
Cauliflower ears	**80%** · cystic swelling of pinnae · responds to early **compressive bandages**
Hitchhiker thumb	**Short 1st MC + radial subluxation of MCP** (abducted)
Cleft palate	**60%** (59%)
Cervical kyphosis	**Severe**, early · often **resolves spontaneously**
Thoracolumbar kyphoscoliosis	Progressive
Feet	**Rigid clubfeet (equinocavovarus)** · **skewfoot** (TMT adductus + hindfoot valgus) · tarsal valgus 43%, MT adductus 37%
Lower limb	**Genu valgum** (± dislocated patellae) · hip + knee contractures · hip dislocation
Hand	Symphalangism (stiff fingers)
Other	**Tracheomalacia** (can be fatal) · atlantoaxial instability (rare)



Hitchhiker thumb



Cauliflower ear

4 Diagnosis & Classification

Key radiology findings

Views AP + lateral **cervical** and **thoracolumbar** spine · standing 3-joint lower-limb films



Cervical kyphosis in diastrophic dysplasia

5 Management

Most observation + supportive

Operative criteria

Problem	Surgery
Atlantoaxial instability + neuro symptoms	**Occipitocervical fusion** (quadriplegia risk)
Cervical kyphosis **not resolving**	Posterior cervical fusion
Progressive TL kyphoscoliosis	ASF / PSF
Foot deformity after casting · severe hip/knee contractures	Soft-tissue release
Progressive valgus, resistant clubfeet, fixed knee flexion, early hip subluxation	Osteotomies (distal femoral extension, valgus intertrochanteric ± acetabuloplasty)
End-stage hip OA	THA

How it's asked

"Newborn hitchhiker thumb, SLC26A2 mutation, carrier parents?"	→ **diastrophic dysplasia** ✗ Wrong options: achondroplasia · OI · hypochondroplasia · EDS
"Pathoanatomy of hitchhiker thumb + cauliflower ear dwarfism?"	→ **sulfate transporter mutation affecting cartilage matrix** ✗ Wrong options: CBFA-1 · lysosomal · lipid storage · FGFR3
"Genetic cause of hitchhiker thumb, cauliflower ears, severe clubfeet?"	→ **sulfate transport protein** ✗ Wrong options: FGFR2 · FGFR3 · CBFA-1 · Gs α

– – Active space – –

"DTDST defect phenotype?"	→ **cauliflower ears + hitchhiker thumb** ✗ Wrong options: blue sclera · MR + cardiac = Down · craniosynostosis + syndactyly = Apert · rhizomelia + frontal bossing = achondroplasia
"Spine region most deformed?"	→ **cervical kyphosis** (often resolves)
"Treatment of cauliflower ears?"	→ **early compressive bandaging**

DISTAL CLAVICLE PHYSEAL FRACTURES

– Pediatrics · 34 / 112 –

1 Basics

Incidence	rare · **5–10%** of pediatric clavicle fractures
Equivalent	childhood **"pseudodislocation"** = adult **AC separation**

Pathoanatomy – intact periosteal sleeve

Injury	clavicle (metaphysis) displaces **superiorly** out of its **periosteal sleeve**, through the **physis** · **inferior periosteal sleeve + epiphysis stay attached to intact CC + AC ligaments** · **AC joint intact**
Healing	new clavicle forms **within the sleeve → "Y-shaped" clavicle**; displaced fragment remodels/resorbs

Anatomy & ossification

Clavicle	**first bone to ossify** (5th week in utero, intramembranous) · **physes last to close**
Medial physis	**80% of growth** · ossifies **18–20 yr** · **last physis to close (20–25 yr) →** teenage "SC dislocations" are **physeal fracture-dislocations**
Lateral epiphysis	ossifies ~18 yr
CC ligaments	vertical stability · **trapezoid** 2 cm, **conoid** 4 cm from AC joint · **AC ligaments** – horizontal stability

2 Mechanism

Fall on side of shoulder / outstretched arm · direct blow · rarely **child abuse**

3 Presentation

Exam	pain, distal clavicle tenderness, deformity, ecchymosis · rarely tenting
Newborn	**pseudoparalysis** with **intact reflexes** (vs brachial plexus palsy)

4 Diagnosis & Classification

Key radiology findings

Views	**AP ± Zanca** (intra-articular extension) · **axillary lateral** (type IV posterior displacement)
Later	new bone from intact sleeve **inferior** to the superiorly displaced fragment

– – Active space – –



Distal clavicle physeal fracture

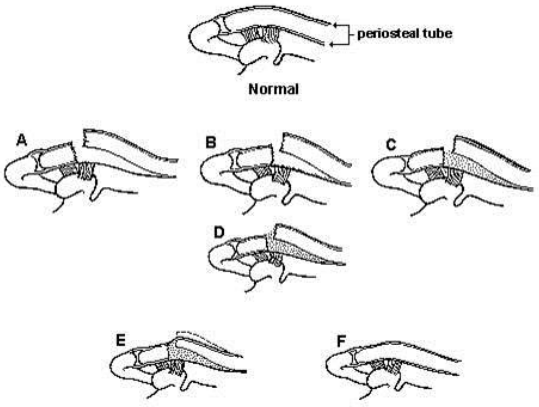
Rockwood (Dameron-Rockwood) classification

Type	Injury
I	AC ligament sprain, **periosteal tube intact**
II	Partial periosteal tube disruption
III	Large split, **superior** displacement
IV	**Posterior** displacement through **trapezius**
V	Complete disruption, through deltoid + trapezius
VI	**Inferior**, below coracoid

5 Management

Pseudodislocation – intact periosteal sleeve & sling treatment

Most (even displaced, elite athletes) → sling immobilization
✗ Wrong options: CRPP · plate · CC reconstruction/suture · stress views · MRI



Remodelling within intact periosteal sleeve

Operative criteria

Absolute (rare)	Relative
Open fracture	Severely displaced, **near physeal closure**
Skin compromise	Fragment entrapped in **trapezius**
Displaced intra-articular extension	Floating shoulder
NV injury needing surgery	Some type III near maturity · **types IV–VI** (open reduction + sleeve repair)

Avoid pins around clavicle (**migration**)

6 Complications

Subclavian vessel laceration — rare · **rapidly expanding hematoma** → vascular repair (thick periosteum protective)

Nonunion — rare, after open reduction → fixation + iliac crest graft

How it's asked

"6-yr distal clavicle fracture, lateral fragment superior?"	→ **sling**
"16-yr hockey player, likely injury pattern?"	→ **physeal separation, intact inferior sleeve, intact CC + AC**
"12-yr elite gymnast, distal clavicle physeal fracture?"	→ **sling** ✗ Wrong options: stress X-ray · MRI · fixation
"MRI of 11-yr distal clavicle fracture shows?"	→ **intact inferior periosteal sleeve** ✗ Wrong options: CC disruption · AC dislocation
"Last physis to close?"	→ **medial clavicle (20–25 yr)**

– – Active space – –

DISTAL FEMORAL PHYSEAL FRACTURES - PEDIATRIC

– Pediatrics · 35 / 112 –

1 Basics

Who	open physes · often **misdiagnosed as collateral ligament sprain** · 2–5% of physeal injuries (<2% OB)
Most common type	**Salter-Harris II** (physis fails on **tension** side, metaphysis on **compression** side → **Thurston-Holland**)

Anatomy

Ossification	single nucleus present **at birth** – **first epiphysis to ossify**
Physis	**undulating** (mammillary processes) → fracture crosses **multiple layers** → high arrest rate
Muscles	gastrocnemius + plantaris arise just proximal → **flex** distal fragment
Ligaments	collaterals + cruciates attach to **epiphysis** → stress transmitted to physis (ligament injury uncommon)
Vessels	**popliteal artery directly posterior**, trifurcates here, poor collaterals → limb-threatening

Growth

Physis	Growth / yr
Proximal femur	3 mm
Distal femur	**9 mm** (Orthobullets) · **10 mm** (TO) · 10–12 mm at peak
Proximal tibia	6 mm
Distal tibia	5 mm

Distal femur	**70% of femoral growth, 37% of limb** (TO) · most lower-limb growth at the **knee**
Epidemiology (TO)	~2% · peak **10–12 yr** · **males 6×** · neonates: birth injury or **NAI** (metaphyseal corner / bucket-handle fractures = abuse)
Growth stops	**14 yr girls, 16 yr boys** (Orthobullets) · fuses 14–16 ♀ / 16–18 ♂ (TO)

Physeal histology – hypertrophic zone

Fracture plane	**hypertrophic zone** (chondrocytes **enlarge**, then **apoptosis** + matrix calcification in zone of provisional calcification) ✗ Wrong options: proliferative (cells ↑ in number, max matrix) · reserve (precursors) · primary spongiosa (woven bone) · secondary spongiosa (lamellar)
Distal femur SH I	**propagates through multiple layers** (undulating physis) (OB)

2 Mechanism

Mechanism	direct trauma + rotation · **valgus** or **hyperextension** force · sports, vehicle
Displacement direction	anterior = hyperextension · posterior = hyperflexion · medial = valgus · lateral = varus

3 Presentation

Exam	pain, can't bear weight · knee held **flexed** (hamstring spasm) · **tenderness over physis** (medial epicondyle region) · effusion · varus/valgus "instability" · popliteal swelling → vascular injury

4 Diagnosis & Classification

Diagnosis – occult Salter-Harris I, MRI & imaging

Initial	**AP, lateral, oblique** knee ✗ Wrong options: MRI first · stress views · CT · notch view
Physeal widening	normal 3–5 mm · compare contralateral
Stress radiographs	**fallen out of favour** (pain, sedation, physeal damage) → **MRI** (or US) replaced them
MRI	**modality of choice to confirm** occult physeal fracture ✗ Wrong options: CT · weight-bearing views · aspiration/NSAID trial
CT	intra-articular extension, fixation planning
Angiography	**ABI <0.9** · wide displacement, posterior spike
Alternative	suspected SH I → long-leg cast + repeat X-ray 2–3 wk

Key radiology findings

Peterson classification (TO)	I metaphyseal transverse into physis · II SH II equivalent · III through physis · IV epiphyseal along physis · V traverses all · **VI part of physis missing**



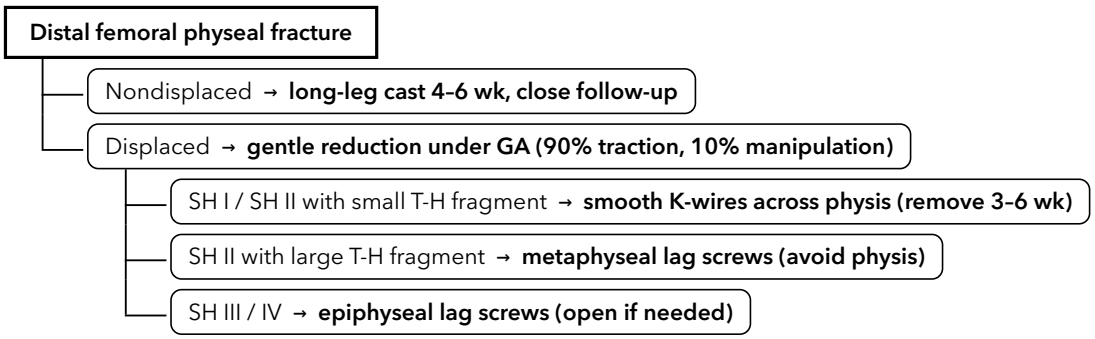
Distal femoral SH II fracture

– – Active space – –

5 Management

Operative criteria

Fracture	Treatment
Nondisplaced (incl. occult SH I)	**Long-leg cast 4–6 wk**, crutches, close follow-up (✗ Wrong options: hinged brace WBAT · immobiliser)
Displaced SH I / II (majority)	**Closed reduction + percutaneous fixation + cast** (unstable even if reduced) (✗ Wrong options: cast alone · traction · plate)
SH III / IV with anatomic closed reduction	CRPP
SH III / IV with articular displacement	**ORIF** (avoid physis)
Irreducible SH I / II	**Open reduction** – remove **periosteum on tension side**



Fixation – smooth pins vs screws (Thurston-Holland fragment)

By SH type (TO)	I: closed reduction, open if periosteum entrapped, smooth K-wires · II: **2 screws in Thurston-Holland fragment** if large, K-wires if small · III/IV: anatomic ORIF, screws · reduction = gentle traction, **knee flexion relaxes gastrocnemius**, recheck NV status
Smooth K-wires across physis are safe (TO)	few passes, low torque, central, avoid perichondrial ring · **growth arrest is from the injury, not the wires**
Pins vs screws	decided by **size of Thurston-Holland fragment** · large → **metaphyseal lag screws** · small / SH I → **smooth K-wires across physis** + supplemental cast
SH I	very unstable → **anatomic reduction + smooth pins + cast** ✗ Wrong options: screws (↑ physeal injury) · blade plate · flexible nails
Avoid	threaded wires, screws, plates across physis (rigid tether)
Pin direction	retrograde (through joint) theoretical ↑ septic arthritis (no proven difference)
Screws	fully vs partially threaded: no difference with lag technique



Smooth-wire fixation of distal femoral physeal fracture

Vascular injury

Pulses only on Doppler / ABI 0.8 → urgent reduction + (percutaneous) fixation, then reassess vascular status
✗ Wrong options: arteriogram/CT/MRI first · cast · fasciotomy first · vascular exploration before reduction · plate across physis

Highest risk	**anterior epiphyseal displacement (hyperextension)** / **posterior metaphyseal spike** → treat like **knee dislocation** · arterial injury up to 20% in some series · compartment syndrome <1–3%

6 Complications

Growth arrest risk – LLD & angular deformity

Most common complication	**physeal arrest → LLD / angular deformity** ✗ Wrong options: nonunion (rare) · ligament injury · stiffness (transient) · AVN
Physeal arrest rate	**30–50%** (Orthobullets, OB "~40%") · **~60%** (OB) · 50–55% / 50–80% (OB) – read the options
LLD >2 cm	**one-third** of cases

SH type	Growth disturbance
I	36%
II	58%
III	49%
IV	**64% – highest** (✗ Wrong options: "type III maximal" (TO false statement))

Pattern	**SH I + V → LLD (non-angular)** · **SH II → angular deformity**
Predictors of complications	SH type · **displacement** · **hardware crossing physis** · articular incongruity · open fracture · surgical treatment · **younger age / less skeletal maturity** (tibial tuberosity ossification stage) · **NOT direction of displacement**

Rate of arrest ↑ with displacement; older age (OB)

LLD calculation

Surveillance (TO)	remove implants **3–4 months** · **MRI at 4–6 months (T1 spoiled gradient echo)** · follow to maturity · growth arrest **30–40%** (TO)

– – Active space – –

Complete arrest LLD = remaining years × ~9–10 mm · e.g. 11-yr girl (stops 14) → **3 cm** · 13-yr boy with 2 cm LLD (stops 16) → 20 + 27 = **~4.7–5 cm**, affected side short

Physeal bar – resection

Post-traumatic distal femoral bar / LLD

- Predicted LLD <2 cm, no angulation → **observe**
- Bar <50%, ≥2 yr (2–2.5 cm) growth left → **bar resection + PMMA / fat**
  (angulation >20° → add osteotomy)
- LLD 2–6 cm → **contralateral distal femur ± proximal tibia epiphysiodesis**
  (>5 cm → ipsilateral lengthening)

11-yr, 5° varus, 25% bar, rest open → bar resection + PMMA · results 62–90% good
✗ Wrong options: observe · osteotomy · guided growth · epiphysiodesis

12-yr girl after fixation (~2 yr growth left) → physeal disturbance – close observation (predicted LLD <2 cm)
✗ Wrong options: contralateral epiphysiodesis · lengthening

Septic arthritis **intra-articular pins**

X vs Y

	Smooth K-wires	Lag screws
Use	**SH I**, SH II with **small T-H** fragment	SH II **large T-H** (metaphyseal), SH III/IV (epiphyseal)
Physis	Cross it (smooth)	**Parallel, avoid physis**
Removal	3–6 wk	–

How it's asked

"NOT true about distal femoral physeal fracture?"	→ **type 3 has maximum LLD risk** (it's type 4) (TO)
"Risk factors for complications – EXCEPT?"	→ **direction of displacement**
"SH II treated with lag screws – least predictive of outcome?"	→ **direction of displacement**
"11-yr girl with complete arrest – LLD?"	→ **3 cm**
"Closed reduction + cast at rural hospital – main complication?"	→ **limb angular deformity**
"Prognosis?"	→ **high risk of premature arrest that frequently causes deformity**
"13-yr fell from truck, SH II – likely complication?"	→ **growth arrest**
"Physeal arrest risk after operative SH II?"	→ **~60%** (OB) · **~40%** (other OB)
"12-yr after fixation – expect and do?"	→ **growth disturbance, observe**
"13-yr with 2 cm LLD, 3 yr left – final LLD?"	→ **~5 cm, affected side short**
"10-yr 'knee sprain', MRI SH I – long-term sequel?"	→ **leg length discrepancy**

"More skeletally mature child (ossified tuberosity)?"	→ lower risk of angular deformity needing surgery
"25% bar, 5° varus, open physis?"	→ bar resection + PMMA
"13-yr displaced SH II?"	→ closed reduction + percutaneous fixation
"10-yr SH I?"	→ anatomic reduction + smooth pins + cast
"Which injury gets smooth K-wires?"	→ SH I
"True about SH II treatment?"	→ Thurston-Holland size decides pins vs screws
"Undisplaced SH I with physeal widening?"	→ cast + close follow-up
"Initial imaging for twisting knee injury in 9-yr?"	→ AP, lateral, oblique X-rays
"Effusion, physeal tenderness, normal X-ray?"	→ MRI
"Suspected SH I with normal films – confirm?"	→ MRI
"Nondisplaced medial distal femoral physeal injury?"	→ long-leg cast + crutches
"SH I of distal femur – fracture location?"	→ propagates through multiple layers
"MRI-confirmed physeal injury – zone?"	→ where chondrocytes grow in size (hypertrophic)
"Zone injured – what happens there?"	→ chondrocyte apoptosis
"Pulses only on Doppler?"	→ reduce + fix fracture, then reassess
"ABI 0.8 with displaced SH II?"	→ reduction + percutaneous fixation, then reassess

– – Active space – –

DISTAL HUMERUS PHYSEAL SEPARATION - PEDIATRIC

– Pediatrics · 36 / 112 –

1 Basics

Also called — **transphyseal distal humerus fracture**

Age — **<3 years** (physis weaker than bone–ligament interface → physeal fracture, not dislocation)

Mechanism — **birth trauma** (vaginal delivery, shoulder dystocia; C-section traction) · **non-accidental trauma** (rotational/twisting shear – **up to 50% battered child**) · FOOSH, fall from bed/couch

Displacement — **posteromedial** (most) · rarely anterior

Age	SH type
<3 yr	**SH I** (pure physeal)
>3 yr	SH II (metaphyseal fragment)
Rare	SH III/IV – mimics lateral condyle fracture

3 Presentation

Exam — **pseudoparalysis** / not using arm · swelling, ecchymosis · NV compromise rare

NAT red flags — unwitnessed, inconsistent history, multiple injuries, burns, bites, bruises (**skin lesions most common NAT finding**)

4 Diagnosis & Classification

Diagnosis & non-accidental trauma association (skeletal survey)

Key radiology findings

Views — **AP + lateral centred on elbow** · stress views may help · **skeletal survey if abuse suspected**
✗ Wrong options: "babygram" (misses it)

Findings — **radius + ulna displaced posteromedially relative to humerus** (may be the only sign in infants) · **forearm not aligned with humeral shaft** · **radiocapitellar alignment maintained** (capitellum stays with radius – definitive if ossified) · effusion, swelling

Capitellum ossifies — **1–2 yr** (first elbow centre)

Ultrasound — uncertain dx, **no sedation** · static (cartilage absent on metaphysis) / dynamic instability

Arthrogram — uncertain dx, in OR with CRPP · **direct posterior portal** (olecranon fossa) in young · shows pin start on capitellum + anterior humeral line

MRI not routine (sedation)

Missed up to **50%** by radiologists · delayed up to 1 wk



Posteromedial displacement of forearm on humerus



Transphyseal distal humerus fracture

5 Management

Transphyseal distal humerus fracture (<3 yr)

- Suspect NAT → **skeletal survey + child protective services**
- Displaced (most) → **arthrogram + closed reduction + percutaneous pinning**
- Nondisplaced / late presentation → **long-arm splint/cast 2–3 wk**
  (late: deformity → osteotomy later)

After immobilisation, 8-month-old → contact child protective services
✗ Wrong options: MRI · CT elbow/head · emergent fixation (not emergent without open/NV injury)

NAT team consult most likely for **distal humerus physeal separation in 16-month-old**
✗ Wrong options: BBFF 5-yr · type II supracondylar 4-yr · femur 10-yr · Tillaux 13-yr

Treatment – closed reduction & pinning

Displaced (even if NAT ruled out) → CRPP under GA with **arthrogram**
✗ Wrong options: observe · sling · rigid immobilisation · compartment monitoring · ORIF / plate · hinged ex-fix

Reduction gentle traction (little force) → correct translation/rotation → flex

– – Active space – –

Acceptable	**no cubitus varus** · **anterior humeral line bisects capitellum** · no malrotation
Pins	**2–3 × 0.062" lateral, divergent**, bicortical, wide spread · test under live fluoro
Post-op	splint · X-ray 1 wk · **remove pins at 3 wk** · follow **2–4 yr** (arrest, deformity, AVN)
Nonop	limited: **nondisplaced** · **late-presenting** (→ later osteotomy)

6 Complications

Cubitus varus	**up to 70%** – **more common than supracondylar** · from **medial condyle AVN**, malunion (missed/lost reduction), growth arrest → **lateral closing-wedge osteotomy**
Medial/lateral condyle AVN	fishtail deformity
Growth disturbance	observe, operate **>5 yr** (bigger limb, cooperative, one-stage)
Loss of motion	rarely functional
NOT associated	**tardy ulnar nerve palsy** (→ lateral condyle nonunion + cubitus valgus)

X vs Y

	Transphyseal separation	Elbow dislocation
Age	**<3 yr**	Older (almost never <3)
Displacement	**Posteromedial**	**Posterolateral**
Radiocapitellar line	**Intact**	Disrupted
NAT	**Strong association**	–

How it's asked

"10-month-old, posteromedial displacement – all EXCEPT?"	→ **tardy ulnar nerve palsy** (✓ abuse, cubitus varus, medial condyle AVN)
"Injury most commonly from child abuse?"	→ **distal humerus physeal separation** ✗ Wrong options: Monteggia · proximal femur · toddler's · supracondylar
"8-month-old after immobilisation – next?"	→ **contact child protective services**
"2-year-old, no injury history, transphyseal separation – true?"	→ **skeletal survey** ✗ Wrong options: posterolateral · nonop reliable · 6–8 yr · incarcerated fragment (medial epicondyle)
"NAT team present – your consult?"	→ **distal humerus physeal separation in 16-month-old**
"7-month-old swollen elbow, forearm displaced on humerus?"	→ **closed reduction + percutaneous pinning**

"12-month-old, NAT ruled out?"	→ **operative percutaneous pinning**
"9-month-old fell down steps?"	→ **CRPP in OR, immobilise 3–4 wk**
"8-month-old displaced transphyseal fracture?"	→ **closed reduction with pinning**

– – Active space – –

– – Active space – –

DISTAL RADIUS FRACTURES - PEDIATRIC

– Pediatrics · 37 / 112 –

1 Basics

Epidemiology – most common paediatric fracture

Most common fracture <16 yr	**distal radius (22.7%)** > hand phalanges (18.9%) > carpal/MC (8.3%) > clavicle (8.1%) > supracondylar (3.3%) > distal femur (1.6%)
Forearm	~40% of pediatric long-bone fractures (OB: distal radius = 40% of all pediatric fractures) · **M 2–3×** · peak **10–12 yr ♀, 12–14 yr ♂** (metaphyseal growth spurt)
Lifetime fracture risk <16 yr	boys 42%, girls 27%

Anatomy & remodelling

Distal radial physis	**75% of radial growth**, **40% of upper limb** · **~5.25 mm/yr**
Location	**metaphysis 62%** > diaphysis 20% > physis
Remodelling	greatest **near physis** + in **plane of joint motion (sagittal – flexion/extension)** · **~1°/month** · **least for rotation** (rotation doesn't remodel)

2 Mechanism

FOOSH, wrist extended · rule out **abuse** (inconsistent history, multiple-age injuries, grip marks)

4 Diagnosis & Classification

Type	Feature
Torus / buckle	**Unicortical**, nondisplaced, compression side
Greenstick	Fails on **tension (convex)** side; plastic deformation on compression side
Plastic deformation	Many microfractures, lower force over time
Complete metaphyseal	Apex volar (Colles-type) / apex dorsal (Smith-type)
Physeal	SH I–V (SH II most common)
Galeazzi	Distal-third radius + **DRUJ** injury
Monteggia	Ulna + radiocapitellar dislocation

Key radiology findings

Views	AP + lateral **wrist** and **forearm** · elbow if tender or diaphyseal · check SL interval, **DRUJ**, ulnar styloid
CT	intra-articular only (radiation)



Torus (buckle) fracture

5 Management

Remodelling potential & acceptable angulation (buckle fractures need no follow-up)

Age	Bayonet	Shaft angulation	Malrotation	Distal radius dorsal angulation
<10 yr	**<1 cm**	15–20°	45°	**30°**
>10 yr	**None**	10°	30°	**20°**

Other cut-offs young child remodels sagittal deformity up to **~20°**, older **~10°** (OB) · operate <9 yr if post-reduction **dorsal angulation >30°** or malrotation · >10 yr: angulation >20°, rotation >30°

Diaphyseal 10° angulation → blocks 20–30° rotation

Distal radius (TO)	Angulation	Displacement
< 8 yr	**≤ 45°**	**≤ 60%**
8–12 yr	**≤ 30°**	**≤ 50%**
Adolescent	**≤ 15°**	**≤ 40%**

TO points remodelling depends on the **physis, not age** · distal radial physis = fastest in forearm, **~75% of forearm growth** · overriding fractures **< 10 yr can fully remodel** · **no re-manipulation after 7 days** (physeal injury) · smooth wire safe if single pass, **< 2.5 mm**, low torque, avoid peripheral physis · fix if floating elbow, median nerve symptoms, compartment syndrome · **follow all for 6–12 months** · arrest → ulnar epiphysiodesis / ulnar shortening / radial osteotomy

Other TO lecture distal forearm acceptable: infants **< 30°**, children **< 15°**, peripubertal needs 2–3 yr growth · buckle = stable (2–3 wk splint) · **greenstick = unstable**, redisplaces → reduce + mould

4-yr, 12° apex volar at 2 weeks → observe – remodels within ~1 year
✗ Wrong options: osteoclasis · re-reduction · 2 years

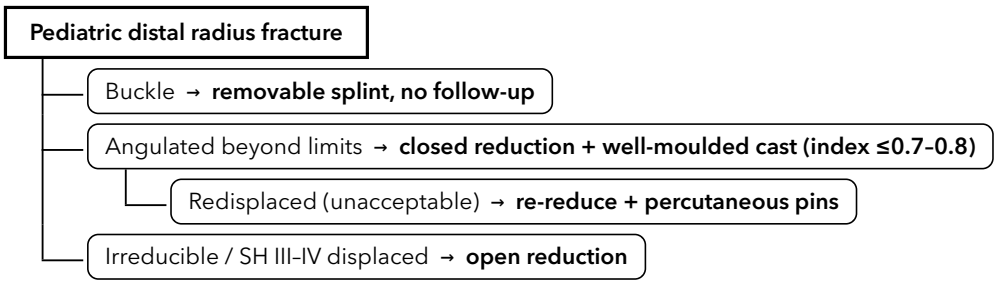
9-yr, 18° unreduced metaphyseal deformity → no long-term deformity or deficit

Buckle (torus) fracture → removable splint, no cast, no follow-up · same pain + refracture as cast
✗ Wrong options: physeal-arrest monitoring · close follow-up

– – Active space – –

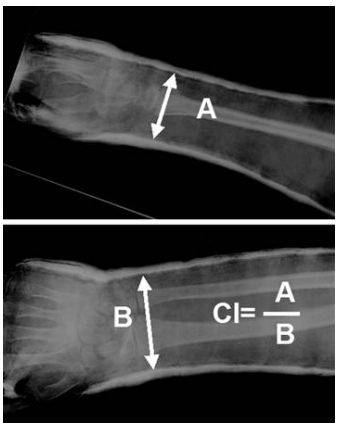
Operative criteria

Situation	Treatment
Unicortical/bicortical, **<10°**, buckle	**SAC 2–3 wk** / removable splint
>10–20° angulation, SH I/II unacceptable	**Closed reduction + cast**
Unstable / **lost reduction** in cast / reduced only in OR / SH I–II with **NV compromise**	**CRPP**
Displaced SH III/IV · **irreducible** (periosteum, **pronator quadratus**)	**ORIF**
Near-mature, redisplaced shaft / bayonet (15-yr)	**ORIF** (✗ Wrong options: accept · remanipulate)



Cast index & loss of reduction – management of redisplacement

Cast index — **sagittal (lateral) ÷ coronal (AP) inner width** · **best predictor** of failure · good ≈ **0.7** · loss of reduction **>0.8** (mean 0.79 in failures) · OB answer **>0.85**
✗ Wrong options: padding, gap, three-point, 2nd MC–radius indices



Cast index = lateral ÷ AP inner width

Short vs long arm cast — well-moulded SAC **= LAC** (no ↑ displacement, fewer missed school days)
✗ Wrong options: LAC/SAC/plaster/sedation as risks

Redisplacement — **20–25%** early · risks: **poor cast index**, ↑ **initial displacement**, incomplete reduction

11-yr, redisplaced to 22° at 1 week despite good mould → closed reduction + percutaneous fixation
✗ Wrong options: accept · recast · flexible nail (too distal) · plate

Physeal reduction — gentle, **one attempt**, **not >1 week** after injury (late reduction → arrest)

Technique — metaphyseal: recreate deformity, use periosteal hinge (traction counterproductive) · intact ulna: ulnar-deviate wrist

Reduction manoeuvre – greenstick fractures

Rotate the palm toward the apex — **apex-volar (supination injury) → pronate** · apex-dorsal (pronation injury) → supinate

CRPP technique

Radial styloid pins — entry just proximal to physis (smooth if transphyseal) · avoid **dorsal sensory radial nerve** · dorsal pin for volar tilt · DRUJ pin → long-arm cast

6 Complications

Cast thermal injury

Risk factor	
Dip water >24 °C (75 °F)	Yes
>8 plaster layers / thick cast	Yes
Resting on pillow while curing (most significant)	Yes
Fibreglass over plaster	Yes
Excess moulding pressure	Yes
Fracture pattern	**not a risk**
22 °C water, thin fibreglass, single padding layer	No

Physeal growth arrest (radius/ulna) – CT & epiphysiodesis

Distal ulnar physis — **most likely to arrest** (**30–40%**) vs distal radius **<5%**

Radial arrest → ulnar-positive variance → **ulnocarpal impaction**, TFCC + DRUJ injury, radial deviation · variance changes visible **3–4 months**

Risk factors — repeated injury, **late presentation/late reduction**, multiple attempts, SH III/IV

Distal radial physeal arrest

- First → **CT (physeal map) to size the bar**
- Bar <50% + growth left → **bar resection**
- Bar >50% / little growth → **complete radial + ulnar epiphysiodesis**
  (± radial lengthening for established discrepancy)
- Skeletally mature, ulnar-positive → **ulnar shortening osteotomy**

10-yr, painful radial deviation after SH II → CT to assess bar first
✗ Wrong options: osteotomy · bilateral epiphysiodesis

14-yr asymptomatic, radial arrest, small discrepancy → ulnar epiphysiodesis
✗ Wrong options: bar resection · lengthening · styloid repair

>50% radial bar, ulnar-positive, immature → distal ulna epiphysiodesis
✗ Wrong options: mature (ulnar shortening) · Kienböck · Madelung (Vickers ligament release)

Other — **malunion (most common complication)** · median neuropathy / acute carpal tunnel · cast-saw burns (cool blade)

– – Active space – –

How it's asked

"4-yr, 12° apex volar at 2 weeks?"	→ **observe — remodels within 1 year**
"9-yr, 18° unreduced?"	→ **no long-term deformity or deficit**
"2-yr buckle fracture — true?"	→ **no cast or follow-up needed**
"Risk of displacement after reduction?"	→ **cast index >0.85** ✗ Wrong options: long/short arm cast · plaster · sedation
"11-yr redisplaced 22° at 1 week?"	→ **closed reduction + percutaneous fixation**
"15-yr redisplaced distal radial shaft, physis closing?"	→ **ORIF**
"Best predictor of failure of closed treatment?"	→ **cast index**
"Isolated pronation reduces?"	→ **apex-volar greenstick (supination injury)**
"Thermal injury risk — EXCEPT?"	→ **fracture pattern**
"Most significant cast-burn risk?"	→ **resting wet cast on pillow**
"9-yr radius + ulna physeal fracture — likely complication?"	→ **distal ulna growth arrest**
"14-yr, radial arrest, asymptomatic?"	→ **ulnar epiphysiodesis**
"10-yr painful deformity after SH II?"	→ **CT to evaluate arrest**
"Best treated by distal ulna epiphysiodesis?"	→ **immature >50% radial bar with ulnar-positive variance**
"Most common fracture <16 yr?"	→ **distal radius**

DOWN SYNDROME

– Pediatrics · 38 / 112 –

1 Basics

Epidemiology

Most common chromosomal abnormality (US) · **1:700** live births

Risk	**advanced maternal age**: 1 in 250 if >35 yr · 1 in 5000 if <30 yr

Genes & lab markers

Genetics	**trisomy 21** (maternal duplication)
Collagen	chr 21 codes **type VI collagen (COL6A1, COL6A2)** → generalised laxity
Labs / medical	**hypothyroidism** (check with SCFE) · cardiac disease **50%** (ASD, PDA) · duodenal atresia · premature aging · Alzheimer's · intellectual disability

Orthopaedic manifestations

Ligamentous laxity + hypotonia · short stature · delayed walking (**2–3 yr**) · **C1–2 instability** · **occipitocervical instability** · **hip instability** · **patellofemoral instability** · scoliosis, **spondylolisthesis (6%)** · **pes planus (50%)** · metatarsus primus varus · **hallux valgus (25%)** · **SCFE**

3 Presentation

Exam	flattened facies · **upward-slanting eyes** · **epicanthal folds** · **single palmar (simian) crease** · laxity · hearing loss

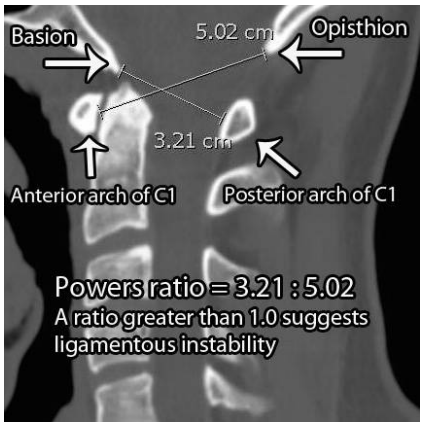
4 Diagnosis & Classification

Diagnosis	**chromosomal testing** (trisomy 21)

Key radiology findings

C-spine	**flexion-extension lateral** – **ADI** normal **<5 mm** (Orthobullets) / up to 4 mm (OB) · 5–10 mm tolerated in Down · **space available for cord (SAC)** threshold **14 mm**
Occipitocervical	**Powers ratio**
Hip	**acetabular retroversion** (crossover + ischial spine signs) · coxa valga
Knee	long-leg film (genu valgum) · sunrise/Merchant (patellar subluxation)

– – Active space – –



Powers ratio

5 Management

Atlantoaxial instability – ADI & sports clearance



Atlantoaxial instability in Down syndrome

Prevalence	**17.5%** (Orthobullets) · ~15% (OB)
Presentation	subtle: change in gait, bowel/bladder
Flexion-extension films	**before intubation** · **Special Olympics requires** C-spine imaging before gymnastics, swimming, high jump etc. ✗ Wrong options: AP pelvis · knee/elbow films · neurology referral · clear without work-up

Atlantoaxial instability in Down syndrome

- ADI 5–10 mm, asymptomatic, SAC >14 mm → **surveillance (neuro exam + repeat imaging); avoid contact sports**
- ADI >5 mm + myelopathy, or ADI >10 mm, or SAC <14 mm → **C1–2 posterior fusion** *(complications up to 50%)*

Asymptomatic, ADI 6 mm → avoid contact sports, high jump, diving, gymnastics, butterfly
✗ Wrong options: clear all · collar · halo · fusion

Asymptomatic, ADI 5 mm → surveillance: symptoms, neuro exam, repeat imaging
✗ Wrong options: O–C2 fusion · anterior fusion · soft collar · traction

Operative criteria

Problem	Surgery when
C1–2 instability	ADI >5 mm + **symptoms/myelopathy** · **ADI >10 mm** · **SAC ≤14 mm** · cord compression on MRI/CT
Occipitocervical	Progressive neuro deficit / myelopathy → occipitocervical fusion (most observed)
Scoliosis	Brace **25–30°** · fuse **>50°** (more complications than idiopathic)
Hip instability	Older symptomatic → capsulorrhaphy + pelvic (**triple / PAO** for retroversion) + femoral varus osteotomy (high complications)
Patellar instability	Symptomatic → lateral release, medial reefing, semitendinosus tenodesis, **TTO if mature**
SCFE	**Pin both hips**
Pes planus	Orthotics if symptomatic; surgery if refractory

Lower-limb problems – hip instability, patellar instability & genu valgum

Hip instability — **5%**, age **2–10 yr** · laxity + **acetabular retroversion**, coxa valga · young, mild subluxation, no bony change → **abduction brace**
✗ Wrong options: observe · NSAIDs · shoe lift · osteotomy

Patellofemoral instability — mature, painless, knock-knees → **observation** · symptomatic → **patellar stabilising brace** → surgery (TTO in mature)
✗ Wrong options: lateral release alone · trochleoplasty first

10-yr with Down facies, laxity, new limp → associated image = **acetabular retroversion**
✗ Wrong options: foramen magnum stenosis (achondroplasia) · absent cruciates (fibular/femoral deficiency) · iliac horns (nail-patella) · absent clavicles (CCD)

How it's asked

"10-yr Down, high jump, ADI 6 mm, asymptomatic?"	**→ avoid contact sports, high jump, diving**
"7-yr Down, ADI 5 mm, asymptomatic?"	**→ surveillance with neuro exam + repeat imaging**
"15-yr gymnast with trisomy 21 — before participation?"	**→ cervical spine imaging**
"17-yr Down, painless knock-knees, dislocated patellae?"	**→ observation**
"3-yr Down, mild hip subluxation, no bony change?"	**→ abduction brace**
"Hip image associated with Down syndrome?"	**→ acetabular retroversion**
"Indication for C1–2 fusion?"	**→ myelopathy with ADI >5 mm, ADI >10 mm, or SAC <14 mm**

– – Active space – –

DUCHENNE MUSCULAR DYSTROPHY

– Pediatrics · 39 / 112 –

1 Basics

Epidemiology

Most common hereditary neuromuscular disease (most common MD) · prevalence **2–3/10,000** (Orthobullets) · ~**1 in 3500 male births** (OB)

Who	**boys only** · onset **2–6 yr**

Genetics – dystrophin, X-linked recessive, carrier mother & offspring risk

Inheritance	**X-linked recessive** ✗ Wrong options: AD · AR · X-linked dominant · mitochondrial · Y-linked
Gene	**dystrophin, Xp21(.2)** · point deletion / nonsense (out-of-frame) → **absent dystrophin** (<5%)
Spontaneous	**one-third** new mutations
Carriers	mother carries · **sons 50% affected**, daughters 50% carriers · **affected male → 0% sons affected** (gives Y), all daughters carriers · **2.5–20% of carrier females symptomatic** (Lyon hypothesis) · **CK ↑ in 80% of carriers** (peak 8–12 yr)
Confirm DMD via family	**test mother's CK** ✗ Wrong options: father (always normal) · unaffected brother
Pathophysiology	dystrophin–glycoprotein complex lost → membrane fragility, Ca^{2+} dysregulation, oxidative damage → necrosis/regeneration cycles → **fibrofatty replacement**

Genes & lab markers

Disorder	Gene / inheritance
Duchenne	**Dystrophin, Xp21** (XLR)
Becker	Dystrophin (in-frame, ↓ amount) (XLR)
Emery-Dreifuss	**Emerin** (X-linked)
FSH dystrophy	**D4Z4 deletion, 4q35 · AD**
Myotonic dystrophy	**DMPK / CNBP**, **chromosome 19**
Limb-girdle	**AR and AD** (LGMD2 / LGMD1) · chr 2p (TO)
Marfan	FBN1
CMT	PMP22 (AD)
Turner	45,XO

CK	**markedly ↑ – 10–200×** (Orthobullets) · 50–100× (TO) · often **>25,000 IU/L** (e.g. 43,000) · **peaks by age 2**, falls with fibrofatty replacement
EMG	myopathic (low amplitude, short, polyphasic), **normal NCV**

Biopsy necrosis, fibrofatty infiltration, **absent dystrophin stain** · DNA confirms ~two-thirds · **dystrophin testing of biopsy NOT 100% confirmatory** (TO)

– – Active space – –

3 Presentation

History delayed walking (e.g. 30 mo) · toe walking, clumsy, **waddling** · difficulty climbing stairs, hopping · falls

Weakness **proximal first, starting with gluteals** · lumbar **lordosis** (gluteal weakness) · wide-based stance

Gowers sign rises by **walking hands up thighs** (gluteus maximus + quadriceps weakness)

Meryon sign child slips through examiner's hands when lifted under arms (shoulder girdle weakness)

Calf pseudohypertrophy fat + fibrous infiltration

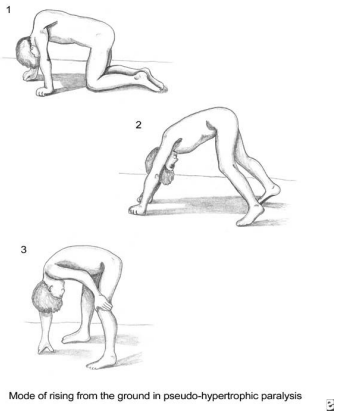
Reflexes **present**
✗ Wrong options: SMA · GBS

Trendelenburg · contractures (elbow, hip, knee, ankle) · **equinovarus** (tibialis posterior retains strength longest) · scoliosis · fractures (falls)

Other cardiomyopathy · mild intellectual impairment · pharyngeal weakness · malignant hyperthermia risk



Calf pseudohypertrophy



Gowers sign

- - Active space - -

4 Diagnosis & Classification

Diagnosis – Gowers sign, CK & Becker variant

First test — **serum CK** (✗ Wrong options: leg X-rays · NCS · brain/spine MRI) · screen asymptomatic at-risk boy with **CK** (✗ Wrong options: DNA, biopsy, Western blot first)

Confirm — **DNA testing** (± biopsy)

Key radiology findings

MRI muscle pattern — **gastrocnemius early** → gluteus max/med, adductor magnus → quads etc. · **spared: sartorius, gracilis, semitendinosus, semimembranosus, tibialis posterior** · upper limb: triceps, biceps, rotator cuff; **deltoid spared**

Echo — **dilated cardiomyopathy in all by late teens/20s**

X-ray — lucent fatty soft tissue, scoliosis, cardiomegaly

Differential

	Duchenne	Becker	SMA	Emery-Dreifuss	Limb-girdle	GBS
Dystrophin	**Absent**	**Decreased/ abnormal**	Normal	Normal (**emerin** absent)	Normal	–
CK	**Very high**	Very high	**Normal**	Near normal	Mildly ↑	Normal
Pseudohypertrophy	Yes	Yes	No	No	No	No
Reflexes	Present	Present	**Absent + fasciculations**	–	–	**Absent**
Onset	2–6 yr	**~8 yr, slower**, lives to 40s	Earlier	Teens; **early elbow + ankle contractures**, **heart block** (pacemaker)	1st–4th decade	Acute, post-infectious
Key test	DNA	DNA	–	ECG	–	**CSF** (lumbar puncture)

Differential – Guillain-Barré syndrome

GBS — acute ascending flaccid weakness 1–6 wk after (GI) infection · areflexia · autonomic + diaphragm involvement · **lumbar puncture (CSF)** best test · IVIG / plasma exchange · 80% walk at 6 mo
✗ Wrong options: C-spine MRI/CT · M-spike · stool

5 Management

Corticosteroids (prednisone / deflazacort)	age **5–7 yr** with progression · prolong ambulation (3–4 yr), ↓ **scoliosis needing fusion 92% → 20%**, delay pulmonary + cardiac decline, ↓ mortality 76% · side effects: **osteonecrosis, osteoporosis**, weight gain, cataracts, short stature · **deflazacort** – fewer glucose/bone effects, far costlier
Exon-skipping (eteplirsen, golodirsen, viltolarsen, casimersen)	convert out-of-frame to in-frame (**DMD → BMD-like**) · hot-spot exons 45–53 (~50%) · costly
Cardiac	ACE inhibitor + β-blocker · echo/ECG every 2 yr to age 10, then yearly (6-monthly with cardiomyopathy)
Other	nocturnal ventilation · PT, power chair · nutrition
Soft-tissue releases (ambulatory)	ITB (abduction contracture) · sartorius, rectus, TFL (flexion) · hamstrings · **TAL + tibialis posterior** lengthening/transfer · mobilise early
Equinovarus	stretching, night AFO → **TAL + PTT transfer + toe flexor tenotomies**

Scoliosis – early fusion (20–30°, FVC), T2-pelvis & preoperative cardiopulmonary work-up

Natural history	**90–95%** develop scoliosis · starts when **full-time sitting (11–13 yr)** · **progresses rapidly** (1–2°/month), **small curves always progress** · **kyphoscoliosis** + pelvic obliquity · **restrictive lung disease** ✗ Wrong options: 50%
Bracing	**ineffective / contraindicated** (impairs respiration) ✗ Wrong options: TLSO · PT · observe a 25° curve

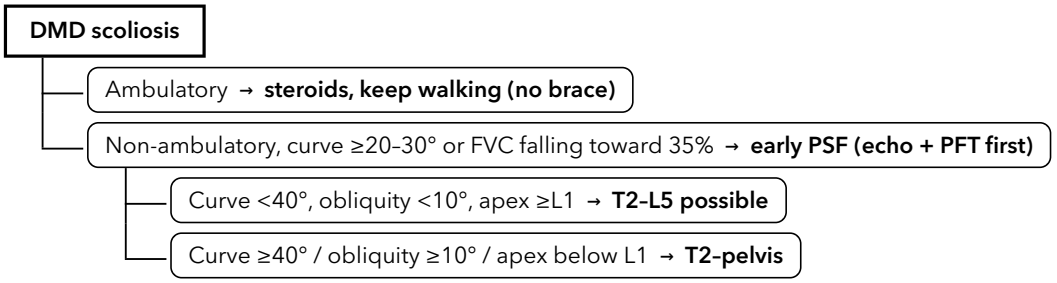
Operative criteria

Criterion	Value
Cobb angle	**20–30°** in non-ambulator (up to ~40° if responding to steroids)
FVC	Operate **before FVC ≤35%** (≤35% = indication / threshold) · **>50% = good outcomes** · <10% contraindication (some)
Other	**Non-ambulatory** · **rapid progression** · poor steroid response (✗ Wrong options: decreased muscle mass)
Fuse to pelvis	Curve **≥40°** · **pelvic obliquity ≥10°** · lumbar apex **below L1**
Construct	**T2-pelvis posterior instrumented fusion** (pedicle screws) (✗ Wrong options: T10–pelvis · growing rods · T2–L5 with obliquity · delay for steroids)
Anterior approach	**Avoid** (lung deflation, ↑ pulmonary complications)

Timing	driven by **cardiopulmonary status, not Cobb angle** · 12-yr, 38°, FVC 50% → **as soon as possible** ✗ Wrong options: wait for FVC 35% · cardiac decline · 50°

– – Active space – –

Pre-op — **echocardiogram + PFTs**, **cardiology consult** · stop ACE-I 2–3 d, taper β-blocker over weeks · TXA · beware anaesthesia-induced **rhabdomyolysis**, hypotensive anaesthesia may not be tolerated
✗ Wrong options: hematology · neurology · rheumatology/biopsy · MRI for ectasia · baclofen trial · bracing trial



Other muscular dystrophies (FSH, myotonic)

FSH dystrophy	**AD**, 4q35 · facial weakness (**can't whistle, purse lips**, wrinkle brow) · **scapular winging** – scapular stabilisers weak while **deltoid (abductors) stay strong** → can't flex/abduct arm · **scapulothoracic arthrodesis** · foot drop (AFO) · open-eyed sleeping ✗ Wrong options: "due to weak abductors"
Myotonic dystrophy	**chromosome 19** (DMPK) · can't relax muscle · most common adult-onset MD · baldness, gonadal atrophy, cardiac, clubfoot, hip dysplasia, scoliosis
Limb-girdle	**AD (LGMD1) and AR (LGMD2)** · pelvic/shoulder girdle · CK mildly ↑
Oculopharyngeal	40–70 yr, eyelids + throat

6 Complications & prognosis

Untreated	walking lost by **~10 yr** (10–12) · wheelchair by **15** · death from respiratory/cardiac failure ~**20** (Orthobullets) / ~25 (OB) / early 30s (TO) · 15-yr survival 27.9% → **78.6% with steroids**

How it's asked

"5-yr, Trendelenburg + calf pseudohypertrophy – genetic cause?"	→ **deletion in Xp21** ✗ Wrong options: FBN1 · D4Z4 4q35 = FSH · DMPK/CNBP = myotonic
"Inheritance of DMD?"	→ **X-linked recessive**
"Most commonly X-linked recessive?"	→ **Duchenne** ✗ Wrong options: CMT · Marfan · Larsen · Turner
"Affected boy's chance of passing to sons?"	→ **0%**
"3-yr large calves, Gowers, CK 43,000?"	→ **X-linked recessive**
"Gowers sign – pedigree?"	→ **X-linked recessive pedigree**
"DMD transmitted by?"	→ **X-linked**
"Gowers + scoliosis + ↑ CPK – protein?"	→ **dystrophin** ✗ Wrong options: type I collagen · neurofibromin · Runx2

"Confirm DMD — test CK in?"	→ **mother**
"8-yr weak, big calves, equinovarus, scoliosis — pedigree?"	→ **X-linked recessive**
"Mother asks about future children — which pedigree?"	→ **X-linked recessive**
"DMD — all EXCEPT?"	→ **dystrophin biopsy testing is 100% confirmatory** (TO)
"Becker characteristics?"	→ **pseudohypertrophy, very high CK, abnormal (present) dystrophin**
"2-yr toe walker, big calves, Gowers — most useful test?"	→ **creatine kinase**
"Asymptomatic boy, uncle with DMD — screen?"	→ **serum CK**
"9-yr wheelchair, FVC 35%, lumbar 35° — indications?"	→ **FVC 35%, non-ambulatory, curve >30° (all)**
"12-yr non-ambulator with cardiomyopathy and scoliosis — true?"	→ **surgery for Cobb >30°** ✗ Wrong options: bracing · steroids eliminate surgery · A/P approach · FVC <35% better
"9-yr DMD, 20° curve?"	→ **PSF before curve passes 30°**
"16-yr for T2–pelvis fusion — pre-op work-up?"	→ **echocardiogram + PFTs**
"Untreated DMD scoliosis — true?"	→ **fusion offered at 20–30°**
"Supports early PSF EXCEPT?"	→ **decreased muscle mass**
"12-yr, 40° curve — before surgery?"	→ **cardiology consult**
"12-yr, 38°, FVC 50% — when?"	→ **as soon as possible**
"85° curve, 20° pelvic tilt, FVC 40%?"	→ **T2–pelvis posterior instrumented fusion**
"14-yr wheelchair, 25° curve, FVC 55%?"	→ **posterior spinal fusion to pelvis**
"Ascending weakness after diarrhoea — test?"	→ **lumbar puncture CSF analysis**
"FSH dystrophy inheritance?"	→ **AD** (TO)
"Can't abduct arms or whistle — FALSE statement?"	→ **"weak shoulder abductors cause it"** (TO)
"Myotonic dystrophy chromosome?"	→ **19** (TO)
"Limb-girdle inheritance?"	→ **both AD and AR** (TO)

– – Active space – –

DYSPLASIA EPIPHYSEALIS HEMIMELICA (TREVOR'S DISEASE)

– Pediatrics · 40 / 112 –

1 Basics

Definition	**osteochondroma arising from the epiphysis → asymmetric (hemimelic) epiphyseal overgrowth** of cartilage
Incidence	extremely rare, **~1:1,000,000** · **male** predilection · young children
Location	**lower extremity**: **knee or ankle** (one side of the epiphysis, usually medial) · usually **one joint** ✗ Wrong options: upper-limb exostoses
Cause	defect in the **groove of Ranvier**

Genes & lab markers

Genetics	**non-hereditary** ✗ Wrong options: AD · AR / non-viable births
Malignancy	**no reported transformation**
Histology	identical to **osteochondroma**; diagnosis rests on **epiphyseal location**

3 Presentation

Symptoms	asymmetric limb deformity · joint symptoms · ↓ motion · joint-line tenderness

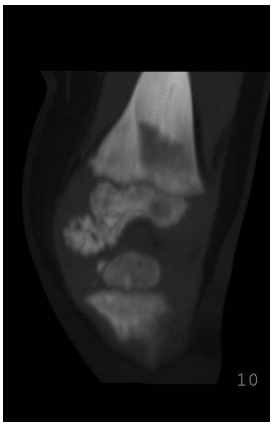
4 Diagnosis & Classification

Key radiology findings

X-ray	irregular calcified mass at joint line / epiphysis
CT	**cortical continuity with the epiphysis**



Trevor's disease of the knee



CT – continuity with epiphysis

5 Management

Stage	Treatment
Asymptomatic, small	Observe
Symptomatic early	**Excision of overgrowth**
Symptomatic late	**Osteotomies**

6 Complications

Recurrence after excision (common) · articular irregularity → **early OA**

How it's asked

"Woman with history of DEH, genetic counselling – consistent feature?"	→ **asymmetric epiphyseal overgrowth** ✗ Wrong options: malignant risk · upper-limb exostoses · AD inheritance
"Most common sites?"	→ **knee and ankle**
"Symptomatic early lesion?"	→ **excision**

– – Active space – –

EHLERS-DANLOS SYNDROME

– Pediatrics · 41 / 112 –

1 Basics

Genes & lab markers

Classic EDS	**COL5A1 / COL5A2** (40–50%) → **type V collagen** (co-expressed with type I; skin fibril + basement membrane assembly) · ↓ type V for fibrillogenesis
Diagnosis	**collagen typing of skin biopsy**

Clinical features & Beighton score

Features	**hyperelastic, fragile skin** · easy bruising, **poor wound healing**, widened atrophic scars · **joint hypermobility + dislocation** (shoulder, patella, ankle) · generalised laxity · early OA · chronic MSK pain (50%)
Other	**mitral valve prolapse** · **aortic root dilatation (up to 1/3)** · DDH · clubfoot · pes planus · **scoliosis** · high palate · gastroparesis · soft-tissue calcification
NOT EDS	**superior lens dislocation → Marfan** (inferior → **homocystinuria**)

Villefranche classification

Type	Inheritance	Former name / feature
Classical	**AD**	I (gravis), II (mitis) · COL5A1/2 · skin + scars + hypermobility
Hypermobile	**AD**	III
Vascular	**AD**	IV
Kyphoscoliosis	**AR**	VI (ocular-scoliotic)
Arthrochalasis	AD	VIIA, VIIB
Dermatosparaxis	AR	VIIC – severe fragility, sagging skin

Berlin types I–XI · **II + III most common, least disabling**

3 Presentation

Beighton-Horan score (/9)

Manoeuvre	Points
Passive **5th MCP hyperextension >90°**	1 each side
Passive **thumb apposition to forearm**	1 each side
Elbow hyperextension >10°	1 each side
Knee hyperextension >10°	1 each side
Palms flat on floor, knees straight	1

Hypermobility **≥5/9** (threshold varies: >3, 5, 6/9) · used for **EDS / joint laxity**
✗ Wrong options: CP · myelodysplasia · arthrogryposis · Kniest

– – Active space – –

Beighton cut-off (TO) adults **≥ 5** · children **≥ 6**

Thumb apposition to forearm	5th finger hyperextension
	

4 Diagnosis & Classification

Key radiology findings

X-ray subluxations/dislocations · kyphoscoliosis

Echocardiogram **mandatory** (aortic root)

5 Management

Mainstay **PT, orthotics, pain support**

Operative criteria

Problem	Surgery
Recalcitrant unstable/painful joint	**Arthrodesis** (soft-tissue stabilisation unlikely to hold)
Progressive scoliosis (esp. kyphoscoliosis type)	**PSF – longer fusion** (junctional problems)

How it's asked

"NOT associated with EDS?"	→ **superior lens dislocation** (Marfan)
"Beighton-Horan scale used for?"	→ **Ehlers-Danlos**
"Image of thumb-to-forearm / 5th finger hyperextension – condition?"	→ **Ehlers-Danlos** (TO; answer not provided by source)
"Gene in classic EDS?"	→ **COL5A1/COL5A2 (type V collagen)**
"Mandatory investigation?"	→ **echocardiogram**

– – Active space – –

ELBOW DISLOCATION - PEDIATRIC

– Pediatrics · 42 / 112 –

1 Basics

Incidence	**3–6% of pediatric elbow injuries** · **M:F 3:1** · peak **10–15 yr** · **very rare <3 yr** (→ think transphyseal fracture + NAT)
Mechanism	**FOOSH** · posterior = hyperextension + valgus + supination · anterior = posterior-to-anterior blow on flexed elbow

Small coronoid in children → can't resist posterior ulnar displacement

Associated	**medial epicondyle avulsion = most common** (may **incarcerate** in joint) · radial head/neck, olecranon, coronoid fractures
Nerves	**ulnar** (with medial epicondyle) = most common neuropraxia · **median** + **brachial artery** stretched / entrapped

Classification

By proximal radioulnar joint position	**posterolateral (most common)** · posteromedial · anterior (rare) · divergent

3 Presentation

Exam	swollen, painful · **held flexed** · forearm looks short · **distal humerus fullness in antecubital fossa** · NV exam (brachial artery, median, ulnar)

4 Diagnosis & Classification

Key radiology findings

Views	AP + lateral (± contralateral) · loss of radiocapitellar + ulnohumeral relationship, **radius–ulna relationship maintained**
Post-reduction	**locate the medial epicondyle** – if not seen in normal position, assume **intra-articular**



Elbow dislocation with medial epicondyle fracture

5 Management

Reduce promptly by the most available provider (ED) under sedation · don't delay for orthopaedics

Technique supine: elbow flexed, forearm **supinated**, gradual traction · prone: forearm hanging, anterior push on olecranon · anterior dislocation: inline traction, posterior push on forearm

Management & surgical indications

Stable after reduction (most) → brief immobilisation 1–2 wk (posterior splint ~8–12 days) → early active ROM
✗ Wrong options: cast 6 wk · hinged ex-fix · MRI

Operative criteria

Absolute	Relative
Open dislocation	Significant instability (rare)
Incarcerated medial epicondyle (or coronoid)	–
Failure to obtain/maintain closed reduction	–

Medial epicondyle fracture – when to fix absolute: **incarcerated fragment**, open, articular extension · relative: **elbow dislocation / instability**, ulnar nerve palsy, **throwing/upper-limb athlete** · nonunion common nonoperatively (often asymptomatic) · fuses **16–18 yr** (last)
✗ Wrong options: buckle radius fracture · <2 mm displacement · radial nerve symptoms · video games

Entrapped medial epicondyle fragment



Incarcerated medial epicondyle after reduction

Incongruous joint after reduction / fragment in joint → open reduction (medial approach) + ORIF of medial epicondyle
✗ Wrong options: re-reduction attempt · lateral epicondyle · coronoid

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6 Complications

Loss of motion after immobilisation

Stiffness – loss of terminal extension	from **immobilisation >2 weeks** ✗ Wrong options: sex · age >9 · ED reduction · no ORIF
Others	**heterotopic ossification** (vigorous reduction) · transient nerve injury (**median entrapment**, **ulnar with medial epicondyle**) · brachial artery (rare) · compartment syndrome (hyperflexion splint) · recurrent instability (coronoid / radial head fractures)

X vs Y

	Elbow dislocation	Transphyseal distal humerus fracture
Age	**10–15 yr**	**<3 yr**
Displacement	**Posterolateral**	Posteromedial
Radiocapitellar	Disrupted	Maintained
NAT	Rare	**Strong association**

How it's asked

"11-yr – treated with posterior splint 8–12 days then ROM?"	→ **acutely reduced elbow dislocation** ✗ Wrong options: lateral condyle · type I supracondylar · olecranon (3–5 wk)
"15-yr, post-reduction joint incongruous with medial epicondyle fracture?"	→ **open reduction**
"10-yr medial epicondyle fracture – indication for surgery?"	→ **elbow dislocated during fall**
"9-yr, fragment in joint after reduction?"	→ **medial epicondyle – ORIF**
"Loss of extension at 4 months?"	→ **splint immobilisation for 4 weeks**

EPIPHYSEAL BRACKET

– Pediatrics · 43 / 112 –

1 Basics

Also called	**longitudinal epiphyseal bracket (LEB)**, delta phalanx / delta bone
Incidence	**2–14%** of congenital hand + foot deformities · **11% in great toe**
Pathology	incomplete primary ossification · **aberrant physeal cartilage runs longitudinally along the diaphysis ("C-shaped")** → tethers growth → short, wide, **triangular / trapezoidal** bone
Bones	short tubular bones (phalanges, metatarsals – esp. **1st MT**)

Genes & lab markers

Genetics	**AD when in hand**, otherwise sporadic
Associated	**hallux varus** (feet) · **clinodactyly**, polydactyly (hands) · Rubinstein-Taybi · fibrodysplasia ossificans progressiva · Apert

3 Presentation

Deformity	angulated, short digit / ray (hallux varus, medial great-toe deviation)

4 Diagnosis & Classification

Key radiology findings

X-ray	**trapezoidal / triangular** bone · **"D-shaped" metatarsal** · wedge-shaped middle phalanx
Ultrasound	shows **pre-ossified** bracket
MRI	**diagnoses** LEB



Longitudinal epiphyseal bracket – C-shaped physis

– – Active space – –

Radiographic staging

Stage	Feature
1	Delta-shaped diaphysis, radiolucent cartilaginous epiphysis
2	Secondary ossification at proximal + distal ends
3	Complete ossification of bracket along one side
4	Physeal closure

5 Management

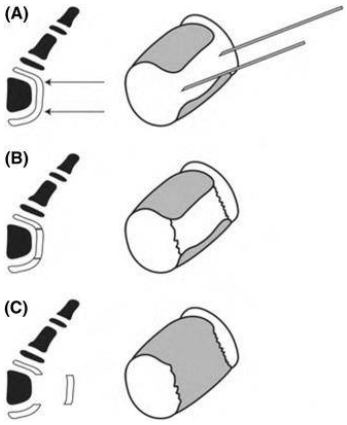
Longitudinal epiphyseal bracket

- Young child, open bracket → **early excision (physiolysis) + fat / PMMA interposition**
- Older, bracket closed → **opening/closing wedge osteotomy ± lengthening**

Treatment - bracket excision

Early open bracket excision + fat graft (or PMMA) interposition — best; restores near-normal longitudinal growth, ↓ deformity
✗ Wrong options: percutaneous epiphysiolysis (open needed) · Syme · ex-fix lengthening as index · fusion (macrodactyly) · observation · ray resection

Osteotomy — **after bracket epiphysis closes** (older) · secondary realignment/ lengthening if growth insufficient



Physiolysis of bracket

6 Complications

Untreated — forefoot deformity, short bone, clinodactyly

Operative — recurrence, incomplete correction

How it's asked

"4-yr with metatarsal LEB — next step?"	**→ bracket excision + fat graft interposition**
"3-yr, first-ray bracket on X-ray?"	**→ excision of the abnormal structure**

EXTERNAL TIBIAL TORSION

– Pediatrics · 44 / 112 –

1 Basics

Definition	abnormal **external rotation of the tibia → out-toeing** · usually **right-sided** if unilateral
Normal development	tibia externally rotates ~15° in childhood · femoral anteversion falls ~25°
Natural history	**worsens** in late childhood/adolescence (tibia rotates externally with growth) · **more often needs surgery than internal tibial torsion**
Associated	**miserable malalignment** (ETT + femoral anteversion) · **Osgood-Schlatter** · **OCD** · early knee DJD · neuromuscular (myelodysplasia, polio)

3 Presentation

Symptoms	out-toeing, tripping · **anterior knee pain**, **patellofemoral instability** (positive apprehension)



External tibial torsion

Out-toeing by age	infancy/toddler → **hip external rotation contracture** · late childhood/adolescence → **external tibial torsion**

4 Diagnosis & Classification

Key radiology findings

Thigh–foot axis (TFA)	**best clinical measure** · prone, knee 90° · infancy ~5° internal · **age 8: ~10° external (−5 to +30°)** · **>20° external** = ETT
Transmalleolar axis	supine · infancy 4–5° internal · adult **23° external** (0–40°)
Foot progression angle	~5° external in late childhood
CT torsional study	tibial version = distal minus proximal axis; femoral version neck minus condyles
Imaging	usually none

– – Active space – –



Thigh–foot axis

5 Management

First line rest, rehab, activity modification (bracing ineffective)

Operative criteria

Source	Indication
Orthobullets	**>8 yr** + ETT **>3 SD (>40° external)**
OB	**≥9 yr** + **TFA >30°** (or < −15°) · normal TFA −5° to 20°

Procedure **supramalleolar (distal tibial) derotation osteotomy** (most common) ± oblique fibular osteotomy · pins or plate · IM nail only if mature

Distal vs proximal distal: easier, safer (↓ NV injury, ↓ compartment syndrome) · proximal: more contact/soft tissue but **peroneal nerve risk** → anterior fasciotomy

12-yr, right ETT 45° (left 32°), normal femoral version → right supramalleolar osteotomy
✗ Wrong options: observe · KAFO · distal tibial guided growth (causes coronal deformity) · femoral osteotomy

X vs Y

Cause of out-toeing	Age
Hip ER contracture	Infant / toddler
External tibial torsion	Late childhood / adolescent

How it's asked

"12-yr, out-toeing, TFA +30°, patellar apprehension, symmetric hip rotation?"	**→ excessive external tibial torsion** ✗ Wrong options: femoral anteversion · ER contracture · internal torsion
"12-yr right ETT 45° on CT?"	**→ supramalleolar osteotomy**
"Best exam measure?"	**→ thigh–foot axis**

FEMORAL ANTEVERSION

– Pediatrics · 45 / 112 –

1 Basics

Definition & measurement

Definition transverse-plane angle between **femoral neck axis** and the **posterior femoral condylar plane** (TO) · femoral anteversion = functional anteversion + condylar twist
✗ Wrong options: lesser trochanter · tibial plateau · distal condylar plane

Pathology ↑ neck anteversion → **compensatory internal rotation of femur → in-toeing** · **packaging** disorder

Demographics **3–6 yr** · **girls 2:1** · hereditary · usually **bilateral** (be wary of asymmetry)

Associated packaging **DDH** · **metatarsus adductus** · **torticollis** · also ↑ in clubfoot, Blount, flatfoot, CP

Normal anteversion

Age	Anteversion
Birth / infancy	**30–40°** (≈39–40°)
2nd year	~31°
To age 10	↓ 1–2°/yr → ~24°
Adult	**~15–16°**
After age 8	Minimal change

Three causes of in-toeing

Cause	Age	Resolves by
Metatarsus adductus	**Infant**	~4 yr
Internal tibial torsion	**Toddler**	6–7 yr
Femoral anteversion	**Early childhood (3–6)**	**~10 yr** (8–13)

Order of resolution **MA → ITT → femoral anteversion**

3 Presentation

History in-toeing gait, tripping, awkward running · **sits in "W" position** · knee pain if with tibial torsion · more symptomatic in neuromuscular / brace-dependent walkers

– – Active space – –

Examination findings & diagnosis

Test (prone)	Femoral anteversion	Normal
Hip IR	**>70°**	20–60°
Hip ER	**<20°**	30–60°
Thigh–foot angle	**Normal** (−5° to +20°)	Infant 5° internal · age 8: 10° external
Heel bisector	Normal (2nd–3rd web)	MA: 3rd–4th web
Patella in gait	**Internally rotated**	–
Trochanteric prominence test	Hip IR at max GT prominence ≈ anteversion	–

Internal tibial torsion **TFA < −10°** (e.g. −20° on right) with normal hip rotation → ITT on that side



W-sitting



Increased hip internal rotation in prone

4 Diagnosis & Classification

Key radiology findings

None usually · **CT or MRI** to measure actual anteversion (pre-op)

5 Management

Natural history & observation

Observation + reassurance remodels in ~99% by 8–10 yr · **bracing, shoe inserts, PT, sitting restrictions do NOT change natural history**
✗ Wrong options: night ER brace · inserts · osteotomy in a young child

Operative criteria

Source	Indication for derotation femoral osteotomy
Orthobullets	**ER <10°** in child **>10 yr** (rare)
Staheli (TO)	**>8 yr** · significant cosmetic + functional disability · **anteversion >50°** · **IR >85° + ER <10°** · family aware of risks

NOT a Staheli indication — **medial (internal) hip rotation >35°** (TO)

Osteotomy — **intertrochanteric** (blade plate / screw-side plate) · subtrochanteric or supracondylar alternatives · correction ≈ **(IR − ER) / 2**

Prognosis — persistent anteversion **not linked to hip/knee OA** (Orthobullets) · TO lists hip OA, patellar malalignment, chondromalacia as associations · **miserable (torsional) malalignment** = anteversion + external tibial torsion

X vs Y

	Femoral anteversion	Internal tibial torsion	Metatarsus adductus
Age	3–6 yr	Toddler	Infant
Key exam	**Hip IR >70°, ER <20°**	**TFA internal (>10°)**	**Convex lateral border**, heel bisector lateral
Patella	**Points in**	Forward	Forward
Resolves	~10 yr	6–7 yr	~4 yr

How it's asked

"6-yr in-toeing – finding favouring femoral anteversion?"	→ **hip IR 80°, ER 15°** ✗ Wrong options: TFA internal (ITT) · convex lateral border (MA)
"5-yr, in-toeing with knees in line with feet – finding?"	→ **increased hip internal rotation**
"6-yr, hips IR 60/ER 40, TFA −20° right?"	→ **internal tibial torsion, right**
"Staheli – NOT an indication?"	→ **medial hip rotation greater than 35 degrees** (TO)
"3-yr, IR 85°/ER 15°, neutral TFA?"	→ **observation** ✗ Wrong options: osteotomy · inserts · brace
"7-yr, IR >70°, ER <20°?"	→ **observation**
"Order of resolution of in-toeing causes?"	→ **metatarsus adductus → internal tibial torsion → femoral anteversion**
"Anteversion measured relative to?"	→ **posterior femoral condylar plane** (TO)

– – Active space – –

FEMORAL SHAFT FRACTURES - PEDIATRIC

– Pediatrics · 46 / 112 –

1 Basics

Epidemiology

Incidence	**1.6–2%** of pediatric fractures · **most common orthopaedic cause of pediatric hospitalisation** · **bimodal: toddlers 2–4 yr + adolescents** · **M 2.6:1**
Mechanism by age	0–2 yr: **NAT** · <10 yr: **falls** · >10 yr: **MVA** (cortex thickens with age)
Associated	OI · neuromuscular osteopenia · tumours

Deforming forces

Proximal fragment	**iliopsoas** (flexion) + **abductors** (gluteus medius/minimus → abduction) + **short external rotators** (piriformis → ER) ✗ Wrong options: adductors · hamstrings · rectus (insert distally)
Distal fragment	**adductors** → shortening + **varus**
Typical malunion	**varus + flexion (procurvatum)** of distal fragment

Stability

Length-stable	transverse, short oblique
Length-unstable	**long spiral (length >2× diameter)**, comminuted
TO rule	**TENS contraindicated / not alone in length-unstable** fractures → 5–11 yr unstable: **submuscular bridge plate or ex-fix** · TENS nail **30–40% of canal each**, same size, don't bend at entry · rigid nail **trochanteric entry only, never piriformis**, > 11 yr / > 50 kg (40–50 kg TO) · before walking age → suspect NAI

3 Presentation

Exam	thigh pain, can't walk, deformity, shortening, swelling
Hemodynamic instability	**isolated pediatric femur rarely causes shock → re-evaluate for other injuries** (fluids if Hb >7, transfuse if <7) ✗ Wrong options: pressors · transfusion first if Hb >7 · fasciotomy · emergent fixation

4 Diagnosis & Classification

Key radiology findings

Views	**AP + lateral femur** + ipsilateral **hip and knee**

Non-accidental trauma

Suspect **<3 yr** (AAOS: evaluate <36 months), **especially non-walkers**

Patterns most predictive **transverse** femoral shaft · **metaphyseal corner fracture (pathognomonic)** · **posterior rib fractures**, scapula, sternum, spinous process · injury inconsistent with history (9-month-old "running")
✗ Wrong options: spiral, long oblique, toddler's

Infant with metaphyseal corner fracture → admit + social services
✗ Wrong options: Pavlik and discharge · spondylometaphyseal dysplasia/syphilis work-up first · infection work-up

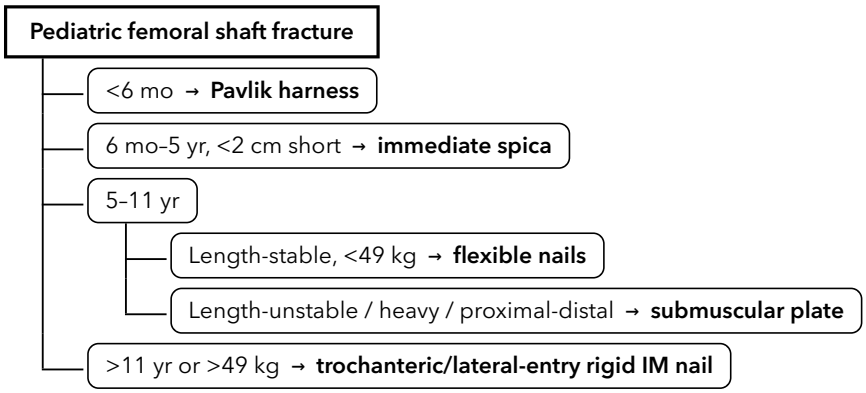
Distractors lytic lesion = pathologic · **blue sclera = OI** · low vitamin D / metaphyseal fraying = **rickets**

5 Management

Treatment selection by age & weight

Age	Pattern / factor	Treatment
<6 months	Any	**Pavlik harness** (or spica)
6 mo – 5 yr	Stable, **shortening <2 cm**	**Immediate spica**
6 mo – 5 yr	Unstable, **>2–3 cm short**, polytrauma, open	Traction → delayed spica · ex-fix
5–11 yr	**Length-stable + <49 kg (100 lb)**	**Flexible (titanium elastic) nails**
5–11 yr	**Length-unstable** (comminuted, long spiral), very proximal/distal, any weight	**Submuscular bridge plate** · stainless Ender nails · ex-fix
>11 yr / >49 kg / near maturity	Diaphyseal	**Antegrade rigid nail – trochanteric or lateral entry** (✗ Wrong options: piriformis)
>11 yr	Proximal/distal, severe comminution	Submuscular plate
Polytrauma, open, vascular repair, damage control	–	**External fixation**

Age ranges vary spica 6 mo–5 yr (Orthobullets) · 6 mo–6 yr / 1–6 yr (OB) · Pavlik <6 mo (Orthobullets) / <1 yr (OB) · trochanteric rigid nail from **~9 yr** (trochanteric apophysis no longer key; earlier → **coxa valga**) (OB)



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Operative criteria

Device	Key criterion
Flexible nails	5–11 yr · **<49 kg / 100–108 lb** · length-stable · open or closed OK · not comminuted
Contraindication to flexible nails	**Weight** (e.g. 146 lb) – not age, height, multiple fractures, open fracture
Submuscular plate	Length-unstable · >49 kg · proximal/distal · comminuted (incl. **open comminuted in 4-yr**)
Rigid IM nail	>11 yr / near maturity · >49 kg · unstable · trochanteric/lateral entry
Spica	Shortening **<2 cm** (loss of reduction doubles per cm: 12% → 24% → 50%) · not polytrauma, open, >2–3 cm
Ex-fix	Polytrauma (damage control), open, vascular injury, segmental

Ipsilateral femur + tibia (floating knee) in 7-yr → flexible nails both
✗ Wrong options: spica · reamed nails (physeal/AVN risk)

Infant fractures - Pavlik harness

<6 months → Pavlik (no anaesthesia, easy diapering, fewer skin problems than spica) · not in **neuromuscular spasticity** or **polytrauma** · 9-month-old → spica
✗ Wrong options: flexible nails, ex-fix, growing rods, traction, plate

Pavlik risk — **femoral nerve compression** with excessive flexion + swollen thigh (↓ quads)

Spica technique

Position — hips **60–90° flexion**, ~**30° abduction**, **external rotation** to correct rotation · mould into **recurvatum + valgus** · single-leg or 1½ spica (special car seat sometimes)

Spica tips (TO) — traction during application · **valgus moulding** (prevents varus collapse) · reinforce the **V-junction** · hip and knee flexion equal · **connecting bar** between limbs

Pavlik for femur (TO) — < 6 mo · **more proximal fracture → more hip flexion** · usually ≤ 90° · ~**2 wk** · watch femoral nerve · confirm union on X-ray



Spica cast

Acceptable — **<10° coronal, <20° sagittal, ≤2 cm shortening, ≤10° rotation**

Union — 4–8 wk · check at 7–10 d

Spica cast complications

Most common needing revision	**loss of reduction** (11%) ✗ Wrong options: delayed/nonunion · cosmetic · LLD
Compartment syndrome risk	**"90/90" technique: short-leg portion first then traction** · **knee flexion >90°** (e.g. 105°) · **firm popliteal mould** · high energy · **NOT** immediate ED casting under sedation, single-leg spica, casting within 24 h → apply **body + thigh first**, lower leg last

Traction

Distal femoral pin (proximal to physis) · **avoid proximal tibial pin → tibial tubercle damage → genu recurvatum** (also from SH V physeal injury) · 2–3 wk then spica · more complications than immediate spica

Flexible nailing technique & complications

Entry	retrograde, medial + lateral, **~2–2.5 cm proximal to distal physis**
Nail size	each = **0.4 × isthmus**, **two equal nails**, **80% canal fill** (e.g. 8-yr, 35 kg → **two 4.0 mm**) · **largest equal nails** → best stability ✗ Wrong options: asymmetric · impacted ends · open reduction routinely · hinged brace
Most common complication	**pain/irritation at insertion site near knee (up to 40%)** → keep protrusion **<25 mm** ✗ Wrong options: shortening · AVN · malunion · rotation
↑ Complications	**≥11 yr or >49–50 kg (>108 lb)** → ↑ nonunion, malunion, complications, reoperation · comminuted/proximal/distal → malunion · ~50% have 15° malalignment

Union 10–12 wk · remove ~1 yr



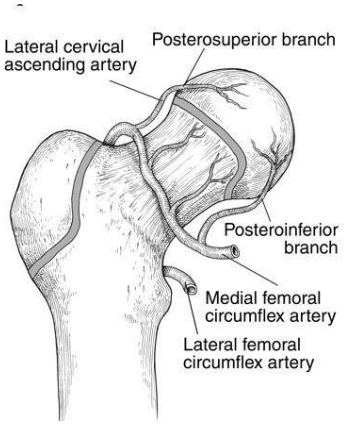
Submuscular bridge plate

Rigid nail entry point & femoral head AVN

Piriformis entry	**contraindicated with open physes** → injures **deep branch of MFCA** (lateral epiphyseal / superior retinacular vessels) → **femoral head AVN** (groin/hip pain months later) · AVN **1–2%** (Orthobullets) · **≥4%** (OB)
MFCA	main head supply **after age 4**
Safest	**lateral entry** (lateral GT) > GT tip · GT entry risks **GT apophyseal arrest → coxa valga**, neck narrowing, hip subluxation

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Rigid vs flexible — rigid: **higher AVN** (with open physes), less angular malunion, better in >11 yr / >49 kg · flexible: more knee pain, more malunion in older/ heavier



Medial femoral circumflex artery near piriformis fossa

Polytrauma, head injury & timing

Polytrauma + emergent laparotomy (13-yr, GCS 6) → external fixation (damage control)
✗ Wrong options: traction · nailing · plate

Early fixation with head injury — **no ↑ pulmonary or CNS complications**, **↓ length of stay**
✗ Wrong options: ↑ ARDS/pneumonia · worse mental function

6 Complications

Overgrowth — **most common in young** · **0.7–2 cm <10 yr** (avg 0.9 cm 2–10 yr) · within **2 yr** → accept shortening ≤2–3 cm

Shortening >2–3 cm — symptomatic → traction/fixation

Malunion — varus + flexion · remodels best in **sagittal plane** · **rotation doesn't remodel**

Refracture — **after ex-fix removal**, esp. **varus** (1.5–21%) · ex-fix: pin-tract infection (~50% of its complications), delayed/nonunion

Plate — refracture after hardware removal

Genu recurvatum — tibial tubercle arrest (traction pin / SH V)

X vs Y

	Flexible nails	Submuscular plate	Rigid trochanteric nail
Age	5–11	5–11+	>11 (≥9 OB)
Weight	<49 kg	Any	>49 kg
Pattern	**Length-stable**	**Length-unstable**, comminuted, proximal/distal	Any
Main complication	Knee pain	Refracture on removal	AVN (piriformis), coxa valga (GT)

How it's asked

"7-yr, 55 lb, isolated stable fracture?"	→ **closed reduction + flexible nails**
"11-yr, 146 lb, open femur — contraindication to elastic nails?"	→ **her weight**
"Best candidate for submuscular plate?"	→ **10-yr, 120 lb, long spiral comminuted midshaft**
"Trochanteric rigid nail appropriate for?"	→ **15-yr with butterfly fragment**
"Immediate spica for?"	→ **26-month-old, <2 cm shortening**
"Titanium elastic nails for?"	→ **10-yr, 35 kg, transverse** ✗ Wrong options: comminuted · 51 kg
"Submuscular plate for?"	→ **7-yr highly comminuted**
"10-yr TEN — appropriate case?"	→ **transverse open, 35 kg**
"7-yr floating knee?"	→ **flexible nails femur + tibia**
"2-yr, 1 cm short?"	→ **closed reduction + spica**
"10-yr, 115 lb, transverse?"	→ **solid nail via greater trochanter**
"10-yr, 75 lb?"	→ **retrograde elastic nails**
"1-yr, 25 lb, no NAT?"	→ **hip spica**
"Best for immediate spica?"	→ **3-yr long oblique, 2 cm short**
"7-yr, 38 kg, length-stable?"	→ **closed reduction + flexible nails**
"4-yr GSW comminuted open femur, artery intact?"	→ **submuscular plating**
"9-yr, 35 kg, transverse, 1 cm short?"	→ **flexible IM nails**
"Pavlik appropriate for?"	→ **4-month-old displaced spiral** ✗ Wrong options: spastic · polytrauma · 9-month-old
"3-month-old, minimally displaced?"	→ **Pavlik harness**
"5-month-old, father fell?"	→ **Pavlik harness** (after excluding abuse)
"2-month-old, NAT negative?"	→ **Pavlik harness**
"3-yr — most predictive of NAT?"	→ **transverse femur fracture**
"3-week-old, distal femoral metaphyseal corner fracture?"	→ **admit + social services**
"9-month-old 'running', spiral femur — strengthens CPS referral?"	→ **posterior rib fractures**
"13-yr polytrauma for laparotomy?"	→ **external fixation**
"Early fixation with head injury?"	→ **no ↑ pulmonary complications**
"10-yr with SDH + ↑ ICP — timing?"	→ **no difference in pulmonary/CNS complications**

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"8-yr isolated femur, hypotensive, Hb 8.5?"	→ re-evaluate for other injuries
"Piriformis nail, groin pain at 6 months?"	→ femoral head osteonecrosis
"Greatest MFCA risk?"	→ piriformis entry nail
"Damage to deep MFCA branch?"	→ piriformis entry
"13-yr hip pain after piriformis nail?"	→ femoral head ischaemic necrosis
"Rigid vs flexible in 11-yr – true?"	→ rigid has higher femoral head AVN
"Concern with piriformis start in immature patient?"	→ femoral head AVN
"Implant NOT to use in adolescents?"	→ piriformis-entry nail
"Tibial traction pin + flexible nails – later deformity?"	→ genu recurvatum
"Most common problem after elastic nails?"	→ insertion-site pain
"100 lb, 3 cm short – optimise TEN stability?"	→ largest diameter nails
"8-yr, 70 lb, two flexible rods – most common complication?"	→ knee irritation
"8-yr, 35 kg – start point + nail size?"	→ 2.5 cm above physis, two 4.0 mm nails
"115 lb vs 85 lb with flexible nails?"	→ heavier child: ↑ nonunion, malunion, complications, reoperation
"Spica in 4-yr – most common complication needing revision?"	→ loss of reduction
"Spica compartment syndrome – EXCEPT?"	→ immediate ED casting under sedation
"Spica technique raising compartment risk?"	→ short-leg portion first, then traction
"Subtrochanteric proximal fragment pulled by?"	→ iliopsoas, abductors, external rotators

FIBULAR DEFICIENCY (ANTEROMEDIAL BOWING)

– Pediatrics · 47 / 112 –

1 Basics

Definition shortening or absence of the fibula (formerly **fibular hemimelia**; postaxial hypoplasia) · involves the **whole limb**

Most common congenital long-bone deficiency · 7.4–20 per million (TO) · bilateral not uncommon

Genetics no known inheritance (mostly sporadic) · linked to **sonic hedgehog** · AD/AR/X-linked reported (TO)

Bilateral **9–52%** (TO)

Embryology (TO) limb bud **4–8 wk** · AER drives longitudinal growth · ossification centres in all long bones by **12 wk** · **vascular theory: absent anterior tibial artery**

Intramedial bowing (TO) fibula absent → loss of lateral tether → tibia drifts **anteromedially**

Three types of tibial bowing

Bowing	Association
Anterolateral	**NF1 / congenital pseudarthrosis** (✗ Wrong options: fibular deficiency (the EXCEPT))
Posteromedial	Physiologic, calcaneovalgus
Anteromedial	**Fibular deficiency** (most common cause)

Associated anomalies

Region	Association
Tibia	**Anteromedial bowing** + **skin dimple** over apex (half) · **fibular anlage** (fibrocartilaginous band, 60% of complete CFD) tethers tibia
Foot	**Equinovalgus** (TO, Orthobullets) · talipes equinovarus less often · **absent lateral rays** · **tarsal coalition (50%)** · **ball-and-socket ankle** → instability
Knee	**ACL/PCL deficiency** · **lateral femoral condyle hypoplasia** → **genu valgum** · hypoplastic tibial spines, shallow notch · patella alta/hypoplasia
Femur / hip	**Femoral shortening (50%)** · **PFFD**, **coxa vara** · **DDH**
LLD	Significant (femur ± tibia)

NOT associated **quadriceps hypotonia** · **congenital vertical talus** · SCFE · scoliosis · polydactyly · MCL insufficiency · cleft hand / insufficient extensor mechanism (→ **tibial** hemimelia)

ACL dysplasia correlates with **lateral femoral condyle hypoplasia** · lateral tibial spine dysplasia · absent fibula · absent lateral rays
✗ Wrong options: medial condyle · anterolateral bowing · PCL · ↓ slope

– – Active space – –

Foot conflict "pes planovalgus not associated; equinovarus is" (one OB explanation) vs **equinovalgus = classic** (Orthobullets, TO) — read the options

3 Presentation

Classic **short limb** · **skin dimple over mid-anterior tibia** · **equinovalgus foot** · missing lateral toes · genu valgum

Ankle sprains + ACL laxity + coalition → look for **LLD** (fibular deficiency)



Fibular deficiency — short limb, dimple, equinovalgus

4 Diagnosis & Classification

Key radiology findings

X-ray absent/short fibula · anteromedial bow · hypoplastic tibial spines · shallow notch · **ball-and-socket ankle** (from tarsal coalition)

Achterman & Kalamchi classification

Type	Fibula	Treatment
IA	Present but short — proximal epiphysis below tibial physis, distal above talus	Heel lift (LLD >2 cm)
IB	**Partial absence (30–50%)**, distal part can't support ankle	Contralateral epiphysiodesis
II	**Complete absence**	Multiple Ilizarov lengthenings **or Syme amputation**

Birch classification by LLD + foot function · **directs treatment**

Coventry & Johnson (TO) 1 short fibula · 2 absent fibula · 3 bilateral · **Stanitski (TO)** — fibular length + ankle (normal / valgus / ball-and-socket) + coalition + number of rays

Birch % LLD (TO)	Implication
< 6%	Minor reconstruction
6–10%	Limited lengthening
11–30%	Multiple lengthenings
> 30%	Consider amputation

Paley classification (TO)

Paley type (TO)	Ankle	Procedure
1	**Stable** ankle	Lengthening
2	**Dynamic valgus** ankle	**SHORDT** (SHortening Osteotomy Realignment Distal Tibia)
3	**Fixed equinovalgus**	**SUPERankle** – release, excise fibular anlage, tibial osteotomy, stabilise ankle
4	**Equinovarus**	Specific reconstruction

Birch = prognosis / counselling · Paley = surgical planning (TO)

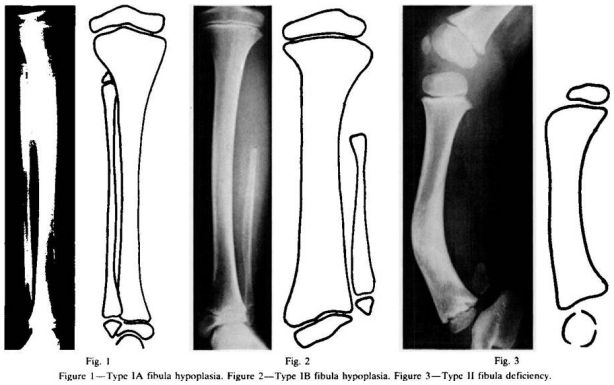


Fig. 1 Fig. 2 Fig. 3
Figure 1—Type 1A fibula hypoplasia. Figure 2—Type 1B fibula hypoplasia. Figure 3—Type II fibula deficiency.

Achterman-Kalamchi types

5 Management

Treatment based on foot/ankle function + LLD — **not** on amount of fibula

LLD natural history

Proportional growth inhibition — LLD **increases slowly, limb stays in proportion** (same % at maturity) → **Paley multiplier** predicts final LLD early
✗ Wrong options: stays 2 cm · equalises · 2 cm/yr · complete failure

Prediction (TO) — **Moseley straight-line graph** · **Green-Anderson** charts · **Paley multiplier** · e.g. 4-yr boy, 4 cm, multiplier ~2 → **~8 cm** at maturity

Lengthening (TO) — ex-fix · over nail · hexapod · magnetic IM nail · **assess ankle + knee stability first** · guided growth for genu valgum

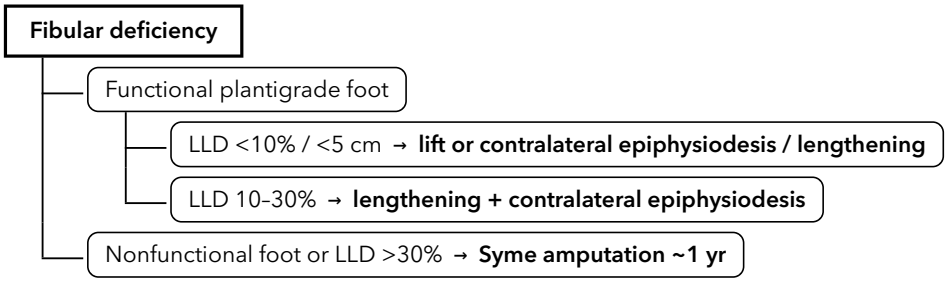
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Operative criteria

Situation	Treatment
Mild LLD, functional foot	Shoe lift / orthosis
LLD **<5 cm or <10%**, stable plantigrade foot	**Contralateral epiphysiodesis**
Plantigrade stable foot, LLD **<10%**	Lengthening (resect **fibular anlage**)
Moderate LLD 10–30%	**Lengthening + contralateral epiphysiodesis**
LLD >30% / **>5 cm at birth** / **nonfunctional unstable foot** (≥3 absent rays) / can't cope with lengthenings	**Syme amputation (~1 yr)** + prosthesis

Syme > Boyd — Boyd bulbous, only ~1 cm longer · satisfaction **88% amputation vs 55% lengthening**

Amputation (TO) — LLD **> 30%** · non-reconstructable foot (**≤ 2 rays**) · multiple high-risk surgeries · **Syme or Boyd** · **avoid if significant upper-limb deficiency**



Fibrodysplasia ossificans progressiva – palovarotene

FOP — **ACVR1 (R206H)** activating BMP type I receptor mutation → progressive heterotopic ossification

Palovarotene — oral selective **RAR-γ agonist** → inhibits BMP/SMAD 1/5/8 → ↓ HO (TO)
✗ Wrong options: FSH dystrophy · fibromuscular dysplasia · OI

How it's asked

"Congenital fibular deficiency associated with all EXCEPT?"	→ **anterolateral bowing** (TO)
"Fibular hemimelia — EXCEPT?"	→ **quadriceps hypotonia** (✓ absent lateral rays, ACL deficiency, ball-and-socket ankle, coxa vara)
"Associated EXCEPT?"	→ **congenital vertical talus**
"Image associated with fibular hemimelia?"	→ **ball-and-socket ankle**
"7-yr congenital limb difference — commonly associated with?"	→ **absent lateral rays**
"Concurrent conditions?"	→ **tarsal coalition + ball-and-socket ankle**
"ACL dysplasia correlates with?"	→ **lateral femoral condyle hypoplasia**

"Recurrent sprains, TC coalition, ACL laxity – also present?"	→ **leg length discrepancy**
"12-month-old, 2 cm tibial shortening – natural history?"	→ **increases slowly, staying proportional**
"Palovarotene treats?"	→ **fibrodysplasia ossificans progressiva** (TO)
"Classification for surgical planning in fibular hemimelia?"	→ **Paley** (Birch = prognosis) (TO)

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FLEXIBLE PES PLANOVALGUS (FLEXIBLE FLATFOOT)

– Pediatrics · 48 / 112 –

1 Basics

Definition	↓ medial arch + **hindfoot valgus** + forefoot abduction **on weight-bearing**, from **ligamentous laxity**
Prevalence	**44%** of school-age children (↓ with age) · 20–25% adults · normal toddler variant
Associated	generalised laxity · **gastrocsoleus contracture (25%)** · rotational problems · accessory navicular (controversial)
Lever arm dysfunction (TO)	planus + **hindfoot valgus** + **forefoot abduction** → foot CORA shifts medially, ↓ plantarflexion lever arm, ↑ energy at push-off
Developmental (TO)	infant fat pad + laxity · **arch develops by ~6 yr** · Staheli: "flatfeet are usual in infants, common in children, unusual in adults" · flatfoot in **90% of infants, 25% of adults** (TO)

Types

Type	Note
Hypermobile flexible	**Most common**, familial, bilateral
Flexible with **tight heel cord**	Stretch
Rigid flatfoot / tarsal coalition	Least common · hindfoot valgus **doesn't correct** on toe-standing

3 Presentation

Symptoms	usually **asymptomatic** · arch or pretibial pain
Exam	**flat only on standing**; arch **reconstitutes on toe-standing, hallux dorsiflexion (Jack test), or hanging** · hindfoot valgus → **varus on toe-rise** · **normal painless subtalar motion** · check **Silfverskiöld** for tight heel cord



Arch reconstitutes on toe standing

History clues (TO) morning stiffness → inflammatory · activity pain → coalition / accessory navicular · recurrent sprains · family: familial flatfoot, **hereditary spastic paraplegia** · progressive → neurological · **progressive unilateral flatfoot = red flag for tethered cord** (spread buttocks: sacral dimple/sinus)

Exam (TO) **toe-walking test** + **Jack's (great-toe dorsiflexion) test** restore the arch in flexible feet · subtalar inversion-eversion normal **20–30°** · **Silfverskiöld**: ↓ DF **> 10°** with knee extended = gastrocnemius tightness · **lock the subtalar joint** (avoid pseudo-DF)

4 Diagnosis & Classification

Key radiology findings

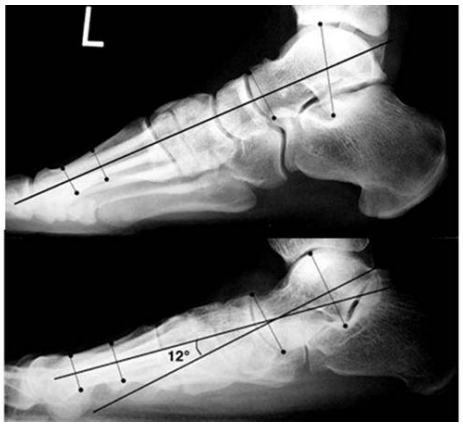
Paediatric foot angle (TO, weight-bearing)	View	Normal	Meaning
Talus–1st MT	AP	**0 to ~7–10°**	Negative = adductus · > 10° = abduction
Talocalcaneal (Kite)	AP	**25–40°** (20–40° < 5 yr)	> 40° hindfoot valgus · < 20–25° varus
Calcaneus–4th MT	AP	–	Midfoot abduction
Calcaneal pitch	Lateral	**10–15°**	High = cavus / calcaneus · low/negative = equinus, flatfoot, CVT
Meary (talus–1st MT)	Lateral	**0–4°**	Apex dorsal = cavus · apex plantar = planus

Principles (TO) standing AP + lateral ("one view is no view") · exam first · **most foot ossification centres by ~5 yr** · **1st MT physis at its BASE**, other MT physes at heads

Indications **painful** flatfoot or **rigid** flatfoot only
✗ Wrong options: painless flexible

Views **WB AP** (talar head coverage, TC angle) · **WB lateral** (Meary's) · **WB oblique** (coalition) · ± plantarflexion lateral (exclude CVT) · ± ankle AP/ lateral (ankle valgus – myelodysplasia)

Meary's angle **apex plantar** · **>4°** = arch collapse



Meary's angle – apex plantar

Differentials tarsal coalition · CVT · accessory navicular

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5 Management

Nonoperative management & observation

Asymptomatic	**observation + education** · arch develops with age ✗ Wrong options: arch supports · X-rays · PT "to help feet develop"
Tight heel cord, painless	**gastrocnemius stretching** ✗ Wrong options: spine MRI (that's cavus) · X-rays · night AFO · UCBL without pain
Orthotics	**do NOT change natural history** · use/stop by **comfort** (no harm, no benefit) · soft arch support for symptoms · **UCBL** for symptomatic advanced cases (rigid → poorly tolerated)
Short-foot exercises	intrinsics, medial arch (TO)
Painful flexible flatfoot (TO)	causes: **obesity**, **tight TA** → weight loss + TA stretching with subtalar locked · asymptomatic: reassurance, **no orthotics**, shoes don't change natural history
Age-based (TO)	**0–3 yr: supervised neglect** · **3–12 yr: orthotics + physiotherapy** · adolescents / adults: guided by clinical + radiological findings · only symptomatic feet need surgery (TO)
Orthotics in childhood	**do not change natural history; no orthotics if asymptomatic** (Orthobullets) · **3–12 yr: orthotics + physiotherapy** (TO) – read the options
EOTTS / sinus tarsi stent (TO)	extra-osseous talotarsal stabilisation · corrects **talar plantarflexion + inversion** · for **selected adolescent flexible flatfeet** (TO)

Operative criteria

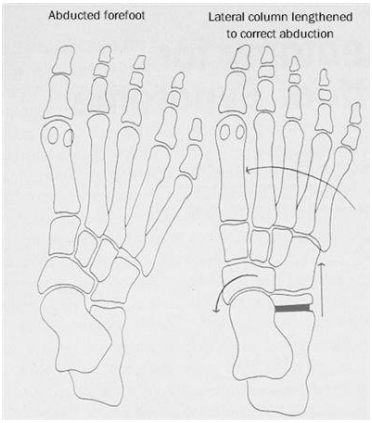
Indication	Procedure
Painful + tight heel cord refractory to stretching	**TAL / gastrocnemius recession**
Refractory pain (talar head, sinus tarsi) after extensive nonop	**Calcaneal (lateral column) lengthening – Evans** ± cuneiform osteotomy, peroneal lengthening
Hindfoot valgus	Medial **sliding calcaneal osteotomy**
Forefoot supination	**Plantar-based closing-wedge osteotomy of medial cuneiform**

Surgical - calcaneal lengthening osteotomy

Evans lateral calcaneal lengthening → flexible pes planovalgus
✗ Wrong options: clubfoot · cavus · hallux valgus · Köhler

12-yr, 2 yr pain, failed UCBL + orthoses, supple → calcaneal neck lengthening osteotomy
✗ Wrong options: lateral slide · PL→PB · 1st MT DF osteotomy · PTT to dorsum (all cavus procedures)

Isolated medial calcaneal slide without cuneiform osteotomy → persistent forefoot supination
✗ Wrong options: persistent heel valgus



Evans calcaneal lengthening

Mosca procedure (TO)	**Evans-modified calcaneal lengthening** with **tricortical iliac crest graft** · restores lateral column, corrects lever-arm dysfunction
Triple C osteotomy (TO)	**cuboid opening wedge + medial cuneiform closing wedge + calcaneal sliding** osteotomy

X vs Y

	Flexible flatfoot	Rigid flatfoot (coalition)	CVT
Toe-rise	Arch + heel varus **return**	**No correction**	Rigid
Subtalar motion	Normal	↓	–
Pain	Rare	Often	–
X-ray	Only if painful	Oblique / CT	Forced PF lateral

How it's asked

"5-yr painless flexible flatfeet, DF neutral knee extended / 15° flexed?"	**→ gastrocnemius stretching**
"2-yr flatfoot, arches reconstitute, normal ROM?"	**→ observation and education**
"3-yr, mother asks about custom inserts?"	**→ continue/discontinue by comfort; resolves with time**
"Evans osteotomy for?"	**→ flexible pes planovalgus**
"12-yr refractory painful flexible flatfoot?"	**→ calcaneal neck lengthening osteotomy**
"Calcaneal slide without medial cuneiform osteotomy?"	**→ persistent forefoot supination**

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FRIEDREICH'S ATAXIA

– Pediatrics · 49 / 112 –

1 Basics

Definition	inherited **spinocerebellar degeneration**
Incidence	**1 in 50,000** · onset **7–25 yr** (Orthobullets) / **7–15 yr** (OB) · earlier with more GAA repeats

Genes & lab markers

Inheritance	**autosomal recessive**
Gene	**GAA trinucleotide repeat** in **frataxin (FXN)**, **chromosome 9q13** · frataxin = mitochondrial protein (iron metabolism, oxidative stress)
Pathology	**dorsal root ganglia neuronopathy** + loss of peripheral sensory fibres + **posterior column** degeneration · not anterior horn α-motor neuron loss (= **SMA**)
EMG	motor + sensory defects, ↑ polyphasic potentials · ↓ NCV in upper limbs

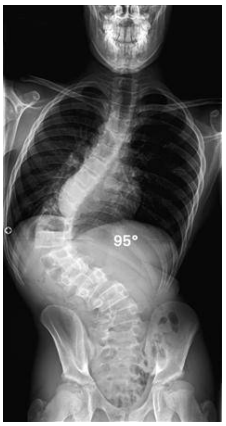
3 Presentation

Classic triad	**ataxia** (wide-based staggering gait – presenting symptom) · **areflexia** · **positive Babinski (extensor plantar)**
Other	weakness · nystagmus · ↓ proprioception · **cardiomyopathy**
Orthopaedic	**cavovarus foot** (very high arch, **rigid**, progressive, claw toes; strong plantar reflex + PTT pull into equinovarus) · **scoliosis**



Cavovarus foot – Coleman block assessment

Most likely condition	**foot deformity** ✗ Wrong options: hip dysplasia (CMT) · malignant hyperthermia (myelomeningocele) · tibia vara (Blount)



Scoliosis in Friedreich's ataxia

4 Diagnosis & Classification

Diagnosis **genetic testing** (frataxin)

Key radiology findings

Standing scoliosis series · AP + lateral foot if cavovarus

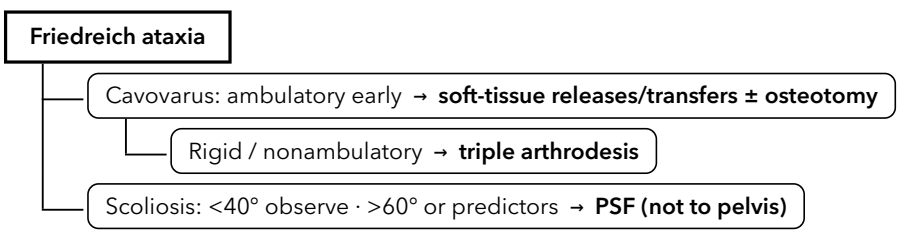
5 Management

Cardiomyopathy **cardiology review before surgery** · coenzyme Q (antioxidant) slows cardiac decline, not ataxia

Operative criteria

Problem	Treatment
Cavovarus, **early, ambulatory**	**Plantar release + tendon transfers ± MT/calcaneal osteotomy** (tenotomies + bracing early)
Cavovarus, **rigid / late / nonambulatory**	**Triple arthrodesis** (bracing/stretching ineffective)
Scoliosis **<40°**, no progression predictors	Observe
Scoliosis **40–60°**	Grey zone
Scoliosis **>60°** or rapid progression with predictors	**PSF + segmental instrumentation**, upper thoracic → lower lumbar (**usually not to pelvis**)

Scoliosis progression predictors **disease onset <10 yr · scoliosis onset <15 yr**



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6 Complications & prognosis

Wheelchair by ~**30** (Orthobullets) / 1st–2nd decade of disease (OB)

Death **cardiomyopathy** (± pneumonia, aspiration), 4th–5th decade (~50)

X vs Y

	Friedreich ataxia	CMT	SMA
Gene	**Frataxin (AR)**	PMP22 (AD)	Survival motor neuron protein
Lesion	DRG + posterior columns + cerebellum	Peripheral myelin	**Anterior horn cells**
Reflexes	Areflexia + **Babinski +**	Areflexia	Areflexia, fasciculations
Foot	Rigid cavovarus	Cavovarus (flexible → rigid)	–
Heart	**Cardiomyopathy**	–	–

How it's asked

"Friedreich's ataxia – all true EXCEPT?"	→ **loss of anterior horn α-motor neurons** (SMA)
"Patients most likely have?"	→ **foot deformity (cavovarus)**
"Scoliosis surgery threshold?"	→ **>60° (observe <40°)**
"Gene + inheritance?"	→ **frataxin GAA repeat, autosomal recessive**

GAUCHER DISEASE

– Pediatrics · 50 / 112 –

1 Basics

Most common lysosomal storage disease · ~1 in 40,000 · Ashkenazi Jewish

Enzyme deficiency

Genes & lab markers

Inheritance **autosomal recessive** · chromosome **1q21** (TO)

Enzyme **β-glucocerebrosidase** deficiency → **glucocerebroside (sphingolipid)** accumulates in macrophages of **liver, spleen, marrow**

Disease	Deficient enzyme
Gaucher	**β-glucocerebrosidase**
Niemann-Pick	Sphingomyelinase
Fabry	α-galactosidase
Tay-Sachs	Hexosaminidase A
Krabbe	Galactocerebrosidase

Labs **anaemia + thrombocytopenia** · **↑ plasma glucocerebrosides** (diagnostic)

Histology marrow: **giant (binucleate) Gaucher storage cells** filled with glucocerebroside

Physis **reserve (resting) zone** (OB, achondroplasia chapter)

Classification

Type	Features	Prognosis
1 (adult) – most common	Bruising, anaemia, fractures, no CNS	**Enzyme replacement** works
2 (infantile)	Brain + organs	**Lethal by ~3 yr**, untreatable
3 (juvenile)	Teen onset, thrombocytopenia, hepatomegaly, fractures, gradual CNS	Non-CNS responds to ERT

3 Presentation

Systemic fatigue (anaemia), bleeding (thrombocytopenia), infection, seizures / delay (CNS) · **hepatosplenomegaly** · murmur

Orthopaedic bone pain · **"bone crisis"** (acute infarct, like sickle crisis) · joint pain/ contracture · **pathologic fractures** · osteomyelitis

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Erlenmeyer flask — **distal femur / proximal tibia** in **~80%** · **most common radiological finding**, **asymptomatic abnormal remodelling**, needs no treatment



Bilateral distal femoral Erlenmeyer flask flaring

Osteopenia — **almost all** (worse after splenectomy)

Osteonecrosis — medullary (often asymptomatic) + corticocancellous: **proximal/distal femur, proximal tibia, proximal humerus** · **10× more likely after splenectomy**

Osteomyelitis — ↑ prevalence

FALSE statement (TO) — "**medullary osteonecrosis is the most common skeletal manifestation**" (✓ Erlenmeyer flask most common radiological; ✓ abnormal remodelling most common asymptomatic; ✓ post-splenectomy ↑ AVN)



Erlenmeyer flask deformity

5 Management

– – Active space – –

Treatment	Indication
Observation / supportive	Unaffected (normal bloods, minimal organomegaly, no MRI bone lesions)
Enzyme replacement (imiglucerase, velaglucerase, taliglucerase)	**All children + symptomatic** · not type 2
Substrate reduction (miglustat)	Milder, can't tolerate IV ERT
Bone marrow transplant	Early → may be curative

Fracture surgery

Pre-op — **optimise with ERT** · blood, clotting factors, platelets available (bleeding) · maintain **oxygenation** (avoid bone crisis) · ↑ infection risk

How it's asked

"Absent glucocerebrosidase causes?"	→ **Gaucher disease** ✗ Wrong options: Niemann-Pick · Fabry · Tay-Sachs · Krabbe
"Gaucher – all correct EXCEPT?"	→ **medullary osteonecrosis is the most common skeletal manifestation** (TO)
"Erlenmeyer flask distal femur?"	→ **Gaucher**
"Most common type?"	→ **type 1 (adult), treatable with ERT**

– – Active space – –

GENU VALGUM (KNOCKED KNEES)

– Pediatrics · 51 / 112 –

1 Basics

Physiologic tibiofemoral angle by age

Age	Alignment
Birth – <2 yr	**Varus** (10–15° at birth, TO)
~2 yr	**Neutral**
3–4 yr	**Peak valgus** – **15–20°** (Orthobullets) · **10° at age 3** (TO answer)
≥7 yr	Adult – valgus **≤12°**, intermalleolar distance **<8 cm** · adult **~7°** (TO) / 4–7° / 5–10° (OB)

Salenius curve (TO) varus birth–1 yr · neutral ~2 yr · valgus 2–6 yr (up to **15°**) · adult alignment by **7–8 yr** · another TO lecture: valgus **2–5 yr (15–20°)**, adult **5–6° valgus by 5–6 yr** – read the options

Clinical definitions (TO) genu valgum: **intermalleolar distance > 8 cm** · genu varum: **intercondylar (medial joint line) distance > 6 cm** · genu recurvatum: hyperextension **> 5°**

Femorotibial angle (TO) normal **5–6° valgus** · **> 6° valgus = valgum** · **< 4° valgus / varus = varum**

Presentation age **3–5 yr** (2–8) · rarely worsens after **7**

Pathologic site **distal femur** most common

Rule (TO) **valgum usually femoral, varum usually tibial** · clinically: valgus **disappearing on knee flexion → femoral** · varus persisting on flexion → tibial · supine knees flexed: **parallel tibiae → femoral**, divergent → tibial

Risk factors prior infection/trauma · **rickets** · obesity · skeletal dysplasia · lysosomal storage

Causes

Bilateral	Unilateral
Physiologic	**Physeal injury** (trauma, infection, vascular)
Renal osteodystrophy, hypophosphatemic rickets	**Proximal tibial metaphyseal fracture (Cozen)**
Morquio (MPS IV) · **SED (tarda)** · **Ellis–van Creveld** · pseudoachondroplasia	**Fibrous dysplasia**, **osteochondroma/MHE**, **enchondroma**
–	Fibular hemimelia

Apparent valgus obese thighs · femoral anteversion · external tibial torsion

Anatomy & mechanics

Normal LDFA / MPTA **85–90°** each

Normal joint angles (TO) **mLDFA 85–90°** · **mMPTA 85–90°** · mLPFA 85–95° · mLDTA 85–95° · LDFA / MPTA ~**87°** · **↓ mLDFA = femoral valgus** · ↓ MPTA = tibial varus

Mechanical axis hip centre → ankle centre should pass knee centre · valgus → **lateral** deviation, ↑ lateral compartment load



Mechanical axis, LDFA and MPTA

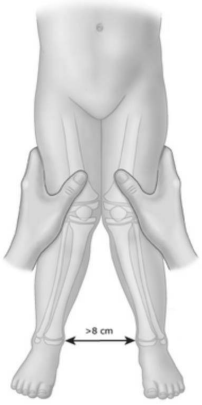
Hueter-Volkmann compression ↓, distraction ↑ growth · growth change mostly from **hypertrophic zone (75%)** vs proliferative (25%)

Patellar instability ↑ Q angle, hypoplastic lateral femoral condyle / shallow sulcus

3 Presentation

Complaints cosmesis (most common) · medial knee pain · circumduction gait · patellar instability

Exam knees touch, **intermalleolar distance >8 cm** · rotational profile · stigmata of rickets/dysplasia (Maffucci)



Intermalleolar distance

4 Diagnosis & Classification

Key radiology findings

Indications asymmetry · beyond physiologic range for age · short stature · trauma/ infection · LLD

View **standing long-leg AP, patellae forward** · lateral MAD · physeal narrowing/ closure · Park-Harris lines

– – Active space – –

Standing scanogram (TO) weight bearing, **patella forward**, foot position irrelevant · **CT scanogram not useful** for alignment

Mechanical axis zones (TO)

Zone (TO)	Axis passes through	Action
1	**Tibial spines**	Normal
2	Tibial condyle	Observe (treat if symptomatic: knee pain, patellar subluxation)
3	Within one knee-joint width from centre	**Pathological → intervene**
4	Beyond one knee-joint width	Severe → intervene

CT/MRI rarely (tumour, physeal bar)

Genes & lab markers

Rickets Ca, PO_4, **25-OH vitamin D**, PTH · **Morquio** – urinary **keratan sulfate** · syndromes – genetic tests

5 Management

Operative criteria

Situation	Treatment
<7 yr, angle **<15°**	**Observe** (bracing ineffective long term)
>15–20° valgus at 7–10 yr · TF angle >15° · **IMD >10 cm after age 10** · rapid progression after 7	**Medial hemiepiphysiodesis**
Age **>10**, mechanical axis in **lateral quadrant** (outer 25%) of plateau	Medial hemiepiphysiodesis
Skeletally mature / insufficient growth / physeal bar	**Osteotomy** at site of deformity

Hemiepiphysiodesis temporary (**8-plate**, staples, Metaizeau screw – most common) or permanent (Phemister) · extraperiosteal, mid-sagittal · one 8-plate or 2 staples per physis · correction ~**7°/yr femur, 5°/yr tibia** · >95% idiopathic / 80% pathologic correct · remove when axis at/just medial of centre

Indications for correction (TO) **FTA > 15°** · **IMD > 10–12 cm** · mechanical axis in **zone 3** · zone 2 with symptoms · **mechanical axis = most reliable indication**

Growth remaining (TO)	Treatment
1–2 yr	**Permanent** hemiepiphysiodesis
> 2 yr	**Reversible** hemiepiphysiodesis (staples / 8-plate)
Physes fused	Corrective osteotomy

Growth to (TO) boys **16 yr**, girls **14 yr** · overcorrect slightly (**rebound**) · **1 month of growth ≈ 1° correction** (TO rule of thumb)

Techniques (TO)	permanent: **Phemister** (physeal destruction) · temporary: staples · **PETS** (percutaneous epiphysiodesis using transphyseal screws) · **8-plate** — small incision, reversible, early mobilisation, faster predictable correction
Remaining growth (TO)	**Sauvegrain** elbow score · open olecranon + phalangeal physes = growth left

8-yr, IMD 12 cm → bilateral medial proximal tibial tension-band plates
✗ Wrong options: 11-yr Drennan 18° (that's Blount → lateral plate) · 4-yr 20° (physiologic) · 7-yr 9° (normal) · 16-yr (near mature → osteotomy)

12-yr bilateral valgus, open physes → temporary hemiepiphysiodesis of bilateral medial distal femurs
✗ Wrong options: HKAFO · osteotomy · unilateral · lateral (worsens)

Treatment - hemiepiphysiodesis vs osteotomy

Mature 18-yr, mLDFA 73° (femoral valgus), normal MPTA → medial closing-wedge distal femoral (varus) osteotomy
✗ Wrong options: tibial osteotomy → oblique joint line

	Lateral opening-wedge DFO	Medial closing-wedge DFO
Pros	Adjustable correction	**Stable**, shorter immobilisation, **no peroneal stretch**
Cons	Graft, less stable, **peroneal nerve stretch**, nonunion, lengthens	Technically demanding, shortens

DFO options for valgum (TO)	**closed wedge preferred** (better union, less loss of correction) · open wedge (graft, collapse) · **dome** (no length loss, quadriceps fibrosis risk) · supracondylar **V** osteotomy · step-cut translational · **oblique wedgeless (Herzenberg)**
Planning (TO)	identify **CORA** · axis to neutral · avoid joint-line obliquity · correct both femur and tibia if both contribute · correction **> 25° → decompress common peroneal nerve**
Windswept deformity (TO)	valgus one knee + varus the other · children: **rickets** · adults: **RA**
Peroneal nerve decompression threshold	correction **> 25°** (TO lecture A) · **> 15–20°** (TO lecture B) — read the options · combined osteotomy + guided growth in severe cases (TO)
Third TO lecture alignment	physiological valgus **14–16° at 2–4 yr** → adult **6–7° valgus by 6 yr** · causes of valgum also: obesity, **MHE**, hypophosphataemic rickets

6 Complications

Rebound	**56%** (≥5° loss after plate removal) · risks: **femoral deformity, younger age, faster correction**, intentional overcorrection → slight overcorrection / operate nearer maturity
Overcorrection	lost to follow-up (12%) · **undercorrection** — little growth left
Peroneal nerve	opening-wedge / acute correction → **decompress** (lateral compartment fascia + anterior-lateral septum) or gradual frame
Nonunion	opening wedge, >20° · **physeal closure** <1% (extraperiosteal, remove within 2–3 yr)
Cozen	post-fracture valgus peaks **12–18 mo**, remodels in **2–4 yr → observe**

– – Active space – –

Cozen (TO)	usually self-corrects **within 18 months** (TO) · remodels in 2–4 yr (Orthobullets) — read the options

How it's asked

"18-yr, mLDFA 73°, MPTA 87°, TF angle 25°?"	→ **medial closing-wedge femoral osteotomy**
"12-yr skeletally immature bilateral severe valgus?"	→ **temporary hemiepiphysiodesis of bilateral medial distal femoral physes**
"Medial proximal tibial tension-band plates — best candidate?"	→ **8-yr with intermalleolar distance 12 cm**
"Normal tibiofemoral axis at age 3?"	→ **valgus of 10°** (TO)
"Most reliable indication for surgery in angular deformity?"	→ **mechanical axis (zone 3–4)** (TO)
"Growth remaining > 2 yr — modality?"	→ **reversible hemiepiphysiodesis (8-plate)**; 1–2 yr → permanent (TO)

HIP SEPTIC ARTHRITIS - PEDIATRIC

– Pediatrics · 52 / 112 –

1 Basics

Epidemiology

Incidence	4–5 / 100,000 children/yr · **50% <2 yr** · **M:F 2:1** · hip and knee each **35%** of septic joints
Risk factors	**prematurity** · **C-section** · NICU · umbilical/venous catheters, heel puncture (transient bacteremia)

Mechanism & pathophysiology

Routes	haematogenous (URI precedes ~80%) · direct inoculation · **contiguous metaphyseal osteomyelitis** – **intra-articular metaphysis: hip, shoulder, elbow, ankle (NOT knee)** · neonates: **transphyseal vessels**
Cartilage destruction	**matrix metalloproteinases (MMPs)** from bacteria, synovium, PMNs, cartilage → damage within **8 hours** (6 h in animal studies) · TLR → NF-κB → MMP production ✗ Wrong options: MMPs anabolic · TLRs inhibit · activated by chelators (chelators inactivate; stromelysin is an MMP)
↑ Intra-articular pressure	→ femoral head **osteonecrosis**
Transphyseal spread (TO)	**< 18 months** → metaphyseal infection enters joint

Causative organisms by age (S aureus, GBS, Kingella, gonococcus)

Age / setting	Organism	Gram stain
Neonate, community (vaginal birth)	**Group B streptococcus**	GP cocci in chains
Neonate, nosocomial	**S. aureus**	–
<4 yr, subacute, mild labs, negative cultures	**Kingella kingae** – grows in **blood-culture bottles** (BACTEC); **PCR (rtxA)** · resistant to **vancomycin + clindamycin**	**GN coccobacillus** (facultative anaerobe)
>2 yr / overall	**S. aureus** – **most common** (catalase + coagulase +) · also causes **impetigo**	**GP cocci in clusters**
Adolescent, sexually active	**N. gonorrhoeae** – **migratory polyarthralgia**, maculopapular rash, tenosynovitis · F:M 3:1 · cultures negative ~50%	**GN intracellular diplococci**
Post-varicella	**Group A β-haemolytic strep**	–
Unvaccinated	H. influenzae (↓ with vaccine) – LP if meningitis signs	GN coccobacillus
MRSA osteomyelitis + DVT	**Panton-Valentine leukocidin** · CRP very high	–

– – Active space – –

Kingella "most common in children of all ages" (Orthobullets) · **S. aureus** most common overall (OB) · no organism in **up to 55%**

3 Presentation

Symptoms **acute** pain (faster than osteomyelitis) · fever, **toxic** · limp / **refusal to bear weight** · antibiotics may mask

Exam hip held in **flexion, abduction, external rotation** (max capsular volume) · pain with passive motion / log-roll · pseudoparalysis · check adjacent joints + spine

4 Diagnosis & Classification

Diagnostic criteria – Kocher

Original Kocher criterion	Threshold
Fever	**>38.5 °C (101.3 °F)**
Non-weight-bearing	✓
ESR	**>40 mm/h**
Serum WBC	**>12,000/mm³**
Modified (Caird) – **CRP**	**>2.0 mg/dL (20 mg/L)**

Criteria met	Probability of septic hip
0	**<0.2%**
1	**3%**
2	**40%**
3	**93%**
4	**99.6%**

Most predictive order **fever > CRP > ESR > refusal to bear weight > WBC** (CRP 2nd-best after fever)

Caird (TO) **fever = strongest predictor** · original Kocher (4 criteria) **does not include CRP**

CRP >2 mg/dL + refusal to bear weight → 74% · **CRP <2 mg/dL (<20 mg/L)** → favours **transient synovitis** · both absent <1%

Kocher criteria apply to the HIP only **not the knee** → knee: **aspirate**

Labs WBC ↑ in only 30–60% (neonates may be leukopenic) · ESR may be normal early · **CRP rises in 6–8 h** · **blood cultures** if febrile (often positive when joint culture negative)

Blood culture (TO) positive in **~60%** · **PCR ↑ yield**

Synovial fluid

Diagnosis	Synovial WBC	PMN
Traumatic effusion	<5,000	–
Transient (toxic) synovitis	5,000–15,000	<25%
Rheumatic fever	10,000–15,000	~50%
JRA	15,000–80,000	~75% (gradual onset, keeps walking)
Septic arthritis	**>50,000**	**>75%**

PMN % more sensitive than total WBC (85% PMN → 88% sensitivity) · glucose ≥50 mg/dL below serum · Gram stain + culture (± synovial tissue)

Key radiology findings

X-ray (AP + frog) often normal early · **joint-space widening** (infant: **lateral displacement of proximal femur**) · subluxation/dislocation · bone lesions

Ultrasound **effusion** (can't tell septic vs sterile) · guides aspiration · **neonate: scan both hips** if any septic joint

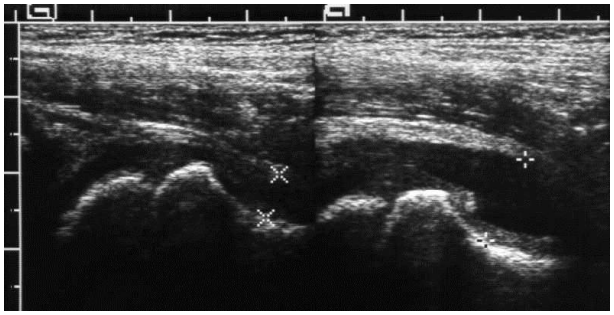
MRI effusion + adjacent osteomyelitis · don't let it delay surgery

MRI (TO) gold standard for marrow oedema / early osteomyelitis · **mandatory in delayed presentation** · detects **obturator internus abscess** (mimics septic hip → drain via **subadductor approach**)

X-ray extras (TO) **lateral subluxation** of head · metaphyseal erosion late



Joint-space widening — lateral displacement of proximal femur

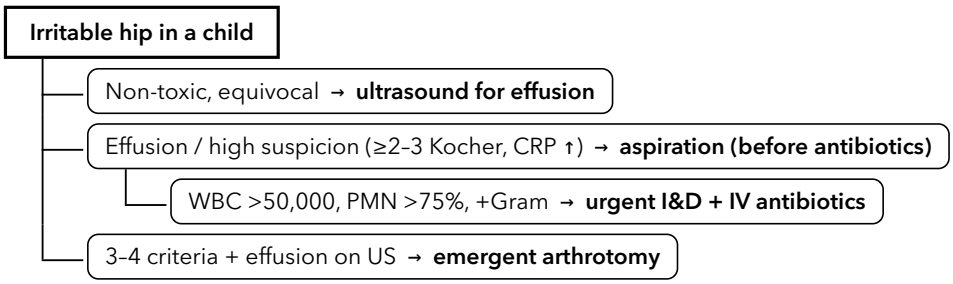


Hip effusion on ultrasound (right) vs normal (left)

- - Active space - -

5 Management

Workup & next step (US, aspiration, arthrotomy)



6-week-old (preterm, C-section), afebrile, widened joint space, painful motion → aspiration
✗ Wrong options: MRI (delays) · CT · observe · Pavlik

Non-toxic 3-yr, normal X-ray, fever, ESR 50, CRP 5 → ultrasound next
✗ Wrong options: NSAID trial · CT · IV antibiotics first · discharge

Fever + refusal + ↑ CRP/ESR → hip aspiration
✗ Wrong options: IV abx · oral abx · ibuprofen · MRI · Lyme Western blot · rheumatology

3-yr, 3/4 criteria + US effusion → emergent arthrotomy + I&D
✗ Wrong options: observe · repeat US · MRI · antibiotics before cultures in a non-septic child

Lyme-endemic hiker, fever 38.2, WBC 14, CRP 5.5 → aspirate (septic more likely than Lyme)

Operative criteria

Urgent I&D = standard of care	**surgical emergency** (removes chondrolytic enzymes, ↓ pressure / epiphyseal ischaemia) · err toward drainage
Arthrotomy indications (TO)	**any septic hip** · any delayed presentation · any obturator internus abscess · drill metaphysis if MRI unavailable · arthroscopic drainage without traction = recent advance
Antibiotics alone	**adolescent gonococcal** arthritis (high-dose penicillin ± aspiration)
Technique	anterolateral or **anterior (Smith-Petersen)** arthrotomy · remove pus, irrigate · ± **1 × 1 cm capsular window** · drain · other joints open/ arthroscopic
Antibiotics	aspirate first, then empiric IV · **2–7 d culture-specific IV → 2–3 wk oral** · stop when CRP/ESR + clinical picture normal

Age	Empiric
<3 mo	1st-gen cephalosporin (GBS, S. aureus, gram-neg)
3 mo–5 yr	2nd/3rd-gen cephalosporin (Kingella, S. aureus, pneumococcus, GAS, H. flu)
5–12 yr	1st-gen cephalosporin
12–18 yr	2nd/3rd-gen cephalosporin (gonococcus)

6 Complications

Sequelae, prognosis & cartilage destruction

Poor prognosis	**age <6 months** · **associated osteomyelitis** · **delay >4 days** (strongest) · **hip** (vs knee)

Sequelae — **complete femoral head/neck destruction** (neglected neonatal sepsis) · **osteonecrosis**, articular destruction · physeal arrest → **LLD**, angular deformity · chronic osteomyelitis · persistent limp · contracture · follow **1–2 yr**

Mechanisms (TO) — toxins + inflammation (chondrolysis) · **tamponade → head ischaemia** · triradiate closure → **acetabular dysplasia** · physeal closure → coxa breva / vara / valga, LLD, **GT overgrowth** · neck fracture → **pseudarthrosis**

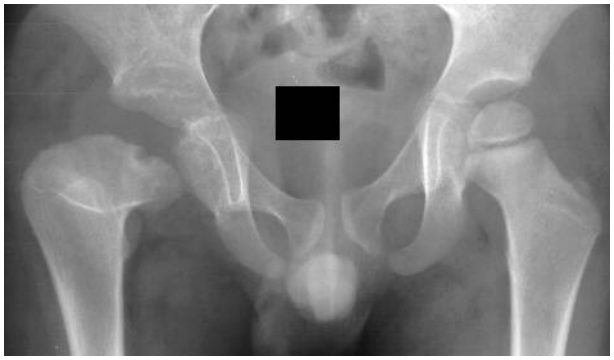
Physis damaged (TO)	Deformity
Inferior	**Coxa vara**
Superior	**Coxa valga**
Anterior	Femoral **anteversion**
Posterior	**Retroversion**

NOT a typical sequel — **chronic posterior instability/dislocation** (esp. after anterior approach)

Salvage — varus/valgus proximal femoral osteotomy · trochanteric osteotomy · modified Albee arthroplasty

Pelvic support osteotomy (TO) — subtrochanteric valgus-extension angulation osteotomy for post-septic hip, neglected DDH, Girdlestone hip, neck pseudarthrosis · ↑ stability, ↓ Trendelenburg · **level = contact point with pelvis on maximum-adduction X-ray** · valgus = max adduction + adduction contracture + **15–30° overcorrection** · **excess valgus → knee valgus + pelvic obliquity** · blade / bent plate (remove ~1 yr) · usually **after skeletal maturity** (TO)

Trochanteric procedures (TO) — relative neck lengthening for trochanteric overgrowth (↑ abductor lever) · **true neck lengthening** (shaft lateralisation + graft) for post-septic sequelae



Destroyed femoral head after untreated septic hip

– – Active space – –

Choi classification & reconstruction of sequelae (TO)

Choi (TO) — most widely used	Features
IA / IB	Near-normal / mild coxa magna
IIA	**Coxa breva** (symmetric physeal closure)
IIB	**Coxa vara / valga** (asymmetric closure) + acetabular dysplasia
IIIA	Severe angular deformity, **excess ante-/retroversion**
IIIB	**Femoral neck pseudarthrosis** (complete epiphyseal slip)
IVA	Loss of head, small remnant articulating
IVB	**Loss of head and neck**

Other classifications (TO) **Hunka I–V** (minimal → total destruction + dislocation) · **Forlin–Milani** (located vs dislocated, head involvement) · **Johari** (Indian, most comprehensive: 1 loss of CFE · 2 unstable without CFE · 3 unstable with CFE · 4 AVN / physeal deformity · 5 pseudarthrosis)

Principles (TO) head coverage + congruity · LLD · abductors · alignment · stability

Problem (TO)	Options
Coverage	Abduction cast / brace · open reduction · femoral or pelvic osteotomy (Salter, Dega, Pemberton, shelf) · osteochondroplasty
LLD	Shoe raise · contralateral epiphysiodesis · ipsilateral ex-fix lengthening
Abductor insufficiency	**GT epiphysiodesis < 7 yr** · **GT advancement / distalisation > 7 yr** · neck lengthening
Absent head (Choi IV)	**Albee** (denuded GT into acetabulum) · **L'Episcopo** (medial half of split remnant into acetabulum) · **modified Harmon** (open-wedge + graft, cartilage cap preserved) · trochanteric arthroplasty
Salvage	Arthrodesis · **pelvic support osteotomy** · **Ilizarov hip reconstruction**

Neck lengthening (TO) **Morscher** needs **congruent head–acetabulum** (osteotomy, shaft lateralisation, GT distal transfer) · **Wagner** also changes neck–shaft angle

Ilizarov hip reconstruction (TO) **subtrochanteric valgus (pelvic support) + distal femoral varus osteotomy + gradual lengthening** → corrects **LLD, instability, abductor lurch** · indications: DDH / traumatic / septic sequelae, absent head or neck, dysplasias · contraindicated **< 12 yr**, young adults fit for THR, paralytic hips

X vs Y

	Septic hip	Transient synovitis
Fever	**>38.5 °C**	Low / none
Weight bearing	Refuses	Often limps
ESR / WBC	**>40 / >12,000**	Normal
CRP	**>2 mg/dL**	**<2 mg/dL**
Synovial WBC	**>50,000, >75% PMN**	5,000–15,000
Treatment	**Urgent I&D**	NSAIDs, observe

How it's asked

"ESR >40, fever, refusal + which other Kocher finding?"	→ **serum WBC >12,000**
"3-yr, febrile, refuses to walk, WBC 8,000, ESR 20, CRP 1.9?"	→ **arthrocentesis best confirms** (2/4 = 40%) ✗ Wrong options: CRP is a Kocher criterion
"8-month-old FABER hip — 2nd best predictor after fever?"	→ **elevated CRP**
"Fever + refusal, WBC 11, ESR 18 — probability?"	→ **26–50%** (2/4 = 40%)
"Afebrile 37.8, WBC 13, ESR 19, refuses — criteria / risk / organism?"	→ **2/4, 40%, S. aureus**
"Favours transient synovitis?"	→ **CRP 15 mg/L** (<20) ✗ Wrong options: ESR 43 · 38.7 °C · WBC 12,500 · leptin (SCFE)
"Lab supporting septic hip?"	→ **CRP 22 mg/L** ✗ Wrong options: WBC 10,000 · ESR 38 · 38.2 °C · synovial 40,000
"6-yr knee effusion, 1/4 Kocher?"	→ **aspirate the knee — Kocher not valid for knees**
"6-week-old preterm, widened joint space?"	→ **aspiration**
"18-month-old, 60,000 WBC, 95% PMN?"	→ **septic arthritis**
"3-yr, WBC 15, ESR 120, CRP 5, effusion?"	→ **emergent arthrotomy + I&D**
"Lyme-area hiker, fever, ↑ WBC/CRP, effusion?"	→ **hip aspiration**
"3-yr, 39 °C, WBC 19,000, ESR 45, effusion?"	→ **hip aspiration**
"5-yr, fever, ↓ hip ROM, normal WBC, ↑ CRP/ESR?"	→ **aspiration of right hip**
"Non-toxic 3-yr, fever, ESR 50, CRP 5, normal X-ray?"	→ **ultrasound left hip**
"Gram stain of main organism in pediatric osteomyelitis/ septic arthritis?"	→ **gram-positive cocci (clusters)**
"8-day-old after vaginal birth?"	→ **group B streptococcus**
"No growth on blood agar, grew in blood-culture medium after 1 wk?"	→ **Kingella kingae**
"Osteomyelitis + DVT, positive blood culture?"	→ **MRSA**
"3-yr septic hip — most common organism's Gram stain?"	→ **gram-positive cocci in clusters**
"6-yr septic hip organism?"	→ **catalase + coagulase positive GP cocci in clusters**
"17-yr with macular rash?"	→ **gram-negative diplococci**
"19-month-old, mild labs, cultures negative at 14 days?"	→ **gram-negative facultative anaerobic coccobacillus (Kingella)**
"18-month-old, mild labs, effusion — micrograph?"	→ **Kingella kingae**

– – Active space – –

"16-yr gonococcal hip – true?"	→ **usually polyarthralgia of multiple joints**
"2-yr knee, PCR positive, culture negative?"	→ **gram-negative coccobacilli (Kingella)**
"Most common organism is also associated with?"	→ **impetigo**
"7-yr septic hip micrograph?"	→ **staphylococcus (clusters)**
"3-yr limp, prior neonatal sepsis, destroyed femoral head?"	→ **residual untreated septic hip**
"Strongest poor-prognosis predictor?"	→ **delay >4 days**
"2-yr limp, neonatal knee infection, absent femoral head?"	→ **untreated neonatal hip infection**
"MMPs – true?"	→ **mediate cartilage destruction in septic arthritis**
"LEAST likely complication of delayed treatment?"	→ **chronic posterior hip instability**
"Strongest single predictor of septic hip (Caird)?"	→ **fever** (TO)
"Choi IIIB?"	→ **femoral neck pseudarthrosis** (TO)
"Procedure correcting LLD, instability and abductor lurch together?"	→ **Ilizarov hip reconstruction** (TO)

HUMERAL SHAFT FRACTURE - PEDIATRIC

– Pediatrics · 53 / 112 –

1 Basics

Incidence	**<10%** of pediatric humerus fractures
Mechanism	neonates: birth (hyperextension/rotation) · adolescents: direct trauma · low energy → **pathologic** lesion · **<3 yr + spiral → abuse** · any fracture in **non-ambulator → NAT** work-up
Growth	**proximal 80% : distal 20%** of humeral length (distal ~3 mm/yr)

3 Presentation

Exam	swelling, deformity · neonate **pseudoparalysis** (reflexes intact) · **radial nerve palsy** (no wrist/finger extension)

4 Diagnosis & Classification

Key radiology findings

Full-length AP + lateral humerus + shoulder and elbow views · transverse / oblique · look for pathologic lesion



Pediatric humeral shaft fracture

– – Active space – –

5 Management

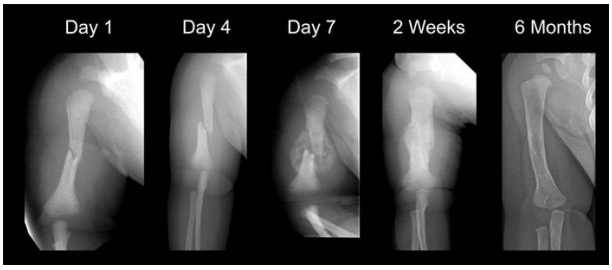
Treatment & prognosis

Almost all → immobilisation (huge remodelling) · neonate: swathe / pin sleeve to shirt · young child: **sling and swathe**, collar-and-cuff · coaptation splint, hanging cast · **Sarmiento brace** (older) · ROM at 2–3 wk

Acceptable alignment	Value
Younger child	**<35–45°** angulation (Orthobullets) · up to 40° (OB)
Older child	**<20°** varus/valgus · **<20°** procurvatum · **<15°** rotation · **<2 cm** shortening

Neonatal birth fracture → callus by 2 weeks, complete remodelling within 6 months
✗ Wrong options: malunion · nonunion · radial nerve risk

8-month-old midshaft fracture, angulated → splint + sling for comfort · consider NAT
✗ Wrong options: ORIF · CRPP · hanging cast (child doesn't sit/stand)



Remodelling of a pediatric humeral shaft fracture

Operative criteria

Indication for fixation
Open fracture
Polytrauma (needs arms for mobilisation)
Ipsilateral forearm fracture – "floating elbow" (✓ the indication)
Associated shoulder injury
Unacceptable / unstable alignment

Not indications mechanism, age, open physes, **radial nerve palsy**

Fixation **flexible IM nails** · K-wires · plate (anterior/anterolateral/posterior) 3.5/4.5 mm

6 Complications

Radial nerve palsy **<5%**, middle + distal third · **neurapraxia → observe** · EMG ± explore if no return by **3–4 months**
✗ Wrong options: immediate exploration

Malunion rarely functional (shoulder ROM) · 20–30° fine · LLD common but harmless · delayed union rare (US stimulator)

How it's asked

"3-day-old birth humeral fracture — prognosis?"	→ **callus by 2 weeks, remodelled by 6 months**
"6-yr distal-third fracture, can't dorsiflex wrist?"	→ **monitor radial nerve + sling and swathe**
"8-month-old midshaft humerus?"	→ **splint and sling for comfort**
"10-yr humerus fracture — best indication for fixation?"	→ **ipsilateral forearm fracture (floating elbow)**
"Distal humeral physis contribution?"	→ **20%**

– – Active space – –

INFANTILE BLOUNT'S DISEASE (TIBIA VARA)

– Pediatrics · 54 / 112 –

1 Basics

Epidemiology

Definition	**progressive pathologic genu varum centred at the proximal tibia** · age **2–5 yr** (Orthobullets) · 0–3 yr (OB) · infantile **<8 yr** vs adolescent >8 yr (TO) – read the options
Risk factors	**obesity** · **early walkers (<1 yr)** · **African American / Hispanic** · never in non-ambulatory children (TO)
Sex / side	**M > F** (Orthobullets) · OB lists female association · **bilateral in 50%**

Pathophysiology

Cause	**mechanical overload** of medial proximal tibial physis in a **genetically susceptible** child → **osteochondrosis** of medial physis + epiphysis → may progress to **physeal bar**
Deformity components	**varus** of proximal medial metaphysis + **procurvatum (apex anterior)** + **internal rotation** (TO) · **external rotation** is NOT a component (TO) · not recurvatum (OB) · one TO explanation says "genu recurvatum" – read the options
Associated	**internal tibial torsion** (✗ Wrong options: metatarsus adductus) · **compensatory distal femoral VALGUS** (✗ Wrong options: distal femoral varus (adolescent))

Physiologic alignment

Age	Orthobullets	TO
Birth–1 yr	**Varus** (bow legs, normal <2 yr)	Varus up to 10–15°
Neutral	**~14 months**	**18–24 months**
Peak valgus	**~3 yr**	**3–4 yr** (up to 15–20°)
Adult valgus	**~7 yr**	5–6 yr (5–6°)

Physiologic bowing	resolves by **36 months** in almost all (OB) · symmetric, femur + tibia · Brooks & Gross: varus to 18 mo → valgus peak 4 yr → adult 5–6 yr (OB)
Tibial torsion (TO)	5° internal at birth → neutral by 4–5 yr → **20–25° external by 8 yr**

3 Presentation

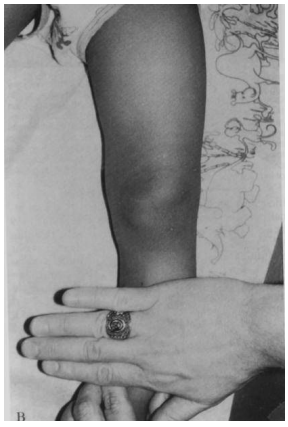
Exam	**genu varum + flexion + internal rotation** deformity · **lateral thrust on walking** (✓ abnormal finding) · internal tibial torsion · LLD · **no** tenderness, effusion or ↓ ROM

Siffert-Katz sign (TO)	**varus thrust during walking** · acute varus **below the knee** · obesity **> 95th percentile** · early walkers
Pathological varum (TO)	persists **> 2 yr** · unilateral · LLD · severe · obesity · causes: rickets, Blount, dysplasia, trauma/infection, hemimelia · **focal fibrocartilaginous dysplasia** – self-limiting, no surgery
Not abnormal	**internal thigh–foot angle 5°** in an infant (OB)

Cover-up test

Technique patella forward · examiner covers lower leg + foot · look at thigh–upper tibia axis (age 1–3 yr) (TO)

Result	Meaning
Positive (bowing in upper tibia)	X-ray · **high sensitivity**
Neutral (straight axis)	X-ray
Negative (slight valgus upper tibia)	Physiologic bowing · **high specificity + NPV**



Cover-up test

When to X-ray

Physiologic, <2 yr, symmetric → clinical observation (OB)
✗ Wrong options: X-rays · vitamin D · genetic screen

Triggers for workup	persists **>2 yr** (OB) · persists **>18 months** or asymmetric (Birch) · severe **>20°** · worsening · asymmetry · **varus thrust** · low height/weight percentile ✗ Wrong options: family history alone
Vitamin D workup	if **<20th percentile** on growth curves (OB)

4 Diagnosis & Classification

Key radiology findings

View	standing AP **with patellae facing forward** (ITT confounds)
Findings	varus focused at proximal tibia · sharp angular deformity · asymmetry · **medial + posterior sloping** of epiphysis · **metaphyseal beaking** ✗ Wrong options: physiologic (symmetric flaring of femur + tibia)
Pathognomonic	**medial metaphyseal fragmentation** → progressive tibia vara (TO)

– – Active space – –

X-ray signs (TO)	metaphyseal flaring · **fibular physis near knee level** · physeal bar · Drennan MDA
MRI (TO)	↑ thickness of medial proximal tibial chondroepiphysis · ↑ height + width of **medial meniscus** · abnormal medial femoral epiphysis
Neglected infantile Blount (14 yr)	**severe depression of medial tibial epiphysis** ✗ Wrong options: adolescent Blount (metaphyseal) · septic/JRA (both sides of joint) · hemophilia (widened intercondylar notch)

Metaphyseal–diaphyseal (Drennan) angle

Measurement line through metaphyseal beaks vs line **perpendicular to tibial long axis**

MDA	Meaning
<10°	**95% resolve** naturally
11–16°	**Close observation** (Orthobullets) · **>11°** warrants observation (TO) · 9–16° → brace if no correction after 24 months (OB)
>16°	**Abnormal** · **95% progress** · ✓ most reliably separates Blount from physiologic
>20°	Operative indication

MDA cut-offs (TO) **< 11° normal · 11–16° doubtful · > 16° Blount** · Levine & Drennan (TO lecture 2): **> 10° suspicious**, **> 16° high likelihood of progression**

Tibiofemoral angle femoral vs tibial long axes



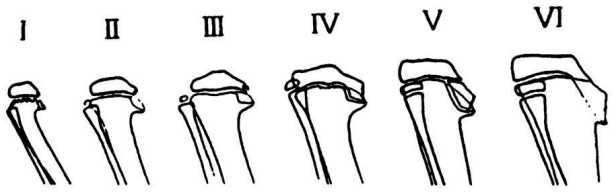
Metaphyseal–diaphyseal angle

Langenskiöld classification

Stage	Feature
I–IV	Increasing medial **metaphyseal beaking + sloping**
V–VI	**Epiphyseal–metaphyseal bony bridge** (physeal bar)

Use prognosis · stage II and IV can correct spontaneously in young children

Langenskiöld (TO)	Feature	Treatment (TO)
I	Medial metaphyseal **beak**	**Mermaid brace × 2 yr**
II	**Depression**	Mermaid brace × 2 yr
III	**Step** formation	Surgery (guided growth / osteotomy)
IV	**Ossification of step**	Surgery
V	**Separate medial fragment**	**Osteotomy + medial tibial plateau elevation**
VI	**Medial physeal bar**	Osteotomy + plateau elevation



Langenskiöld stages

CT for physeal bar (TO)

Order CT pre-op if age **>5 yr** · medial physeal slope **50–70°** · **Langenskiöld IV** · weight **>95th percentile** · black girls meeting the above

Differential (pathologic genu varum)

List persistent physiologic varus · **rickets** · OI · MED · SED · metaphyseal dysostosis (Schmid, Jansen) · focal fibrocartilaginous defect · TAR · physeal injury (radiation, infection, trauma)

5 Management

Infantile Blount

- <3 yr, Langenskiöld I–II → **KAFO** *(improvement within 1 yr; wear ~2 yr)*
- Brace failure / progression → **proximal tibia-fibula valgus osteotomy**
- ≥3–4 yr, stage III–VI or MDA >20° → **osteotomy**
  - physeal bar (V–VI) → **add bar resection / hemiepiphysiodesis**

Bracing (knee-ankle-foot orthosis, KAFO)

Indication **stage I–II in children <3 yr** (Orthobullets) · <3 yr **and unilateral** stage I–II (TO) · TO calls it "mermaid splints"

Duration ~**2 yr** for bony resolution · improvement should show **within 1 yr** · fails after **12 months** in 2–3 yr old → osteotomy (OB)

Poor results / failure risk **obesity (>90th percentile)** · **bilateral** · **varus thrust** · age **≥3 yr** at start · Langenskiöld **III** · progressive deformity · age >4 (TO)

Evidence **no strong consensus** on bracing efficacy (OB) · night KAFO poorly tolerated · (✗ Wrong options: AFO (doesn't address knee) · serial casting) ✗ Wrong options: "effective" · "ineffective" · needs PT

– – Active space – –

Proximal tibial osteotomy

Operative criteria

Indication for proximal tibia/fibula valgus osteotomy
Stage I–II in child **>3 yr**
Stage III, IV, V, VI
Age ≥4 yr (all stages)
Failed brace / progressive deformity
MDA >20°
Morbidly obese (OB)

Timing	**before age 4** → recurrence significantly ↓ · early osteotomy (**2–4 yr**) = best results, least recurrence (TO) · best results **<4 yr** (OB)
Technique	**below the tibial tubercle** · **acute correction** · distal fragment into **valgus + external rotation + lateral translation** · staged for IV–VI · epiphysiolysis in V–VI · medial plateau elevation if depressed · **prophylactic anterior compartment fasciotomy**
Osteotomy types (TO)	closing / opening wedge · **Rab's osteotomy** (biplanar oblique + derotation) · chevron · dome · adolescent: osteotomy, fixator realignment, lateral hemiepiphysiodesis
Focal fibrocartilaginous dysplasia (TO)	varus **< 30° observe** (resolves) · **40–50° → osteotomy**
Overcorrection	**10–15° valgus** (Orthobullets, TO) · **~10° valgus** (OB) · at least **5° valgus** (OB60) · ✓ medial physeal abnormality persists ✗ Wrong options: correction to neutral · undercorrection
Site	**tibia, not femur** ✗ Wrong options: distal femur osteotomy
Renal osteodystrophy	**postpone** until renal status stable (osteotomy short-lived otherwise)
Late-onset with distal femoral varus	**LDFA >92°** → distal femoral osteotomy or guided growth (OB)

Guided growth & Hueter-Volkmann

Principle	**↑ compression → slows longitudinal growth** · ↑ tension → stimulates growth
Lateral proximal tibial hemiepiphysiodesis	tension band plate / staples (extraperiosteal plates) · **prevents worsening, does NOT fully correct** (OB) · less morbid than osteotomy · not **medial** hemiepiphysiodesis (worsens varus)
Caution	young child, severe stage III → relies on injured medial physis → **severe shortening** → prefer osteotomy (OB) · lateral tibial epiphysiodesis after age **9** (TO)
Unilateral LLD	epiphysiodesis of uninvolved leg (TO)

Physeal bar

Bar resection	needs **≥4 yr growth remaining** · with osteotomy · interposition **fat or PMMA** · **hemiepiphysiodesis if bar >50%**

Hemi-plateau elevation — severe uncorrected or **relapsed** disease

6 Complications

Compartment syndrome — after high tibial/fibular osteotomy → prophylactic anterior compartment release

Recurrence — **medial physeal bar** → progressive varus after well-done osteotomy → lateral hemiepiphysiodesis or bar resection
✗ Wrong options: overcorrection · plate malposition · age 5 · vitamin D

Screen for SCFE (TO) — all Blount patients (obesity)

X vs Y

	Infantile Blount	Adolescent Blount
Age	**2–5 yr**	**>10 yr**
Side	**50% bilateral**	Usually unilateral
Risk	Early walking, large stature, obesity	Obesity
Classification	**Langenskiöld**	None radiographic
Physeal disturbance	More severe	Less severe
Femur	**Compensatory distal femoral valgus**	**Distal femoral varus** + distal tibial valgus
Natural history	Stage II/IV may resolve	Never resolves
Treatment	Bracing + surgery	Surgery only

	Physiologic bowing	Infantile Blount
Age	**<2 yr**	**>2 yr**
Shape	Symmetric, femur + tibia	Asymmetric, **proximal tibia**, sharp
MDA	**<10–11°**	**>16°**
X-ray	Symmetric flaring	**Metaphyseal beaking**, medial sloping
Gait	No thrust	**Lateral thrust**

How it's asked

"NOT a component of Blount's?" → **external rotation** (TO)
✗ Wrong options: varus · procurvatum · internal rotation

"Varus thrust during gait in Blount's – eponym?" → **Siffert-Katz sign** (TO)

"1-yr, bowing, normal MDA?" → **within normal limits** (TO)
✗ Wrong options: bracing · surgery · vitamin D

"17-month-old, symmetric 10° varus, no thrust?" → **clinical observation**
✗ Wrong options: X-rays · vitamin D · guided growth

– – Active space – –

"Physiologic bowing corrects by?"	→ **36 months**
"12-month-old just standing with bowed legs?"	→ **normal development** ✗ Wrong options: rickets · NAT · bracing
"Most reliable sign separating Blount from physiologic?"	→ **MDA >16°** ✗ Wrong options: walking at 15 months · normal ICD
"3-yr progressive bowing — abnormal exam?"	→ **lateral thrust** ✗ Wrong options: TFA 5° internal · forefoot adduction · recurvatum · distal femoral varus
"14-yr, severe medial epiphyseal depression?"	→ **neglected infantile Blount** ✗ Wrong options: adolescent Blount · hemophilia
"1.5-yr, acute bowing below knee, MDA 14°/12°?"	→ **Blount's / tibia vara — mermaid splints** (TO) ✗ Wrong options: physiologic · osteotomy · vitamin D
"30-month-old, Langenskiöld II?"	→ **KAFO** ✗ Wrong options: AFO · osteotomy
"2½-yr, Langenskiöld I?"	→ **KAFO**
"Counsel on bracing efficacy?"	→ **no strong consensus** ✗ Wrong options: effective · ineffective
"4-yr, 90th percentile, bilateral, thrust?"	→ **predisposed to failing nonoperative care** ✗ Wrong options: undercorrect · medial hemiepiphysiodesis · KAFO
"4-yr obese, MDA 18°, medial articular depression?"	→ **osteotomy correcting to 10° valgus** (TO) ✗ Wrong options: neutral axis · bar resection
"32-month-old, worsening despite 1 yr bracing?"	→ **bilateral proximal tibial osteotomies** ✗ Wrong options: femoral · medial hemiepiphysiodesis
"Valgus osteotomy with 10° overcorrection — which patient?"	→ **4-yr obese, Langenskiöld IV** ✗ Wrong options: renal osteodystrophy · 18-month MDA 11°
"4-yr, 12 months KAFO, MDA 25°, worsening?"	→ **proximal tibia/fibula osteotomy**
"3 yr 10 months, worsening?"	→ **proximal tibia valgus osteotomy** ✗ Wrong options: KAFO · distal femur
"Obese 3-yr, 30° unilateral varus, stage 3?"	→ **proximal tibiofibular osteotomy, acute correction** ✗ Wrong options: lateral hemiepiphysiodesis (shortening)

"3½-yr progressive bowing, medial sloping?"	→ **tibia and fibula osteotomy** ✗ Wrong options: hemi-plateau elevation · epiphysiodesis
"4-yr, Langenskiöld IV, progressive?"	→ **tibial osteotomy** ✗ Wrong options: bar resection · bracing
"3½-yr, MDA >20°, advanced changes?"	→ **bilateral tibia and fibular osteotomies** ✗ Wrong options: KAFO · Denis-Browne bar
"4-yr, Langenskiöld II?"	→ **proximal tibia/fibula valgus osteotomy** (age ≥4) ✗ Wrong options: KAFO · distal femur
"How does a lateral staple work?"	→ **↑ compression slows growth** (Hueter-Volkmann)
"5-yr leg bowing — next step?"	→ **bilateral lateral proximal tibial epiphysiodesis, extraperiosteal plates** ✗ Wrong options: medial · night KAFO · casting
"4-yr, MDA 25°, expected outcome of lateral hemiepiphysiodesis?"	→ **prevents worsening** ✗ Wrong options: complete correction
"Varus recurs 2 yr after osteotomy?"	→ **medial tibial physeal bar** ✗ Wrong options: overcorrection · plate malposition

– – Active space – –

INTERNAL TIBIAL TORSION

– Pediatrics · 55 / 112 –

1 Basics

Most common cause of in-toeing in toddlers (**1–3 yr**) · often bilateral · if unilateral → **left** more often (TO) · intrauterine moulding

Age-wise in-toeing infant: **metatarsus adductus** · toddler: **internal tibial torsion** · early childhood: **femoral anteversion** · OB: MA ± ITT most common at 18 months

3 Presentation

History noticed on walking · legs "turn in" · tripping

Rotational profile evaluation

Staheli rotational profile (5) **foot progression angle** · **hip IR/ER** · **thigh–foot axis** · **transmalleolar axis** · **heel bisector**
✗ Wrong options: **Q angle** (not part)

Measure	Normal	ITT
Foot progression angle	−5° to +20°	Negative (internal)
Thigh–foot angle (prone)	Infant 5° internal (−30 to +20) · age 8: 10° external (−5 to +30) · −10° to +20° (OB)	**>10° internal**
Transmalleolar axis	Infant 0 to −10° → 15–23° external	**>15° internal**
Hip IR / ER	IR ≤ 70°	Normal
Heel bisector	2nd web space	Normal

Thigh–foot axis	Transmalleolar axis
	

Red flags for work-up (X-rays) **pain** · **LLD** · **progression** · family history of rickets / dysplasia / MPS · rotational profile **>2 SD**
✗ Wrong options: **metatarsus adductus** (benign)

In-toeing cause	Key finding
Metatarsus adductus	Abnormal heel bisector, normal hindfoot
Internal tibial torsion	**TFA >10° internal**
Femoral anteversion	**IR >70°, ER <20°**

4 Diagnosis & Classification

Key radiology findings

X-rays	not needed unless red flags · CT/MRI for surgical planning

5 Management

Natural history & observation

Observation + parental education	resolves by **age 3–4** (Orthobullets, OB) · 6–8 yr (one OB explanation) · **bracing, Denis Browne bar, twister cables, casting, shoes don't help**

18-month-old triplets with bowing + ITT → observation (packaging)
✗ Wrong options: vitamin D · serial alignment films · genetics

Operative criteria

Supramalleolar derotation osteotomy	**>6–8 yr** (adolescent) + **functional problems** + **TFA >15°** internal · fewer complications than proximal · plate/K-wires (IM nail if mature)

How it's asked

"2-yr, negative foot progression angle (L > R)?"	→ **internal tibial torsion** (TO)
"In-toeing – prompts further work-up EXCEPT?"	→ **metatarsus adductus**
"Part of rotational profile EXCEPT?"	→ **Q angle**
"4-yr, TFA −10° left, normal hips?"	→ **observation + education** ✗ Wrong options: tibial/femoral osteotomy · brace · casting
"Toddler ITT – expected outcome?"	→ **resolves by 3–4 yr without treatment**
"18-month-old, TFA 30° internal, normal hips?"	→ **no treatment – resolves**
"18-month-old triplets, bowlegs + in-toeing?"	→ **observation**

– – Active space – –

KOHLER'S DISEASE

– Pediatrics · 56 / 112 –

1 Basics

Natural history & etiology

Definition	idiopathic **avascular necrosis (osteochondrosis) of the tarsal navicular** in young children
Demographics	age **4–7 yr** (Orthobullets) · **2–9 yr** (OB) · **boys 4:1** (80%) · **bilateral up to 25%**
Blood supply	intraosseous flow **centripetal → central one-third = watershed** · primary supply = perforating branches of **dorsalis pedis** · revascularised from the **perichondral ring** → excellent prognosis
Theory	navicular is the **LAST tarsal bone to ossify → compressed between the ossified talus and cuneiforms** → central ischemia (✓ theorised etiology) ✗ Wrong options: first
Mimic	often misdiagnosed as **infection**

3 Presentation

Symptoms	**dorsomedial midfoot pain** + **limp** · may be asymptomatic · swelling, warmth, redness possible · point tenderness over navicular
Exam	antalgic gait, weight on **lateral border** of foot · no fever, no constitutional symptoms

4 Diagnosis & Classification

Key radiology findings

X-ray	**sclerosis, fragmentation, flattening** of navicular · diagnosis on **plain films alone** ✗ Wrong options: cysts · widening
Bone scan	**↓ uptake** · unnecessary
MRI	not needed initially · if pain **persists >6 months** or diagnosis unclear → exclude **(cartilaginous) tarsal coalition**
Labs / biopsy	unnecessary when no illness or constitutional symptoms ✗ Wrong options: ESR/CRP/cultures · biopsy
Classification	none widely used



Kohler's — sclerotic, flattened navicular



Navicular sclerosis and fragmentation

5 Management

Kohler's disease

- Mild / no severe pain → **observation · NSAIDs · arch support orthotic**
- Pain with activity / failed orthotic → **short-leg WALKING cast 4–6 wk**
  (surgery never indicated)

Treatment - short leg cast

First line	**NSAIDs** short course · activity limitation · **shoe modification / arch support**
Cast	**short-leg walking (weight-bearing) cast for 4–6 wk → shortens duration of symptoms · 2 wk splint is insufficient** (OB)
Weight bearing	**safe**, does not affect outcome · OB6130 accepts "restriction of weight bearing or short leg cast" · NWB 6 wk also relieves symptoms (OB211605) ✗ Wrong options: NWB cast (unnecessary, hard to enforce)
Long vs short cast	excellent function **regardless of casting technique** ✗ Wrong options: long leg cast quicker
Surgery	**not indicated** ✗ Wrong options: core decompression · triple / subtalar fusion · curettage + graft · medial column lengthening · ORIF · I&D
PT	does not aid recovery

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Prognosis

Course	**self-limiting** · intermittent activity symptoms for **1–3 yr** (✓ resolution in 1–3 yr; ✗ Wrong options: 6 months)
X-ray	improves **6–48 months** from onset · navicular **reconstitutes** ✗ Wrong options: complete absorption
Long term	**no long-term disability** · no midfoot collapse · **no talonavicular arthritis** · residual deformity almost always asymptomatic

X vs Y

	Kohler's disease	Infection (mimic)
History	No illness, **no constitutional symptoms**	Recent illness, constitutional symptoms
X-ray	**Sclerosis, flattening, fragmentation** – diagnostic	–
Workup	**X-ray alone** (✗ Wrong options: ESR/CRP/WBC · biopsy)	Bloods, cultures
Treatment	**Walking cast / NSAIDs** (✗ Wrong options: antibiotics)	Antibiotics

How it's asked

"5-yr, medial foot pain, failed orthotics?"	→ **short-leg walking cast** ✗ Wrong options: core decompression · fusion · I&D
"6-yr, limp, navicular sclerosis – treatment?"	→ **short leg walking cast** ✗ Wrong options: long leg NWB cast · biopsy
"Splinted 2 wk, still painful – next step?"	→ **weight-bearing cast** ✗ Wrong options: MRI · bone scan · NWB cast
"5-yr, navicular tenderness + swelling?"	→ **symptomatic: restrict weight bearing or short leg cast** ✗ Wrong options: antibiotics · medial column lengthening
"5-yr, limp, normally active, mild pain?"	→ **observation or orthotic arch support** ✗ Wrong options: MRI · aspiration
"Redness, warmth over midfoot + flattened navicular?"	→ **weight-bearing short leg cast** ✗ Wrong options: antibiotics · excision
"Kicked at soccer, 2 months dorsal pain?"	→ **walking cast** ✗ Wrong options: ORIF · bloods · MRI
"Natural history untreated?"	→ **resolution in 1–3 yr** ✗ Wrong options: 6 months · collapse · TN arthritis

"Most accurate statement about etiology?"	**→ compression between ossified talus and cuneiforms** ✗ Wrong options: first to ossify · cysts + widening	– – Active space – –

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LARSEN'S SYNDROME

– Pediatrics · 57 / 112 –

1 Basics

Incidence ~**1 in 100,000** live births

Genes & lab markers

Inheritance	Gene
Autosomal dominant	**Filamin B**
Autosomal recessive	**Carbohydrate sulfotransferase 3** deficiency

Diagnosis family history + clinical/radiographic features · **no commercial genetic test** (many mutations)

Orthopaedic manifestations

Dislocations **hips** · **knees (bilateral)** · shoulders · **elbows (radial heads, bilateral)**

Other **clubfeet** (equinovarus / equinovalgus) · scoliosis · **cervical kyphosis** (hypoplastic cervical vertebrae) · extra carpal ossification · hand deformities

Non-ortho laryngeal / tracheal problems · heart valve or aortic lesions · hyperelasticity → hypotonia

3 Presentation

Face **flattened (depressed) nasal bridge** · **prominent forehead** · hypertelorism / telecanthus

Hands long cylindrical **non-tapering fingers** · wide distal phalanx of thumb

Intelligence **normal**

Hypotonia uncommon · may signal **cervical cord compression** · myelopathy → extremity weakness



Larsen facies

4 Diagnosis & Classification

Cervical kyphosis screening

C-spine AP + lateral	**in the first year of life** · and **before any intubation** or manipulation → avoid iatrogenic cord injury
MRI	cervical kyphosis or myelopathy
Findings	vertebral **hypoplasia** · **cervical kyphosis with subluxation** · hip dislocation
Hips	AP pelvis + lateral · **ultrasound if <3 months**



Cervical kyphosis in Larsen syndrome

5 Management

Priority	**stabilise / image the cervical spine BEFORE any hip or knee procedure** ✗ Wrong options: hip open reduction first · knee procedures first

Operative criteria

Problem	Treatment
Cervical kyphosis, **no deficit**	**Posterior cervical fusion** in **first 18 months**
Cervical kyphosis **with deficit**	**Anterior/posterior decompression + fusion**
Hip dislocation	Closed reduction rarely works → **open reduction** (failed closed, contractures, unilateral) · bilateral = controversial, do early and only once
Knee dislocation	Closed reduction + casting rarely works → **open reduction + femoral shortening + collateral ligament excision**

How it's asked

"Bilateral radial head + knee dislocations, flat facies – must check?"	→ **cervical kyphosis** ✗ Wrong options: vertical talus · trigger thumb · tibial hemimelia

– – Active space – –

"Larsen child, hip pain, knee dislocations — next step?"	→ **cervical spine radiographs** ✗ Wrong options: hip open reduction · VDRO · tibial osteotomy · ACL reconstruction
"Cervical kyphosis without deficit — timing of fusion?"	→ **posterior fusion in first 18 months**

LATERAL CONDYLE FRACTURE - PEDIATRIC

– Pediatrics · 58 / 112 –

1 Basics

Epidemiology

Incidence	**2nd most common pediatric elbow fracture** (after supracondylar) · 17% of distal humerus fractures · **11%** of elbow injuries · peak age **6 yr**
Signature	**highest nonunion rate** about the pediatric elbow · ↑ malunion, AVN
Physeal pattern	usually **Salter-Harris IV** (Orthobullets, OB) · TO: Milch II "is really SH II", Milch I = SH IV – read the options

Pathoanatomy

Mechanism	**pull-off** (avulsion by common extensors) · **push-off** (FOOSH → radial head impacts lateral condyle)
Fracture line	starts **posterior** distal humeral metaphysis → distal + anterior across physis + epiphysis into joint · may extend into **trochlear groove** → unstable, prone to dislocation
Posteromedial dislocation	rare association · FOOSH, slight flexion + **adduction** · **LCL stays intact**, attached to condylar fragment + radial neck ✗ Wrong options: LCL repair

Blood supply - avoid posterior dissection

Supply	distal humerus mainly from **posterior** anastomotic vessels · brachial artery anterior
ORIF rule	**avoid posterior dissection → AVN of lateral condyle** (and trochlea)

Ossification ages

Centre	Appears (yr)	Fuses (yr)
Capitellum	**1**	12–14
Radial head	**3**	14–16
Internal (medial) epicondyle	**5**	16–18
Trochlea	**7**	12–14
Olecranon	**9**	15–17
External (lateral) epicondyle	**11**	12–14

OB ages	capitellum first 6 months · radial head 3–6 · medial epicondyle 5–7 · trochlea 8 · olecranon 10 · **lateral epicondyle last, 12–14 yr in males** (OB answer) · **females ~1 yr earlier**
Trap	appearance and fusion are independent events

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3 Presentation

History	FOOSH · lateral elbow pain + swelling · may be subtle
Exam	**no obvious deformity** · lateral tenderness · pain with resisted wrist flexion/extension · crepitus ✗ Wrong options: supracondylar look
Lateral ecchymosis	tear of **brachioradialis aponeurosis** → **unstable** fracture

4 Diagnosis & Classification

Key radiology findings

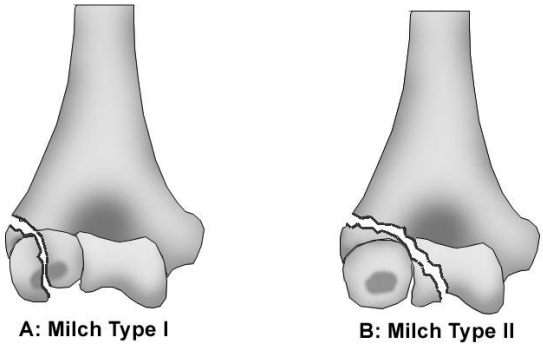
Views	**AP, lateral + INTERNAL oblique** · internal oblique shows **maximum displacement** (fracture is **posterolateral**) ✗ Wrong options: external oblique (TO EXCEPT trap)
Findings	capitellum displaced laterally vs radial head · **posterior Thurston-Holland** fragment on lateral
Arthrogram	minimally displaced fractures · assess cartilage when epiphysis unossified · dynamic · **preferred over MRI**
MRI	cartilaginous trochlear integrity · planning of delayed/nonunion · expensive, needs sedation
CT	rarely, only if fracture type unclear · stress views **not** recommended



Internal oblique view shows true displacement

Milch classification

Type	Fracture line	Stability
I	**Lateral** to trochlear groove	Stable · less common
II	Into **trochlear groove**	**Unstable** · **more common** (95%, TO)



Milch classification

Weiss classification & treatment

Type	Displacement	Cartilage	Treatment
1	**<2 mm**	Intact hinge	**Long arm cast**
2	**2–4 mm** (Orthobullets) · >2 mm (TO)	**Intact** on arthrogram	**CRPP**
3	**>4 mm** (Orthobullets) · >2 mm (TO)	**Disrupted**	**Open reduction + fixation**

Prognosis (TO) type 3 complications **>3×** type 2 · type 3 = **highest nonunion risk** (✓ even after K-wires)

Jakob & Song classifications (TO)

Jakob (TO)	Feature
I	**< 2 mm**, intact hinge
II	**> 2 mm**, no rotation
III	Displaced **with rotation**

Song (TO) **most useful** · uses the **internal oblique view** · stages **1–5** by stability + displacement · **stage 3 = complete and unstable even if minimally displaced**

Milch (TO) not predictive of treatment or outcome · **arthrogram** via lateral soft spot or posterior olecranon fossa assesses the cartilage hinge → CRPP vs ORIF · large metaphyseal fragment in older child → **screw** · pins out at union (~4–6 wk)

Other TO lecture **< 2 mm cast · 2–4 mm CRPP · > 4 mm or rotated → ORIF** · ~15% of elbow fractures · ↑ nonunion and AVN

– – Active space – –

Pediatric elbow injury frequency

Injury	%	Peak age	Needs OR
Supracondylar	41%	7	Majority
Radial head subluxation	28%	3	Rare
Lateral condyle	**11%**	**6**	**Majority**
Medial epicondyle	8%	11	Minority
Radial head/neck	5%	10	Minority
Elbow dislocation	5%	13	Rare
Medial condyle	1%	10	Rare

5 Management

Lateral condyle fracture

- <2 mm all views, hinge intact → **long arm cast 4–6 wk, weekly X-ray ×3 wk**
- 2–4 mm, cartilage intact → **CRPP**
- >4 mm / incongruity → **open reduction + fixation**
  (anterolateral approach, no posterior dissection)

Operative criteria

Cast — only if **<2 mm in ALL views** + **intact medial cartilaginous hinge** · elbow ~90° · **weekly X-rays (incl. internal oblique) for 3 wk** · occasionally >6 wk

Surgery — **>2 mm on any view** = unstable → fixation

Treatment (ORIF displaced)

Approach — **anterolateral** (blood supply posterior) · dissection often done by injury · **see joint reduction directly** · metaphysis may not fit perfectly (plastic deformation)

Fixation — **2 smooth pins** (3 if comminuted), parallel or **divergent** (most stable) · **smooth pins** because fixation **crosses the physis** · single **cannulated screw** for large fragment / nonunion → compression + early motion · no plate · 3–6 wk above-elbow cast

Closed reduction — rarely succeeds for type 3 · CRPP can't inspect articular surface
✗ Wrong options: CRPP in widely displaced

With posteromedial dislocation — **open reduction + percutaneous or internal fixation**
✗ Wrong options: LCL repair · cast

6 Complications

Nonunion, cubitus valgus & tardy ulnar palsy

Stiffness — **most common** · may be early sign of delayed/nonunion · 90% motion by 24 wk, full by 48 wk

Delayed union — not healed after **6 wk** · risk: seen **>2 wk** after injury

Nonunion causes — **pull of wrist extensors** (fragment rotates up to 180°, TO) · **intra-articular synovial fluid** · poor metaphyseal circulation / small metaphyseal fragment · instability in cast · risk ↑ **nonoperative care**

Nonunion scenario	Treatment
Asymptomatic, isolated	**Observation**
Painful / unstable / valgus + ulnar palsy	**ORIF of nonunion** (minimal posterior stripping) ± bone graft (✗ Wrong options: fragment excision → valgus)
Painless nonunion (OB3302)	**In situ screw + bone graft**
Untreated 4 months, painful	**ORIF** (✗ Wrong options: late percutaneous pins · cast · MRI)
Well-established (TO)	**Osteotomy** (varus) · avoid arthrotomy / articular realignment → stiffness, AVN

Nonunion goal — union of **metaphyseal fragment**, not joint surface

Cubitus valgus — from **nonunion** (most common) or lateral physeal arrest · ~**10%** · less common than cubitus varus

Tardy ulnar nerve palsy — slow **stretch** palsy from cubitus valgus · **lateral condyle nonunion = most common cause** (TO) · counsel **ring + small finger numbness** · also after cubitus varus (nerve dislocates anterior to medial epicondyle, FCU band, TO) · treat: **supracondylar osteotomy after maturity + ulnar nerve transposition**

AVN — 1–3 yr after fracture · from **posterior dissection**, most often in treatment of nonunion/delayed union

Fishtail deformity — gap between medial + lateral ossification centres fails · usually no dysfunction · supracondylar osteotomy if needed



Cubitus valgus after nonunion

"Fracture of necessity" (TO) — displaced (> 2 mm) lateral condyle **always needs fixation** · **Cocker's / Kocher lateral approach**, ≥ 2 parallel or divergent K-wires · **cubitus valgus** from "hyperaemic overgrowth of lateral column" (TO) → **tardy ulnar palsy 20–25 y later**

Lateral spur/overgrowth

Incidence — up to **50%** regardless of treatment → **counsel families in advance**

Facts — size **correlates with initial displacement** · **independent of age** · **does NOT affect final ROM** · gives **pseudo-varus** look, not true varus · no trimming asymptomatic spur · lateral periosteal realignment prevents it
✗ Wrong options: caused by surgery

– – Active space – –

Growth arrest rare · young → bar resection or osteotomy · older → complete epiphysiodesis + osteotomy

Prognosis worse than supracondylar (articular, missed diagnosis, nonunion) · with dislocation: slower ROM, same final ROM

X vs Y

	Lateral condyle	Medial condyle (TO)
Frequency	11%	**1%**
Main complication	**Nonunion** → cubitus **valgus**	**Missed diagnosis** (mistaken for medial epicondyle)
Late deformity	Valgus + tardy ulnar palsy	Nonunion → cubitus **varus**, trochlear AVN
Pitfall	–	Trochlea unossified until ~8 yr · complication rate up to 33%
Displaced	ORIF	ORIF

How it's asked

"View showing maximum displacement?"	→ **internal oblique** ✗ Wrong options: external oblique
"6-yr, displaced intra-articular fragment?"	→ **ORIF with K-wires** ✗ Wrong options: CRPP · plate
"6-yr, displaced – best treatment?"	→ **ORIF with smooth pins or cannulated screw** ✗ Wrong options: cast 3–8 wk · plate
"2 mm on internal oblique, casted – injury with highest nonunion rate?"	→ **lateral condyle** ✗ Wrong options: medial condyle · transphyseal separation
"3-yr, lateral condyle + posteromedial dislocation?"	→ **open reduction + fixation** ✗ Wrong options: LCL repair · CRPP
"ORIF – avoid dissection where?"	→ **posterior**
"Tardy ulnar palsy seen after?"	→ **nonunion of lateral condyle** (TO) ✗ Wrong options: malunion · gunstock
"Treatment of well-established nonunion?"	→ **varus osteotomy** (TO) ✗ Wrong options: ORIF + graft · observation
"Weiss type most likely to go to nonunion?"	→ **type 3** (TO)
"Nonunion associated with?"	→ **ulnar nerve palsy** ✗ Wrong options: radial nerve · cubitus varus
"36-yr claw hand, old untreated condyle fracture – elbow shows?"	→ **cubitus valgus** ✗ Wrong options: AVN · fishtail · normal carrying angle

"7-yr, painful lateral condyle nonunion, normal alignment?"	→ **internal fixation of nonunion** ✗ Wrong options: excision · ulnar decompression
"Injured 4 months ago, untreated, daily pain?"	→ **ORIF** ✗ Wrong options: percutaneous pins · MRI
"Untreated displaced fracture — expected deformity?"	→ **cubitus valgus** ✗ Wrong options: HO · PIN palsy · synostosis
"Painless nonunion with valgus, observed — counsel about?"	→ **ring + small finger numbness** ✗ Wrong options: medial spurring
"True about lateral spur?"	→ **final ROM not affected by spur size** ✗ Wrong options: caused by surgery · age-dependent · trim it
"Lateral epicondyle ossifies in males at?"	→ **12–14 yr** (OB)
"All EXCEPT (lateral condyle)?"	→ **external oblique helps assess displacement** (TO)
"Most common complication of medial condyle fracture?"	→ **missed diagnosis** (TO) ✗ Wrong options: nonunion · trochlear AVN

– – Active space – –

LEG LENGTH DISCREPANCY (LLD)

– Pediatrics · 59 / 112 –

1 Basics

Incidence **2 cm LLD in up to 2/3** of the population · most <2 cm, asymptomatic, no long-term consequence

Hemihypertrophy & congenital causes

Group	Causes
Congenital	Hemihypertrophy · dysplasias · **PFFD** · DDH · **unilateral clubfoot**
Paralytic	Spasticity (CP) · polio
Physeal disruption	**Infection** · **trauma** · tumor

Unilateral clubfoot **entire involved limb hypoplastic** (✓) → LLD, mostly **tibia** · (✗ Wrong options: hemihypertrophy of other limb) · **no bone-age difference** · **posterior tibial artery dominant** (anterior tibial hypoplastic) · casting first line

Beckwith-Wiedemann hemihypertrophy + **macroglossia** → up to **10% tumor (Wilms)** → **abdominal + pelvic US and AFP 3–4×/yr until age 8**
✗ Wrong options: thyroid · echo · spine MRI

Associations **back pain** · **OA of long-leg hip** (↓ head coverage, 84%) · functional scoliosis · inefficient gait · ankle equinus · observation of small LLD not linked to scoliosis or back pain (Orthobullets)

Classification

Type	Example	Behaviour
Static	Malunion of femur / tibia (extra-physeal trauma)	Doesn't change
Progressive	**Physeal arrest** · congenital	Absolute LLD ↑ · **congenital: proportion stays the same**

3 Presentation

Measurement & evaluation

Block test blocks under short leg until **pelvis level** · **best initial screening** · accurate, simple, cheap

Tape **ASIS → medial malleolus** (inaccurate with contracture)

Contractures hip, knee, ankle alter apparent length · **hip adduction contracture → apparent shortening of adducted side**

4 Diagnosis & Classification

Key radiology findings

Method	Technique	Key fact
Teleoradiograph	Single exposure, both legs, long film	Good for **angular deformity** · **magnification error** (beam parallax)
Orthoradiograph	3 exposures (hip, knee, ankle)	Avoids magnification
Scanogram	3 exposures, small cassette	Avoids magnification · less error if patient moves
CT scanogram	AP scout	**Most accurate with hip / knee / ankle contractures**
Lateral CT scanogram	Lateral scout	**Best with knee flexion contracture** (others are coronal)

Primary initial imaging (Sabharwal, OB) full-length standing AP, pelvis level, **magnification marker**

Bone age hand X-ray



CT scanogram

5 Management

Predicting LLD from growth arrest (physeal growth rates)

Arithmetic (Menelaus) method growth stops **boys 16 yr, girls 14 yr** · LLD added = **physeal rate × years left** · easiest, least accurate, best for **final years** of growth and exams

Physis	Orthobullets / most OB	Other OB values
Proximal femur	**3 mm/yr** (1/8 in)	4 mm/yr
Distal femur	**9 mm/yr** (3/8 in)	**10 mm/yr** · 9–11
Proximal tibia	**6 mm/yr** (1/4 in)	–
Distal tibia	**5 mm/yr** (3/16 in)	4 mm/yr
Whole leg	**23 mm/yr**, **knee 15 mm/yr**	–

– – Active space – –

Other physes (OB) **distal femur fastest** · proximal humerus 7 · distal radius 5–6 · distal tibia 4–5 mm/yr

Dimeglio knee grows **2 cm/yr** (femur 1.1 + tibia 0.9) · growth stops at **bone age 15.5 (boys) / 13.5 (girls)**

Method	Feature
Green-Anderson tables	Limb length for age
Moseley straight-line graph	Graph form · accounts for **skeletal vs chronologic age** · averages serial measurements
Multiplier (Paley)	Current LLD × age/sex factor · **most accurate for congenital LLD** · uses **chronological age**
Half of final leg length	**Girls at 3 yr · boys at 4 yr**

Best predictor in adolescence **skeletal age**
✗ Wrong options: chronological age · Tanner · father's height · Paley

Worked examples (OB)

Scenario	Answer
14-yr boy, distal femoral arrest	2 yr × 10 = **2 cm**
9-yr boy, distal femoral arrest	7 yr × 9 = **~6 cm**
12-yr boy, closed distal femur, LLD 2 cm now	2 + 4 × 10 = **6 cm**
8-yr girl, closed distal femur, LLD 1 cm	6 yr × 9 + 1 ≈ **6–7 cm, uninjured (left) longer**, no angulation (central arrest)
14-yr boy, DF + PT epiphysiodesis	(10 + 6) × 2 = **3.2 cm**
14-yr boy, PT arrest, LLD 3 mm	3 + 6 × 2 = **15 mm → observe**

Epiphysiodesis indications & timing

Projected LLD at maturity	Treatment
<2 cm	**Observation ± shoe lift**
2–5 cm (OB: 2–6)	**Epiphysiodesis** of long side (femur, tibia or both) · shortening if already mature
>5 cm (OB: >5–6)	**Lengthening** of short side ± contralateral epiphysiodesis/ostectomy
>20 cm / non-reconstructable	**Amputation + prosthesis**
Historical (OB)	<1.5 cm none · 1.5–5 cm epiphysiodesis · >5 cm lengthening

Decide on LLD at maturity, not at presentation

Physeal bar excision bar **<50%** of physis + **≥2 yr growth** left

DF + PT epiphysiodesis removes **1.5 cm/yr** · 13-yr boy with 4.6 cm = most equal (3 yr left)

Timing (13-yr boy, puberty 1 month ago, 4 cm) **femur AND tibia** at **13 yr 4–6 months** (Menelaus 13 y 4 m · Dimeglio 13 y 6 m)

Closed physis on short side + LLD already 2.5 cm — close **both** long-side DF + PT
✗ Wrong options: DF only (no differential left) · PT only

Persistent LLD after epiphysiodesis — most common cause = **error in timing**
✗ Wrong options: overgrowth · joint penetration

Small asymptomatic LLD + acute injury — treat the injury (NSAIDs, rest)
✗ Wrong options: epiphysiodesis

Lengthening & distraction osteogenesis

Corticotomy — **metaphyseal** · periosteal supply matters more than endosteal (lengthening over nails works)

Step	Value
Latency	**5–7 days** (no lengthening in week 1)
Rate	**1 mm/day** (0.25 mm ×4)
Too fast (>1 mm/day)	Painful neuropathy, soft-tissue problems, poor regenerate
Poor regenerate	**Slow the rate** → then **accordionization**
Consolidation	Fixator stays **as many days as lengthened**

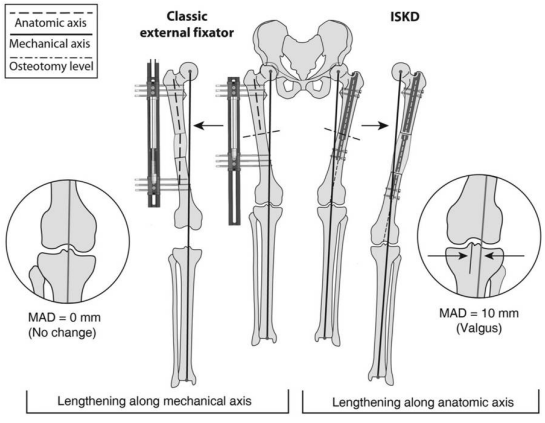
Which bone — lengthen the **femur** when the discrepancy is femoral (✗ Wrong options: tibia (asymmetric joint lines)) · 15-yr with 5.5 cm → **femoral lengthening** (✗ Wrong options: shoe lift · late epiphysiodesis · 6.5 cm femoral shortening (weak muscles))

Pre-op stability — need stable joints above + below · **cruciate deficiency → knee subluxation** · **hip CE angle <20° → hip subluxation**
✗ Wrong options: Achilles contracture · 8 cm LLD

Mechanical axis deviation

Technique	MAD
Lengthen along **mechanical axis** (ex-fix)	**None**
Lengthen along **anatomic femoral axis** (nail)	**Lateral** MAD
Shorten along anatomic axis	**Medial** MAD

Angle — ~**7°** between limb mechanical axis and femoral anatomic axis



Lengthening along anatomic vs mechanical axis

– – Active space – –

6 Complications

Epiphysiodesis incomplete arrest / **angular deformity** (open or percutaneous) · **persistent LLD from timing error**

Lengthening pin-site infection · fracture · delayed union · premature cessation · **joint subluxation/dislocation**

Complication	Clue	Treatment
Incomplete corticotomy	Turning gets harder in **first week** → sudden **pop** + pain	–
Premature consolidation	Harder to turn at **~6 wk** · **bowed wires**, mature regenerate · femur + fibula most	Keep distracting until bridge ruptures · osteoclasis · repeat corticotomy
Causes of premature consolidation	Long latency, slow rate, exuberant bone, compliance, device failure	–

X vs Y

	Incomplete corticotomy	Premature consolidation
When	**First days** of distraction	**Weeks** in (~6 wk)
Event	Acute pop as osteotomy completes	Device seizes, wires bow

How it's asked

"14-yr boy, complete distal femoral arrest – LLD?"	→ **2 cm**
"9-yr boy, distal femoral arrest?"	→ **6 cm**
"Distal femur stops growing in boys at?"	→ **16 yr**
"12-yr, 2 cm LLD, closed distal femur – projected?"	→ **6 cm**
"8-yr girl, central distal femoral arrest – consequence?"	→ **7 cm LLD, left femur longer** ✗ Wrong options: varus · valgus · recurvatum
"Same girl – LLD at maturity?"	→ **6 cm**
"Fastest-growing physis?"	→ **distal femur** ✗ Wrong options: proximal humerus
"14-yr boy, DF + PT epiphysiodesis – correction?"	→ **3.2 cm**
"14-yr boy, PT arrest, LLD 3 mm, bone age 14?"	→ **observation**
"17-yr girl, 10 mm block levels pelvis?"	→ **observation** ✗ Wrong options: CT scanogram · epiphysiodesis
"Epiphysiodesis at 14 with 4.2 cm, still 3 cm at 18 – why?"	→ **error in timing of surgery**

"13-yr boy, puberty 1 month ago, projected 4 cm?"	→ **epiphysiodesis of left femur and tibia in 6 months** ✗ Wrong options: immediate lengthening
"11-yr boy, femoral LLD 4 cm at maturity?"	→ **timed epiphysiodesis of longer leg**
"13-yr, left DF closed, LLD 2.5 cm?"	→ **right DF + proximal tibia/fibula epiphysiodesis**
"DF + PT percutaneous epiphysiodesis — most equal result?"	→ **13-yr boy with 4.6 cm**
"Best predictor of remaining growth in adolescents?"	→ **skeletal age** ✗ Wrong options: Paley · Tanner
"Distraction rate?"	→ **1 mm/day**
"15-yr, 5.5 cm femoral LLD, closed physis?"	→ **right femoral lengthening** ✗ Wrong options: tibial lengthening · femoral shortening
"Lengthening along mechanical axis vs anatomic axis nail?"	→ **no MAD vs lateral MAD**
"Turning harder, pop on day 9 of lengthening?"	→ **incomplete corticotomy with spontaneous completion**
"Harder to turn at 6 wk, bowed wires?"	→ **premature consolidation**
"Lengthening in first week after osteotomy?"	→ **none** (latency)
"Congenital short femur — risk factor for complication?"	→ **cruciate deficiency** ✗ Wrong options: Achilles contracture · LCEA 25°
"JRA knee flexion contracture — most accurate LLD method?"	→ **CT scanogram** ✗ Wrong options: teleoradiograph
"Best LLD method with knee flexion contracture?"	→ **lateral CT scanogram** ✗ Wrong options: blocks · tape
"Gymnast, 1.2 cm LLD, acute knee strain?"	→ **NSAIDs, rest 1 wk** ✗ Wrong options: epiphysiodesis · cast
"Unilateral clubfoot — true?"	→ **whole limb hypoplastic → LLD** ✗ Wrong options: hemihypertrophy · bone-age difference · anterior tibial dominant
"Hemihypertrophy + large tongue?"	→ **abdominal/pelvic US + AFP** (Beckwith-Wiedemann)

– – Active space – –

LEGG-CALVE-PERTHES DISEASE

– Pediatrics · 60 / 112 –

1 Basics

Epidemiology – age

Definition	**idiopathic AVN of the proximal femoral epiphysis** in children · self-limiting, heals + remodels ✗ Wrong options: adult osteonecrosis
Age	**4–8 yr** (Orthobullets) · TO answer **5–10 yr** – read the options ✗ Wrong options: 4–8 yr in the TO MCQ; text says 4–10
Other hip ages (TO)	SCFE boys **12–15**, girls **11–13** · transient synovitis **4–10** (mean 6)
Demographics	**1 in 10,000** · **M:F 5:1** · urban · low socioeconomic class · high latitude · Caucasian > East Asian, African American
Bilateral	**12%** (Orthobullets) · 11% (OB) · **asymmetric + asynchronous** · symmetric, same stage → think **MED**
TO epidemiology	age **4–12 yr** (lecture A) / **5–10 yr** in India (lecture B) · **M:F 4–5:1** · bilateral **10–15%** · ~**3%** of paediatric hip disorders · higher incidence in **south-western belt of India**
Exclusion rule (TO)	if a cause is found (sickle cell, coagulopathy, hypothyroidism, dysplasia) it is **not Perthes** · extremes of age or **symmetric bilateral** → look for secondary cause
Associations (TO)	likely: trauma in susceptible child · **passive smoking (strongest correlation)** · hyperactivity · coagulopathy · heredity · unlikely: endocrinopathy, urban environment, synovitis
Predisposed child (TO)	**bone age lags 2–4 yr** · skeletal standstill during active disease · worse delay in bilateral · carpal bones delayed: unilateral **lunate, triquetrum** · bilateral **trapezoid**
Risk factors	family history · **low birth weight** · abnormal birth presentation · **second-hand smoke** · Asian, Inuit, Central European

Pathophysiology

Sequence	infarction → revascularisation → **resorption → collapse** → creeping substitution → remodeling
Theories	**coagulopathy** (protein C/S deficiency; thrombophilia 50%, **some coagulopathy up to 75%**) · repeated subclinical trauma (multiple-infarction theory) · smoking aggravates
Associated	**ADHD 33%** · **delayed bone age 89%** · hypercoagulability (TO)
Histology	disorganised cartilage, hypercellularity, fibrillation in epiphysis + physis

Femoral head blood supply in childhood

Age (TO)	Supply
Birth → 4–7 months	**MFCA + LFCA + artery of ligamentum teres** (retinacular vessels cross freely)
4–7 months → 14–18 yr	**MFCA ALONE** (physis blocks LFCA) · **superior retinacular (lateral epiphyseal) artery = main**
Adult	Mainly MFCA · small LFCA (anterior retinacular) + ligamentum teres

18 months → MFCA alone
✗ Wrong options: MFCA + LFCA · + ligamentum teres

3 Presentation

Symptoms	insidious **painless limp** · intermittent **hip, groin, thigh or knee** pain
Exam	loss of **internal rotation + abduction** · **Trendelenburg** (collapse → GT rises → slack gluteus medius) · **Catterall sign** = obligatory ER on hip flexion (TO) · LLD late, worsened by adduction contracture

4 Diagnosis & Classification

Key radiology findings

X-ray	**AP pelvis + frog lateral**
Earliest X-ray sign	**medial joint space widening** (teardrop → ossific nucleus)
Other	small, sclerotic ossific nucleus · **crescent sign** = subchondral fracture · X-ray may stay normal **3–6 months**
Imaging (TO)	X-ray sensitivity ~**97%**, may lag symptoms **up to 8 months** · **MRI = gold standard for early diagnosis** (extent of necrosis, cartilage) · **contrast-enhanced MRI** when unclear (vs TB hip) · bone scan: early, sensitive + specific
Signs (TO)	**sagging rope sign** = coxa magna + neck shortening · **dimple sign** – normal opposite hip, not pathological · **V sign** – medial epiphyseal defect from pressure overload
Bone scan	**cold** (↓ uptake) before X-ray changes · extent of involvement
MRI	early / occult disease · **more sensitive** than X-ray
Perfusion studies	predict max lateral pillar involvement · **dynamic arthrogram** → coverage, containment, hinge abduction

– – Active space – –

Waldenström stages

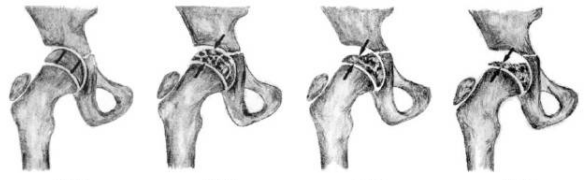
Stage	X-ray	Note
Initial	Small sclerotic epiphysis, **medial joint space widening**	Occult 3–6 months
Fragmentation	**Crescent sign**, head fragments/ dissolves	**Most symptomatic** · **lateral pillar graded here** · 6 months–2 yr
Reossification	New bone replaces necrotic bone	Up to 18 months
Healing / remodeling	Head remodels to maturity	Trabeculae return

Elizabethtown sub-stage (TO)	Feature
1a	Sclerosis, **no loss of height**
1b	Sclerosis **with loss of height**
2a	Early fragmentation
2b	Advanced fragmentation, **no new lateral bone**
3a	Early new bone, porotic, **< ⅓ width**
3b	Mature new bone, **> ⅓ width**
4	Healed, remodelled

Lateral pillar classification timing

Lateral pillar (Herring) **best agreement, most predictive** · graded in **early fragmentation**, **~6 months after symptom onset**
✗ Wrong options: at first visit

Group	Lateral pillar	Prognosis
A	**Full height**, no density change	Consistently good
B	**>50%** height	Poor if **bone age >6** (Orthobullets) · >6 yr age (TO)
B/C	Narrow (2–3 mm) / poorly ossified, ~50%	Added for consistency
C	**<50%** height	**Poor in all**, regardless of treatment



The Herring classification showing: a normal head (a); group A – the height of the lateral pillar is normal (b); group B – reduction to between 50% and 100% of the original height (c); group C – reduction to less than 50% (d).

Lateral pillar (Herring) classification

Catterall classification & head-at-risk signs

Group	Head involvement
I	Anterior epiphysis only
II	Anterior + **central sequestrum**
III	Only a small part spared
IV	**Total head**

Head-at-risk signs (poor) **Gage sign** (lateral V-shaped lucency) · **calcification lateral to epiphysis** · **metaphyseal cyst** · **lateral subluxation** · **horizontal physis**





Gage sign

Salter-Thompson crescent sign **<½** head (A) vs **>½** (B)

Stulberg **gold standard for residual head shape + congruence** at maturity · poor inter/intra-observer reliability

Stulberg (TO)	Head	Outcome
I	Spherical, congruent (normal)	Excellent
II	Spherical, minor deformity (coxa magna, short neck, steep acetabulum)	Good
III	**Aspherical, congruent**	Fair
IV	**Aspherical, incongruent** (flat)	Poor
V	**Flat head, incongruent** (normal acetabulum)	Very poor

Catterall % (TO) I 25% · II 50% · III 75% · IV 100% · head-at-risk signs = need for **containment**, not mandatory surgery (TO)

Differential - bilateral disease workup

Differential infection (septic hip, osteomyelitis, pyomyositis) · transient synovitis · **MED** · **SED** · sickle cell · Gaucher · hypothyroidism · **Meyers dysplasia**

Symmetric bilateral changes → X-rays of knees (MED) and spine (SED)
✗ Wrong options: PT · brace · Salter

5 Management

Goals symptom relief (NSAIDs, traction, crutches) · **restore ROM** (PT, muscle lengthening, Petrie casts) · **containment** (head seated in acetabulum)

– – Active space – –

Perthes disease

- <8 yr (bone age <6) or pillar A → **observation, activity limits, ROM PT**
- >8 yr (bone age >6), pillar B or B/C → **femoral varus and/or pelvic osteotomy**
- Pillar C → **poor regardless of treatment**
- Hinge abduction → **valgus (abduction-extension) ± shelf osteotomy**

Treatment (osteotomy, containment, observation)

Scenario	Treatment
<8 yr (bone age <6) · **pillar A**	**Observation**, activity restriction / protected weight bearing, **ROM PT** (60% need no surgery)
>8 yr (bone age >6) · **pillar B, B/C**	**Proximal femoral varus osteotomy** and/or **pelvic osteotomy** (Salter, triple)
Lateral subluxation	**Shelf** → prevents lateral epiphyseal overgrowth
Hinge abduction	**Valgus (abduction-extension) osteotomy** ± shelf – moves hinge away from acetabular edge, fixes adduction shortening, improves abductor length
Head no longer containable	Shelf or **Chiari**
Healed LCPD with FAI	Hip arthroscopy (emerging)
Arthrodiastasis (ex-fix distraction)	Controversial

Bracing / casting for containment — **no benefit** in large prospective study

Containment (TO)	
Contain	Stage **1a, 1b, 2a** · 2b with good ROM · age **> 6 yr** · **extrusion > 20%** · **≥ 50%** epiphysis involved
Do not contain	Stage **3a, 3b, 4** · severe stiffness / **hinge abduction** · < 6 yr, minimal involvement, no extrusion

Reimers migration index (TO) — uncovered head ÷ head width × 100 · **> 20% → contain**

Stiff hip first (TO) — **traction 2–3 wk** · adductor tenotomy · **Petrie cast** / abduction brace (Scottish Rite, Petrie, Nevington) · **Snyder sling** for weight relief

VDRO rules (TO) — only with adequate ROM · varus **≤ 20°** · final NSA **≥ 105°** · **GT epiphysiodesis** · **hinge abduction = absolute contraindication** → valgus osteotomy / shelf

Age-based (TO lecture A) — < 6–8 yr any pillar: conservative · ≥ 8 yr pillar A or C: conservative · **≥ 8 yr pillar B or B/C: surgical containment**

Other options (TO) — arthrodiastasis · trochanteric advancement · Salter, shelf, Steel, Chiari (salvage)

Surgery evidence — pillar A + B <8 yr do well regardless · surgery improves **B and B/C >8 yr** · earlier surgery before deformity may be best

Varus derotation osteotomy — early extrusion → re-seats head

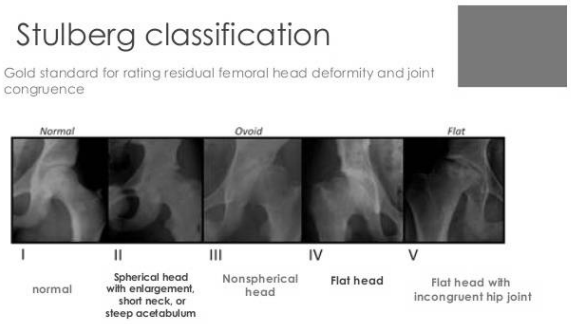
Never	core decompression · hip aspiration (no infection)
TO false statement	"**early containment surgery before age 6 prevents arthritis**" is false · ✓ true: loss of abduction + IR · ✓ age at presentation = most important prognostic factor · ✓ 7-yr with maintained pillar → activity modification
Valgus femoral osteotomy is for	**Perthes with painful hinge abduction** ✗ Wrong options: CP MI 40% (VDRO) · DDH AI 35° at 4 yr (pelvic osteotomy; AI should be <25° by 6 months) · coxa vara HE 35° (observe) · post-SCFE FAI (osteochondroplasty)

6 Complications

Head deformity	**coxa magna** (wide) · **coxa plana** (flat) · Stulberg grades it
Lateral subluxation (extrusion)	poor prognosis → **hinge abduction**
Premature physeal arrest	**trochanteric overgrowth** · **coxa breva** · mild LLD
Sequelae surgery (TO)	coxa breva: **distal GT transfer**, relative/absolute neck lengthening · coxa magna: acetabuloplasty, **femoral head reduction osteotomy** (via surgical dislocation; for severely aspherical head), osteochondroplasty · shelf: hinge abduction, incongruent, coxa magna
Other	acetabular dysplasia · labral tear / FAI · OCD (loose bodies)

Prognostic factors

Best good prognostic factor	**bone age <6 yr at presentation** · <6 yr with pillar A/B → **80% good outcome** (spherical, congruent)
Clinical predictors	**age at presentation + hip ROM** ✗ Wrong options: LLD · pain duration
Poor	**female** · adduction contracture / ↓ abduction · heavy · long course · stiffness · head-at-risk signs · **pillar C** · bilateral · extensive subchondral fracture · age >9 worst (TO)
Arthritis predictor	**asphericity** of head at maturity → OA after 40 ✗ Wrong options: lateral calcification
Natural history	most fine until **5th–6th decade** · **~½ get premature OA** (aspherical head) · Stulberg I + most II do well lifelong · 2–5 yr course



Stulberg classification

– – Active space – –

X vs Y

	Perthes (bilateral)	MED / Meyers dysplasia
Hips	**Asymmetric, asynchronous**, different stages	**Symmetric**, same stage
Next test	–	**Knee + spine X-rays**

	Femoral varus osteotomy	Valgus osteotomy
Purpose	**Containment**	Relieve **hinge abduction**
When	Early, containable head, >8 yr, B/BC	Late, deformed lateral head

	Perthes (TO)	Idiopathic chondrolysis of hip (TO)
First symptom	**Limp** (painless)	**Pain** (then stiffness, limp later)
Sex	Male	**Female 3–4×**
Age	4–12 yr	**11–12 yr**
Pathology	AVN of epiphysis	**Rapid cartilage loss**, no infection/trauma
X-ray	Sclerosis, fragmentation	**Early joint-space narrowing**, head initially normal
Rx	Containment	Non-ambulatory protection → protected walking · late **THA** often

How it's asked

"Lateral pillar classification possible how long after onset?"	→ **6 months**
"All worsen outcome EXCEPT?"	→ **presentation at 5 yr** ✗ Wrong options: pillar <50% · Gage sign · lateral calcification
"Prognostic factor most linked to OA in 5th–6th decade?"	→ **femoral head asphericity**
"Clinical factors predicting long-term outcome?"	→ **age at presentation + hip ROM**
"4-yr, pillar B – prognosis?"	→ **80% chance of good outcome**
"FALSE about Perthes?"	→ **early containment surgery before 6 prevents arthritis** (TO)
"9-yr, 6 months painless limp, pillar B?"	→ **femoral or pelvic osteotomy** ✗ Wrong options: core decompression · aspiration · bracing
"5-yr Caucasian boy, painless limp?"	→ **physical therapy and range-of-motion exercises** ✗ Wrong options: Salter · VDRO · shelf · Chiari
"10-yr, IR 5°, abduction 10°, Trendelenburg, pillar B?"	→ **femoral varus and/or pelvic osteotomy; group B**

"Valgus proximal femoral osteotomy indicated for?"	→ **9-yr Perthes with painful hinge abduction**
"5-yr, symmetric bilateral changes, symptoms resolved?"	→ **knee and spine X-rays**
"Femoral head blood supply at 18 months?"	→ **MCFA (MFCA) alone** (TO)
"Absolute contraindication to VDRO in Perthes?"	→ **hinge abduction** (TO)
"Extrusion (Reimers) threshold for containment?"	→ **> 20%** (TO)
"Idiopathic chondrolysis — presenting symptom?"	→ **pain before limp** (TO)
"Most common age group for Perthes?"	→ **5 to 10 years** (TO) ✗ Wrong options: 4–8 in that MCQ

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LOCAL GIGANTISM

– Pediatrics · 61 / 112 –

1 Basics

Definition overgrowth of some or all tissue elements of a digit / limb · rare · mostly **nonhereditary** · classically **one toe, often the 2nd**, with **normal nails**

Type	Causes
Congenital	**Neurofibromatosis** · **Proteus syndrome** · **Klippel-Trenaunay-Weber**
Acquired	Amyloidosis · elephantiasis (filariasis) · AV malformation · tumor · acromegaly

Growth plate layers resting · proliferative · hypertrophic

Macrodactyly (TO)

Macrodactyly (TO) hamartomatous enlargement of toes · associations: **NF**, **congenital lipofibromatosis**, **Proteus**, **Maffucci**

Type (TO)	Feature
Macrodystrophia lipomatosa	Overgrowth of **all mesenchymal elements**
Macrodactyly simplex congenita	**Phalanges enlarged, metatarsals spared**

Focal gigantism case (TO) **PIK3CA-related overgrowth** / vascular malformation (Klippel-Trenaunay; on **sirolimus**) · X-ray: metatarsals AND phalanges enlarged → growth arrest would need many tiny physes · symptomatic → disarticulation at PIP

3 Presentation

Symptoms pain · **can't fit shoes** · enlarged digit or limb

Diagnosis clinical · X-rays only for **surgical planning** · **no biopsy** needed
✗ Wrong options: antifungals (not fungal)



Local gigantism of a toe

– – Active space – –



Second toe gigantism

5 Management

Treatment - ray amputation

Situation	Treatment
Initial / mild	**Observation**, monitor growth
Open physes	**Epiphysiodesis** ± bony + soft-tissue debulking
Closed physes	**Bony reduction and/or ray amputation**
Recurrence prevention	**Complete ray amputation ↓ recurrence**

Goals cosmesis + function (shoe wear) · individualised by etiology, site, growth left, severity

Options (TO) soft-tissue debulking (**high recurrence**) · **staged debulking (Kotwal and Paru)** · bony shortening · **ray excision** for severe

Cosmesis after ray resection often **poor** · Syme amputation = excessive
✗ Wrong options: "favorable"

How it's asked

"14-yr, closed physes, enlarged 2nd toe, shoes don't fit – true?"

→ **complete 2nd ray amputation ↓ recurrence**
✗ Wrong options: biopsy · antifungals · cosmesis favorable · epiphysiodesis

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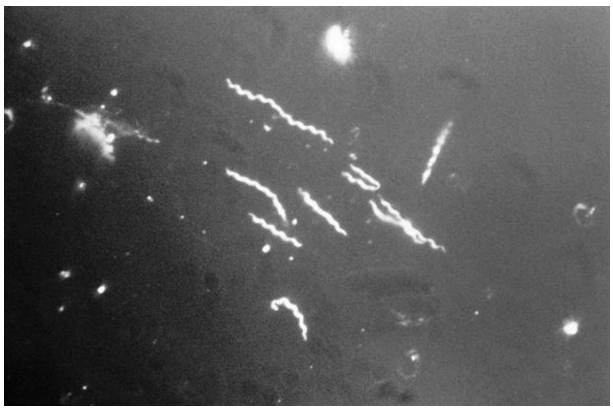
LYME DISEASE

– Pediatrics · 62 / 112 –

1 Basics

Organism - spirochete

Organism — ***Borrelia burgdorferi*** · **spirochete** · tick-borne (**Ixodes** deer tick) · zoonosis (mice, squirrels, shrews)



Borrelia burgdorferi spirochete

Transmission — tick must stay attached **48–72 h** · bites **May–November**, most cases **June–August**

Epidemiology — **most common tick-borne illness in US** · bimodal: **children 5–9**, adults 55–59

Geography — **Northeast** (Maryland → Maine, **Connecticut**, Rhode Island, Catskills) · Great Lakes / Midwest (I. scapularis) · Pacific Northwest (I. pacificus) · deer areas

Joint pathology — synovial hypertrophy · vascular proliferation · mononuclear infiltrate · immune complexes · **MMPs from chondrocytes** → cartilage damage

Stages

Stage	Timing	Features
1 Early localized	**1–30 days**	**Erythema migrans** (hallmark) · flu-like illness
2 Early disseminated	**Weeks–months** · 15–20% untreated	**CN VII** (bilateral 50%, unlike Bell's) · **CN VI** · **lymphocytic meningitis** · migratory poly/monoarthritis · **carditis (heart block)** – rare
3 Late	**Months–years** · **60%** untreated	**Arthritis, usually knee** · intermittent / chronic monoarthritis · **acrodermatitis chronica atrophicans**

3 Presentation

Erythema migrans — **60–80%** · expanding **>5 cm** bull's-eye · **1–3 wk** after bite (Orthobullets) · 1 wk–1 month (OB) · itch/burn · fades ~1 month · **axillary / gluteal folds**, hairline, elastic bands

Lyme arthritis	**knee** effusion, **large and relatively painless** · **swelling > tenderness** · near-full ROM · **can bear weight** · recurrent, migratory · rash often already resolved
Acrodermatitis chronica atrophicans	**"cigarette-paper" skin** · dorsum hands/feet/knees/elbows · older patients



Acrodermatitis chronica atrophicans

Neuro	headache, neck stiffness, encephalitis · **polyradiculoneuropathy**



Erythema migrans

4 Diagnosis & Classification

Presentation & diagnosis (Western blot)

Serology (two-tier)	**ELISA** (sensitive, not specific) → **Western blot** (specific, ✓ confirmatory) ✗ Wrong options: Northern (RNA) · Southern (DNA) · blood/synovial culture
Serology pitfalls	negative early (weeks to seroconvert) · stays positive after old infection · **vaccination: ELISA +, Western blot –**
Endemic area + EM → treat without blood tests	
Blood	WBC **normal or mildly ↑** · **ESR/CRP ↑**
Synovial fluid	**10,000–25,000 WBC/mm³** (lower than septic) · PMN predominance
CSF	↑ protein + **lymphocytic pleocytosis**
Other	skin-edge punch biopsy of EM → culture on **Barbour-Stoenner-Kelly** medium · PCR

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MRI	**large knee effusion with no other pathology** ✗ Wrong options: hip AVN · HAGL with effusion · vertebral osteomyelitis · Kienböck
Organism images (distractors)	S. aureus (GPC clusters) · Pseudomonas · P. acnes · AFB (TB) · Corynebacterium (Chinese letters) · E. coli · gonococcus (GN diplococci) · Kingella (GN coccobacilli, <3 yr)

5 Management

Treatment - doxycycline mechanism

Group	Treatment
Adult / ≥8 yr, mild	Oral **doxycycline** · amoxicillin · cefuroxime
Child <8 yr	**Amoxicillin** · cefuroxime (**no tetracyclines**)
Neuro, arthritis, failed oral therapy	**IV ceftriaxone** / cefotaxime / penicillin G
Chronic arthritis failing IV antibiotics	**Synovectomy**

Duration	doxycycline **10 days** (Orthobullets) · **28 days** for arthritis (OB) · single **4-wk** oral course (OB) · 1–2 wk (OB) – read the options
Doxycycline mechanism	binds **30S** ribosomal subunit → blocks **tRNA binding** → ↓ protein synthesis (minor 50S)

Mechanism	Drugs
30S	**Tetracyclines (doxycycline)**
50S	Macrolides, chloramphenicol, clindamycin, linezolid, streptogramins
Cell wall	Penicillins, cephalosporins, **vancomycin (D-ala-D-ala)**, carbapenems, aztreonam, bacitracin
Folate	Sulfonamides, trimethoprim (TMP-SMX)
DNA gyrase	Fluoroquinolones
RNA synthesis	Rifampin

6 Complications

Persistent joint pain / swelling despite therapy · **chronic Lyme disease** (disabling MSK pain, neurocognitive symptoms, fatigue) · chronic arthritis rare

Transfusion screening

Not screened in donated blood	**Lyme** · malaria · babesiosis · Chagas
Screened	HBV · HCV · syphilis · HIV-1/2 · **West Nile virus**

X vs Y

	Lyme arthritis	Bacterial septic arthritis
Weight bearing	**Can bear weight**	Usually can't
Serum WBC	**Normal**	↑
Fever / CRP	Low-grade, mild ↑	High
Synovial WBC	**10–25k**	Higher
History	Rash + endemic travel 1–2 wk before	–
Treatment	**Antibiotics**	**I&D**

How it's asked

"MRI best supporting B. burgdorferi?"	→ **knee with large isolated effusion**
"Bull's-eye rash, fever – organism?"	→ **Borrelia burgdorferi** ✗ Wrong options: Bartonella · Kingella · Brucella
"10-yr, painless swollen knee, target rash in Connecticut?"	→ **Lyme arthritis** ✗ Wrong options: septic · rheumatic fever · transient synovitis
"Confirmatory test after positive ELISA?"	→ **Western blot** ✗ Wrong options: Northern · Southern · culture
"Catskills camp, intermittent painless knee swelling – organism image?"	→ **spirochete (Borrelia)** ✗ Wrong options: S. aureus · P. acnes · AFB
"Connecticut camping, WBC 25,000 in knee, walking?"	→ **Borrelia burgdorferi**
"Appalachian trip, rash, warm knee, bears weight – organism?"	→ **spirochete** ✗ Wrong options: GPC clusters · GN diplococci · GN coccobacilli
"13-yr, Rhode Island, EM + tense knee – oral drug + mechanism?"	→ **doxycycline, 30S ribosomal subunit** ✗ Wrong options: 50S · cephalexin
"Polyarthralgia + EM – mechanism of first-line drug?"	→ **blocks tRNA binding at 30S** ✗ Wrong options: 50S translocation · DNA gyrase · D-ala-D-ala
"Not tested in routine blood screening?"	→ **Lyme disease** ✗ Wrong options: West Nile · HCV · syphilis

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MARFAN SYNDROME

– Pediatrics · 63 / 112 –

1 Basics

Incidence **1 in 10,000** · no ethnic or sex predilection

Genes & lab markers

Inheritance **autosomal dominant** · **~30% sporadic**

Gene **fibrillin-1 (FBN1)**, chromosome **15q21** · fibrillin = glycoprotein of **elastin-containing microfibrils**

TGF-β misfolded fibrillin **sequesters TGF-β poorly** → ↑ tissue TGF-β (lungs, valves, aorta) → **aortic dilatation**

Aneurysm mechanism **TGF-β + renin-angiotensin system** act synergistically (AngII → wall remodeling + TGF-β via Smad) → **losartan (ARB)** proposed (mixed results vs β-blockers)
✗ Wrong options: CRP · interferon-γ

Genetic testing FBN1 testing **neither sensitive nor specific** → diagnosis by **Ghent nosology** (clinical + family history; diagnoses/excludes 86%, caution in young children)

Genetics & pathophysiology (fibrillin, TGF-beta)

Defective protein	Disease
Fibrillin-1	**Marfan**
Elastin	Supravalvular aortic stenosis, AD cutis laxa
FGFR3	Achondroplasia 10
Type I collagen (COL1A1/2)	OI, Ehlers-Danlos
Type II collagen	SED, achondrogenesis, Stickler
COMP	Pseudoachondroplasia
Neurofibromin (NF1)	Neurofibromatosis 324

Other molecule (distractors)	Role
BMP-2	Osteoblast differentiation, bone growth
Osteocalcin	↑ insulin, glucose uptake
Indian hedgehog	Chondrocyte proliferation

Clinical features

Orthopaedic	Non-orthopaedic
Arachnodactyly	**Aortic root dilatation / dissection**
Scoliosis (50%) – often first diagnosed	**Mitral valve prolapse**
Protrusio acetabuli (15–25%)	**SUPERIOR lens dislocation (60%)** (✗ Wrong options: inferior (homocystinuria))
Dural ectasia (>60%) · meningocele	**Spontaneous pneumothorax**
Ligamentous laxity, recurrent dislocations (patella, shoulder, fingers), ankle sprains	Myopia
Pes planovalgus	Skin striae, recurrent hernias
Pectus excavatum / carinatum	–

Dural ectasia also in **NF1** · complicates instrumentation

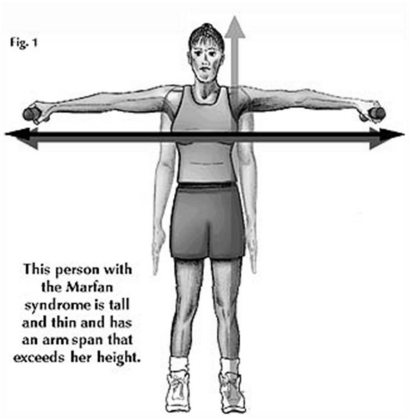
3 Presentation

Dolichostenomelia **arm span > height (ratio >1.05)**

Steinberg thumb sign thumb tip extends beyond ulnar border of hand when clasped under fingers

Walker-Murdoch wrist sign thumb + small finger distal phalanges **overlap** around opposite wrist

Sudden death aortic dissection / MVP → **exertion or pregnancy** · cardiovascular causes nearly all early deaths



Arm span to height ratio



Positive wrist and thumb signs

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4 Diagnosis & Classification

Key radiology findings

X-ray	scoliosis series → scoliosis, kyphosis
MRI spine before surgery	**dural ectasia** (→ CSF leak risk)
CT before surgery	pedicle / bony adequacy for fixation
Echo + cardiology	**before ANY surgery** (dissection, anticoagulation for prosthetic valves)
Protrusio acetabuli	CE angle **>40°** + medial wall past **ilioischial line** · differential: Marfan, Paget, RA, psoriatic, AS, infection (TB), osteomalacia, fibrous dysplasia, idiopathic



Dural ectasia

5 Management

Cardiovascular evaluation

Rule	orthopaedist often first to suspect → refer **cardiology** (± genetics) · **cardiovascular evaluation before conception** and before surgery ✗ Wrong options: genetic testing first · spine/brain MRI first
Medical	**β-blockers** ↓ aortic dilatation · laxity → observation + orthotics

Scoliosis management

Bracing	early mild curves · **less effective than AIS**, usually refractory ✗ Wrong options: "often effective if started early"
Surgery	**PSF ± ASF** · rapidly progressive curve in immature patient · large curve in mature · **extend construct** (avoid adding-on) · fuse to pelvis for obliquity / poor distal fixation · growing rods if young · **maximise fixation points**, slow distraction/compression · **avoid overcorrection** (cardiovascular complications) · treating kyphosis risks vertebral subluxation

Early surgery routinely needed for **progressive scoliosis**
✗ Wrong options: kyphosis · ankle instability · protrusio · tibial pseudarthrosis (NF1)

Complication vs AIS	Marfan
Fixation failure (most common – thin laminae/pedicles, osteopenia)	↑
Infection	↑
Pseudarthrosis	↑
Dural tear / CSF leak (8%)	↑
Decompensation, reoperation	↑
Blood loss	Same
Neurologic deficit	**Same** (✗ Wrong options: increased)
Length of stay	Same

Acetabular protrusio

Situation	Treatment
Asymptomatic, stable	Observation
Progressing or painful, immature	**Triradiate cartilage closure** (controversial)
Mature, young, minimal OA	**Valgus osteotomy**
Older, symptomatic OA	**THA**

X vs Y

	Marfan	Homocystinuria
Lens	**Superior** dislocation	**Inferior** dislocation

How it's asked

"Marfan – defective gene encodes?" → **fibrillin**
✗ Wrong options: elastin · FGFR3 · collagen I/II

"Finding most linked to FBN1 mutation?" → **dural ectasia**
✗ Wrong options: Sprengel · sacral agenesis · fibular hemimelia · Lisch nodules

"Misfolded glycoprotein, TGF-β accumulation, dural ectasia – associated?" → **positive thumb and wrist signs**
✗ Wrong options: spinal stenosis · odontoid hypoplasia · enlarged calves

"Fibrillin-1 defect → diagnosis?" → **Marfan syndrome**
✗ Wrong options: pseudoachondroplasia (COMP)

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"Two factors synergistic in Marfan aortic aneurysm?"	**→ TGF-β and renin-angiotensin system**
"Elevated TGF-β associated with?"	**→ aortic dilatation** ✗ Wrong options: impaired chondrocyte proliferation
"All associated with Marfan EXCEPT?"	**→ inferior lens dislocation**
"23-yr, bilateral protrusio, wants to conceive?"	**→ cardiovascular evaluation** ✗ Wrong options: genetic testing · pelvic CT
"Tall basketball player, pectus, flatfeet – refer for?"	**→ cardiovascular evaluation** ✗ Wrong options: ophthalmology · genetics
"Marfan scoliosis – all true EXCEPT?"	**→ bracing often effective if started early**
"Vs AIS surgery – all true EXCEPT?"	**→ increased neurologic deficits**
"Routinely needs early surgery in Marfan?"	**→ progressive scoliosis**

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MEDIAL CLAVICLE PHYSEAL FRACTURE

– Pediatrics · 64 / 112 –

1 Basics

Definition — **pseudodislocation of the SC joint** · childhood equivalent of adult **SC separation** · rare · usually **SH I or II**

Mechanism — FOOSH · direct blow (tackle) · abuse rare

Pathoanatomy — physis = **weakest link** · periosteal sleeve + costoclavicular / SC ligaments usually intact → remodels · Lee et al: of 40 operated posterior SC injuries, **50% physeal fractures, 50% true dislocations**

Physeal fusion age

Clavicle — **first bone to ossify** (5th week in utero) · central intramembranous centre <5 yr · lateral epiphysis ossifies ~18 yr

Medial physis — **~80% of clavicular growth** · ossifies **18–20 yr** · fuses **20–25 yr** (Orthobullets, OB) · **~20–21 yr** (TO, Schmeling) · **last physis to close** (body – Orthobullets; upper extremity – OB) → "SC dislocation" in teens/young adults is often a physeal fracture

TO — medial clavicle unites last with humeral head, acromion, ischial tuberosity · males 1–2 yr later than females

Schmeling stage (TO)	Male (yr)	Female (yr)
1	11–14	12–15
2	15–20	14–18
3	16–36	15–26
4	21–40	21–37
5	26–40	26–39

3 Presentation

	Anterior displacement	Posterior displacement
Exam	**Palpable bump**, sharp metaphysis under skin · SCM pulls fragment → head tilts to affected side	Swelling, depression of medial clavicle
Danger	Low	**Dyspnea, dysphagia, stridor**, ↓ distal pulses, venous congestion, **paresthesias**
At risk	–	**Innominate artery + vein, IJV, phrenic + vagus nerves, trachea, esophagus**

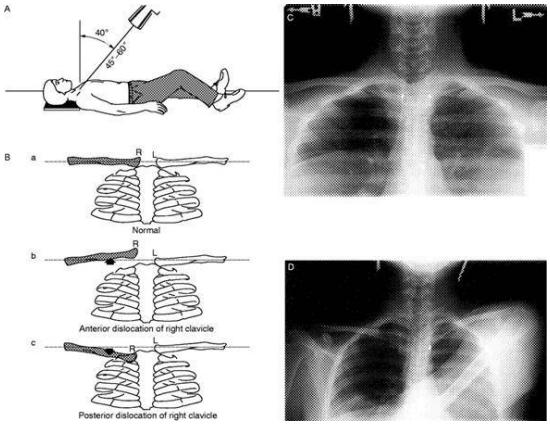
General — pain, ↓ arm ROM · prominence ↑ with abduction / elevation

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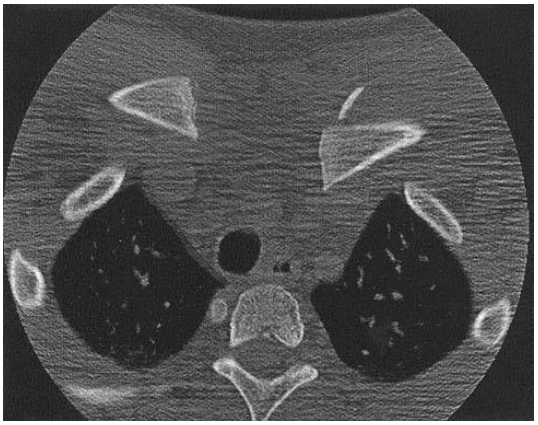
4 Diagnosis & Classification

Key radiology findings

AP X-ray	unreliable
Serendipity view	beam **40° cephalic** · **anterior** = affected clavicle **above** opposite · **posterior** = **below**
Axial CT	**study of choice** · distinguishes physeal fracture from **SC dislocation** · shows mediastinal injury



Serendipity view



Axial CT of medial clavicle physeal fracture

5 Management

Management

Scenario	Treatment
Asymptomatic · **anterior** displacement	**Immobilization for comfort / observation** (remodels)
Posterior, **no mediastinal injury**	Observation
Acute posterior + airway, esophageal or NV compromise	**Closed reduction under GA, thoracic surgeon available**
Failed closed reduction · **chronic symptomatic posterior**	**ORIF**
Late-presenting posterior	No closed reduction (clavicle stuck to mediastinal vessels)

Anterior reduction supine, bolster, traction both arms, posterior pressure ± towel clip · fails → sling

Posterior reduction supine, bolster, traction with **shoulder adducted**, posterior force on shoulder, **towel clip pulls medial clavicle forward** · fails → open

ORIF horizontal incision · **sutures** clavicle → sternum / epiphysis (allow future MRI) · **no pins (migration)** · no plate · CT to confirm stability

6 Complications

Persistent instability rare (children remodel)

Subclavian artery/ vein laceration rare · **rapidly expanding hematoma** · thick periosteum protective → vessel repair

Pin migration avoid pins around clavicle

How it's asked

"9-yr tackled, CT 2 cm anterior displacement, no distress?"	→ **immobilization for comfort** ✗ Wrong options: ORIF · closed reduction with thoracic backup · pinning
"Medial clavicle physis – true?"	→ **fusion around 20–21 yr** (TO) ✗ Wrong options: fusion 15–16 · appears 10–11
"10-yr medial clavicle physeal injury – fuses when?"	→ **20–25 yr**

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MEDIAL EPICONDYLAR FRACTURES - PEDIATRIC

– Pediatrics · 65 / 112 –

1 Basics

Epidemiology

Frequency	**3rd most common** pediatric elbow fracture · up to **20%** of elbow fractures (Orthobullets, OB) · 12% (OB5909) · **75% boys 9–14 yr** · rising with youth sport
Mechanism	**avulsion**: valgus stress + **flexor-pronator contraction** (throwers) · direct blow
Elbow dislocation	present in **50–60%** · **most common fracture with pediatric elbow dislocation** (33% of dislocations, Rasool) · fragment stays **incarcerated in ~15%**

Anatomy & ulnar nerve risk

Medial epicondyle	**apophysis** (no longitudinal growth) · **last distal humeral centre to fuse** · origin of **flexor-pronator mass + UCL** · lies **posteromedial**, fragment displaces **ANTERIORLY** (flexor-pronator pull)
Flexor-pronator mass	**pronator teres** · FCR · palmaris longus · FDS · FCU ✗ Wrong options: pronator quadratus · brachioradialis · ECRL · brachialis
Blood supply	anterior: inferior ulnar collateral a. · posterior: superior + inferior ulnar collateral aa.
Ulnar nerve	**C8–T1**, **medial cord** · splits two heads of FCU → runs **between FDP and FCU** · Guyon's canal, **superficial** to transverse carpal ligament · sensation **palmar + dorsal ulnar 1½ digits** · injury → ↓ wrist palmar flexion (FCU) + ring/small finger numbness

Centre	Appears (Orthobullets)	Fuses
Capitellum	1	12–14
Radial head	3	14–16
Medial epicondyle	**5**	**16–18**
Trochlea	7	12–14
Olecranon	9	15–17
Lateral epicondyle	11	12–14

CRITOE (OB214100 variant)	1, 3, 4, 5, 7, 9 yr · same text: medial side not ossified before ~5, trochlea complete ~9
Trochlea seen, medial epicondyle not (TO)	CRITOE order broken → **avulsed medial epicondyle with entrapment** until proven otherwise

3 Presentation

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Exam	medial pain · valgus instability · ecchymosis · **document ulnar nerve motor + sensory** · generalised swelling → elbow probably dislocated
Classification	none routine · acute (nondisplaced, minimally displaced, displaced, **entrapped**, through apophysis) vs chronic (tension stress)

4 Diagnosis & Classification

Displacement assessment & imaging

Views	AP + lateral + **internal oblique** · **distal humeral axial view** (beam **25° anterior** to humeral axis) = **most accurate X-ray** for displacement
Pitfall	AP, lateral, IR views **underestimate**; fragment invisible on lateral if <10 mm
CT	most accurate, more radiation
Arthrogram	before ossification · e.g. **displaced medial condyle fracture in a 4-yr-old** ✗ Wrong options: nondisplaced medial epicondyle at 6 · radial head at 5
Young child (<5 yr)	unossified epicondyle → **irreducible dislocation = incarcerated epicondyle until proven otherwise** ✗ Wrong options: capitellum / radial head fracture (would be visible)



Distal humeral axial view technique



Medial epicondyle incarcerated in joint after reduction

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5 Management

Medial epicondyle fracture

- Incarcerated fragment / open / medial condyle extension → **ORIF (absolute)**
- Thrower or gymnast >2–5 mm, ulnar nerve dysfunction, dislocation → **consider ORIF (relative)**
- Others → **long arm cast 1–3 wk at 90°, then early motion**

Incarcerated fragment & post-reduction incongruity

Post-reduction incongruent ulnohumeral joint → interposed medial epicondyle
✗ Wrong options: annular ligament · brachialis · ulnar nerve

Roberts maneuver — **forearm supination + wrist/finger extension + valgus load** → may pull fragment out
✗ Wrong options: Patterson (radial neck)

Persistent incarceration — **absolute indication for ORIF**
✗ Wrong options: repeat closed reduction · CRPP · immobilization

With ulnar paresthesia — **ORIF**
✗ Wrong options: observation · nerve transposition (not usually needed) · UCL reconstruction

Operative indications & union (displacement, throwers)

Absolute	Relative
Incarcerated fragment	**Ulnar nerve dysfunction**
Medial condyle / articular extension	Displacement **>2–15 mm** (controversial) · 2–10 mm (OB)
Open fracture	**>2–5 mm in throwers / gymnasts** (valgus stability)
–	Associated dislocation · valgus instability

Non-op — long arm cast **1–3 wk**, elbow **90°** (AROM within ~1 wk) · historically **<5 mm** · RCTs: mean ~7 mm with equal outcomes · grip + ROM equal by 3 months

Non-op vs ORIF — **functional outcome equivalent** (displacement <15 mm) · **↑ nonunion** without surgery · **union odds 9× with surgery**
✗ Wrong options: cubitus varus · fishtail · ↑ ulnar neurapraxia

Throwing athlete, 7 mm or large displaced fragment → ORIF (highest union, valgus stability)
✗ Wrong options: excision (destabilises) · CRPP (ulnar nerve risk)

ORIF technique — **medial approach**, supine, arm abducted 90° (or prone) · interval **brachialis (musculocutaneous) / triceps (radial)** · find + protect **ulnar nerve proximal → distal** · **cannulated screw** ± washer · K-wires for small fragments / young children · avoid comminution · bicortical screw risks radial nerve
✗ Wrong options: triceps/brachioradialis (lateral condyle) · brachialis/brachioradialis



Screw fixation of medial epicondyle

6 Complications

Asymptomatic fibrous union

Nonunion	common after non-op · **majority asymptomatic** · poor correlation with function ✗ Wrong options: Tinel · valgus pain · malalignment
Ulnar nerve injury	**10–16%** (~10% OB) · neurapraxia after dislocation usually resolves · delayed palsy from missed fracture recovers if treated promptly
Missed incarceration	esp. young child with unossified epicondyle
Stiffness	**most common**: loss of a few degrees of **extension** · ↑ with prolonged immobilization
Prognosis	good to excellent with either treatment

X vs Y

	Medial epicondyle	Medial condyle
Frequency	12–20%	**1–2%**
Age	9–14 yr	Younger (trochlea unossified)
Articular	**Extra-articular** apophysis	**Intra-articular**, often SH IV
Risk	Incarceration, ulnar nerve	**Nonunion**, growth arrest, **AVN** (posterior terminal supply), misdiagnosis
Treatment	Mostly non-op	**ORIF** (anatomic) · arthrogram if unossified

How it's asked

"11-yr, elbow reduced but ulnohumeral joint incongruent?"	**→ interposed medial epicondyle fragment**
"11-yr monkey bars, fragment in joint after reduction?"	**→ ORIF** ✗ Wrong options: hinged brace · sling · pinning

– – Active space – –

"ORIF indicated because?"	→ **incarcerated fragment in ulnohumeral joint** ✗ Wrong options: >5 mm · prevent cubitus valgus
"8-yr gymnast, incarcerated after reduction — approach?"	→ **ORIF through brachialis/triceps plane** ✗ Wrong options: triceps/ brachioradialis
"12-yr, incarcerated + ring/little finger numbness?"	→ **ORIF** ✗ Wrong options: nerve transposition · splint and observe · UCL reconstruction
"Roberts maneuver fails, fragment still in joint?"	→ **ORIF** ✗ Wrong options: Patterson · CRPP
"Which X-ray most likely needs surgery?"	→ **post-reduction entrapped medial epicondyle**
"4-yr, irreducible posterolateral dislocation, no fracture seen?"	→ **incarcerated medial epicondyle** ✗ Wrong options: capitellum · radial head
"5 mm displaced, cast — vs surgery?"	→ **↑ nonunion** ✗ Wrong options: worse function · cubitus varus
"15-yr pitcher, 7 mm displaced — highest union?"	→ **ORIF** ✗ Wrong options: brace · excision
"10-yr, displaced rotated medial condyle fracture?"	→ **ORIF** ✗ Wrong options: casting
"17-yr baseball player, significantly displaced?"	→ **ORIF** ✗ Wrong options: excision · CRPP
"Direction of displacement + best view?"	→ **anteromedial; distal humeral axial view**
"Arthrogram most useful for?"	→ **displaced medial condyle fracture in a 4-yr-old**
"Muscle causing avulsion?"	→ **pronator teres**
"Structure at risk in repair?"	→ **ulnar nerve**
"Post-op wrist flexion weakness + ring/small numbness — nerve?"	→ **medial cord, between FDP and FCU**
"5-yr medial condyle ORIF — C8–T1 nerve injured, numbness where?"	→ **palmar ulnar 1½ digits**
"Fracture most associated with pediatric elbow dislocation?"	→ **medial epicondyle**
"Pitcher, casted 3 wk, radiographic nonunion?"	→ **asymptomatic**

METATARSUS ADDUCTUS

– Pediatrics · 66 / 112 –

1 Basics

Definition forefoot **adduction at the tarsometatarsal joint** with **normal hindfoot** · **packaging** disorder (intra-uterine position)

Epidemiology **1 in 1,000** · M = F · **bilateral ~50%** · ↑ late pregnancy, **first pregnancy**, **twins**, **oligohydramnios**

Associated **DDH (15–20%)** · torticollis

3 Presentation

Diagnosis

History parents notice **in-toeing in first year**

Exam adducted forefoot · **convex lateral border** · **medial soft-tissue crease → more rigid** · **normal hindfoot + subtalar motion** · tickle foot to test **active** correction
✗ Wrong options: straight

TO case 5-month-old, **no medial crease**, gap between 1st/2nd toes, adduction correctable to 10° abduction · options: metatarsus adductus / constriction band / metatarsus primus varus / positional MA · **answer not provided by the source** · no crease + correctable = flexible deformity

Causes of intoeing

Condition	Key finding
Metatarsus adductus	Forefoot medially deviated, **hindfoot normal** · heel bisector lateral
Internal tibial torsion	**Thigh-foot angle >10° internal** (normal 0–20° external)
Femoral anteversion	Prone **IR >70°** (normal 30–60°), **ER <20°** · patella points in

Work up intoeing if pain · LLD · progression · family history of rickets / dysplasia / MPS · rotational profile **>2 SD** from normal

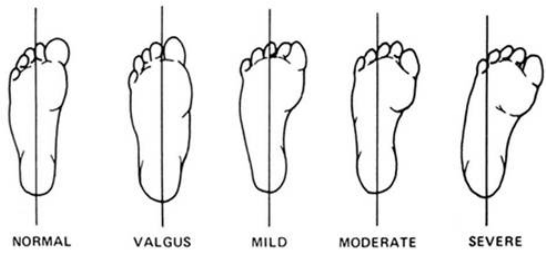
4 Diagnosis & Classification

Bleck classification (heel bisector)

Grade	Heel bisector passes through
Normal	**2nd–3rd web space**
Mild	3rd toe
Moderate	3rd–4th web space
Severe	4th–5th web space

– – Active space – –

Heel bisector grades (TO) normal 2nd–3rd toe · mild **3rd toe** · moderate **4th toe** · severe **lateral to 4th toe** · types: flexible · rigid · **dynamic (tibialis anterior overactivity)**



Heel bisector line

Berg classification

Type	Features
Simple MTA	MTA
Complex MTA	MTA + **lateral shift of midfoot**
Skew foot	MTA + **valgus hindfoot**
Complex skew foot (**serpentine**)	MTA + lateral shift + valgus hindfoot

Serpentine foot TMT adductus + **talonavicular lateral subluxation** + hindfoot valgus · **casting ineffective**
✗ Wrong options: like simple MTA

X-rays only in older children

Other foot differentials clubfoot · **atavistic great toe** (congenital hallux varus) · skewfoot

5 Management

Treatment - observation & stretching

Natural history **90% resolve by age 4** · another 5% in early walking years · flexible: **90–95% spontaneous correction**

Flexibility	Treatment
Actively correctable to midline	**No treatment – reassurance + observation**
Passively correctable	**Serial stretching by parents**
Rigid, medial crease	**Serial casting** (goal straight lateral border) · **before 6 months** (OB)



Rigid metatarsus adductus with medial crease

Shoes, braces, splints efficacy not established · (✗ Wrong options: Ponseti (clubfoot) · Achilles lengthening unless tight heel cord)
✗ Wrong options: dynamic AFO

Residual deformity – tarsometatarsal capsulotomy

Indication	Surgery
Age 2–4 yr, failed non-op	**TMT capsulotomies**
Age >5 yr, resistant (medial crease), pain / shoe problems	**Lateral column shortening (cuboid closing wedge) + medial column opening (cuneiform) wedge** · multiple metatarsal osteotomies · abductor hallucis recession (atavistic toe)
Symptomatic **serpentine** foot	Calcaneal osteotomy + midfoot osteotomies + **multiple metatarsal osteotomies with pinning + TMT release (Hamen)** · often accepted / observed

Surgery timing (OB) >3 yr or rigid unresponsive to casting · medial release if symptomatic beyond 4 yr

Campbell algorithm (TO) mild: observe · moderate: stretching + serial casting · severe **2–4 yr: transmetatarsal (TMT) release/osteotomy** · **> 4 yr: multiple metatarsal osteotomies (Berman and Gartland)**

Indications (TO) pain · objectionable appearance · **difficulty fitting shoes** after the age for stretching/casting has passed

Prognosis residual MTA **not linked to pain or ↓ function** · late **medial cuneiform obliquity**
✗ Wrong options: hallux valgus

X vs Y

	Metatarsus adductus	Clubfoot	Skewfoot
Forefoot	Adducted	Adducted	Adducted
Hindfoot	**Normal**	**Varus + equinus**	**Valgus**
Ankle	Full DF/PF	Equinus	–
First line	Observe / stretch / cast	Ponseti	Often accept; osteotomies if symptomatic

– – Active space – –

How it's asked

"Newborn MTA, corrects actively and passively?"	→ **parental reassurance and observation** ✗ Wrong options: stretching · casting · Ponseti
"Newborn mild–moderate MTA — initial treatment?"	→ **observation with possible stretching** ✗ Wrong options: serial casting · medial release · dynamic AFO
"3-yr, moderate MTA, failed stretching/casting, shoe problems?"	→ **tarsometatarsal capsulotomies** (TO) ✗ Wrong options: observation · more casting · metatarsal osteotomies

MONTEGGIA FRACTURE - PEDIATRIC

– Pediatrics · 67 / 112 –

1 Basics

Definition	**radial head dislocation** + proximal ulna fracture **or plastic deformation** of ulna
Age	peak **4–10 yr**
Rule	**isolated radial head dislocation almost never occurs in children** → look for ulnar plastic deformation
Pattern	Bado **I most common in children** · II more common in adults · children do better than adults

3 Presentation

Exam	pain, swelling, deformity · **palpate radial head with every ulna fracture** (radial head often relocates spontaneously)

4 Diagnosis & Classification

Key radiology findings

Rule	**elbow X-rays for every forearm / ulna fracture** · **forearm X-rays for every radial head dislocation** · orthogonal films of whole forearm, wrist + elbow
Radiocapitellar line	line along radial neck axis passes through **centre of capitellum on every view** · any deviation = Monteggia

Ulnar shaft fracture after ED reduction → dedicated elbow radiographs before anything else
✗ Wrong options: cast · fixation



Radiocapitellar line

– – Active space – –

Bado classification

Type	Ulna apex	Radial head	Cast position
I	**Anterior**	**Anterior** (most common in children)	**110° flexion + full supination**
II	Posterior	Posterior	Full extension
III	**Lateral**	**Lateral**	Full extension + **valgus mold**
IV	**Radius + ulna** fractured same level	**Anterior** (1–11%)	– operative



Bado type I

5 Management

Pediatric Monteggia

- Length-stable ulna (plastic / greenstick / buckle) + stable RC joint → **closed reduction + long arm cast**
- Length-unstable ulna or RC unstable → **fix ulna (IM nail or plate) + closed RC reduction**
- RC joint irreducible after anatomic ulna → **open reduction (annular ligament block)**
- Bado IV · open · ≥10 yr unstable · closed physes → **operative**

Acute treatment - restore ulnar length

First consideration — **reduce + maintain ulnar length and alignment** → radial head usually reduces itself
✗ Wrong options: open RC reduction first · annular ligament reconstruction · RC pin

Closed reduction first — traction · Type I: **flexion** is main maneuver (+ supination) · prompt reduction → delay makes it difficult/impossible
✗ Wrong options: plate · pins · splint without reduction

Why supination + 110° flexion
tightens interosseous membrane + **relaxes biceps** · annular, quadrate, oblique + interosseous ligaments tighten in supination
✗ Wrong options: tightens biceps

Operative indication	
Types I–III with **unstable radial head** after reduction	Yes
Ulnar length not maintained (complete, comminuted, displaced)	Yes
Acute Bado IV (even if head reduces)	Yes
Open fracture · ≥10 yr unstable · closed physes (adult ORIF)	Yes
Loss of RC reduction on follow-up	Yes
Incomplete ulna fracture + reduced radial head	No (cast)

Ulna fixation choice
transverse / short oblique → IM flexible nail · **long oblique / comminuted → open reduction + plate** (nails won't prevent shortening)

After failed closed treatment (Bado I, lost reduction at 5 days)
fix ulna + closed RC reduction + cast in supination
✗ Wrong options: open RC · annular / LCL repair (rarely needed)

Comminuted ulna, RC won't hold
closed RC reduction + ORIF plate ulna
✗ Wrong options: RC pin (breaks) · IM nail · repeat closed reduction



Intramedullary nailing of ulna

Irreducible radial head - annular ligament

Block to RC reduction
annular ligament (most common, trapped between capitellum and radial head) · rare: biceps, PIN, capsule

Annular ligament function
stabilises the proximal radioulnar joint
✗ Wrong options: PIN · LUCL · MUCL · biceps

Check
after anatomic ulna, examine radial head in pronation **and** supination · still out → **open reduction**

Chronic missed Monteggia

Presentation
pain, ↓ motion, progressive valgus after missed / delayed diagnosis

Treatment
open reduction of radial head (lateral approach, radial head still concave) ± **annular ligament repair/reconstruction**, **ulnar osteotomy**, radial osteotomy · cast in **supination + flexion**
✗ Wrong options: closed reduction · bracing

Acute
annular ligament reconstruction almost never needed

– – Active space – –

X vs Y

	Acute Monteggia	Chronic (>2–3 wk) missed
Key step	**Restore ulnar length** → RC reduces closed	**Open RC reduction**
Annular ligament	Rarely addressed	Repair / reconstruct
Extra	IM nail or plate ulna	**Ulnar osteotomy**

How it's asked

"6-yr, comminuted ulna + anterior RC dislocation, RC won't stay reduced?"	→ **closed RC reduction + ORIF plate of ulna** ✗ Wrong options: RC pin · IM nail
"Surgical indication EXCEPT?"	→ **incomplete ulna fracture with lateral dislocation successfully reduced**
"Why cast at 110° flexion + full supination?"	→ **tightens interosseous membrane** ✗ Wrong options: tightens biceps
"Lost ulnar + RC reduction at 5 days?"	→ **fix ulna, closed RC reduction, cast in supination** ✗ Wrong options: annular / LCL repair
"5-yr monkey bars Monteggia – first consideration?"	→ **reduce and maintain ulnar fracture** ✗ Wrong options: open RC · annular reconstruction
"5-yr trampoline Monteggia – next step?"	→ **closed reduction** ✗ Wrong options: plate · splint
"Ulnar shaft fracture post-reduction – next?"	→ **dedicated elbow radiographs**
"Failed closed RC reduction – blocking structure?"	→ **annular ligament** ✗ Wrong options: biceps · PIN
"Interposed structure's function?"	→ **stabilises proximal radioulnar joint**
"3-month-old radial head dislocation, closed reduction fails?"	→ **open reduction + cast in supination and flexion**

MUCOPOLYSACCHARIDOSES

– Pediatrics · 68 / 112 –

1 Basics

Definition **13 lysosomal storage disorders** · missing/faulty lysosomal enzymes → **glycosaminoglycans (GAGs)** accumulate in chondrocytes, cartilage ECM, synovium, organs

Most common **Morquio + Sanfilippo** (Orthobullets) · Morquio (OB)

Genes & lab markers

Syndrome	Enzyme	Urine GAG	Inheritance	Intelligence	Cornea	Prognosis
Morquio (MPS IV)	A: galactosamine-6-sulfatase · B: β-galactosidase	**Keratan sulfate**	AR	**Normal**	**Cloudy**	Adulthood (A worse)
Hurler (MPS I)	**α-L-iduronidase**	**Dermatan** (+ heparan, TO)	AR	Progressive MR	**Cloudy**	**Death 1st decade** · most severe
Sanfilippo (MPS III)	Several	**Heparan sulfate**	AR	MR	**Clear**	Death 2nd decade
Hunter (MPS II)	Sulpho-iduronate sulfatase	Dermatan + heparan	**X-linked recessive**	MR	**Clear**	Death by 2nd decade

Tests urine **toluidine blue spot** test · **skin fibroblast** enzyme assay · **chorionic villous sampling**

Clinical features (CTS, contractures)

Orthopaedic **proportionate dwarfism** · **C1–C2 instability** · **carpal tunnel syndrome** (esp. MPS I Scheie; thick retinaculum) · trigger finger · **progressive joint contractures** (✓ most common early MSK sign — wrist, digits, hips, knees) · delayed hip dysplasia · abnormal epiphyses · **bullet-shaped phalanges/ metacarpals** · **genu valgum** (lateral proximal tibial metaphyseal dysplasia) · thoracolumbar kyphosis (L1–L2 beaking, gibbus)

Non-orthopaedic complex sugars in urine · visceromegaly · corneal clouding · cardiac disease · deafness · **MR (except Morquio)** · enlarged skull · coarse facies

Not in MPS **dural ectasia** (Marfan, EDS, NF1, AS) · Klippel-Feil triad · sacralization of L5 · congenital ACL absence (NF1, fibular hemimelia) · block vertebrae (VACTERL) · congenital radial head dislocation

Lysosomal CTS Hurler, Hunter, Scheie, pseudo-Hurler · Phalen/Tinel unreliable in children · **transverse carpal ligament release**

3 Presentation

Morquio syndrome

Onset **18–24 months** · ~1 in 40,000 · short-trunk dwarfism

– – Active space – –

Features	**normal intelligence** · waddling gait · **genu valgum** · thoracic kyphosis · corneal clouding · skin + joint laxity · hip dysplasia
X-ray	thick skull · wide ribs · **vertebral beaking** · **platyspondyly** · coxa vara with unossified head · bullet-shaped metacarpals · **odontoid hypoplasia → C1–C2 instability**



Genu valgum in Morquio

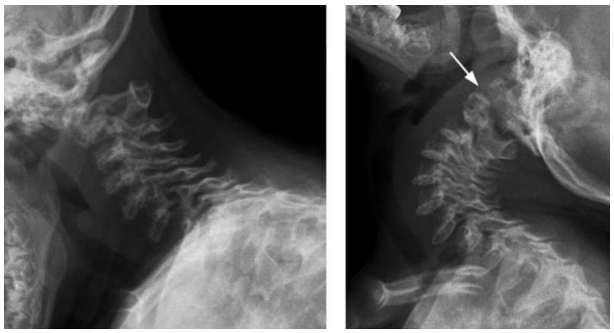
Hurler syndrome (gargoylism)

Features	**gibbus (thoracolumbar kyphosis) — often first sign, ~6 months** (TO) · C1–C2 instability (25%) · severe global spinal stenosis · CTS + trigger fingers · severe hip dysplasia · genu valgum · coxa valga → subluxation · J-shaped sella, thick calvarium, platyspondyly (TO)
Survival	median **6.8 yr** untreated (TO)

4 Diagnosis & Classification

Spine screening & atlantoaxial instability (Morquio, Hurler)

Odontoid hypoplasia + AAI	**Morquio** most classic · OB220568 + TO: Hurler also has odontoid hypoplasia — read the options ✗ Wrong options: Hurler per OB2836 (Hurler instability "not from odontoid hypoplasia")
Morquio pre-op	**C-spine flexion–extension X-rays** (intubation risk) · GAG behind dens → stenosis
Hurler pre-op	**MRI of whole spine** ✗ Wrong options: Kniest · Schmid · Turner
Others with odontoid hypoplasia	pseudoachondroplasia (normal urine) · diastrophic dysplasia (sulfate transporter, cervical kyphosis, cauliflower ears, hitchhiker thumb) · SED · Down (no hypoplastic dens, abnormal intelligence)
Pre-op C-spine NOT needed	Friedreich ataxia · Gaucher · MED · cleidocranial dysplasia



Odontoid hypoplasia with atlantoaxial instability

5 Management

Morquio	**decompression + C1–C2 fusion** for myelopathy with instability · realignment osteotomies (malalignment recurs) · guided growth (little evidence)
AAI surgery triggers	unsteady gait, limb weakness, dysesthesias, urinary retention

Treatment – bone marrow transplant

HSCT / BMT	↑ **life expectancy** (Hurler if done in 1st year · Sanfilippo) · **does NOT change orthopaedic manifestations** (poor enzyme penetration of cartilage/bone)
Age	HSCT **<24 months** → better development · **<2 yr severe MPS I → HCT preferred** (TO)
Graft failure	**relatively HIGH (~30%)** + GVHD (TO) ✗ Wrong options: "low incidence"
Enzyme replacement (ERT)	↓ visceral + cardiac GAG, ↑ joint mobility, endurance, respiration · **can't cross BBB** → no help for cognition or **cord compression** · little effect on valves, bone, cornea · approved MPS I, II, IVA, VI, VII

X vs Y

	MPS	MED	Achondroplasia
Dwarfism	**Proportionate**	Short-limbed, disproportionate	Rhizomelic
Urine	**GAGs**	Normal	Normal
Spine	AAI, kyphosis	**Normal**	Stenosis

How it's asked

"Lysosomal storage disease – increased rate of?"	→ **carpal tunnel syndrome** ✗ Wrong options: syringomyelia · diastematomyelia
"Morquio – all EXCEPT?"	→ **low hairline + webbed neck + ↓ C-spine ROM (Klippel-Feil)**

– – Active space – –

"6-yr on enzyme replacement — all EXCEPT?"	→ **dural ectasia** ✗ Wrong options: kyphosis · genu valgum · CTS · AAI
"MPS I — most common MSK abnormality?"	→ **progressive joint contracture**
"Odontoid hypoplasia AAI — which MPS?"	→ **Morquio** ✗ Wrong options: Hurler · Hunter
"Normal intelligence, urinary keratan sulfate, unstable C1–C2?"	→ **Morquio** ✗ Wrong options: Sanfilippo · pseudoachondroplasia · Down
"Needs C-spine X-rays pre-op?"	→ **Morquio syndrome** ✗ Wrong options: MED · Gaucher · cleidocranial
"Genu valgum growth modulation — whole-spine MRI first?"	→ **Hurler syndrome**
"FALSE about MPS treatment?"	→ **low incidence of graft failure and GVHD after HSCT** (TO)

MULTIPLE EPIPHYSEAL DYSPLASIA (MED)

– Pediatrics · 69 / 112 –

1 Basics

Definition failure of formation of **secondary ossification centres** → irregular, **delayed epiphyseal ossification** at multiple sites → secondary articular cartilage deformity

Epidemiology ~**1 in 10,000** · presents **5–14 yr**

Sites **proximal femur + proximal humerus** most · **hips** = most common joint

Diagnosis & genetics (COMP)

Inheritance	Gene	Note
Autosomal dominant (most common)	**COMP**, chromosome **19**	**Most common + most severe**
AD	**Type IX collagen** (COL9A1/2/3)	Link protein for type II collagen → articular cartilage
AD	**Matrilin 3**	–
Autosomal recessive	**SLC26A2** (diastrophic dysplasia sulfate transporter)	≥1 finding at birth: clubfoot, cleft palate, clinodactyly

Up to ½ of clinical MED have none of the typical mutations

3 Presentation

Features **short-limbed, disproportionate dwarfism** (mild) · joint pain · **waddling gait** / bilateral gluteus medius lurch · valgus knees · early OA · contractures · **short stubby fingers + toes**

Normal **spine** · neuro exam · **intelligence** · facies

4 Diagnosis & Classification

Key radiology findings

General **irregular, delayed ossification at multiple epiphyses** · multiple **OCD** lesions

Hip **bilateral, symmetric** proximal femoral epiphyseal flattening · **early acetabular changes** · **no metaphyseal cysts** · acetabular dysplasia · may have concurrent AVN

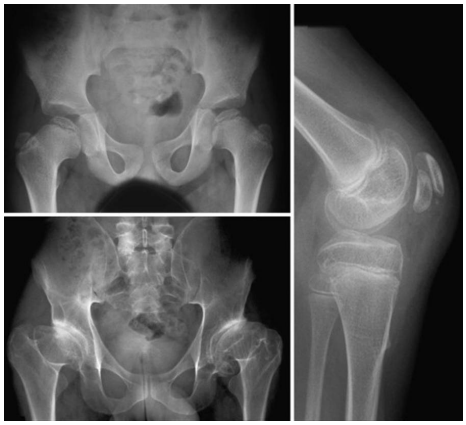
Knee valgus · flattened condyles · **double-layer patella**

Hand / foot short stunted metacarpals · hyperextensible fingers · short metatarsals

– – Active space – –



MED – bilateral symmetric femoral head changes



Double-layer patella and bilateral hip involvement

Workup - skeletal survey

Bilateral symmetric femoral head flattening → skeletal survey
✗ Wrong options: bone scan · MRI · CT · CBC/CRP

Unilateral flattening → Perthes (no survey needed)

Symmetric bilateral + family history of early OA + myopia → skeletal survey + COL2A1 genetic screening (type II collagen disorder mimicking LCPD)
✗ Wrong options: MRI for AVN · steroids for JIA · Perthes care

Skeletal survey not for CP windswept hips · SCFE · hamstring avulsion · unilateral LCPD

5 Management

Problem	Treatment
Early OA	**NSAIDs + PT**
Childhood acetabular dysplasia	Often resolves by maturity
Progressive genu varum / valgum	**Realigning osteotomy or hemiepiphysiodesis**
Severe hip arthritis (**by age ~30**)	**THA**

X vs Y

	MED	Bilateral Perthes	SED
Hips	**Symmetric, same stage**	**Asymmetric, different stages** (bilateral 10–15%)	Involved
Acetabulum	Early changes	Late	–
Metaphyseal cysts	**No**	Yes (head-at-risk)	–
Spine	**Normal**	Normal	**Involved**, AAI, myelopathy, sharp curve
Gene	**COMP**, type IX collagen	–	**Type II collagen**

How it's asked

"9-yr short, valgus knee, waddling, COMP mutation?"	→ **multiple epiphyseal dysplasia** ✗ Wrong options: achondroplasia · diastrophic · Perthes
"Which X-ray needs a skeletal survey?"	→ **symmetric bilateral femoral epiphyseal flattening (MED)** ✗ Wrong options: unilateral Perthes · SCFE
"5-yr, mild short stature, bilateral gluteus medius lurch, flat femoral heads?"	→ **skeletal survey**
"6-yr, symmetric bilateral head irregularity, family early OA + myopia?"	→ **skeletal survey + COL2A1 sequencing**

– – Active space – –

MYELODYSPLASIA (MYELOMENINGOCELE, SPINAL BIFIDA)

– Pediatrics · 70 / 112 –

1 Basics

Definition — failure of fetal neural tube closure → anatomic anomalies + neurologic deficit · incidence **0.1–0.2%**

Prevention - folic acid

Risk factors — **folate deficiency** (supplement ↓ risk **70%**) · maternal hyperthermia · maternal diabetes · **valproic acid**

Supplement — **folic acid**
✗ Wrong options: vitamin D · B12 · niacin · thiamine

Genes & lab markers

Chromosomal — up to **10%** of fetuses: **trisomy 13, trisomy 18**, triploidy, single-gene mutations

Maternal AFP — **↑ in 75%** of open spina bifida · **2nd trimester** · fetal ultrasound diagnoses in utero

Latex allergy

Mechanism — **IgE-mediated type 1 hypersensitivity** → anaphylaxis · **20–70%** of patients · from repeated latex exposure · **latex-free OR** · non-spina bifida latex allergy = type 4 contact dermatitis
✗ Wrong options: IgM · IgA · complement · T cells

Associations

Neurosurgical — **Chiari II** (most common associated anomaly) · **hydrocephalus 70%** · **tethered cord**

Other — neurogenic bladder · pathologic fractures · scoliosis / kyphosis · hip dislocation + contractures · tibial torsion · foot deformity

Split cord malformation & dysraphism types

Form	Feature
Spina bifida occulta	Arch defect, **cord + meninges confined**
Meningocele	Sac **without** neural elements
Myelomeningocele	Sac **with neural elements** · **most common dysraphism** (TO)
Rachischisis	Neural elements **exposed**, no cover

Open vs closed (TO) open = skin defect, neural tissue exposed · closed = skin-covered, ± subcutaneous mass (lipomyelocele, lipomyelomeningocele, meningocele, myelocystocele)

Split cord (diastematomyelia, TO)	Feature
Type 1	**Two dural tubes**, **rigid bony / cartilaginous septum** · more symptoms
Type 2	**One dural tube**, ± non-rigid fibrous septum, **no bony septum**
Associated	Scoliosis, tethered cord, **hairy tuft**

Exam (TO) gaps between spinous processes / absent bony prominences in sacral triangle → **spina bifida** · step-off → spondylolisthesis · becs de perroquet → spinal TB



Rachischisis

4 Diagnosis & Classification

Functional prognosis - motor level

Level named by **lowest functioning level** · **functional motor level = most important predictor of outcome**
✗ Wrong options: sensory level · hip dysplasia · hydrocephalus

Level	Ambulation	Key motion (muscle)	Foot	Orthosis
L1–L2	**Non-ambulatory**	–	Equinovarus	**HKAFO**
L3	Marginal household · **high hip dislocation risk**	**Hip flexion + adduction** (iliopsoas, adductors)	Equinovarus	**KAFO**
L4	Household + · **key level (quads work)**	**Knee extension** (quads) · ankle DF + inversion (TA)	**Cavovarus**	**AFO**
L5	**Community**	Toe DF (EHL/EDL) · **hip extension + abduction** (gluteus med/min)	**Calcaneovalgus**	AFO
S1	Normal	Ankle PF (gastroc-soleus)	–	Shoes
S2	Normal	Toe PF (FHL)	–	–
S3–4	Normal	Bowel + bladder	–	–

– – Active space – –

OB descriptions L1: hips ER + flexed, equinovarus, HKAFO · L3: hips adducted + flexed, **knee recurvatum**, equinovarus, KAFO

Key radiology findings

X-ray monitor scoliosis / kyphosis, hips, fractures · standing AP ankle → **tibiotalar valgus / talar tilt**

MRI **any change in neuro exam or rapid curve progression → urgent MRI for tethered cord**

5 Management

Hip dislocation management & functional level

Cause paralytic: unopposed flexors + adductors · most common at **L3** · can be teratologic

Rule function depends on **neurologic level, NOT hip status** · open reduction → high redislocation + complications, no functional gain · reduced hips may even hurt more

Scenario	Treatment
Thoracic / L3 and above	**Leave dislocated** · observation + **ROM to prevent contractures** (✗ Wrong options: Pavlik · spica (promote contractures) · open reduction)
L4	Most controversial · may consider if **unilateral** (OB) · L4 household ambulator with full ROM → **no intervention**
Contractures	**Contracture release** (improves ROM → ambulation)

Hip abduction contracture pelvic obliquity, scoliosis → **Ober-Yount** (proximal fascia lata + distal ITB release) if limits sitting/bracing

Hip flexion contracture high lumbar / thoracic → anterior release (iliopsoas, sartorius, rectus, TFL) if **>40°**

Knee

Problem	Treatment
Weak quads	**KAFO**
Flexion contracture **>20°**	**Hamstring lengthening ± posterior capsulotomy** · older / failed → **supracondylar extension osteotomy**
Extension contracture	**Serial casting** to 90° flexion
Tibial torsion	**<5 yr** observe + orthotics · **>5 yr distal tibial derotation osteotomy**

Foot & ankle deformity

Incidence **60–90%** (low nerve roots)

Clubfoot **30%**, most common · **rigid + insensate** · serial casting first (high complication rate) → **posteromedial-lateral release at 12–18 months** · rigid clubfoot, no muscles below knee → **soft-tissue releases** for plantigrade AFO foot · releases beat transfers
✗ Wrong options: tendon transfers (nothing to transfer) · Botox · triple

Calcaneus / dorsiflexion (L5, sacral)	unopposed tibialis anterior · bulky heel pad · **posterior transfer of tibialis anterior** if bracing can't get neutral ✗ Wrong options: triple arthrodesis
Planovalgus	distal tibial valgus (mature) → **supramalleolar osteotomy** · hindfoot valgus → **calcaneal lengthening + TAL** · also calcaneal slide, subtalar arthrodesis, Grice · **tibiotalar valgus missed pre-op → poor surgical result** ✗ Wrong options: equinus · calcaneovalgus · forefoot abduction
Arthrodesis	avoid (skin sores, Charcot) · triple only for severe deformity with **sensate** feet



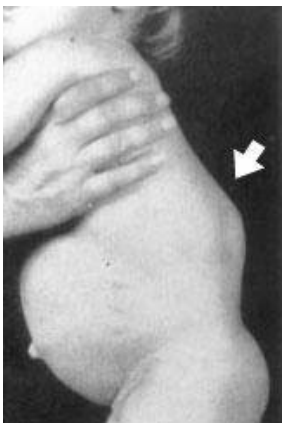
Calcaneus foot in L5 myelomeningocele

Spinal deformity & kyphectomy

Scoliosis	neurogenic or congenital · **100% with thoracic level** · rapid progression → **tethered cord** · **bracing ineffective → ASF + PSF with pelvic fixation** (anterior needed – dysplastic posterior elements) · high **pseudarthrosis** · **infection 15–25%** (poor soft tissue)
Congenital kyphosis	**10–15%** · progressive · **gibbus → recurrent skin breakdown** sitting · **kyphectomy + fusion + posterior instrumentation** · **check shunt function first** (shunt failure intra-op can kill)

Both curves progressive → ASF + PSF with pelvic fixation + kyphectomy
✗ Wrong options: kyphectomy alone · bracing

Multidisciplinary team	**↓ postoperative complications** ✗ Wrong options: ↓ deformity progression · ↓ revision



Gibbus deformity

– – Active space – –

6 Complications

Pathologic fracture & neuropathic joints

Fractures	**disuse osteopenia** · ↑ with higher level · hip + knee, age 3–7 · **fever, warmth, erythema, swelling → mimics infection / cellulitis / osteomyelitis → get X-rays** · not immunocompromised · treat with short **well-padded** splint/cast · **avoid long casting** (↑ osteopenia, refracture)
Prognosis	untreated infants 90–100% mortality · L3 and above mostly wheelchair · L5 good community ambulation

How it's asked

"8-yr mid-lumbar MMC, bilateral hip dislocation – true?"	→ **long-term function not associated with hip status** ✗ Wrong options: reduction improves ambulation
"4-yr L3, unilateral dislocation, walks with walker?"	→ **observation** ✗ Wrong options: open reduction · pelvic osteotomy
"Newborn thoracic MMC, unstable hips?"	→ **observation + ROM** ✗ Wrong options: Pavlik · spica · open reduction
"13-yr, knee extension but no hip extension/abduction, dislocated hip?"	→ **L4; no intervention**
"6-yr, walks, strong DF, bulky heel pad, calcaneus?"	→ **L5; posterior transfer of tibialis anterior** ✗ Wrong options: triple
"18-yr, planovalgus, distal tibia 15° valgus?"	→ **supramalleolar osteotomy**
"18-month, rigid clubfeet, nothing below knee?"	→ **soft-tissue releases** ✗ Wrong options: transfers · Botox
"Pre-op deformity predicting poor foot surgery result?"	→ **tibiotalar valgus**
"Thoracic MMC, progressive scoliosis + kyphosis, skin ulcers?"	→ **ASF + PSF with pelvic fixation + kyphectomy**
"Benefit of multidisciplinary team?"	→ **reduced postoperative complications**
"L1 MMC, fall, febrile warm thigh?"	→ **pathologic femur fracture from disuse osteopenia** ✗ Wrong options: cellulitis · immunocompromise
"Non-ambulatory, painless red swollen knee after PT, osteopenic tibia?"	→ **myelomeningocele image**
"Latex allergy mechanism?"	→ **IgE antibodies**
"Most important predictor of function?"	→ **functional motor level**
"Supplement ↓ neural tube defects?"	→ **folic acid**

"Most common spinal dysraphism?"	→ **myelomeningocele** (TO)
"Gaps between sacral spinous processes?"	→ **spina bifida** (TO)
"Split cord, no bony septum?"	→ **split cord malformation type 2** (TO)

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NEUROFIBROMATOSIS

– Pediatrics · 71 / 112 –

1 Basics

Incidence NF1 **1 in 3,000** (Orthobullets) · 1 in 4,000 (OB) · **~50% new mutations** · **most common genetic disorder from a new single-gene mutation** · neural crest origin

Genetics & inheritance (NF1, neurofibromin)

Inheritance **autosomal dominant**

Gene **NF1 on chromosome 17q11.2 → neurofibromin**
✗ Wrong options: 11q17.2

Pathway neurofibromin **negatively regulates Ras** → deficiency → ↑ Ras → ↑ **Ras-dependent MAPK** (osteoclast function/survival) · ↓ osteoblast activity → osteopenia

Selumetinib oral **MEK1/2 inhibitor** · FDA 2020 for **symptomatic inoperable plexiform neurofibromas in children with NF1**

Gene / protein (distractors)	Disease
Neurofibromin (NF1)	**NF1**
FGFR3	Achondroplasia 10
GNAS / Gs-α	**McCune-Albright** (coast-of-Maine spots, fibrous dysplasia, precocious puberty)
SPRED1	**Legius** (café-au-lait + freckling, **no neurofibromas**, no MPNST)
COMP	MED
Fibrillin	Marfan
COL2A1	SED
RUNX2/CBFA1	Cleidocranial dysplasia
EWS	Ewing (t11;22)
EXT1	Multiple hereditary exostoses
p53 / Rb	Li-Fraumeni / retinoblastoma

Type	Feature
NF1 (von Recklinghausen)	Most common · all orthopaedic features
NF2	**Bilateral vestibular schwannomas** · meningiomas, ependymomas · **no scoliosis** · no pseudarthrosis
Segmental	NF1 features in one body segment

3 Presentation

Diagnostic criteria & clinical features

NIH 1987 – ≥2 of

Criterion	Detail
Café-au-lait	**≥6**, **>5 mm** prepubertal / **>15 mm** postpubertal · smooth "**coast of California**" border
Neurofibromas	**≥2 any type** or **1 plexiform**
Freckling	**Axillary or inguinal** (✗ Wrong options: plantar)
Optic glioma	(✗ Wrong options: glaucoma)
Lisch nodules	**≥2** iris hamartomas
Osseous lesion	**Sphenoid dysplasia** · thin long-bone cortex ± pseudarthrosis
1st-degree relative	With NF1

Orthopaedic — **anterolateral tibial bowing / pseudarthrosis** (most common presentation, more than forearm) · **forearm (radius/ulna) bowing + pseudarthrosis** · **hemihypertrophy** / limb overgrowth · scoliosis, kyphosis · **atlantoaxial instability** · verrucous hyperplasia

NOT NF1 — **talipes equinus** · **posteromedial tibial bowing** · **pectus excavatum** (Marfan) · radial head subluxation (MHE) · glenoid retroversion (brachial plexus palsy) · ACL absence (fibular hemimelia) · DDH · **hemivertebrae**

Crawford (588 pts) — spinal deformity 21% · plexiform neurofibroma 25% · LLD 7.1% · congenital tibial dysplasia 5% · pectus 4.3% · hemihypertrophy 1.4%



Café-au-lait spots

Neoplasia

Plexiform neurofibroma — **pathognomonic** · 4% (Orthobullets) · 25% (Crawford) · limb overgrowth · → **neurofibrosarcoma / MPNST**

MPNST — **50% occur in NF1** · enlarging painful mass · resection · 5-yr survival ~30%

Other — Wilms tumor · **aortic stenosis ~2%** → new fatigue/dyspnea needs urgent cardiology · hypertension · life expectancy ↓ **8–10 yr**

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4 Diagnosis & Classification

Spinal deformity (dystrophic scoliosis, rib penciling, dural ectasia)

Spine — **most common skeletal site** in NF1 (10–60%) · not in NF2

	Nondystrophic	Dystrophic
Curve	Long, gentle, like **AIS**	**Short (4–6 vertebrae), sharp, angular kyphoscoliosis** (thoracic; thoracolumbar per OB)
Recognised	Later	**Earlier**
Bracing	Treat like AIS	**Ineffective**
Surgery	Like AIS	**Early ASF + PSF** (<7 yr if dystrophic)

X-ray (dystrophic) — **vertebral scalloping** (posterior) · **rib penciling** · **enlarged foramina** · rotation

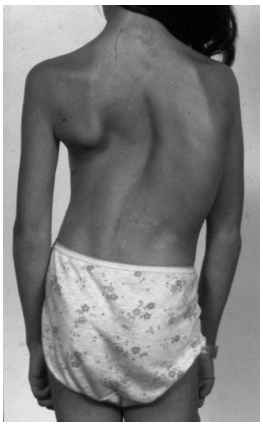
Rapid progression ("modulation") — **penciling of ≥3 ribs** = single strongest predictor (87% progress)
✗ Wrong options: scalloping · curve >50° · rotation

MRI before surgery — always · **dural ectasia** + **dumbbell lesion** (intraforaminal neurofibroma) · paraspinal masses separate from AIS
✗ Wrong options: syrinx · hemivertebra · diastematomyelia

Spinal manifestations — dural ectasia, dumbbell lesions, scalloping
✗ Wrong options: **hemivertebrae** (TO)

Fusion — **PSF alone 40% pseudarthrosis** · **ASF + PSF 10%** · ± repeat iliac crest graft at 6 months · anterior fusion also prevents crankshaft

Cervical kyphosis — **anterior + posterior fusion**
✗ Wrong options: in situ posterior · halo + posterior · collar



Dystrophic scoliosis

5 Management

Congenital pseudarthrosis of tibia

Association — **50% of anterolateral bowing have NF1** (55% OB) · **10% of NF1** have it · pseudarthrosis = NF1 (✗ Wrong options: NF2) · dysplastic **constricted diaphysis** (✗ Wrong options: abuse · OI II · osteopetrosis)

Pathology hamartomatous fibrous tissue in pseudarthrosis · cortical thinning → fracture in first year of walking

– – Active space – –

Tibial bowing	Association
Anterolateral	**NF1 / congenital pseudarthrosis**
Anteromedial	**Fibular hemimelia**, absent lateral rays
Posteromedial	Intrauterine packaging · **calcaneovalgus** · LLD

Anterolateral tibial bowing (NF1)

- No fracture / pseudarthrosis → **total contact (clamshell) orthosis**
  (no remodeling expected; osteotomy contraindicated)
- Fracture / pseudarthrosis → **resect + bone graft + fixation (IM rod / free fibula / Ilizarov)**
- 3 failed surgeries → **Syme amputation (preferred over BKA)**

Infant with fractured pseudarthrosis → genetics referral + brace
✗ Wrong options: abuse workup · ORIF first · cast if not walking

IM rod (Charnley-Williams) resect, antegrade through site then retrograde through heel · **<4 yr → into calcaneus** · **5–10 yr → into talus** · rod pushed proximal ~2 yr later to free ankle

Free fibula usually **contralateral** (ipsilateral abnormal) · **Ilizarov** alternative

Failed bracing/casting with pseudarthrosis → resection + bone graft + IM nailing
✗ Wrong options: IM nail alone (hamartoma left) · bone stimulator

Amputation after **3 failed surgeries** · **Syme** (calf atrophic and scarred)



Anterolateral bowing – congenital pseudarthrosis

Forearm pseudarthrosis

Features bowing, **medullary canal obliterated in distal third**, little trauma · **~50% have NF** · early excision of fibrous tissue + bone replacement (free vascularised fibula)

Forearm X-ray distractors radial club hand (VACTERL, Fanconi, Holt-Oram) · congenital radial head dislocation · Madelung (Leri-Weill) · radioulnar synostosis (Apert, Carpenter, arthrogryposis, Klinefelter)

– – Active space – –

X vs Y

	NF1 café-au-lait	McCune-Albright
Border	**Smooth — coast of California**	**Rough — coast of Maine**
Gene	NF1	GNAS

How it's asked

"NF1 criteria — all EXCEPT?"	→ **plantar freckling**
"3-yr radial bowing + Lisch nodules?"	→ **NF1** ✗ Wrong options: fibrous dysplasia · OI · Ollier
"NF1 features — EXCEPT?"	→ **talipes equinus**
"Associated with NF — EXCEPT?"	→ **posteromedial tibial bowing**
"4-yr anterolateral bowing — exam finding NOT expected?"	→ **pectus excavatum**
"Healed-but-dysplastic tibia, axillary freckles — also expect?"	→ **multiple café-au-lait spots** ✗ Wrong options: posteromedial bowing · glaucoma
"Untreated refugee: tibial pseudarthrosis + dystrophic scoliosis — also?"	→ **hemihypertrophy** ✗ Wrong options: DDH · ACL absence
"NF1 — radiograph most associated?"	→ **congenital forearm pseudarthrosis** ✗ Wrong options: radial club hand · Madelung
"Inheritance + mutation?"	→ **AD; NF1 gene → neurofibromin** ✗ Wrong options: FGFR3 · Gs-α
"Plexiform neurofibroma of hand — gene?"	→ **neurofibromin** ✗ Wrong options: COMP · RUNX2
"Popliteal mass + café-au-lait — gene?"	→ **NF1** ✗ Wrong options: EXT1 · EWS
"Freckling + iris hamartomas — inheritance + leg?"	→ **AD; anterolateral bowing**
"Infant, café-au-lait, scoliosis-operated mother — genetics?"	→ **AD NF1 on 17q11.2**
"Enlarging thigh mass, parent with neurofibromas, MPNST?"	→ **NF1** ✗ Wrong options: Legius · McCune-Albright · NF2
"Pseudarthrosis seen in?"	→ **NF1** (TO)
"3-yr anterolateral bowing, no fracture?"	→ **total contact orthosis** ✗ Wrong options: IM nail · free fibula
"9 months KAFO + cast, now pseudarthrosis?"	→ **resection + bone graft + fixation** ✗ Wrong options: stimulator · Syme

"2-yr constricted tibial diaphysis fracture?"	→ **neurofibromatosis** ✗ Wrong options: abuse
"6-month-old fractured tibial pseudarthrosis – with brace?"	→ **genetics referral** ✗ Wrong options: abuse workup
"6-yr NF1, failed brace + cast?"	→ **resection + bone graft + IM nailing** ✗ Wrong options: IM nail alone
"NF spinal manifestations – EXCEPT?"	→ **hemivertebrae** (TO)
"True about NF scoliosis?"	→ **rib penciling predicts rapid progression** ✗ Wrong options: NF2 · dystrophic like AIS
"Pre-op MRI must assess?"	→ **dural ectasia** ✗ Wrong options: syrinx · diastematomyelia
"15-yr, axillary freckling, short sharp curve?"	→ **dystrophic scoliosis in NF1** ✗ Wrong options: NF2 · Klippel-Feil
"10-yr Risser 0, 50° short thoracic curve, progressing?"	→ **combined anterior + posterior fusion with instrumentation** ✗ Wrong options: TLSO · in situ PSF
"Sign of impending modulation?"	→ **penciling of ≥3 ribs**
"7-yr NF, severe cervical kyphosis?"	→ **anterior + posterior fusion**
"13-yr, axillary freckles, AIS-like curve?"	→ **nondystrophic scoliosis in NF1**
"56-yr NF1, fatigue + dyspnea – urgent?"	→ **aortic stenosis** ✗ Wrong options: MPNST · SMA syndrome
"Selumetinib indicated for?"	→ **inoperable plexiform neurofibromas in NF1 children** ✗ Wrong options: CML (imatinib) · PVNS (pexidartinib) · GCT (denosumab) · desmoid (tamoxifen)

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NURSEMAID'S ELBOW

– Pediatrics · 72 / 112 –

1 Basics

Epidemiology & recurrence

Definition	radial head subluxation / **annular ligament interposed between radial head and capitellum**
Age	**1–4 yr**, average **28 months** (Orthobullets) · peak **2–3 yr** (TO) · TO text also says peaks in first 2 yr · **rare after 5** ✗ Wrong options: "peak 3–5 yr" · "common in infants"
Why rare after 5	thicker, stronger distal annular ligament attachment (Orthobullets) · radial head ossified (TO)
Sex / side	**slightly more common in girls** (✓ TO true statement) · left arm (TO)
Mechanism	**sudden longitudinal traction on hand, elbow EXTENDED + forearm PRONATED** (✗ Wrong options: supinated) · lifting / swinging by the hand · occasionally a fall · annular ligament tears or slips over radial head (✗ Wrong options: LCL tear)
Recurrence	**up to 5%** (TO true statement) · 5–39% reported (TO) · esp. **<3 yr** · capsule stays stretched for months ✗ Wrong options: "recurrence is rare"

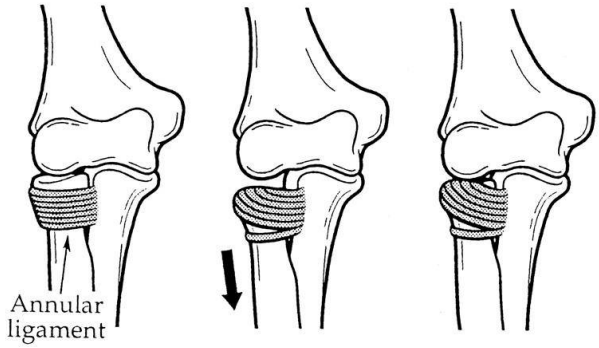
3 Presentation

History	**click** felt by the person pulling · child cries, **refuses to use arm**
Exam	**elbow slightly flexed, forearm pronated** · lateral elbow tenderness · full flexion/extension · **pain / restricted supination**

4 Diagnosis & Classification

Key radiology findings

X-rays	**not needed** with classic history + age ≤5 + typical exam · when taken: **normal** · **25%** radiocapitellar line slightly lateral to capitellar centre
Ultrasound	when mechanism unclear or exam inconclusive · ↑ echo-negative area between capitellum + radial head · **sensitivity 64.9%, specificity 100%** · no radiation
Diagnosis of exclusion	supracondylar, olecranon, radial neck, lateral condyle fractures · contusion · septic arthritis · congenital radial head dislocation · synostosis

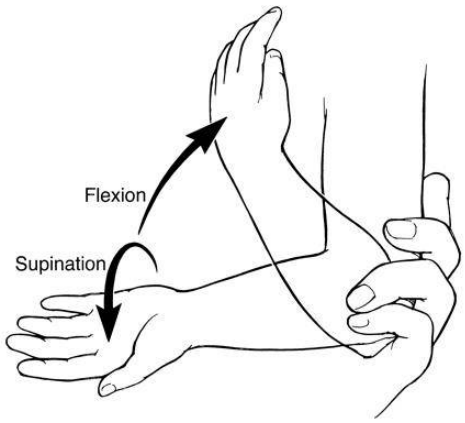


Annular ligament interposed between radial head and capitellum

5 Management

Reduction - supination & flexion

Rule	exclude fracture before manipulating
Supination technique	supinate forearm → **flex elbow** (maximally per Orthobullets · 60–90° OB · 90° TO) · thumb on radial head → **click**
Hyperpronation technique	hyperpronate forearm in flexion
After reduction	child uses arm within minutes · **no immobilization after first episode** ✗ Wrong options: long arm cast (stiffness) · pinning · MRI · observation · longitudinal traction
Recurrent	**cast in flexion + neutral or supination** (lets annular ligament heal)
Open reduction	rare: chronic irreducible symptomatic subluxation



Supination–flexion reduction

Misfiled TO – Weil osteotomy

Weil osteotomy	distal oblique metatarsal head osteotomy (dorsal-distal → proximal-plantar, ~30°, screw) for **metatarsalgia** ✗ Wrong options: Perthes · SCFE · genu valgum
Indications	long / plantarflexed metatarsal · transverse-plane MTP deformity · MTP subluxation/dislocation · recalcitrant metatarsalgia
Complications	**dorsiflexed (floating) toe** · painful/↓ MTP dorsiflexion · PIP contracture

– – Active space – –

How it's asked

"2-yr, elbow flexed + pronated, normal X-rays?"	→ **supination and flexion reduction maneuver** ✗ Wrong options: + long arm cast · CRPP · MRI
"3-yr, arm pulled crossing street, guarding?"	→ **closed reduction via supination and flexion** ✗ Wrong options: aspiration · longitudinal traction · splint
"Pulled elbow – true?"	→ **recurrence in 5% of cases** (TO) ✗ Wrong options: peak 3–5 yr · supinated forearm · LCL tear
"Nursemaid's elbow – true?"	→ **more common in girls** (TO) ✗ Wrong options: infants · recurrence rare
"Weil osteotomy used for?"	→ **metatarsalgia** (TO)

OBSTETRIC BRACHIAL PLEXOPATHY (ERB'S, KLUMPKE'S PALSY)

– Pediatrics · 73 / 112 –

1 Basics

Definition birth stretch injury of brachial plexus, usually difficult vaginal delivery · rare after C-section

Incidence **1–4 per 1,000** (Orthobullets) · 0.9/1,000 · 0.38–1.56/1,000 (OB) · 0.1–0.4% (TO) · falling with better obstetric care · often right-sided or bilateral

Risk factors

Greatest risk **shoulder dystocia**
✗ Wrong options: macrosomia · gestational diabetes · race

Others **macrosomia** (LGA) · multiparous · difficult presentation · **forceps** · **breech** (→ lower plexus) · **prolonged labor** · maternal diabetes · induced labor · prior affected child · **50% have no risk factor**

Other risk factors (TO) **bicornuate uterus** (in utero) · **macrosomia > 4.5 kg** · birth asphyxia · vacuum · humeral / clavicular fracture · **C-section reduces but does not eliminate** risk

Anatomy & injury patterns (Erb upper trunk)

Narakas group	Roots	Deficit	Frequency (TO)	Spontaneous recovery (TO)
I – Erb-Duchenne	**C5–C6**	**Deltoid + biceps** paralysed · wrist/digit flex/ext intact	**46%** (most common)	**>80%**
II – Intermediate / extended Erb	**C5–C7**	+ **wrist and digit extension** (wrist drop)	30%	~60%
III – Total	C5–T1	**Flail**, no Horner	20%	Shoulder/elbow 30–50%
IV – Total + Horner	C5–T1	Flail + **Horner**	–	**Worst**
Klumpke	C8–T1	Claw hand	**<1%** (TO) · 2–3% (TO)	Poor

Erb frequency elsewhere 50–60% (TO) · >80% (OB) – read the options

– – Active space – –

OB212897 variant most common injury = deltoid + biceps paralysis **+ weak wrist/digit extension** (C6) · isolated deltoid + biceps = C5 only

Root	Muscles (OB618)
C5	**Deltoid**, **biceps/brachialis** (C5–6), ECRL/ECRB (C5–6)
C6–7	Triceps
C7–8	FDP
T1	Interossei
CN XI	Trapezius

Elbow flexion loss → upper trunk (musculocutaneous fibres from C6)
✗ Wrong options: lateral cord · posterior cord · C4



Waiter's tip (Erb palsy)

2 Mechanism

Erb	**lateral flexion of head away + depression of ipsilateral shoulder** (shoulder dystocia)
Klumpke	**traction on abducted arm** / arm presentation (breech)
Total	stretch, rupture + avulsion

3 Presentation

Erb (waiter's tip)	shoulder **adducted + internally rotated**, elbow **extended**, forearm **pronated**, wrist flexed · C5: axillary (deltoid, teres minor), suprascapular (SS, IS), musculocutaneous (biceps, brachialis) · C6: radial (brachioradialis, supinator)
Klumpke	**claw hand**: wrist extended (unopposed extensors), MCP hyperextended, IP flexed · intrinsics (ulnar + median) lost · often **preganglionic + Horner**
Total	flaccid arm, motor + sensory loss · **CXR for hemidiaphragm (phrenic)**
Exam tools	Moro, asymmetric tonic neck, Vojta reflexes · pain on gentle shaking of flail arm → **pseudoparalysis** (infection / fracture)
Associated	clavicle + humerus fractures (no prognostic value) · torticollis



Klumpke claw hand

Assessment tools

Tool	Use
Toronto Test Score	Surgical indications, post-reconstruction assessment
Hospital for Sick Children Active Movement Scale	Whole plexus, **15 movements** · must reach **4 (full ROM gravity eliminated)** before higher scores
(Modified) Mallet	**Most common outcome measure** · weighted to ER · Abzug added hand-to-navel (IR)
Wechsler (WISC-IV)	Not used — neuropsychological IQ test (TO)

4 Diagnosis & Classification

Key radiology findings

X-ray clavicle/humerus fractures · little use in infant shoulder (unossified) · **axillary view** in older child for subluxation

Ultrasound glenohumeral subluxation/dislocation in infants

Myelography / CT myelography / MRI **root avulsion (preganglionic) vs extraforaminal rupture** · MRI may show C6 avulsion

EMG/NCV poor reliability, **underestimates** severity · baseline at 4 wk, repeat at 3 months if no recovery (TO)

Preganglionic (root avulsion) signs

Sign	Nerve
Horner (ptosis, miosis, anhidrosis)	Sympathetic chain / T1
Elevated hemidiaphragm	Phrenic
Winged scapula	Long thoracic
Absent rhomboids	Dorsal scapular
Absent rotator cuff	Suprascapular
Absent latissimus	Thoracodorsal

- - Active space - -

5 Management

Obstetric brachial plexus palsy

- All → **observation + daily passive ROM by parents (first 3 months)**
- Antigravity biceps by 3 months → **continue therapy, expect full recovery**
- No biceps by 3 months, postganglionic → **neuroma excision + sural graft (3–9 months)**
- Preganglionic (Horner, phrenic, winging, rhomboids) → **nerve transfer at ~3 months**

Prognosis (biceps recovery by 3 months, Horner)

Overall	**90% recover** without surgery · recovery up to **2 yr** · most with elbow flexion by 3 months normal by 18 months
Good	**Erb palsy** (absent shoulder abduction + ER, intact wrist/hand = best) · **antigravity biceps + deltoid by 3 months** · **early biceps twitch** · elbow flexion by 3 months → recovery up to 100% · partial antigravity strength in first 2 months
Key indicator	**active elbow flexion (biceps, musculocutaneous) at 3 months** ✗ Wrong options: shoulder abduction · ER · elbow extension · axillary at 1 yr
Poor	**no biceps by 3 months** · **preganglionic injury (worst)** · **Horner → <10% recover** motor function · **C7 involvement** · **Klumpke** · total plexus ✗ Wrong options: 30–40%
No prognostic value	clavicle fracture · age at diagnosis · head circumference

Indications for microsurgery / nerve transfer

Lesion	Surgery	Timing
Postganglionic (rupture, neuroma-in-continuity)	**Neuroma excision + sural nerve grafting** (shorter grafts <10 cm better · no direct repair)	**3–9 months** (Gilbert: biceps < MRC 3 at 3 months → surgery at 4–6 months, TO)
Preganglionic (avulsion)	**Nerve transfer / neurotization** (grafting impossible)	**~3 months** (or less)
Global palsy + Horner	Microsurgery	~3 months (TO) · flail + Horner at 1 month (OB)

Nerve transfer	fascicles from one nerve into another nerve's motor branch · **neurotization** = fascicles directly into neuromuscular junction
Transfers	**ulnar fascicle → biceps branch (Oberlin)** + **median fascicle → brachialis branch** · accessory → suprascapular (abduction) · intercostals → musculocutaneous (elbow flexion) · triceps (long head) / thoracodorsal → axillary · contralateral C7 in total avulsion
Most important activity for timing surgery	**return of elbow flexion** (TO)

13-yr traumatic plexus injury, 5 months, no Horner, Tinel at clavicle → exploration + nerve grafting (TO)
✗ Wrong options: neurotization (preganglionic only) · tendon transfer (later)

Urgent sub-specialist referral at 7 weeks → Horner (miosis, anhidrosis)
✗ Wrong options: absent biceps (normal at this age) · Moro present

– – Active space – –

Cookie test (TO) normal: brings food to mouth with elbow flexion + **shoulder adduction** · **failed at 9 months → surgery**

Intervene (TO) **no biceps recovery at 3 months** or **failed cookie test at 9 months**

Glenohumeral dysplasia & IR contracture

Cause strong internal rotators vs weak external rotators → **IR contracture** (most common problem needing treatment) → **glenohumeral dysplasia in 70%** · by age 2

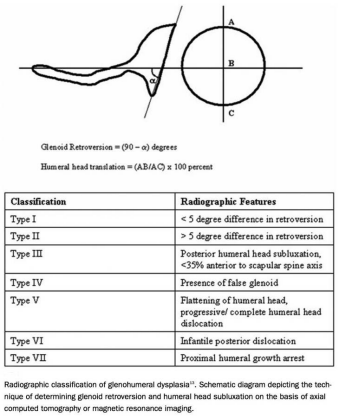
Deformity **glenoid hypoplasia + RETROVERSION** · **posterior humeral head subluxation** · **humeral head flattening** · pseudoglenoid ·
✗ Wrong options: anteversion · anterior-superior escape · anterior subluxation

Zancolli (TO) most common: **IR-adduction contracture with joint deformity + posterior subluxation/dislocation** (>60%)

Putti sign prominent scapula on adduction

Before shoulder reconstruction **MRI of shoulder**
✗ Wrong options: EMG · brain MRI

Waters type	Feature	Treatment (OB5599)
I	<5° retroversion difference	–
II	>5° retroversion difference	**Latissimus + teres major transfer**
III	Posterior subluxation (<35% anterior to scapular spine axis)	Transfer ± open/arthroscopic reduction + anterior release
IV	**False glenoid**	Reduction + release, or **humeral derotation osteotomy**
V	Flattened head, progressive / complete dislocation	**Humeral derotation osteotomy**
VI	Infantile posterior dislocation	–
VII	Proximal humeral growth arrest	–



Classification	Radiographic Features
Type I	< 5 degree difference in retroversion
Type II	> 5 degree difference in retroversion
Type III	Posterior humeral head subluxation, <35% anterior to scapular spine axis
Type IV	Presence of false glenoid
Type V	Flattening of humeral head, progressive/ complete humeral head dislocation
Type VI	Infantile posterior dislocation
Type VII	Proximal humeral growth arrest

Radiographic classification of glenohumeral dysplasia[15]. Schematic diagram depicting the technique of determining glenoid retroversion and humeral head subluxation on the basis of axial computed tomography or magnetic resonance imaging.

Waters classification

– – Active space – –

Secondary reconstruction (tendon transfer, osteotomy, elbow contracture)

Hoffer (latissimus + teres major → posterior rotator cuff)	IR contracture / ER weakness **without dysplasia** (congruent joint) · ± pectoralis major / subscapularis lengthening · arthroscopic release
Hoffer contraindications	**convex or absent glenoid** · **complete posterior dislocation** · **humeral head flattening** · latissimus/teres major <4/5 · deltoid <2/5 · ✗ Wrong options: IR contracture (that's the indication) · no subscapularis function · positive hornblower
Wickstrom humeral derotation osteotomy	IR contracture / ER weakness **with dysplasia** (Waters IV–V; grade V TO) ✗ Wrong options: glenoid osteotomy
Shoulder arthrodesis	non-functional deltoid, good hand

Elbow flexion contracture	Treatment
<40° (Orthobullets) · <35° (OB)	**Serial nighttime extension splinting** (prevents progression, doesn't correct)
>40°	**Serial extension casting**
Severe persistent	Anterior capsular release + biceps/brachialis lengthening (recurs)

Cause of elbow contracture	persistent **triceps (C7) weakness** vs biceps (C5–6)
Forearm supination contracture	unopposed biceps (Klumpke / mixed) · **biceps rerouting** (supinator → pronator) if passive pronation intact · + **forearm osteotomy** if passive pronation limited · ± interosseous membrane release (TO)
Wrist / hand transfers	wrist drop: **PT → ECRB** · finger extension: **FCR or FCU → EDC** · thumb abduction: **EIP → APB** · force ∝ cross-section, amplitude ∝ length
Co-contraction (TO)	partially recovered NBPP; antagonists fire together · **trumpet sign** = shoulder abducts + forearm pronates when bringing hand to mouth · **Botox** for co-contractions
Secondary OBPI procedures (TO)	**external rotation osteotomy** · **triangular tilt surgery** · **Mod Quad** (LD, teres major, subscapularis, pectoralis release)

6 Complications

Phrenic palsy	persistent → **diaphragm plication**
Shoulder tendon transfers	radial + axillary nerve palsy

X vs Y

	Preganglionic	Postganglionic
Site	Root avulsed from cord (proximal to DRG)	Distal to DRG (rupture, neuroma)
Signs	Horner, phrenic, winging, rhomboids	–
Recovery	**None spontaneous**	Possible
Surgery	**Nerve transfer ~3 months**	**Graft 3–9 months**

	Erb	Klumpke
Roots	C5–C6	C8–T1
Posture	Waiter's tip	Claw hand
Mechanism	Head-shoulder distraction	Abducted arm traction
Prognosis	**Best**	Poor, Horner

How it's asked

"Most common root combination?"	→ **C5–C6** (TO)
"Most common pattern in Erb palsy?"	→ **C5–C6** (TO)
"C5 plexopathy – muscle affected?"	→ **biceps** ✗ Wrong options: trapezius · triceps · interossei
"Waiter's tip – true?"	→ **most common pattern** ✗ Wrong options: Horner · 75% need surgery · Klumpke
"Can't flex elbow at 4 months – structure?"	→ **upper trunk**
"Most common injury – deficits?"	→ **deltoid + biceps paralysis, weak wrist and digit extension**
"Most common presentation of the limb?"	→ **pronated forearm and flexed wrist** ✗ Wrong options: claw · benediction
"Greatest risk factor?"	→ **shoulder dystocia**
"Poor prognosis for spontaneous recovery?"	→ **no biceps function by 3 months** ✗ Wrong options: clavicle fracture · postganglionic
"Total palsy + Horner – recovery by 3 months?"	→ **<10%**
"Best functional outcome?"	→ **absent shoulder abduction + ER, intact wrist/digits**
"Flail at birth, elbow flexion at 10 wk – at 18 months?"	→ **normal exam**
"Passive stretching appropriate for?"	→ **1-month-old with partial antigravity biceps** ✗ Wrong options: winging + elevated diaphragm · post-transfer
"Poorest prognosis at 2 months?"	→ **persistent ptosis, miosis, anhidrosis**
"Favorable sign in flail arm?"	→ **biceps return by 3 months**
"Finding at 3 months predicting full recovery?"	→ **active elbow flexion**
"Muscle + timeline for good prognosis?"	→ **musculocutaneous; 3 months**
"True about outcomes?"	→ **full recovery possible if biceps + deltoid antigravity by 3 months** ✗ Wrong options: twitch unfavourable · 50% resolve

– – Active space – –

"13-yr, RTA, 5 months, no Horner?"	→ **exploration and nerve grafting** (TO)
"Determines surgical timing?"	→ **return of elbow flexion** (TO)
"Preganglionic Erb's – procedure?"	→ **nerve transfer at 3 months** (TO)
"4-month-old, winging, rhomboid atrophy, elevated diaphragm, flail?"	→ **median/ulnar fascicles → biceps + brachialis branches** (TO) ✗ Wrong options: intercostal graft after neuroma excision
"3-month-old, winging, no rhomboids, cuff, latissimus?"	→ **nerve transfer to musculocutaneous branches** ✗ Wrong options: sural graft · latissimus transfer
"3-month-old, MRI C6 avulsion, no elbow flexion?"	→ **nerve transfer(s)** ✗ Wrong options: observe · graft C6
"7-week-old – needs early surgical consideration?"	→ **miosis + anhidrosis**
"Zancolli – most common shoulder sequela?"	→ **IR-adduction contracture with posterior subluxation** (TO)
"Most common long-term sequel?"	→ **glenoid hypoplasia with retroversion** (TO)
"Untreated IR contracture – LEAST likely?"	→ **anterior-superior escape**
"Most common long-term consequence – image?"	→ **posterior subluxation on dysplastic retroverted glenoid** ✗ Wrong options: cleidocranial · winging
"Before shoulder reconstruction?"	→ **MRI of shoulder**
"Persistent imbalance – shoulder changes?"	→ **retroversion, posterior subluxation, flat head, IR contracture**
"Ptosis + IR contracture at 3 months – long term?"	→ **glenohumeral dysplasia**
"Untreated Erb – dysplasia pattern?"	→ **glenoid retroversion, posterior subluxation**
"Congruent joint – procedure?"	→ **latissimus dorsi + teres major transfer to rotator cuff** (TO)
"Contraindication to Hoffer?"	→ **convex glenoid**
"Relative contraindication EXCEPT?"	→ **IR contracture**
"6-yr, Putti sign, posterior subluxation on MRI?"	→ **humeral osteotomy** ✗ Wrong options: glenoid osteotomy · isolated transfer
"6-yr, elbow 35–130°?"	→ **serial nighttime extension splinting** ✗ Wrong options: biceps tenotomy · capsular release
"NOT an OBPP assessment tool?"	→ **Wechsler Scale for Children-IV** (TO)
"Failed cookie test at what age mandates surgery?"	→ **9 months** (TO)

"Child abducts shoulder + pronates forearm bringing hand to mouth?" → **trumpet sign (co-contraction)** (TO)

– – Active space – –

– – Active space – –

OLECRANON FRACTURES - PEDIATRIC

– Pediatrics · 74 / 112 –

1 Basics

Epidemiology **uncommon** · <5% of pediatric fractures (US) · peak **5–10 yr**

Mechanism

Mechanism	Result
FOOSH, elbow FLEXED (most common)	Triceps + brachialis tension → **transverse** fracture
FOOSH, elbow extended	Varus/valgus → **longitudinal** lines · varus → **radial head dislocation** · valgus → **radial neck fracture**
Direct blow (least common)	Shear → anterior tension failure, distal fragment anterior, posterior periosteum intact

Location **metaphyseal** (most common) · physeal · apophyseal (intra- or extra-articular)

Association with osteogenesis imperfecta

OI **olecranon (avulsion) fracture after trivial trauma = highly suspicious** ("pathognomonic" per OB) for **OI**
✗ Wrong options: NF · abuse · ABC

Ossification

Olecranon apophysis appears **9 yr** · fuses **15–17 yr** · fuses **anterior → posterior** · **partial closure mimics fracture**

Age 5 girl capitellum, radial head, medial epicondyle ossifying · trochlea, olecranon, lateral epicondyle still cartilaginous → any olecranon disruption = fracture

3 Presentation

Symptoms posterior swelling · pain · **can't actively extend elbow**

Exam palpable gap / crepitus · **gapping → posterior periosteum torn → unstable** · abrasion suggests direct blow

4 Diagnosis & Classification

Key radiology findings

Views AP + lateral

Physis vs fracture physis is **oblique**; fracture usually **transverse** · **patella cubiti** (secondary ossification centre) ≠ fracture

Look for radial neck, lateral condyle, distal radius injuries

– – Active space – –



Oblique apophysis vs transverse fracture

5 Management

Treatment - tension band

Fracture	Treatment
Stress fracture / apophysitis (throwers)	**NSAIDs, rest, immobilise, avoid resisted extension**
Minimally displaced, extensor mechanism intact	Long arm splint/cast **3–4 wk** · repeat X-ray at **7 days**
Displaced · lost posterior periosteum · comminuted	**ORIF**

Fixation

Option	Note
Tension band wiring (AO, axial K-wires)	✓ standard for displaced · **wires stronger** for high loads · **high hardware removal** rate
Tension band suture (PDS / FiberWire) ± oblique lag screw	Metaphyseal fractures
Plate + screws	**Comminuted** with partly fused centres / near maturity
Axial screw ± wire	Risk of physeal disruption with IM lag screw

Never in children ESIN · CRPP · casting a displaced fracture · **fragment excision + triceps advancement** (elderly, low-demand only)



Transverse olecranon fracture

– – Active space – –

6 Complications

List nonunion · delayed union · compartment syndrome · **ulnar neurapraxia** (pseudarthrosis, poor fixation) · loss of reduction · stiffness

How it's asked

"Olecranon fracture after minimal trauma — suspect?"	→ **osteogenesis imperfecta** ✗ Wrong options: abuse · NF
"5-yr, displaced olecranon fracture?"	→ **ORIF with tension band** ✗ Wrong options: ESIN · cast · plate
"10-yr skateboard FOOSH, displaced?"	→ **tension band construct with wires** ✗ Wrong options: IM lag screw (physis) · CRPP · excision + triceps advancement

OSTEOGENESIS IMPERFECTA

– Pediatrics · 75 / 112 –

1 Basics

Genetics & type I collagen defect

Type I collagen — main structural protein of bone, skin, tendon, dentin, sclera · triple helix = **2 α1 chains (COL1A1) + 1 α2 chain (COL1A2)** · **glycine every 3rd residue** allows helix

Mutation — **COL1A1 / COL1A2** in **90%** (Orthobullets) · **85%** (OB) · **missense glycine substitution** in procollagen → abnormal cross-linking · **dominant-negative** effect → severe phenotype
✗ Wrong options: nonsense

Pathophysiology — **↓ collagen secretion** + **abnormal collagen → insufficient osteoid** by physeal + **periosteal osteoblasts → bone can't remodel**
✗ Wrong options: ↑ secretion · "uninterrupted remodeling"

Quantitative vs qualitative — **quantitative (half normal amount, haploinsufficiency) → mild (type I)** · **qualitative (glycine substitution) → severe (II, III, IV)**

Inheritance — **AD most common** · AR forms (VI, VII, CRTAP / LEPRE1 — severe/lethal, no primary collagen defect) · **not X-linked recessive** · AR forms more severe

Defect (distractors)	Disease
Glycine substitution in procollagen	**OI**
Gsα	Fibrous dysplasia
Vitamin C (hydroxylation)	Scurvy
Sulfate transporter	Diastrophic dysplasia
Homogentisic acid oxidase	Alkaptonuria
Osteoclast (carbonic anhydrase / M-CSF)	Osteopetrosis 360
Activating FGFR3	Achondroplasia 10
Type V collagen (basement membrane)	Ehlers-Danlos (OB)
N-Ac-Gal-6-sulfate sulfatase	Morquio
EXT (prehypertrophic chondrocytes)	MHE
COMP / type II collagen	Pseudoachondroplasia / MED, SED

– – Active space – –

Sillence classification

Type	Inheritance	Collagen	Sclera	Features
I	AD	**Quantitative**	**Blue**	**Mildest, most common** · preschool (tarda) · **hearing loss 50%** · A (no DI) / B (DI)
II	AD (Orthobullets, TO) · AR (OB)	Qualitative	Blue	**Perinatal lethal**
III	AD (Orthobullets, TO) · AR (OB)	Qualitative	**Normal**	**Most severe survivable** · fractures at birth · progressive short stature
IV	AD	Qualitative	**Normal**	**Moderate** · bowing, vertebral fractures · hearing normal · A/B by teeth
V	AD	**No collagen defect** (IFITM5)	–	**Congenital anterolateral radial head dislocation** · **hypertrophic callus** · **IOM ossification** (forearm, leg)
VI	AR	No collagen defect (SERPINF1)	–	Like IV · ↑ ALP · fish-scale lamellae · responds poorly to bisphosphonates → denosumab (TO)
VII	AR	No collagen defect (CRTAP)	–	Rhizomelia, coxa vara · severe

90% fall into types I + IV · TO adds VIII (LEPRE1), IX (PPIB), X (SERPINH1), XI (FKBP10, Bruck), XV (WNT1)

Recurrent fractures + hearing loss in a child → type I (TO) · hearing loss 50–92%, rare in type IV

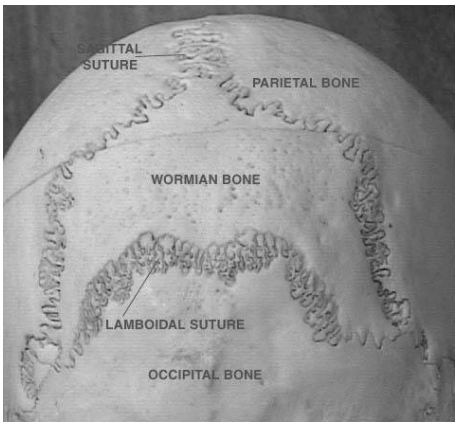
3 Presentation

Orthopaedic — fragility fractures (heal normally, **don't remodel** → bowing, **saber shin**) · ligamentous laxity · short stature · scoliosis · **codfish vertebrae** · **basilar invagination** · **coxa vara (10%)** · congenital radial head dislocation

Olecranon apophyseal avulsion — **most common first presenting sign** · minor trauma (e.g. pushing a sibling) · **bilateral** → work up OI · high contralateral risk · ORIF if displaced
✗ Wrong options: abuse · Noonan · vitamin D

Non-orthopaedic — **blue sclera** · triangular facies · **hearing loss** (50% adults; conductive, sensorineural, mixed) · **dentinogenesis imperfecta** (brown/blue opalescent teeth, primary > secondary) · **wormian bones** · **hypermetabolism → malignant hyperthermia risk** · thin skin, bruising · MVP, aortic regurgitation

Fracture course — fewer with age, usually stop after puberty



Wormian bones

4 Diagnosis & Classification

Radiographic recognition

X-ray	**thin cortices** · generalized osteopenia · **saber shins** / bowing · **wormian bones** on skull · coxa vara
Bisphosphonate zebra lines	**dense parallel horizontal metaphyseal bands** (growth arrest lines) after **8–10 wk** of cyclical therapy · migrate to diaphysis, vanish after physeal closure ✗ Wrong options: from growth hormone (TO false statement)
Diagnosis	**family history + clinical + radiographic** · no commercial test · labs normal (mildly ↑ ALP) · **skin biopsy / fibroblast culture** for collagen (diagnostic per TO; positive in 80% of type IV) · iliac crest biopsy · **prenatal PCR on chorionic villi at 8–12 wk** (TO)
Histology	↑ Haversian canal + lacunar diameter · replicated cement lines · ↑ osteoblasts + osteoclasts · ↓ trabeculae · thin cortex · woven bone, crowded osteocytes (TO)



Bisphosphonate growth arrest lines

5 Management

Bisphosphonates

Indication	most OI · ↓ fracture rate, ↓ pain (1–6 wk), ↑ ambulation, ↑ lumbar BMD

– – Active space – –

Mechanism — **inhibit osteoclast resorption** (N-bisphosphonate → **FPP synthase** → osteoclast apoptosis) · endosteal resorption ↓ while periosteal formation continues → **cortex thickens**
✗ Wrong options: ↑ mineralization · better collagen · ↑ turnover · RANKL antibody

Effect — cortical diameter **↑88%** · cancellous volume **↑46%** · **no effect on scoliosis**

Pamidronate — 1 mg/kg/day × 3 days every 2–4 months · zoledronic acid every 6 months (TO)

IM rodding — **bisphosphonate-free period** around surgery (impairs **osteotomy** healing >> fracture healing)

Others — growth hormone (adjunct) · BMT (some success) · denosumab (type VI, refractory) · teriparatide **adults only** (osteosarcoma risk) · early bracing to limit deformity

Fracture management - intramedullary rodding

Situation	Treatment
<2 yr	Treat as non-OI child (closed)
>2 yr	**Telescoping rods** (Bailey-Dubow, Sheffield, Fassier-Duval) – grow with child · **preferred** (TO)
Deformity beyond tolerance after reduction	**Load-sharing IM device**
Severe bowing	**Sofield-Miller (shish-kebab / "seekh kebab") multiple osteotomies + IM rod**

IM rod over plate — plates → **stress shielding + periprosthetic fracture** · OI **not** at higher nonunion risk

Teen forearm (14–15 yr) — >10° won't remodel → **open reduction + IM rod**
✗ Wrong options: plate · cast · realignment osteotomy (that's for chronic deformity)

Non-telescopic — Rush, Williams rods



Sofield-Miller shish-kebab osteotomy

Spine (basilar invagination, scoliosis)

Scoliosis curves early + progress · VC falls to **40% at 60°**

Curve	Treatment
<45°	Observe
>45° mild OI	**PSF**
>35° severe OI (thin bones)	**PSF early**
Bracing	**Ineffective, not recommended** (rib fragility)

PSF allograft (little autograft) · large blood loss · **ASF only in very young** (crankshaft) · no anterior approach (pulmonary) · not telescoping rods (long bones)

Basilar invagination odontoid into foramen magnum · **severe forms** · 3rd–4th decade (teens possible) · **apnea, altered consciousness, ataxia, myelopathy** (finger escape, Hoffmann) · → brainstem compression, hydrocephalus, syrinx · **transoral decompression + posterior fusion** · also Klippel-Feil, achondroplasia, Morquio, SED
✗ Wrong options: AAI (Down)

X vs Y

	OI	Non-accidental trauma
Sclera / teeth	Blue / DI	Normal
Family history	Fractures	–
Signature fracture	**Olecranon apophysis**	Metaphyseal corner, posterior ribs
Bones	Osteopenic, wormian	Normal

How it's asked

"Skin biopsy helps diagnose?"	→ **osteogenesis imperfecta** (TO)
"Recurrent fractures + hearing loss in a child?"	→ **OI type I** (TO)
"Severe COL1A1 phenotype vs type I – mechanism?"	→ **glycine missense, dominant-negative** (TO) ✗ Wrong options: haploinsufficiency
"COL1A1/2 mutation – EXCEPT?"	→ **webbed neck**
"Normal sclera, dentinogenesis imperfecta, fractures?"	→ **qualitative type I collagen defect (type IV)**
"Gene causing OI?"	→ **COL1A1/COL1A2** ✗ Wrong options: FGFR3 · COMP
"Glycine substitution in procollagen?"	→ **OI** ✗ Wrong options: scurvy · diastrophic
"COL1A1/2 mutations cause?"	→ **OI** ✗ Wrong options: osteopetrosis · MED

– – Active space – –

"FALSE about OI?"	→ **X-linked recessive inheritance**
"11-yr olecranon apophysis fracture, blue sclera, normal teeth?"	→ **quantitative type I collagen defect**
"14-yr, low-energy fractures, blue sclera — protein?"	→ **collagen**
"Majority of OI caused by?"	→ **type I collagen genes**
"AD, no collagen mutation, hypertrophic callus, radial head dislocation?"	→ **type V**
"Glycine substitution, olecranon avulsion — pathophysiology?"	→ **insufficient osteoid by periosteal osteoblasts**
"COL1A1/2 + blue sclera — associated abnormality?"	→ **glycine substitution in a precursor protein**
"Decreased functional type I collagen — X-ray?"	→ **bowed osteopenic long bones** ✗ Wrong options: cleidocranial · rickets
"Dense parallel metaphyseal bands at knees?"	→ **OI on bisphosphonates** ✗ Wrong options: Caffey · JIA
"Bilateral olecranon apophysis fractures?"	→ **OI** ✗ Wrong options: XLH · osteopetrosis
"Displaced olecranon apophyseal fracture associated with?"	→ **OI** ✗ Wrong options: abuse
"Common presenting injury in OI?"	→ **olecranon fracture** ✗ Wrong options: corner fracture · rib
"Sofield procedure used for?"	→ **OI** (TO)
"14-yr OI type I forearm, residual deformity after casting?"	→ **IM rod fixation** ✗ Wrong options: plate
"15-yr OI, bayonet, 20° after reduction?"	→ **open reduction + IM rod + cast** ✗ Wrong options: plate · accept malunion
"OI — incorrect statement?"	→ **brace effective for scoliosis** (TO)
"COL1A1 adolescent, unsteady gait, finger escape, Hoffmann?"	→ **basilar invagination** ✗ Wrong options: AAI · OPLL
"Severe OI, 42° curve progressing?"	→ **posterior spinal fusion with instrumentation** ✗ Wrong options: brace · ASF
"OI — NOT true?"	→ **zebra stripes from growth hormone** (TO)
"Pamidronate mechanism?"	→ **FPP synthase inhibition → osteoclast apoptosis** (TO) ✗ Wrong options: RANKL antibody · Wnt
"Pamidronate thickens cortex because?"	→ **inhibits osteoclast resorption**

OSTEOMYELITIS - PEDIATRIC

– Pediatrics · 76 / 112 –

1 Basics

Epidemiology

Incidence — **1 in 5,000** children <13 yr · mean age **6.6 yr** · **boys 2.5×** · common in **healthy** children · first decade (rich metaphyseal supply, immature immunity) · mortality 50% → **<1%** with antibiotics

Route — **hematogenous seeding of the METAPHYSIS**
✗ Wrong options: epiphysis · direct inoculation

Trauma history — **30%** (Orthobullets) · **20–40%** (OB) · bacteremia source: skin, otitis, pharyngitis, tooth brushing / dental care

Risk factors — diabetes · hemoglobinopathy · JRA · chronic renal disease · immune compromise · **varicella**

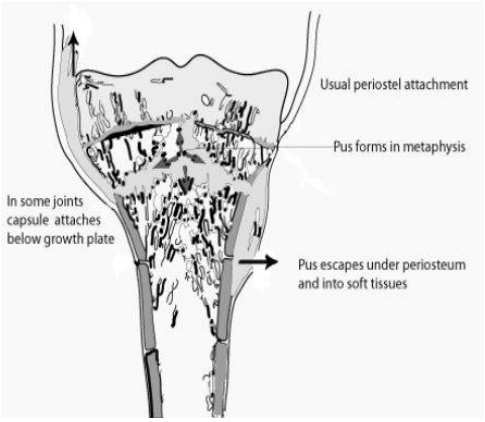
Pathophysiology

Metaphysis — capillaries make **sharp turns** into venous sinusoids → **sluggish, turbulent flow** · low pH + low O2 near physis · spread via Haversian + Volkmann canals

Pus through metaphyseal cortex → subperiosteal abscess

Pus through INTRA-ARTICULAR metaphysis → septic arthritis · hip, shoulder, elbow, ankle ("SHEA") · not **knee** (extra-articular metaphysis)

Infants — transphyseal vessels → epiphyseal osteomyelitis + septic arthritis · **<1 yr** (Orthobullets) · **<2 yr** (TO)



Metaphyseal osteomyelitis spreading to joint and subperiosteal space

Duration (TO) — acute **< 2 wk** · subacute **2–6 wk** · chronic **> 6 wk** with sequestrum · ~5% of paediatric orthopaedic operative cases · 75% part of systemic sepsis

Neonate (TO) — sluggish metaphyseal flow, few phagocytes, **hairpin bends** of nutrient vessels · **< 2 yr transphyseal spread** to epiphysis + joint · **> 2 yr physis blocks** → subperiosteal abscess · **circumferential abscess** devitalises periosteum → sequestrum, pathological fracture

– – Active space – –

Intra-articular metaphysis & septic arthritis spread

Bone (intra-articular metaphysis/physis)	Joint
Proximal femur	Hip
Proximal humerus	Shoulder
Proximal radius / radial neck	Elbow
Distal fibula / distal lateral tibia	Ankle
✗ Wrong option: distal femur, proximal tibia	**Knee — least likely**
✗ Wrong option: distal radius	**Wrist — not** (TO)

Organisms & antibiotics (MRSA, PVL, GAS, Pseudomonas)

Setting	Organism
All children	***S. aureus*** (90% Song & Sloboda · 70% Blyth)
Neonate	**Group B strep**
Young child	***Kingella kingae*** (↑) · most common SA organism 3 months–5 yr · PCR-negative cases
Puncture wound of foot (through sneaker)	***Pseudomonas*** (100% in Puffinbarger)
After varicella	**Group A β-hemolytic strep**
Sickle cell	***Salmonella***
Unvaccinated	*H. influenzae* (now rare)
TB	Extrapulmonary in children · spine 50%, large joints 25%, long bones 11%

PVL (Panton-Valentine leukocidin) CA-MRSA (also some MSSA) cytotoxin · **<5% of S. aureus** · complex/multifocal infection, prolonged fever, **abscess, DVT, sepsis**, necrotizing pneumonia · lyses neutrophils (indirect)
✗ Wrong options: Strep · S. epidermidis · Klebsiella

Coagulase S. aureus factor that **activates prothrombin → thrombin → fibrin clot** → DVT (✓ direct cause; PVL indirect)
✗ Wrong options: adhesins · eDNA (biofilm) · vWF binding protein

Other virulence (distractors) superantigens (toxic shock) · biofilm · protein A (binds IgG Fc)

MRSA predictors **T >38°C · WBC >12,000 · Hct <34% · CRP >13 mg/L** → MRSA probability **4/4 = 92%** · 3 = 45% · 2 = 10% · 1 = 1%

Organisms (TO) **Pasteurella**: animal bites · **Eikenella**: human bites · **Salmonella**: sickle cell · fungal: NICU graduates · preterm: nosocomial gram-negative + fungal · term infants: CA-MRSA · **K. kingae**: culture-negative, **PCR**, CRP barely ↑

3 Presentation

Symptoms **limp / refusal to bear weight** · **usually not toxic** · ± fever · prior antibiotics mask

Exam warm, swollen, **point tenderness** (check pelvis, spine) · motion limited by pain · pelvic compression pain (Gaenslen) → pelvic osteomyelitis

Neonate failure to thrive, **multifocal** (shoulder + hips) → **ultrasound** of hip + shoulder
✗ Wrong options: MRI first · Pavlik

4 Diagnosis & Classification

Workup - aspiration negative, MRI & US

Labs

Test	Fact
CRP	**↑ in 98%** · rises in **6 h** · **best to monitor response** · should fall by **48–72 h** · normalises ~1 wk (Orthobullets) / 1 month (OB)
ESR	↑ in **90%** · peaks **3–5 days** · falls too slowly (normal ~3 wk) · unreliable in neonates + sickle cell
WBC	↑ in only **~25%** (✓ "often normal")
Procalcitonin	↑ in **58%** · falls with effective treatment
Blood culture	+ in **30–50%** (✗ Wrong options: 75–80%)
Bone aspirate culture	+ in **50–70%** (51–73%, 50–85% OB)

Most likely abnormal test → CRP

Imaging

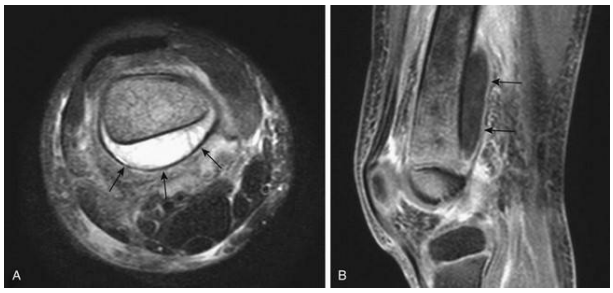
Modality	Fact
X-ray	Early normal / soft-tissue edema · periosteal new bone **5–7 days** · osteolysis **10–14 days** · metaphyseal rarefaction 1–2 wk
MRI	**Gold standard** · **T1 ↓, T2 ↑, gadolinium ↑** · sensitivity **88–100%** · shows abscess · plan surgery
Bone scan	**92%** sensitive · non-focal toddler, **multifocal** · **cold scan = aggressive / osteonecrosis**
CT	Later bone changes (✗ Wrong options: study of choice)
US	Neonate effusion, subperiosteal collection

Fever + ↑ CRP + negative joint aspirate (e.g. 1,000–9,000 cells) → MRI (osteomyelitis / subperiosteal abscess)
✗ Wrong options: arthrotomy · observe · oral antibiotics

Septic joint excluded synovial WBC **<50,000** essentially rules out (Kocher 2/4 ≈ 40%)

Suspected SA of shoulder/hip/elbow/ankle **pre-op MRI ↓ repeat surgery** · don't delay unstable patient
✗ Wrong options: MRSA low risk · negative PCR rules out

– – Active space – –



MRI subperiosteal abscess

TO lab / imaging	ANC > 12,000 · **CRP > 20 mg/L** sensitive, good NPV, **best marker of response** · ESR > 20 slow to normalise · blood culture before antibiotics · X-ray: soft-tissue swelling **48 h**, periosteal reaction **7 days** · **MRI gold standard** early; **rim sign** (hyperintense rim) · CT maps sequestrum · **FDG-PET highest accuracy** when MRI limited (hardware) · "culture all biopsies, biopsy all cultures"
Gledhill classification (TO)	radiological patterns of **subacute** OM (metaphyseal lucency, cortical hyperostosis, onion skin)
Epiphyseal osteomyelitis (TO)	primary epiphyseal, around knee · **TB commonest in India** · minimal angular deformity risk

Subacute osteomyelitis vs tumour - biopsy

Subacute	bone pain + X-ray changes, **no systemic signs** · ↑ host resistance / ↓ virulence / prior antibiotics · afebrile, mild pain, **normal WBC, ↑ ESR**
Classification	IA/IB lucency · II metaphyseal with cortical loss · III diaphyseal · IV onion skin · V epiphyseal · VI spinal
Biopsy + culture	when diagnosis unclear / rule out tumour (Ewing, osteosarcoma, EG, lymphoma) · send tissue for pathology at every debridement

Brodie abscess – subacute osteomyelitis

Type	**subacute** (TO) · age **10–20 yr** · **proximal tibia** most common · metaphysis (tibia, femur) · S. aureus
X-ray	solitary well-circumscribed **lucency with sclerotic rim** at metaphyseal-epiphyseal junction
MRI	**penumbra sign**: rim of vascular granulation tissue, **higher T1 signal** than cavity (TO) ✗ Wrong options: hydatid · hemangioma
Labs	normal or mildly ↑ ✗ Wrong options: "elevated blood counts" (TO false statement)
Treatment	**surgical debridement / curettage (± graft) + antibiotics** ✗ Wrong options: wide resection + chemo · empiric antibiotics alone · cast

TB osteomyelitis

Features	constitutional symptoms, night sweats · normal WBC, ↑ ESR · **PPD +** · osteopenia + subchondral erosions · polyostotic
Diagnosis	biopsy: **caseating granulomas, giant cells** · **Ziehl-Neelsen** ("red snappers") · **Löwenstein-Jensen** culture
Treatment	multi-agent antibiotics **≥1 yr** · rarely debridement (sinus risk)

5 Management

Pediatric acute hematogenous osteomyelitis

- No abscess → aspirate / biopsy for culture → **empiric IV antibiotics**
  (improves within 48 h → no surgery; IV → oral)
- Subperiosteal / intraosseous abscess, pus aspirated → **I&D + antibiotics**
- No improvement 48–72 h or CRP not falling → repeat MRI → **I&D**

Surgical indications & subperiosteal abscess

Nade's principles (TO)	antibiotics work **before abscess** · **abscess needs drainage** · antibiotics prevent recurrence · surgery must not damage normal bone · continue antibiotics after surgery
Technique (TO)	**oval cortical window** (drill holes) ↓ fracture · medullary decompression with infant feeding tube · pan-medullary abscess: **Gigli-saw intramedullary decompression** · IV antibiotics 4–6 wk (TO)
Surgery	**subperiosteal or deep abscess** · **pus on aspiration** · **failure of appropriate antibiotics** (48–72 h) · **chronic infection / sequestrum** · adjacent septic arthritis
Not failure	improving at 24 h · on wrong drug (e.g. gentamicin for S. aureus) · no trial yet

Worsening at 2–3 days on antibiotics → subperiosteal abscess
✗ Wrong options: CRMO · atypical organism · Brodie

Subperiosteal abscess sites	femur 23–29%, tibia 19–26%

Septic shock + MRSA + abscess → fluids + antibiotics + urgent I&D
✗ Wrong options: wait for cultures

Septic hip with adjacent osteomyelitis	straight to **OR debridement** (cultures intra-op) · complication rate 38% vs <1% isolated SA
Technique	evacuate pus, debride, drill intraosseous collections, culture + pathology · drains or pack, re-look in 2–3 days · remove sequestrum in chronic
Antibiotics before cultures	only if septic / unstable

Antibiotic choice

Situation	Drug
Low MRSA prevalence	**Nafcillin / oxacillin** (PBP binders)
High MRSA prevalence	**Clindamycin** (50S) or **vancomycin**
Sepsis / severe / MRSA	**Vancomycin** – binds **D-Ala-D-Ala**
Gram-negative rods	+ 3rd-gen cephalosporin
<1 month (Copley)	Ampicillin/sulbactam + gentamicin
1–3 months	Vancomycin + ceftriaxone

Duration	IV **4–6 wk** (Orthobullets, controversial) · 3–4 wk (OB) · 20 days = 30 days for MSSA/strep · MRSA: 24–48 h IV → 4–6 wk oral (OB)

– – Active space – –

Oral vs PICC IV — **equal cure** · PICC → **more ED visits / rehospitalisation**, ↑ cost

Monitoring response - CRP

Best early indicator — **serial CRP** · peaks ~day 2 post-op then falls
✗ Wrong options: ESR · WBC · MRI · X-ray · procalcitonin rise

6 Complications

DVT with MRSA osteomyelitis

Risk factors — **CRP >6 mg/dL** · **surgical treatment** · **age >8 yr** · **MRSA**
✗ Wrong options: fever >38.5°C

Organism — **MRSA** · PVL endothelial injury + coagulase
✗ Wrong options: MSSA · Kingella · TB

Action — calf pain / swelling → **venous duplex US** · therapeutic anticoagulation · MRSA often needs repeat I&D

Chronic osteomyelitis (sequestrum, involucrum)

Beit CURE (TO)	Feature	Rx
A	**Brodie abscess**	Drilling + curettage
B1–B3	Sequestrum, involucrum adequate	**Sequestrectomy** + curettage
B4	Sequestrum, **inadequate involucrum**	Sequestrectomy + **stabilisation**
C	Sclerosis	Drainage + long-term antibiotics

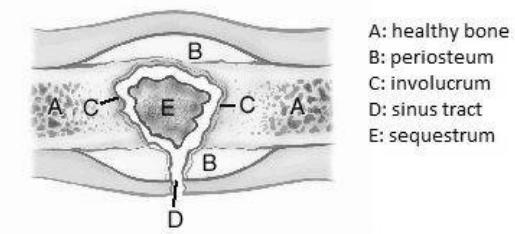
Cierny-Mader anatomic (TO) — A medullary · B superficial · C localised · D diffuse

Chronic OM surgery (TO) — debride to **paprika sign** · **wait for adequate involucrum** before sequestrectomy · dead space: **Papineau** grafting, VAC, flaps · biofilm → high-dose local antibiotics · **Masquelet** induced membrane (cement spacer, 2-stage) · **absorbable calcium sulphate** carriers (no 2nd surgery, heat-sensitive antibiotics)

Sequestrum — **necrotic bone walled off from blood supply = nidus** of chronic infection

Involucrum — **new periosteal bone** around sequestrum

Brodie abscess — chronic abscess walled by sclerosis + fibrous tissue



Sequestrum and involucrum

Late growth disturbance

Physeal damage → LLD + angular deformity (∝ physeal destruction) · LLD apparent ~**9 yr** average · **follow to maturity** with **standing full-length X-rays**

Others septic arthritis (I&D) · meningitis · pathologic fracture · AVN · chronic / multifocal infection

X vs Y

	Acute osteomyelitis	Subacute (Brodie)
Systemic	Fever, ↑ CRP	Afebrile, normal WBC, ↑ ESR
Age	Mean 6.6	10–20
X-ray	Normal early	Lucency + sclerotic rim
Treatment	IV antibiotics ± I&D if abscess	Curettage + antibiotics

Infant mimic (TO)	Clue
Caffey disease	Infantile cortical hyperostosis, irritable, self-limiting
CRMO	Multifocal / bilateral, normal markers, NSAIDs
CIPA	Insensitivity to pain + anhidrosis → hypertrophic callus
Scurvy	Irritable, metaphyseal white line, dramatic response to vitamin C

How it's asked

"10-yr distal femoral tenderness, fever, knee aspirate 1,700 cells?"	→ **MRI of knee**
"7-yr febrile, CRP 11, knee aspirate 2,500 cells?"	→ **MRI** ✗ Wrong options: IV antibiotics first · arthrotomy
"10-yr bike fall, knee aspirate 1,000 WBC?"	→ **MRI of lower extremity**
"Posterior hip pain, pelvic compression pain, hip aspirate 9,000?"	→ **MRI of pelvis**
"4-day-old, swollen shoulder, stiff hips, ↑ CRP?"	→ **ultrasound of hip and shoulder**
"6-yr, swollen patella region, mild ↑ CRP?"	→ **MRI of knee** ✗ Wrong options: biopsy first · debridement
"Suspected shoulder SA – true?"	→ **pre-op MRI ↓ repeat surgery** ✗ Wrong options: Kingella infrequent · negative PCR rules out
"14-yr, 6 months after sprain, distal tibial lesion, ↑ ESR only?"	→ **biopsy and culture**
"7-yr, elbow + wrist swelling, night sweats, lymphadenopathy, caseating granulomas?"	→ **polyostotic mycobacterial infection** ✗ Wrong options: LCH · multicentric GCT
"8-yr, worsening tibial pain 6 days after twist, ↑ CRP, MRI marrow signal?"	→ **percutaneous biopsy with culture and antibiotics** ✗ Wrong options: chest CT + oncology

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"13-yr, fever, ↑ ESR/CRP, distal tibial marrow edema?"	**→ admit, aspirate for culture, IV antibiotics, serial ESR/CRP**
"Vertebral osteomyelitis — true?"	**→ WBC often normal** ✗ Wrong options: GBS · ESR best monitor
"Brodie abscess — all true EXCEPT?"	**→ wide resection + chemotherapy**
"10-yr ankle, CT biopsy — diagnosis?"	**→ osteomyelitis** ✗ Wrong options: Ewing · EG · osteosarcoma
"Which tibial osteomyelitis needs debridement?"	**→ lethargic, CRP 20, no better after 1 wk nafcillin + vancomycin** ✗ Wrong options: improving at 24 h · gentamicin
"Distal fibula osteomyelitis + subperiosteal abscess?"	**→ I&D**
"MSSA bacteremia, humeral MRI — surgical feature?"	**→ subperiosteal abscess** ✗ Wrong options: metaphyseal osteomyelitis alone
"Cold bone scan + shoulder MRI sepsis?"	**→ drain GH joint + debride humerus** ✗ Wrong options: CT · PICC only
"2-yr, humeral abscess on MRI?"	**→ I&D of humerus**
"14-yr 1st metatarsal abscess, worse on oral antibiotics?"	**→ I&D + IV antibiotics**
"12-month-old shoulder aspirate 63,000, 90% PMN?"	**→ I&D** ✗ Wrong options: MRI first
"16-yr afebrile, Brodie abscess?"	**→ surgical debridement + IV antibiotics** ✗ Wrong options: empiric IV alone
"Septic shock, MRSA, subperiosteal abscess?"	**→ fluids + antibiotics + urgent I&D**
"Improves day 1 then worse, CRP static?"	**→ subperiosteal abscess** ✗ Wrong options: CRMO · atypical organism
"Persistent fever on pip-tazo, GPC in blood — image?"	**→ subperiosteal abscess MRI**
"4-yr, all 4 Kocher, hip effusion, proximal femoral metaphyseal signal?"	**→ surgical debridement with cultures** ✗ Wrong options: antibiotics before cultures · drain only
"Best early test of response after debridement?"	**→ CRP**
"Most likely abnormal test in AHO?"	**→ CRP** ✗ Wrong options: WBC · blood culture
"Response indicator after decompression?"	**→ decrease in CRP** ✗ Wrong options: procalcitonin rise · ESR
"Nail through sneaker → calcaneal osteomyelitis ↑ rate of?"	**→ Pseudomonas**

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"Hypotensive 8-yr, tibial osteomyelitis — empiric drug mechanism?"	→ **D-Ala-D-Ala binding (vancomycin)** ✗ Wrong options: PBP (oxacillin) · 30S
"Fever + shoulder after varicella — organism?"	→ **group A β-hemolytic strep**
"T 38.8, WBC 16k, Hct 30%, CRP 20, GPC clusters?"	→ **vancomycin** ✗ Wrong options: oxacillin · cephalexin
"Oral vs PICC IV for 6 wk — true?"	→ **PICC → more rehospitalisation**
"Osteomyelitis + popliteal DVT — virulence factor?"	→ **Panton-Valentine leukocidin**
"Abscess on day 2 despite antibiotics — organism?"	→ **GPC clusters with PVL** ✗ Wrong options: PYR-positive strep
"PVL produced by?"	→ **Staphylococcus aureus**
"DVT risk factors — EXCEPT?"	→ **fever >38.5°C**
"Osteomyelitis + DVT — organism?"	→ **MRSA**
"13-yr MRSA osteomyelitis, septic — surveil for?"	→ **DVT** ✗ Wrong options: septic arthritis of knee · Marjolin
"3-yr MRSA femoral abscess, calf pain — true?"	→ **venous duplex US**
"Factor activating prothrombin → DVT?"	→ **coagulase** ✗ Wrong options: PVL
"Direct extension to joint least likely?"	→ **knee** (TO)
"True about physis?"	→ **may allow spread to epiphysis <2 yr** (TO)
"Physeal infection → septic arthritis EXCEPT?"	→ **wrist** (TO)
"Direct metaphyseal spread — EXCEPT?"	→ **knee**
"Sequestrum is?"	→ **necrotic bone, nidus of chronic infection**
"Blue arrowheads around dead bone (chronic osteomyelitis)?"	→ **involucrum**
"After treating distal femoral AHO + abscess — late study?"	→ **standing full-length bilateral lower extremity X-ray**
"AHO — true?"	→ **trauma history in 20–40%** ✗ Wrong options: GAS most common · epiphysis · CT · cultures 75–80%
"Brodie abscess type?"	→ **subacute** (TO)
"Penumbra sign on MRI?"	→ **Brodie abscess** (TO)
"19-yr proximal tibial Brodie — NOT true?"	→ **elevated blood counts** (TO)

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OSTEOPETROSIS

– Pediatrics · 77 / 112 –

1 Basics

Definition "marble bone disease" · **defective osteoclastic resorption** → dense but **brittle** bone, obliterated medullary canals

Prevalence AD **1 in 20,000** · AR **1 in 200,000** · AD = **most common** · 75% penetrance, may skip generations

Pathophysiology (carbonic anhydrase II, osteoclast acidification)

Core defect osteoclast **cannot acidify Howship's lacuna** → no HCl → hydroxyapatite not dissolved → bone stays thick, disorganised
✗ Wrong options: ↑ remodeling · ↓ osteoblasts · ↓ hydroxyapatite

Carbonic anhydrase II $CO_2 + H_2O \rightarrow H^+ + HCO_3^-$ · H+ pumped across **ruffled border**

Causes **carbonic anhydrase II** loss-of-function or **chloride channel** dysfunction (Orthobullets) · CA II "most common" (OB4418, OB219284) · proton pump 60%, CLCN7 12%, CA II <5% (OB4814) – read the options

Osteoblasts normal

Defect (distractors)	Disease
CA II / Cl channel / proton pump (osteoclast)	**Osteopetrosis**
Gsα	Fibrous dysplasia
↓ Type I collagen	OI
↓ Osteoid mineralization	Rickets
Renal 1α-hydroxylase	Vitamin D-dependent rickets type I
Proximal tubule phosphate loss	Hypophosphatemic rickets
FGFR3	Achondroplasia 10
FGFR2	Apert
RUNX2	Cleidocranial dysplasia
PMP22	CMT
COMP	MED
DTD sulfate transporter	Diastrophic dysplasia
COL2A1	SED

Classification

Type	Inheritance	Defect	Course
Malignant (infantile)	**AR** (11q13)	**Proton pump / Cl channel**	Fatal early without **BMT**
Intermediate	AR	**Carbonic anhydrase II** / Cl channel	–
Benign	**AD** (most common)	**Cl channel**	Normal life · **type II = Albers-Schönberg** (anemia, fractures, early OA) · **most seen by orthopaedists**

ADO I vs II I: diffuse sclerosis, esp. cranial vault · **II: rugger jersey spine, bone-in-bone iliac wings, skull base**

TO lists AD, AR **and X-linked recessive** forms

3 Presentation

Clinical features & complications

AR (malignant) **severe anemia** from infancy (marrow encroachment) → bleeding, infections · **hepatosplenomegaly** (extramedullary hematopoiesis) · macrocephaly · progressive **blindness + deafness** · dental abscess, **mandibular osteomyelitis** · death from bleeding / infection

AD (benign) often asymptomatic until **low-energy (transverse) fracture** · anemia · **low back pain** (↑ spondylolysis) · early **hip OA** · normal height · **high risk of cranial nerve palsy**

Cranial nerve palsies foraminal overgrowth · **optic > auditory > trigeminal > facial**

Fractures lifelong **40–50%** (40–60% OB) · **LE > UE > axial** · transverse · delayed union, malunion

Associated **coxa vara** (femoral neck nonunion / stress fractures) · **osteomyelitis** (poor marrow vascularity, ↓ WBC function) · carpal tunnel · Caffey disease (listed by Orthobullets) · seizures

NOT associated **vertebral compression fractures**

4 Diagnosis & Classification

Key radiology findings

General ↑ cortical thickness · **↑ overall density** · loss of medullary canal · **bone-in-bone (endobones – pathognomonic, TO)**

Signs **"rugger jersey" / sandwich spine** (dense endplate bands) · **Erlenmeyer flask** distal femur + proximal humerus · block femoral metaphysis · coxa vara · thick skull base, poor sinuses

Not osteopetrosis **dripping candle wax** (melorheostosis) · Paget pelvis · bone infarct

Labs AR: ↑ acid phosphatase, ± ↑ PTH, ± ↑ Ca · anemia ± thrombocytopenia · AD: normal

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Histology osteoclasts **lack ruffled border + clear zone** · islands of **calcified cartilage** in trabeculae · empty lacunae, plugged Haversian canals

Rugger jersey spine	Osteopetrotic femur
	

Erlenmeyer flask deformity – differential

Definition constricted diaphysis + flared metaphysis (undertubulation)

Causes **osteopetrosis** · **Gaucher**, Niemann-Pick · thalassemia, sickle cell · **Pyle (metaphyseal dysplasia)** · craniotubular dysplasias · **achondroplasia** · fibrous dysplasia · **Ollier** · MHE · rickets · RA · lead toxicity

NOT **Langerhans cell histiocytosis** · Sprengel · melorheostosis · DEH (TO)

5 Management

Medical

Form	Treatment
AR malignant	**Bone marrow transplant** (only durable cure, TO) · **high-dose calcitriol**
AD	**Interferon γ-1b**
Supportive (TO)	Erythropoietin (darbepoetin), vitamin D, corticosteroids
✗ Wrong option: **alendronate**	**Inappropriate** – bisphosphonate further suppresses osteoclasts

Cranial nerve impingement neurosurgical decompression

Fracture management

Fracture	Treatment
Diaphyseal long bone, upper extremity	**Non-op**: prolonged casting, NWB · slow healing, limited remodeling → malunion, refracture
Proximal femur	**ORIF with plate + screws** – **avoid IM devices** (can't pass reamer) · ↑ union, ↓ coxa vara

Technique **slow steady drilling, constant cooling, frequent drill-bit changes** · ↑ hardware failure

Arthroplasty THA: cannulated reamers under fluoro, **short-stem uncemented** femur, small sharp acetabular reamers + screws · TKA: consider navigation

6 Complications

Refracture (brittle bone, hardware failure) · **infection** (↓ vascularity) · malunion · nonunion

X vs Y

	Osteopetrosis	Melorheostosis
X-ray	Diffuse density, bone-in-bone, rugger jersey	**Dripping candle wax** cortical hyperostosis

How it's asked

"Mechanism of intermediate osteopetrosis?"	→ **inactive osteoclast carbonic anhydrase** ✗ Wrong options: Gsα · RUNX2
"22-yr distal radius fracture, dense pelvis, rugger jersey spine — etiology?"	→ **defective osteoclast function** ✗ Wrong options: FGFR3 · 1α-hydroxylase
"Carbonic anhydrase mutation — which X-ray?"	→ **Erlenmeyer flask femur** ✗ Wrong options: Paget · melorheostosis · infarct
"Cause of osteopetrosis?"	→ **loss-of-function of carbonic anhydrase II** ✗ Wrong options: Gsα · FGFR3
"Can't acidify Howship's lacuna?"	→ **Albers-Schönberg disease**
"Why is the bone thicker?"	→ **↓ acidification of Howship's lacuna**
"Fractures, CN palsies, back pain — cause?"	→ **osteoclast can't create acidic Howship lacunae**
"10-yr, seizures, anemia, dense bones — defect?"	→ **carbonic anhydrase II** ✗ Wrong options: PMP22 · COMP
"5-yr, fractures since birth, rugger jersey — inheritance?"	→ **autosomal recessive**
"10-yr, rib fracture, HSM, deafness, caries — NOT appropriate?"	→ **alendronate** (TO) ✗ Wrong options: interferon γ · calcitriol · darbepoetin
"NOT a feature of osteopetrosis?"	→ **dripping candle wax** (TO)
"18-yr, hearing + vision loss, couldn't pass reamer — spine X-ray?"	→ **rugger jersey**
"3-yr abnormal X-rays — risk of all EXCEPT?"	→ **vertebral compression fractures**
"Erlenmeyer flask in all EXCEPT?"	→ **Langerhans cell histiocytosis** (TO)
"Erlenmeyer flask of femoral metaphysis?"	→ **osteopetrosis** (TO)

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OVERLAPPING TOE

– Pediatrics · 78 / 112 –

1 Basics

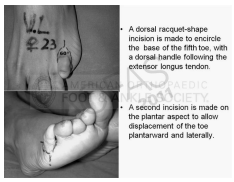
Definition congenital overlap of a toe from **extensor digitorum longus contracture** · rare

Epidemiology **5th toe** most common · usually **bilateral** · M = F · often **familial**

3 Presentation

Symptoms pain with **shoe wear**

Exam toe **adducted + slightly externally rotated**, overlapping 4th · **MTP dorsiflexed** ("cock-up") · **smaller nail plate**

Overlapping fifth toe	Butler procedure
	

5 Management

Surgical treatment

Situation	Treatment
First line	**Passive stretching + buddy taping**
Failed, still symptomatic	**Butler procedure**
Alternatives	EDL tenotomy · dorsal capsulotomy · **syndactylization to 4th toe (McFarland)**
Skeletally mature only	**Proximal phalanx resection** (induces secondary deformity)

Butler **dorsal + plantar racquet-shaped incision**, **EDL tenotomy**, **circumferential MTP release**, toe translated plantarly · addresses chronic MTP dislocation (needs plantar release) · main problem = **recurrence**
✗ Wrong options: tenotomy alone · V-Y + dorsal capsulotomy

How it's asked

"9-yr, painful cock-up overlapping 5th toe with shoes?" → **Butler procedure: racquet incision, EDL tenotomy, circumferential MTP release**
✗ Wrong options: EDL tenotomy alone · phalangeal resection

PATELLA SLEEVE FRACTURE

– Pediatrics · 79 / 112 –

1 Basics

Definition	cartilage **"sleeve" pulled off the ossified patella** ± small bone fleck · unique to skeletally immature
Epidemiology	<1% of pediatric fractures but **>50% of pediatric patella fractures** · **boys 5:1** · **8–12 yr** (ossification nearly complete)
Mechanism	**indirect**: powerful **quadriceps contraction on a flexed knee** (stumble, pop) · not a direct blow · superior pole: patella pushed down → intra-articular dislocation (OB)
Anatomy	largest sesamoid · ossifies **3–5 yr** · lateral facet larger · cartilage up to 1 cm · quadriceps 3 layers (rectus superficial, vasti middle, intermedius deep) · **blood supply mainly inferior pole** (geniculate ring)
Location	**inferior pole most common** · superior pole least common

3 Presentation

Symptoms	severe pain, can't bear weight, **painful pop**
Exam	hemarthrosis / tense effusion · **high-riding patella or palpable gap** = extensor mechanism disrupted · **extensor lag / can't straight-leg raise**

4 Diagnosis & Classification

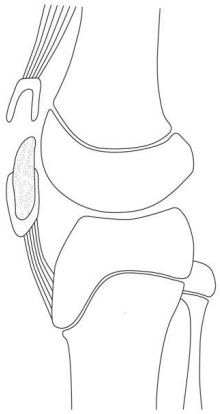
Key radiology findings

Views	AP, lateral, tangential
Findings	**small bone flecks** at superior or inferior pole (easily missed) · **inferior sleeve → patella ALTA** · **superior sleeve → patella BAJA** + slight anterior tilt of superior pole
Rule	fragment is **much larger than it looks** (mostly cartilage)
MRI / ultrasound	when diagnosis unclear (soon after injury)



Inferior patellar sleeve fracture

– – Active space – –



Superior patellar sleeve fracture

5 Management

Treatment - ORIF

Situation	Treatment
Nondisplaced, extensor mechanism intact (rare) · gap <2 mm	**Cylinder cast 6 wk**
>2–3 mm displacement or step-off · disrupted extensor mechanism	**ORIF**

ORIF — medial parapatellar approach · repair retinaculum / tendon · **transosseous sutures through bone tunnels** · **modified tension band** · suture anchors · screws · K-wires if fragment large · cylinder cast in extension **2–3 wk**

Superior sleeve — suture between bony fragments through **tunnels in the larger distal fragment** (fracture is distal to quadriceps insertion)
✗ Wrong options: quadriceps tendon repair · immobiliser

Never — **excision of the fragment** (cartilage is large) · patellar tendon repair to tubercle · partial patellectomy · knee arthroscopy · PT alone · cast with extensor lag

6 Complications

List — patella alta · extensor lag · quadriceps atrophy · malunion · nonunion · painful hardware · worse with comminution + displacement

How it's asked

"10-yr fell from bike, extensor lag, inferior pole avulsion?"	→ **open reduction + suture fixation** ✗ Wrong options: plate + cerclage · patellectomy · tendon midsubstance repair
"13-yr superior pole sleeve, can't SLR, gap on US?"	→ **ORIF** ✗ Wrong options: hinged brace · patellar tendon repair

"11-yr basketball, pop, can't extend — distal pole fleck?"	→ **open reduction + tension band** ✗ Wrong options: excision · cast · tendon to tubercle
"12-yr kicked, displaced sleeve?"	→ **ORIF of patella** ✗ Wrong options: tibial tubercle · cylinder cast
"10-yr tripped, pop, extensor lag?"	→ **ORIF** ✗ Wrong options: cast in extension · arthroscopy
"Arthrogryposis, superior pole rim avulsed, can't straighten?"	→ **ORIF with sutures through patellar bone tunnels** ✗ Wrong options: quadriceps tendon repair · cast

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PEDIATRIC ABUSE

– Pediatrics · 80 / 112 –

1 Basics

Most common presentations & epidemiology

Burden **>1 million** substantiated victims/yr (US) · 15–42 per 1,000 children · >1,200 deaths/yr · **~⅓ of abused children see an orthopaedic surgeon**

Mortality **2nd most common cause of death in children >1 yr** (after accidental injury) · after SIDS at 1–5 months (OB) · death usually from **head trauma**

Age **90% of abuse fractures <5 yr** · **50% of fractures <1 yr** · **30% of fractures <3 yr** (Orthobullets) · 10–15% <3 yr (Kempe, TO) · most common <3 yr

Femur **NAT = most common cause of femur fracture in non-ambulatory infant** · NAI up to **35% of femur fractures <3 yr**, **40–90% <1 yr** (OB)

Type (overlap)	%
Neglect (✓ most common)	**78%**
Physical	18%
Sexual	9% (~10% TO)
Psychological	8%

Presentation	Rank
Skin lesions (bruises in various stages, burns, cuts)	**Most common** (✗ Wrong options: fractures)
Fractures	**2nd**
Head injury	**Most long-term morbidity** (✗ Wrong options: burns)

Fracture location **humerus > tibia > femur** · **diaphyseal 4× more common than metaphyseal** · single transverse diaphyseal fracture classic (TO)
✗ Wrong options: metaphyseal 4×

Risk factors

Child **first-born** · unplanned · **premature** · **disability (CP, neurocognitive syndromes, Down)** · **step-child** · age <3 yr

Parent / social single parent · recent stressor (**job loss**, move) · unemployment · drugs · abused as child · low SES · no support · age <20 · Medicaid (⅔)

NOT risk factors **female sex** · **no siblings**

3 Presentation

Red flags inconsistent / no history · **delay in seeking care** · **long-bone fracture in a non-ambulatory child** (e.g. "fell running" at 6 months) · unwitnessed injury · multiple carers · bruises in various stages · **pinna bruising, cigarette burns** (highly specific)

Fracture patterns suggestive of abuse

Specificity	Fractures
High	**Classic metaphyseal lesion (corner + bucket handle)** · **posteromedial rib** · **long bone in non-ambulatory child** · **transphyseal distal humerus** (moderate per OB220897) · scapula · sternum · spinous process
Moderate	**Multiple fractures in various stages of healing** · vertebral body · digital · complex skull
Low	**Toddler's fracture** (non-displaced spiral tibia, ambulatory 9 months–3 yr) · clavicle · supracondylar humerus · SCFE

Classic metaphyseal lesion through **primary spongiosa** at metaphysis-physis junction · **torsion / traction-shear** (pull, twist, shake) · **corner fracture** = discrete metaphyseal avulsion · **bucket handle** = same seen en face · present in **40–50%** of abused infants (TO) · "almost pathognomonic" (TO) · **most suspicious femur pattern**
✗ Wrong options: midshaft spiral/transverse

Femur NAI vs accident **mid-diaphyseal most common in both** · NAI children **younger**
✗ Wrong options: corner fracture most common in NAI

Rib fractures **posteromedial** · highest specificity (OB) · 70% chance of abuse · 51% of fractures in fatal abuse · anterior rib not pathognomonic

Transphyseal distal humerus separation rotatory shear · ≤3 yr · radiocapitellar line preserved (vs dislocation) · arthrogram/US/MRI if unossified · CRPP like supracondylar

Shaken baby syndrome **subdural hematoma + retinal hemorrhage + posterior rib fractures** (TO) · chest ecchymosis · seizures

No fracture is pathognomonic judge age, ambulation, pattern, mechanism, psychosocial picture

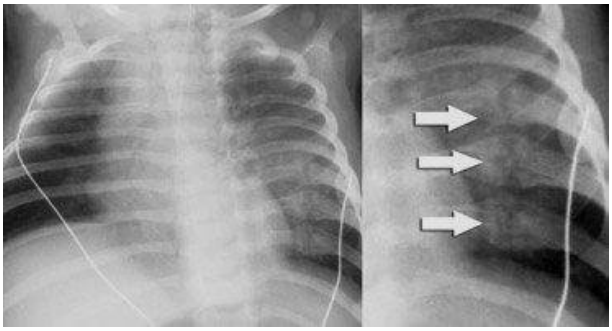


Metaphyseal corner fracture

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Bucket handle fracture



Posterior rib fractures

4 Diagnosis & Classification

Key radiology findings

Skeletal survey	**study of choice / gold standard** · AP appendicular + AP/lateral axial (skull, spine, ribs) · **mandatory <2 yr** with suspected abuse (AAP) · **repeat at 2 wk** for healing injuries ✗ Wrong options: SPECT · CT CAP · brain/spine MRI · isolated forearm film
Add	lateral thoracolumbar <12 months · **head CT <1 yr** or neuro signs · ophthalmology if shaking suspected
Bone scan	adjunct >1 yr · ↑ sensitivity for ribs, subtle shaft fractures, periosteal elevation · **useless for metaphyseal + skull fractures** · less efficient than survey
OI vs abuse (TO)	**DEXA 4–6 months apart** (density rises without metabolic disease, not in OI) · fibroblast culture / collagen analysis if unclear

Differential

List	accidental injury · **OI** · osteopenia of prematurity · scurvy · copper deficiency · **Menkes** · disuse osteopenia · chronic renal/liver disease · rickets
X-ray mimics	rickets (physeal widening, metaphyseal fraying) · wormian bones (OI, rickets; normal in 10% Caucasians, 80% Asians) · bisphosphonate zebra lines (vs Park-Harris arrest lines – persist, parallel to physis) · **Caffey disease**

Differential – Caffey disease (infantile cortical hyperostosis)

Features	infant · **solid wavy periosteal new bone** along long-bone shafts (ulna, femur), **clavicles, mandible**, ribs, scapulae · **resolves spontaneously** within ~a year
TO stems	infant with forearm cortical hyperostosis (OI / congenital syphilis / Caffey / infantile osteomyelitis) · diffuse ulnar thickening + clavicle + femur, resolved in a year (scurvy / Ewing / Garré / Caffey) · **answers not provided by the source**

5 Management

Management - skeletal survey & CPS

Rule	**report reasonable suspicion → child protective services** · **admit** for safety + multidisciplinary evaluation (pediatrics, social work, CPS, radiology) · document + photograph
Order	CPS/NAT work-up **before definitive fracture plan or discharge** ✗ Wrong options: cast and send home · next-week clinic · metabolic labs first · genetic testing first · talk home safety · restrain father / police
Fracture care	usually splint/cast until callus (Pavlik / spica for infant femur) · surgery rarely needed · still treat appropriately (e.g. forearm: **closed reduction + cast + skeletal survey + CPS**)
Toddler's fracture alone	not abuse · **plus corner fracture or suspicious skin lesions → admit / report to CPS**
If missed	**30–50% repeat abuse**, **5–10% death**

Legal duty to report

Physicians = mandated reporters · need only **reasonable suspicion** · **immune from civil + criminal liability in good faith** · **failure to report can be a crime**
✗ Wrong options: prove cause · call daycare first

Sexual abuse disclosure (minor) → child protective services
✗ Wrong options: parents · school board · athletic director · counselor

Medical child abuse - multidisciplinary evaluation

Munchausen by proxy (caregiver-fabricated illness)	usually biologic mother with medical knowledge · repeated doctor shopping, unwarranted tests/surgery · normal exam + X-rays → **early multidisciplinary evaluation** (medical, psychosocial, CPS, legal) ✗ Wrong options: CT · casting · coalition excision · another opinion

Developmental milestones

Milestone	Age
Sits	**9 months** (not sitting at 15 months → work up)
Walks	Up to **15 months** still normal
Hand dominance	**3rd year (~30 months)** · strong preference <2 yr → neuro deficit
Shoes on correct feet, buttons	4 yr
Hops on one foot, heel-to-toe walk	5 yr

– – Active space – –

Cast saw burns

Reduce burns	vacuum · **thinner cast** · **"up and down" technique** · check blade often · **avoid fiberglass** (more heat than plaster)

How it's asked

"2nd most common presenting sign of abuse?"	→ **fractures**
"Most common presenting sign?"	→ **skin cuts or bruises** ✗ Wrong options: fractures · head injury
"Two most common lesions?"	→ **skin lesions and fractures**
"Correct statement on child abuse?"	→ **2nd most common cause of death >1 yr** ✗ Wrong options: >5 yr · burns most morbidity · metaphyseal 4×
"Social risk factors – EXCEPT?"	→ **child with no siblings**
"10-month-old femur – NOT a NAT risk factor?"	→ **female gender**
"FALSE about NAT?"	→ **fractures most common initial finding**
"Most common form of maltreatment?"	→ **neglect**
"9-month-old – most common manifestation?"	→ **multiple bruises in various stages of healing**
"Metaphyseal bucket handle fracture seen in?"	→ **battered baby syndrome** (TO)
"Pathognomonic bone lesion of abuse?"	→ **metaphyseal corner fracture of distal femur** (TO) ✗ Wrong options: anterior rib · greenstick
"Isolated spiral femur – suspect abuse when?"	→ **not yet walking**
"NOT a common abuse pattern?"	→ **anterior femoral neck translation (SCFE)**
"Highly suggestive EXCEPT?"	→ **non-displaced spiral tibia in a toddler**
"6-month-old, corner + rib fractures + physeal separation, sibling with bone disorder?"	→ **non-accidental trauma** ✗ Wrong options: OI
"Which X-ray most concerning?"	→ **corner fracture**
"2-yr transphyseal distal humerus separation?"	→ **child abuse**
"X-ray characteristic of NAT?"	→ **rib fractures** ✗ Wrong options: rickets · wormian · Caffey
"Femur NAI vs accidental?"	→ **midshaft most common in both; NAI younger**

"8-month-old, seizures, femur fracture, chest bruises, retinal hemorrhages?"	→ **shaken baby syndrome**
"8-month-old healing supracondylar femur fracture, multiple carers?"	→ **non-accidental trauma**
"Acute femur fracture in 5-month-old?"	→ **non-accidental trauma** ✗ Wrong options: birth trauma
"Femur fracture most suggestive of NAT?"	→ **metaphyseal corner fracture**
"Foster parents, skin lesion — matching X-ray?"	→ **metaphyseal corner lesion** ✗ Wrong options: zebra lines · anterolateral bowing
"Most specific isolated fracture (3-yr)?"	→ **transphyseal distal humerus**
"10-month-old femur + bruises — study?"	→ **skeletal survey**
"6-month-old corner lesions, vague history?"	→ **admit + abuse work-up**
"12-month-old physeal separation + healing femur?"	→ **CPS + skeletal survey**
"Non-ambulatory infant with diaphyseal fracture — must do?"	→ **CPS consultation ± admission**
"10-month-old distal humeral separation with callus?"	→ **skeletal survey + contact authorities** ✗ Wrong options: CRPP first
"6-month-old spiral tibia, vague parents?"	→ **skeletal survey**
"11-month-old, elbow + ankle bucket handle, 3 AM?"	→ **admit to hospital**
"9-month-old forearm fracture off changing table?"	→ **reduction + cast + skeletal survey + CPS**
"6-month-old femur, aunt says fell from highchair?"	→ **immediate CPS notification** ✗ Wrong options: Pavlik first
"2-yr toddler's fracture + humeral corner fracture on survey?"	→ **cast + admit for NAT work-up**
"16-month-old tibia + proximal corner fracture?"	→ **CPS consultation**
"4-month-old with syndrome, femur fracture from bouncer?"	→ **additional imaging + social work**
"10-month-old Down syndrome, femur fracture, normal labs?"	→ **immediate CPS notification**
"6-month-old 'fell running after dog' — femur?"	→ **additional imaging + social work**
"3-yr toddler's fracture + pinna bruises + cigarette burn?"	→ **contact state CPS** ✗ Wrong options: call daycare
"12-yr discloses coach touching him?"	→ **inform child protective services**
"30-month-old — milestone present?"	→ **hand dominance established**
"15-month-old not walking — worrying if not?"	→ **sitting**
"Nurse mother, repeated sprains, demands coalition surgery?"	→ **early multidisciplinary evaluation**
"Cast saw burn — NOT protective?"	→ **fiberglass cast material**

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PEDIATRIC TRAUMA EVALUATION & MANAGEMENT

– Pediatrics · 81 / 112 –

1 Basics

Epidemiology trauma = **most common cause of death in children >1 yr** · polytrauma mortality **~20%** · falls + MVAs most common

Morbidity **CNS injury = highest overall morbidity + mortality** · **spine fractures = highest among MSK injuries** · C-spine injury more common **<8 yr** (restraints don't fit)

Anatomy & physiology vs adults

System	Child
Airway	Smaller (obstructs easily) · large tongue, floppy epiglottis · **large occiput flexes neck supine** · **larynx higher + anterior (C2–C3 vs C6–C7)**
Breathing	**Hypoventilation = most common cause of arrest** · horizontal ribs · diaphragmatic · fewer type I fibres → tire fast · RR varies with age
Circulation	Relatively larger but absolutely small blood volume · lower SVR · **hypotension is LATE** (25–30% loss) · small veins → **IO access**
Disability	Open sutures/fontanelle · thin skull · **large head** → head/neck injury
Exposure	High BMR + surface area → **heat loss** · ↑ glucose need, small glycogen stores

5 Management

Resuscitation (airway, IO access, blood volume, transport)

Transport **<8 yr: spine board with occipital cut-out or raised chest** (✓ modified board for suspected spine injury)
✗ Wrong options: routine pelvic binder · routine IO · nasal airway in head injury

ATLS Airway → Breathing → Circulation → Disability → Exposure · fix each before moving on

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Intubate — **GCS <8** / can't protect airway (✗ Wrong options: GCS <12) · e.g. GCS 6 + SpO2 83% → **intubation first** (✗ Wrong options: splint ankle · CT · close laceration)

Formula	Value
Broselow tape	Drug doses, equipment size, defibrillation energy
ETT (uncuffed)	**(age/4) + 4** or (age + 16)/4
Systolic BP	**80 + (2 × age)**
Chest tube	**4 × ETT**
Blood volume	**75 mL/kg** (Orthobullets) · **75–80 mL/kg** (OB)
First bolus	**20 mL/kg NS**
After 2 boluses	**10 mL/kg PRBC**

Shock signs — narrowed pulse pressure at **~15%** loss · hypotension after **25–30%** · 100 mL in 5 kg child ≈ 26.7% of volume

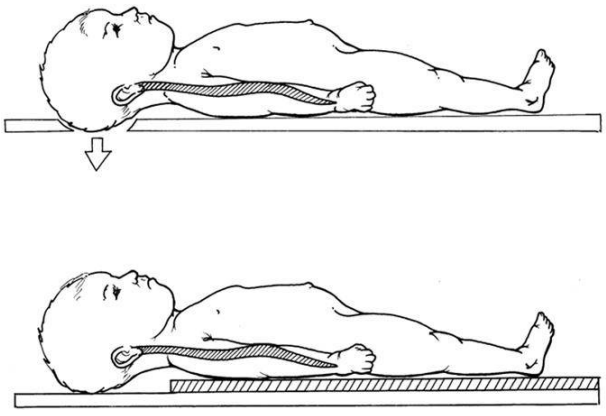
IV access fails (even normotensive) → intraosseous line
✗ Wrong options: PICC · cutdown · subclavian · keep trying

Triad of death — **acidosis + hypothermia + coagulopathy** (inadequate resuscitation)

Pediatric scoring systems

PTS	Mortality
<0	**100%**
1–4	40%
5–8	7%
≤8	**Send to pediatric trauma centre**

Pediatric GCS <8 → ↑ mortality · **SpO2 at presentation + GCS at 72 h** predict long-term neuro recovery



Pediatric spine board with occipital cut-out

– – Active space – –

Table 1. Pediatric Trauma Score

Patient features	+2	+1	-1
Size (kg)	> 20	10-20	< 10
Airway	Patent	Maintainable	Non maintainable
Systolic BP	> 90	50-90	< 50
Mental status	Awake	Obtunded	Comatose
Open wound	None	Minor	Major
Extremity fracture	None	Close	Open or multiple

Pediatric trauma score

Injury patterns & outcomes (TBI, multiorgan failure)

TBI — **most common ATV injury** (✗ Wrong options: open fractures · pneumothorax) · **greatest long-term morbidity** (✗ Wrong options: spine · abdomen · open fracture) · pain ↑ ICP → **fracture fixation ↓ ICP** · **HO more common → ↑ serum ALP heralds HO** → NSAID prophylaxis

Peripheral nerve injury — mostly with **closed fractures** · **EMG if no return by 2–3 months**

Multiorgan failure — **early after admission, all organs simultaneously** · adults: from 48 h, sequential starting with lungs · acute lung injury ~6× lower in children ✗ Wrong options: 48 h · sequential

Femur fixation in polytrauma with head injury

Timing — fix once **cerebral perfusion pressure stable** · early fixation eases transport, ↓ malunion/shortening, ↓ ICP

9-yr TBI + transverse femoral shaft → retrograde flexible IM nails
✗ Wrong options: spica · 90-90 traction (spasticity) · piriformis reamed nail (AVN risk)

Prognosis — children recover from TBI better than adults

How it's asked

"8-yr fell from tree, GCS 6, SpO2 83%?"	→ **intubation**
"Blood volume per kg in a 5-yr?"	→ **75–80 mL**
"5-yr polytrauma, no peripheral IV, BP 110/70?"	→ **intraosseous device**
"Transporting 4-yr with fluctuating consciousness?"	→ **modified spine board** ✗ Wrong options: intubate if GCS <12 · routine binder
"4-yr polytrauma – greatest long-term morbidity?"	→ **traumatic brain injury**
"Multiorgan failure in children?"	→ **early, all organs simultaneously**
"6-yr high-speed ATV – most likely injury?"	→ **brain contusion**
"9-yr TBI + transverse femur fracture?"	→ **retrograde flexible IM nails**

PELVIS FRACTURES - PEDIATRIC

– Pediatrics · 82 / 112 –

1 Basics

Incidence **1–2%** of pediatric fractures · acetabulum 1–15% of these · avulsions almost only in **adolescents**

Mechanism ring: **high energy** (MVA, pedestrian struck) · avulsion: low energy, **explosive** activity (sprint, jump, kick)

Associated **CNS + abdominal visceral injury >50%** (energy needed to break elastic pelvis) · chest injury ~63%, long bones ~50%, spine ~25% · femoral head fracture/dislocation with acetabular fracture · **GU injury with Torode IV**

Child vs adult pelvis

Child	Reason
More **lateral compression** (adult: APC)	–
More **single-bone ring fractures**	Elastic bone, cartilage absorbs energy, elastic SI + symphysis, **thick periosteum**
"Dislocations" heal like fractures	Periosteal tube
Less hemorrhage	Small vessels vasoconstrict · pelvic volume rarely ↑
Acetabulum: **transverse** > both column	Triradiate injury with less force → growth arrest

Anatomy

Primary centres (endochondral) – ilium 9 wk · ischium 16 wk · pubis 20 wk · fuse at triradiate **12 (girls) / 14 (boys)**

Acetabulum enlarges by **interstitial growth of triradiate cartilage** · concavity from femoral head pressure · secondary centres appear 8–9, fuse 17–18: **os acetabuli (anterior wall)**, acetabular epiphysis (superior), ischial secondary centre (posterior wall)

Apophysis	Appears	Fuses
Iliac crest (Risser)	13–15	15–17
Ischial	15–17	**19–25**
AIIS	14	16

– – Active space – –

Apophyseal avulsion fractures

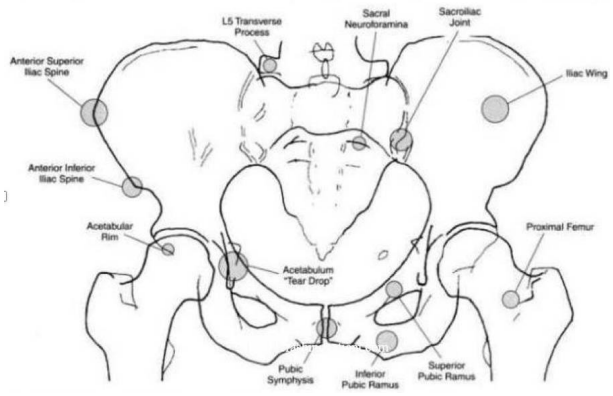
Avulsion site (TO)	Muscle	Frequency
Ischial tuberosity	Hamstrings	**Most common**
AIIS	Rectus femoris	Common
ASIS	Sartorius	Common
Pubic symphysis / iliac crest	Abdominals	Rare
Lesser trochanter	Iliopsoas	**Least common**

Child pelvis (TO) elastic, cartilaginous · more single-bone fractures than ring disruptions · **less haemorrhage** (vasoconstriction, mobile vessels) · **visceral injury without obvious fracture** · complications: **premature triradiate closure** → acetabular dysplasia, LLD

Site	Muscle	%
Ischial tuberosity	**Hamstrings + adductors**	**54%**
AIIS	**Rectus femoris**	22%
ASIS	**Sartorius**	19%
Pubic symphysis	Abdominals	3%
Iliac crest	Abdominals (external oblique)	1%
Lesser trochanter	**Iliopsoas**	–

Ischial avulsion age **13–17** as tuberosity ossifies (apophysis 2–5× weaker than tendon) · hip extension + knee flexion weakness · **X-rays negative → MRI**
✗ Wrong options: CT (for displacement) · EMG · bone scan · steroid

AIIS groin pain, snap, **pain on resisted knee extension** · **ASIS** – hockey check / eccentric sartorius



Pelvic apophyseal avulsion sites

Iliac crest apophyseal avulsion (TO) adolescent (e.g. 16-yr-old cricketer), sudden crest pain · unfused apophysis pulled off by **core muscles** = **type I physeal injury** · often missed

3 Presentation

– – Active space – –

Exam ATLS · **rectal + GU exam** (open fracture) · log-roll for ecchymosis · palpate ASIS, crests, SI, symphysis · iliac crest compression / posterior pressure pain · excess mobility = serious injury

4 Diagnosis & Classification

Key radiology findings

Views **AP pelvis** · **Judet** (45° obliques – acetabulum) · **inlet/outlet** (35° caudal/cranial – ring)

CT X-rays miss **~50%** of pediatric pelvic fractures · SI tenderness, negative films with suspicion, planning, spine injury

MRI occasionally for apophyseal avulsions

Torode/Zieg classification (pediatric ring)

Type	Injury	Treatment
I	**Avulsion**	Protected WB if <2 cm · ORIF >2–3 cm (historically >3 cm)
II	**Iliac wing**	Protected WB if <2 cm · ORIF >2–3 cm
III	Ring, **no segmental instability**	WBAT 6 wk · ORIF if acetabulum displaced >2 mm
IV	Ring **with segmental instability**	**<2 cm → bed rest** · unstable / >2 cm → ORIF

Tile A stable · B rotationally unstable, vertically stable · C unstable both



Torode/Zieg type IV

Bucholz classification (acetabulum / triradiate)

Type	Physeal	Features
Shearing	SH I / II	Blow to pubis/ischium/femur · triangular medial Thurston-Holland fragment (SH II)
Crushing / impaction	**SH V**	Hard to see · narrowed triradiate · **premature closure** · earlier = worse deformity

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5 Management

Pelvic ring injury management

Non-op	symphysis/SI "dislocations" heal by periosteum · types I–II <2 cm: protected WB 2–4 wk → stretch/strengthen 4–8 wk → sport after 8 wk · type III WBAT 6 wk

Stable type IV open book (symphysis 1.2 cm, SI 0.7 cm, stable) → bed rest + progressive mobilisation
✗ Wrong options: ex-fix · binder · fixation

Hemodynamically unstable + rotationally unstable (APC) ring → pelvic binder · **ex-fix** for instability / ↑ volume · **angiography** if still bleeding
✗ Wrong options: iliac wing · rami · avulsions

Adolescent with displaced symphysis + SI widening → anterior symphyseal plate + percutaneous SI screw
✗ Wrong options: non-op · anterior only · ex-fix

Operative indications	displaced posterior SI · iliac fracture >2 cm · anterior disruption with vertical/rotational instability · older children/adolescents do better with fixation
Acetabulum ORIF	**displacement >2 mm** · incongruity · **intra-articular fragments** · instability (medial/posterior subluxation) · central fracture-dislocation · open · comminution not improved by traction
Technique	ilioinguinal / Stoppa · **physeal-sparing** · if needed **smooth pins across triradiate 4–6 wk, removed early**

Apophyseal avulsion management

First line	crutches, **protected WB**, surveillance X-rays (e.g. minimally displaced lesser trochanter avulsion in hurdler)
Differential on hip films	SCFE → in situ screw · FAI → PT · compression femoral neck stress fracture → NWB (screw if >50% width) · **tension-sided** → cannulated screws

6 Complications

Triradiate premature closure	**<5%** · risk: **age <10 yr**, **Bucholz crush (SH V)** · thick medial wall → **shallow acetabulum, lateral subluxation**, hypoplastic hemipelvis · bar excision (may still close) · late **pelvic osteotomy (Chiari)** · follow to maturity
Others	LLD (vertical displacement >2 cm) · malunion (well tolerated) · nonunion (rare) · NV injury · HO · **femoral head AVN** (acetabular fracture + dislocation) · hip OA (displaced acetabular) · death (head/visceral)
Prognosis	complications rare · surgery more often needed after triradiate closure

How it's asked

"8-yr hit by car, SBP 48 — which pelvis benefits from binder?"	**→ AP compression (open book) ring injury**
"6-yr, symphysis 1.2 cm + SI 0.7 cm, stable?"	**→ bed rest + progressive mobilisation**
"14-yr, displaced symphysis + SI widening?"	**→ anterior plate + percutaneous SI screw**

"14-yr soccer snap, pain on resisted knee extension — muscle?"	→ **rectus femoris (AIIS)** ✗ Wrong options: sartorius · iliopsoas
"14-yr gymnast, posterior thigh, negative X-rays?"	→ **MRI** ✗ Wrong options: CT · EMG
"16-yr hockey check, ASIS avulsion — muscle?"	→ **sartorius**
"14-yr hurdler — which injury gets crutches + repeat films?"	→ **lesser trochanter avulsion** ✗ Wrong options: SCFE · tension stress fracture

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PHYSEAL CONSIDERATIONS

– Pediatrics · 83 / 112 –

1 Basics

Pediatric bone — **elastic** → buckle + greenstick fractures · **remodeling**: faster **in plane of joint motion** (sagittal at wrist) and at **most active physes** · intra-articular fractures still need reduction

Most active physes — UE: **proximal humerus, distal radius** ("away from the elbow") · LE: **distal femur, proximal tibia**, distal tibia ("at the knee")

Skeletal age — differs from chronologic age by **>1 yr in 26%** · maturity ~**14 girls / 16 boys**

Laws of the physis

Law	Statement
Hueter–Volkmann	↑ **compression slows** longitudinal growth
Delpech	↑ **tension speeds** growth
Wolff	Bone adapts to the loads placed on it
✗ Wrong option: **Maxwell**	Not a physeal law (TO)

Physeal zones & fracture plane (hypertrophic zone)

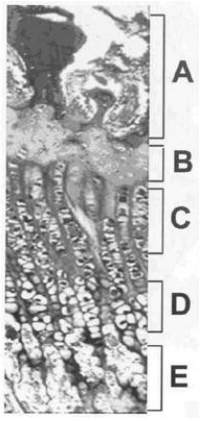
Zone	Function	Disease
Reserve	Stores lipid, glycogen, proteoglycan · **low O2**	**Gaucher** · Kniest · pseudoachondroplasia
Proliferative	Chondrocyte stacking → **longitudinal growth** · **most matrix production** · ↑ O2 inhibits calcification	**Achondroplasia** · gigantism
Hypertrophic – maturation, degeneration (cells ×5), **provisional calcification**	Chondrocyte death → calcium release → matrix calcifies · **type X collagen** · PTHrP / Indian hedgehog	**SCFE (non-renal)** · **rickets (provisional calcification)** · enchondroma · MPS · Schmid · **physeal fractures**
Primary spongiosa	Vascular invasion, osteoblasts on cartilage bars → woven bone	**Metaphyseal corner fracture (abuse)**
Secondary spongiosa	Internal remodeling (lamellar bone) + funnelization	**Renal SCFE**

Fracture plane — physeal fractures run through the **hypertrophic zone**, specifically the **zone of provisional calcification** (weakest in tension; calcified/uncalcified transition) · applies to SH I, II and III (OB) · chronic gymnast wrist injury → same zone

Reserve + proliferative zones stay with epiphysis, blood supply preserved → good prognosis in SH I–II · **TO: SH III + IV involve the reserve zone** (germinal cells) → ↑ arrest risk · SH V crush of germinal layer

Perichondral ring (Ranvier, LaCroix)

Structure	Role
Groove (zone) of Ranvier	Fibrous ring in 1st year · **↑ mechanical strength** · **appositional growth** · source of **osteochondroma**
Perichondral ring of LaCroix	Dense fibrous limiting membrane · **peripheral stability** · up to **50% of shear resistance** · injured in SH VI
Perichondral artery	Major nutrition of physis

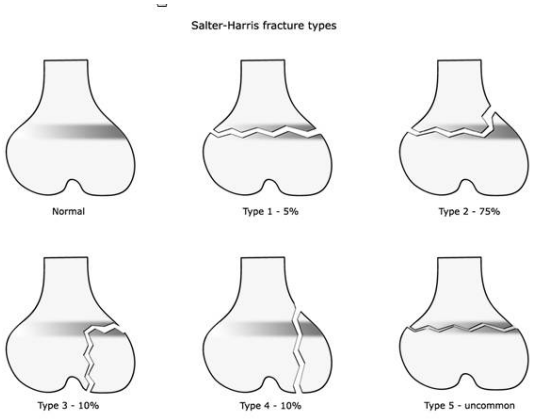


Physeal zones histology

4 Diagnosis & Classification

Salter-Harris classification

Type	Path
I	Through physis (separation)
II	Physis → exits **metaphysis** · **Thurston-Holland fragment** · **most common (>50%)** · periosteum torn on convex (tension) side, intact on concave
III	Physis → exits **epiphysis** (intra-articular)
IV	Epiphysis → physis → metaphysis
V	**Crush** of physis · diagnosed late



Salter-Harris classification

– – Active space – –

Physeal vs ligamentous ankle injury

Pediatric inversion injury, normal X-ray → lateral ligament sprain on MRI (more common than SH I)
✗ Wrong options: physeal widening · bone bruise · peroneal tear

Bone age estimation

Method	Film / basis
Greulich-Pyle / Tanner-Whitehouse	**Left hand + wrist** (right more often injured; anthropology convention) (TO)
TW-III RUS	Distal radius, ulna, short bones · correlates with PHV
Sauvegrain	**Left elbow** · 27 points: lateral condyle, trochlea, olecranon apophysis, proximal radial epiphysis
Olecranon method	6-monthly age: girls 11–13, boys 13–15
Risser	Iliac apophysis 0–4 ossification, 5 fused · variable
Peak height velocity	Better than Risser/age · **8 cm/yr girls, 9.5 cm/yr boys**
Sanders	Phalanx, metacarpal, distal radius epiphyses · predicts scoliosis progression

Hand dominance — **2–3 yr** (third year) · strong preference <2 yr → neuro deficit · persistent ambidexterity normal

5 Management

Treatment — closed reduction vs CRPP vs ORIF by pattern · **intra-articular must be anatomic**

Displaced SH III/IV distal tibia (11-yr) → ORIF with fixation PARALLEL to physis (epiphyseal or metaphyseal screw) + NWB cast · ankle physeal arrest up to **50%** → long follow-up
✗ Wrong options: transphyseal lag screw · closed manipulation

Growth contribution by physis

Physis	Growth
Distal femur	**9–10 mm/yr** (✓ greatest LLD if arrested)
Proximal tibia	6 mm/yr
Distal radius	5–6 mm/yr
Proximal femur	3 mm/yr
Distal humerus	2–3 mm/yr
Humerus	**Proximal 80% · distal 20%**

12-yr boy distal femoral arrest → 4 yr × 10 mm = **40 mm**

6 Complications

Growth arrest - location, mechanism & follow-up

Mechanism fracture through all zones (SH III/IV) → **vascular invasion across physis** → osteoblasts → **bony bar**
✗ Wrong options: LaCroix proliferation · Wolff · Hueter-Volkmann

Pattern complete arrest → **shortening** · partial → **angulation**

Most common site of post-traumatic arrest **distal (medial) tibia** · also high: distal femur, distal ulna · rare: distal radius, proximal humerus

Implant violation **<7% of physeal area unlikely to arrest** (Mäkelä: 7% arrested, 3% didn't) · retrograde nail through 5% central physis, screws above physis → **injury unlikely**

Follow-up **≥2 yr** after physeal injury · multiple X-rays needed
✗ Wrong options: one film · stop at 14/16

Bar	Treatment
<50% of physis + **>2 yr or 2 cm growth left**	**Bar resection + interposition** (fat / wax)
>50%	**Complete ipsilateral arrest** ± contralateral epiphysiodesis / ipsilateral lengthening



Angular deformity from partial arrest

Little League elbow outcome – Panner disease

Panner (osteochondrosis of capitellum) **age 5–10** · repetitive valgus → ischemia of whole capitellar epiphysis → flattening, fragmentation, reossification (like Perthes) · **symptomatic** (pain, swelling, ↓ motion)

Outcome non-op succeeds · **full motion returns** · X-ray irregularity / sclerosis often **persists**
✗ Wrong options: anatomic restoration · incidental · spontaneous · high failure

vs capitellar OCD focal lesion, **>10 yr**

TO points physeal injuries ~**20%** of skeletal injuries · SH frequency **II > I > III > IV > V** · injury near physis may **stimulate overgrowth** · ligaments stronger than bone in children · **plastic deformation** = microscopic osteon realignment without cortical break · remodelling ↑ with younger age, proximity to physis, angulation in plane of joint motion · **rotation does not remodel**

– – Active space – –

How it's asked

"Which physeal injury involves the reserve zone?"	→ **both type III and IV** (TO)
"Law NOT pertaining to growth plate?"	→ **Maxwell's law** (TO)
"Zone affected in SH II?"	→ **hypertrophic zone**
"5-yr SH II distal femur, injury through which layer?"	→ **lower hypertrophic**
"5-yr SH III ankle → bar a year later — layer?"	→ **hypertrophic zone, provisional calcification**
"12-yr gymnast chronic wrist pain — location?"	→ **zone of provisional calcification**
"10-yr SH I distal femur — zone?"	→ **hypertrophic** ✗ Wrong options: primary spongiosa
"11-yr FOOSH physeal fracture — zone?"	→ **hypertrophic (D)**
"Correct match?"	→ **SH III: provisional calcification** ✗ Wrong options: Gaucher-proliferative · renal SCFE-degenerative · achondroplasia-reserve
"7-yr large Thurston-Holland — zone?"	→ **hypertrophic zone (D)**
"12-yr SH II distal tibia — region damaged?"	→ **where matrix calcification occurs** ✗ Wrong options: LaCroix · most matrix production
"Ranvier + LaCroix — true?"	→ **↑ mechanical strength of physis**
"Distal humerus physis contributes?"	→ **20%**
"Arrest causing greatest LLD at 12?"	→ **distal femur**
"11-yr displaced SH IV ankle?"	→ **ORIF with fixation parallel to physis + NWB cast**
"Post-traumatic arrest most common at?"	→ **distal tibia**
"Mechanism of premature arrest?"	→ **vascular invasion across physis**
"Retrograde nail violated 5% of physis — counsel?"	→ **physeal injury unlikely**
"SH IV distal femur, normal at 6 months — follow-up?"	→ **at least 2 yr**
"Twisted ankle, normal X-ray — MRI?"	→ **lateral ligament complex injury**
"8-yr pitcher, Panner disease — true?"	→ **full elbow motion usually returns**
"Hand dominance by?"	→ **2–3 yr**
"Tanner-Whitehouse bone age — radiograph?"	→ **left wrist** (TO)

POLYDACTYLY OF FOOT

– Pediatrics · 84 / 112 –

1 Basics

Incidence	**1 in 500** (Orthobullets) · postaxial ~1 in 200 (OB) · **postaxial (lateral) most common** · African American > Caucasian · foot less common than hand
Cause	failure of differentiation of the **apical ectodermal ridge** in the 1st trimester
Incidence	**1 in 500** (Orthobullets) · **~2 per 1000** (TO) · **most common isolated congenital foot anomaly** (TO) · mainly cosmetic / footwear problem

Inheritance & associations

Inheritance	**autosomal dominant** (family history) ✗ Wrong options: X-linked · AR · mitochondrial · spontaneous only
Isolated postaxial with normal exam	**normal motor milestones** · **no cardiac / renal work-up** · no systemic association

Venn-Watson classification

Type	Patterns
Postaxial (lateral)	**Y** metatarsal · **T** metatarsal · wide MT head · complete duplication
Central (added)	Duplicated 2nd, 3rd or 4th toe
Preaxial (medial)	Short block 1st MT · wide MT head

Y vs T metatarsal (TO) **Y = equal** duplication · **T = unequal** duplication



Venn-Watson classification

4 Diagnosis & Classification

X-ray	MTs visible at birth · **delay films** until phalanges ossify for planning

– – Active space – –



Postaxial polydactyly radiograph

5 Management

Situation	Treatment
Well-aligned postaxial / central	**Observation** (unless foot much wider)
Malaligned toe, esp. preaxial	**Ablate extra (usually border) digit at 9–12 months**
Small skin tag	Remove in newborn nursery
Cosmesis / shoe wear	Elective, any age per family wishes (OB)

Natural history shoe-fitting problems, angular toe deformity

Excision (TO) supernumerary toe + **capsular reconstruction** · **modified Farmer procedure** for duplicated hyperplastic toe: remove extra ray, rotational flap, **reinsert adductor hallucis**

How it's asked

"10-yr postaxial foot polydactyly, father similar, normal exam?"	→ **normal milestones, no further work-up** ✗ Wrong options: cardiac/renal work-up · spontaneous mutation
"7-month-old postaxial polydactyly – inheritance?"	→ **autosomal dominant**

POPLITEAL CYST IN CHILDREN

– Pediatrics · 85 / 112 –

1 Basics

Definition herniated posterior knee capsule synovium between **semimembranosus** and **medial head of gastrocnemius** · common soft-tissue mass · **unlike adults, usually NOT linked to meniscal tears**

3 Presentation

Findings usually **painless**, mobile, soft · posteromedial, distal to knee crease · **most prominent with knee extended** · **transilluminates**



Popliteal cyst

4 Diagnosis & Classification

Key radiology findings

X-ray normal · **US** – cystic · **MRI** – fluid-filled simple cyst (confirms)

5 Management

Treatment - observation

Observation mainstay · Dinham: of 70 untreated cysts, **51 resolved spontaneously** over a mean **1 yr 8 months**

Aspiration may work but **high recurrence** · **excision** only if significant discomfort or failure to resolve

✗ Wrong options: radical excision · arthroscopic resection first · CT · bone scan

– – Active space – –

How it's asked

"10-yr, soft painless posteromedial mass, MRI simple cyst?" → **observation**

POSTEROMEDIAL TIBIAL BOWING

– Pediatrics · 86 / 112 –

1 Basics

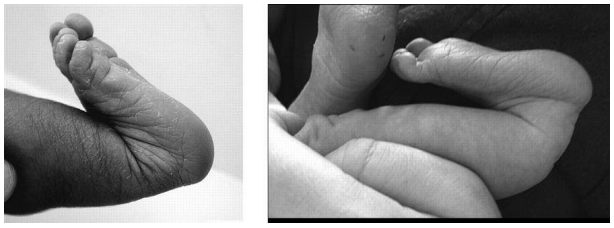
Associated calcaneovalgus foot & aetiology

Cause	**intrauterine positioning** · no genetic / familial pattern ✗ Wrong options: AR · HOXD10 (vertical talus) · Shh (anteromedial bowing/fibular hemimelia) · fracture nonunion (anterolateral)
Site	**middle + distal third** of tibia (distal more common) · usually unilateral · posterior bow more severe than medial · angulation 25–70° (TO)
Associated	**calcaneovalgus foot** (✓ most common association; together or separately) · ankle dorsiflexion contracture · distal tibial epiphyseal valgus wedging, fibular hypoplasia · shortening present at birth
Fracture / pseudarthrosis	**rare** (unlike anterolateral bowing)

Tibial bowing	Association
Posteromedial	Packaging · **calcaneovalgus** · **LLD**
Anteromedial	**Fibular hemimelia** (Shh)
Anterolateral	**NF1, congenital pseudarthrosis**

3 Presentation

At birth	short, bowed leg · dorsum of foot may rest on anterior shin
Exam (TO)	**skin dimple over apex** · **underdeveloped calf** · **↓ ankle plantarflexion** · apex **posterior on lateral, medial on AP** · differentiate from positional calcaneovalgus
Apex	posteromedial bowing apex in **distal tibia** · calcaneovalgus apex at **ankle** (Orthobullets) / heel (TO)
Calcaneovalgus vs vertical talus	**plantar-flexion lateral foot X-ray**: in CVT the talar axis stays plantar to (not collinear with) the 1st MT
TO case	neonate, convex posteromedial bow, calf atrophy, lateral dimple, thick anterolateral cortices, normal canal · options: constriction band / CPT / posteromedial bowing / fibular hemimelia · **answer not provided by the source**



Posteromedial bowing vs calcaneovalgus

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Views AP + lateral tibia · posteromedial bow obvious, normal bone quality



Posteromedial tibial bowing X-ray

5 Management

Natural history & residual LLD

Bowing **spontaneously corrects over 5–7 yr** (~80% by age 5) · **observe** · calcaneovalgus → observation + parental stretching
✗ Wrong options: osteotomy · bracing / clamshell KAFO · serial casting (no effect; casting only for severe unresolving contracture) · genetic testing

LLD = main problem proportional, ↑ with growth · ∝ amount of bow

Remodelling (TO) **rapid in first year** · mainly by **physeal reorientation** · growth inhibition persists → predictable LLD · average **3.5 cm** (TO) · **no corrective osteotomy for the bow** · Rx LLD: contralateral epiphysiodesis or ex-fix lengthening

Source	Residual LLD
Orthobullets	**Average 3–4 cm** · residual 2–5 cm
OB	3–8 cm · ~4 cm · 2–5 cm
TO	3–7 cm
Di Gennaro (OB)	4.3 cm

Surgery for LLD **>50%** of patients (Wright 53% lengthening)

– – Active space – –

LLD (TO)	Residual angle	Ankle	Treatment
<2 cm	Resolved	Normal	None
2–4 cm	Resolved	Normal	**Contralateral tibial epiphysiodesis**
>4 cm	Resolved	Normal	**Tibial lengthening**
>2 cm	**Residual angulation**	–	Metaphyseal lengthening + diaphyseal osteotomy
Any	Any	**Ankle valgus**	Supramalleolar osteotomy / medial hemiepiphysiodesis

Most common complication needing surgery → LLD (TO)
✗ Wrong options: calcaneovalgus · bowing · pseudarthrosis fracture

Major deformity needing correction → LLD (TO)
✗ Wrong options: posterior bow · medial bow

How it's asked

"Newborn posteromedial bowing, normal neuro + hips?"	→ **observation** (TO) ✗ Wrong options: casting · early osteotomy · bracing
"Major deformity needing correction?"	→ **LLD** (TO)
"Most common complication requiring surgery?"	→ **limb length discrepancy** (TO)
"Infant posteromedial bow – associated condition?"	→ **residual LLD** ✗ Wrong options: pseudarthrosis · knee instability · coalition
"2-yr posteromedial bowing – complication?"	→ **LLD** ✗ Wrong options: fibular hemimelia · NF
"Healthy newborn – next step?"	→ **observe for resolution + possible LLD** ✗ Wrong options: AFO · NF gene counselling
"Newborn isolated posteromedial bow?"	→ **observe; bow resolves, LLD remains** ✗ Wrong options: Syme · casting
"3-yr – counsel family about?"	→ **limb-length discrepancy** ✗ Wrong options: NF · renal
"2-month-old posteromedial bow?"	→ **observe, revisit in 4 months** ✗ Wrong options: clamshell · casts
"2-yr, prognosis?"	→ **resolves by ~5, will likely need LLD surgery**
"Most common associated deformity?"	→ **calcaneovalgus foot** (TO)
"3-month-old tibial deformity – foot deformity?"	→ **calcaneovalgus** ✗ Wrong options: equinovarus · MA

– – Active space – –

"Posteromedial bow + calcaneovalgus – etiology?"	→ **intrauterine positioning**
"2-week-old bowed leg – other finding?"	→ **calcaneovalgus foot** ✗ Wrong options: café-au-lait · limb lengthening

PROXIMAL FEMORAL FOCAL DEFICIENCY

– Pediatrics · 87 / 112 –

1 Basics

Definition	defect of the **primary ossification centre (cartilage anlage)** of proximal femur · spectrum: **absent hip · femoral neck pseudarthrosis · absent femur · short femur**
Epidemiology	**bilateral 15%** · mostly **sporadic** · rare AD form (dysmorphic facies)

Genes & lab markers

Gene	**sonic hedgehog** (limb bud)

Associated ACL deficiency & knee instability

Associated	**fibular hemimelia (50%)** · **ACL deficiency / absent cruciates** · underdeveloped tibial spines · **coxa vara** · knee flexion contractures · acetabular dysplasia · ball-in-socket ankle

ACL deficiency → knee subluxation during femoral lengthening → examine with **Lachman**
✗ Wrong options: dial · Galeazzi · Thessaly · varus stress

Not associated	clubfoot · café-au-lait · webbed neck (Turner) · scapular winging (Sprengel)
Associations (TO)	**fibular hemimelia ~45%** · **clubfoot** (TO lists it – conflicts with the "not associated" line above; read the options) · congenital heart disease · spinal and facial dysplasia
ACL deficiency most likely in	**PFFD** ✗ Wrong options: cleidocranial (RUNX2) · Apert (FGFR2) · achondroplasia (FGFR3) · radial head dislocation

3 Presentation

Exam	severe shortening · **% shortening constant with growth** · short bulky thigh **flexed, abducted, externally rotated** · feet usually normal
Exam (TO)	**knee flexion contracture** · foot often at level of **contralateral knee** · **skin dimple over greater trochanter**

– – Active space – –



PFFD clinical appearance

4 Diagnosis & Classification

Aitken classification

Class	Femoral head	Acetabulum
A	Present	Normal
B	Present	Mildly dysplastic
C	**Absent**	Severely dysplastic
D	**Absent**	**Absent**

Femoral head ossifies ~6 months → defer hip decisions until then

Aitken – TO version A: head present, short femur · B: **femoral neck pseudarthrosis** · C: **absent head** · D: **absent head + acetabulum** · **Paley**: group A congenital short femur → B–D increasing discontinuity

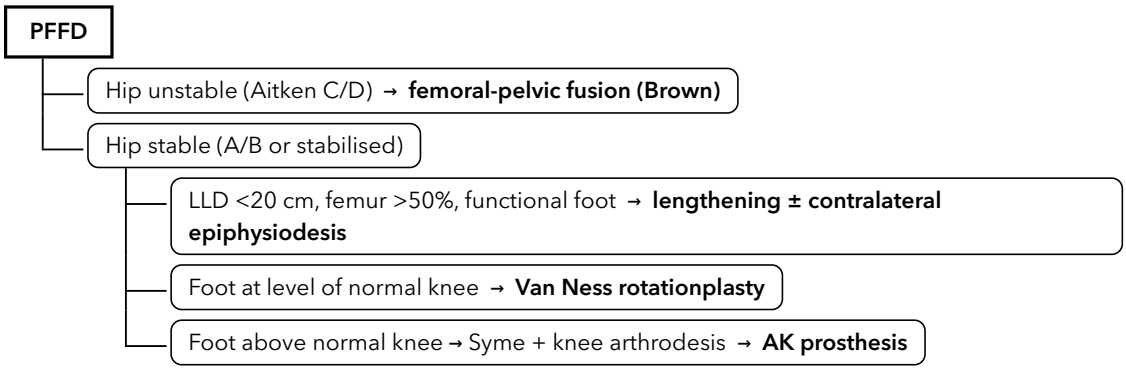
Options (TO) orthosis · lengthening (selected Paley A; hip + knee instability makes it hard) · valgus osteotomy for coxa vara · **knee fusion + extension prosthesis** · rotationplasty · amputation



Femoral neck pseudarthrosis

5 Management

Individualise by final LLD · foot · musculature · **proximal (hip) stability**



Operative criteria

Procedure	Indication
Lengthening ± contralateral epiphysiodesis	Predicted LLD **<20 cm** · stable hip + functional foot · **femur >50%** of other side · **head present (Aitken A/B)** · not with unaddressed coxa vara, neck pseudarthrosis, acetabular dysplasia
Knee arthrodesis + foot ablation (Syme)	Foot **proximal to opposite knee** · prosthetic knee won't sit below opposite knee · Syme at 10–14 months, knee fusion 3–4 yr (OB)
Femoral-pelvic fusion (Brown)	**Femoral head absent (Aitken C/D)**
Van Ness rotationplasty	**Foot at level of opposite knee** · ankle **>60% motion** · absent head (C/D) · **180° rotation** → ankle DF becomes knee flexion → BK prosthesis
Amputation	Femur **<50%** · keep length · **through joint** (avoid overgrowth) · prosthesis after 1 yr
Observation / extension prosthesis	Bilateral PFFD

Treatment - rotationplasty

11-week-old, ankle will reach normal knee level → delayed Van Ness rotationplasty + hip stabilisation at 1 yr if head deficient
✗ Wrong options: immediate femoral-pelvic fusion (head status unknown) · Syme + knee fusion (too short) · immediate lengthening

How it's asked

"ACL most likely deficient in?"	→ **PFFD**
"PFFD – test for condition causing knee subluxation with lengthening?"	→ **Lachman**
"6-yr PFFD – other finding?"	→ **knee ligamentous laxity** ✗ Wrong options: clubfoot · café-au-lait
"Infant PFFD, ankle at normal knee level at maturity?"	→ **delayed Van Ness rotationplasty, hip stabilisation at 1 yr if needed**

– – Active space – –

PROXIMAL FEMUR FRACTURES - PEDIATRIC

– Pediatrics · 88 / 112 –

1 Basics

Epidemiology	**<1%** of pediatric fractures · **boys 2.5:1** · bimodal: **<2–3 yr (NAT)** and adolescents (MVA, ATV) · **high energy 75–80%** · low energy → pathologic bone
Associated injuries	**30–85%** (head/face, spleen, retroperitoneal bleed, perineum, pelvis/acetabulum, hip dislocation, femur)

Anatomy

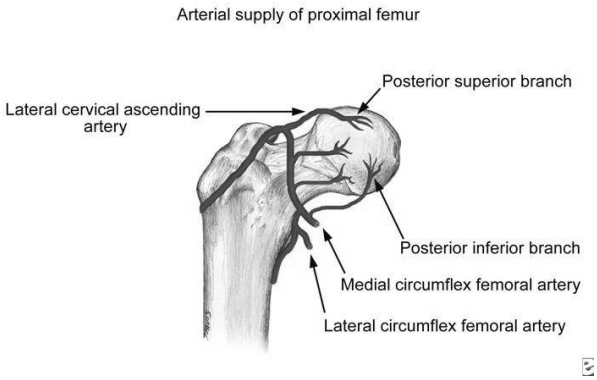
Proximal femoral physis	ossifies 4–6 months (girls), 5–7 (boys) · **3 mm/yr** · **13–15% of leg length**, 30% of femur · arrest → LLD >2 cm rare (only very young)
Trochanteric apophysis	ossifies ~4 yr · appositional GT growth · **GT injury/arrest → coxa VALGA** · **GT overgrowth → coxa VARA**
Superior gluteal nerve (L4–S1)	gluteus medius + minimus

Vascular anatomy (MFCA)

Age	Femoral head supply
<4 yr (OB: <18 months)	**MFCA + LFCA + ligamentum teres** (+ metaphyseal <3 yr)
~18–36 months	Ligamentum teres stops first (Trueta)
≥4 yr (dominant by 6, Trueta)	**MFCA** via **posterosuperior + posteroinferior retinacular branches**
3 to 14–17 yr	Physis blocks metaphyseal vessels
>14–17 yr	Metaphyseal-epiphyseal anastomoses

AVN after fracture → retinacular branches of **MFCA**
✗ Wrong options: LFCA · ligamentum teres

Piriformis-entry nail contraindicated → damages **posterosuperior MFCA** branches → AVN · use **lateral/trochanteric** entry for IM nails



Blood supply of the proximal femur

Deforming forces

Subtrochanteric proximal fragment	**flexion** (iliopsoas: L1–3 / femoral n.) · **abduction** (gluteus medius/minimus – **superior gluteal nerve**) · **external rotation** (short rotators – sciatic)

3 Presentation

Exam	severe hip pain, can't bear weight · **shortened + externally rotated** leg · <2–3 yr type I → think **NAT** (skeletal survey)

4 Diagnosis & Classification

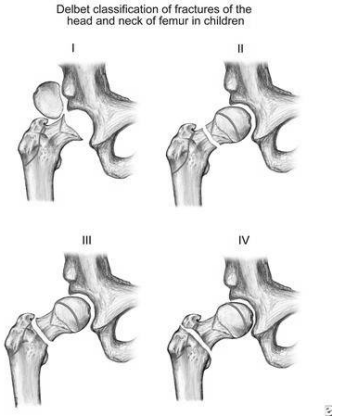
Key radiology findings

X-ray	AP + cross-table lateral · broken trabeculae at Ward triangle = undisplaced/impacted
Occult / stress fracture	**MRI best** (TO) · low-signal line + T2 edema · CT alternative · **US in infants** (hemarthrosis, epiphyseal mobility)
Differential	Perthes · transient synovitis · hemarthrosis · infection

AVN risk by fracture type (Delbet)

Type	Fracture	Incidence	AVN	Nonunion
I	**Transphyseal** (IA no dislocation / **IB with epiphyseal dislocation**)	<10%	**38–100%** (IB ~100%)	–
II	**Transcervical** (most common)	40–50%	**28%** (0–61% OB)	15%
III	**Cervicotrochanteric / basicervical**	30–35% (25–35% TO)	**18%** (27% OB · 20–30% OB)	15–20%
IV	**Intertrochanteric** (best outcome)	10–20% (6–15% TO)	**5%** (14% OB)	5%

AVN ranking	**IB > IA > II > III > IV** · I/II/III = 15×/6×/4× type IV risk · overall ~30%



Delbet classification

- - Active space - -

5 Management

Treatment & fixation

Situation	Treatment
Nondisplaced I (no dislocation), II, III, IV **and <4 yr**	**Closed reduction + abduction spica** (rare) · OB: spica only <6 yr types III/IV
Displaced / >4 yr, types I–III	**CRPP** (smooth pins <4 yr · cannulated screws >4 yr)
Can't get anatomic closed reduction	**Open reduction + screws** (Watson-Jones)
Type IV displaced / >4 yr	**Pediatric (or adult) sliding hip screw** (lateral approach)
Open fracture · vessel repair · **epiphyseal / hip dislocation (IB)** · marked displacement	**Emergent ORIF + capsulotomy / aspiration**

Implant by age (OB) 0–3 yr smooth wires · 4–10 yr 4.5–6.5 mm cannulated screws · >10 yr 6.5–7.3 mm screws
✗ Wrong options: **cephalomedullary nail** · immediate WB

Physis **short of physis** <4–6 yr, most type III (less stable) · **transphyseal** >12 yr, little metaphysis, or needed for stability (**type II: transphyseal** – poor head purchase otherwise) · "easier to treat LLD than nonunion" · within 5 mm of subchondral bone, avoid anterolateral epiphysis + posterior neck

Implant age cut-off smooth pins **< 4 yr** (Orthobullets) · **< 8 yr** smooth pins, **≥ 8 yr** cannulated screws (TO) – read the options · avoid multiple passes across physis (TO)

TO principles **vascular emergency** · gentle reduction under GA · **capsular decompression** (tamponade by intracapsular haematoma) · internal fixation + spica · after **age 3** physis blocks metaphyseal supply → **lateral epiphyseal artery (MFCA)** main; **3–12 yr most vulnerable** · Delbet **II most common** (TO) · **nonunion up to 10%** (TO)

Reduction gentle traction + **abduction + internal rotation** · weekly X-rays ×3 wk if casted · accept type II **<2 mm translation, <5°, no malrotation** · III/IV **<10°**

Post-op spica (abduction + IR) **6–12 wk** if <4 yr or screws short of physis

Minimising AVN - anatomic reduction

Most important factor **anatomic reduction** · open reduction OK if it gets anatomy
✗ Wrong options: timing · closed vs open · capsulotomy · transphyseal screw

Timing reduction **<24 h** may ↓ AVN (Orthobullets) · surgery **<48 h** ↓ AVN (OB4584) · delay to 24 h doesn't ↑ complications; literature conflicting (OB) – fix with rested experienced team

Capsulotomy favoured by some, unproven · **skeletal traction ↑ intracapsular pressure** (avoid)

6 Complications

AVN	**most common** · kinked/torn vessels, intracapsular tamponade · risk: **type (I highest)**, displacement, **age (×1.14 per year; >10 yr)**, delay >24 h, poor reduction · presents **2–12 months** (groin/knee pain) · Ratliff I whole head, II partial, III neck · **core decompression**, vascularised fibula · annual X-rays to maturity (≥2 yr in older teens)

Coxa vara (NSA <120°) — **2nd most common** · non-op treatment, types I–III (**25% type III**; ~15% OB) · 0–3 yr remodels if NSA >110° · **GT apophyseal arrest** (mild, <6–8 yr) · **valgus osteotomy** (nonunion, Trendelenburg, FAI, older)

Coxa valga	type IV in young child → premature GT apophysis closure
Nonunion	non-op II/III, occult infection, malreduction → **ORIF** (spica if young) ± graft
Physeal arrest	implants across physis, AVN (Ratliff II/III) · LLD: lift <2 cm, **contralateral distal femur ± proximal tibia epiphysiodesis 2–5 cm**
Chondrolysis	rare, with AVN or hardware penetration · TO: "**not a sequel**" in its MCQ (✓ answer) despite listing it as rare
Order (TO)	**osteonecrosis > coxa vara > physeal arrest/LLD > nonunion > infection > chondrolysis**
Poor function	head trauma · amputation · nerve damage · AVN

How it's asked

"Pediatric hip fracture with maximum AVN?"	→ **type I (transphyseal)** (TO)
"8-yr basicervical fracture fixed — most common complication?"	→ **AVN** ✗ Wrong options: chondrolysis · LLD
"14-yr displaced transcervical fixed — risk?"	→ **AVN** ✗ Wrong options: coxa vara
"Order AVN risk IA, IB, II, III, IV?"	→ **IB > IA > II > III > IV**
"Highest osteonecrosis incidence?"	→ **13-yr displaced transcervical femoral neck** ✗ Wrong options: Garden I · Jones
"4-yr basicervical — AVN risk?"	→ **25%**
"10-yr sled vs tree, transphyseal without dislocation — complication?"	→ **osteonecrosis**
"Delbet II — most likely complication?"	→ **osteonecrosis**
"13-yr ATV type III — age raises risk of?"	→ **osteonecrosis**
"13-yr femoral neck — true?"	→ **closed reduction + transphyseal cannulated screws acceptable** ✗ Wrong options: cephalomedullary nail · capsulotomy contraindicated
"14-yr Delbet II?"	→ **transphyseal cancellous screws** ✗ Wrong options: physis-sparing · DHS · spica
"11-yr, non-anatomic closed reduction?"	→ **open reduction + screws**

– – Active space – –

"Femoral epiphysis supply <18 months?"	→ **MFCA + LFCA + ligamentum teres**
"Juvenile intertroch/subtroch — preserve which artery?"	→ **MFCA**
"11-yr Delbet IB → AVN — vessel injured?"	→ **MFCA**
"Why no piriformis nail in a child?"	→ **osteonecrosis from posterosuperior MFCA damage**
"Most important in minimising AVN (type III)?"	→ **anatomic reduction** ✗ Wrong options: fixation within 2 h · traction · capsulotomy
"Fixed at 28 h, anatomic — biggest AVN factor?"	→ **anatomic reduction**
"Subtrochanteric proximal fragment abduction — nerve?"	→ **superior gluteal**
"NOT a sequel of pediatric neck fracture?"	→ **chondrolysis** (TO)
"Best test for occult pediatric neck fracture?"	→ **MRI** (TO)

PROXIMAL HUMERUS FRACTURE - PEDIATRIC

– Pediatrics · 89 / 112 –

1 Basics

Epidemiology <5% of pediatric fractures · **peak age 15**

Pattern by age **<5 yr: SH I · 5–12 yr: metaphyseal · >12 yr: SH II**

Deforming forces epiphysis **abducted + externally rotated** (rotator cuff) · shaft **anterior, adducted, shortened** (pectoralis major, deltoid)

Humeral growth contribution

Proximal : distal humerus = 80 : 20 (highest ratio; femur 30 : 70) · proximal physis ~7 mm/yr → **huge remodeling**

Ossification epiphysis 6 months · greater tuberosity 1–3 yr · lesser tuberosity 4–5 yr · closes 14–17 (girls), 16–18 (boys)

3 Presentation

Exam shoulder pain, deformity, ecchymosis · check neck, SC joint, elbow · brachial plexus + vascular exam

4 Diagnosis & Classification

Key radiology findings

Views AP, lateral, **axillary / scapular Y** · ± contralateral · assess max angulation · **US in newborn** (unossified)

Differential **Little Leaguer's shoulder** (throwers: physeal widening, metaphyseal change — not a fracture) · pathologic fracture (**UBC, ABC**)



Little Leaguer's shoulder — physeal widening

– – Active space – –

Neer-Horowitz classification

Type	Displacement
I	**<5 mm**
II	**<⅓** shaft width
III	**⅓–⅔**
IV	**>⅔**



Neer-Horowitz type IV

5 Management

Treatment by age & displacement

Age	Acceptable alignment (non-op)
<10 yr	**Any angulation**
10–12 yr	Up to **60–75°**
>12 yr	Up to **45°** or **⅔ displacement**

Rockwood (OB) reduce ± fix if **>50% displaced in >11 yr**

Acceptable (TO) **1–5 yr: 70°, 100%** · **5–12 yr: 50°, 50%** · **> 12 yr: 30°** – read the options · sling or sugar-tong splint

Non-op sling ± swathe, immobiliser, coaptation splint · 7-yr Neer-Horowitz IV → **sling + PT** · 9-yr displaced → **sling + follow-up X-rays**
✗ Wrong options: CRPP · ORIF (higher complications)

Older adolescent (≥15, <1 yr growth) Neer-Horowitz III/IV → closed reduction + percutaneous pinning · 13-yr marked varus SH II → **closed/open reduction + pins** (varus = lost abduction)
✗ Wrong options: shoulder immobiliser · hanging arm cast · spica (redisplaces) · hemiarthroplasty

Outcome by age older patients with III/IV treated non-op → **↑ limitations in daily activities** vs younger · long-term (Lähdeoja, 13 yr) no impairment vs controls, op = non-op
✗ Wrong options: nonunion · more displacement

Reduction traction + **90° abduction + ER**

Fixation 2–3 lateral threaded pins (avoid axillary n. laterally, musculocutaneous anteriorly) · retrograde elastic nails · cannulated screws in older teens

Open reduction	irreducible from **interposed long head of biceps (most common)**, capsule, periosteum, deltoid · **open fracture** · **vascular injury** · intra-articular displacement · deltopectoral approach

– – Active space – –

6 Complications

Loss of reduction	unstable fracture reduced without pins
Axillary nerve	<1% from injury (neurapraxia, **medially displaced shaft**) · ↑ with pins → lateral pin below nerve (~5 cm below acromion in adults; = 2× humeral head height)
Others	varus malunion → impingement · shortening (<3 cm tolerated <12 yr) · growth arrest rare · hypertrophic scar · **pin-site infection 5–22%**
Prognosis	excellent (remodeling + shoulder ROM)

How it's asked

"9-yr skateboard, displaced proximal humerus?"	→ **sling + follow-up X-rays** ✗ Wrong options: CRPP · ORIF
"16-yr, <1 yr growth — which needs CRPP?"	→ **Neer-Horowitz III physeal fracture** ✗ Wrong options: type I · 10° metaphyseal
"13-yr hockey SH II, marked varus — best motion?"	→ **closed or open reduction + percutaneous pins** ✗ Wrong options: spica
"15-yr displaced proximal humerus?"	→ **closed reduction + percutaneous pins** ✗ Wrong options: hanging cast · hemiarthroplasty
"7-yr Neer-Horowitz IV?"	→ **sling, graduated PT, monitoring**
"12-yr gymnast, non-op — vs younger child?"	→ **↑ risk of activity limitations**
"Proximal : distal humeral growth?"	→ **80 : 20**

– – Active space – –

PROXIMAL TIBIA EPIPHYSEAL FRACTURES - PEDIATRIC

1 Basics

Epidemiology	<1% of pediatric fractures · **12–14 yr** · high energy · varus/valgus or **hyperextension**
Associated	part of **type III tibial tubercle fracture** · **popliteal artery injury (5%)** – **hyperextension** · peroneal nerve (5%) · **ligament injury up to 40% in SH III/IV** · **compartment syndrome 3–4%**

Anatomy

Physis	proximal tibia **6 mm/yr** · knee = 15 of 23 mm/yr leg growth · closes **posterior → anterior**, **medial → lateral**, **posteromedial → anterolateral**
Buttresses	**superficial MCL inserts below physis** (medial buttress) · **LCL to fibula** (lateral buttress) · **patellar tendon** (resists posterior displacement) → physeal fractures are rare
Popliteal artery	**tethered** to posterior proximal tibia by septa · **trifurcation at physeal level** → injury with displacement · geniculate arteries around popliteus

3 Presentation

Exam	can't bear weight · physeal tenderness · step-off · varus/valgus instability · **thorough neurovascular exam**

Normal X-ray + exquisite physeal tenderness after high-energy hit → may have displaced + reduced → **splint, admit, frequent neurovascular / compartment checks** (treat like knee dislocation) · MRI/CT to define
✗ Wrong options: cylinder cast + discharge · arthroscopy · prophylactic fasciotomy

4 Diagnosis & Classification

Key radiology findings

X-ray	AP, lateral ± obliques · stress views risk physis
CT	displacement · **best for SH III / IV**

Salter-Harris classification

Type	Features
I	Mean age 12 · usually displaced (>50%)
II	Mean age 14 · usually displaced (>67%) · **medial gapping + lateral Thurston-Holland + proximal fibula fracture**
III	Exits epiphysis · usually **tibial tubercle** fractures
IV	Physis + metaphysis + epiphysis



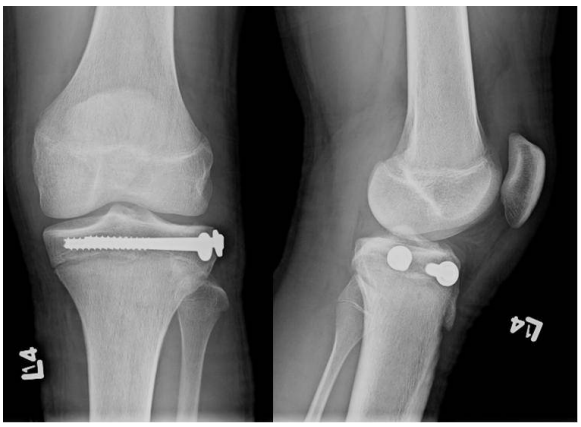
Proximal tibial SH II

5 Management

Fracture	Treatment
Non-displaced (<2 mm), stable SH I/II	**Long leg cast, slight flexion, 6 wk** · redisplacement common
Unstable SH I/II · redisplaced	**CRPP**
Irreducible (periosteal flap) · **displaced >2 mm SH III/IV** · **vascular injury**	**ORIF**

Fixation **crossed smooth pins**: **transphyseal** for SH I/II with small Thurston-Holland · **extraphyseal** for SH III/IV · **cannulated screws parallel to physis** for SH II with large metaphyseal fragment or III/IV · cast in slight flexion 4–6 wk

Approach anterior midline (patella → tubercle) · **medial approach if vascular injury**



ORIF proximal tibial physeal fracture

– – Active space – –

Popliteal artery injury & vascular monitoring

Hyperextension physeal fracture → popliteal artery most at risk
✗ Wrong options: tibial n. · peroneal n. · PCL

6 Complications

Growth disturbance 25% (LLD, angulation; ↑ open fractures) · loss of reduction · compartment syndrome · ligament instability

How it's asked

"9-yr hyperextension proximal tibial physeal fracture — at risk?"	**→ popliteal artery**
"11-yr sled hit, normal X-rays, tender physis, NV intact?"	**→ splint, admit, frequent neurovascular checks**

PROXIMAL TIBIA METAPHYSEAL FRACTURES - PEDIATRIC

– Pediatrics · 91 / 112 –

1 Basics

Epidemiology **3–6 yr** (2–10 yr OB) · "Cozen fracture"

Mechanism low-energy **valgus** force → **incomplete (greenstick/torus)** medial fracture · torsion · classic: **down a slide on an adult's lap**, leg extended · fibula fracture = higher energy, less stable

Anatomy posterior tibial artery → nutrient (inner ⅔) + periosteal vessels

3 Presentation

Exam pain, refuses to bear weight · little swelling / deformity · check compartments

4 Diagnosis & Classification

X-ray AP + lateral · complete vs incomplete · displacement · usually **valgus** · fibula fracture



Proximal tibial metaphyseal fracture

- - Active space - -

5 Management

Initial treatment & counselling

Fracture	Treatment
Nondisplaced	**Long leg cast in EXTENSION with VARUS mold** (slight overcorrection) 4–6 wk · WB at 2–3 wk
Displaced	**Reduction under sedation** (complete the greenstick) → same cast, three-point varus mold
Irreducible (rare)	**Limited medial open reduction** – remove interposed **periosteum, pes tendons, MCL** ± crossed K-wires

Counsel before treatment valgus may develop despite perfect care but **often resolves spontaneously**
✗ Wrong options: knee immobilizer · prophylactic surgery · "always needs surgery"

6 Complications

Post-traumatic valgus (Cozen) - natural history & observation

Cozen phenomenon **late valgus** · etiology unknown (medial physeal overgrowth / stimulation, lateral tether, pes anserinus loss) · **occurs regardless of treatment** · not physeal arrest

Feature	Value
Incidence	**50–90%** (14–100% OB)
Appears	**5–15 months**
Maximum	**12–18 months**, average **18°**
Resolves	By **~3 yr**, residual ~6° (can leave S-shaped tibia)
Risk	Incomplete reduction · physeal injury · infolded periosteum · pes injury

Treatment **observation 12–24 months** · surgery only **>15–20° near skeletal maturity** → **medial proximal tibial hemiepiphysiodesis** or varus proximal tibia/fibula osteotomy
✗ Wrong options: KAFO · hemiepiphysiodesis early · osteotomy · ring fixator · MRI · bar resection

LLD affected tibia **longer** (~9 mm) · **doesn't resolve** · usually no treatment



Cozen valgus

How it's asked

"3-yr, 6 months after metaphyseal fracture, valgus?"	→ **follow-up X-ray in 6 months** ✗ Wrong options: hemiepiphysiodesis · osteotomy · KAFO
"Which can be observed alone?"	→ **5-yr with 15° valgus 8 months after fracture** ✗ Wrong options: Blount MDA 18° · SCFE · AIS 80°
"5-yr trampoline proximal tibia — counsel about?"	→ **valgus**
"6-yr, 15° valgus 1 yr after healed fracture?"	→ **continued observation**
"4-yr knock-knee 9 months after fracture?"	→ **observation** ✗ Wrong options: stapling · bar resection
"5-yr proximal tibia — deformity + treatment?"	→ **valgus; observation**
"Deformity within a year — initial treatment?"	→ **observation**
"13 months post-injury, leg looks funny?"	→ **maximum ~18 months, resolves over ~3 yr** ✗ Wrong options: shorter leg · short immobilisation
"Valgus 12 months after perfectly reduced injury?"	→ **late valgus from proximal tibial metaphyseal fracture** ✗ Wrong options: vitamin D · physiologic bowing
"4-yr proximal tibia fracture — tell parents?"	→ **reduce + long leg cast; deformity may develop but often resolves**

– – Active space – –

PSOAS ABSCESS - PEDIATRIC

– Pediatrics · 92 / 112 –

1 Basics

Epidemiology	**more common in the tropics** · primary form favours children
Organism	***S. aureus*** most common
Spread to hip	via **psoas bursa** (communicates with hip joint in **15%**) · or directly between iliofemoral + iliopubic ligaments → septic arthritis
Associated	septic hip · **tuberculous spondylitis** · bowel source (secondary)

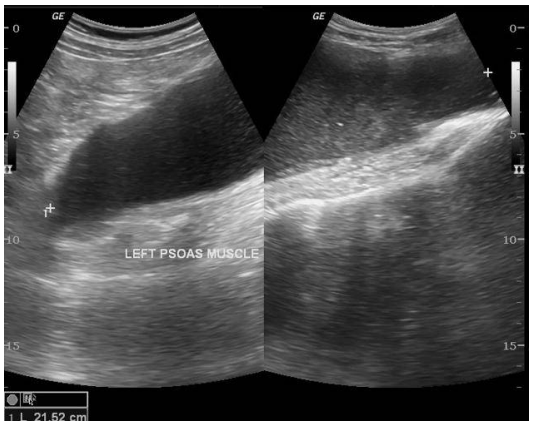
3 Presentation

Mimics septic hip	**fever, hip pain**, limp / refuses weight
Exam	**hip held in flexion** · tender iliac fossa mass · **psoas sign**: pain with **extension + internal rotation**

4 Diagnosis & Classification

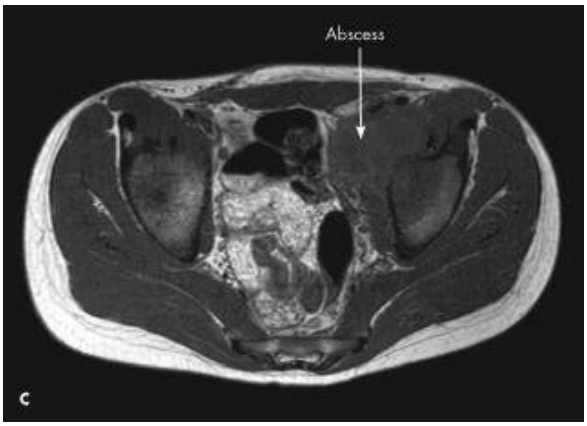
Key radiology findings

X-ray	AP + frog pelvis to exclude septic hip · enlarged / blurred psoas shadow
Ultrasound	**study of choice** (imaging section) · summary says MRI generally required – read the options
MRI	early changes · hard to get quickly
CT with contrast	mass with **ring enhancement** · may be negative early · radiation
Bone scan	helpful



Psoas abscess on ultrasound

– – Active space – –



Psoas abscess on MRI

5 Management

First line	**percutaneous US/CT-guided drainage + IV antibiotics**
Open (extraperitoneal) drainage	**secondary abscess** (e.g. bowel) to address source

6 Complications

Septic arthritis of hip · **sepsis**

X vs Y

	Psoas abscess	Septic hip
Hip X-ray	No effusion widening	Effusion / widening
Exam	Psoas sign, iliac fossa mass	Pain with all motion
Treatment	Percutaneous drainage	Arthrotomy / I&D

How it's asked

"17-month-old, fever, hip flexed, pain on extension + IR, CT iliopsoas mass?" → **psoas abscess**
✗ Wrong options: septic arthritis · sacroiliitis · transient synovitis

– – Active space – –

RADIAL HEAD AND NECK FRACTURES - PEDIATRIC

1 Basics

Epidemiology	**5–10% of pediatric elbow fractures** · 1% of all · 3rd most common elbow fracture (OB) · median age **9–10** · M = F
Site	**90% physis or metaphysis (neck)** · ~50% metaphyseal, 50% SH I/II · articular head fracture only 4–7% with open physes (52–96% after closure)
Mechanism	FOOSH with **extension + valgus** · elbow dislocation
Associated (30–50%)	**elbow dislocation** · **medial epicondyle** · olecranon · condyles · **forearm compartment syndrome**
Anatomy	radial head ossifies **3–5 yr** (may be bipartite) · fuses 16–18 · **annular ligament covers epiphysis + physis** → stress riser at metaphysis → **neck fractures** · thick epiphyseal cartilage resists articular extension · head blood supply mainly **metaphyseal**

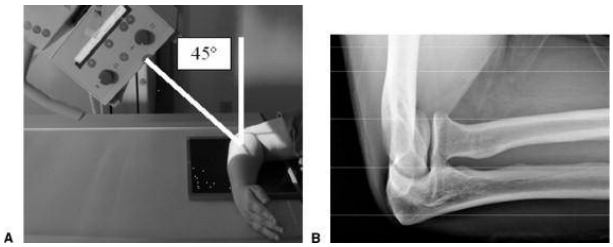
3 Presentation

Exam	lateral swelling · pain with **pronation/supination** · may refer to wrist · **watch for forearm compartment syndrome** (rising forearm pain overnight)

4 Diagnosis & Classification

Key radiology findings

Views	AP + lateral · **radiocapitellar (Greenspan) view**: elbow 90°, thumb up, beam 45° proximal



Greenspan (radiocapitellar) view technique

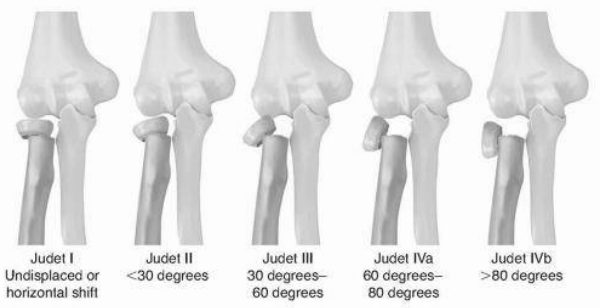
Lines	radiocapitellar line (unreliable in young, esp. AP) · **lateral humeral line** (tangent to lateral condyle, lies lateral to radial neck) – more reliable in young children
Pitfalls	nondisplaced hard to see · **radial head flipped 180°** · posterior fat pad always pathologic · **neck partly extra-articular → effusion / fat pad may be absent**
CT	comminuted head in older child · **MRI** – rare (unossified elbow, anterior head compression with ligament injury)

Classification

O'Brien	Angulation
I	<30°
II	30–60°
III	>60°

Judet	Angulation
I	Undisplaced
II	<30°
III	30–60°
IVa	60–80°
IVb	>80°

Chambers group 1 primary radial head displacement (most common; valgus: SH I/II, SH III/IV, metaphyseal; dislocation: reduction / dislocation injury) · group 2 radial neck displacement (**Monteggia variant**) · group 3 stress (**OCD**)



Judet classification

5 Management

Pediatric radial neck fracture

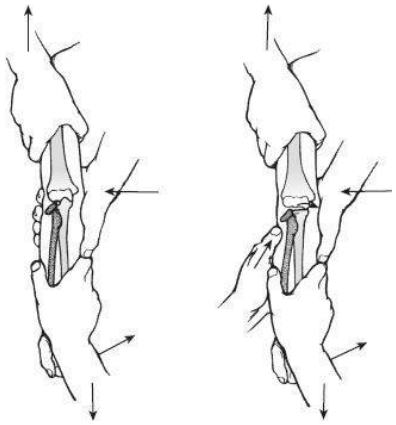
- <30° and <3 mm translation → **long arm cast/splint, no reduction (7–21 days, then ROM)**
- >30° → **closed reduction (Patterson, Israeli, Neher-Torch, Esmarch)**
  (stable once reduced – fixation usually not needed)
- Residual >30° / >3–4 mm / <45° pro-supination → **percutaneous reduction (K-wire joystick, Metaizeau) ± pin**
- Can't get <45° → **open reduction (lateral approach, forearm pronated)**

Treatment by angulation (closed, percutaneous, open)

Closed reduction 45° or 35° angulation → **attempt closed reduction** · >60° unlikely to close-reduce (OB) · some tolerate up to 60° (Radomisli) · >50° often needs fixation (Nicholson & Skaggs)
✗ Wrong options: cast without reduction · percutaneous first · open

– – Active space – –

Techniques — **Patterson**: elbow extended, traction, supination, **varus** + direct radial head pressure · **Israeli (Kaufman)**: elbow 90°, **pronate** the supinated forearm with thumb on head · **Neher-Torch**: extension, supination, traction, varus; assistant pushes shaft lateral, surgeon pushes head medial · **Esmarch** wrap → spontaneous reduction



Patterson maneuver

Percutaneous — **K-wire joystick** (push posterolateral / lever) · **Metaizeau**: retrograde flexible nail across fracture, rotate to reduce · hardware out at ~8 wk · after failed ED reduction with >30° (e.g. 62°) → **percutaneous reduction ± pin / ESIN**
✗ Wrong options: cast 6 wk · CT · ORIF first · plate · epiphysiodesis

Open — widely displaced head fracture with radial subluxation → **ORIF** · **lateral approach**, **pronate forearm to protect PIN** · fix only if unstable · no **transcapitellar pins**

Outcome — better if younger, less angulation, isolated neck · open reduction → more stiffness, **AVN, synostosis**

6 Complications

Complications & prognosis

Complication	Fact
↓ ROM	**Most common (10–31%)** · pronation > supination
Radial head overgrowth	20–40% · rarely matters
Osteonecrosis	10–20% · **up to 70% after open reduction**
PIN palsy	~15% after surgery
Physeal arrest	→ cubitus valgus
Synostosis	**Most serious** · open reduction, extensive dissection, delay
Malunion (intra-articular)	Radiocapitellar instability, pain, arthritis
Forearm compartment syndrome	Increasing pain + swelling overnight

Worse outcome — **age >10** · intra-articular (50% complications, 25% revision vs 0% extra-articular)
✗ Wrong options: BMI · dominance · sex

TO points

~5% of elbow fractures, **3rd commonest** (after supracondylar, lateral condyle) · head entirely cartilaginous, metaphyseal blood supply · > 30° → eccentric head, cam impingement at radiocapitellar joint · residual **< 45° after reduction may be accepted** to avoid open surgery · other methods: Jeffery, Kaufman, Monson, Orthokids · **PIN crosses ~3 finger-breadths distal to neck**; safe zone posterolateral, **pronation ↑ safety** · open reduction via **anconeus-ECU** interval, results inferior

Angulation (other TO lecture)	Treatment
< 15°	Posterior slab
15–30°	Early orthopaedic referral
> 30°	Reduction · preferred fixation **Metaizeau** elastic nail

PIN injury

PIN

branch of radial nerve wrapping radial neck · **loss of thumb IP extension (EPL)**, finger extension incl. **small finger (EDM)**, partial wrist extension · supplies posterior compartment (except anconeus) + ECRB · **pins posterolateral with arm in maximal pronation**

How it's asked

"10-yr, 45° physeal radial neck – first step?"	→ **attempt closed reduction**
"12-yr, >30° after ED reduction?"	→ **further reduction + percutaneous ESIN/pin** ✗ Wrong options: cast 6 wk · CT · ORIF
"10-yr, 35° within 2 hours?"	→ **closed reduction in ER** ✗ Wrong options: CRPP · open
"9-yr, 62° residual after sedated reduction?"	→ **percutaneous reduction ± pin**
"8-yr displaced radial head fracture + anterior subluxation after reduction attempt?"	→ **ORIF** ✗ Wrong options: annular ligament reconstruction · repeat closed
"6-yr radial neck fracture, increasing forearm pain overnight?"	→ **forearm compartment syndrome**
"Factor for poor surgical outcome?"	→ **age >10 yr**
"7-yr after percutaneous pinning, can't extend thumb IP – nerve origin?"	→ **radial nerve (PIN)** ✗ Wrong options: median · ulnar
"14-yr open reduction, nerve palsy – deficit?"	→ **small finger extension** ✗ Wrong options: thumb IP flexion (AIN) · opposition

– – Active space – –

RETT SYNDROME

– Pediatrics · 94 / 112 –

1 Basics

Incidence **1 in 10,000** · **girls** · males die before term

Genetics - X-linked dominant MECP2

Inheritance **X-linked dominant**
✗ Wrong options: AR · AD · mitochondrial · XLR

Gene **MECP2** (methyl-CpG-binding protein 2, DNA methylation) · **95% de novo** · on paternal X · affects grey matter

Associated developmental delay · **seizures (80%)** · screaming, loss of speech · constipation · spasticity + contractures · ataxia · hypotonia · **bruxism** · chorea-like hand movements · cold blue extremities · **coxa vara**

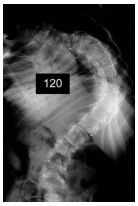
3 Presentation

Course normal to **6–18 months** → rapid regression to age 3 → plateau to 10 → **half can't walk after 10** · **first sign: apraxia (loss of purposeful hand use)**

Exam ↓ head circumference (from 5–6 months) · abnormal gait, breathing · dementia-like slowing · **diagnosis** clinical + developmental history (genetic testing may miss some)

4 Diagnosis & Classification

Scoliosis **C-shaped (long) curve** · by **~age 10** · **>50%**

C-shaped curve in Rett syndrome	Rett syndrome
	

5 Management

Non-op supportive (feeding, hygiene, seizures) · **PT to prevent hand contractures**

Scoliosis **bracing ineffective** · **PSF** when thoracic curve hurts sitting/balance · **include entire scoliotic + kyphotic curve**

How it's asked

"14-yr girl, seizures, delay, C-shaped curve, MECP2 – inheritance?" → **X-linked dominant**

SACRAL AGENESIS

– Pediatrics · 95 / 112 –

1 Basics

Clinical features & maternal diabetes

Definition	partial/complete absence of sacrum + lower lumbar spine · part of **caudal regression syndrome**
Incidence	1–2.5 per 100,000
Risk factor	**maternal diabetes** (poor glucose control early pregnancy) ✗ Wrong options: folate deficiency (myelodysplasia) · alcohol · drugs
Neurology	**motor deficit matches the level** · **protective sensation usually intact** (sensory doesn't match level) → **fewer decubitus ulcers** ✗ Wrong options: like myelodysplasia
Progression	neurologic deficit present at birth + static (OB6072) · OB2199: neuro + MSK manifestations may be progressive – read the options
Associated	**imperforate anus** / GI · GU · cardiac · LE deformities · **progressive kyphosis**

3 Presentation

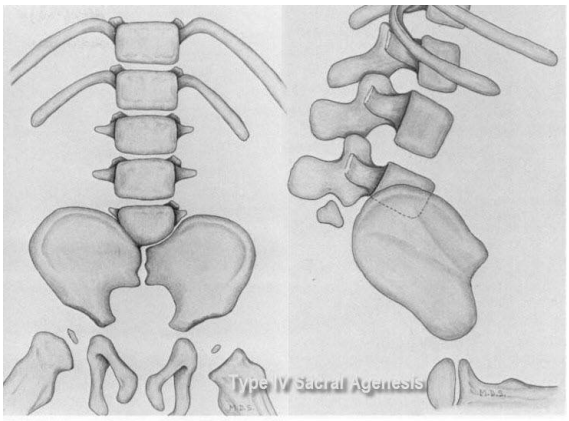
Exam	prominent last vertebra (T12) · **buttock dimpling** · **"sitting Buddha" posture** · **popliteal webbing**, extended or flexed knees, flexed hips, **rigid equinovarus** feet · motion at spine-pelvis junction

4 Diagnosis & Classification

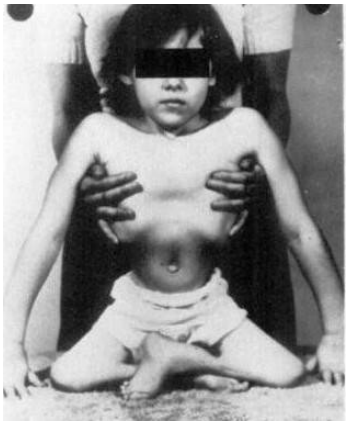
Renshaw classification

Type	Defect
I	Partial/total **unilateral** sacral agenesis
II	Partial sacrum, **bilaterally symmetric** defect
III	**Ilium articulates with sides of lowest vertebra**
IV	Lowest vertebral endplate **rests above fused ilia** / iliac amphiarthrosis

– – Active space – –



Renshaw type IV



Sitting Buddha posture

5 Management

Type	Treatment
I–II	**PT** · most become community ambulators · treat foot/knee deformity
III–IV with progressive kyphosis/scoliosis	**Spinopelvic stabilisation (fusion) → sitting balance + frees hands** (✗ Wrong options: goal of ambulation)
Non-functional limbs	Amputation → better mobility

Untreated complete absence → progressive spinopelvic kyphosis → **must use hands to sit**
✗ Wrong options: progressive paralysis · hip dislocation

How it's asked

"6-yr, colorectal surgery at birth, sits only with hands — true?"	**→ motor deficits match level, sensory deficits don't** ✗ Wrong options: protective sensation absent · fusion for ambulation · folate
"2-yr complete sacral + lower lumbar absence, no stabilisation — outcome?"	**→ can't sit without hand support**
"Newborn, knee webbing, rigid clubfeet, Buddha posture, absent lumbar spine — maternal condition?"	**→ diabetes mellitus**

SEVER'S DISEASE

– Pediatrics · 96 / 112 –

1 Basics

Definition	**calcaneal apophysitis · most common cause of heel pain in growing children** (TO) · traction apophysitis + repetitive microtrauma (like Osgood-Schlatter) · ± para-apophyseal stress fracture (OB)
Who	running/jumping athletes (soccer, basketball, volleyball) · **just before / during peak growth** · ages **9–12** · boys ~12, girls ~11 · **bilateral ~60%** (TO)
Forces	heel-strike impact + opposing pull of **Achilles (gastrocsoleus)** and plantar fascia

3 Presentation

Diagnosis

Symptoms	posterior heel pain with activity / impact, relieved by rest · ± warmth, erythema, swelling
Exam	**tight Achilles** (↓ dorsiflexion) · **positive squeeze test** (medial-lateral calcaneal compression) · apophyseal tenderness · pain on triceps surae stretch · difficulty single-leg heel raise · Thompson negative

4 Diagnosis & Classification

Key radiology findings

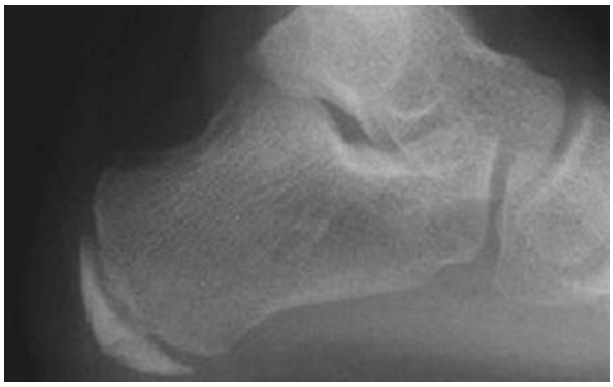
X-ray	diagnosis is **clinical** · **sclerosis** in both affected and normal · **fragmentation** more frequent in Sever · rules out osteomyelitis, cysts



Sever's disease – sclerotic fragmented apophysis

MRI	localises apophyseal inflammation · excludes stress fracture, lytic lesion, osteomyelitis (usually unnecessary)
Bone scan	↑ apophyseal uptake, not helpful
Incidental calcaneal cyst	treat only if full width or ⅓ of calcaneal length

– – Active space – –



Calcaneal apophysis fragmentation

5 Management

Treatment - activity modification

Non-op	**activity modification** · **Achilles stretching** (↓ recurrence) · ice · **heel cups/ pads** · NSAIDs · **short leg cast** if persistent
Never	**surgery** · custom orthotics not indicated · curettage of incidental cyst ✗ Wrong options: calcaneal epiphysiodesis · Achilles lengthening
Prognosis	**self-limiting**, resolves with apophyseal closure · recurrence common

X vs Y

Heel pain	Clue
Sever	Squeeze test +, apophysis, 9–12 yr athlete
Achilles rupture	**Thompson +**
Os trigonum	Posterior impingement, ballet
Gastrocnemius strain	Proximal myotendinous junction
Plantar fasciitis	Anterior heel pad, no squeeze pain
Baxter nerve	Arch pain like plantar fasciitis
Haglund	Posterolateral prominence, women 15–35 in heels
Kohler	Dorsomedial midfoot, navicular
Coalition	Recurrent ankle sprains

How it's asked

"9-yr player, activity heel pain relieved by rest?"	→ **Sever's disease** (TO)
"12-yr soccer, warm tender calcaneal tuberosity, imaging?"	→ **Sever's disease** ✗ Wrong options: stress fracture · osteomyelitis · Kohler
"12-yr, DF 5°, squeeze +, Thompson −?"	→ **calcaneal apophysitis** ✗ Wrong options: Achilles rupture · os trigonum

"11-yr volleyball, squeeze +, etiology?"	→ **Achilles traction apophysitis** ✗ Wrong options: Baxter nerve · plantar fascia
"14-yr peak growth, red swollen heel, squeeze +?"	→ **Sever's disease** ✗ Wrong options: Haglund · retrocalcaneal bursitis
"11-yr basketball heel pain — image?"	→ **fragmented calcaneal apophysis** ✗ Wrong options: Kohler · Iselin · coalition
"10-yr soccer, tight Achilles, 2-cm anterior calcaneal cyst?"	→ **observation with reduced activity** ✗ Wrong options: curettage · bone scan
"10-yr year-round soccer, heel pain every step?"	→ **activity modification** ✗ Wrong options: MRI · orthotics · epiphysiodesis

– – Active space – –

SI JOINT INFECTION - PEDIATRIC

– Pediatrics · 97 / 112 –

1 Basics

Epidemiology rare · **1–2% of septic arthritis** · children >10 yr more than adults

Adult risk factors IV drug use · immunosuppression · pregnancy · trauma · endocarditis · sepsis · TB

Causative organism

Organism	Note
S. aureus	**Most common** (✗ Wrong options: GBS (neonatal osteomyelitis) · H. influenzae (vaccine))
M. tuberculosis	Skeletal TB = 3–5% of TB; ~10% of that at SIJ
Salmonella	Sickle cell disease

3 Presentation

Symptoms progressive **low back + buttock pain** · **can't bear weight** on that side · **fever**

Exam pain with **iliac wing compression** + SI palpation · **normal hip ROM** · **FABER strongly positive**

4 Diagnosis & Classification

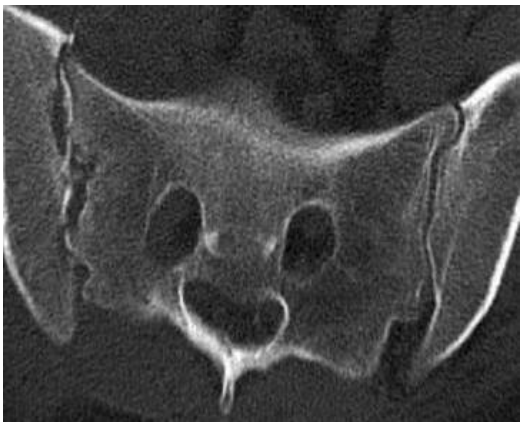
Key radiology findings

X-ray AP/lateral pelvis ± inlet/outlet · late: destruction, fusion, periarticular osteopenia, widening, cavitation

MRI **most sensitive** (> CT) · acute: fluid/gas, subchondral edema, enhancement · chronic: marrow reconversion, pannus, erosion, sclerosis, ankylosis

Labs ↑ WBC · **ESR >50** · **CRP >70** · **blood cultures before antibiotics** (often guide treatment; SIJ aspiration usually unnecessary)

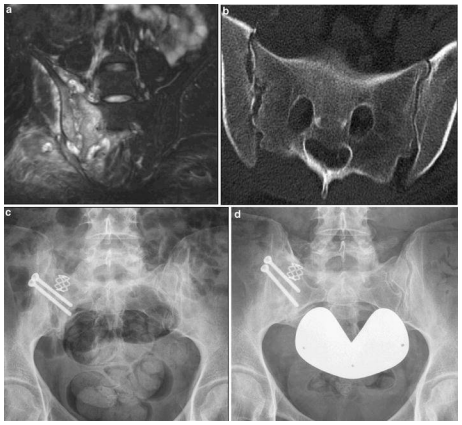
– – Active space – –



SI joint infection on MRI

5 Management

Situation	Treatment
First line	**Organism-targeted IV antibiotics** (cover S. aureus if cultures negative)
Abscess · failed antibiotics · sequestrum	**Debridement** (rare in children)
Chronic · sequestrum with joint destruction · recurrent	**Debridement + SI arthrodesis** (longer, more complications)



SI joint arthrodesis

6 Complications

List chronic pain · recurrence · wound healing delay · hardware problems · sepsis

How it's asked

"Most common organism in acute pediatric SI septic arthritis?" → **Staphylococcus aureus**
✗ Wrong options: GBS · TB · Salmonella

– – Active space – –

SLIPPED CAPITAL FEMORAL EPIPHYSIS (SCFE)

– Pediatrics · 98 / 112 –

1 Basics

Epidemiology

Incidence	**most common adolescent hip disorder** · 10 per 100,000
Who	**obese** (single greatest risk) · **males** (2:1.4 Orthobullets; ~2× TO) · **African American**, Pacific Islander, Latino · rapid growth **10–16 yr** · boys **12–13.4**, girls **11.2–12.2** · **left hip** more common (2× TO)
Bilateral	17–50%, average **25%** (Orthobullets) · 25–40% (TO) · 20–30% at presentation (TO) · second slip usually within **12–18 months**
Risk factors	obesity · **acetabular + femoral retroversion** (↓ anteversion, ↓ neck-shaft angle) · prior **radiation** · **↑ leptin (4.9× odds, independent of obesity)** · Down syndrome · open triradiate

Pathophysiology

Slip	**metaphysis moves ANTERIOR + externally rotates** · epiphysis stays in acetabulum, **posterior/inferior** to neck
Plane	through the **hypertrophic zone** (✓) · renal SCFE through **secondary spongiosa**
Why adolescents	**perichondral ring thins** · mammillary processes unlock · physis still vertical (160° birth → 125° maturity) → shear · epiphyseal tubercle pivot
vs SH I fracture	antecedent epiphysiolysis, slower, periosteum intact (chronic) · type I transepiphyseal fracture = high energy, young child, M = F (TO)

Histopathology

Chronic SCFE histology	**granulation tissue between columns in hypertrophic zone** · ↓ perichondral ring thickness · ↓ physeal undulations · abnormal proteoglycan accumulation · ↑ hypertrophic zone height ✗ Wrong options: vascular invasion of proliferative zone · thick ring

Endocrine & metabolic associations

Endocrine	**hypothyroidism (most common non-idiopathic; ↑ TSH)** · **renal osteodystrophy** (↑ BUN, creatinine) · **GH deficiency** · **panhypopituitarism** (pituitary adenoma) · Loder: hypothyroidism 40%, GH deficiency 25%, all hypothyroid → bilateral
Endocrine work-up if	**<10 yr** · **weight <50th percentile** · prepubertal / short · labs: **TSH, free T4, BUN, creatinine**
Atypical SCFE	<10 or >16 yr (TO) · renal failure, radiation, hypogonadism, Down

Blood supply

Lateral epiphyseal artery (MFCA) = main supply · LFCA anterior ring · ligamentum teres insignificant · **posterosuperior** hardware endangers vessels

3 Presentation

Presentation & diagnosis (knee pain, hip films)

Pain	hip 52%, **thigh 35%**, groin 14% · **knee pain 15–50%** (23% Matava; obturator nerve referral) · hip pain still more common than knee ✗ Wrong options: "knee more often than hip"
Delay in diagnosis	**knee pain** · stable slip · Medicaid · severity · 88% of unstable slips had ~42 days of prior symptoms
Exam	antalgic / Trendelenburg / out-toeing gait · **obligatory ER with hip flexion (Drehmann sign)** · ↓ **IR**, abduction, flexion · sits with leg crossed · tender Scarpa triangle (TO)

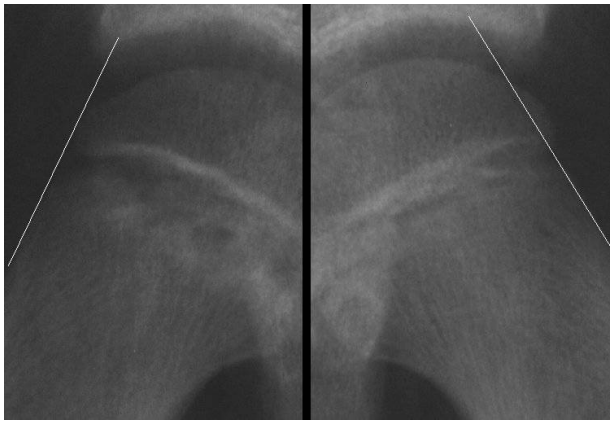
Obese teen with knee pain + normal knee X-rays → examine hip + AP pelvis/frog-lateral
✗ Wrong options: knee MRI · arthroscopy · labs · PT

Unmanipulated severe SCFE on AP → leg lies in **external rotation**

4 Diagnosis & Classification

Key radiology findings

Views	**AP + frog-lateral of both hips** · lateral best for subtle slip · **unstable → cross-table lateral** (not frog)
Klein line	along superior neck · normally cuts lateral epiphysis · SCFE: intersects less / not at all · ≥2 mm asymmetry improves sensitivity 40% → ~80% (TO)
Other signs	**metaphyseal blanch sign of Steel** (double density, AP) · **S sign** broken on frog · **epiphysiolysis** (early widening)
MRI	**pre-slip**: physeal widening + **metaphyseal edema** (T1 ↓, T2 ↑) · CT: physeal closure, osteotomy planning
Age pattern (TO, radiology)	hip pain without sport at **11–12 yr → SCFE** · **7–8 yr → Perthes** · frog-leg shows irregular physis · MRI: **physeal widening + fluid**, effusion, head slipped backwards



Klein line

– – Active space – –

"Metaphysis moves, not the epiphysis" (TO) metaphysis displaces **anterosuperiorly**, epiphysis stays in acetabulum · weak **hypertrophic zone** + thinned **perichondrial ring** · shear ↑ by obesity, vertical physis, **femoral retroversion**

Signs (TO) Klein line = **Trethowan sign** · **blanch sign of Steel** (chronic) · widened irregular physis · broken Shenton · risk: **prior radiotherapy** · **< 10 yr or thin → endocrine work-up**

Loder AVN (TO) stable **< 10%** · unstable **up to 40%** · stability predicts AVN better than slip angle

Loder classification (stability)

Type	Definition	AVN
Stable (85–95%)	Can walk ± crutches	**<10%** (~0%)
Unstable (5–15%)	**Can't walk even with crutches**	**24–47%** (Orthobullets) · 47–50% · 25–50% · 15–50% · up to 60% (OB) · 2–5× stable

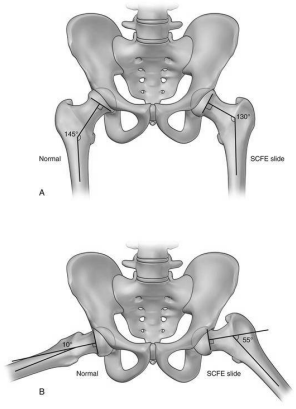
Temporal acute <3 wk · chronic >3 wk · acute-on-chronic (no prognostic value)

Slip severity measurement

Southwick (epiphyseal-shaft) angle **severity** · frog lateral (± AP) · affected minus normal · bilateral reference
145° AP / 10° lateral (Orthobullets) · 12° (TO)
✗ Wrong options: presence · AVN prognosis · chronicity

Severity	Orthobullets	OB / TO
Mild	**<30°**	<30°
Moderate	**30–50°**	30–60°
Severe	**>50°**	>60°

% displacement grade I 0–33% · II 34–50% · III >50%



Southwick angle

Differential septic hip, transient synovitis, osteomyelitis, Perthes, DDH, trauma (adductor strain, AIIS avulsion) · breaststroke swimmer, open physes, normal head-neck → **external rotator (obturator externus) tear**

5 Management

SCFE

- Stable → **single central screw in situ (no reduction), WB after**
- Unstable → urgent bed rest → **in situ fixation ± 2nd screw, ± decompression/capsulotomy, TDWB**
- High-risk contralateral → **prophylactic pinning**
- Residual deformity → **osteochondroplasty (<30°) / Imhauser osteotomy (>30–45°)**

In situ fixation technique

Rule	**percutaneous in situ fixation for stable AND unstable** · **no forceful reduction** (↑ AVN) · serendipitous positional reduction OK
Screw	**single cannulated screw** usually (↓ AVN vs multiple) · **2 screws more stable** (some for unstable) · not inverted triangle (adult neck fracture, ↑ penetration + AVN)
Placement	**start on ANTERIOR femoral neck → perpendicular to physis → centre-centre** of epiphysis on AP + lateral · **not medial to intertrochanteric line** (screw head impinges, labral tear) · severe slip → more oblique from intertrochanteric region
Depth	**≥5 threads across physis** (<5 → 41% progression vs 0%) · tip ~5 mm from subchondral bone · partially threaded: 40–60% of threads in epiphysis strongest · **approach-withdraw** fluoroscopy to exclude penetration
Post-op	stable WB · unstable TDWB · young child: short-thread screw not crossing physis (TO)

Stable mild slip in 13-yr → single-screw in situ fixation
✗ Wrong options: spica (chondrolysis) · ORIF · NWB · closed reduction

Stable SCFE seen only on MRI → in situ fixation
✗ Wrong options: observation · labral repair



In situ screw fixation

Unstable SCFE & AVN

Unstable slip → strict bed rest + urgent surgery (within hours; 8 h OB, 24 h OB) on affected hip
✗ Wrong options: MRI · elective · NSAIDs trial

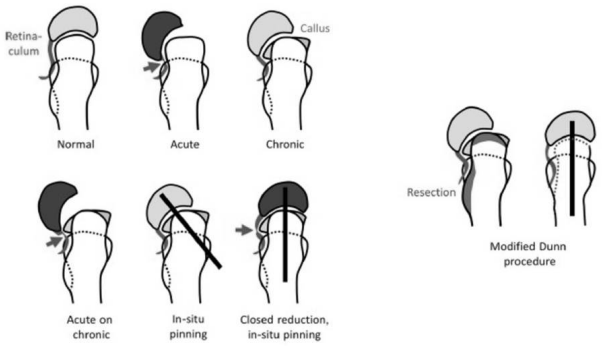
AVN risk factors	**instability (greatest)** · initial displacement · **forceful / over-reduction**, reduction of chronic slip · **posterosuperior (superolateral) hardware** · multiple pins · femoral neck osteotomy ✗ Wrong options: age, sex, obesity · single screw · in situ without reduction

– – Active space – –

Capsulotomy / decompression — unstable ICP ~2× normal · may ↓ AVN (low evidence) · TO advocates routine capsulotomy in acute unstable · **ICP probe monophasic waveform → decompress hip, then capsulotomy if still monophasic**

Modified Dunn (surgical dislocation, capital realignment) — unstable + severe · posterior retinacular flap carries blood supply · remove posterior callus · AVN **~26%** vs 24% in situ · complications 37% · steep learning curve

Most dreaded complication — **AVN** (TO); MFCA injury



Modified Dunn

Contralateral prophylactic pinning

Indication	NOT an indication
Age <10 (girls <10, boys <12–12.5)	Unstable presenting slip
Open triradiate cartilage	Chronicity / duration of symptoms
Endocrinopathy (hypothyroid, renal, GH, pituitary adenoma)	Height >95th percentile
Obesity (BMI >95th) / obese males	African American ethnicity
Posterior sloping angle >12° (TO) / >14.5° (OB)	Elite athlete · DI · weight 40th percentile
Modified Oxford score 16–18 (~98% risk; 24 = 0%)	Closed triradiate
Weight <10th or >90th percentile · unreliable follow-up	Continued slipping after fixation (TO)

Contralateral slip — **most common complication after unilateral fixation** · 20–80% · up to 36% high-risk · ↓ with weight loss · **more common in YOUNGER children**
✗ Wrong options: older (TO false statement)

Prophylactic pinning risk — 2–6% complications

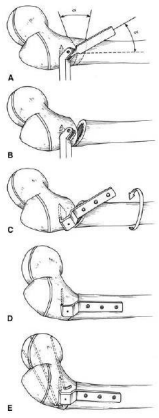
Both hips involved / silent grade 1 on other side / pre-slip on MRI → pin both

Chronic deformity, osteotomy & natural history

Residual deformity ↑ α-angle, **pistol-grip**, cam FAI · sequelae: **shortening, ↓ flexion, ↓ IR**, out-toeing

Deformity	Procedure
Mild–moderate (<30°) cam FAI	**Osteochondroplasty** (arthroscopy, limited anterior arthrotomy, surgical dislocation)
Severe (>30–45°), painful, no OA/ AVN	**Imhauser intertrochanteric osteotomy** – **flexion + internal rotation + valgus** (most used; AVN 2–7%)
Base-of-neck (TO)	**Kramer** closing wedge (lesser → greater trochanter, posterior cortex intact) – away from physis, ↓ AVN · subcapital: **Fish**, Dunn, modified Dunn · intertrochanteric "good indication" for chronic SCFE **25–50°** (TO)
	Southwick subtrochanteric osteotomy
Greatest correction	**Femoral neck (subcapital) cuneiform osteotomy** – AVN 37%, OA 37% · sole indication **Southwick >60°** (TO)

Natural history most well tolerated · **~5% THA at 20 yr** · realignment not proven to prevent OA · closed reduction worse
✗ Wrong options: 25%



Imhauser osteotomy

6 Complications

Complications - chondrolysis, impingement, slip progression

Chondrolysis 0–2% · joint space **<3 mm** (normal 4–6) + ↓ ROM (TO) · **pin penetration, ANTEROSUPERIOR quadrant** · **spica cast** · CT best for penetration
✗ Wrong options: grade of slip · no reduction

Labral tear / impingement **anterior 2nd screw medial to IT line** (in-out-in)

Slip progression 1–2% with single screw · prevented by **two (fully threaded) screws**, ≥5 threads, WB restriction
✗ Wrong options: spica · bone graft epiphysiodesis

Other AVN (op-related 4–6%) · infection 0–2% · chronic pain 5–10% · knee pain 5–10% · LLD · OA · femoral neck fracture (extra drill holes)

– – Active space – –

X vs Y

	SCFE	Delbet I (transepiphyseal) fracture
Energy	Minor / none	High
Age	10–16	Young child
Sex	M > F, obese	M = F
Course	Insidious ± acute-on-chronic	Acute

How it's asked

"11-yr 70 kg, Scarpa tenderness, obligatory ER?"	→ **SCFE** (TO) ✗ Wrong options: Perthes · type I fracture
"SCFE — all true EXCEPT?"	→ **knee pain more often than hip pain**
"14-yr overweight, knee pain, normal knee films?"	→ **AP pelvis + frog-lateral** ✗ Wrong options: knee MRI
"12-yr knee pain 6 months, growth chart obese?"	→ **hip + pelvis X-rays**
"Severe SCFE, unmanipulated AP — hip position?"	→ **external rotation**
"12-yr limping, knee pain, normal knee?"	→ **examine the hip**
"Presentation that delays diagnosis?"	→ **knee pain**
"11-yr, BMI 95th, knee pain, limp?"	→ **X-ray of left hip**
"Southwick angle associated with?"	→ **SCFE** (TO)
"Southwick angle purpose?"	→ **severity of slip**
"Method for radiographic severity?"	→ **Southwick angle image** ✗ Wrong options: Klein · Shenton · CE angle
"Hypothyroidism associated with?"	→ **SCFE**
"10-yr hypothyroid girl after unilateral pinning — likely problem?"	→ **contralateral SCFE**
"11-yr hypothyroid obese girl, unable to walk?"	→ **in situ pinning of both hips**
"12-yr, TSH 11, low T4, SCFE?"	→ **pin both hips** ✗ Wrong options: endocrine referral first
"8-yr hypothyroid, stable slip + widened other physis?"	→ **in situ pinning both hips**
"Pituitary adenoma + unstable slip — indication for prophylaxis?"	→ **endocrine disorder** ✗ Wrong options: unstable presentation · age 13
"Indication for same-admission contralateral fixation?"	→ **renal osteodystrophy** ✗ Wrong options: chronicity · height

"11-yr BMI 32, knee pain, out-toeing — true?"	→ **bilateral treatment with endocrinopathy**
"Elevated leptin — X-ray?"	→ **SCFE**
"SCFE — all EXCEPT?"	→ **contralateral slips more common in older children** (TO)
"NOT predictive of contralateral slip?"	→ **continued slipping after fixation** (TO)
"Relative indications EXCEPT?"	→ **African American ethnicity**
"9-yr unstable slip — NOT a contralateral risk?"	→ **height >95th percentile**
"10-yr hypothyroid obese — prophylaxis EXCEPT?"	→ **symptoms for one week**
"12-yr 95th percentile, bilateral stable slips?"	→ **in situ pinning of both hips**
"14-yr, prior left pinning, intermittent right pain, MRI physeal edema?"	→ **in situ fixation of right hip**
"12-yr left SCFE + silent grade 1 right, open triradiate?"	→ **pin both hips**
"9-yr BMI 42, grade 1 left slip?"	→ **percutaneous screws both hips**
"Factor favouring contralateral pinning?"	→ **age <10** ✗ Wrong options: Oxford 24 · closed triradiate
"↓ AVN in unstable SCFE?"	→ **in situ single screw, no reduction**
"13-yr acute-on-chronic, no endocrine findings?"	→ **single in situ screw left hip only**
"Stable SCFE on MRI, X-rays read normal?"	→ **in situ screw fixation**
"Correct screw start point?"	→ **anterior femoral neck**
"13-yr mild stable SCFE?"	→ **single screw in situ**
"12-yr 4 months thigh pain, stable?"	→ **in situ pinning** ✗ Wrong options: reduction · surgical dislocation
"Most dreaded complication of SCFE surgery?"	→ **AVN** (TO)
"8-yr — vessel supplying most of head?"	→ **MFCA**
"Risk factor for post-op AVN?"	→ **hardware in posterosuperior neck** ✗ Wrong options: <5 threads (progression)
"8 months post-op complication?"	→ **AVN**
"Most likely to cause AVN?"	→ **unstable SCFE** ✗ Wrong options: sex · obesity
"14-yr can't bear weight even with crutches?"	→ **bed rest, urgent surgery on left hip only**
"Closed reduction + pinning → head collapse — vessel?"	→ **MFCA**
"ICP probe monophasic waveform?"	→ **decompress hip ± capsulotomy, reassess**

– – Active space – –

"Osteonecrosis 6 months after severe SCFE — association?"	→ **inability to bear weight pre-op**
"Labral tear after fixation — technique?"	→ **anterior two-screw fixation**
"Slip progression after fixation — prevent with?"	→ **two fully threaded screws**
"Chondrolysis 4 yr after pinning — cause?"	→ **pin tip in anterosuperior quadrant**
"Sole indication for subcapital osteotomy?"	→ **Southwick >60°** (TO)
"Long-term limb + rotation after unstable SCFE?"	→ **shortening, ↓ flexion, ↓ IR**
"Missed SCFE, ER deformity, FAI pain?"	→ **Imhauser osteotomy: flexion, IR, valgus** ✗ Wrong options: Pauwels · PAO
"Natural history after pinning?"	→ **mostly well tolerated, ~5% THA at 20 yr**
"Chronic SCFE histology?"	→ **granulation tissue between hypertrophic columns**
"Zone involved (no renal disease)?"	→ **hypertrophic zone**
"15-yr breaststroke swimmer, hip pain, MRI?"	→ **external rotator muscle tear** ✗ Wrong options: SCFE · stress fracture

SPINAL MUSCULAR ATROPHY

– Pediatrics · 99 / 112 –

1 Basics

Epidemiology **most common genetic cause of death in childhood** · 1 in 10,000

Pathology progressive loss of **α-motor neurons in the anterior horn** (✓ SMN protein deficiency) · weakness **proximal > distal**, **LE > UE**, symmetric

Genetics & pathophysiology (SMN1, SMN2)

Inheritance **autosomal recessive**

Gene **homozygous telomeric deletion of SMN1**, chromosome 5 · in **90%** (Orthobullets) · **95–98%** (OB) · SMN protein: RNA metabolism, apoptosis

Severity all lack SMN1 protein · **number of functional SMN2 copies sets severity** (✓ SMN2 correlates with severity) · type I has 1 SMN2 copy

Gene (distractors)	Disease
SMN1	**SMA**
Dystrophin (↑ CPK, DTRs present, calf pseudohypertrophy)	Duchenne
PMP22 duplication	CMT (70–80%)
Androgen receptor CAG repeat	Spinobulbar (Kennedy)
Frataxin GAA repeat	Friedreich ataxia
β-galactosidase	Morquio B
COMP / COL1A1	Pseudoachondroplasia / OI
Sphingolipid (glucocerebroside)	Gaucher
Fetal ACh receptor, non-progressive contractures	Arthrogryposis 37

One SMN1 deletion → any of the 3 types (✓ "any of the above")

Types

Type	Name	Onset	Function	Prognosis
I	**Acute Werdnig-Hoffmann**	**<6 months**	Floppy, poor head control, feeding difficulty, **tongue fasciculations**, absent DTRs	**Death by ~2 yr**
II	Chronic Werdnig-Hoffmann	**6–12 months**	**Sits but can't walk** · LE worse	To **5th decade**
III	**Kugelberg-Welander**	**2–15 yr**	Walks as child, wheelchair as adult · proximal weakness	**Normal** life expectancy ± respiratory support

– – Active space – –

3 Presentation

Exam	**absent DTRs** · **fasciculations** · hypotonia · normal CPK + aldolase ✗ Wrong options: Duchenne (present)

4 Diagnosis & Classification

Diagnosis	**DNA testing** (positive test usually avoids biopsy) · muscle biopsy · prenatal possible
X-ray	scoliosis series · pelvis

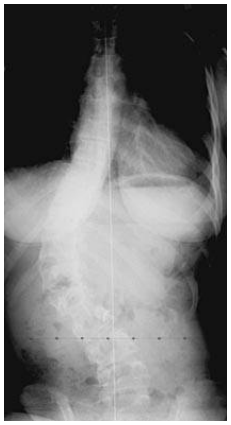
5 Management

Medical therapy - nusinersen

Nusinersen	**antisense oligonucleotide** · alters **SMN2 splicing to include exon 7** → full-length SMN · **intrathecal** (doesn't cross BBB) · FDA-approved
Others	**risdiplam** (same mechanism, **oral** small molecule) · **Zolgensma** (onasemnogene, **IV gene replacement of SMN1**) · albuterol (adjunct) ✗ Wrong options: prednisone (Duchenne)

Scoliosis management

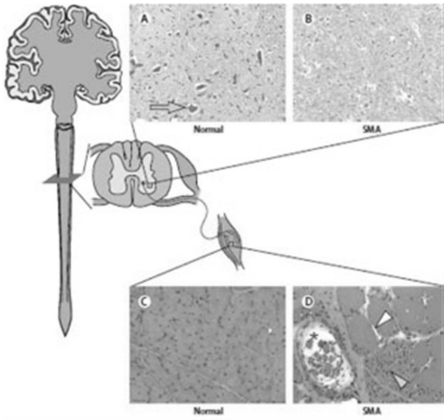
Curve	**almost universal** (100% type II) · from **age 2–3** · **long C-shaped thoracolumbar**, **very flexible** · progresses despite brace · curve size correlates with ambulatory status
Bracing	may **delay, not prevent** surgery (<10 yr)
Surgery	**PSF to pelvis** for progressive curve (>30–40°, OB; >50–60° with sitting/ pulmonary decline, OB) · goals: **sitting balance, free the hands, slow pulmonary decline** · **release hip / LE contractures first** for seating · leave a **lower lumbar laminectomy window unfused** for intrathecal nusinersen
Anterior + posterior	only **curves >100°** or **very young (crankshaft)** · avoid with poor lungs · **typically NOT needed** · growing rods only if young ✗ Wrong options: anterior alone (secondary progression)
Post-op	**temporary ↓ upper-limb function** (e.g. ↓ self-propelling wheelchair) ✗ Wrong options: better strength · walking · contracture improvement



SMA scoliosis

Hip subluxation - observation

Hip instability	subluxation/dislocation in **62% of type II**, less in III · usually **painless** · open reduction redislocates
Treatment	**observation (leave dislocated)** = standard ✗ Wrong options: adductor/psoas release · varus osteotomy · PAO · spica · abduction brace
Contractures	hip + knee (± ankle equinus in non-walkers) · PT · release controversial (rarely helps non-walkers, recurs)



Anterior horn cell loss

X vs Y

	SMA	Duchenne
Lesion	Anterior horn cell	Muscle (dystrophin)
DTRs	**Absent**	Present
CPK	Normal	**Very high**
Fasciculations	Yes (tongue)	No
Inheritance	AR	X-linked recessive

How it's asked

"Homozygous SMN1 deletion results in?"	→ **any of types I–III**
"3-month-old floppy, absent DTRs, tongue fasciculations – pathology + prognosis?"	→ **anterior horn cell loss, unlikely to live past 2** ✗ Wrong options: dystrophin · CP · CMT
"Newborn tongue fasciculations, hypotonia, no DTRs?"	→ **SMN1 mutation**
"12-month-old sits not walks, normal CPK?"	→ **SMN-1 gene** ✗ Wrong options: dystrophin · PMP22 · frataxin
"SMN protein deficiency leads to?"	→ **anterior horn cell loss, progressive weakness**
"Gene correlating with severity?"	→ **SMN2**

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"Gene implicated in SMA?"	→ **SMN-I**
"Intrathecal SMN2 splicing modifier?"	→ **nusinersen** ✗ Wrong options: risdiplam (oral) · Zolgensma (IV gene therapy)
"13-yr non-ambulatory, flexible 68° curve progressing in brace?"	→ **posterior spinal fusion** ✗ Wrong options: AP fusion · growing rods · hip reduction
"Expected change after PSF?"	→ **diminished ability to self-propel wheelchair**
"SMA scoliosis — INCORRECT?"	→ **anterior release + posterior fusion typically required**
"12-yr type 3, 60° curve, night ventilation?"	→ **posterior spinal fusion** ✗ Wrong options: AP · anterior alone
"7-yr type II, painless hip popping, subluxation?"	→ **observation**
"11-yr non-ambulatory, painless full ROM, dislocated hip?"	→ **observation**

SPONDYLOEPIPHYSEAL DYSPLASIA (SED)

– Pediatrics · 100 / 112 –

1 Basics

Definition abnormal **type II collagen** → vertebrae + epiphyses involved · rare

Genes & lab markers

Gene **COL2A1**, chromosome **12**

Form	Inheritance	Severity
SED congenita	**Autosomal dominant**	More severe · LE angular deformity · myelopathy from AAI
SED tarda	**X-linked recessive**	Milder · no LE angular deformity · nephrotic syndrome
–	Random mutation in **50%**	–

3 Presentation

Symptoms **cervical myelopathy** (AAI) · respiratory insufficiency (thoracic dysplasia) · **myopia / retinal detachment** · hip pain (coxa vara) · ↓ walking distance

Exam **short stature** (short trunk) · **flat facies** · kyphoscoliosis · ↑ lumbar lordosis · **coxa vara** · genu valgum · waddling gait, ↓ hip ROM

4 Diagnosis & Classification

Key radiology findings

C-spine AP, lateral, **open-mouth** + **flexion–extension** · **upper cervical instability** · **odontoid hypoplasia / os odontoideum**

Thoracolumbar **platyspondyly** · incomplete ossification fusion · endplate irregularity, narrow discs · kyphoscoliosis · lordosis

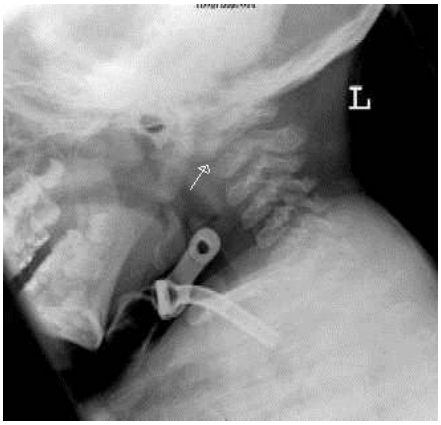
Hips horizontal acetabular roofs · delayed pubic + femoral head ossification · wide Y cartilage · **coxa vara**

MRI instability / myelopathy → cord signal

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Platyspondyly in SED



SED with cervical instability

SED vs MED

	SED	MED
Spine	**Involved** (✓ distinguishing)	Normal
Gene	COL2A1 (type II collagen)	COMP / type IX collagen
Shared	Irregular epiphyses, genu valgum, early hip OA, short stature	Same
Neck	**AAI, myelopathy**	–

Differential achondroplasia · diastrophic dysplasia · MED

5 Management

Non-op (all) multidisciplinary: PT, OT, **yearly eye exam**, pulmonology, bracing mild scoliosis

Operative criteria

Problem	Indication	Procedure
Atlantoaxial instability	**≥8 mm** or **myelopathy**	**Posterior atlantoaxial fusion**
Scoliosis	**>50°**	Growing rods (young) / PSF (older)
Coxa vara	**NSA <100°**, progressive, symptomatic OA	**Valgus (+ extension) intertrochanteric osteotomy**

Atlantoaxial instability & myelopathy

SEDC child, new unsteadiness, hyperreflexia, Hoffmann, ↓ dexterity → cervical myelopathy from AAI
✗ Wrong options: scoliosis · tethered cord · coxa vara gait

6 Complications

List cervical instability · spinal deformity · eye problems · hip deformity · early OA

How it's asked

"10-yr SEDC, new myelopathic signs?"	**→ cervical myelopathy due to atlantoaxial instability**
"Feature distinguishing SED from MED?"	**→ spine involvement** ✗ Wrong options: irregular epiphyses · genu valgum · short stature

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SPRENGEL'S DEFORMITY

– Pediatrics · 101 / 112 –

1 Basics

Diagnosis & associations (Klippel-Feil)

Definition	**congenital elevation of a small, hypoplastic scapula** from **failure of caudal migration** · **most common congenital shoulder anomaly**
Demographics	**M:F 1:3** (Orthobullets) · equal (TO) · **left** more common (TO) · **bilateral 10–30%** (10% TO)
Cause	interrupted **embryonic subclavian supply** (subclavian, internal thoracic or suprascapular) · Poland = subclavian interruption proximal to internal thoracic, distal to vertebral
Embryology (TO)	arrest of descent between **9th–12th weeks** · hypoplasia of **trapezius, rhomboids, serratus anterior**
Omovertebral connection	cartilage/fibrous/bony bar from superomedial scapula to **lower cervical spine** · **30–50%** (Orthobullets) · ~30% (OB) · ~½ (TO) · limits abduction to ~90°
Associations	**Klippel-Feil** (most common associated finding: **cervical spine anomalies**) · congenital scoliosis · fused/cervical ribs · torticollis · diastematomyelia · myelomeningocele · kidney disease · UE anomalies · iniencephaly (TO)

Overlap	Rate
Klippel-Feil patients with Sprengel	~⅓ (Orthobullets) · 35% (OB) · 20% (TO)

Klippel-Feil (TO)

Triad (~50%)	short neck · low posterior hairline · ↓ neck motion · fusion **C2–C3 most common (71%)** · gene chromosome 8 · fetal alcohol → 50% fusions
Associated	**scoliosis 60%** (most common orthopaedic) · renal anomaly · VSD · deafness · **synkinesis** · respiratory · occipitalization, hemivertebrae, basilar impression · symptomatic → posterior C3–C7 fusion ± decompression

3 Presentation

Exam	scapula **high-riding, medially rotated** (inferior pole medial) · small equilateral-triangle shape, loss of long medial border · **abduction most limited** + forward flexion · short, broad, webbed-looking neck (Turner-like) · often referred for "scoliosis" ✗ Wrong options: adduction · rotation
Differential	FSHD (no omovertebral bone) · Turner · torticollis



Sprengel deformity

4 Diagnosis & Classification

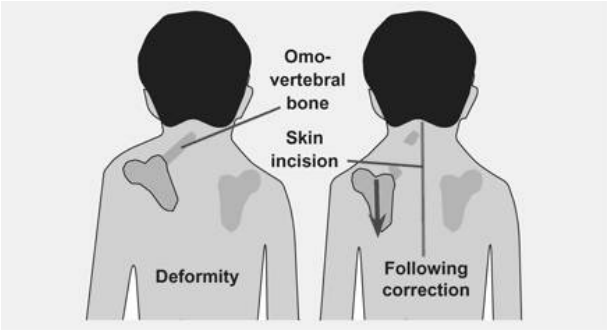
Classification & surgery (Cavendish, Woodward)

Cavendish 4-grade **cosmetic** classification of **congenital elevation of scapula**
✗ Wrong options: torticollis · radial head dislocation · clavicle pseudarthrosis

Cavendish (TO)	Severity
I	Very mild
II	Mild
III	Moderate
IV	Severe – scapula **near the occiput**

Rigault (TO)	Superomedial angle
1	Below T2, above T4 transverse process
2	Between C5 and T2
3	Above C5

Pre-op imaging **MRI or CT for omovertebral bar**



Omovertebral connection

- - Active space - -

5 Management

Operative criteria

Observe — no significant cosmetic or functional problem

Operate — severe cosmesis or **abduction <110–120°** · ideal **age 3–8** (soon after 3) · **>8 yr → brachial plexus risk**
✗ Wrong options: defer until after 8 (TO false statement) · wait for maturity

Surgical age — **3–8 yr** (Orthobullets) · **3–6 yr** (TO lecture) – read the options · **Woodward preferred** (TO) · omovertebral bar excision · **partial scapulectomy**

Procedure	Essence
Woodward	Detach + reattach medial parascapular muscles (**trapezius + rhomboid origins shifted distally at spinous processes**) → scapula descends, rotates to abduction · modified: + superomedial border resection
Schrock / Green	Extraperiosteal detachment of muscles at scapular insertion, scapula pulled down with traction, reinserted
Clavicle osteotomy / morcellation	**Severe deformity**, before scapular descent → protect **brachial plexus** (2 cm midclavicular if >8 yr, 1 cm younger) (✗ Wrong options: mild deformity)
Bony resection	Extraperiosteal superomedial prominence (cosmesis)
Mears	Most abduction gain (TO)

Outcome — **abduction +40–50°** (Orthobullets) · ~40° (TO) · +38° (Farsetti)

Complication — **brachial plexus palsy = most severe and most common** (TO) · SSEP monitoring

Other complications (TO) — **keloid** · shoulder stiffness · one TO lecture names **radial nerve** as the most commonly affected nerve (TO – verify: brachial plexus palsy is the standard answer)

How it's asked

"4-yr Klippel-Feil, elevated left scapula – most limited motion?"	→ **abduction**
"Failure of scapular caudal migration?"	→ **Sprengel deformity** ✗ Wrong options: Poland
"5-yr elevated, medially rotated scapula – commonest imaging association?"	→ **cervical spine abnormalities** ✗ Wrong options: absent clavicle · horseshoe kidney
"6-yr bilateral, omovertebral bone on CT, can't raise arms?"	→ **Sprengel deformity** ✗ Wrong options: FSHD · Turner
"Woodward procedure used for?"	→ **Sprengel shoulder** (TO)
"6-yr, abduction 100°, surgery planned – NOT true?"	→ **defer 2 yr because plexus palsy is rare after 8** (TO)
"Cavendish classification for?"	→ **congenital elevation of scapula** (TO)

"True about Sprengel?"	→ **brachial plexus palsy is the most severe and common surgical complication** (TO) ✗ Wrong options: delay to maturity · morcellation for mild · right more common	– – Active space – –

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SUPRACONDYLAR FRACTURE - PEDIATRIC

– Pediatrics · 102 / 112 –

1 Basics

Epidemiology most common pediatric elbow fracture (50–80%) · age **5–7** · **M = F** · FOOSH

Type **extension 95–98%** (98% TO) · **flexion <5%** (2–4%) – fall on olecranon

Associated **ipsilateral distal radius / forearm 5–10%** (TO) · vascular compromise **5–17%** · nerve injury 10–20%

Epidemiology (TO) ~**60%** of paediatric elbow injuries · age **4–10 yr**

Site (TO) just proximal to **olecranon fossa** (thinnest part of distal humerus)

Elbow ossification order

Centre	Appears (yr)	Fuses (yr)
Capitellum	1	12–14
Radial head	3	14–16
Medial (internal) epicondyle	5 (6 in Orthobullets text)	**16–18 – LAST to fuse** (16–19 OB)
Trochlea	7	12–14
Olecranon	9	15–17
Lateral (external) epicondyle	**11 – last to appear**	12–14

3 Presentation

Nerve injuries (AIN, radial, ulnar)

Pattern	Nerve	Deficit
Extension (overall most common)	**AIN** (median branch)	**Can't flex thumb IP (FPL) + index DIP (FDP)** · no "OK" sign · PQ
Extension, **posteromedial** displacement	**Radial** (2nd most common)	Wrist drop, no thumb/MCP extension (PIN: EPL, ECU) · dorsal hand numbness
Extension, **posterolateral** displacement	**Median** nerve, **brachial artery**	Volar index sensory loss
Flexion	**Ulnar** (C8–T1)	**Intrinsics**: can't cross fingers / abduct index (FDI) · small-finger numbness
Medial pin	**Ulnar** (iatrogenic)	Can't cross fingers

Rates extension neurapraxia 11% · flexion 17% · pre-op deficit ~20% in displaced extension (OB)

Management examine **before** any reduction · **observe** · recovery 6–12 wk, complete 3–6 months (2–3 months OB)
✗ Wrong options: EMG · exploration · neurolysis · nerve repair

Recovery (TO) radial 100% · median 88% · **ulnar only 25% fully recover**
✗ Wrong options: "majority recover in 3 months" (TO false statement)

Obese children more pre- and post-op nerve palsy (TO)

Vascular exam

Assess pulse (palpation, Doppler) + **perfusion** (warm, pink vs **cold, pale, CRT >2 s**) · **decision to explore is based on perfusion, not pulse** · radial pulse absent at presentation 7–12%

Brachialis sign antecubital ecchymosis, **dimpling/puckering**, palpable bone → proximal fragment buttonholed through brachialis → more arterial injury, swelling, harder reduction

Bedside nerve tests (TO) median / AIN: **OK sign** · radial: **thumbs up**, finger extension · ulnar: **finger abduction**, fist

4 Diagnosis & Classification

Classification & radiographic assessment

Gartland	Description	Treatment
I	**Nondisplaced** (beware medial comminution → varus)	Long arm cast **3–4 wk**, X-ray at 1 wk
II	Displaced in 1 plane, **posterior cortex/ hinge intact** (IIA no rotation, IIB rotation)	**CRPP** (US standard) · cast only if AHL hits capitellum, little swelling, no medial comminution
III	Displaced in 2–3 planes, no cortical contact	**CRPP** ± open
IV	**Periosteum completely disrupted** – unstable in flexion AND extension (found under EUA)	CRPP ± open
Medial comminution	Medial column collapse, ↓ Baumann	**CRPP** (valgus force) – else **cubitus varus**
Flexion type	Distal fragment **anterior**	**CRPP**, **more often open**

Fat pads **posterior fat pad** = occult fracture · **anterior "sail" sign** (TO) = effusion · most common cause: supracondylar (child), radial head (adult)

Anterior humeral line through middle ⅓ of capitellum **>5 yr** · **<5 yr: anterior and middle ⅓ about equally**, posterior ⅓ in a few normal kids · capitellum visible from ~1 yr · extension type → capitellum **posterior** to line · compare other side
✗ Wrong options: always fracture

Baumann angle humeral shaft vs lateral condylar physis on AP · **70–75°** · deviation **>5–10°** unacceptable

Baumann angle (TO) normal **64–82°** · every **5° change = 2° change in carrying angle** · **↑ Baumann** in cubitus varus · key **intra-operative guide**

Baumann angle – other TO figure normal **8–28°** (TO – verify: Orthobullets **70–75°**; TO figure presumably measured as the complement / other reference)

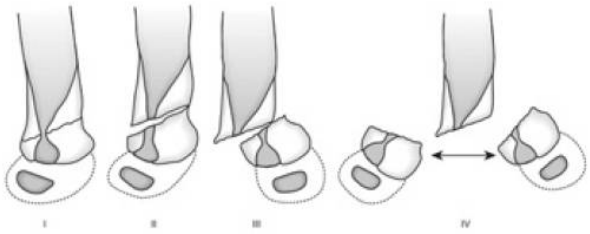
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Anterior humeral line	middle ⅓ of capitellum **> 5 yr** (Orthobullets) · **> 4 yr** (TO) – read the options
Other lines (TO)	**shaft-condylar angle ≈ 40°** (lateral) · **teardrop sign** (lateral) · **crescent sign** = cubitus varus (overlap of capitellum and radial head) · AP: metadiaphyseal angle, humeroulnar (carrying) angle
Views (TO)	AP + lateral elbow + **entire humerus and forearm** (floating elbow)

Other classifications (TO)

Classification	Feature	Significance
Gartland IIA (TO)	Posterior hinge intact, **no rotation**	Cast / CRPP
Gartland IIB (TO)	Posterior hinge intact **with rotation**	**CRPP**
Leitch type IV (TO)	**Unstable in flexion and extension** – global instability	CRPP
De Boeck (TO)	**Medial column comminution**	High risk of **cubitus varus**

Angiography not indicated (lesion is at fracture)



Gartland classification



Anterior humeral line

5 Management

Treatment indications - CRPP

Supracondylar fracture – timing by neurovascular status

- Pulseless + POORLY perfused (cold, pale) → **EMERGENT CRPP (hours, override NPO)**
- Pulseless + well perfused, sensory deficit, swelling, brachialis sign, open, floating elbow → **URGENT (same day, 6–8 h)**
- Warm, perfused, neuro intact (± isolated AIN) → **next day (splint 30–40°, elevate)**

Operative — **Gartland II, III, IV** · **flexion type** · **medial comminution** · displaced type II with medial impaction (3-yr) → **CRPP**
✗ Wrong options: cast (varus) · hyperflexion cast (compartment syndrome)

Floating elbow (supracondylar + forearm / distal radius) → **pin BOTH** (✗ Wrong options: casting either (compartment risk up to **33%**)) · watch for compartment syndrome · ↑ nerve injury (✗ Wrong options: ↑ vascular · ↑ open)

Isolated AIN palsy → **not** a reason for overnight surgery

Pin configuration & iatrogenic ulnar nerve injury

Reduction (extension type) — traction, correct coronal/rotation, **hyperflexion** · posteromedial → **pronate** · posterolateral → **supinate** (try opposite if fails)

Reduction order (TO) — longitudinal traction → correct varus-valgus → flexion + **olecranon pressure** → pronation / supination by periosteal hinge

Brachialis sign fracture (TO) — **brachialis milking technique** · **Tansen joystick technique** (Nepal)

Stable lateral pinning rules (TO) — **≥ 2 cortices** purchase · pin separation **≥ ⅓ humeral width** at fracture · **two 2 mm pins better than three thin pins**

Indications for a medial pin (TO) — **medial oblique** fracture · **intercondylar extension** · **high** supracondylar fracture · **internal rotation instability**

Construct	Use
2 lateral pins	Type II · test stability
3 lateral pins	**Type III / IV, comminution (medial column)** · = crossed-pin stability · **construct of choice** (AAOS)
Crossed (medial + lateral)	**Strongest in torsion** (experimental) · **ulnar nerve injury 3–8%**

AAOS — **2–3 lateral pins**, avoid medial pin (weak recommendation) · **no Level 1 evidence that lateral is superior** (RCT: equal outcomes) – the INCORRECT statement · lateral = fewer ulnar injuries, equal radiographic + ROM outcomes

Pin tips — **maximise separation at fracture**, engage both columns, cross **proximal** to fracture · 3rd lateral pin ↑ stiffness with medial comminution or residual IR

Elbow position for pinning — **flexion** for extension type · **extension for flexion type**

Medial pin safety — place with **elbow extended** (ulnar nerve subluxes anteriorly in flexion) · small incision · anterior-to-posterior sagittal angle ↑ injury

Medial pin flexion (TO) — insert with elbow **< 45° flexion** (ulnar nerve) · wires **outside skin if removed < 3 wk, buried if > 6 wk** (TO)

Post-op ulnar palsy after crossed pins → **observe + avoid hyperflexion >90°** (TO) · resolves 1–7 months
✗ Wrong options: remove medial pin now · exploration/transposition

Open reduction — failed closed · **open fracture** · irreducible with gap (entrapped vessel/nerve) · **flexion type more often** · **coronal displacement >7 mm** predicts it · approach **anterior** (pulseless / median injury) or medial/lateral where periosteum torn · **never posterior** (AVN)

– – Active space – –



Lateral pin construct

Pulseless hand & vascular management

Situation	Action
Pulseless + cold/pale at presentation	**Emergent CRPP** → reassess (✗ Wrong options: angiogram · splint and wait · explore first)
Pink pulseless (warm, brisk CRT ± Doppler)	**Urgent CRPP** · after: if still pink (± triphasic Doppler) → **long arm splint + admit 48 h observation** · pulse returns **24–48 h** in most
Pulseless + still cold/pale after CRPP	**Explore antecubital fossa** (✗ Wrong options: observe · re-pin · fasciotomy)
Perfused hand becomes pale/ pulseless after CRPP	**Remove pins, re-displace, reassess** → still bad → **open exploration** (✗ Wrong options: medial pin · CT angiogram · warm and wait)
Irreducible + anterior gap + poor perfusion	**Vascular exploration** (entrapped artery)

Pulseless + poorly perfused after CRPP → brachial artery injury up to 82%

After CRPP (TO) pale, non-perfused → **immediate exploration** · pink, perfused, no deficit → **close observation** · **pink + poor Doppler or nerve palsy → exploration** · **prophylactic fasciotomy** if prolonged ischaemia

Emergency surgery (TO) open fracture · **skin tenting** · neurovascular compromise · floating elbow · compartment syndrome · otherwise splint, operate next morning



Gartland III with vascular risk

Postoperative care (pin removal, PT, pin infection)

Pins remove at **3 (–4) wk** in clinic, stop immobilisation
✗ Wrong options: 3 more wk cast

ROM	~**94% by 6 months** · **no routine PT** (no functional or motion benefit)
Pin infection	1–2.4% · **only proven risk: age <4.5 yr** · oral antibiotics ✗ Wrong options: delay >24 h · no antibiotics · late follow-up; daily pin care ↑ infection
Pin tract infection	**1–2.4%** (Orthobullets) · **≈ 4%** (TO) – read the options

6 Complications

Compartment syndrome

Rate	0.1–0.3% (TO) · <1% · ↑ with **floating elbow**, splint >90°, type III, vascular injury, delay
Signs in children	**3 A's: anxiety, agitation, ↑ analgesic need** · rising narcotic need overnight → **emergent surgery ± fasciotomy** ✗ Wrong options: rely on 6 P's
Volkmann ischemic contracture	deep volar compartment · **casting in >90° flexion**

Cubitus varus

Cubitus varus (gunstock)	**most common complication** (OB) · Orthobullets lists **pin migration (~2%)** as most common · **malreduction / malunion** (medial collapse, IR, extension) – **NOT growth arrest** ✗ Wrong options: medial physeal arrest · lateral overgrowth
Effects	mainly **cosmetic**, function normal · posterolateral rotatory instability · forearm IR deficit · **tardy ulnar palsy** (anterior nerve subluxation most common, then FCU bands)
Osteotomy	for **cosmesis** · **lateral closing wedge → lateral prominence** · reverse V, step-cut, dome, medial opening with medialisation avoid it
Cubitus valgus	malunion → tardy ulnar palsy · **recurvatum** – non-op type II/III
Cubitus varus = 3-plane deformity (TO)	coronal **varus** · sagittal **extension** · axial **internal rotation**
Rotational tests (TO)	**Yamamoto test** · **Bhende test** · radiology: ↑ Baumann · crescent sign · fishtail (progressive)
Remodelling (TO)	**sagittal** deformity may remodel · **varus does NOT remodel**
Surgery indications (TO)	cosmesis · function · **PLRI** · **ulnar nerve instability** · **recurrent lateral condyle fractures**
Surgery criteria (TO lecture 2)	varus **> 5–10°** · age **> 5 yr** · **≥ 1 yr after trauma** · flexion/extension component · other sequelae: **triceps snapping**, tardy ulnar palsy · options: medial open wedge, lateral closing wedge (most common), step-cut, dome, **French osteotomy** (corrects rotation), guided growth in young · **PLRI → LUCL reconstruction + osteotomy**
Oblique lateral closing wedge osteotomy (TO)	**current gold standard** · apex near original fracture · equal-limb · **no lateral prominence** · K-wires or plate · options: simple closing wedge · step-cut · dome

– – Active space – –

TRASH lesions (TO)

TRASH	"The Radiographic Appearance Seemed Harmless" · unossified **medial condyle** fracture · **transphyseal** distal humerus fracture · entrapped **medial epicondyle** · osteochondral fracture-dislocation · **radial head compression** fracture · **Monteggia variants**
Fishtail deformity (TO)	**lateral trochlear ossification centre fails/resorbs** (AVN of posterior vessels to **lateral** trochlea, "type A") → inverted-V distal humerus · hard to see before **~9 yr** · after supracondylar (most), lateral/medial condyle, SH I · (✗ Wrong options: "medial trochlea blood supply") · **Hegemann** = trochlear osteochondrosis ✗ Wrong options: epicondylar

X vs Y

	Extension type	Flexion type
Frequency	95–98%	<5%
Distal fragment	Posterior	**Anterior**
Nerve	**AIN** (then radial)	**Ulnar**
Pin elbow position	Flexion	**Extension**
Open reduction	Less	**More**

How it's asked

"Arrow on anterior elbow fat pad?"	→ **sail sign** (TO)
"6-yr, displaced, posterior cortex intact?"	→ **Gartland II**
"FOOSH injury needing surgery?"	→ **Gartland II supracondylar** ✗ Wrong options: angulated BBFF · proximal humerus <12
"Fully displaced, no cortical contact?"	→ **Gartland III**
"22-month-old normal AHL position?"	→ **anterior ⅓ more often than posterior ⅓**
"Supracondylar – all EXCEPT?"	→ **ulnar injuries in flexion type mostly recover in 3 months** (TO)
"Flexion type – motor deficit of commonest nerve?"	→ **hand intrinsic weakness**
"Commonest nerve innervates all EXCEPT?"	→ **EPL**
"NOT an urgent/emergent indication?"	→ **can't flex thumb IP (isolated AIN)**
"Commonest nerve controls?"	→ **thumb IP flexor**
"Small-finger numbness – which X-ray?"	→ **flexion-type supracondylar**
"Can't flex thumb IP, pulses present?"	→ **AIN**
"5-yr, no thumb IP / index DIP flexion – nerve management?"	→ **observation**

"8-yr 100% displaced, no thumb IP / index DIP flexion?"	→ **AIN palsy** ✗ Wrong options: Volkmann
"No wrist/thumb extension — initial management?"	→ **CRPP, expect nerve recovery** ✗ Wrong options: explore radial nerve
"Posteromedial displacement, dorsal hand numbness — muscles?"	→ **EPL + ECU**
"Flexion type — exam finding?"	→ **can't abduct index finger**
"Flexion type — nerve + management?"	→ **ulnar; observation**
"Extension type — nerve + management?"	→ **AIN; observation**
"Most common deficit muscle?"	→ **FPL** ✗ Wrong options: interossei · EPL
"Puckering + pink hand + deficit?"	→ **thumb flexion**
"Flexion type commonest nerve roots?"	→ **C8–T1**
"Partial ulnar palsy after crossed pins?"	→ **observe, avoid hyperflexion >90°** (TO)
"Crossed vs lateral pins — disadvantage?"	→ **ulnar nerve injury risk**
"Advantage of crossed pins?"	→ **greater experimental biomechanical stability**
"6-yr type III, NV intact — AAOS?"	→ **closed reduction + 2–3 lateral pins** ✗ Wrong options: crossed
"7-yr type III, OK/thumbs up/cross fingers intact?"	→ **CRPP with lateral pins**
"Type III + medial comminution — construct?"	→ **3 lateral pins**
"4-yr Gartland III — preferred pins?"	→ **3 lateral**
"INCORRECT about lateral pinning?"	→ **Level 1 evidence of superiority**
"Medial pin (A) palsy sign?"	→ **can't cross fingers**
"Adding 3rd lateral pin?"	→ **↑ stiffness with medial comminution**
"Pin with elbow EXTENDED for?"	→ **fracture displaced anteriorly (flexion type)**
"22-yr dorsoulnar numbness after varus malunion?"	→ **anterior ulnar nerve subluxation**
"10-yr pulseless cold hand?"	→ **emergent CRPP** ✗ Wrong options: angiogram · exploration
"Pink pulseless after CRPP — outcome?"	→ **pulse returns in 24–48 h**
"Pink, triphasic Doppler, no palpable pulse after CRPP?"	→ **keep pins, long arm immobilisation, observe**
"Cool pale hand after anatomic CRPP (pulseless at start)?"	→ **explore antecubital fossa**
"Lost pulse after CRPP with fracture gap?"	→ **remove pins + displace fracture**
"3-yr pink pulseless → pink with Doppler after CRPP?"	→ **splint + admit for observation**

– – Active space – –

"10-yr type III, perfused, no pulse?"	→ **closed reduction + pinning**
"7-yr pulseless after ED realignment?"	→ **CRPP in OR, reassess** ✗ Wrong options: angiogram
"3-yr pale hand, diminished pulses?"	→ **CRPP**
"8-yr floating elbow, pink pulseless?"	→ **compartment monitoring + urgent surgery**
"Pulse lost after CRPP of perfused hand?"	→ **remove pins, re-displace, reassess**
"6-yr pale pulseless at end of call?"	→ **emergent CRPP tonight** ✗ Wrong options: vascular surgery first
"5-yr cold pulseless at 9 pm — timing?"	→ **emergent, as soon as OR allows**
"Became pale after anatomic CRPP?"	→ **remove pins, reassess**
"Exploration indicated when?"	→ **irreducible, anterior gap, poorly perfused**
"Still cold after pin removal?"	→ **open exploration + vascular**
"7-yr supracondylar + 25° distal radius?"	→ **CRPP of both fractures**
"3-yr type II with medial impaction?"	→ **CRPP**
"7-yr displaced, AHL misses capitellum?"	→ **CRPP** ✗ Wrong options: cast · plate
"7-yr type II, antecubital ecchymosis, pink hand?"	→ **CRPP and discharge** ✗ Wrong options: arteriography
"Gymnast — supracondylar + BBFF — ↑ complication?"	→ **compartment syndrome**
"Floating elbow overnight — prompts emergent surgery?"	→ **↑ anxiety + narcotic need**
"Fishtail deformity seen in?"	→ **old supracondylar humerus fractures** (TO)
"Fishtail — all EXCEPT?"	→ **disturbed medial trochlear blood supply** (TO)
"Current standard osteotomy for cubitus varus?"	→ **oblique lateral closing wedge** (TO)
"Which cubitus varus component does NOT remodel?"	→ **varus** (TO)
"Correcting mild gunstock — result?"	→ **improved cosmesis**
"Etiology of cubitus varus?"	→ **malreduction** ✗ Wrong options: medial physeal arrest
"8-yr varus after splint — cause?"	→ **malreduction → malunion**
"Osteotomy causing lateral prominence?"	→ **lateral closing wedge**
"Healed in varus, untreated?"	→ **poor appearance, normal function** ✗ Wrong options: PMRI · tardy AIN
"Routine hardware removal at 3–4 wk?"	→ **supracondylar CRPP pins**
"ROM >90% by?"	→ **6 months**

"3 wk post-CRPP, clean pins?"	→ **remove all pins, stop immobilisation**
"When is PT indicated?"	→ **no routine PT**
"Twins, one gets PT?"	→ **no difference**
"Pin infection — highest risk factor?"	→ **age <4.5 yr**
"Flexion type vs extension — counsel?"	→ **higher odds of open reduction**
"7 mm coronal displacement ↑ risk of?"	→ **open reduction**
"Last elbow apophysis to fuse?"	→ **medial (internal) epicondyle**

– – Active space – –

SYNDACTYLY OF THE TOES

– Pediatrics · 103 / 112 –

1 Basics

Incidence **1 in 2,000** · most often **2nd–3rd toes**

Cause incomplete / absent **apoptosis** between digits

Inheritance mostly **autosomal dominant**

Syndromic associations

Associated familial syndactyly · **Down** · **Klippel-Feil** · **Smith-Lemli-Opitz**

Smith-Lemli-Opitz **AR** · **7-dehydrocholesterol reductase** deficiency (cholesterol synthesis) · **microcephaly, cleft palate, cardiac defect (ASD)**, postaxial polydactyly, **2–3 toe syndactyly**
✗ Wrong options: Beckwith-Wiedemann (hemihypertrophy, macroglossia) · Prader-Willi (chr 15, obesity) · FSHD · Rett

Type	Fusion
Simple	Soft tissue only
Complex	Bony fusion

3 Presentation

Symptoms painless, cosmetic · no motor delay

X-ray AP foot

Toe syndactyly	After release
	

5 Management

Type	Treatment
Simple	**Observation** (cosmetic) · surgery risks contracture, painful scar · no toe spacers
Complex	Digit release

Release (TO) **staged** release to prevent skin necrosis · **osteotomy** for complex syndactyly · often syndromic

How it's asked

"Microcephaly, cleft palate, ASD + 2–3 toe syndactyly?"	→ **Smith-Lemli-Opitz syndrome** ✗ Wrong options: Beckwith-Wiedemann · Prader-Willi · Rett

– – Active space – –

TARSAL COALITION

– Pediatrics · 104 / 112 –

1 Basics

Prevalence **1–2%** (symptomatic ~1%) · **~75% asymptomatic** · bilateral ~50% · 2 coalitions in 10–20% (OB) · extra coalition on CT ~5% (Orthobullets)

Embryology - failure of segmentation

Cause **failure of mesenchymal differentiation + segmentation** (✓ both CN and TC) · bridge fibrous → cartilage → bone
✗ Wrong options: failure of formation · ischemic necrosis · microtrauma · peroneal hypertonicity

Interface	Name
Fibrous	**Syndesmosis**
Cartilaginous	**Synchondrosis**
Bony	**Synostosis**

Acquired (less common) – trauma, degeneration, infection

Genes & lab markers

Non-syndromic **autosomal dominant** (sibling history)

Syndromic **fibular hemimelia** · carpal coalition · **FGFR1/2/3 craniosynostosis** (**Apert**, Pfeiffer, Crouzon, Jackson-Weiss, Muenke) · TN coalition: Apert, Nievergelt (TO)

Site	Incidence	Symptom age	Interface
Calcaneonavicular (CN)	**Most common** (Orthobullets) · ~45% (TO)	**8–12 yr** (10–14 OB)	Synchondrosis / syndesmosis
Talocalcaneal (TC), middle facet	~45% (TO)	**12–15 yr** (12–16 TO)	Synchondrosis → **synostosis**
Talonavicular	1–2%	Infancy	Synostosis
Calcaneocuboid · naviculocuneiform	<1%	Variable / adult	–

Symptoms start when the bar ossifies (CN earlier than TC)

2 Mechanism

Mechanics subtalar joint normally internally rotates ~10° in stance → blocked → **flat arch, forefoot abduction, hindfoot valgus, peroneal spasm** ("**peroneal spastic flatfoot**" – coalition = leading cause)

Pain sources ossification of bar · microfracture at interface · chondral damage · stress on other joints

3 Presentation

Presentation & diagnosis (imaging, anteater sign)

History	**recurrent ankle sprains** · activity pain · calf pain (peroneal spasm) · uneven ground
Pain site	**sinus tarsi / below fibula → CN** · **distal to medial malleolus / medial foot → TC**
Exam	**rigid pes planus** · hindfoot valgus that **doesn't swing into varus on toe-rise** · arch doesn't reconstitute · **↓ subtalar motion** · heel cord contracture · reverse Coleman block for subtalar rigidity
Deformity association	**rigid pes planus** ✗ Wrong options: cavus · rocker-bottom · varus
Ball-and-socket ankle	from large **neonatal TC coalition** (develops by ~5 yr)

4 Diagnosis & Classification

Key radiology findings

Views AP · standing lateral · **45° internal oblique (best for CN)** · Harris axial heel

Coalition	Sign
CN	**"Anteater (nose)" sign** – elongated anterior calcaneal process (lateral/oblique) · **"comma" sign** (TO) · ± talar beaking
TC	**"C-sign"** – talar dome to posteroinferior sustentaculum (lateral) · **talar (talonavicular) beaking** – dorsal talar lip remodeling from altered motion, not OA · irregular middle facet on Harris · **dysmorphic, rounded sustentaculum** (Oestreich)

CT	**pre-op**: size, location (TC % of posterior facet on coronal), **rule out 2nd coalition**, arthritis · gold standard in adults
MRI	**fibrous / cartilaginous coalition** (normal X-rays + suspicion) · STIR for inflammation
C-sign extras (TO)	continuous posterior talus + sustentaculum on lateral X-ray = **diagnostic of talocalcaneal coalition** · MRI: cysts, hypertrophy, stress oedema · **CT classification of TC coalition: types 1–5** (TO)
Anterolateral (anterior) process fracture of calcaneum (TO)	look for an **underlying calcaneonavicular coalition** · CN coalition may be **fibrous or osseous** (TO)

Normal X-rays, rigid valgus heel, sinus tarsi pain, repeated sprains → suspect coalition → MRI
✗ Wrong options: Sever · Kohler · Iselin · flexible flatfoot surgery

– – Active space – –



Anteater sign



C-sign



45° oblique view of CN coalition

TO imaging points CN coalition best on **45° external (standard) oblique** · TC: Harris view (beam 45° along middle facet), **Saltzman** for hindfoot valgus · CT: sagittal (CN; rules out a **second coalition ~20%**), **coronal 3 mm cuts** to measure % posterior facet · MRI: fibrous/cartilaginous coalitions (stress oedema) · **peroneal spastic flatfoot**

5 Management

Symptomatic tarsal coalition

- First → **short-leg walking cast 4–6 wk + NSAIDs (then orthosis / UCBL)**
- Recurs – CN → **resection + EDB or fat interposition**
- Recurs – TC <50% → **resection + fat / split FHL interposition ± deformity correction**
- TC >50%, no arthritis → **flatfoot reconstruction (OB) / subtalar fusion (Orthobullets)**
- Failed resection / diffuse arthritis → **triple arthrodesis**

Initial nonoperative treatment - cast immobilisation

First line (every first presentation) — **below-knee walking cast 4–6 wk** (6 wk Orthobullets · 4 wk OB) ± NSAIDs · up to **30% pain-free** (Orthobullets) · very high short-term relief (OB) · inserts may hurt rigid feet · **UCBL** after
✗ Wrong options: resection first · arthrodesis · CT/MRI first if diagnosis clear · sinus tarsi injection

Pain recurs after good cast → surgery
✗ Wrong options: repeat cast · NWB cast · orthotic · more PT · quit sport

Calcaneonavicular coalition resection with interposition

Approach (TO) — **Ollier incision** + **fat graft** interposition · TC: poor outcome if **hindfoot valgus > 16°** (correct with calcaneal lengthening) or coalition **> 50%** (consider subtalar fusion)

Procedure — **resect bar + interpose EDB muscle or fat** (or bone wax) · resection alone ↑ recurrence · **lateral oblique incision over sinus tarsi between EXTENSOR and PERONEAL tendons** · protect superficial peroneal + sural nerves · ~1 cm gap · NWB cast 3–4 wk

Add — calcaneal osteotomy / lengthening for marked valgus · heel cord lengthening if DF < neutral · mild flatfoot → no osteotomy

Outcome — **80–85% pain relief** · **full return to sport (87%, Mubarak)** · **↑ subtalar motion (74%)**, ↑ plantar flexion (82%) · repeat resection ~5%
✗ Wrong options: subtalar fusion · arthroereisis · manipulation

Contraindications — advanced degeneration · multiple coalitions → arthrodesis

Talocalcaneal coalition - resection vs arthrodesis/reconstruction

Operative criteria

TC situation	Treatment
≤50% of subtalar width (coronal CT), acceptable alignment, no OA	**Resection + fat / split FHL interposition** ± valgus correction
>50%, joints healthy	**Bony flatfoot reconstruction WITHOUT resection** (OB220914) · **subtalar arthrodesis** (Orthobullets, OB3016) — read the options
>50% + **heel valgus >16°**	Poor results with resection (Wilde)
Failed resection, >50%, or **diffuse TN/CC arthritis**	**Triple arthrodesis**

TC approach — **medial**, curved incision over **sustentaculum**, between **FDL and neurovascular bundle** (Orthobullets) · between FDL and FHL (OB) · find normal anterior + posterior facets, resect with burr, check inversion/eversion · NWB cast 3 wk

Subtalar fusion — avoid in young (10–12 yr) — limits foot growth (OB)

Poor outcomes — resecting **>50%** · **uncorrected hindfoot valgus** · degenerative change

Adults (TO) — usually ossified + secondary arthritis → CT · accommodative orthosis, AFO, injections · resection only if small bar and normal joints · otherwise subtalar / double / triple arthrodesis ± gastrocnemius recession

– – Active space – –

6 Complications

List incomplete resection · **recurrence** · residual pain / stiffness (malalignment, arthritis, **missed 2nd coalition** → pre-op CT)

X vs Y

	CN coalition	TC coalition
Age	8–12	12–15
Pain	Sinus tarsi, lateral	Below medial malleolus
X-ray	**Anteater**, 45° oblique	**C-sign**, talar beak, Harris view
Interposition	**EDB** / fat	Fat / **split FHL**
Approach	Lateral (extensor/peroneal interval)	Medial (sustentaculum)

How it's asked

"Identify the pathology (anteater sign)?"	→ **calcaneonavicular coalition** (TO)
"18-yr rigid flatfoot — most common site, age, interface?"	→ **TC middle facet; 12–16 yr; synchondrosis or synostosis** (TO)
"Lateral sign of TC coalition kinematics (dorsal TN lip)?"	→ **talonavicular beaking** (TO)
"14-yr bilateral sinus tarsi pain, heel stays valgus on rise?"	→ **recurrent ankle sprains** ✗ Wrong options: PTT insufficiency · supination in swing
"13-yr ↓ subtalar motion, C-sign + coronal CT?"	→ **rigid pes planus**
"11-yr recurrent sprains, valgus heel, normal X-rays?"	→ **tarsal coalition; MRI**
"13-yr ↓ subtalar motion, talar beak, C-sign, MRI fibrous bar?"	→ **talocalcaneal coalition** ✗ Wrong options: osteochondroma
"57-yr ball-and-socket ankle?"	→ **significant neonatal talocalcaneal coalition**
"12-yr sinus tarsi pain, brother had foot problem?"	→ **CN coalition; autosomal dominant**
"15-yr 'anteater sign' — what's resected?"	→ **anomaly between calcaneus and navicular** ✗ Wrong options: os trigonum · accessory navicular
"Primary embryological failure?"	→ **failure of mesenchymal differentiation + segmentation** (TO)
"CN vs TC pathology?"	→ **both failure of segmentation**
"14-yr pain below medial malleolus, C-sign — cause?"	→ **failure of mesenchymal segmentation** ✗ Wrong options: glycine substitution

"10-yr 4 wk foot pain, CN coalition?"	→ **4 wk short leg cast** ✗ Wrong options: excision · arthrodesis
"13-yr first visit, sinus tarsi pain, CN coalition?"	→ **cast immobilization**
"23-yr medial pain, C-sign?"	→ **immobilization** ✗ Wrong options: CT · MRI · injection
"16-yr 1-yr sprains, CN bar?"	→ **activity modification ± immobilization**
"16-yr anteater sign, first visit?"	→ **calcaneus–navicular; short immobilization**
"13-yr first evaluation, CN on oblique?"	→ **cast or boot 6 wk**
"24-yr cartilaginous CN, <⅓ joint, no arthritis?"	→ **resection + EDB or fat** (TO) ✗ Wrong options: arthroereisis · MDCO + FDL
"14-yr failed bracing/therapy, CN?"	→ **resection + EDB interposition**
"10-yr failed 5-wk cast, isolated CN, neutral heel?"	→ **resection + EDB** ✗ Wrong options: sustentaculum excision · triple
"16-yr after CN resection — likely?"	→ **full return to sport** ✗ Wrong options: ↓ subtalar motion · repeat surgery
"11-yr failed 6-wk cast, CT clear?"	→ **resection with fat or muscle**
"12-yr softball, pain back after cast?"	→ **excision of coalition**
"15-yr TC <50%, fibrocartilaginous, relief only with cast?"	→ **surgical resection** ✗ Wrong options: repeat cast · fusion
"15-yr CN, failed non-op — best for sport?"	→ **bar resection with tissue interposition**
"11-yr gymnast, failed cast + orthotics?"	→ **surgical treatment**
"13-yr failed WB cast, PT, orthosis — CN?"	→ **resection CN + fat/muscle** ✗ Wrong options: Broström
"Interval for CN resection?"	→ **between extensor and peroneal tendons**
"14-yr failed TC resection, near-complete subtalar involvement?"	→ **triple arthrodesis**
"11-yr — isolated TC resection best when?"	→ **40% width, no ↑ valgus** ✗ Wrong options: 80% · 25° valgus
"Appropriate treatment for indication?"	→ **TC <50% → resection + graft**
"18-yr TC 70% posterior facet, healthy joints?"	→ **bony flatfoot reconstruction without resection** ✗ Wrong options: subtalar fusion · triple

– – Active space – –

TIBIAL DEFICIENCY

– Pediatrics · 105 / 112 –

1 Basics

Definition	congenital **longitudinal deficiency of the tibia** (tibial hemimelia), partial → complete absence

Genes & lab markers

Inheritance	**autosomal dominant** → genetic counseling
Associated MSK anomalies (75%)	**ectrodactyly** · **preaxial polydactyly** · **ulnar aplasia**
Tibial vs fibular hemimelia (TO)	tibial: very rare, **AD**, foot **equinovarus + toe duplication** · fibular: most common, sporadic, **missing rays**

3 Presentation

Deformity	short limb · **anterolateral tibial bowing** · prominent fibular head
Knee	usually flexed, **flexion contracture** · check **active knee extension** (quadriceps) · stability in all planes → **treatment based mainly on knee stability**
Foot	**rigid equinovarus + supination** · sole faces perineum



Tibial hemimelia, clinical

4 Diagnosis & Classification

Key radiology findings

X-ray	AP + lateral tib/fib → tibial deficiency · early films: small, poorly ossified **distal femoral epiphysis**

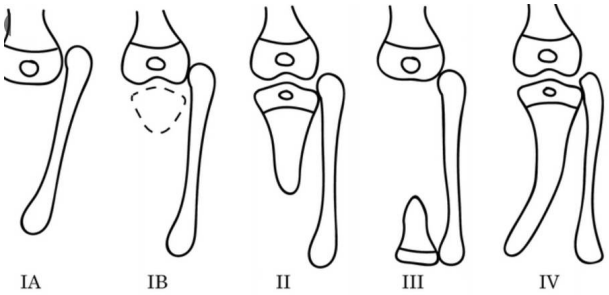
Jones classification

Type	Findings
1a	**No proximal tibia** on X-ray · **extensor mechanism absent** · hypoplastic distal femoral epiphysis
1b	Proximal tibia ossifies later, extensor mechanism works · distal femoral epiphysis normal
2	Proximal tibia present at birth, short tibia
3	Diaphysis + distal tibia present, **proximal tibia absent**
4	Short tibia, fibula migrated proximally, **distal tib-fib diastasis**

Alternative Kalamchi and Dawe (types 1–3)

Jones (TO version) I complete absence · II proximal present, distal absent · III distal present, proximal absent · IV tibiofibular diastasis

Weber classification (TO) **A: tibial anlage present** · **B: anlage absent** · subtypes: proximal / distal diastasis, focal, bifocal aplasia, complete absence · includes **double fibula**



Jones classification

5 Management

Operative criteria

Finding	Procedure
Complete tibial absence · **no active knee extension** (most cases)	**Knee disarticulation + prosthesis**
Proximal tibia present, **intact extensor mechanism**, minimal flexion contracture	**Tibiofibular synostosis + modified Syme**
Ankle diastasis	**Syme / Boyd amputation**
Brown (fibula centralized under femur)	**No longer recommended** – high failure
Weber patellar arthroplasty (TO)	Complete / proximal tibial aplasia: **patella fused to proximal fibular epiphysis** as articular surface

Principles (TO) stable weight-bearing limb · **align ankle first, lengthen later** · **knee extensor mechanism decides reconstruction vs amputation**

Syme amputation prognosis

Good below-knee result **full functional quadriceps** + **no knee flexion contracture** · otherwise knee disarticulation

✗ Wrong options: 35° flexion contracture · absent synostosis · bilateral · family history

– – Active space – –

How it's asked

"13-mo tibial hemimelia — best predictor of good Syme outcome?"	**→ full functional quadriceps strength** ✗ Wrong options: 35° flexion contracture · absent tibiofibular synostosis · bilateral

TIBIAL EMINENCE FRACTURE

– Pediatrics · 106 / 112 –

1 Basics

Definition	intra-articular **avulsion of the bony ACL attachment** (tibial spine) · **ages 8–14** · 2–5% of pediatric knee injuries with effusion
Why bone fails	eminence **incompletely ossified** → weaker than ACL · fails through **deep cancellous bone** · may extend into plateau (medial most common)
Anatomy	non-articular area between plateaus · ACL → **medial spine** (lateral aspect) · insertion 9 mm behind intermeniscal ligament, beside anterior horns · **PCL doesn't attach to spines**
Associated injuries (~40%)	meniscal · collateral ligament · capsular · osteochondral · MRI: 15–37% intra-articular pathology

Adult equivalent - ACL avulsion

Equivalent in skeletally mature	**ACL tear** · with fracture → Segond (lateral capsule) ✗ Wrong options: PCL tear · patellar tendon rupture · lateral meniscus tear
Adult tibial ACL avulsion	bony fragment on MRI + arthroscopy ✗ Wrong options: midsubstance ACL tear · PCL avulsion · bucket-handle

2 Mechanism

Mechanism	rapid deceleration, **hyperextension / rotation** (sport) · fall from bike (hyperextension) · same mechanism as adult ACL tear

3 Presentation

Findings	**immediate effusion (hemarthrosis)** · knee held **flexed** · can't bear weight · +anterior drawer / Lachman · motion block after analgesia → meniscus or entrapped fragment

4 Diagnosis & Classification

Key radiology findings

X-ray	AP · **lateral (best for displacement)** · intercondylar notch · oblique (plateau extension)
CT	pre-op planning · displacement unclear on X-ray
MRI	ligament / meniscal injury (better than CT) · not routine once X-ray confirms ✗ Wrong options: MRI as next step

– – Active space – –

Classification & closed reduction in extension

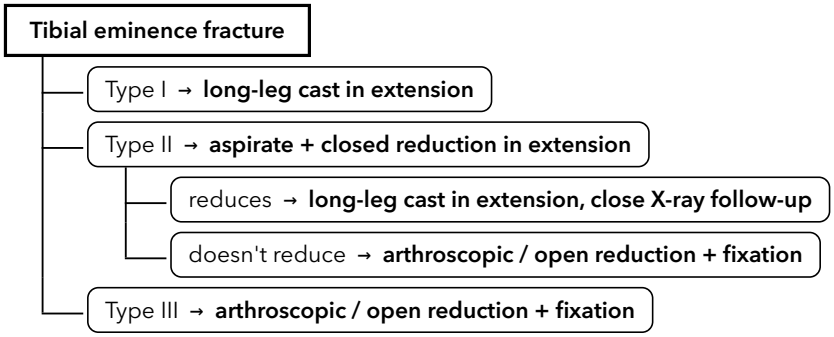
Modified Meyers & McKeever	Description
I	**Nondisplaced (<3 mm)**
II	**Anterior displacement, intact posterior hinge**
III	**Completely displaced**
III+	Type III + rotation
IV	Completely displaced, rotated, comminuted

OB variant IIIa comminution · IIIb rotation

Modified Meyers and McKeever Classification	
Type I	Nondisplaced (<3mm)
Type II	Minimally displaced with intact posterior hinge
Type III	Completely displaced
Type III+	Type III fracture with rotation
Type IV	Completely displaced, rotated, comminuted

Meyers and McKeever classification

5 Management



Closed reduction aspirate tense hemarthrosis + lidocaine · **full extension / hyperextension** · cast at **0°** (or 10–20° flexion – debated) · **3–4 wk** (Orthobullets) · 4–6 wk then hinged brace (OB) · lateral X-ray to confirm + serial films · CT/MRI if unclear · **<5 mm displacement → closed** (OB)

Traps cast in **extension** (✗ Wrong options: 90° flexion) · reduce before immobilizing (✗ Wrong options: knee immobilizer + NWB) · 1–2 wk cast too short
✗ Wrong options: immediate WB · ACL reconstruction · fragment excision (ACL attached)

Meniscal entrapment blocking reduction

Block to reduction **anterior horn of medial meniscus (most common)** > intermeniscal ligament > anterior horn lateral meniscus
✗ Wrong options: ACL · PCL · fat pad

Kocher (80 cases) medial meniscus 36 · intermeniscal 6 · lateral meniscus 1 · entrapment in **26% type II**, **65% type III** · meniscal tear 3.8%

Lowe lateral meniscus + ACL pull fragment in opposite directions (not entrapment)

Associated with type II **meniscal entrapment**
✗ Wrong options: compartment syndrome (tibial tubercle fracture) · patellar dislocation · growth arrest

Operative fixation

Operative criteria

Indications **type III** · **irreducible type II** · block to extension · failed closed reduction · late displacement

Treatment **arthroscopic (or open) reduction + internal fixation** of eminence · anatomic reduction
✗ Wrong options: ACL reconstruction · tibial tubercle ORIF

	Suture	Screw
Pros	Minimal physeal damage · growth breaks non-absorbable sutures · **biomechanically superior** in type II (OB)	Easier · earlier motion
Cons	Technically demanding	Needs large fragment · impingement · iatrogenic comminution · physeal damage · **remove only if malpositioned**

Post-op cast in extension 7–10 days (variable) → early controlled ROM



Screw fixation

6 Complications

Outcomes (laxity, arthrofibrosis)

Loss of motion very common, esp. **extension** (fragment impinges in notch)

Arthrofibrosis **more common after surgery** · adolescents / adults → counsel stiffness · **concomitant ACL injury = independent risk factor for return to OR for MUA**
✗ Wrong options: nonunion · physeal closure · compartment syndrome

ACL laxity **38–100%**, more after surgery · **common but rarely symptomatic** (even after anatomic fixation — ligament stretched at injury) · ACL reconstruction later 15–25%

Growth arrest iatrogenic at surgery

Prognosis good · **85% return to prior sport**

– – Active space – –

How it's asked

"Which may be treated with closed reduction in extension?"	→ **McKeever type II** ✗ Wrong options: SH II distal femur · type III · SH IV proximal tibia · tibial tubercle type IV
"12-yr type II, tense effusion — next?"	→ **closed reduction; cast if successful, surgery if not** ✗ Wrong options: MRI · immobilizer
"7-yr successful closed reduction, +Lachman — next?"	→ **immobilization in near full extension + frequent monitoring** ✗ Wrong options: arthroscopic fixation · ACL reconstruction
"14-yr gymnast, moderately displaced eminence?"	→ **long-leg cast after successful fluoroscopic closed reduction** ✗ Wrong options: 1–2 wk cast · ACL reconstruction
"9-yr bike fall — type and next step?"	→ **type II; closed reduction + long-leg cast in extension** ✗ Wrong options: 90° flexion
"Failed reduction — structure entrapped under fragment?"	→ **anterior horn of medial meniscus** ✗ Wrong options: lateral meniscus · ACL · fat pad
"More likely with type II tibial spine fracture?"	→ **meniscal entrapment** ✗ Wrong options: compartment syndrome · growth arrest
"14-yr obese, surgery planned — interposed structure?"	→ **medial meniscus** ✗ Wrong options: intermeniscal ligament
"12-yr soccer, failed ER reduction — block?"	→ **medial meniscus**
"12-yr failed closed reduction?"	→ **arthroscopic reduction + fixation of tibial eminence** ✗ Wrong options: ACL reconstruction
"12-yr displaced eminence, Lachman lax?"	→ **anatomic reduction + internal fixation** ✗ Wrong options: casting · ACL reconstruction
"Adult equivalent of tibial spine fracture?"	→ **ACL tear**
"22-yr wakeboard, bony fragment on MRI?"	→ **tibial avulsion of ACL** ✗ Wrong options: midsubstance ACL
"19-yr arthroscopic suture fixation — likely complication?"	→ **arthrofibrosis** ✗ Wrong options: hardware removal · SONK
"ACL laxity after eminence fracture?"	→ **common, infrequently symptomatic**

"Concomitant ACL injury + all-epiphyseal reconstruction – outcome?"	**→ return to OR for MUA** ✗ Wrong options: physeal closure · nonunion	– – Active space – –

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TIBIAL SHAFT FRACTURES - PEDIATRIC

– Pediatrics · 107 / 112 –

1 Basics

Epidemiology	**3rd most common pediatric long-bone fracture** · **15% of pediatric fractures** · boys > girls · **mean age 8 yr** · **39% mid-diaphysis** · fibula fractured in 30%
NAT	**2nd most common bone fractured in non-accidental trauma** · suspect abuse in non-ambulatory child
Anatomy	triangular, apex anterior · anteromedial border subcutaneous · distal flare mostly cancellous · **posterior tibial a.** → nutrient + periosteal vessels · **anterior tibial a.** at risk through interosseous membrane · fibula carries 6–17% of load

2 Mechanism

Adolescent	**pedestrian vs vehicle (50%)** · direct blow
Toddler	low-energy twist / fall → spiral or oblique ("toddler's fracture")

3 Presentation

Findings	pain, bruising · **refuses to bear weight** · tenderness · **pain on ankle dorsiflexion** · always think **compartment syndrome**

4 Diagnosis & Classification

Key radiology findings

X-ray	**AP + lateral tib/fib** · **image ipsilateral knee + ankle** · ± contralateral · toddler's fracture may be **normal initially** → repeat at **2 wk** for callus / periosteal reaction
CT	physeal / intra-articular extension (distal third) · pathologic lesion
MRI	pathologic / stress fracture · occult fracture
Bone scan	occult fracture

Pattern	Description
Incomplete	**Greenstick** of tibia ± fibula
Complete	Tibia ± fibula fracture or plastic deformation
Toddler's fracture	**Nondisplaced spiral / oblique tibia, intact fibula, age <2.5 yr**

Toddler's fracture

Typical afebrile · normal WBC / ESR / CRP · distal tibial tenderness · X-rays often negative · usually unwitnessed

Treatment **long leg cast** · heals **3–4 wk**
✗ Wrong options: MRI · aspiration · vitamin D / metabolic labs · child services · chromosomal analysis

Not osteomyelitis / subperiosteal abscess (afebrile, normal labs) · NOF · NAT (age + story fit)



Toddler's fracture

5 Management

Operative criteria

Acceptable alignment translation **50%** (Orthobullets, OB: <50% = >50% cortical overlap) · shortening **<1 cm** · angulation **<5–10°** sagittal + coronal (Orthobullets) · **<10°** (OB210163) · **<5°** (OB218688) – read the options

Method	Indication
Long leg cast	Toddler's · greenstick
Closed reduction + long leg cast	**Most traumatic fractures** with acceptable reduction
External fixator	Extensive soft-tissue injury · length unstable · polytrauma
Flexible IM nails	Open / closed in skeletally immature · multiple long bones · **floating knee**
Percutaneous pins	Noncomminuted unstable oblique (± cast)
Rigid IM nail	**At / near skeletal maturity** (✗ Wrong options: open physes)
Plate	Physeal / articular extension · length unstable · nonunion / malunion

Reasons to operate (Ho) open fracture · obesity · compartment concern · segmental · polytrauma · floating knee · failed non-op

Treatment - casting vs flexible nailing

Casting technique sedation / GA · cast **to groin, knee flexed 30°**, molded · ± bivalve · loss of reduction → **cast wedging**

– – Active space – –

Early weight bearing — WB at **1 wk** in long leg cast (knee ~10°) is **safe and may speed healing** (Jenkins: +0.3 days healing per day delayed)
✗ Wrong options: NWB 4–6 wk · short leg cast · flexible nails first

Distal tibia fractures — **high secondary displacement → follow-up within 1–2 wk** · most non-op
✗ Wrong options: >1 month

Casting vs flexible IMN — only real difference: **casting = prolonged immobilization** (>3 months) · **compartment syndrome more with IMN** (up to 20%) · union, malunion, LLD similar

Elastic nails — **period of casting / splint + NWB** after (nails don't control rotation, bending) · **no 50-kg contraindication** (unlike femur) · open fractures OK
✗ Wrong options: soft dressing + WBAT · added ex-fix

Complication	Rate
Ex-fix nonunion	~2% · **refracture** · pin tract infection
Flexible nail nonunion	~10%

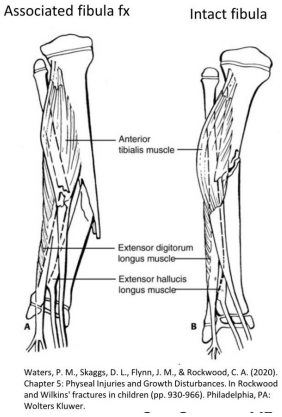


Long leg cast after reduction

Valgus deformity

Fibula	Drift
Intact	**Varus**
Fractured (both bones)	**Valgus + recurvatum (procurvatum, OB)**

Why valgus — lateral compartment pull + pes anserinus · **gentle varus mold** (too much → varus malunion) · residual → wedging



Tib-fib fracture drifts into valgus

6 Complications

– – Active space – –

Compartment syndrome

Compartment syndrome	less common than adults · open + closed · signs in kids: **3 As — analgesia ↑, anxiety, agitation** · firm leg with pulses → **4-compartment fasciotomy** ✗ Wrong options: more narcotics · nerve block · PCA · observe
LLD	age <10 · comminution → overgrowth · pins → growth arrest, **recurvatum from tibial tubercle arrest**
Angular deformity	complex · **valgus + apex posterior** · physeal extension · osteotomy for rotational malunion / symptomatic
Nonunion	**25% of open fractures** · ↑ age, wound severity · hypertrophic → graft + rigid fixation
Healing	**3–4 wk toddler's** · **6–8 wk** others

How it's asked

"23-mo refuses to bear weight, normal labs, X-rays negative?"	→ **long leg cast** ✗ Wrong options: MRI pelvis · knee aspiration · vitamin D
"3-yr won't bear weight, afebrile, normal CRP/ESR — diagnosis?"	→ **nondisplaced oblique / spiral tibia, intact fibula** ✗ Wrong options: osteomyelitis · NAT · NOF
"2-yr 11-mo fall, spiral distal tibia?"	→ **long leg cast** ✗ Wrong options: child services · MRI
"Casting vs flexible IMN — casting has more?"	→ **prolonged immobilization** ✗ Wrong options: compartment syndrome · nonunion · LLD
"Titanium elastic nailing — TRUE?"	→ **may need a period of casting** ✗ Wrong options: higher union · >50 kg contraindicated · avoid in open
"13-yr ATV, elastic nails — post-op?"	→ **NWB in splint or cast** ✗ Wrong options: soft bandage + WBAT
"7-yr tree fall — maximize healing?"	→ **closed reduction + long leg cast, WB at 1 wk** ✗ Wrong options: NWB 4–6 wk · short leg cast · nailing
"11-yr distal tibia, long leg cast — TRUE?"	→ **follow-up within 1–2 wk (secondary displacement)** ✗ Wrong options: most need surgery · rigid nail
"9-yr tib + fib fracture in cast — likely deformity?"	→ **valgus** ✗ Wrong options: varus · procurvatum

– – Active space – –

"3-yr, 30 h after fracture, firm leg, pulses present, pain despite morphine?"	**→ 4-compartment fasciotomy** ✗ Wrong options: nerve block · PCA · observe

TIBIAL TUBERCLE FRACTURE

– Pediatrics · 108 / 112 –

1 Basics

Epidemiology	<1% of pediatric fractures · **males >> females** · **12–15 yr** (near skeletal maturity) · basketball, football, sprinting, high jump · growth hormone weakens physis
Ossification	**2 centers**: primary (proximal tibial physis) + secondary (tubercle apophysis, patellar tendon insertion)
Physeal closure	**posterior → anterior, proximal → distal** · **tubercle last to fuse** → vulnerable in older adolescents
Vessel	**recurrent anterior tibial artery** at **lateral border of tubercle** → lacerated → retracts under fascia → **anterior compartment syndrome**

2 Mechanism

Mechanism

Mechanism	**knee flexion + quadriceps contraction** · concentric quad contraction at **jump take-off** · eccentric quad contraction in forced knee flexion ✗ Wrong options: hamstring · gastrocnemius · direct blow

3 Presentation

Findings	sudden pain at jump / sprint start · can't walk · effusion / hemarthrosis (type III) · tubercle tenderness · **extensor lag / can't straight-leg raise** (II–III; retinaculum may allow some extension) · **patella alta** · palpable defect · watch **anterior compartment**
Associated	**compartment syndrome** · **meniscal tears with type III** · cruciate / collateral / patellar injuries uncommon (OB7708, OB210101) · patellar / quadriceps tendon avulsion, ligament tears to assess (OB1487)

4 Diagnosis & Classification

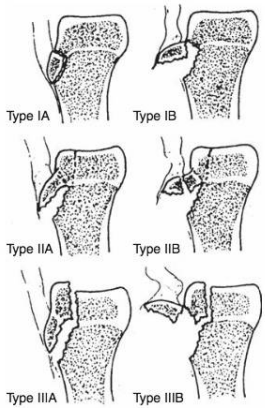
Key radiology findings

X-ray	AP + lateral · **internal rotation view** (tubercle in profile) · contralateral comparison (younger) · widening / hinging of apophysis · **patella alta** · anterior swelling only → type V sleeve
CT	intra-articular / posterior extension · arteriogram if popliteal concern ✗ Wrong options: delay fasciotomy
MRI	not routine · nondisplaced type II extension or type V

– – Active space – –

Ogden classification (modified Watson-Jones)

Type	Fracture
I	Secondary ossification center near patellar tendon insertion
II	Propagates proximally **between primary and secondary centers**
III (most common)	**Coronal**, extends posteriorly **across primary center → intra-articular**
IV	**Through entire proximal tibial physis**
V	**Periosteal sleeve avulsion** of extensor mechanism from secondary center
Modifier	**A** nondisplaced · **B** displaced



Ogden classification

5 Management

Operative criteria

Situation	Treatment
Type I / displacement **<2 mm** / acceptable after closed reduction	**Long leg cast in extension 6 wk**
Displaced, types II–IV	**ORIF** ± arthrotomy / arthroscopy (joint surface, meniscus)
Type V	**Soft-tissue (periosteal) repair**
Compartment syndrome	**Emergent fasciotomy + ORIF** (± prophylactic fasciotomy)

Treatment - ORIF

ORIF	midline incision · **remove interposed periosteum** (blocks closed reduction) · anatomic reduction · **4.0 cancellous partially threaded screws** (or headless compression) · smooth K-wires if >3 yr from maturity · cannulated screws **better than CRPP** (OB) · NWB long leg cast / brace **4–6 wk** · sport **≥3 months**
Arthrotomy / arthroscopy	intra-articular (type III) → visualize joint, evacuate hematoma, meniscus repair
Soft-tissue repair (V)	heavy suture of periosteum to secondary center · cast **8–10 wk**

Traps

✗ Wrong options: long leg cast for displaced · patellar tendon bearing cast · CRPP · ligament reconstruction · spanning ex-fix (temporizing only) · patellar tendon repair (tendon intact, deforming) · fragment excision · MRI / CT if X-ray clear



Screw fixation

6 Complications

Compartment syndrome - anterior tibial recurrent artery

Rate

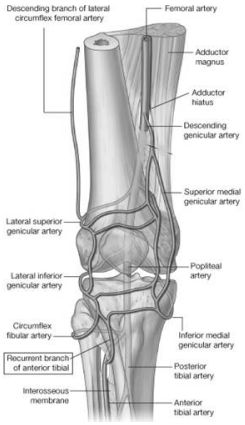
4% (Orthobullets, OB) · 4–5% (OB) · up to **10%** (OB) — read the options · high with **type III**

Vessel

anterior tibial recurrent artery
✗ Wrong options: peroneal · posterior tibial · middle geniculate (ACL) · medial / lateral inferior geniculate · popliteal (knee dislocation)

Signs

rising analgesic need, anxiety, distress, swelling overnight → **emergent fasciotomies + ORIF**
✗ Wrong options: observe · vascular exploration · ORIF alone



Recurrent branch of anterior tibial artery

– – Active space – –

Other stiffness · refracture

Complication	Fact
Bursitis	**Most common after surgery** · screw prominence → removal
Recurvatum	Anterior arrest, posterior growth → ↓ slope · **more common than LLD** (LLD low – age) · unusual (near maturity)
Popliteal artery injury	Over distal metaphyseal fragment (physeal types)

How it's asked

"15-yr ski jump, type III tubercle – watch for?"	→ **compartment syndrome** ✗ Wrong options: ACL disruption · popliteal artery · patella baja
"Compartment syndrome after tubercle fracture – artery?"	→ **anterior tibial recurrent** ✗ Wrong options: peroneal · middle geniculate · saphenous
"15-yr ramp jump, intra-articular tubercle, ↑ leg pain in 3 h?"	→ **anterior tibial recurrent artery** ✗ Wrong options: inferior geniculates
"220-lb basketball, tubercle fracture – complication?"	→ **compartment syndrome** ✗ Wrong options: LLD · varus · procurvatum
"Displaced type III, noncontact – more likely?"	→ **compartment syndrome** ✗ Wrong options: ACL + MCL · patella dislocation · peroneal palsy
"13-yr ↑ analgesia – vessel in figure?"	→ **recurrent anterior tibial artery (lateral border of tubercle)**
"13-yr admitted for morning ORIF, ↑ narcotics, swollen leg?"	→ **emergent fasciotomies + ORIF** ✗ Wrong options: ORIF only · observe · vascular exploration
"12-yr basketball, anterior tibial pain at 6 h – artery?"	→ **recurrent branch of anterior tibial artery** ✗ Wrong options: popliteal
"14-yr volleyball, type III, anxious overnight?"	→ **recurrent anterior tibial artery**
"14-yr can't straight-leg raise, displaced tubercle?"	→ **ORIF** ✗ Wrong options: long leg cast · PTB cast · CRPP
"13-yr displaced intra-articular tubercle?"	→ **ORIF** ✗ Wrong options: MRI · cast · fasciotomy
"10-yr cheerleader, palpable defect, patella elevated, ligament retracted >2 cm?"	→ **ORIF** ✗ Wrong options: MRI · CT · excision · cast
"14-yr CT shows articular extension – definitive?"	→ **ORIF** (± arthrotomy) ✗ Wrong options: spanning ex-fix · percutaneous screws · tendon repair

"17-yr can't SLR, screw ORIF — which image?"	→ **type III tibial tubercle fracture** ✗ Wrong options: quadriceps rupture · Osgood-Schlatter · ACL tear
"Mechanism of tubercle avulsion?"	→ **knee flexion + quadriceps contraction**

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TILLAUX FRACTURES

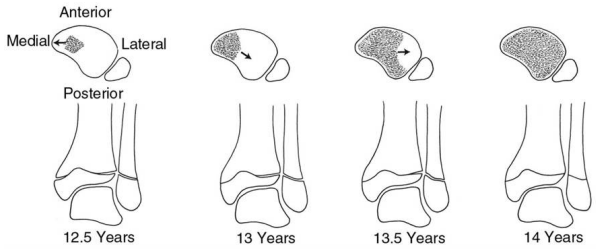
– Pediatrics · 109 / 112 –

1 Basics

Definition	**Salter-Harris III** of the **anterolateral distal tibial epiphysis** · **transitional** fracture (physis closing) · entirely epiphyseal
Epidemiology	**3–5% of pediatric ankle fractures** · **more common in girls** · **12–14 yr** (12–15 OB) · within ~1 yr of complete distal tibial closure (18-month transitional window) · **older than triplane** (triplane ~10–12, OB)
Associated	distal fibula fracture (SH I/II) · ipsilateral tibial shaft fracture

Distal tibial physeal closure pattern

Order	**central → anteromedial → posteromedial → lateral** (Orthobullets, TO, OB) · central → posterior → medial → anterolateral (OB3880) · central → medial → lateral (OB) · central → anteromedial → posterolateral (TO17124838)
Last to fuse	**anterolateral** (TO) → open lateral physis avulsed by AITFL ✗ Wrong options: anteromedial · posteromedial · posterolateral
Distal tibial physis	35–40% of tibial growth · 15–20% of lower-limb growth · 3–4 mm/yr · growth until **14 girls / 16 boys**



Pattern of distal tibial physeal closure

Anatomy - AITFL avulsion

Ligament	**anterior inferior tibiofibular ligament (AITFL)** · Chaput tubercle (anterolateral tibial epiphysis) → **Wagstaffe tubercle** (anterior distal fibula) · part of **syndesmosis** (+ PITFL) ✗ Wrong options: PITFL · ATFL (ankle sprain) · PTFL · CFL · deltoid
Same ligament injured in	**high ankle sprain** (syndesmosis) ✗ Wrong options: low ankle sprain (ATFL) · Weber A (below syndesmosis) · Gosselin (V-shaped plafond fracture) · subtalar dislocation
Axial CT	displaced fragment = **anterolateral region (AITFL)** ✗ Wrong options: posterolateral · medial malleolus



AITFL on the Tillaux fragment

2 Mechanism

Mechanism

Mechanism	**supination / eversion – external rotation** → AITFL tension → anterolateral avulsion ✗ Wrong options: pronation/eversion–ER (SH II lateral tibia + suprasyndesmotic fibula) · supination–plantarflexion (posterior tibia) · axial compression (SH V) · supination–inversion (ankle sprain)
vs triplane	**no coronal fracture in posterior metaphysis** = Tillaux

3 Presentation

Findings	pain · can't bear weight · slight swelling · **focal anterolateral joint-line tenderness** · deformity rare (fibula blocks displacement)

4 Diagnosis & Classification

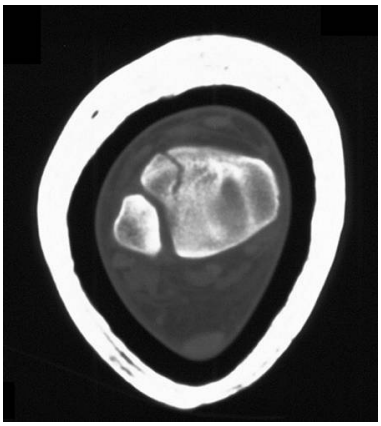
Key radiology findings

X-ray	AP · lateral · **mortise = best view** · SH III anterolateral epiphysis
CT	fracture pattern · **articular displacement** · intramalleolar / medial variants · **after closed reduction → CT for residual step-off** ✗ Wrong options: discharge without CT



Tillaux on mortise view

– – Active space – –



Axial CT

5 Management

CT & surgical indications (>2 mm step-off)

Operative criteria

After closed reduction	Treatment
<2 mm displacement (rare)	**Long leg cast** 3–4 wk → short leg cast / boot 2–4 wk
>2 mm residual step-off	**CRPP or ORIF**

Not indications — age, sex, any step-off <2 mm
✗ Wrong options: age >12 · male · any step-off

Closed reduction — **internal rotation** of foot (or dorsiflex pronated foot then IR) · long leg cast controls **rotation** · early X-rays

CRPP — K-wire joystick · fluoro / arthrogram · K-wire or cannulated screw · **crossing physis OK** (little growth left) · good outcome if residual **<2.5 mm**

ORIF — **anterolateral approach** · reduce with **internal rotation** · intraepiphyseal or **transphyseal** wires / screws · long leg cast 3–4 wk → short leg walking cast 2 wk

Post-reduction care — low compartment risk · long leg > short leg cast
✗ Wrong options: admit + splint

6 Complications

Growth arrest — **rare** (physis already closing) · ↓ with anatomic reduction

Early arthritis — ↑ with articular displacement · degenerative change as early as 4 yr

X vs Y

	Tillaux	Triplane
Salter-Harris	**III**	**IV** (as described by OB)
Age	**Older**, nearer maturity	Younger (~10–12, OB)
Posterior metaphyseal coronal fracture	**Absent**	Present

How it's asked

"Juvenile Tillaux — avulsed by?"	→ **anterior inferior tibiofibular ligament** ✗ Wrong options: PITFL · ATFL · CFL
"Axial CT — Tillaux fragment region + ligament?"	→ **anterolateral region; AITFL** ✗ Wrong options: posterolateral · deltoid
"Ligament displacing Tillaux fragment?"	→ **anterior tibiofibular** ✗ Wrong options: deltoid · calcaneofibular
"14-yr girl, displaced fragment — attached ligament?"	→ **AITFL** ✗ Wrong options: ATFL · long plantar
"14-yr eversion injury — ligament in figure?"	→ **AITFL** ✗ Wrong options: PITFL · PTFL · CFL
"Tillaux — same ligament involved in?"	→ **high ankle sprain** ✗ Wrong options: low ankle sprain · Weber A · Gosselin
"13-yr SH III anterolateral distal tibia — mechanism?"	→ **supination/eversion–external rotation** ✗ Wrong options: pronation/eversion–ER · axial compression
"12-yr after closed reduction — next to optimize outcome?"	→ **CT scan of ankle** ✗ Wrong options: discharge in cast · pinning before CT
"Strongest indication for fixation after reduction?"	→ **articular step-off >2 mm** ✗ Wrong options: age · male sex · any step-off
"Last part of distal tibial physis to fuse?"	→ **anterolateral** (TO)
"Correct pattern of distal tibial physeal closure?"	→ **central → anteromedial → posteromedial → lateral** (TO)

– – Active space – –

TRANSIENT SYNOVITIS OF HIP

– Pediatrics · 110 / 112 –

1 Basics

Definition nonspecific inflammation + hypertrophy of synovium · **most common cause of pediatric hip pain** · diagnosis of exclusion vs **septic arthritis**

Epidemiology **3% of children 3–10 yr** · peak **4–8 yr** · **M:F 2:1** · hip most common joint · **recurrence up to 20%**

Diagnosis & preceding viral illness

Cause largely unknown · **viral upper respiratory infection** (most cited) · bacterial (**poststreptococcal toxic synovitis**) · trauma · ↑ interferon · allergy

Preceded by **infectious respiratory illness**
✗ Wrong options: hematogenous bacterial inoculation (septic arthritis) · hypothyroidism (SCFE) · secondhand smoke (Perthes) · LLD (painless limp)

Genes & lab markers

Lab	Transient synovitis
WBC	Normal / slightly ↑
CRP	**<20 mg/L – most important to RULE OUT septic arthritis** · CRP >20 = independent risk factor for sepsis
ESR	Usually <20 mm/h (can be ↑ – 35 with normal CRP + rapid improvement still synovitis)
Synovial WBC	Near normal · **>50,000 → treat as septic**

3 Presentation

History recent URI or trauma · mild / no fever · acute or **insidious groin / thigh pain** · refuses to bear weight → **better later in day** (limps by evening) · muscle spasm

Exam hip in **flexion, abduction, external rotation** (lowest intracapsular pressure) · not toxic · **↓ internal rotation = most sensitive** ROM sign · painless arc → synovitis > sepsis · log-roll guarding · spine + knee non-tender · toe-walking / cavus / claw toes → neurological limp

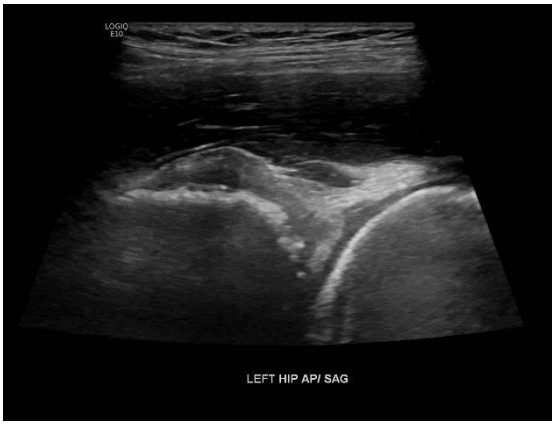
4 Diagnosis & Classification

Key radiology findings

X-ray AP + frog-leg / lateral · ± spine · **usually normal**

US effusion (accurate) · synovial thickening · **can't separate synovitis from sepsis**, but **no effusion → infection unlikely**

MRI	myositis / osteomyelitis suspicion · markedly ↑ labs
Aspiration	if septic suspicion (clinical / Kocher criteria)
Differential	septic hip · osteomyelitis (point tenderness, swelling, ↑ markers) · **SCFE** (10–16 yr, visible on AP) · **Perthes** (visible on X-ray) · leukemia (normal CBC rules out) · JIA (rarely hip) · fracture (no trauma)



Hip effusion on ultrasound



Normal pediatric pelvis

5 Management

Low septic suspicion	**IV / PO NSAIDs + close observation 24 h** · minimize walking · ± traction · **improves on NSAIDs → synovitis**
Course	**marked improvement 24–48 h** · resolves **<1 wk** (Orthobullets treatment) · 1–2 wk (prognosis)

6 Complications

Recurrence	**~20%**
Perthes	0–10% (controversial; most series no sequelae)

– – Active space – –

X vs Y

	Transient synovitis	Septic arthritis
Fever	Mild / none	Higher
CRP	**<20 mg/L**	>20
Course	Rapid improvement on NSAIDs	Doesn't improve
Synovial WBC	Near normal	>50,000

How it's asked

"8-yr hip pain, CRP 1.8, ESR 35, walking after 24 h ketorolac?"	→ **transient synovitis** ✗ Wrong options: septic arthritis · leukemia · JIA
"5-yr recent URI, CRP 9, ESR 20, limited IR?"	→ **transient synovitis** ✗ Wrong options: septic · SCFE · Perthes
"5-yr limp, normal labs – what preceded?"	→ **infectious respiratory illness** ✗ Wrong options: bacterial inoculation · hypothyroidism · secondhand smoke
"2-yr improving, afebrile, normal CRP, US effusion?"	→ **toxic synovitis** ✗ Wrong options: septic · osteomyelitis · leukemia

TRAUMATIC HIP DISLOCATION - PEDIATRIC

– Pediatrics · 111 / 112 –

1 Basics

Epidemiology — **80% posterior** · **more common than hip fracture** in children

Mechanism — **<10 yr: low-energy** sport / trip and fall (lax ligaments, cartilaginous acetabulum; <6 yr minimal force, OB) · **>10 yr: high-energy** (MVA)

Associated — acetabular fractures **less common than adults** (posterior wall most common) · femoral head / neck / proximal femoral physis fracture · missed ipsilateral femoral shaft fracture (OB) · **AVN ↑ if not reduced within 6 h**

3 Presentation

Direction	Limb position
Posterior (most common)	**Slight flexion, adduction, internal rotation** · LLD (large posterior wall fragment → short without malalignment)
Anterior	Slight flexion, **abduction, external rotation**
Inferior	External rotation + extension

Nerves — **sciatic or gluteal** palsy (rare) · usually resolves after reduction

4 Diagnosis & Classification

Key radiology findings

X-ray — **AP pelvis** (most diagnosed) · **lateral** (anterior vs posterior; check femoral neck before reduction) · **post-reduction films → joint incongruity / nonconcentric reduction**

MRI — **study of choice** for abnormal post-reduction films (joint widening) · less radiation · **best for interposed soft tissue, entrapped labrum / capsule**

CT — **2nd choice** · **osteochondral fragments** (older children) · soft-tissue interposition hard to see

– – Active space – –

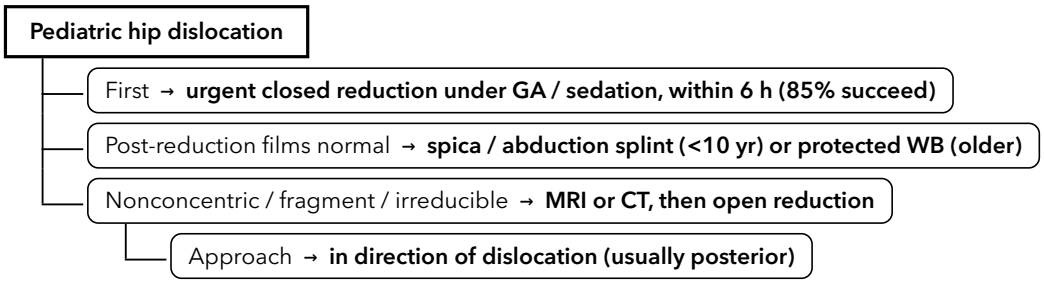


Nonconcentric reduction



Entrapped labrum — subtle asymmetry

5 Management



Closed reduction — **adequate anesthesia mandatory** (avoid displacing unrecognized **proximal femoral epiphyseal fracture**) · **under fluoroscopy** (epiphyseolysis risk) · traction in flexion + gentle rotation · test stability before waking · post-reduction imaging

Aftercare — spica / bed rest with abduction splint **4 wk** if <10 yr · brace + **6–12 wk** protected WB in older

Operative criteria

Open reduction indications
Nonconcentric reduction
Intra-articular fragment
Unstable acetabular rim fracture · femoral head / neck fracture
Irreducible closed

Non-concentric reduction - arthrotomy

Persistent incongruity soft-tissue interposition (**torn labrum + lateral acetabular apophysis**) · osteochondral fragment → **hip arthrotomy via POSTERIOR approach** (posterior structures already torn)
✗ Wrong options: anterior approach · repeat closed reduction · traction 1 month · WBAT

6 Complications

AVN with delayed reduction

AVN **3–15%** (Orthobullets) · 0–43% (OB) · ↓ under age 5 · less than adults (without neck fracture) · **only complication linked to delay in reduction** · >24 h → 66% (OB)
✗ Wrong options: coxa magna · redislocation · chondrolysis · HO (not time-dependent)

Complication	Fact
Coxa magna	**20%**, radiographic, no functional limit
Redislocation	Rare · prolonged immobilization → capsulorrhaphy if recurrent
Nerve	Sciatic / gluteal, usually resolves
Other (OB)	Femoral head deformity · premature physeal fusion · degenerative arthritis · chondrolysis

Prognosis good when treated promptly · most return to high-demand activity

How it's asked

"10-yr post-reduction CT — joint widened, fragment?"	→ **hip arthrotomy via posterior approach** ✗ Wrong options: anterior approach · repeat closed reduction · traction
"3-yr MVC hip dislocation — delayed concentric reduction ↑ risk of?"	→ **avascular necrosis** ✗ Wrong options: coxa magna · recurrent dislocation · chondrolysis · HO

– – Active space – –

TRIPLANE FRACTURES

– Pediatrics · 112 / 112 –

1 Basics

Definition complex **Salter-Harris IV** pattern in **3 planes** · **transitional** fracture (physis closing) · 2, 3 or 4 parts

Epidemiology **5–15% of pediatric ankle fractures** · **males** · ages 10–17 · **mean 13 yr** · **younger than Tillaux**

Physeal closure central → anteromedial → posteromedial → lateral (Orthobullets) · central → medial → lateral (OB) · 18-month transitional period

Associated **fibula fracture 50%** (spiral, above physis · unstable after tibial fixation → **fix fibula**) · ipsilateral tibial shaft fracture

2 Mechanism

Mechanism

Pattern	Mechanism
Lateral triplane	**Supination–external rotation** (like Tillaux)
Medial triplane	**Adduction**

Asked **external rotation**
✗ Wrong options: internal rotation · supination–adduction · pronation–abduction · axial load

3 Presentation

Findings pain · can't bear weight · swelling · focal tenderness · deformity

4 Diagnosis & Classification

Radiographic fracture pattern

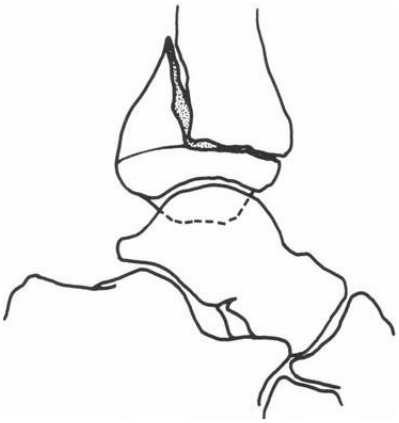
Plane	Structure	Seen on
Sagittal	**Epiphysis** (lateral)	**AP X-ray → SH III**
Axial (transverse)	Physis	–
Coronal	**Posterior metaphysis**	**Lateral X-ray → SH II**

Overall SH IV pattern, but **no true SH IV component** on either view
✗ Wrong options: SH II on AP · SH III on lateral

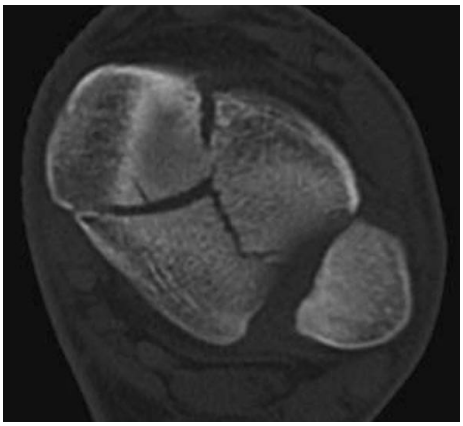
von Laer SH III epiphyseal component on AP depends only on **physeal maturity**, not mechanism · biplane vs triplane type I (metaphyseal line ends in physis) vs type II (into epiphysis + joint)

Key radiology findings

X-ray	AP · lateral · **mortise = best for displacement**
CT	usually **required** (pattern + articular congruity) · after closed reduction · all 3 planes · **"Mercedes-Benz sign"** · 4-part only seen on CT



Triplane on AP and lateral



Mercedes-Benz sign

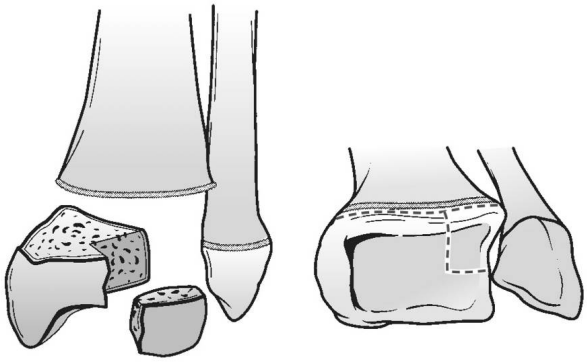
Classification by parts

Parts	Fragments
2-part	(1) anterolateral + posterior epiphysis with posterior metaphyseal fragment · (2) anteromedial epiphysis with rest of tibia
3-part	(1) anterolateral epiphysis · (2) posterior epiphysis + posterior metaphyseal fragment · (3) anteromedial epiphysis + rest of tibia
4-part	Comminuted · **CT only** separates from 3-part

– – Active space – –

Classification by pattern

Pattern	Epiphysis	Metaphysis	Note
Lateral	**Sagittal**	**Coronal**	**Most common** · looks like Tillaux on AP → SH II on lateral distinguishes
Medial	Coronal	Sagittal	–
Intramalleolar	I – intra-articular, weight-bearing · II – intra-articular, non-weight-bearing · III – extra-articular	–	–



3-part triplane

5 Management

Surgical indications & articular gap outcomes

Operative criteria

Displacement	Treatment
<2 mm	**Closed reduction + long leg cast 3–4 wk** → short leg cast / boot 2–4 wk · ideal for **2-part**
>2 mm (Orthobullets) · >2.5 mm (OB220849)	**CRPP or ORIF**

Closed reduction — **reduce fibula first** · lateral → **internal rotation** · medial → **eversion** (closed) / external rotation (operative) · post-reduction CT

ORIF — anterolateral approach (lateral) · anteromedial (medial) · arthroscopic assist described · K-wires / cannulated screws · **epiphyseal screws parallel to physis** · metaphyseal screw if large · long leg cast 3–4 wk → short leg walking cast 2 wk · hardware irritation, transient neuropathy

Why operate — **restore articular congruity → ↓ posttraumatic arthritis** (near maturity)
✗ Wrong options: avoid growth arrest · osteonecrosis · rotation deformity · tarsal tunnel

Residual gap — **>2.5 mm → worse functional outcome**
✗ Wrong options: observation fine to 5 mm · bracing / WB protocol matters more · arthritis risk negligible · physeal closure dominates

6 Complications

Growth arrest	**7–21%** · ↑ **pronation–abduction** vs SER · ↑ residual displacement · usually insignificant · watch if **>2 yr growth left**
Ankle pain / degeneration	↑ with **step-off >2 mm**

X vs Y

	Triplane	**Tillaux**
Age	**Younger** (mean 13)	Older
Sex	Males	Girls (Orthobullets)
Incidence (peds ankle)	5–15%	3–5%
Lateral X-ray	**Coronal posterior metaphyseal fracture (SH II)**	None – entirely epiphyseal
Surgery	>2 mm	>2 mm

How it's asked

"15-yr skateboarder, triplane – mechanism?"	→ **external rotation** ✗ Wrong options: supination–adduction · axial load · pronation–abduction
"Most common triplane – fracture lines and views?"	→ **sagittal epiphyseal on AP + coronal metaphyseal on lateral**
"Most common triplane – TRUE?"	→ **SH III component seen on AP** ✗ Wrong options: SH II on AP · SH IV on either view
"14-yr failed closed reduction – main reason for surgery?"	→ **↓ posttraumatic arthritis** ✗ Wrong options: avoid growth arrest · osteonecrosis
"13-yr post-op CT 3 mm gap – counsel?"	→ **residual gap >2.5 mm → worse function** ✗ Wrong options: observe to 5 mm · arthritis negligible

– – Active space – –

CLEFT FOOT (LOBSTER FOOT)

– Pediatrics · TO notes –

1 Basics

Definition	**partial adactyly with central cleft** of the foot · "lobster foot" · split foot (TO)
Key features	**1st and 5th rays usually present** · variable cleft depth (TO)

4 Diagnosis & Classification

Blodgett and Borisch classification (TO)

Type	Metatarsals
1	**Five** metatarsals, normal
2	Five **hypoplastic** metatarsals
3	**Four** metatarsals
4	**Three** metatarsals
5	**Two** metatarsals
6	**Single** metatarsal

5 Management

Principle	**convert the cleft into a syndactyly** for better cosmesis and shoe wear (TO)
Wood and Shook technique (TO)	**alternating rectangular skin flaps** · excision of fibrous tissue · **closing-wedge osteotomies** · **crossed K-wire** fixation

How it's asked

"Principle of cleft foot surgery?"	→ **convert cleft into syndactyly** (TO)
"Cleft foot – rays usually preserved?"	→ **1st and 5th** (TO)